Tone in Yongning Na

Lexical tones and morphotonology

Second edition

Alexis Michaud

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À mon père

Contents

Preface to the second edition

> That looked good. Yes, that looked very good. In fact it went on looking better and better, straight along — until by-and-by it grew into positive *proof*. And then he put the matter at once out of his mind, for he had a private instinct that a proof once established is better left so.
>
> (Mark Twain, *The man that corrupted Hadleyburg*, 1899.)

I have not put this book out of my mind – as a settled, authoritative account – since its publication in April 2017. Data collection and analysis have continued, leading to improvements in the dictionary, which recently reached version 2 (Michaud et al. 2024), as well as in the online collection of audio and video materials, where the number of fully transcribed and translated narratives reached twenty-nine in 2024. In the course of continuing documentation and description work, the analysis presented in the book has undergone further verification. The findings have held up well, and the volume's core analytical content remains unchanged. This second edition is a wonderful opportunity to incorporate some refinements.

To facilitate navigation between Na texts and their linguistic analysis, this edition introduces one-click links from the examples discussed in the book to the online corpus of Yongning Na hosted in the Pangloss Collection. This enhancement promotes direct consultation of the primary data, encouraging readers to engage more deeply with the material.

The discussion of intonation was expanded, aiming to provide a cogent, detailed argument about this important and delicate topic (in Chapters 8 and 9). In the first edition of this book, there was no independent chapter on intonation. Observations about Na intonation were contained within a chapter entitled *From surface phonological tone to phonetic realization*, which covered both intonation and the phonetic implementation of tone. While this title was not technically incorrect – given the broad definition of intonation adopted here, whereby intonation encompasses all aspects of the concrete phonetic realization of speech that are not directly predictable from phonemic contrasts –, it was nonetheless misleading. The phrasing suggested that the entire chapter would be devoted to the phonetic implementation of tonal sequences, whereas its scope was broader.

The decision to avoid the term *intonation* in the chapter title reflected a pragmatic concern about the work's reception. In a scholarly landscape where the autosegmental-metrical approach enjoys widespread acceptance, including in the field of language documentation and description (Himmelmann & Ladd 2008), I wished to contribute to a broader scholarly dialogue by raising fundamental questions anew and examining several frameworks for intonation analysis. The aim is not only to explore diverse perspectives but also to make a case for an alternative approach, arguing that the autosegmental-metrical model is not the most suitable for the study of intonation in tonal languages. However, I was aware that a critical engagement with the mainstream model could be perceived as contentious, potentially jeopardizing a manuscript's acceptance for publication. A degree of caution thus seemed advisable.

The second edition of this book provides an opportunity to devote two full chapters to intonation. Chapter 8 engages with the theoretical landscape of intonation studies, establishing theoretical and typological foundations through an open discussion of the key issues at stake. Chapter 9 turns to the specifics of Yongning Na, presenting the detailed empirical work delving into its intonation system as well as issues of phonetic implementation.

The journey of this book since its initial publication has been enriched by feedback from its readers. Through personal correspondence and published reviews (Alves 2017, Konoshenko 2018, Gros 2018, Rialland 2018, Lǐ Zǐhè 2023), scholars have pointed out areas where further clarification would be helpful. Their precious suggestions have been gratefully addressed through targeted revisions.

Finally, the entire volume has undergone a fresh round of editorial polishing. Writing about tone in an accessible manner is no easy task; the complexity of the subject matter makes clarity all the more essential. The bulk of the effort in preparing this second edition has therefore been devoted to improving fluency, precision, and stylistic consistency.

As I present this revised edition, I only wish I had made the topic even more accessible to a wider audience, and I apologize to readers who may still find the volume a demanding read.

Acknowledgments

> [Writing a grammar] takes an individual who loves language in general, the target language in particular, and is trained and happy to spend time doing this work.
>
> (Nurse 2011: xxiv)

I am deeply grateful to my teachers for their patience and inspiration over the years. Laurent Danon-Boileau, my first linguistics tutor, encouraged students to explore lesser-documented languages. I knew at once that this was what I wanted to do. In 1994, reading Michel Launey's newly published description of Classical Nahuatl as "an omnipredicative grammar" (Launey 1994), my imagination was fired by his mention (p. 16) of the professional and personal coincidences that led him to study a language remote in time and space, ultimately yielding groundbreaking insights into human language. That story resonated with me, planting the idea that I, too, might contribute by exploring an unfamiliar linguistic and cultural world. I finally had the chance to experience this wonderful blend of discovery, excitement and fulfilment in my fieldwork in Southwest China.

A special thanks to Jacqueline Vaissière, who took on the challenge of raising a novice steeped in the cloudiest romanticism to the status of Doctor in phonetics, and who has been a wise and supporting mentor ever since.

My work on Yongning Na began in 2006, the same year I joined the *Langues et Civilisations à Tradition Orale* (LACITO) research centre within *Centre National de la Recherche Scientifique* (CNRS). LACITO, with its deep commitment to immersion fieldwork, has been an intellectually stimulating and supportive home base. I am grateful to CNRS for making it possible for me to carry out extended fieldwork in China in 2011–2012, with a temporary affiliation at the *Centre d'Études Français sur la Chine contemporaine* (CEFC). My thanks also to the Institute of Linguistics at Academia Sinica for hosting me for three months in early 2011. From late 2012 to mid-2016, I was based at the International Research Institute MICA in Hanoi, in an environment that facilitated close collaboration with colleagues from across Asia and beyond. I especially appreciated the support and encouragement of the institute's directors, Phạm Thị Ngọc Yến (succeeded in 2015 by Nguyễn Việt Sơn) and Eric Castelli.

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My deepest gratitude goes to Latami Wangyong 拉他咪王勇 (Latami Dashi 拉他咪达石), who welcomed me into his world and supported my work with his mother, Mrs. Latami Daeshilamu 拉他咪打史拉姆 (**lɑ˧tʰɑ˧mi˥ ʈæ˧ʂɯ˧-ɬɑ˩mv̩˩**). My main language consultant, Mrs. Latami Daeshilamu, and the other Yongning Na speakers who collaborated with me showed extraordinary patience and generosity. Their contribution to this project is immeasurable. I also thank Picus Ding for introducing me to Latami Wangyong.

I am indebted to many fellow researchers for stimulating discussions and invaluable feedback on drafts: Lamu Gatusa 拉木·嘎吐萨 (pen-name: Shi Gaofeng 石高峰), Liberty Lidz, Christine Mathieu, Pascale-Marie Milan and Ho Sana 何撒娜. My heartfelt thanks to Roselle Dobbs, for her sharp editorial eye and deep dives into linguistic nuance, and to Nathan Hill for a careful revision of the entire manuscript. I also extend my appreciation to the three anonymous reviewers, whose wonderfully thorough and helpful feedback helped sharpen this book. Many others provided insightful comments on draft chapters: Chen Yen-ling 陳彥伶, Katia Chirkova, Denis Creissels, Stéphane Gros, Guillaume Jacques, Martine Mazaudon, Boyd Michailovsky, Frédéric Pain, Phạm Thị Thu Hà, Annie Rialland, Martine Toda, and Meng Yang.

A special note of thanks to Séverine Guillaume, the engineer in charge of the Pangloss Collection, for her assistance in archiving the annotated audio recordings that underpin this work. I am also grateful to Luise Dorenbusch for converting the original draft to LaTeX, as well as to Guillaume Jacques, Thomas Pellard, and Sebastian Nordhoff for their LaTeX expertise. Jérôme Picard kindly created the map.

Many thanks to the series editor, Martin Haspelmath, and to Language Science Press coordinator Sebastian Nordhoff, for taking care of the whole editorial process, and for making the experience of publishing a second edition uncannily simple and straightforward. I am also deeply appreciative of the many volunteers who proofread the volume (the list of names appears in the colophon: the legal notices page). Any remaining errors are mine alone.

Many thanks to my wife and my daughter for their patience and support.

I am also grateful to readers of the first edition who pointed out passages in need of improvement. Their keen observations helped refine this second edition.

Fieldwork on Yongning Na was funded through two grants from the *Agence Nationale de la Recherche* (ANR, France): "Phylogenetic assessment of Southern Qiangic" (PASQi, ANR-07-JCJC-0063) and "Himalayan corpora" (Himalco, ANR-12-CORP-0006). The present book is also a contribution to the work packages "Evolutionary approaches to phonology: New goals and new methods" and "Lab-Field: Bringing the lab to the field" of the Labex project "Empirical Foundations of Linguistics" (ANR-10-LABX-0083).

More broadly, the research presented here took shape over a period during which I was involved in several other ANR-funded projects, including "Computational Language Documentation by 2025" (CLD2025, ANR-19-CE38-0015-04), "Probing neural representations for typological signal" (DeepTypo, ANR-23-CE 38-0003), and "Glottalization in the light of Machine Learning" (Glot-TAL, ANR-24-CE38-3766) as well as an NSF project, "DEL: Discovering and demonstrating linguistic features for language documentation" (DEL, NSF-1761548). Although the second edition of this book was not an intended outcome of these projects, they provided a stimulating research environment, fostering discussions, collaborations, and methodological advances that indirectly contributed to its development. Many thanks to Martine Adda, Gilles Adda, Katia Chirkova, Elisabeth Delais-Roussarie, Séverine Guillaume, Guillaume Jacques, Aimée Lahaussois, Graham Neubig, Claire Pillot-Loiseau, Rachid Ridouane, Trần Thị Thủy Hiền and Guillaume Wisniewski for having me on board their projects.

So many people have contributed to this project that I am sure I have inadvertently left someone out. My apologies to those whose names should be here.

Abbreviations and conventions

Interlinear glosses

Glosses follow the Leipzig Glossing Rules (Comrie et al. 2008), with some additions.

A	agent marking (adposition)
ABILITIVE	abilitive (suffix)
ABL	ablative (adposition)
ADJ	adjective
ADVB	adverbializer (suffix)
AFFIRM	affirmative (particle)
ALL	allative
ASSOCIATIVE	associative plural
AUG	augmentative
CAUS	causative
CERTITUDE	a use of the copula described by Lidz (2010:497) as “an epistemic strategy that marks a high degree of certitude”
CLF	classifier
CNTR	contrastive
COM	comitative
COMPLETION	completion (suffix)
COP	copula
DAT	dative
DEM	demonstrative
DESIDERATIVE	desiderative (suffix)
DIM	diminutive
DISC.PTCL	discourse particle
DIST	distal
DU	dual
DUR	durative
EXCL	exclusive
EXIST	existential (verb)

FUT	future
IMM.FUT	immediate future
IMMINENCE	imminence (prefix): the event is imminent
INCL	inclusive
INTERROG	interrogative (particle or pronoun)
INTJ	interjection
INTS	intensifier
N	noun
NEG	negation
NMLZ	nominalizer
NUM	numeral
O	object
OBLIGATIVE	obligative (suffix)
PERMISSIVE	permissive
PFV	perfective
PL	plural
POSS	possessive
PROG	progressive
PROH	prohibitive
PROX	proximal
PST	past
RECP	reciprocal
REDUPL	reduplication
REL	relativizer
REP	reported-speech particle
S	subject
SG	singular
TOP	topic marker (suffix)
V	verb
1	first person
2	second person
3	third person

Verbs are glossed in the infinitive, as ‘to fly’, ‘to say’, ‘to lead’, ‘to go’, and so on, in order to clarify that the intended English gloss is the verb, not the noun.

Other abbreviations and symbols

cs	centisecond: a unit of time equal to 0.01 seconds
F	focalization of the word that precedes (through local intonational modification of tone: see §9.2.2)
F_0	fundamental frequency (a standard abbreviation in phonetics)
F1, F2…	Female language consultant number 1, 2… (this is a standard convention in phonetics; the numbering is chronological, referring to the set of Naish recordings that have been collected since 2002)
H	High tone
L	Low tone
M	Mid tone
M1, M2…	Male language consultant number 1, 2…
N/A	no answer: used in tables for combinations that are not possible
p.c.	personal communication
=	clitic boundary
-	affix boundary
.	syllable boundary
σ	syllable (a standard convention in phonology)
|	tone group boundary (see Chapter 7)
◊	boundary preceding extrametrical syllables (see §7.3)
~	reduplication (example: /**wɤ˩**~**wɤ˩˥**/)
≈	free variation: variation that is not conditioned by phonological or morphosyntactic parameters (example: [**tɕɥe**] ≈ [**tɕɥi**]). In this volume, the tilde ~, commonly used in linguistics as a symbol for free variation, is reserved for reduplication.
↑	emphatic stress on syllable that follows (see §9.2.1)
#	word boundary (see Chapter 2); by extension: the boundary of the entire expression to which a tone pattern is associated (see Chapters 3-6)
$	a symbol used in H$, one of the lexical categories of H tones

–	morpheme break, used in the representation of the tone pattern of an expression made up of two (or more) morphemes. Thus, for a dimorphemic compound noun, –L refers to an L tone that attaches after the morpheme break (i.e. on the second noun: the head noun), and LM–L indicates that the first noun gets LM tone and the second gets L. (In my publications on Yongning Na before 2016, the symbol used was a superscript circle °, but this use of the symbol conflicted headlong with Africanist usage, in which L° refers to a *level* L tone: an L tone followed by a floating H tone which results in the absence of a phonetic falling contour on the L tone. See e.g. Hyman 1993: 84 and Akumbu 2016: 4.)
//**ʐwæ˥**//	underlying phonological form (a vertical bar is more usual, but in this volume the vertical bar is used for tone group boundaries)
/**ʐwæ˧**/	surface phonological form
[**ʐwæ˧**]	phonetic realization
***ʐwæ˧**	a reconstructed form
† **ʐwæ˧**	a form that is predicted on the basis of regular rules, but unattested
‡ **ʐwæ˧**	incorrect (ungrammatical) form; note that the asterisk is not used, to preclude confusion with reconstructed forms
::	phonological correspondence between two languages or dialects (a standard convention in historical linguistics)

References to texts, and links to online annotated recordings of Yongning Na

Examples from the online texts are referred to in the following format: text identifier followed by a dot followed by the sentence number. For instance, *Sister.27* refers to sentence 27 in the narrative referred to for short as 'Sister'. For some of the stories, several versions were recorded. In these cases, the version number is indicated after the text identifier, without a separator: thus, *Dog2.35* refers to sentence 35 in the second version of the narrative 'Dog'.

One-click links from examples cited in this book to the audio recordings available online (about which see §1.5) have been added for the present (second) edition. These hyperlinks use the Digital Object Identifiers assigned to all the resources of the Pangloss Collection since 2020 (Vasile et al. 2020). The link leads straight to the relevant sentence (in texts) or word (in word lists) inside the document.

The typographical conventions for inserting these links into the PDF document follow the example set by Guillaume Jacques's grammar of Japhug (Jacques 2021: 3). Only the informative part of the URL (Uniform Resource Locator) is visible, omitting the prefix shared by all the documents in the Pangloss Collection, namely https://doi.org/10.24397/pangloss-; thus, {0004531#S119} links to sentence 119 inside the text that bears number 0004531 in the Pangloss Collection. The embedded link contains the complete URL: https://doi.org/10.24397/pangloss-0004531#S119.

Here is a list of texts, providing the correspondences between identifiers and full titles. Links to the online texts are also provided for the convenience of readers of the electronic version.

Short title	**Title**	**Link**
Agriculture	Agriculture: agricultural activities over the course of the year	{0004440}
Benevolence	Benevolence: about attitudes that were valued among the Na	{0004582}
BuriedAlive	Buried alive: how a young woman ran into great trouble because of her greed	1: {0004536} 2: {0004538}
Caravans	Caravans: about the trade which flourished in the area in the second quarter of the twentieth century	{0004530}
ComingOfAge	Coming of age: the ritual performed at age thirteen	2: {0004588}
Dog	Dog: how dog and man exchanged their lifespans	1: {0004442} 2: {0004555}
Elders	Elders: elders and ancestors	1: {0004442} 2: {0004532}
FoodShortage	Food shortage: how parents set out to sell children, and then changed their mind	1: {0004557} 2: {0004442}
Funeral	Funeral: how funeral rites used to be conducted	{0004571}
Healing	Healing: how diseases used to be treated through rituals	{0004540}

Short title	Title	Link
Housebuilding	Housebuilding: the process of building a house	1: {0004448} 2: {0004549}
Lake	Lake: how the Lake was created	3: {0004348} 4: {0004350}
Mountains	Mountains: some beliefs associated to the mountains around Yongning	{0004573}
Mushrooms	Mushrooms: which ones are collected for cooking and for medicine	{0004616}
Renaming	Renaming: how one used to change a child's name to give it a happier start in life	{0004534}
Reward	Reward: how the heavens rewarded an honest man	{0004446}
Seeds	Seeds: how mankind obtained seeds and learnt to grow crops	1: {0004547} 2: {0004542}
Sister	Sister: the sister's wedding	1: {0004341} 3: {0004344}
Tiger	Tiger: how the tiger attacked a woman and her daughter	1: {0004444} 2: {0004545}
TraderAndHisSon	Trader and his son: how a trader taught his son how to handle the ups and downs of commerce	{0004632}

For elicited phonological and morphotonological data, the correspondences between identifiers and full titles are as follows.

Short title	Title	Link
AccompPfv	Verbs illustrating the various tone categories, in the frame ACCOMPLISHED+VERB+PERFECTIVE	{0004518}
ApicalizedVowels	Words illustrating the opposition between two apicalized high front vowels following alveolo-palatal initials	{0004470}

Short title	Title	Link
CoordCompounds	Coordinative compounds, 1	{0004561}
CoordCompounds2	Coordinative compounds, 2: pairs of numerals in association with ‘year’, ‘month’ or ‘day’	{0004662}
CoordCompounds3	Coordinative compounds, 3: pairs of numerals in association with ‘year’, ‘month’ or ‘day’	{0004869}
DemClf	Demonstrative-plus-classifier phrases, 1	{0004830}
DemClf2	Demonstrative-plus-classifier phrases, 2	{0004832}
DemClf3	Demonstrative-plus-classifier phrases, 3	{0004834}
DetermCompounds1to4	Determinative compound nouns, 1-4: body parts of animals	{0004450}
DetermCompounds5	Determinative compound nouns, 5: body parts of animals (verifications)	{0004452}
DetermCompounds6	Determinative compound nouns, 6: body parts of animals (verifications)	{0004454}
DetermCompounds7	Determinative compound nouns, 7: body parts of animals (extensive set)	{0004456}
DetermCompounds8to10	Determinative compound nouns, 8-10: body parts of animals (supplements)	{0004458}
DetermCompounds11	Determinative compound nouns, 11: body parts of animals (a few verifications)	{0004460}
DetermCompounds12	Determinative compound nouns, 12: body parts of animals	{0004462}
DetermCompounds13	Determinative compound nouns, 13: body parts of animals (compounds with the noun ‘sheep’)	{0004464}

Short title	Title	Link
DetermCompounds14	Determinative compound nouns, 14: body parts of animals in possessive constructions	{0004466}
DetermCompounds15	Determinative compound nouns, 15: body parts of animals (a few compounds with the noun ‘cat’)	{0004468}
DetermCompounds16	Determinative compound nouns, 16: cultural artefacts and peoples	{0004491}
LocativePostp	Nouns followed by locative (spatial) postpositions: ‘beside’, ‘behind’, ‘to the left’, ‘to the right’	{0004563}
NounsEven	Nouns followed by ‘even’	{0004565}
NounsInFrame	Disyllabic nouns placed in a carrier sentence: ‘This is (a/the) N’, in order to bring out their tone patterns	{0004529}
NumClf (41 documents)	The titles of all 41 documents follow the same format: “Numeral-plus-classifier phrases. Tone: *T*. Classifier: *C*. Range: *n* to *m*”, where *T* is the lexical tone, *C* the class of objects to which the classifier applies, *n* the start of range (either 1 or 30), and *m* the end of range (either 30 or 100).	{0004357} {0004357} {0004361} {0004365} {0004371} {0004377} {0004379} {0004387} *+33 more*
ObjectVerb	Data illustrating the tone patterns of object-plus-verb combinations, 1	{0004472}
ObjectVerb2	Data illustrating the tone patterns of object-plus-verb combinations, 2	{0004474}
ObjectVerb3	Data illustrating the tone patterns of object-plus-verb combinations, 3	{0004567}
OnlyAnd	Nouns followed by the morpheme ‘only’ (homophone: ‘and’)	{0004569}

Short title	Title	Link
PossessPro	Possessive constructions with pronouns, without an intervening particle	{0004567}
SpatialOrientation	Spatial orientation: combinations of verbs with prefixes (or adverbials) indicating spatial orientation	{0004559}
SubjectVerb	Data illustrating the tone patterns of subject-plus-verb combinations	{0004476}
VerbDurative	Verbs illustrating the various tone categories, in the frame DURATIVE+V+PROGRESSIVE	{0004520}
VerbProhib	Verbs of all tone categories preceded by the PROHIBITIVE, 1	{0004522}
VerbProhib2	Verbs of all tone categories preceded by the PROHIBITIVE, 2	{0004524}
VerbReduplObj	Reduplicated verbs (tones: M, H, L, and MH) preceded by an object	{0004526}
VerbReduplObj2	Reduplicated verbs of all tone categories preceded by an object	{0004516}

Photographs and figures

All photographs and figures are my own.

Translation of citations

Unless a reference to a published translation is provided, English translations of citations are my own.

For quick reference: Lexical tones and main tone rules

This volume is organized in analytical order, setting out facts and then gradually advancing towards an analysis. This mode of exposition mirrors the process of analysis during fieldwork, working upwards from surface observations. The aim is to allow the reader to evaluate the analysis step by step and to consider possible alternatives, rather than presenting a complete analysis from a top-down perspective. A drawback of this approach is that the key facts of the tone system are not gathered in a single location, which may make them difficult to look up. The present reference section serves as a lookup guide, providing a place to recheck (i) the inventory of the lexical tones of nouns and verbs, (ii) the meaning of the custom notations used for Yongning Na tone categories (e.g. H#, #H), (iii) the list of phonological tone rules, and (iv) three sets of tables presenting tone combination rules in determinative compound nouns, subject-plus-verb combinations, and object-plus-verb combinations.

The tones of nouns and verbs

The lexical tone of a noun is determined using the copula and the possessive as diagnostic contexts, since their surface tone varies according to the tone category of the target word. The use of different elicitation contexts can be likened to the application of chemical agents in the development of photographic films: just as a blend of chemicals is required to render a latent image visible, several contexts need to be used to reveal underlying tone. Together, the copula and the possessive suffice to bring out all the categories of nouns. For nominal classifiers (Table 3), nine tonal categories were identified by examining tonal patterns in association with numerals. For verbs (Table 4), seven categories were arrived at by piecing together evidence from four contexts: in isolation, with a preceding NEGATION prefix /**mɤ˧-**/, with a preceding ACCOMPLISHED prefix /**le˧-**/, and in association with the object 'a bit' (/**ɖɯ˧-kʰwɤ˥$**/).

Table 1: The lexical tone categories of monosyllabic nouns. From §2.4.2.

analysis	in isolation	+COP	+POSS	//example//	meaning
// LM //	LH	L+H	L+H	**bo˩˧**	pig
// LH //	LH	L+H	L+H	**ʐæ˩˥**	leopard
// M //	M	M+L	M+M	**lɑ˧**	tiger
// L //	M	L+LH	L+M	**jo˩**	sheep
// #H //	M	M+H	M+M	**ʐwæ˥**	horse
// MH# //	MH	M+H	M+H	**ʈʂʰæ˧˥**	deer

Table 2: The lexical tone categories of disyllabic nouns. From §2.4.2.

analysis	in isolation	+COP	+POSS	//example//	meaning
// M //	M.M	M.M+L	M.M+M	**po˧lo˧**	ram
// #H //	M.M	M.M+H	M.M+M	**ʐwæ˧zo#˥**	colt
// MH# //	M.MH	M.M+H	M.M+H	**hwɤ˧li˧˥**	cat
// H$ //	M.H	M.M+H	M.M+M	**kv̩˧ʂe˥$**	flea
// H# //	M.H	M.H+L	M.H+L	**hwæ˧ʈʂæ˥**	squirrel
// L //	L.LH	L.L+H	L.L+H	**kʰv̩˩mi˩**	dog
// L# //	M.L	M.L+L	M.L+L	**dɑ˧ʝi˩**	mule
//LM+MH#//	L.MH	L.M+H	L.M+H	**õ˩dv̩˧˥**	wolf
//LM+#H//	L.H	L.M+H	L.M+M	**nɑ˩hĩ#˥**	Naxi
// LM //	L.H	L.M+L	L.M+M	**bo˩mi˧**	sow
// LH //	L.H	L.H+L	L.H+L	**bo˩ɬɑ˥**	boar

Table 3: One example of each of the nine tonal categories of monosyllabic classifiers. From §3.1.2.

classifier	tone	description: classifier for...
ɖwæ˥$_\mathbf{a}$	H_a	steps (of stairs)
ɲi˥$_\mathbf{b}$	H_b	days
hɑ̃˧˥$_\mathbf{a}$	MH_a	nights
kv̩˧˥$_\mathbf{b}$	MH_b	people, persons
nɑ˧$_\mathbf{a}$	M_a	tools
dzi˧$_\mathbf{b}$	M_b	pairs of non-separable objects, e.g. shoes
dze˩$_\mathbf{a}$	L_a	pairs of separable objects, e.g. pots, bottles
dzi˩$_\mathbf{b}$	L_b	trees, bamboos
ʐɤ˩$_\mathbf{c}$	L_c	lines, patterns (in weaving or drawing)

Table 4: The seven tonal categories of monosyllabic verbs: analysis into H, M, L, and LH tones. From §6.1.1.

tone	example	in isolation	NEG	ACCOMP	V+'a bit'
H	**dzɯ˥** 'to eat'	M	M.H	M.H	M.M.M
M_a	**hwæ˧**$_\mathbf{a}$ 'to buy'		M.M	M.M	M.H.L
M_b	**tɕʰi˧**$_\mathbf{b}$ 'to sell'				M.M.M
M_c	**bi˧**$_\mathbf{c}$ 'to go'			M.L	N/A
L_a	**dze˩**$_\mathbf{a}$ 'to cut'	LH	M.L		M.M.H
L_b	**ʈʰɯ˩**$_\mathbf{b}$ 'to drink'				M.M.MH
MH	**lɑ˧˥** 'to strike'	MH	M.MH	M.MH	M.H.L

Notational conventions for tones

L Low tone.

M Mid tone.

H High tone.

Word boundary; by extension, the boundary of the entire expression to which a tone pattern is associated (e.g. a compound noun or a numeral-plus-classifier phrase).

H# Final High tone: this H gets anchored on the last syllable of the expression to which it is associated.

#H Floating High tone: this H gets anchored *after* the word boundary, i.e. on a following morpheme, if one is available and can serve as a host. In some contexts, the floating H tone does not surface (for want of syllabic anchoring) but instead lowers following tones to L.

MH# Final Mid-to-High tone, realized either as a rising contour on the last syllable of the expression or, where the morphotonological context allows, as an M tone on the last syllable of the expression and an H tone on (the first syllable of) the following morpheme.

H$ A type of H tone (analyzed in §2.3.5) exemplified by the noun ‘flea’ and nicknamed the ‘flea’ tone or ‘hopping’ tone. When a word carrying this tone is pronounced in isolation, the H tone associates to its last syllable: /**kv̩˧ʂ̮e˥**/ ‘flea’ surfaces with an M.H tone sequence. When followed by the copula, the H associates with the copula: /**kv̩˧ʂ̮e˧ ɲi˥**/ ‘is (a/the) flea’. Finally, when the noun is followed by the possessive, no H tone reaches the phonological surface: the observed form is /**kv̩˧ʂ̮e˧=bv̩˧**/ ‘of (a/the) flea’, with M tone throughout.

– In the representation of tones in expressions containing two or more morphemes (e.g. compound nouns), this symbol marks the morpheme boundary. Thus, *–L* refers to an L tone that attaches after the morpheme break, i.e., in the case of a compound noun, to the head noun. *LM–L* indicates that the morpheme before the break (i.e., in a compound, the determiner) receives LM tone, while the noun following the break receives L tone.

The seven phonological tone rules (from §7.1.1)

Rule 1: L tone spreads progressively ("left-to-right") onto syllables unspecified for tone.

Rule 2: Syllables that remain unspecified for tone after the application of Rule 1 receive M tone.

Rule 3: In tone-group-initial position, H and M are neutralized to M.

Rule 4: The syllable following an H-tone syllable receives L tone.

Rule 5: All syllables following an H.L or M.L sequence receive L tone.

Rule 6: In tone-group-final position, H and M are neutralized to H if they follow an L tone.

Rule 7: If a tone group only contains L tones, a postlexical H tone is added to its final syllable.

Combination rules in determinative compounds, S+V combinations, and O+V combinations

Table 5: The underlying tonal categories of σ+σ compound nouns. Leftmost column: tone of determiner; top row: tone of head. From §4.1.4.

tone	LM; LH	M	L	H	MH
LM; LH	LH	LM	LH	LM+#H	LM+MH#
M	–L	#H	–L	#H	MH#
L	L				
H	#H–	#H			–L
MH	H#			H$	

Table 6: The underlying tonal categories of σσ+σ compound nouns. Leftmost column: tone of determiner; top row: tone of head. From §4.1.4.

tone	LH; LM	M	L	H	MH
M	–L	#H	–L	#H	–L
#H	H#	#H			
MH#		MH#			H#
H$	#H–	#H	H$	#H	H#–
L	L+H#	L			L+H#
L#	L#–				
LM+MH#	LM+MH#–	LM+MH#	LM+H$		
LM+#H		LM+#H	LM+H#	LM+#H	LM+H#
LM	LM–L	LM	LM–L		LM+MH#
LH	LH				
H#	H#–				

Table 7: The underlying tonal categories of σ+σσ compound nouns. Leftmost column: tone of determiner; top row: tone of head. From §4.1.4.

tone	M	#H	MH#	H$	L	L#	LM+MH#; LM+#H; LM; LH	H#
LM; LH	LM	LM+#H	LM+MH# / L+#H–	LM+H$	L+#H–	L+#H– / L+H#	L+#H–	LM+H# / L+H#
M	M	#H	MH#	H$	–L	–L#	–L	H#
L	L			L+H#	L	L+H#	L+#H–	L+H#
#H	H#	#H	#H–	–L / H#	#H–	H#	#H–	H#
MH			MH#	#H–				#H

Table 8: The underlying tonal categories of σσ+σσ compound nouns. Leftmost column: tone of determiner; top row: tone of head. From §4.1.4.

tone	M	#H	MH#	H$	L	L#	LM+MH#; LM+#H; LM; LH	H#
M	M	#H	MH#	H$ / #H–	–L	–L#	–L	H#
#H	H#		#H–	H$ / #H– / H#	#H–	H#	#H–	
MH#		MH#		#H– / H#			MH#–	
H$		#H	#H– / H#–				#H–	
L	L+H#	L	L+H#	L+H#	L+#H–	L+H#	L+#H–	L+H#
L#	L#–							
LM+MH#	LM+H#	LM+#H	LM+MH#–	LM+MH#–/ H#	LM+MH#–	LM+H#	LM+MH#–	LM+H#
LM+#H				LM–H$	LM+#H–			
LM	LM–		LM+MH#		LM–L	LM–L#	LM–L	
LH	LH							
H#	H#–							

Table 9: The tone patterns of subject-plus-verb combinations, in surface phonological transcription. From §6.10.1.

	tone of verb					
tone of noun	H	M_a	M_b	L_a	L_b	MH
LM, LH	L.H	L.M+M	L.M+M	L.H	L.H	L.MH
M	M.M+L	M.M+M	M.M+M	M.L	M.L	M.MH
L	M.M+L	L.L	M.M+M	L.L	L.L / M.L	L.L
H	M.M+L	M.M+L	M.M+L	M.MH	M.MH	M.L
MH	M.H	M.H	M.H	M.MH	M.MH	M.H
M	M.M.M+L	M.M.M+M	M.M.M+M	M.M.L	M.M.L	M.M.MH
#H	M.M.M+L	M.M.M+L	M.M.M+L	M.M.MH	M.M.MH	M.M.L
MH#	M.M.MH	M.M.MH	M.M.MH	M.M.MH	M.M.MH	M.M.H
H$	M.M.M+L	M.M.M+L	M.M.M+L / M.M.M+H	M.M.MH	M.M.MH	M.H.L
L	L.L.L	L.L.L	L.L.L	L.L.L	L.L.L	L.L.H
L#	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L
LM+MH#	L.M.M+L	L.M.M+L	L.M.M+L	L.M.MH	L.M.MH	L.M.H
LM+#H	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.MH	L.M.MH
LM	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.L	L.M.MH
LH	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L
H#	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L

Table 10: The tone patterns of object-plus-verb combinations. From §6.8.1.

	tone of verb					
tone of noun	H	M_a	M_b	L_a	L_b	MH
LM	L.M+L	L.M+M	L.M+M	L.M+L	L.M+L	L.MH
LH	L.L / L.H	L.H	L.L / L.H	L.H	L.H / L.L	L.MH
M	M.M+L	M.M+M	M.M+M	M.L	M.L	M.MH
L	L.L	M.M+M	M.M+M / L.L	L.L	L.L / M.L	L.L
H	M.M+L	M.L	M.M+L	M.H	M.MH	M.L
MH	M.H	M.H	M.H	M.H	M.MH	M.H
M	M.M.M+L	M.M.M+M	M.M.M+M	M.M.L	M.M.L	M.M.MH
#H	M.M.M+L	M.M.L	M.M.M+L	M.M.H	M.M.MH	M.M.L
MH#	M.M.MH	M.M.H	M.M.MH	M.M.H	M.M.MH	M.M.H
H$	M.M.M+L	M.H.L	M.M.M+L	M.M.H	M.M.MH	M.H.L
L	L.L.L	L.L.H	L.L.L	L.L.H	L.L.L	L.L.H
L#	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L
LM+MH#	L.M.M+L	L.M.H	L.M.M+L	L.M.H	L.M.MH	L.M.H
LM+#H	L.M.M+L	L.M.L	L.M.M+L	L.M.H	L.M.L / L.M.MH	L.M.L
LM	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.L	L.M.MH
LH	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L
H#	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L

1 Introduction

The aim of this book is to provide a description and analysis of the tone system of Yongning Na (also known as Mosuo), a Sino-Tibetan language spoken in Southwest China.

The richness of this system becomes apparent even from a single sentence. Example (1a) is the first one that I transcribed. At the time, I had just arrived at my future teacher's house; my luggage had been left at another house along the main road, about fifty metres away. I asked my teacher's son, who speaks fluent Mandarin, to translate an explanation for me: "I have brought a lot of stuff; I have to go back [to the main road] and pick it up now." This yielded (1a). Later, I elicited (1b) as a simpler form.

(1) a.

njɤ˧	ʐi˩	bi˩	-zo˩	-ho˥.
1SG	to_take	to_go	OBLIGATIVE	DESIDERATIVE

'I have to go and take [my luggage] now.' (Field notes, 2006)

b.

njɤ˧	bi˧	-zo˧	-ho˩.
1SG	to_go	OBLIGATIVE	DESIDERATIVE

'I have to go. / I'm afraid I have to leave.' (Field notes, 2006)

The contrast in lexical tone on the main verb (/**ʐi**˩/ 'to take' in 1a, versus /**bi**˧/ 'to go' in 1b) affects the tones of the following syllables, extending all the way to the end of the sentence.

This book offers an analysis of the underlying system, encompassing lexical tone categories, phonological rules that operate within tone groups, and morphotonological rules that govern combinations within various types of phrases. As a preview of the results concerning lexical tones, examples (1a–1b) are provided below (as 2a–2b) with morpheme-level transcriptions. These transcriptions indicate lexical tones using tone symbols supplemented by subscript letters $_{a}$, $_{b}$, and $_{c}$, which differentiate subcategories of lexical tones. The following chapters examine the system in depth, detailing how the tone sequences of entire sentences obtain from the lexical tones.

(2) a. njɤ˧ ʑi˩ bi˩-zo˩-ho˥.
njɤ˩ ʑi˩$_{a}$ bi˧$_{c}$ -zo˧$_{a}$ -ho˩
1SG to_take to_go OBLIGATIVE DESIDERATIVE
'I have to go and take [my luggage] now.'

b. njɤ˧ bi˧-zo˧-ho˩.
njɤ˩ bi˧$_{c}$ -zo˧$_{a}$ -ho˩
1SG to_go OBLIGATIVE DESIDERATIVE
'I have to go. / I'm afraid I have to leave.'

To set the stage for the analyses developed in this volume, the introduction provides essential background on the Na language and its study.

Section 1.1 presents the language and its dialects, followed by a review of previous research in Section 1.2 (with complementary historical and ethnological perspectives in Appendix B). The research programme behind the present study is set out in Section 1.3, while Section 1.4 introduces the language consultants and the data elicitation methods. Section 1.5 then presents the online primary documentation on Yongning Na – comprising audio and video recordings with their accompanying philological apparatus – alongside the dictionary and other resources, all conceived as part of a coherent Open Science environment.

Finally, a concise grammatical sketch in Section 1.6 summarizes key linguistic properties, such as the structure of noun and verb phrases.

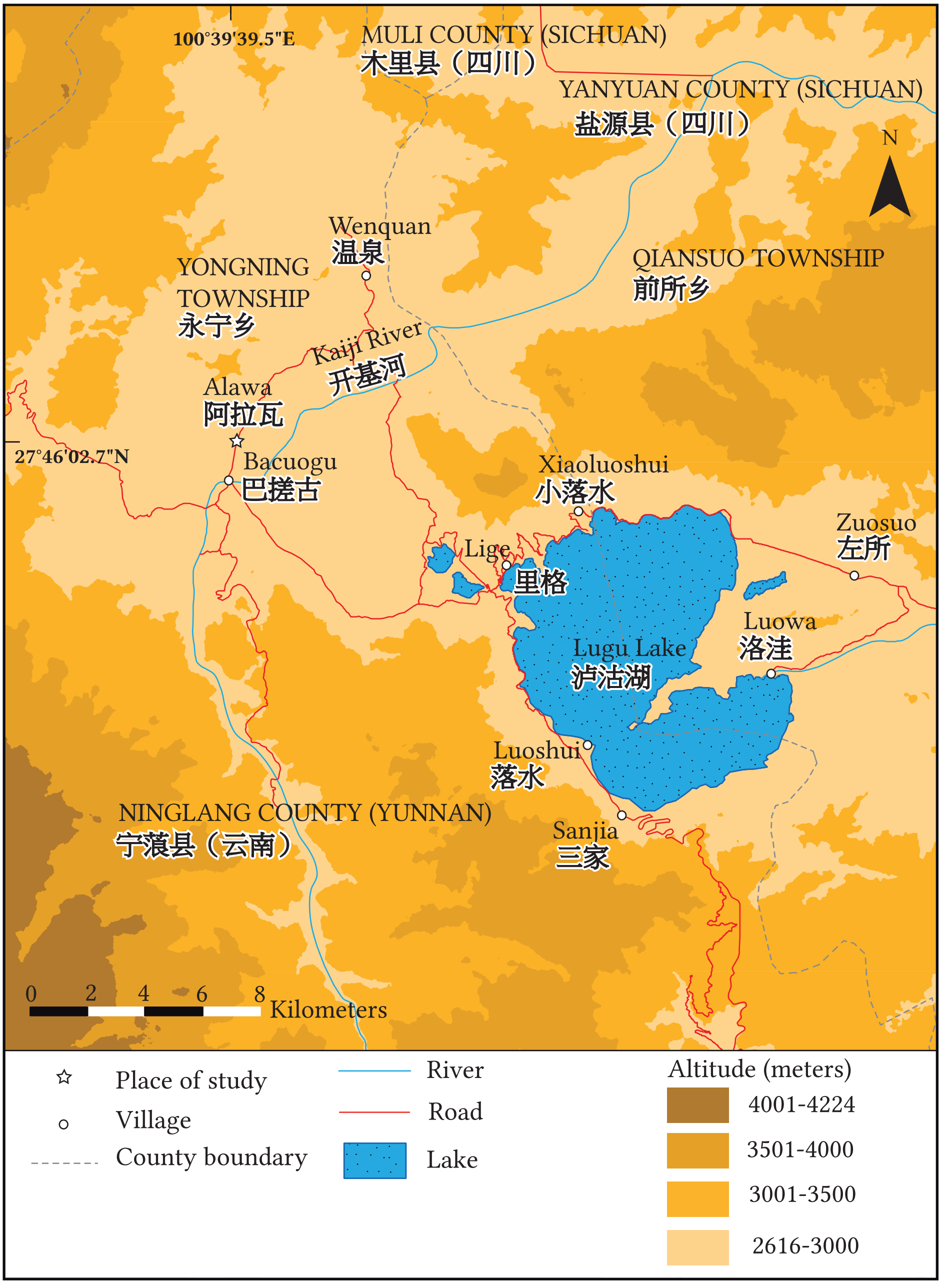
1.1 The Na language

1.1.1 Endonym and exonyms

Yongning Na is a Sino-Tibetan language spoken in Southwest China, straddling the border between the provinces of Yunnan and Sichuan, at a latitude of 27°50' N and a longitude of 100°41' E (see Map 1.1). Speakers refer to the language as /**nɑ˩ʐwɤ˥**/, meaning 'Na language'. The structure of this compound, made up of a noun and a verb, is shown in (3).

(3) nɑ˩ʐwɤ˥
nɑ˩˧ ʐwɤ˩$_{b}$
endonym: Na to_speak
'the language of the Na', i.e. 'the Na language'

The name 'Yongning Na' was introduced by Liberty Lidz (2006), combining the people's endonym with the name of the region where the language is spoken. This naming aligns with principles such as: (i) "language names (like city



Map 1.1: A sketch map of the Yongning area. *Designed by Jérôme Picard. Sources: Geofabrik, ASTER GDEM (a product of METI and NASA), and OpenStreetMap.*

Photo 1.1: The Yongning plain seen from the west, with Gemu Mountain (in Na: **kɤ˧mv̩˥**) in the background. The Lake is behind the pass on the right-hand side. Autumn 2006.

names) are loanwords, not code-switches" (Haspelmath 2017), and (ii) "language names may have a modifier-head structure", whereby 'Yongning Na' conveys the intended meaning of 'the Yongning variety of the Na language'.

Yongning 永宁 is a plain near Lugu Lake 泸沽湖, a lake covering approximately fifty square kilometres (see Map 1.1). The lake fosters a microclimate that makes the area suitable for agriculture despite the high altitude (about 2,800 metres above sea level).

Ethnonymy reflects the high degree of ethnic, cultural and linguistic intricacy of the Sino-Tibetan borderlands (Gros 2023). Table 1.1 presents (i) two endonyms, (ii) the name by which the Naxi 纳西 (a closely related ethnic group) refer to the Na, and (iii) a Chinese exonym found in various sources, under various avatars, for close to two thousand years, and which currently enjoys renewed favour for reasons discussed in Appendix B (§B.2).

The most likely interpretation of the endonym **nɑ˩˥** is that it means 'black'. Use of ethnonyms meaning 'black' or 'white' is widespread in the area. In Yongning, the Na coexist with the Pumi 普米, who call themselves **ʈʰóŋmə** 'white people'.

Table 1.1: The names of the Na: endonyms and exonyms.

transcription	language	romanized equivalents	Chinese equivalents	meaning
nɑ˩˧	Na	Na (Cai 1997, Lidz 2010)	*Nà* 纳 (Yáng Fúquán 2006)	'black'
ɬi˧-hĩ˧	Na	Hli-khin (Rock 1963), Hli-hing (Shih 1993)	*Lǐxīn* 里新 (Shih 2008: 15)	'People of the Centre'
ly˧-çi˧	Naxi	Lü-khi (Rock 1963)	*Lǚxī* 吕西 (Guō Dàliè & Hé Zhìwǔ 1994: 8)	as above: 'People of the Centre'
origin not yet established	Chinese	Moso (Cordier 1908, Nishida 1985, Shih 1993, McKhann 1998, Luo 2008), Mo-So, Mosuo (Knödel 1995)	*Móshā* 摩沙, *Mósuò*[a] 磨些, *Mósuò* 麽些, *Mósuò* 摩些, *Mósuō* 摩娑, *Mòsuò* 末些, *Móhuò* 磨获, *Mòsuān* 莫狻, *Mósuō* 摩梭	*not yet established*

[a]The character 些, standardly read *xiē*, is read *suò* in this context, as also in other place names from the same period: see Pelliot (1904: 170n9).

> The designation **ʈʰóŋ** 'white' sets the Pumi apart from some surrounding ethnic groups whom they designate as **nʲæ̌** 'black': the **goŋnʲæ̌** 'Nuòsū (Yí) 彝' ('black skin') and the **nʲæmə̌** 'Na (Mósuō) 摩梭' ('black person'). (Daudey 2014: 2)

Among the Yi 彝 (formerly known as 'Lolo'), there is a distinction between 'black' and 'white' castes. "The Nasoid groups are also known as Black Lolo, and the assimilated groups connected with them – either Nasoid groups who have become Sinicized, or others who have become assimilated to the Nasoid groups, often by capture or conquest – are called 'white' to denote the fact that they do not 'fit' in the Nasoid clan structure" (Bradley 1979: 53).

> In northeastern Yunnan and western Guizhou, the designation *Nasu* (the black ones) refers to a group of Yi who were the overlords of a series of feudal kingdoms between the 9th and the 20th centuries; they were often contrasted to other, subordinate groups who referred to themselves as white. In the Liangshan region of southwestern Sichuan, on the other hand, Black bones (*Nuoho*, called Black Yi in Chinese), and White bones (*Quho*, called White Yi in Chinese), refer to the aristocratic and commoner castes into which the society is divided – the term *nuo*, or 'Black' also means 'heavy', 'important', or 'serious'. At the same time, the aristocratic caste is also associated with darker colored clothing (…). What is most important here is to realize that the association of people and colors in this region has little or nothing to do with the imagined color of the people themselves, but rather is part of a complex symbolic system that both reflects and is reflected in the styles and colors of people's clothing. (Harrell 2009: 102)

The Na name for Yongning is **ɬi˧di˩**. Shih (2010: 23) interprets it as meaning 'the peaceful land', connecting it to the verb **ɬi˥** 'to rest, to relax'. This folk etymology fits nicely with the author's celebration of Na society's ideals of harmony, reflected in the title of his work: *Quest for harmony*: *the Moso traditions of sexual union and family life*. However, phonetic correspondences with Naxi do not support this interpretation. Instead, they indicate that the Na name for Yongning, **ɬi˧di˩**, means 'the central land, the heartland'. The linguistic evidence is as follows.

Yongning is called **ly˧dy˩** in Naxi (Hé Jírén et al. 2011: 201). This cannot be a recent borrowing from Na, as Na does not have a rounded close front vowel /**y**/. If the contemporary Na form were borrowed into Naxi, it would be interpreted as **li˧di˩** by Naxi ears, with a straightforward correspondence for /**d**/ and /**i**/ (shared by both languages) and a reinterpretation of Na /**ɬ**/ as Naxi /**l**/, since Naxi does not have an unvoiced lateral. The presence of the vowel /**y**/ in Naxi **ly˧dy˩** strongly suggests that the word is cognate with its Na counterpart. While it could be a calque – a root-for-root translation from Na to Naxi by a bilingual speaker who understood the word's morphological structure –, it is not a phonetic loanword.

The second syllable of the noun is easily analyzed: it means 'earth, place, land' (Na: **di˩˥**, Naxi: **dy˩**), a root commonly found in place names across both languages, where it serves as a locative nominalizer (Lidz 2010: 559). The first syllable, however, is less straightforward. Based solely on Na data, it could have a number of different interpretations. It might indeed be connected to the verb 'to rest', **ɬi˥**, as in the folk etymology of Yongning as 'the land of rest, the peaceful land' adopted by Shih Chuan-kang.[1] But it could equally be related to 'moon', **ɬi˧** (as in the disyllable **ɬi˧mi˧** 'moon'), to 'ear', **ɬi˧** (as in **ɬi˧pi˩** 'ear'), to 'middle', **ɬi˧** (as in **ɬi˧gv̩˧**, 'middle part'), or to 'Bai' (an ethnic group), through truncation of the disyllable **ɬi˧bv̩˧** 'Bai'. Any of these roots, combined with **di˩˥** 'earth, place', would yield the form **ɬi˧di˩** by application of regular tone rules.

Moreover, the search needs to be extended further in view of the existence of some tonal irregularities. Certain disyllabic words exhibit tone patterns that deviate from those predicted by synchronically productive rules (as discussed in §4.2.11 and §5.1.3.5). This suggests that tonal constraints should be relaxed when investigating the etymology of **ɬi˧di˩**. Broadening the search to **ɬi** roots across all tonal categories reveals additional candidates: nouns such as 'roebuck' (**ɬi˩**), 'turnip' (in **ɬi˩bi˩**), 'trousers' (in **ɬi˩qʰwɤ˩**), and 'wrath, anger' (in **ɬi˩ʁɑ˩**), as well as verbs like **ɬi˩** 'to measure' and **ɬi˧˥** 'to dry in the sun'. Language-internal evidence thus allows for a wide range of hypotheses: does **ɬi˧di˩** mean 'the peaceful land, the land of rest', 'the land of the moon', 'the land of ears', 'the land of the middle', 'the land of the Bai people', 'the land of the roebuck', 'the land of turnips', 'the land of trousers', 'the land of wrath', 'the land of measurements', or 'the land of sun-drying'?

Semantic plausibility alone cannot resolve this question. A study of toponymy in various languages of China (Yáng Lìquán & Zhāng Qīnghuá 2011) demonstrates its remarkable creativity, with meanings that are often surprising.

Decisive evidence comes from comparison with Naxi. Among all the possibilities, only one is supported by the existence of a cognate in Naxi. The root **ly˥** in Naxi means 'centre', as does **ɬi˧** in Na. This leads to the conclusion that **ɬi˧di˩** (and Naxi **ly˧dy˩**) can be interpreted as 'the central land, the heartland'.

[1]The main language consultant reports this folk etymology in the document *Folk Etymology*, available online {0004606}.

Crucially, this interpretation rests on linguistic evidence provided by the comparative method. By identifying cognates between Na and Naxi words, a definitive conclusion can be reached. While **ɬi˧di˩** can have more than ten different interpretations in Na, and **ly˧dy˩** in Naxi can likewise be interpreted in a range of different ways (such as ‘land of grain’, ‘land of Asian crabapple, *Malus asiatica*’, ‘central land’, ‘land of watchfulness’, or ‘quaking land’), historical comparison points decisively to the ‘central land’ hypothesis.

‘The Centre, the Central Land’, is an apt designation given the geographic specificity of the Yongning basin, set within a rugged landscape of steep slopes and high mountains. It is also compelling in light of the basin’s historical significance for Naish peoples (see Appendix B, in particular §B.1).

From a linguistic perspective, the greatest diversity within the Naish language group – the lower-level subgrouping to which Na belongs (§1.1.2) – is indeed concentrated in and around the Yongning basin, within a radius of less than 100 kilometres. The name ‘People of the Centre’, **ɬi˧-hĩ˧**, originally referred to the inhabitants of the Yongning plain. Ironically, this name is no longer commonly used in the dialect under study, despite the dialect being located squarely within the Yongning plain. Instead, it survives among a community of speakers who relocated from Yongning to the peripheral region of Shuiluo 水落 (in the neighbouring county of Muli 木里) several centuries ago.

1.1.2 Dialect classification

The language spoken in Yongning was investigated in 1979 by linguists He Jiren 和即仁 and Jiang Zhuyi 姜竹仪, who classified it among the Eastern Naxi dialects (Hé Jírén & Jiāng Zhúyí 1985: 4, 104–116). The division of Naxi into Eastern and Western dialects was initially proposed cautiously as a working hypothesis, based on relatively brief fieldwork conducted in 1956 and 1957 as part of the national survey of languages within the borders of the People’s Republic of China.

> From our analysis and comparison of the available linguistic and cultural materials, we propose a preliminary division between two dialects, Western and Eastern. But due to the very short amount of time [that could be devoted to this research] and the shortcomings of our experience, it is difficult to tell whether this division tallies with the actual language situation… (Hé Jírén & Hé Zhìwǔ 1988: 120)
>
> *Original text:* 我们从现有的语言和人文材料来加以分析和比较，将纳西语初步分划为西部和东部两个方言。不过时间短促，经验不足，这样分划不知是否符合客观现实情况……

The Western dialect area thus proposed by and large corresponds to the area ruled by the Naxi chieftains of Lijiang from the 14^{th} to the 18^{th} centuries (see Appendix B, §B.1.2). The Eastern dialect area is located to its east and north-east, across the Yangtze River, in parts of the present-day counties of Ninglang 宁蒗, Yanyuan 盐源, Muli 木里, and Yanbian 盐边. Within the Eastern dialect area, three sub-dialects were distinguished by He and Jiang: Yongning 永宁, Guabie 瓜别, and Beiquba 北蕖坝.

This classification became the standard in Chinese scholarship. It was also adopted by the Summer Institute of Linguistics in its inventory of languages, which serves as an international standard (ISO 639-3): *Ethnologue*: *Languages of the World* (Gordon 2005). Naxi initially appeared in Ethnologue under the language code NBF, encompassing all the dialects mentioned above (i.e. giving the name "Naxi" the same extension as in Chinese scholarship). In 2010, the "Eastern" dialects of Naxi were given a separate entry under the romanized name "Narua" (code: NRU). The original language code NBF was retired, following the split into (i) Naxi proper (new code: NXQ), corresponding to "Western Naxi" in Chinese scholarship, and (ii) "Narua" (code: NRU), corresponding to "Eastern Naxi". The division of dialects within "Narua" remains as proposed by He and Jiang for "Eastern Naxi".

By contrast, the Glottolog database adopts an innovative approach, setting itself the ambitious goal of "improv[ing] on the ISO 639-3 language identifiers in terms of quality, transparency and anchoring" (Hammarström & Forkel 2022: 918). It does so by structuring language labels from the level of the *doculect* – "a linguistic variety as it is documented in a given resource" (Good & Cysouw 2013: 342) – into progressively broader categories, such as subdialects, dialects, languages, subfamilies, and families (Nordhoff & Hammarström 2011). This approach is based on the insight that language data ultimately originates from idiolects of specific speakers and is recorded in specific sources. In version 5.1 of Glottolog, Yongning Na is listed under the code *yong1288*, with a distinct entry for the Na dialect spoken in Lataddi, on the eastern shore of Lugu Lake (Glottocode: *lata1234*). These two dialects together form a grouping labelled 'Narua' (Glottocode: *yong1270*). While the label 'Narua' – a romanization of /**nɑ˩ʐwɤ˥**/ 'Na language', shown in (3) – is thus shared by Ethnologue and Glottolog, its characterization differs between the two databases.

The total number of speakers of Na ("Eastern Naxi") was estimated at around 40,000 in early surveys (Hé Jírén & Jiāng Zhúyí 1985: 107), a figure reiterated by Yang (2009). The Ethnologue entry reports a slightly higher figure of 47,000 as of 2012, based on the Summer Institute of Linguistics' sources (Lewis et al. 2016). The figure of 47,000 speakers includes groups that do not identify their native language by the name /**nɑ˩ʐwɤ˥**/, which highlights a drawback of the name ‘Narua’.

No large-scale dialectal comparison was conducted in the half-century that followed the first dialect survey. The list of “subfamilies” (*zhīxì* 支系) of the “Naxi nationality” (*Nàxīzú* 纳西族) provided by Guō Dàliè & Hé Zhìwǔ (1994: 5–9) could serve as a useful reference for such a survey, keeping in mind that this list was essentially based on ethnological criteria rather than linguistic data. Reliable descriptions of the language varieties that these authors grouped under the label “Naxi” are required for fine-grained dialectological and comparative research. The present volume aims to contribute to this long-term endeavour by offering a synchronic description of the tone system of one specific language variety.

1.1.3 The position of Na and Naxi within Sino-Tibetan

The position of Naxi and Na within Sino-Tibetan is a topical issue in Sino-Tibetan historical linguistics. Naxi was classified as a member of the “Lolo branch” (Yi) by Shafer (1955); however, Shafer clarified that this language, to which he referred as “Mosso”, was among those for which there was “[t]oo little data or too irregularly recorded” (p. 103, note 37). His classificatory proposal for Naxi, as for the other languages placed in the “Unclassified” set within the “Lolo branch”, was thus tentative.

Bradley (1975b) revisited the issue based on advances in the comparative study of Lolo (Yi) languages. He noted that “[w]hile a large proportion of Nahsi vocabulary is plausibly cognate to Proto-Burmese-Lolo (*BL) and Proto-Loloish (*L) forms reconstructed in Bradley 1975a, there is only limited systematic regularity of correspondence. Moreover, the tonal and other developments postulated for *BL and *L by Matisoff are not reflected in Nahsi”. The lack of regular correspondences, and the absence in Naxi of shared innovations deemed characteristic of Loloish and Burmese-Lolo, led Bradley to conclude that Naxi is “certainly not a Loloish language, and probably not a Burmish language either” (p. 6).

Some scholars, especially in mainland China, nonetheless maintain the classification of Naxi as a member of the Yi/Lolo group. Gài Xìngzhī & Jiāng Zhúyí (1990) base this renewed claim on the high percentage of phonetically similar words between Naxi and Yi/Lolo, though without verifying the regularity of sound correspondences. Lama (2012) includes Naxi among the set of thirty-seven Lolo-Burmese languages among which he proposes subgroupings using two methods: (i) searching for candidates for the status of shared innovations, and (ii) conducting automated computation. The latter approach consists of performing Bayesian inference of phylogeny using MrBayes, and computing phylogenetic networks by means of SplitsTree. These two programmes are applied to a 300-word list from the thirty-seven languages. No judgments of cognacy are passed on the 300 word sets fed as input to the computational procedure, which apparently assumes cognacy in all cases as a default hypothesis. By definition, these methods generate subgrouping proposals without questioning the premise that the languages at issue all belong to the same branch.

In light of the conclusion reached by Bradley (1975b), if one looks outside Lolo-Burmese for languages most closely related to Naxi and Na, suggestive evidence comes from comparison with the neighbouring languages Shixing 史兴语 (also known as Xumi; see Huáng Bùfán & Rénzēng Wàngmǔ 1991 and Chirkova et al. 2013) and Namuyi 纳木依语 (Lāmǎ 1994; see also Sūn Hóngkāi 2001, Yáng Fúquán 2006, and Lakhi et al. 2010). However, full-fledged comparison has not been carried out yet, and the state of phonological erosion in these languages is a major impediment to comparative studies.

Sun Hongkai included Shixing (Xumi) and Namuyi within a "Qiangic" language group which he defined on the basis of typological similarities. Other languages in the group include Qiang (Rma), rGyalrongish, Pumi, Muya (Minyag), Queyu and Zhaba.

> Naxi is often regarded as a language of the Yi branch, yet it is widely acknowledged that it does not fit neatly within this classification (Bradley 1979: 14). Reciprocal constructions in Naxi verbs are fully identical to those of the Qiang branch. Furthermore, Naxi dialects exhibit a number of phonological, lexical, and grammatical features that align with those of Qiangic languages. Naxi is thus a 'borderline' language between the Qiang and Yi branches, possessing characteristics from both. That is, in genealogical classification, Naxi has traits of both language groups: some of its features are specific to the Yi branch, while others are distinctive of Qiangic languages. Among these, the system of reciprocal constructions stands out as a key grammatical feature linking Naxi to Qiangic. (Sūn Hóngkāi 1984: 14)
>
> *Original text:* 纳西语经常被认为是彝语支的语言，大家知道，纳西语在彝语支中是不太合套的一种语言（详情请参阅David Bradley《Proto-Loloish》第14页），纳西语动词的互动范畴和羌语支完全一致。此外纳西语方言中还有一些语音、词汇和语法现象，与羌语支语言相一致。纳西语是羌语支和彝语支之间的"临界"语言。即在语言谱系分类上兼有两种语言集团的不同的特征。也就是说，纳西语同时兼有彝语支和羌语支所特有的某些特征，而动词互动范畴，则是纳西语兼有羌语支的一种重要的语法特征。

Sun Hongkai thus proposes that Naxi (understood as encompassing both Naxi and Na) is an "intermediate language" (*línjiè yǔyán* 临界语言) between Loloish and Qiangic (Sūn Hóngkāi 1984). This compromise view projects the presence of Yi-like and Qiangic-like typological features into the indefinite past of Naxi. By a similar reasoning, the presence of words of Sinitic (Chinese), Tai-Kadai, and Mon-Khmer origin in Vietnamese could lead to its classification as an intermediate language straddling the divide between these three language families. Historical linguists, however, favour an approach in which borrowings and other changes in the language are gradually identified, layer after layer, ultimately resulting in a detailed account of the language's evolution that includes the influences to which it was subjected through the ages. Thus, Maspero, in his study on Vietnamese, identified Chinese elements as belonging to a later layer than Tai-Kadai and Mon-Khmer elements:

> Pre-Annamite was born out of the fusion of a Mon-Khmer dialect with a Tai dialect; the fusion may even have involved a third language, which remains unidentified. At a later period, the Annamite language borrowed a huge number of Chinese words. (Maspero 1912: 118)
>
> *Original text:* Le préannamite est né de la fusion d'un dialecte mon-khmer, d'un dialecte thai et peut-être même d'une troisième langue encore inconnue, et postérieurement, l'annamite a emprunté une masse énorme de mots chinois.

Four decades later, Haudricourt attempted to disentangle the Tai-Kadai and Mon-Khmer components, emphasizing that the notion of "language fusion" can be misleading:

> If we admit that there is no such thing as "fusion" between languages, and that genealogical relatedness must be assessed on the basis of core vocabulary and grammatical structure, we are led to consider that the modern form of a language is not determined by its genealogical origin, but by the influences to which it is subjected in the course of its history. (Haudricourt 1953: 121–122)
>
> *Original text:* Si l'on admet qu'il n'y a pas de « fusion » de langues, et que l'apparentement généalogique doit être fondé sur le vocabulaire de base et la structure grammaticale, on sera conduit à penser que ce qui donne sa forme moderne à une langue n'est pas son origine généalogique, mais les influences qui s'exercent sur elle au cours de son histoire.

Haudricourt identified a greater proportion of Mon-Khmer words in basic vocabulary as opposed to words of Tai stock, and concluded that Vietnamese is a Mon-Khmer language. Importantly, this proposed phylogenetic affiliation in no way constitutes a denial of the considerable influence of language contact in the course of history – a point underscored by Dimmendaal (2011: 268-271), citing Manessy (1990).

> It is important to realise that there is no principled way in which one can argue that language x has become "mixed", i.e. embedded with foreign language material, to an extent where it should be classified as non-genetic, or multi-genetic. There are scales or degrees of borrowing, and it is precisely for this reason that it is *not* a useful taxonomic principle to talk about non-genetic or multi-genetic developments. (Dimmendaal 2011: 271)

There is no clear dividing line between cases of contact that are considered to result in language replacement – where the vocabulary inherited from an earlier language is regarded as a substratum, comprising piecemeal vestiges of a language that has been replaced by another (e.g. Basque or Celtic elements in Romance languages) – and cases where the earlier component remains substantial enough to justify the classification of the modern language as belonging to that earlier component's language family, such as the Mon-Khmer component in Vietnamese.

Mixed languages with split ancestry are nonetheless recognized as a special case, and handled accordingly in historical linguistics. "A real challenge to the Tree model is the existence of mixed languages with split ancestry (Bakker 2017), such as Michif (verbs and verbal morphology from Cree, nouns and nominal morphology from French) or Media Lengua (grammar from Quechua, lexicon from Spanish). Mixed languages are exceptional, and given their rarity, the Tree model remains generally applicable as long as such cases are correctly identified" (Pellard et al. 2026). Neither Vietic languages nor Naic languages fall into this special category.

Returning to Naxi and Na, the traditional tools of comparative-historical phonology appear to be the most reliable means of unravelling this language's history and clarifying its relationship to other languages within the Sino-Tibetan family. There is widespread agreement regarding this method: "it is only by searching for lexical and morphological parallels on all sides and by establishing the phonetic equations for such parallels that we can finally decide the genetic relationship of a doubtful group" (Shafer 1955: 98).

The observation on which Sun Hongkai places considerable emphasis as an indicator of phylogenetic relatedness – namely, the use of reduplication in reciprocal constructions in Naxi as well as in the languages he groups together as "Qiangic" – does not carry any significant weight as a diachronic argument. Similar constructions are found in a variety of languages, and there is no barrier to such a construction being borrowed in a situation of language contact or being independently innovated from a pre-existing use of reduplication to convey intensification. This latter possibility is hypothesized, for instance, in the case of Tok Pisin, a creole language spoken throughout Papua New Guinea (Fedden 2003: 6).

Figure 1.1 presents a family tree proposed in a preliminary comparative study (Jacques & Michaud 2011), based on Yongning Na, Lijiang Naxi, and Laze – a language spoken in Muli County. Within the Sino-Tibetan family, waves of mutual influence across languages are so pervasive that concerns regarding the applicability of the tree model have been expressed for decades (Benedict 1972, Matisoff 1978). However, the use of a tree representation does not imply a disregard for the importance of areal diffusion.

In a context of continuing debates about models and methods (see e.g. François 2014, Jacques & List 2019), it is essential to highlight the usefulness of the tree model as one of the tools in the historical linguist's toolbox. It provides a framework for formulating working hypotheses about degrees of phylogenetic closeness between languages – hypotheses that form the basis for comparative efforts and for proposing reconstructions at various historical depths (Pellard et al. 2026).

The aim of proposing a tree model is not to advance classifications for their own sake but to clarify the assumptions underlying historical comparison. The central objective is to document the evolution from the hypothesized common ancestor of a language group to the attested language varieties.

> We may assert or hypothesize a genetic relation on the basis of [regular sound correspondences]. But the proof of the linguistic pudding remains in the follow-up, the systematic exploitation, the full implementation of the comparative method, which alone can demonstrate, not just a linguistic genetic relationship, but a linguistic history. (Watkins 1990: 295)

The proposal that Yongning Na, Lijiang Naxi, and Laze form a Naish lower-level subgroup is supported by shared innovations, notably structural similarities in the tones of numeral-plus-classifier phrases that cannot have been acquired through contact (for details, see Michaud 2011). It is further proposed that the Naish subgroup joins with Shixing (Xumi) and Namuyi into a Naic subgroup. At

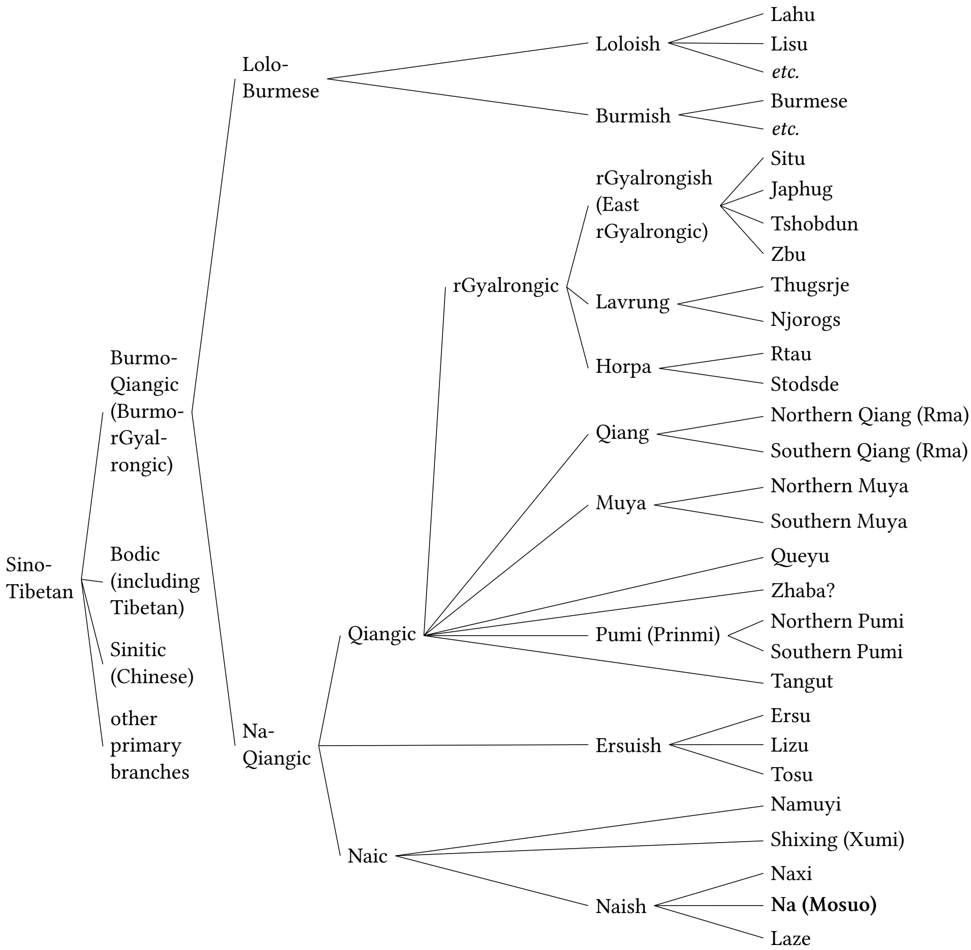


Figure 1.1: A family tree showing the position of Yongning Na within Sino-Tibetan as hypothesized by Jacques & Michaud (2011). Yongning Na is highlighted (second from bottom).

a third, more speculative level, Naic is hypothesized to join with Ersuish (referred to as Ersuic in the historical study of Yu 2012; see also the reference grammar of Ersu by Zhang 2013) and with a broad 'Qiangic' group that includes rGyalrongic, Rma (Qiang), Muya, Pumi, Tangut, and more. The proposed grouping, termed 'Na-Qiangic',[2] is further connected to Lolo-Burmese to form a higher-level grouping referred to as 'Burmo-Qiangic' or 'Burmo-rGyalrongic', which is

[2]The label 'Na-Qiangic' reflects a Na-centred perspective, arising in the context of reconstruction work focusing on Naish languages. In broader contexts, this choice warrants revision. The 'Qiangic' component in the label 'Na-Qiangic' also risks causing confusion due to the

placed on a par with Bodic, Sinitic, and other primary branches. This working hypothesis encourages systematic searches for cognates between Naic languages, Ersuish, and Qiangic. Comparison with Lolo-Burmese languages is also essential for advancing the historical study of Naish languages. All the languages listed in Figure 1.1 are uncontroversially related, as members of the Sino-Tibetan family, therefore the comparative analysis of data from all these languages makes sense.

In the present era, computational approaches offer a useful addition to the historical linguist's toolbox (Wu et al. 2020). These approaches, which encompass computational methods for language phylogeny, have seen significant advances, including the development of techniques for automating cognate detection and conducting Bayesian phylogenetic inference (Rama & List 2019). Accordingly, it would seem advisable to subject the representation in Figure 1.1 – which is based on Guillaume Jacques's expert intuitions about the Sino-Tibetan family – to computational phylogenetic testing. However, the eight languages at the bottom of Figure 1.1 (including Yongning Na) were intentionally excluded from a large-scale study of fifty Sino-Tibetan languages that aimed to produce dated phylogenies (Sagart et al. 2019). This exclusion stemmed from the fact that the historical phonologies of these languages have not yet been sufficiently developed to permit reliable cognate identification and the establishment of sound correspondences with the rest of the family (Guillaume Jacques, p.c.).

Nonetheless, another study published in the same year (Zhang, Yan, et al. 2019) incorporated six of these more challenging languages – Yongning Na, Naxi, Xumi (Shixing), Namuyi, Lyuzu (Lizu), and Ersu – among a set of 109 Sino-Tibetan languages. This study produced a "clade credibility tree" estimating divergence times for the emergence of the major Sino-Tibetan branches and their subsequent diversification. A portion of this tree is presented in Figure 1.2, with modifications: the figure excludes groupings outside Burmo-rGyalrongic (such as Sinitic, Bodic, Karen, Kuki-Chin-Naga, and Himalayish/Kiranti), and omits some details about lower-level branches not directly relevant to the discussion at hand.

varied uses made of the exonym 'Qiang' 羌. Historically, this term has been used in Chinese records with a broad meaning and now serves as the designation for an officially recognized ethnic group, speakers of the Rma language (Sims 2016, 2024). Given the significance of the rGyalrongic group for Sino-Tibetan historical linguistics (Hill 2019, Zhang, Jacques, et al. 2019), recent scholarship has adopted terminology that replaces 'Qiang' with 'rGyalrong' in intermediate phylogenetic labels. For instance, Guillaume Jacques and colleagues (Jacques & Pellard 2021) refer to the node at which Lolo-Burmese joins the grouping containing rGyalrongic languages as 'Burmo-rGyalrongic', instead of the label 'Burmo-Qiangic' used in Figure 1.1 and retained by Hill (2021).

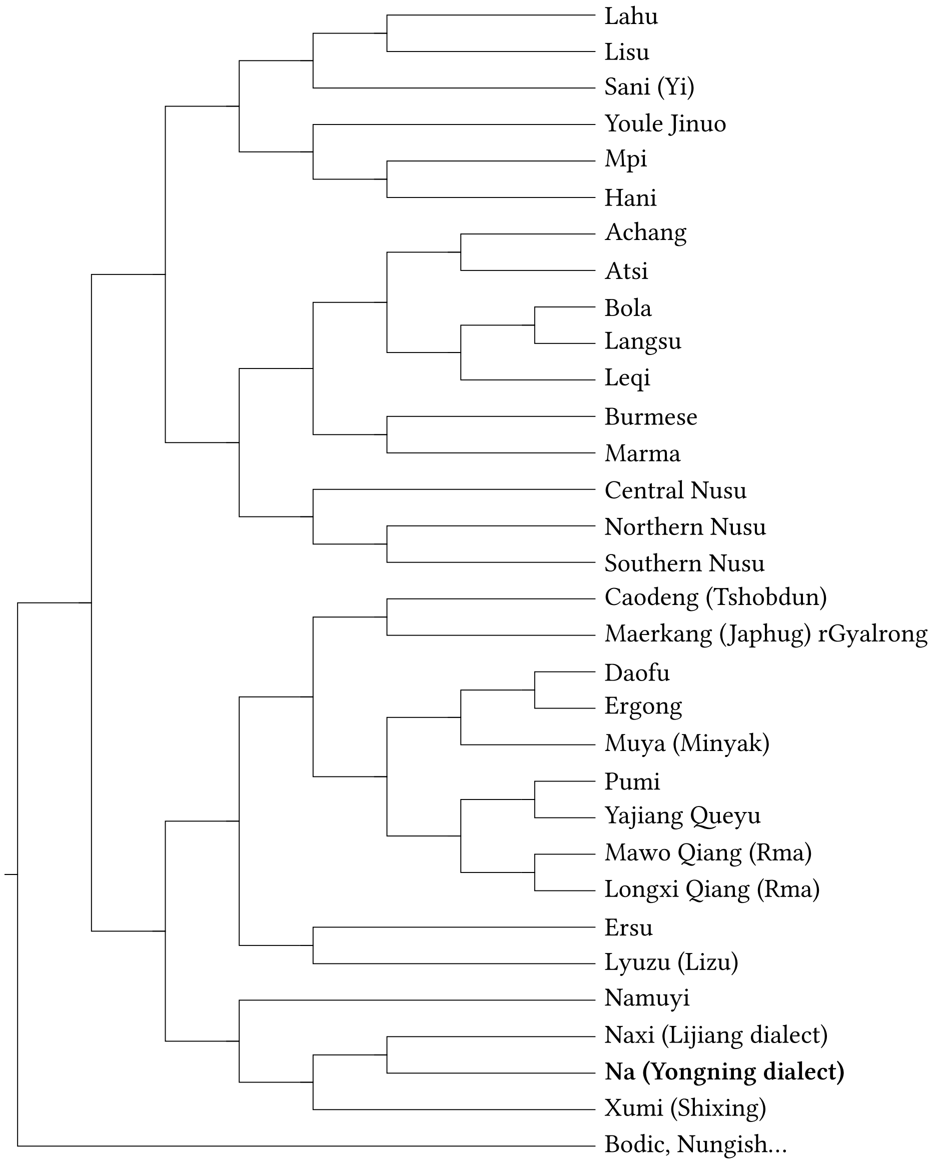


Figure 1.2: A portion of the maximum clade credibility tree for 109 Sino-Tibetan languages shown in Figure 1 of Zhang, Yan, et al. (2019). Yongning Na is highlighted (third from bottom).

A technical difference between Figure 1.1 and Figure 1.2 lies in the branching structure employed by Zhang, Yan, et al. (2019). The binary branching model adopted in their study allows for a more fine-grained resolution at nodes where Figure 1.1 features polytomies (forks with three or more branches). For instance, while Figure 1.1 places the Naish subgroup (Lijiang Naxi and Yongning Na[3]) in a node alongside Namuyi and Xumi (Shixing), Figure 1.2 resolves this differently: Xumi (Shixing) branches off first, followed by Namuyi at the subsequent binary node. Despite this slight difference, the overall representation in Figure 1.2 closely mirrors that in Figure 1.1.

The remarkable similarity between the two trees raises questions about its underlying cause. One possibility is that Zhang et al.'s classification draws on that proposed in Glottolog 3.1 (Hammarström et al. 2017), which incorporates Jacques & Michaud (2011) as one of its sources. While the computational part of Zhang et al.'s study can be reproduced using the article's accompanying data files, a significant limitation is the lack of access to their full set of proposed cognates. This absence makes it impossible to scrutinize the etymological groupings proposed by the authors or to evaluate how they addressed the practical challenges of diachronic comparison for the Naic languages included in their analysis.

Systematic comparison between Naic and Lolo-Burmese holds great potential for clarifying the extent to which their typological similarities result from (i) inheritance, (ii) parallel changes – unrelated developments, from a typologically similar starting point: e.g. the development of retroflex consonants from initial consonant clusters –, or (iii) language contact. This research agenda, outlined in a review of Naish language studies (Lǐ Zǐhè 2015), has been implemented in a comparative monograph (Lǐ Zǐhè 2021), which seeks to reexamine the phylogenetic position of Naish languages based on a new reconstruction of proto-Naish.

Li Zihe's study is based on six dialects, three of which are documented first-hand: Baoshan 宝山 (宝山乡石头城), Malimasa 玛丽玛萨 (塔城镇汝柯村), and Ninglang 宁蒗 (宁蒗县大兴镇堰坝村). The other three – Lijiang 丽江, Bowan 波湾, and Yongning 永宁 (the dialect described in this book) – are incorporated from secondary sources. In total, 730 words are reconstructed for proto-Naish and compared with (i) 830 reconstructed proto-Loloish forms and (ii) 2,000 words from Japhug rGyalrong (Lǐ Zǐhè 2021: 175-204). The study finds that Naish and Loloish share more cognates than Naish shares with rGyalrongish or Loloish shares with rGyalrongish. On this basis, the author concludes that Naish and Loloish form

[3] A detail is that Yongning Na is referred to in Zhang, Yan, et al. (2019) as "Naxi Yongning", mirroring the convention of the Chinese source used in the Sino-Tibetan Etymological Dictionary and Thesaurus (STEDT) database, from which the data is extracted. The name is rewritten in Figure 1.2 as "Na (Yongning)".

a single branch, to which rGyalrongish connects at a higher taxonomic level (Lǐ Zǐhè 2021: 206).

Despite the evident care and effort invested in Li Zihe's comparative study, a striking discrepancy in the materials used for Lolo-Burmese and rGyalrong raises concerns about the robustness of the conclusions. On the one hand, the Lolo-Burmese data draws on reconstructions at the level of the entire subgroup; on the other hand, the comparison with rGyalrongic relies exclusively on synchronic data from Japhug, a single modern variety. This imbalance complicates the interpretation of the findings. A more balanced comparison, using reconstructions for rGyalrongish, might yield different results, but such an undertaking is currently hindered by the lack of a comprehensive reconstruction of Proto-rGyalrongish (a.k.a. East rGyalrongic) or rGyalrongic. There has been recent progress in this field: in 2023, a lexical database for historical comparison within rGyalrongic was released (Lai & List 2023), providing cognacy judgments for words (or parts of words) across the subgroup. However, as of version 0.2, the database does not include reconstructed forms for the 303 words (concepts) it contains, leaving this critical piece of the puzzle incomplete. Further advances in this area are eagerly anticipated, as the stakes are high for Sino-Tibetan historical linguistics as a whole.

In summary, there remains considerable room for progress in the historical-comparative study of Naish languages within their broader Sino-Tibetan context. Greater clarity in the phylogenetic relationships among these languages will depend on continued progress in documentation and reconstruction.

1.2 A review of studies on Yongning Na

Historical sources in Chinese offer fascinating glimpses into the language spoken in Yongning centuries ago. The *Yuan Yi Tongzhi* 《元一统志》, a book dated 1286, records the Chinese phonetic equivalents for present-day Lijiang and Yongning as 样渠头 and 楼头 (present-day Mandarin: *yàngqútóu* and *lóutóu*), respectively. In the variety of Chinese recorded in the 14th-century rhyme table *Zhongyuan Yinyun* 《中原音韵》, the initial of 头 is unvoiced. However, using the reconstruction of 'Phags-pa[4]) by Coblin (2007), it is interpreted as ***dəw**, i.e. with the same voicing feature as in present-day Na and Naxi. The name 楼头 reconstructs as

[4]'Phags-pa is an alphabetic script devised during the 1260s by the Tibetan monk Drogön Chogyal Phagpa, at the request of emperor Kublai Khan, to transcribe the main languages of the empire: Mongolian, Chinese, Tibetan, and Uyghur (as well as Sanskrit). 'Phags-pa was used from 1269 until the mid-14th century, so that a fair amount of parallel-text Chinese-'Phags-pa materials were produced, which constitute important materials for the study of Chinese historical phonology.

*[**ləw dəw**](Jacques & Michaud 2011: 487), which is evidently cognate with the current names of Yongning in Na (**ɬi˧di˩**) and Naxi (**ly˧dy˩**), discussed above (§1.1.1). This historical record establishes that the name dates back at least eight centuries. Moreover, it provides evidence on a disputed point of Chinese historical phonology, suggesting that the standard dialect of Yuan dynasty Chinese (Northern Mandarin) retained voiced obstruents (Jacques & Michaud 2011: 487).

Regarding tone, however, these early records offer little insight. 'Phags-pa does not indicate tone: segmentally identical syllables under the same tone are spelt the same way (Coblin 2007: 13). The limited data and the lack of detailed knowledge about tone in both Yuan dynasty Chinese and Yuan-era Naish languages leave many questions unanswered. It is plausible that tone was simply disregarded in the process of selecting Chinese equivalents for local terms. Similarly, the notes of explorers of the turn of the 20th century provide little information about the language and less about tone. For discussion of such early sources, readers are referred to Michaud & Jacques (2010) and references therein. This review focuses on contemporary linguistic research on Na.

1.2.1 Information about Yongning Na in *A brief description of the Naxi language*

He Jiren & Jiang Zhuyi's 1985 *A brief description of the Naxi language* focuses essentially on the dialects spoken in the Lijiang plain, i.e. Naxi proper, not Na. However, the volume includes a word list for Yongning Na, along with observations on phonology, syntax, and dialectal diversity (pp. 107–116; see also Jiāng Zhúyí 1993).

The transcription in this work is not phonemic and may lack consistency. Only four tones are transcribed over monosyllables: LM (˩˧), M (˧), ML (˧˩), and H (˥). In contrast, this volume identifies six categories (LM, LH, M, L, H, and MH: see Chapter 2). He & Jiang's analysis relied heavily on their knowledge of Naxi, a language He Jiren spoke natively and Jiang Zhuyi learnt as a second language. Naxi has a four-way tonal distinction over monosyllables: High, Mid, Low (realized phonetically as low-falling), and Rising. When analyzing Yongning Na, He & Jiang failed to differentiate between its low-rising and mid-rising tones. They also reported a distinction between M and H tones in word-initial position, which does not exist.

Although these discrepancies might be due to dialectal variation, it seems unlikely, given the phonetic similarity between their word list and the data presented here, that the tone system of their consultants differed considerably from that of the consultants who collaborated with me. More plausibly, these differences reflect the challenges of tonal analysis based on brief fieldwork. As noted by Matisoff (2004: 329), phonemic and tonal analyses from short-term data collection often differ substantially across researchers due to incomplete phonemicization.

In the Alawua dialect of Yongning Na, as described in this volume, the phonological distinction between M and H tones only becomes apparent in specific contexts, such as when the word is followed by the copula (see Chapter 2). In isolation, the M and H tones are neutralized to M (see §2.3.2); phonetic realizations of this tone, which can be understood as an architoneme in a Praguean framework, fluctuate freely within the upper half of the speaker's tonal range. This variability can lead an investigator working under the assumption of a contrast between H- and M-tone monosyllables to perceive pitch differences that seem to corroborate such a hypothesis. Such a process of pre-existing assumptions affecting linguistic observation happened to me during the early stages of fieldwork: my initial transcriptions distinguished H-tone words and M-tone words, but it turned out later that there was no such opposition in isolation.

Consequently, the distinction between H and M tones made by He and Jiang based on words pronounced in isolation is unwarranted (unless, as mentioned above, there is a considerable gap between the tone systems of the dialects at issue). For instance, 'field', glossed as 'earth' (*dì* 地), is transcribed by He and Jiang with a High tone: /**l̩v˥**/. However, this word actually carries a Mid tone in Alawua: //**l̩v˧**// (the double slashes are used in this volume to distinguish lexical forms from surface phonological ones). Similarly, 'man' is transcribed as /**xĩ˧**/, with a Mid tone, which corresponds to its realization in isolation. However, its behaviour in other contexts demonstrates that its lexical tone is in fact H: //**hĩ˥**//.

The distribution of Low(-falling) and Low-rising tones in He & Jiang's data is a puzzle to me. In the Alawua variety, no monosyllables are realized with Low tone in isolation. Examples from their data include 'plain, flatlands', 'water', and 'goose', transcribed as /**dy˨˩**/, /**dʑi˨˩**/ and /**o˨˩**/, respectively. My own findings reveal distinct underlying tones for these items:

- LH for 'earth, plain' (//**di˩˥**//, realized as such in isolation: /**di˩˥**/);
- L for 'water' (//**dʑɯ˩**//, realized in isolation as M: /**dʑɯ˧**/);
- LM for 'goose' (//**ɑ˩˧**//, realized in isolation as LH: /**ɑ˩˥**/).

My best guess is that, in the speech of He & Jiang's consultants, L was a (relatively infrequent) free variant of LH in citation forms. Another possibility is that their word list combines data from several speakers, and is not dialectally homogeneous.

Such heterogeneity is evident in their Naxi data. Although the authors attribute this material to the Qinglong 青龙 (now Changshui 长水) dialect, which was spoken by Jiang Zhuyi's teacher He Zhiwu 和志武, He Jiren clarified (p.c. 2002) that he contributed some data based on his native dialect, Yangxi 漾西. In the absence of indications regarding the source of each piece of data, disentangling dialectal variation becomes exceedingly difficult.

He & Jiang's transcription of vowels and consonants also has some issues, as is to be expected in initial field notes. Nasality is marked in only two syllables: /**xĩ**/ (as in 'man', transcribed /**xĩ**˧/; my data: //**hĩ**˥//) and /**ɣə̃r**/ (the sole example is 'bone', transcribed /**ʂa**˧**ɣə̃r**˧/; my transcription: //**ʂæ**˩**ɻ̩̃**˩//). By contrast, the present study identifies eight distinct nasal rhymes. Another notable difference is that He & Jiang do not transcribe the uvular consonants reported in Appendix A of this volume. Such discrepancies may reflect genuine phonological differences between the variety they studied and the one documented here. However, it is not implausible that certain contrasts were overlooked.

Conversely, some of the vowel distinctions transcribed by He & Jiang may not correspond to actual phonemic contrasts. Their word list includes both /**li**/ (as in /**li**˧/ 'to look') and /**lie**/ (as in /**lie**˩˧/ 'tea'). In my data 'tea' and 'to look' share the same initial and rhyme. The vowel /**i**/ is slightly diphthongized towards [**e**], approaching [**lie**], which could account for the variation in perception before the investigator's ear became attuned to the vowel system of Yongning Na. Once again, it remains theoretically possible that these two words contained different phonemes in the dialect investigated by He & Jiang.

1.2.2 A linguist's study of kinship terms: Fù Màojì (1980)

The linguist Fu Maoji 傅懋勣 visited Yongning in May and June 1979 together with He Jiren and Jiang Zhuyi. He collected data in the village of Jjabbu /**dʑɤ**˩**bv̩**˧**ʁwɤ**˩/ (Jiabowa 甲波瓦) for a study on kinship terms, which he presented at the 12th International Conference on Sino-Tibetan Languages and Linguistics (Paris, 1979) and later published in both Chinese and French translation (Fù Màojì 1980, 1983). His paper is a testimony to the broader appeal of Na family structure beyond the circle of professional anthropologists (see Appendix B, §B.3). The article includes an appendix with notes on phonetic transcription, which, when examined in the light of a comprehensive phonemic analysis, reveal some groundbreaking observations – most notably, the recognition of the uvular phonemes /**q**/, /**qʰ**/ and /**ʁ**/. Fu Maoji also documented the approximant /**ɹ**/, noting its occurrence both as an initial consonant before a vowel and as a syllabic element in its own right. This aligns with the analysis proposed in this volume (see Appendix A), where the notation adopted is as a retroflex, /**ɻ**/.

However, alongside these perceptive insights, Fu Maoji also proposed some more puzzling analyses. For instance, his inventory of uvular consonants includes the fricative /**χ**/, which closer scrutiny of the target dialect would likely reveal as non-contrastive with velar realizations, consistent with findings in all recorded Naxi and Na dialects to date. If, during their joint fieldwork, He Jiren and Jiang Zhuyi examined Fu Maoji's notations, they may well have been skeptical about the inclusion of /**χ**/ in the inventory. Such doubts could have led them to question Fu Maoji's correct identification of /**q**/ and /**qʰ**/ as distinct phonemes. Several factors may have contributed to their reservations. First, the uvular sounds [**q**] and [**qʰ**] are also found in Naxi, where allophonic variation of velar stops extends well into the uvular region. This could lead speakers of Naxi to assume that uvulars in other Naish varieties are likewise allophonic variants of velars. Moreover, in Yongning Na, the phonological environments where /**k**/ and /**kʰ**/ contrast with /**q**/ and /**qʰ**/ are relatively restricted (for details, see Appendix A, §A.3.4).

Other problematic aspects of Fu Maoji's notation include: (i) a distinction between plain and laryngeally constricted /**u**/, /**v̩**/ and /**z̩**/ rhymes, (ii) a proposed set of three rhotic vowels in addition to the approximant rhyme [**ɹ**], and (iii) an analysis in which /**i**/ contrasts with /**e**/ but is realized as [**e**] after apical and apical-dental consonants. These notes remain some way from constituting a functional phonemic notation.

As for tone, Fu Maoji's classification into three categories – mid-rising, high-flat, and mid-falling – demonstrates a commendable effort to establish the tonal system on its own terms rather than interpreting it through the lens of the Naxi system. However, his analysis did not fully reach the point where the relevant categories would clearly emerge. The fragmentary nature of his report may explain why He Jiren and Jiang Zhuyi did not draw upon his data in their 1985 book. Nevertheless, it is important to acknowledge that conducting fieldwork in Yongning in 1979 was, in itself, a significant achievement.

1.2.3 An outline of Yongning Na by Yang Zhenhong (2009)

An outline of Yongning Na was published by Yang Zhenhong 杨振洪, a native speaker of the language from Abbuwua /**ə˧bv̩˧-ʁwɤ˧**/ village (Abuwa 阿布瓦村), near the present-day site of Yongning high school. The original version, written in Chinese, appeared as Yáng Zhènhóng (2006), followed by an English translation by Liberty Lidz, incorporating improvements made in consultation with the author, and published as Yang (2009).

The structure of this outline largely follows that of He & Jiang's description of Naxi. Some aspects of its treatment of phonetics and phonology leave room for further refinement. Among consonants, uvular and retroflex stops are not granted phonemic status. As for tone, the analysis is based on the four-tone system of Naxi, which imposes certain limitations. Informal discussions with the author in 2011 suggest that the dialect in question in fact possesses a greater number of tonal categories.

Like many other researchers, Yang Zhenhong employs descriptive tools developed for syllable-tone systems such as those of Sinitic languages, which do not provide an entirely satisfactory framework for describing tone in Na, as explained in §2.4.4. A similar challenge arose in the study of Pumi (also known as Prinmi), where significant advances were only achieved when researchers with expertise in other types of tonal systems – such as those of Japanese dialects – became involved (see Ding 2001, 2006, Jacques 2011a).

1.2.4 Lexical materials

An anthology of everyday words and expressions in the Mosuo language (Zhíbā Ěrchē & Xǔ Ruìjuān 2013) presents vocabulary and expressions organized by semantic field. The second author clarified (p.c. 2024) that the book was intended for a general readership rather than as a contribution to linguistic research. Given this stated objective, it would be inappropriate to assess the work from a strictly phonetic or phonological perspective or to hold it to the standards of specialized linguistic studies. For the purpose of the present review of Na lexicography, it suffices to note that the volume was not designed as a reference work for academic research.

Dictionaries on neighbouring Naxi provided a brilliant example to be emulated (Hé Jírén et al. 2011, Pinson 2012). The *Na (Mosuo) – English – Chinese dictionary* produced as part of the same language documentation suite as the present work is presented in section §1.5.2.

1.2.5 Collections of oral literature: Na ritual texts

In the Yongning plain, Tibetan Buddhism (of the Gelugpa school) coexists with a local tradition of ritual practitioners known as *Ddabe* **dɑ˧pɤ˧**. Unlike among the Naxi, rituals are not written down, although certain written characters are used for calendrical calculations in divination (Yáng Xuézhèng 1985; Lǐ Dázhū 2015: 163-189). The absence of a written tradition partly explains why Na rituals have received less scholarly attention than those of the Naxi. The relatively small size and limited socio-political influence of Yongning, especially in comparison with Lijiang, as well as the difficulty of access to the area until the late 20th century, have further contributed to the lack of documentation.

A bilingual collection published by a native speaker (Āzémíng Cìdázhū 2013) presents ritual texts in a fourfold parallel format: (i) a phonetic approximation of each syllable by means of a Chinese character, (ii) a transcription in the International Phonetic Alphabet, (iii) a syllable-for-syllable Chinese gloss, and (iv) a full-line translation into Chinese. (Most lines contain five, seven, or eight syllables.) This volume were best reserved for readers already familiar with the language, who will be able to identify part of the glosses by making allowance for nonstandard transcription habits influenced by the *Pinyin* system used for romanizing Chinese. For instance, /**p**/, /**t**/, and /**k**/ appear to represent the aspirated phonemes /**pʰ**/, /**tʰ**/, and /**kʰ**/, as in *Pinyin*. Thus, the word for 'white', transcribed as /**puə**/ in this book (p. 4), in fact has an aspirated initial in Na. Tone is not indicated.

During the 2010s, Latami Wangyong (Latami Dashi) launched a documentation programme with the goal of publishing an extensive collection of translated and annotated rituals, accompanied by video recordings. This work has resulted in the online release of a set of rituals in 2025 in the Pangloss Collection ({0009179} through {0009216}).

1.2.6 Liberty Lidz (2010), *A descriptive grammar of Yongning Na (Mosuo)*

By far the most comprehensive description and analysis of Yongning Na to date is Liberty Lidz's Ph.D. dissertation, *A descriptive grammar of Yongning Na (Mosuo)* (Lidz 2010). This work focuses on the variety of Yongning Na spoken in the village of Loshu **lo˧ʂv̩˩** (Luoshui 落水), on the shore of Lugu Lake. Based on in-depth fieldwork, the dissertation provides a detailed account of the morphosyntax of the language and includes 150 pages of transcribed and annotated narratives.

Regarding tone, it has been noted that "[t]he tonal system of Luoshui Narua calls for further analysis. Surface phonological tones are transcribed employing three tonal levels, but a reanalysis in terms of two levels would seem possible in many cases. Mention is made of prolific tone sandhi processes, but tantalisingly, these processes are not elaborated on" (Dobbs & La 2016: 69). It is not unusual for reference grammars to leave open some issues of prosody, such as the status of tone or stress (Zeitoun 2007: 26). In the case of Yongning Na, a tacit division of labour has emerged between Liberty Lidz and myself, whereby the task of conducting a detailed investigation into tone has been left to me. The present volume is dedicated to this endeavour.

1.3 The research programme and its theoretical underpinnings

1.3.1 The aim: Describing a level-tone system of East Asia

Tonal alternations permeate numerous aspects of the morphosyntax of Yongning Na. Crucially, they are not the outcome of a small set of phonological rules, but rather of a complex system of morphotonological patterns that are restricted to specific morphosyntactic contexts. As Guillaume Jacques (p.c. 2009) has observed, in Yongning Na – as in other Naish languages and in Pumi –, irregular morphology mostly consists of irregular morphotonology. The richness of this aspect of the language warrants a book-length study, applying what Scherer (1885: 152) described as "the old philological virtue of exactitude" to this intricate system, in pursuit of a comprehensive and rigorous account.

A search for book-length descriptions of similar systems in other languages suggests that such reference works remain relatively scarce, even within the extensive Bantu branch of the Niger-Congo language family, renowned for its rich morphotonology.

> Theoretical linguistics is primarily concerned with advancing the theoretical enterprise, and tends to produce short pieces – chapters, articles, squibs. It does not have the writing of grammars as a priority, and few of the theoretical grammars of African languages written during the heyday of transformational theory during the 1960s and 1970s have stood the test of time. (…) Are there enough grammars of sub-Saharan African, especially Bantu, languages? The answer is no. (…) The overwhelming impression is that of the small number of real grammars, and the number is not increasing. (Nurse 2011: xxiii–xxiv)

Beyond the issue of sheer numbers, there is also a question of breadth and depth of coverage in existing book-length descriptions. Linguistic fieldwork consists in "going into a community where a language is spoken, collecting data from fluent native speakers, analysing the data, and providing a comprehensive description, consisting of grammar, texts and dictionary" (Dixon 2007: 12). This holistic approach entails decisive advantages for understanding a language as a whole, as it functions in its social setting. However, the breadth of such studies can occasionally come at the expense of depth in specific domains, such as tone. "No variety of Bambara has heretofore been the object of a systematic tonological description aiming at full coverage"[5] (Creissels 1992: 199; see also Clements 2000; Hyman 2005). "Even the 'well described' languages often suffer from a lack of examples, by which to test the descriptive or theoretical claims" (Nurse 2011: xxiii). Similar concerns arise in studies of tone systems from various regions of the world. In Mesoamerican linguistics, for instance, the following call for more extensive documentation of tonal morphology underscores the urgency of such research:

> We need full paradigms in grammars of tonal languages, not just rules, abstract representations or examples of how a given form is used in a natural context. This is a cordial invitation to descriptive linguists to enrich the field with new data on inflection. It matters. It matters in a time when most languages with complex morphology are dying. By doing so, we will be paying tribute both to the languages and to the field of linguistics, because in a hundred years from now, when all of us are gone, it will only be our data that shall remain for future linguists to continue increasing our understanding of our human languages. (Palancar 2016: 134)

In the field of Sino-Tibetan studies, tone systems have occasionally been misrepresented, as noted by Sun (2003b) for Tibetan and by Post (2015: 188) for Tani and languages of Northeast India more generally. Fortunately, languages with level-tone systems and rich morphotonological structures are now receiving increasing scholarly attention. Two grammars of Pumi, a language employing two tonal levels, have been published (Daudey 2014, Ding 2014). The present study of Yongning Na tone is intended as a contribution to this broader movement within Sino-Tibetan linguistics.

At this stage, the primary goal is to establish a precise description, which serves as the necessary basis for further research. Consequently, this volume includes numerous tables that lay out the paradigms in full. There is room for some progress in terms of economy of description: for instance, by identifying a set of morphotonological combination patterns as default for a given category of morphemes and reformulating the data for other categories as *identical to the standard pattern, except for…* rather than presenting all paradigms in tabular form. The long-term research agenda includes modelling the morphotonology of Yongning Na through computer implementation (finite-state modelling), a prospect discussed in the conclusion (Chapter 13).

[5] *Original text:* aucun parler bambara (...) n'a jusqu'ici fait l'objet de ce qui mériterait d'être considéré comme une description tonologique systématique visant à l'exhaustivité.

1.3.2 Theoretical backdrop

This study is *theory-informed*, not *theory-driven*: the primary objective is not to use selected data to engage in current phonological or morphological debates, but rather to provide an in-depth description of the language as it functions. This aim is widely shared among linguists, and is ultimately more significant than theoretical divergences. A key guiding principle is the utmost caution in avoiding Procrustean models. The theoretical backdrop to the present research is intended to remain as unobtrusive as possible, in keeping with the priorities of language description and analysis. Linguistic models will be invoked only as necessary.

In essence, the method used here adheres to the fundamental principles of classical structural-functional phonology, as set out in standard handbooks of phonological description (e.g. Martinet 1956: 15, 34–47). It is difficult to extract a single quotation that would neatly summarize the core tenets of this approach. Martinet devoted an entire volume to articulating *A functional view of language* (Martinet 1962). The following excerpt is drawn from another of his English-language works, *The internal conditioning of phonological systems*.

> A dynamic conception of language presupposes that we do not deal with it as we would with a dead body in the morgue, but try to look at it as a means of satisfying some of the human needs, and essentially that of communication. In other terms, it derives from a functional view of language (…). [E]xperience has shown that even if language is often used for the satisfaction of other needs as, for instance, that of communion, it is, in the last analysis, mutual understanding that determines the choices of the speakers. (…) At every point in time, with every speaker, what is said and how it is said will show a balance between the desire to communicate, and inertia, be it individual, i.e. reduction of energy, or social, i.e. preservation of traditional forms at the expense of personal comfort and communicative efficiency. (Martinet 1996: 2–3)

A major source of linguistic change lies in the constant competition between two opposing forces: phonological integration, on the one hand, and phonetic economy, on the other. Phonological integration tends to fill structural gaps within phonological systems, whereas phonetic economy tends to create such gaps. A straightforward example can be drawn from tone. A system with five level tones (Top, High, Mid, Low, Bottom) could be considered phonologically economical, as it maximizes the use of tone to convey lexical contrasts, while also enabling immense possibilities for tonal morphology and morphotonology. However, such a system is phonetically uneconomical, as distinguishing among a large number of level tones is perceptually difficult – for instance, differentiating sequences such as a Top followed by a High, a Top followed by a Mid, and a High followed by a Mid.

In synchronic description, attention to the conflicting factors that are constantly at play in speech communication leads to the adoption of the method advocated by Martinet under the name of *dynamic synchrony* (Martinet 1990). The primary focus remains synchronic, but the approach is enriched by observations on tensions within the system, assessing which of the competing variants are innovative and which are conservative.

At present, little is known about the diachrony of level-tone systems in Sino-Tibetan. Case studies in dynamic synchrony can yield converging evidence on diachronic tonal changes and their conditioning factors. Ultimately, what is needed is an approach that formulates generalizations about sound change that transcend any single language or language family. The goal is to establish an inventory of types of sound change and to arrive at an improved understanding of the conditions under which they arise. Haudricourt characterizes such an approach as *panchronic* (Haudricourt 1940, 1973; see also Hagège & Haudricourt 1978). Panchronic phonology derives its general laws inductively from a typological survey of precise diachronic events whose analysis reveals the common conditions under which they occur. In turn, panchronic laws can shed light on individual historical developments. The premise is that, within the broader set of potential changes, the direction of evolution in a given language is partly conditioned by the structure of its phonological system: which phonemic oppositions it maintains, the phonotactic constraints governing them, their role in morphophonology, and so on. (For examples, see Jacques 2011a, Lai 2023.)

A challenging mid- to long-term objective for historical linguistics will be to model the origins and evolution of level-tone systems with the same degree of precision as in studies of classical tonogenetic processes in Sinitic, Austroasiatic and Tai-Kadai. Tonogenetic developments in these families constitute a success story of panchronic phonology, demonstrating how a voicing opposition can be transphonologized into a phonation-type opposition, a tonal contrast, or a vowel quality distinction (Haudricourt 1965, Ferlus 1979). In Mon, for instance, the voicing contrast on initial consonants gave rise to two phonation types; in Vietnamese, it led to an increase in the number of lexical tones; while in Khmer, it resulted in a proliferation of vowel qualities.

Phonation-type oppositions have several phonetic cues. In addition to phonation type proper, they include differences in pitch, as well as differences in vowel articulation. This was already noted for a conservative variety of Khmer by Henderson (1952), and is confirmed by experimental studies of Mon (Abramson et al. 2015 and references therein). One or the other of these phonetic cues can become dominant: either vowel quality stabilizes as the new distinctive property (as in Khmer), or the distinctions become tonal (as in Vietnamese). A major structural factor in this bifurcation is whether the language already possesses tones at the time of transphonologization. If the language already has tones, the loss of voicing oppositions induces a split within the tonal system; otherwise, it leads to a vowel quality contrast, producing a two-way split in the vowel inventory. Comparative studies of numerous East and Southeast Asian languages corroborate this model while also offering opportunities for refinements through the study of tonogenetic processes in progress (e.g. Brunelle (2012) on Cham; Kirby (2014) on Khmer; Yang et al. (2015) on Lalo; and Pittayaporn & Kirby (2017) on Cao Bằng Tai).

Panchronic phonology holds similar promise for the study of the area of Sino-Tibetan to which Na belongs. A landmark study of tonogenesis in Rma (Qiāng 羌), Pumi (Prinmi), and Tangut concludes that “far from being an exotic case of tonogenesis without concomitant segmental change (…), [the primary tone split found in Rma, Prinmi, and Tangut] is strikingly similar to the classic model described for the development of tones from Old Chinese to Middle Chinese (Sagart 1999) and also from proto-Vietic to Vietnamese (Haudricourt 1954) in which final stops induced low-rising tones and smooth syllables developed high tones” (Sims 2021: 8).

Panchronic phonology is close – at least in my view – to *evolutionary phonology*, a theoretical framework that seeks to integrate insights from historical phonology and experimental phonetics. This approach aims to establish “a general link between neogrammarian discoveries, advances in modern phonetics, and phonological theory” (Blevins 2004: xiii). Its emphasis on the phonetic bases of change, following Ohala (1989), encourages a mutually enriching dialogue between experimental phonetics and historical phonology. Like panchronic phonology, evolutionary phonology represents a long-term research programme that holds promise of an increasing degree of precision and explicitness in modelling historical change.

While some differences exist in their stated principles and methodologies (see Labov 1994: 601, Andersen 2006, Iverson & Salmons 2006, Mazaudon & Michailovsky 2007 and Smith & Salmons 2008), I would argue that their common objective – explaining synchronic states in terms of the processes that give rise to them and formulating general laws of sound change – is ultimately more significant than their theoretical divergences. In practice, scholars working within panchronic phonology, evolutionary phonology, or other approaches to the typology of sound systems and sound change share the same fundamental goals.

This may seem far too much diachronic background for a synchronic monograph. However, I believe that synchronic descriptions such as the present study benefit from being situated within a diachronic research agenda. In historical linguistics, as in phonetics/phonology, "the devil is in the detail" (Nolan 2003), and the patient, fine-grained analysis of tonal patterns can constitute a good basis for developing a feel for the diachronic evolution of tone systems. Chapter 10 is devoted to examining synchronic variation and diachronic change from this perspective.

1.4 Field trips and collaboration with consultants

The present results are based on data collected since 2006. Four field trips to the village of Alawua were conducted from 2006 to 2009. Excluding time spent on travel and organizational tasks, the duration of these stays was 50 days in 2006, 35 days in 2007, 58 days in 2008, and 40 days in 2009. From July 2011 to October 2012, I had the wonderful opportunity of staying in China for long-term fieldwork. As my main consultant had by that time moved to the town of Lijiang to care for a granddaughter, I was based in this town as well, working with her for an average of two hours a day. In 2013, 2014, 2015, 2016, 2018, and 2024, I made short field trips to Lijiang and Yongning, continuing to work mainly with the same consultant.

Standard procedures for data collection, as reflected in the 'Method' section of papers in phonetics journals, tend to avoid any mention of personal familiarity between the investigator and the subjects. Such mentions are not merely deemed irrelevant but even suspect, since exchanges with consultants beyond providing instructions are viewed as a contaminant: a threat to the objectivity of the experiment. "The subjects were unaware of the purpose of the experiment" is considered a commendable state of affairs. However, linguists with experience in collaborating with consultants, whether in a language lab or in the field, are well aware of how profoundly the relationship established with the consultant influences research. A close examination of data collection in language laboratories suggests that key aspects in consultant selection and the formulation of instructions are often overlooked (Niebuhr & Michaud 2015).

Of course, different research purposes call for different data collection methods, and it would be thoroughly unreasonable to expect all investigators to develop personal familiarity with the subjects of their research. Nevertheless, in the process of describing a language, mutual understanding between the investigator and consultants is essential. In this light, the personal details presented here are not mere fieldwork anecdotes: they constitute relevant information on data collection.

1.4.1 First steps in the search for consultants

On my first field trip, Mr. Latami Wangyong 拉他咪王勇 (Latami Dashi 拉他咪达石), a native speaker of Na and a researcher in ethnology based in the Ninglang county seat, to whom I had been introduced by Picus Ding, accompanied me to his family's house, where I was invited to reside during all my stays in Yongning. He volunteered to work as a consultant (his code in the database of Naish speakers is M18). Mr. Latami has near-native command of both Southwestern Mandarin and Standard (Beijing) Mandarin. It was immediately obvious to both of us, when we began an elicitation session, that his long years of daily practice of Mandarin had taken their toll on his proficiency in Yongning Na.

He offered to help find a speaker who had a relatively homogeneous linguistic experience, having lived continuously in Yongning since childhood. For my part, I wanted to work with a male speaker for a technical reason: spectrographic and electroglottographic analysis, two techniques that I planned to use, are easier to perform on data from male speakers. We also agreed to look for a speaker whose age ranged between 35 and 65 years old. Younger speakers have limited command of the language; as for the oldest speakers, they are often the most proficient in Na, but at a certain age, speech becomes less audible and communication with strangers more difficult.

Mr. Latami therefore set out to find a suitable consultant in the neighbourhood. The procedure, as he narrated it to me, was as follows. He invited a candidate over to his place after dinner, treated him to liquor and sunflower seeds, and launched a conversation about how the language was being lost by the younger generation. He then explained that there was currently a foreigner staying in the house who wanted to study and record everyday language, and he asked if the person would agree to work as a language consultant.

Several acquaintances were thus invited, said they would consider the proposal, and eventually declined. There may have been a number of reasons for this. Mr. Latami's perspective was that, being unfamiliar with the nature of linguistic fieldwork, they were suspicious of potential misuse of the information they would provide and wary of the reputational damage they might suffer if their name became associated with debatable materials about Na language and culture. There are enough examples of ludicrously simplified depictions of Na culture produced to cater to the tourist industry (as reviewed in Appendix B, §B.3.5) to justify their cautious stance.

On the other hand, these individuals had known Mr. Latami since he was a child and had reason to trust that he would not collaborate in a research project that might harm the community's image. In his own research on Na culture, Mr. Latami takes care to gather viewpoints from a range of relatives and acquaintances. After finalizing a draft of one of his books, he circulates copies to Na people literate in Chinese and solicits their feedback. Only after incorporating their criticisms, corrections, and comments into the final version does he proceed with publication. While this offers no absolute guarantee against resentment among community members concerning the book's contents, it may help to allay suspicions.

The reluctance of these Na speakers to participate in data collection provides insight into traditional Na society as a highly conservative agricultural society where deviant behaviour is met with sharp reproach. This has linguistic consequences: "the strong networks typical of rural life" (Milroy & Milroy 1985: 379) not only favour archaism but also foster innovations that increase morphological complexity – a process opposite to what occurs in cases of creolization. For instance, reflecting on the development of person agreement marking on complementizers in Bavarian German (as reported by Bayer 1984), Peter Trudgill speculates that such developments occur only in tightly-knit rural communities. The example given is (4), where *ob* 'whether' receives the second-person agreement marker *-st* (Trudgill 2011: 82, 112–113).

(4) … obst du noch Minga kummst
whether you-SG to Munich come
'whether you are coming to Munich'

Returning to Yongning Na, the development of the rich morphotonology described in the present volume may have been favoured by the same social factors that initially led potential consultants to decline sharing their knowledge of Na with me. (Some initial observations on the social dynamics of Yongning Na morphotonology are set out in Chapter 10.)

Mr. Latami Dashi's mother supported her son's efforts to reassure potential consultants, explaining that they need not be intimidated by the tasks proposed to them. She clarified that the goal was not to collect folklore but to study the everyday language, and that the initial stage of the work was as simple as saying words such as 'head' and 'hand'. While she did not succeed in persuading others, she eventually convinced herself and volunteered as a consultant.

1.4.2 Main language consultant

My Na language teacher is Mrs. Latami Daeshilamu /**lɑ˧tʰɑ˧mi˥ ʈæ˧ʂɯ˧-ɬɑ˩mv̩˩**/. (Her code in the database of speakers of Naish languages is F4.) She was born in 1950 into a family of commoners – the majority group among the Na, distinct on the one hand from the chieftain's family, which constituted the nobility, and on the other hand from the serfs. Her birthplace is the hamlet called Alawua /**ə˧lɑ˧-ʁwɤ˧**/, close to the monastery of Yongning, called *dgra med dgon pa* in Tibetan, a name rendered in Chinese as *Zhāměisì* 扎美寺. The full address is: Yúnnán province, Lìjiāng municipality, Nínglàng Yí autonomous county, Yǒngníng district, Ālāwǎ village (云南省丽江市宁蒗彝族自治县永宁乡阿拉瓦村). The founding of this hamlet is recounted in the narrative *Elders3* {0004532} (for information on the narratives and other online resources, see §1.5.1). My teacher later established a home of her own in a neighbouring hamlet, slightly closer to the road leading to the Yongning marketplace.

My teacher is attached to traditions, closely associating them with the teachings of her grandmother, whom she remembers as an outstanding character who tactfully managed a large household. In narratives, she refers to her grandmother as /**ə˧si˧**/, 'great-grandmother, ancestor of the third generation' – the term of address used by her own children, thereby pointing her out as a model to the next generation. Locally, my teacher is regarded as a connoisseur of Na customs: over the past two decades or so, villagers experiencing doubts about ceremonial procedures have come to her for guidance.

At the same time, she is acutely aware of the profound transformations Na society has undergone since her childhood and does not cling to a bygone past. She is an open-minded character, gaily deriding in retrospect the prejudices that once prevailed in Na villages. For instance, in an account of the introduction of vegetables such as courgettes and eggplants in Yongning, she recalls that distrustful and indignant villagers would warn, "Don't eat those: they are grown in shit!" Yet, these new crops were eventually adopted, along with traditional Chinese methods for fertilizing soil. (See the narrative *Housebuilding2* {0004549}.)

In her childhood, my teacher was one of the actors in a film about the Na and their unusual matrilineal and matrilocal family structure (discussed in Appendix B, §B.3): *'A-zhu' marriage among the Naxi of Yongning.*[6]

[6] I was not able to access this film. Chinese title: 《永宁纳西族的阿注婚姻》. Black and white. Duration: about 60 minutes (six film portions of ten minutes each). Production date: approximately 1966. Advisor: Qiū Pǔ 秋浦. Scenario: Zhān Chéngxù 詹承绪 and Yáng Guānghǎi 杨光海. Director: Yáng Guānghǎi 杨光海. Photography: Yuán Yáozhù 袁尧柱. Sound recording: Zhào Déwàng 赵德旺. Animations: Zhèng Chéngyáng 郑成杨. Narration: Zhōu Qìngyú 周庆瑜. Summary: "Before Liberation, the Naxi of the people's commune of Yongning, in Ninglang Yi Autonomous

Later, one of her sons became an anthropologist specializing in Na society, bringing her into contact with many of his colleagues. She has thus witnessed how the Na of Yongning have become an object of curiosity and fantasy, and how Na culture has been folklorized for the promotion of the tourist industry.[7] Her experiences and reflections have shaken some of the beliefs passed down to her by her grandmother, such as Buddhist faith. While conscientiously performing the prescribed rituals in her daily life, her belief in Buddhist teachings such as reincarnation has waned, though this has not diminished her commitment to the ideals of benevolence and respect for others. The narratives recorded reflect her awareness of the cultural relativity of the customs and traditions of the Na, to which she nonetheless remains deeply attached.

Unlike more traditional parents, who viewed the monastery as the most prestigious prospect for boys, she encouraged her children – girls as well as boys – to study in the Chinese school system, which she considered a better gateway to a life free from daily toil in the fields. All four of her children have since found employment outside Yongning: one in the county town of Ninglang, two in Lijiang, and one in faraway Shenzhen (Guangdong). Although she lived continuously in the village from birth and hardly ever left the Yongning plain (and, indeed, seldom left her own village) until she moved to Lijiang in 2010 to care for a newborn granddaughter, she always had a keen awareness of the wider world.

Her belief that, beneath differences in local customs, the human heart is the same everywhere occasionally emerges in the narratives she agreed to record along the course of our collaboration. She often draws parallels between the situations depicted in her stories and those of the present day. For instance, reflecting on the aspirations of apprentice monks to find a good master, she pointed out the analogy with my own study of Na, which likewise depended on the guidance of dedicated teachers – just as her grandson required good instructors at his university in Kunming. Following her instruction, I have always addressed

County, lived in a feudal society, but they preserved distinctive characteristics of primeval matrilineal societies. They had matrilineal households in which the maternal side constituted the core of the family. They retained a style of marriage (‘A-zhu’ marriage) in which men did not take wives into their family, and women did not marry into another family. This documentary film records this form of matrimony according to the facts.” *Original text:* 在云南省宁蒗彝族自治县永宁公社的纳西族，解放前处于封建领主社会，但长期以来还保存着原始母系社会特征，保存着以母系为核心的母系家庭，保存着男不娶，女不嫁的“阿注婚姻”。男阿注到女阿注家过夜，晚上来白天走的“半同居”婚姻生活。本片对这种婚姻形式，特点和母系家庭作了如实的记录。 The film is part of a series about “ethnic minorities” initiated in 1957: 少数民族社会历史科学纪录片.

[7] One example among many is a report for the French tabloid *Paris Match*. The magazine’s special issue “China is changing” (« La Chine change », May 2001) contained no fewer than ten pages dedicated to the “Mosuo” (pp. 52–61), including an interview with Latami Dashi and his mother, Latami Daeshilamu (p. 60).

Photo 1.2: The main language consultant, Mrs. Latami Daeshilamu (**lɑ˧tʰɑ˧mi˥ ʈæ˧ʂɯ˧-lɑ˩mv̩˩**), shopping at the Yongning marketplace. Spring 2008.

her as ‘mother’ (**ə˧mɑ˧**) and have been the grateful recipient of her affectionate care throughout my stays. She is a model of tact, expertly fine-tuning relations both within the family and beyond. Despite her declining health and the heavy workload of a mother and grandmother, she has been a patient and encouraging teacher. While she may well take pride in having raised four children under harsh circumstances, she possesses a sharp sense of humour and has never been one to pose as a guardian angel, muse, or Madonna (to borrow Baudelaire’s impassioned wording: “l’Ange gardien, la Muse et la Madone”).

She was a stutterer in adolescence but later overcame this difficulty. I might not have noticed had I not been informed, but I now interpret the rare cases of stammering in recorded audio documents as a remnant of this earlier difficulty. She is also known in the community for speaking fast. Finally, although she has never suffered from any major otorhinolaryngological ailments, she has noticed changes in her voice over the years and can no longer sing the high-pitched songs of the Na. The reason is partly social: in local custom, singing is regarded as an activity for young people. The voices of singers over forty (or fifty at the latest) are deemed unattractive, and it is unusual for women past fifty to sing songs. When my teacher agreed to perform a song – understanding its value for language documentation and recognizing that most younger Na speakers no longer learn traditional songs – it became apparent that lack of practice over the years had left her unable to sing them.

She never travelled to other Na villages beyond the Yongning plain, such as Pujjo **pʰv̩˧dʐo˧** (Labai 拉伯) or those in Muli county. She speaks a little Southwestern Mandarin and incorporates common Chinese words into her Na speech, but her proficiency remained very limited until she moved to Lijiang in 2010 and found herself in a predominantly Chinese-speaking environment. Apart from the initial vocabulary elicitation, which was conducted in Chinese, fieldwork was carried out entirely in Yongning Na, with the consultant providing explanations in her own language without translation into Chinese. While this inevitably slowed the elicitation process compared to working with a bilingual consultant, monolingual fieldwork has its own advantages, allowing me to develop a better command of the language.

In addition to phonological materials, the set of texts by consultant F4 expanded from a single traditional story to over one hundred monologues, addressed to the investigator, on topics of her own choosing. She does not consider herself a skilled performer of oral literature, having never been formally trained as a custodian of oral traditions – a role reserved for men in the local ritualist tradition (called *Ddabe* **dɑ˧pɤ˧**, and related to the Naxi *Dobbaq* **to˧mbɑ˩**: see Fāng Guóyú & Hé Zhìwǔ 1995, Lǐ Dázhū 2015). However, she yielded to the investigator's eagerness for narratives and audio recordings in general. Aware of my interest in history, ethnology, and sociology, she spoke about life in Yongning in *the old times*, a conveniently vague era encompassing her childhood and youth. The result is a collection of monologues that belong to ordinary, casual speech as opposed to codified performance of oral literature (following the distinction set out in Dournes 1990).

At the same time, she is keenly aware of the stakes involved in narrating oral history (Milan 2024), including the didactic and mobilizational power of personal stories in "an oral history regime" (Bulag 2010: 100-105). She therefore exercises caution, avoiding topics that she perceives as sensitive. This results in a degree of stylistic evenness and uniformity, limiting both the range of speaking styles and the diversity of content. No claim is made that this corpus (presented in some detail in §1.5.1) constitutes a balanced dataset in any sense. From the perspective of morphotonology, the primary objective was to assemble a sufficiently rich dataset to confirm at least some of the tonal combinations obtained through elicitation, cross-checking them against materials that, in the phonetician's admittedly coarse classification, fall under "spontaneous speech" as opposed to phonological elicitation.

1.4.3 Other language consultants

During my first field trip (2006), as I began working with Mrs. Latami (consultant F4), I reflected on possibilities to extend the study to additional speakers, mindful of the textbook arguments for working with multiple informants:

> Data from varied sources can guard against distortions resulting from dressage, the observer's paradox, faulty questioning, or prescriptive influences of one individual's idiolect. Working with several speakers will provide the researcher with points of comparison so that he or she can learn to distinguish between reliable and unreliable data. (Chelliah & de Reuse 2011: 180–181)

In the household where I stayed, two family members lived in Yongning year-round: Mrs. Latami and her daughter-in-law Gisso, the wife of her second son. Gisso (born in 1973) agreed to assist with basic elicitation. As shown in a documentary film by Franck Guillemain about my linguistic fieldwork in Yongning (Guillemain 2020), Gisso excels at summarizing ongoing activities in a single word or short phrase and at describing the use of farming tools and implements. Her precise, confident elocution is a real asset to the linguist. She does not hesitate to correct my grammatical errors, enabling me to move beyond basic intelligibility to a more precise command of the language.

However, Gisso declined to record continuous texts such as narratives or dialogues, explaining that she did not feel up to the task. Unlike Latami Daeshilamu, who, after a few days of vocabulary elicitation, agreed to record a narrative, Gisso maintained her initial decision, providing only short responses to my questions, apart from a few brief songs recorded in 2007.

Both Mrs. Latami and Gisso had limited spare time, so I would work with one or the other depending on who was available. One day in December 2007, when both were occupied, Gisso asked her niece to "replace" her for a work session, offering an opportunity to get insights into the speech of a younger-generation speaker. Qiddeu (F6), born in 1987, was a high school student in the Ninglang county seat, returning home only for holidays. Vocabulary elicitation revealed that her lexicon was limited and that her phonological system was highly simplified, closely aligning with Mandarin Chinese. These observations are discussed in a joint book chapter with Latami Dashi: "A description of endangered phonemic oppositions in Mosuo (Yongning Na)" (Michaud & Latami 2011; Chinese translation: Michaud & Lātāmī 2010).

In 2008, two additional speakers were recorded. The first was Mr. Ho Jjacee **ho˧dʐɤ˧tsʰe˥** (He Jiaze 何甲泽), hereafter M21, born in 1942. A retired cadre (*gànbù*

干部), he had spent two years in Kunming and three in Yongsheng. However, elicitation tasks proved challenging due to hearing difficulties – he reported complete deafness in one ear and severely reduced sensitivity in the other. His experience of various dialects also contributed to linguistic variability beyond what would have been ideal for phonological consistency. In particular, eliciting stable tone patterns for compound nouns proved unfeasible: patterns given in one session were sometimes rejected in another, only to be reaffirmed later with full confidence.

Later, M21's youngest son, Ddeezzhi **ɖɯ˩ɖʐɯ˧** (He Duzhi 何独知), born in 1974, kindly agreed to participate. His command of Na appears comparable to that of F5. Like F5, M21 and M23 declined to record anything beyond vocabulary and isolated sentences.

1.4.4 Examination of transcribed texts and direct elicitation

"Texts are the lifeblood of linguistic fieldwork. The only way to understand the grammatical structure of a language is to analyse recorded texts in that language" (Dixon 2007: 22). Following classical methods in linguistic fieldwork, observations from continuous speech (in this study, primarily narratives) are verified and further investigated through elicitation. A fair amount of transcribed narrative has been collected and transcribed – more than four hours in total – but a considerably larger corpus would be needed to study the language's tone system on the basis of these texts alone. Not all possible tonal combinations in compound nouns occur in the texts, and no amount of continuous speech would suffice to obtain all the combinations of numerals and classifiers required for the study of numeral-plus-classifier phrases (Chapter 3). Systematic elicitation was therefore carried out to investigate one area after another of Yongning Na's tonal grammar.

Larry Hyman, reflecting on his study of the tones of another Sino-Tibetan language (Thlangtlang Lai), makes the following observation:

> Clearly the speaker had never heard or conceptualized noun phrases such as "pig's friend's grave's price", "chief's beetle's kidney basket" (…). It would not impress any psychologist, and it would definitely horrify an anthropologist. (…) However, when I need to get 3×3×3×3=81 tonal combinations to test my rules, the available data may be limited, or the language may make it difficult to find certain tone combinations. I am personally thankful that speakers of Kuki-Chin languages are willing to entertain such imaginary notions. It is most significant that the novel utterances are produced with the appropriate application of tone rules. (Hyman 2007a: 34)

Using this method, L. Hyman completed the elicitation of tonal data for Thlangtlang Lai within six hours (Hyman 2007a: 9). Such swift progress is possible when the linguist is fortunate to work with consultants who possess strong metalinguistic abilities, engage with linguistic reasoning, and collaborate as colleagues. A well-known example is François Mandeville, a speaker of Chipewyan (Athabaskan family) who "possessed the extraordinary ability to dictate texts and to explain forms with lucidity and patience" (Li 1964: 132).

The relationship with consultants in Yongning Na was somewhat different. Over the course of our collaboration, they developed an understanding of a linguist's interests and aims. In particular, after dozens of hours of joint work, the main consultant became familiar with the investigator's body language. She would understand, for instance, a repetition-beseeching glance upward from the laptop screen, or volunteer an explanation when a lengthy pause suggested that the investigator was experiencing doubt. However, she did not develop metalinguistic intuitions beyond recognizing full homophony between two words. She did not reach the stage of identifying a word's tone with another word's, much less of naming a tonal category. Nor did she become a collaborator in the sense of learning to transcribe her language or engaging in the inventory and analysis of phonemes and tonemes, i.e. gaining training in linguistics. As a result, she did not provide feedback on analytical choices or on the transcriptions.

This clarification is provided in light of the recommendation that "linguists should make it a practice to explicitly indicate whether the tonal categories have been recognised by native speaker consultants and whether words cited have been confirmed in those categories by native speakers, or whether those categories are the result of the linguist's own analysis" (Morey 2014: 638). I would be delighted to work with native speakers as fellow linguists and hope that there will be opportunities for this in the future.[8]

1.4.5 The issue of cross-speaker differences

One of the findings from the systematic study of compound nouns (reported in Chapter 4) was the high degree of cross-speaker difference.

It is a general observation that tone is particularly susceptible to dialectal variation. Existing reports suggest that the greater Yongning area has not only

[8] I was fortunate to co-supervise the M.A. thesis of a native speaker who conducted a comparative study (Ā Huì 2016) on the tone systems of her own dialect – Shèkuǎ 舍垮 – and that of Alawua (Ālāwǎ 阿拉瓦), examined in the present volume. This was, of course, of the highest interest for me. However, A Hui's work did not include verification of my analyses for Alawua: she worked out the tone categories of her own dialect and compared them with transcriptions I provided, without questioning my notations. The two systems are fairly different: Shèkuǎ has only two tone levels (High and Low), whereas Alawua distinguishes High, Mid and Low.

some of the richest tone systems within the Naish-speaking area but also the greatest dialectal diversity. In this light, differences observed within the same village, and even within the same family, did not come as a great surprise.

Another dimension of diversity is the gap between age groups created by the ongoing shift to Chinese. Mrs. Latami's four children are more proficient in Chinese than in Yongning Na. All four left Yongning for work. Two married Han Chinese spouses with no command of Na; one married a Na speaker from a different dialect area (Luggu **lo˧gv̩˩**, known in Chinese as Běiqúbà 北渠坝), but mutual comprehension difficulties led the couple to communicate in Chinese instead. Another married a Na spouse from Yongning (Gisso, F5) but worked in Shenzhen (Guangdong) and rarely returned home. The next generation – her four grandsons and granddaughters – has even less command of Na, despite being cared for by Mrs. Latami in their early years.

There are also notable tonal differences between Mrs. Latami and her daughter-in-law Gisso, despite their living under the same roof. Documenting the tone system of multiple speakers is not simply a matter of verifying data: each speaker's system must be analyzed independently, with comparisons drawn only at a later stage. In addition to this analytical requirement, the investigation faced a practical limitation: transcribed texts are necessary to corroborate at least part of the patterns obtained through systematic elicitation, but consultants F5, M21, and M23 declined to record narratives. Even if they had agreed, assembling a sizeable collection of texts for each of the four speakers would have represented a formidable workload – not to mention the difficulty for the investigator of keeping the four systems distinct.

This raised a fundamental choice: whether to prioritize tonological depth or sociolinguistic breadth.

1.4.6 Tonological depth vs. sociolinguistic breadth

It was clearly not feasible to explore all areas of the tone system in equal detail for all four speakers. This led to the following dilemma: either limiting the scope of investigation – for instance, focusing on compound nouns, or on numeral-plus-classifier phrases – but eliciting data from a broad sample of speakers to achieve decent sociolinguistic coverage; or extending the study to more parts of the linguistic system, aiming for a comprehensive picture of the tonal grammar of a single speaker, with occasional extensions to other speakers.

The second approach was preferred: attempting as complete a description as possible of the linguistic system. Work with Mrs. Latami (consultant F4) appeared more promising because of her much stronger proficiency in Na compared to the two younger consultants (F5 and M23) and her relatively stable linguistic trajectory, in contrast to M21, whose long exposure to various Na dialects and to Chinese had introduced additional complexities. Basing the study primarily on data carefully verified with Mrs. Latami provided a reliable foundation for later comparative work, including cross-speaker and cross-dialect studies.

The ultimate research goal, conceived as a collective endeavour, is to document in fine detail the synchronic tone systems of a number of research locations within the Na-speaking area and, on this solid empirical basis, to conduct comparative analyses and gradually reconstruct the historical evolution of these tone systems. Ideally, this would lead to a full account of the origin and development of tone systems in Naish, shedding light on the stages that led from a non-tonal system to each of the present-day varieties. This is, of course, a long-term undertaking. The immediate task is to provide a detailed synchronic description of the tone system of one dialect.

The analyses of Yongning Na tone presented in this volume are therefore based on data from Mrs. Latami, unless otherwise mentioned.

1.5 Online corpus, dictionary, and other tools

A guiding principle in the present research is that a close association between documentation and research is highly beneficial to both (Garellek et al. 2020). If, as suggested by Whalen (2004), "the study of endangered languages has the potential to revolutionize linguistics" and "the vanguard of the revolution will be those who study endangered languages", then it is all the more regrettable that "enormous amounts of data – often the only information we have on disappearing languages – remain inaccessible both to the language community itself, and to ongoing linguistic research" (Thieberger & Nordlinger 2006; see also Woodbury 2003, 2011).

"[L]anguage documentation as a paradigm in linguistic research" has been recognized as offering several key benefits:

> (i) making analyses accountable to the primary material on which they are based;
>
> (ii) providing future researchers with a body of linguistic material to analyse in ways not foreseen by the original collector of the data; and, equally importantly,
>
> (iii) acknowledging the responsibility of the linguist to create records that can be accessed by the speakers of the language and by their descendants. (Thieberger et al. 2016: 1)

The necessity of making primary data available has been emphasized specifically in relation to the study of tone at a symposium on "Cross-linguistic studies of tonal phenomena":

> The point to be made is extremely simple: declare the status of the primary data for what it is, and allow an evaluation of the data and its subsequent interpretation to take place in view of the declared status of the data. (…) I in no way imply that the wealth of impressionistic data which we have on tonal phenomena are inherently wrong or that they misrepresent the situation in any particular language. (…) What I do, however, argue for is an unashamed scientific approach to the handling of tonal data, abiding with generally accepted criteria of scientific practice. This will not only raise the credibility of analyses, but may even lead to objective empirical testing of particular theories. (Roux 2001: 366)

Accordingly, the recordings conducted in Yongning have been made freely available online, document by document, since 2011.

1.5.1 Transcribed narratives and phonological materials

These recordings consist mainly of narratives and lexical or phonological elicitation sessions. They are accompanied by metadata (providing information about the recordings) and, as far as possible, by full transcriptions and translations. A list of documents is provided in the Abbreviations section, with one-click links to these resources.

The data is hosted by the Pangloss Collection,[9] a language archive developed at the *Langues et Civilisations à Tradition Orale* (LACITO) research centre within *Centre National de la Recherche Scientifique* (CNRS) (Jacobson et al. 2001, Michailovsky et al. 2014, Adamou et al. 2025). The goal of this archive is to preserve and disseminate oral literature and other linguistic materials in (mainly) endangered or poorly documented languages, providing simultaneous access to sound recordings and text annotation.

The Na media files are mostly audio, although a few video recordings were made in 2018. Some of the audio files are accompanied by an electroglottographic signal, which allows for high-precision measurement of the voice's fundamental

[9] Address of the interface since 2021: https://pangloss.cnrs.fr/. Direct link to the Yongning Na corpus: https://pangloss.cnrs.fr/corpus/Yongning_Na.

frequency, as well as other glottal parameters.[10] Documents that include an electroglottographic signal are marked with a special icon in the list of resources.

Figure 1.3 shows a passage from one of the documents as displayed on the web interface, featuring transcription, translation, and time-aligned audio. The original files can also be downloaded without requiring login.

The Na documents are archived with provisions for long-term conservation and will remain accessible regardless of any future changes to the web address of the Pangloss Collection's interface. Direct links to specific locations within a text are provided "to offer readers the means to interact instantly with digital versions of the primary data, indexed by transcripts" (Thieberger 2009). Seamless navigation between linguistic descriptions and data is the main improvement introduced in this second edition, fulfilling a hope expressed in the first edition ("It is hoped that, by the time the next edition of this volume is released, tools for resolving a multimedia document's identifier will be all set up and working, allowing for links from the digital book that will direct the user straight to the relevant passage in the online data").

The availability of these audio and electroglottographic data with synchronized transcriptions allows interested readers to develop a first-hand sense of the data and facilitates further research into a broad range of phonetic topics. Given the substantial time investment required to carry out a state-of-the-art experimental phonetic study, it is simply not feasible for linguists engaged in the description of an entire language (or multiple languages) to conduct a dedicated phonetic investigation to substantiate and refine every observation they make. However, it is possible to collect a sufficiently extensive dataset to enable colleagues with a specialized interest in phonetics to undertake such studies. The electroglottographic signal has not yet been systematically analyzed, apart from occasional use in auditory verification (the pitch can sometimes be perceived more clearly when listening to the electroglottographic signal than when listening to the audio recording). This signal could serve in the future as the basis for a phonetic study of tone implementation in Yongning Na (see §9.4).

Importantly, the documents are also open to entirely different uses, including aesthetic appreciation of a voice captured through the wonder of high-fidelity recording, and preserved unaltered through the magic of digital storage. I, for one, am not insensitive to the luxury of listening to the Yongning Na recordings at leisure. Once a recording session has concluded, the qualms and concerns of

[10] For an introduction to electroglottography, see the original report on its invention: Fabre (1957); a synthesis: Baken (1992); some caveats: Orlikoff (1998); discussions on measurable parameters: Henrich et al. (2004), Michaud (2004a), Herbst (2020); and applications to the study of specific linguistic issues, e.g. Brunelle et al. (2010) and Kuang & Keating (2014).

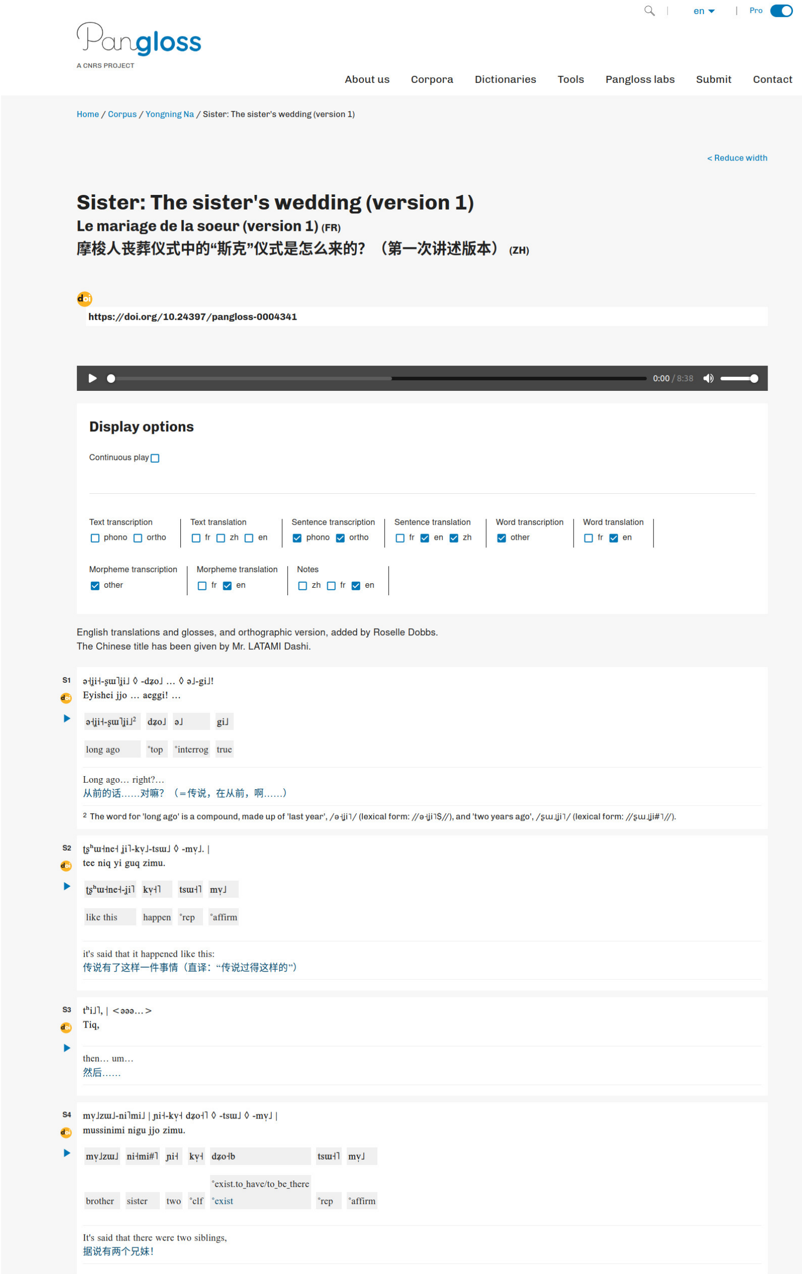


Figure 1.3: A passage from one of the documents as displayed on the web interface: transcription, translation, and time-aligned audio.

fieldwork recede, leaving room for "the fecund miracle of communication within solitude", to borrow Proust's evocative description of reading.[11]

A dictionary of Yongning Na is also shared over the Internet, as well as a group library aiming to facilitate the sharing of publications relating to Naish languages and cultures.

1.5.2 Dictionary

In the classical tradition of linguistic fieldwork, a language description should include a dictionary, alongside a grammar and a collection of texts. Together, these three components constitute what is often referred to as the "Boasian trilogy" (Foley 1999), in reference to Franz Boas's foundational work collecting North American languages (Boas 1902, Boas & Swanton 1911).

The lexicographic data collected during fieldwork on Yongning Na were progressively organized into a dictionary of Yongning Na (Michaud et al. 2024), available both as a PDF and in XML database format. It is designed not only as a tool for academic research but also as a gateway to Na culture and language. In addition to the version tailored for English-speaking readers,[12] separate versions have been prepared for Chinese-speaking[13] and French-speaking[14] audiences. Successive versions of the dictionary are numbered, providing a stable point of reference while allowing for continuous refinement over time. Like this book, and like the materials in the Pangloss Collection, the dictionary is freely available online under a Creative Commons license.

1.5.3 Group library

An online bibliography of Naish studies – covering research on Naxi, Na, Laze, and related languages – was initiated in 2015 as a Zotero group (Duong 2010) under the name 'Naish languages and cultures'.[15] The primary focus is on linguistic research, but the team of contributors also aims to include relevant ethnological, anthropological, historical, and sociological studies.

This bibliography is publicly available, and those interested in contributing to its enrichment and maintenance are encouraged to get in touch.

[11] *Original text:* la lecture, dans son essence originale, dans ce miracle fécond d'une communication au sein de la solitude (…) (*Journées de lecture*, in *Pastiches et mélanges*, Paris: Gallimard, 1919, p. 257).

[12] https://shs.hal.science/halshs-01204638/

[13] https://shs.hal.science/halshs-01744420

[14] https://shs.hal.science/halshs-01204645

[15] https://www.zotero.org/groups/395957/naish_languages_and_cultures/

1.6 A grammatical sketch of Yongning Na

This final introductory section provides a brief grammatical sketch of Yongning Na, outlining structural properties such as the organization of noun and verb phrases. These elements serve as background for the discussion of morphotonology in the chapters that follow. More substantial sketches can be found in Lidz (2016, 2019).

There are no stem alternations in Na, unlike in conservative Sino-Tibetan languages such as rGyalrong (Gong 2018, Zhang 2020). Suppletion is attested only in the verb ‘to go’, which has distinct forms for different grammatical contexts: nonpast /**bi˧**$_\mathbf{c}$/ (5), past /**hɯ˧**/ (6), past perfective /**hɤ˩**$_\mathbf{a}$/ (7), and imperative /**hõ˧**/ (8). Note that the nonpast form in (5) is translated as a preterite, a common tense for narratives in written English.

(5) ɖɯ˧-v̩˧ lɑ˧ **bi˥** | pi˧-dʑo˩ | ʐv̩˩-kv̩˩ tɕɯ˥-kv̩˩ mæ˩!
ɖɯ˧-v̩˧ lɑ˧ **bi˧**$_\mathbf{c}$ pi˥ -dʑo˥ ʐv̩˩-kv̩˩ tɕɯ˧˥
one-CLF.individual only **to_go.NONPAST** to_say TOP four-CLF to_lead
-kv̩˧˥ mæ˧
ABILITIVE OBVIOUSNESS
‘If only one [man] went, he could lead four [horses]! / If a man went alone, he would lead up to four horses!’ *(Caravans.119)* {0004531#S119}

(6) ɻ̩˩ʈʂʰe˧-ɖɯ˩mɑ˩ | tsʰi˧ɲi˧ | ə˧tse˧ **hɯ˧**-ɻ̩˩?
ɻ̩˩ʈʂʰe˧-ɖɯ˩mɑ˩ tsʰi˧ɲi#˥ ə˧tse$˥ **hɯ˧**$_\mathbf{c}$ -ɻ̩˩
given_name today INTERROG:why **to_go.PST** INCEPTIVE
‘Why has Erchei Ddeema gone away today? / How come Erchei Ddeema has left the house?’ *(BuriedAlive2.14)* {0004537#S14}

(7) “ɳæ˧=ɻ̩˩-se˩ | ə˧mv̩˩ le˩-ʝi˩~ʝi˩-ze˩! | se˧kʰɯ˩-ʁo˩ni˩ le˩-po˩ ʝi˩~ʝi˩-ze˩!” | pi˧-kv̩˩ mæ˩, | ho˧di˧ **hɤ˧**-dʑo˥!
ɳæ˧=ɻ̩˩ -se˧ ə˧mv̩˩ le˧- ʝi˧$_c$ ~ -ze˧ se˧kʰɯ˩-ʁo˩ni˩
2PL TOP elder_brother ACCOMP to_come RED PFV satin_headdress
le˧- po˧˥ ʝi˧$_c$ ~ -ze˧ pi˥ -kv̩˧˥ mæ˧
ACCOMP to_bring to_come RED PFV to_say ABILITIVE OBVIOUSNESS
ho˧di˧ **hɤ˩**$_\mathbf{a}$ -dʑo˧
Sichuan **to_go.PST.PFV** PROG
‘When [one’s brothers] had gone away [on a caravan] to Sichuan, [people in the village] would say: “You, your brother is going to come back! [He] will bring back a satin headdress [for you]!” ’ *(Caravans.35)* {0004531#S35}

(8) no˧ | wɤ˩˥ | tsʰo˧qʰwɤ˩ **hõ˩**!
no˩ wɤ˩˥ tsʰo˧qʰwɤ˩ **hõ˧**
2SG again tonight **to_go.IMPERATIVE**
'Go again tonight!' *(Reward.63)* {0004447#S63}

Reduplication of verbs and adjectives is widely used to convey reflexivity and intensification (see §6.2).

In the lexicon, a major force driving the creation of disyllables is the pressure of homophony, resulting from the dramatic phonological erosion of the Naish languages compared with proto-Sino-Tibetan. For instance, [**ɻi˧**] can mean 'jar', 'ox', 'to do', 'to draw', or 'to inform'. The word 'jar' is typically disyllabic: [**ɻi˧mi˧**], incorporating the morpheme for 'mother', which has grammaticalized functions as both a feminine suffix and an augmentative suffix (see §5.1). In this case, the disyllabic form has entirely supplanted the original monosyllable.

1.6.1 Word order

Word order is S+O+V, or, more precisely: "[i]n unmarked, non-idiomatic, pragmatically neutral constructions, subject-object-verb word order is most common" (Lidz 2007: 48). Adverbials can appear in various positions. Agent marking is not obligatory: "agents are unmarked more frequently than they are marked, both in conversation and narrative" (Lidz 2011: 51). Like the agent adposition /**ɳɯ˧**/, the dative suffix /**-ki˧**/ is optional. These morphemes allow for deviations from the standard word order; they also serve a disambiguating function, as illustrated in (9). In this example, the presence of a noun phrase marked as dative clarifies that the verb /**so˩$_a$**/ is to be understood as 'to teach' rather than 'to study; to imitate', as the latter meaning does not take a dative particle.

The context of (9) is as follows: the speaker saw me sitting ready for an elicitation session, silently hoping that someone would spare some time for me. There was genuine uncertainty as to who that person would be, as I was working with two consultants within the household at the time. No subject is indicated in the first part of the sentence ('After feeding the pigs'), since it is self-evident from the situation that the speaker will perform this action. By contrast, in the second part of the sentence, both semantic roles are explicitly mentioned, providing a thorough response to the addressee's implicit concerns.

(9) bo˩-hɑ˧ | le˧-ki˧ le˧-se˩-ze˩, | njɤ˧ ɳɯ˧ | no˧-ki˧ | nɑ˩ʐwɤ˧ so˩-bi˩!
bo˩˧ hɑ˥ le˧- ki˧$_a$ le˧- se˩$_a$ -ze˧ njɤ˩ ɳɯ˧ no˩ -ki˧
pig food ACCOMP to_give ACCOMP to_finish PFV 1SG A 2SG DAT

nɑ˩ʐwɤ˥ so˩$_a$ -bi˧
Na_language to_teach/to_study IMM.FUT
'After feeding the pigs, I will teach you Na (myself) / I shall come over for a language lesson!' (Sentence transcribed on the fly in 2008)

As is well-attested across the Himalayas, nominals, including pronouns, need not be overtly expressed if they can be inferred from the discourse context. This applies to any definite argument of the verb, the head of a relative clause, or the head of a complex noun phrase. Another areal characteristic is that topic markers are among the most frequently occurring morphemes.

1.6.2 Tense, aspect and modality

Tense, aspect and modality are expressed through a number of postverbal particles and some preverbal particles, such as the DURATIVE /**tʰi**˧-/, the ACCOMPLISHED /**le**˧-/, the PERFECTIVE /-**ze**˧/ and the EXPERIENTIAL /-**dʑɯ**˧/. The verb 'to go' /**bi**˧$_c$/ has grammaticalized into an immediate future marker, while /**se**˩$_a$/ 'to complete' functions as a COMPLETION marker – both occurring in postverbal position.

Na stands at a great typological distance from languages such as Wolof, where a single predicative marker associated to the verb encodes the bulk of the grammatical content (Creissels & Robert 1998, Guérin 2015). Instead, Na patterns with Loloish (Yi) languages such as Lahu (Matisoff 1973) and Lalo (Björverud 1998), where particles have rich combinatorial potential. The interplay among these particles gives rise to a wealth of highly dialect-specific nuances, best appreciated through contextualized examples.

1.6.3 Question formation

In yes/no questions, the verb is preceded by an interrogative particle, /-**ə**˩/, as illustrated in (10). In *wh*-questions, exemplified by (11), the interrogative pronoun occupies the same slot as a noun phrase would in a statement.

(10) njɤ˧ mv̩˩ | ə˩-ɲi˩˥?
njɤ˩ mv̩˩˥ ə˩- ɲi˩
1SG daughter INTERROG COP
'Is this my daughter? / Are you my daughter?' *(BuriedAlive3.107)*
{0004538#S107}

(11) ɲi˩ ɳɯ˥ | ə˧mɑ˧-qɑ˧ tɕʰo˧ bi˧˥?
ɲi˩ ɳɯ˧ ə˧mɑ˧ -qɑ˧˥ tɕʰo˩ bi˧c
INTERROG.who A mother COMITATIVE together to_go
‘Who is going to go together with mama? / Who is going to accompany mama?’ *(FoodShortage2.17)* {0004657#S17}

1.6.4 Existential sentences

Na, like many other Sino-Tibetan languages, has several existential verbs. One is preferred for animate referents, as illustrated in (12). Another is “used with things that stand, protrude, or are perpendicular to a plane” (Lidz 2010: 358), as in (13). A third one applies to “objects within a container” (Lidz 2010: 361) – including abstract containers such as narratives, as shown in (14).

(12) hĩ˧ ɑ˥ʁo˩ dʑo˩
hĩ˥ ɑ˩ʁo˧ dʑo˩b
person home EXIST.ANIMATE
‘there are people at home’ *(Reward.11)* {0004446#S11}

(13) tɕʰi˧ tʰi˧-di˥-ɲi˩ mæ˩!
tɕʰi˥ tʰi˧- di˩a -ɲi˩ mæ˧
thorn DUR EXIST CERTITUDE OBVIOUSNESS
‘There are some thorns [on barley], not? / [barley] has beard, hasn’t it!’ *(FoodShortage.43)* {0004557#S43}

(14) dɑ˧pɤ˧ qʰwæ˧-qo˩ | qv̩˧ɻ̍˧ | tʰi˧-ʑi˥-kv̩˩ mæ˩!
dɑ˧pɤ˧ qʰwæ˧ -qo˧ qv̩˧ɻ̍#˥ tʰi˧- ʑi˥
priest_of_local_religion message inside name_of_a_mountain DUR EXIST
-kv̩˧˥ mæ˧
ABILITIVE OBVIOUSNESS
‘Mount Gheu’er **qv̩˧ɻ̍#˥** is mentioned in the tales of the Daba priests!’
Literally: ‘Mount Gheu’er **qv̩˧ɻ̍#˥** exists in the tales of the Daba priests’ *(Mountains.120)* {0004573#S120}

In contrast to these, there is an existential for diffuse entities, namely those that can neither be counted nor contained (and thus quantified), illustrated in (15). Finally, there is a generic existential verb, exemplified in (16).

(15) ə˧tso˧-mɤ˧-ɲi˩ | le˧-ʂe˧, | le˧-ʝi˥!
ə˧tso˧-mɤ˧-ɲi˩ le˧- ʂe˧a le˧- ʝi˥
all_sorts_of_things ACCOMP to_get ACCOMP EXIST
'We get all sorts of things (all the necessary paraphernalia for a ritual, a feast…) [so that] we have it (at hand for when we need it) / We get all sorts of things ready (for the ritual / the feast)!' (Field notes.)

(16) dzɯ˧-di˧ mɤ˧-dʐo˦
dzɯ˥ -di˩ mɤ˧- dʐo˧b
to_eat NMLZ NEG EXIST
'there was nothing to eat / there was no food left' *(FoodShortage2.4)* {0004657#S4}

Lidz (2010: 359) reports an additional existential verb in the Luoshui dialect, "used for the passing of time": "/**ku33**/ EXIST.T [Existential: Used with past existence of time] seems to have something of a connotation of 'pass,' and may be a fairly recent grammaticalization from a lexical verb". In the Alawua dialect (studied in the present volume), this verb, /**gv̩˧c**/, retains the lexical meaning 'to flow, to go by, to elapse (of time); to take place, to occur (of an event)'.

1.6.5 Sentence-final particles

Among sentence-final particles, several function as epistemic or evidential markers: /**tsɯ˦**/ for hearsay, /**mv̩˧**/ for affirmation, /**le˩**/ for exclamation, and /**mæ˧**/ to convey obviousness. Combinations of these particles allow for a considerable range of nuances. Their meaning in context arises through an interplay with intonational cues: the three-level tone system does not impose strong constraints on a sentence's intonation, leaving ample room for the expression of nuances of doubt, surprise, and other attitudes and emotions through intonational means, such as overall raising of the pitch register, pitch range expansion, lengthening, and changes in phonation type. (This topic is taken up in Chapter 9.)

1.6.6 Noun and noun phrase

The classifier, preceded by a numeral or a demonstrative, follows the head noun, as illustrated in (17) with the phrase 'a dumb person'. A numeral and a classifier can also form a well-formed phrase on their own. Such a phrase may receive a suffix, as in 'a family' in the same example.

(17) ɖɯ˧-ʑi˩=ɻ̍˩-dʑo˩ | zo˧bæ˩ | ɖɯ˧-v̩˧ dʑo˩ tsɯ˩ | mv̩˩!
ɖɯ˧ ʑi˩b =ɻ̍˩ -dʑo˥ zo˧bæ˩ ɖɯ˧ v̩˧
one CLF.households ASSOCIATIVE TOP dumb_man one CLF.individual
dʑo˩b tsɯ˦ mv̩˧
EXIST REP AFFIRM
'It is said that, in a household, there was a dumb man.' *(Lake3.3)* {0004348#S3}

The counting system is decimal.

The system of classifiers is extensive, with many items still transparently related to nouns. Some classifiers function as 'self-classifiers', occurring without a head noun; for instance, /**ɖɯ˧-kʰv̩˦**/ 'one year' is acceptable, whereas ‡ **kʰv̩˦ ɖɯ˧-kʰv̩˦** ('year one-year') is not. Classifiers are discussed in Chapter 3.

The order in noun compounds follows a *determined+determiner* pattern. By contrast, a small number of lexicalized compounds involving adjectives have the reverse order, e.g. 'dry field' /**pv̩˧lv̩˧**/ ('dry'+'field'). Compounds are discussed in Chapter 4.

There is no grammatical gender or agreement.

The demonstrative system distinguishes between proximal and distal forms. The proximal demonstrative, /**ʈʂʰɯ˥**/, is homophonous with the third-person singular pronoun. The pronominal system has inclusive and exclusive first-person pronouns, as well as associative forms referring to 'me and my kin', 'you and your kin', etc. Syntactically, pronouns in Na behave like nouns – as in languages such as Chinese, Japanese, Bambara, and Zarma (Creissels 1995: 29) – rather than functioning as pronominal indices.

The clause relativizer is /-**hĩ˥**/, as illustrated in (18).

(18) go˧mi˧ | tʰi˧-ki˧-hĩ˧ | ʈʂʰɯ˧-ʑi˥...
go˧mi˧ tʰi˧- ki˧a -hĩ˥ ʈʂʰɯ˥ ʑi˩b
younger_sister DUR to_give REL DEM.PROX CLF.households
'the family to which the younger sister pledged herself… / the household which the woman entered upon marriage…' *(Sister3.18)* {0004344#S18}

1.6.7 Verb and verb phrase

Directionality is marked before the verb, using /**gɤ˩**-/ 'upward', /**mv̩˩**-/ 'downward', or disyllabic expressions (§6.3.3).

Verb serialization is employed in various constructions, including resultatives, equivalents of Chinese '*de* 得' constructions, and expressions of movement (Lidz

2010: 397-405). Imperatives are conveyed solely through intonational means, except for the verb 'to go', which has a distinct imperative form. The syntactic structure is identical to that of statements, except in the case of the verb 'to go', which has a distinct imperative form, as discussed at the outset of this section (§1.6).

Adjectives function as stative verbs, as they "can take aspect marking, be negated, and can be modified by the intensifier" (Lidz 2010: 362). The intensifiers are /**ɖwæ˦**/, which precedes the adjective ('very ADJ') or verb ('V a lot'), and /**ʐwæ˩**/, which means 'extremely' and occurs exclusively with adjectives. The copula is not used with adjectives but may be added after any verb to express certainty (Lidz 2010: 354).

In addition to examining the lexical tones of nouns (Chapter 2) and verbs (§6.1), the following chapters address various syntactic phrase types in which morphotonological adjustments occur. These include phrases containing nominal classifiers (Chapter 3); compound nouns (Chapter 4); combinations of content words with grammatical elements – nouns in Chapter 5, and verbs and adjectives in §6.2-6.7 –; object and verb (§6.8-6.9); and subject and verb (§6.10).

2 The lexical tones of nouns

> The fundamental problem of a tonological description is to arrive, for each tonal type in the language, at an underlying tonal structure that does not merely record what distinguishes the realizations of the different types in a more or less arbitrarily chosen context, but that makes it possible to predict the full set of combinatorial properties characterizing each type.
>
> *Original text:* Le problème fondamental d'une description tonologique est de déterminer, pour chaque type tonal existant dans la langue, une structure tonale sous-jacente qui ne se borne pas à enregistrer ce qui différencie la réalisation des différents types que connaît la langue dans un contexte plus ou moins arbitrairement choisi, mais qui permette de prédire l'ensemble des propriétés combinatoires qui caractérisent chaque type.
>
> (Creissels 1994: 179)

This chapter examines the lexical tones of nouns. It is customary in tonal studies to proceed "from the tones of nouns to the general organization of the system" (Rialland & Sangaré 1989; see also Hyman 2014: 526-527). In Yongning Na, tonal oppositions are partially neutralized when words are spoken in isolation. This casts a subtle veil over the tonal categories, hiding some of them from casual observation. These cases of neutralization make it necessary to consider how tones behave in context, making this chapter a gateway to the Yongning Na tone system as a whole.

The chapter follows an analytical progression. It starts out from a static inventory of tone patterns across domains of varying lengths and gradually moves towards a systematic analysis. This approach mirrors, as closely as possible, the course of analysis during fieldwork: building up from surface data. The aim is to allow the reader to assess the argumentation step by step and to consider possible alternative analyses, rather than presenting a complete analysis from a top-down perspective.

2.1 A static inventory of tone patterns

Words spoken in isolation (often termed *citation forms*) provide the starting point in the earliest stages of fieldwork. Table 2.1 presents an overview of the tone patterns found in monosyllabic nouns spoken in isolation. Due to the relatively small number of monosyllabic nouns in the language, a strict minimal set (nouns distinguished solely by tone) could not be identified.

Table 2.1: Tone patterns attested over monosyllabic nouns in isolation.

phonetic realization	preliminary label	example
non-rising, non-low	M ? H ? HM ?	/**ʐwæ**/ 'horse'
low-rising	LM ? LH ?	/**bo**/ 'pig'
mid-rising	MH ?	/**ʈʂʰæ**/ 'deer'

At this initial stage, the essential information is that provided in the leftmost column in Table 2.1, describing the three patterns as follows: a non-rising, non-low pattern; a low-rising pattern; and a mid-rising pattern. The second column offers preliminary labels for these patterns, using level tones: L(ow), M(id), H(igh), and their combinations. These labels merely suggest, at this point, how the categories might be conceived in terms of phonological levels. Justification for employing a level-tone analysis comes from morphophonological alternations in which these tones participate; evidence for this will be provided later on in the course of the analysis. The question marks in the 'preliminary label' column emphasize that these possible labels were proposed on a first pass.

The three surface patterns are the same for monosyllables that belong to other word classes, such as verbs. The restrictions on the tones of monosyllables spoken in isolation are the following.

(i) There are no examples of contrastive falling contours.

(ii) There is no opposition between a high tone and a mid tone: only one type of non-low, non-rising tone is observed. Its realizations span the entire upper part of the tonal space, varying from mid to high, with a flat or falling contour.

(iii) There is only one contour that begins on a low pitch. In terms of level-tone labels, this observation can be stated as follows: there is no opposition between LM and LH.

(iv) There are no examples of low, non-rising tones.

Over disyllabic nouns, seven patterns are observed, as shown in Table 2.2. Since lexical roots are monosyllabic, disyllables result from various processes, such as addition of the female gender suffix /**-mi˩**/, found in 'dog' and 'sow' (this suffix is examined in detail in §5.1.2). At this point, the aim is to propose a static inventory covering all attested tonal categories of disyllabic nouns, irrespective of their internal structure.

Table 2.2: Tone patterns attested over disyllabic nouns in isolation.

1st syllable	2nd syllable	preliminary label	example
non-low	low	M.L ?	/**dɑ.ɻi**/ 'mule'
non-low	low-rising	‡ M.LM	–
non-low	mid-rising	M.MH ?	/**hwɤ.li**/ 'cat'
non-low	mid	M.M ?	/**po.lo**/ 'ram'
non-low	high	M.H ?	/**hwæ.ʈʂæ**/ 'squirrel'
low	low	‡ L.L	–
low	low-rising	L.LM ? L.LH ?	/**kʰv̩.mi**/ 'dog'
low	mid-rising	L.MH ?	/**õ.dv̩**/ 'wolf'
low	mid (or high)	L.M ? L.H ?	/**bo.mi**/ 'sow'

Table 2.2 includes two unattested combinations, marked with a double dagger (‡) in the *preliminary label* column. If the tone of the first syllable is non-low, there are four observed tonal patterns on the second syllable: low, mid, high, and mid-rising. Conversely, if the tone of the first syllable is low, three patterns are attested on the second syllable: low-rising, mid, and mid-rising.

The restrictions on the distribution of tones on disyllables can be described in static terms as follows:

(i) Only two tones contrast on the first syllable: low and non-low. There can be no contour on the first syllable.

(ii) A non-low tone cannot be followed by a low-rising tone.

(iii) A disyllable cannot be low throughout.

(iv) There is no contrast between a low+mid pattern and a low+high pattern.

There are also strong limitations on tone patterns in trisyllabic nouns: only twelve patterns are attested. The data in Table 2.3 is based on trisyllabic nouns whose degrees of lexical integration vary more widely than those of the disyllables in Table 2.2. These trisyllabic nouns range from transparent compounds –

such as ‘Year of the Dragon’ and ‘Year of the Snake’ – to fully indecomposable words, such as ‘lips’. (For decomposable compounds, a hyphen is placed between the two parts.) As with disyllables, the focus here is on a static inventory; the rules relating the tones of compounds to those of their components will be analyzed later (in Chapter 4).

Table 2.3: Tone patterns attested over trisyllabic nouns in isolation.

1st σ	2nd σ	3rd σ	preliminary label	example	meaning
non-low	mid	mid	M.M.M ?	**ɖʐ̩ɤ.qʰwɤ.ʈʂe**	awl
non-low	mid	low	M.M.L ?	**mv̩.gv̩-kʰv̩**	Year of the Dragon
non-low	mid	high	M.M.H ?	**njo.bi.li**	lips
non-low	mid	mid-rising	M.M.MH ?	**bv̩.ʐv̩-kʰv̩**	Year of the Snake
non-low	low	low	M.L.L ?	**mo.jo.mi**	owl
non-low	high	low	M.H.L ?	**æ.tse.pʰæ**	kneebone
low	low	mid	L.L.M ?	**tʰo.kʰv̩.mi**	male dog
low	low	low-rising	L.L.LM ?	**dʑɯ.nɑ.mi**	wilderness
low	mid	mid	L.M.M ?	**tʰɑ.ʐwæ.mi**	donkey
low	mid	high	L.M.H ?	**æ.li.pʰæ**	mirror
low	mid	mid-rising	L.M.MH ?	**bi.pʰv̩-dʑɯ**	flood
low	mid	low	L.M.L ?	**bæ.bv̩-bv̩**	ladybird

Since there is a three-way opposition in tonal levels on the second syllable, these levels are labelled as ‘low’, ‘mid’, and ‘high’, whereas for the first syllable, where there is no opposition between mid and high, the two levels are simply labelled ‘low’ and ‘non-low’.

Based on the information in Tables 2.1-2.3, the following generalizations can be proposed:

(i) A non-low tone can be followed by one of four tones: low, mid, high, or mid-rising.

(ii) A low tone can be followed by low, low-rising, mid, or mid-rising.

(iii) A high tone can only be followed by a low tone.

(iv) Non-final syllables never carry a contour.

(v) An entire word cannot carry low tone on all of its syllables. L.L.L is not permitted in trisyllabic nouns, any more than L.L is found in disyllabic nouns or L in monosyllabic nouns – even though L.L is found in syllables 1 and 2 in some trisyllabic nouns, and in syllables 2 and 3 in others.

(vi) There can never be a trough: that is, a tone surrounded by higher tones (for example non-low followed by low followed by mid).

From the above data alone, it is not yet possible to ascertain whether these generalizations apply at the level of the word, the phrase, or the entire sentence, since "words produced in isolation are minimal utterances showing both lexical and utterance-level (postlexical) features" (Himmelmann 2006: 164). To preview later analysis, the relevant domain is a unit between the word and the utterance, which might be termed a phonological phrase. The term adopted here is *tone group* because the defining characteristic of this unit is its role as the domain for tonal processes. Tone groups are discussed in detail in Chapter 7.

A dynamic approach to the tone categories of nouns sheds light on the above generalizations from static inventories.

2.2 A dynamic view, bringing out the tonal categories

A dynamic perspective brings out six tonal categories for monosyllabic nouns and eleven for disyllabic nouns, as the initial patterns branch into subsets when considering contextual variations.

Readers are encouraged to consult Tables 2.8 and 2.9 for a synthetic overview of the noun tone system as it emerges from the analysis. These tables are also reproduced in the 'Quick reference' section at the beginning of this volume.

2.2.1 Monosyllabic nouns: Six tonal categories

As noted earlier, there are three surface patterns for monosyllables spoken in isolation: low-rising, non-low, and mid-rising. However, further examination reveals that these sets are not homogeneous. Consider, for example, the behaviour of /**jo**/ 'sheep', /**ʐwæ**/ 'horse', and /**lɑ**/ 'tiger', all of which are realized with the same non-low tone in isolation. When combined with the copula, they yield three distinct tonal patterns:

- /**jo˩ ɲi˩˥**/ 'is (a/the) sheep', with a low tone on the noun and a rising tone on the copula;

- /**ʐwæ˧ ɲi˥**/ ‘is (a/the) horse’, with a mid tone on the noun and a high tone on the copula;
- /**lɑ˧ ɲi˩**/ ‘is (a/the) tiger’, with a mid tone on the noun and a low tone on the copula.

Since the morphosyntactic context is the same, these differences indicate that these words belong to three distinct lexical tone categories, which all neutralize to non-low when spoken in isolation.

Likewise, the set of nouns realized with a low-rising tone in isolation, such as /**ʐæ**/ ‘leopard’ and /**bo**/ ‘pig’, is not homogeneous. Although differences between these nouns do not emerge when they are combined with the copula (as they do for non-low nouns), they become apparent in other contexts, such as object-plus-verb constructions. For example, ‘has bought leopards’ is realized as /**ʐæ˩ hwæ˧-ze˩**/, with an L tone on the perfective suffix /**-ze**/, whereas ‘has bought pigs’ is /**bo˩ hwæ˧-ze˧**/, with an M tone on the same suffix.

Among the three surface patterns observed for monosyllables spoken in isolation, only one (MH) corresponds to a single phonological set: all words realized with MH tone in isolation have the same tonal pattern in a given morphosyntactic context. In contrast, the two others constitute the neutralization of two or more lexical categories: the low-rising contour corresponds to two distinct lexical categories, while the non-low realization corresponds to three categories (see Table 2.8). In summary, a dynamic view reveals six tonal categories for monosyllables.

2.2.2 Disyllabic nouns: Eleven tonal categories

The same procedure as described above was applied to disyllabic nouns, i.e. examining the behaviour of nouns in different morphosyntactic contexts in order to find out how many tone categories need to be distinguished.

It was discovered that nouns realized with an M.M pattern in isolation belong to two distinct categories: one in association with which the copula bears an L tone, and one with which the copula bears an H tone. One set is illustrated by /**po˧lo˧**/ ‘ram’, yielding /**po˧lo˧ ɲi˩**/ ‘is (a/the) ram’; the other is exemplified by /**ʐwæ˧zo˧**/ ‘colt’, yielding /**ʐwæ˧zo˧ ɲi˥**/ ‘is (a/the) colt’.

Likewise, nouns realized with an M.H pattern in isolation form two distinct sets: one in association with which the copula bears an L tone, and one with which the copula bears an H tone. The first set is illustrated by /**kv̩˧ʂe˥**/ ‘flea’, /**kv̩˧ʂe˧ ɲi˥**/ ‘is (a/the) flea’, the other by /**hwæ˧ʈʂæ˥**/ ‘squirrel’, /**hwæ˧ʈʂæ˥ ɲi˩**/ ‘is (a/the) squirrel’.

Table 2.4: Examples illustrating the existence of three tone categories of nouns neutralized to L.H (≈L.M) in isolation. (First-pass tonal notations, to serve as a basis for further analysis.)

in isolation	meaning	with copula	with possessive
nɑ˩hĩ˥ (≈**nɑ˩hĩ˧**)	Naxi	**nɑ˩hĩ˧ ɲi˥**	**nɑ˩hĩ˧ = bv̩˧**
bo˩mi˥ (≈**bo˩mi˧**)	sow	**bo˩mi˧ ɲi˩**	**bo˩mi˧ = bv̩˧**
bo˩ɬɑ˥ (≈**bo˩ɬɑ˧**)	boar	**bo˩ɬɑ˧ ɲi˩**	**bo˩ɬɑ˧ = bv̩˩**

Finally, the nouns realized with an L.M pattern in isolation (which can also be transcribed as L.H, since there is no opposition between an L.M pattern and an L.H pattern, as noted above) fall into no fewer than three categories. These three categories are distinguished by intersecting evidence from two contexts: one in which a copula is added, and one in which a possessive is added, as shown in Table 2.4. The addition of the copula sets apart a category exemplified by the Na word for 'Naxi', after which the copula receives an H tone. The addition of the possessive sets apart a category exemplified by 'boar', which depresses the tone of the possessive to L, in contrast to its realization as M for the other words. Although the evidence used to bring out these tone categories is morphotonological – based on the behaviour of nouns in context –, the differences in the surface phonological tone strings shown in Table 2.4 must be ascribed to differences between the lexical items, and hence to differences in lexical tone category.

In total, this analysis yields eleven tonal categories for disyllabic nouns.

2.3 Phonological analysis of the tone categories of nouns

As outlined in the preceding paragraphs, a number of tonal categories were brought out on the basis of their different behaviour in various morphosyntactic contexts. The phonological analysis of these categories is up against an issue of circularity, since the tone categories of the simplest units – monosyllabic nouns – can only be revealed through examination of their combinations with other morphemes whose tone categories have not yet been independently analyzed. In practice, however, bootstrapping is often required when analyzing a new language variety: groping for a correct analysis through trial and error.

A step forward in the analysis of the tones of nouns was made possible by making progress in the analysis of tones on other morphemes. Through an analytic process described below, it was realized that the copula carries a lexical L tone, while the possessive carries a lexical M tone. This finding provided a basis for proposing a phonological analysis for each of the tone categories of nouns.

For example, consider the two tonal categories illustrated by /**lɑ˧ ɲi˩**/ 'is (a/the) tiger' and /**ʐwæ˧ ɲi˥**/ 'is (a/the) horse'. In the case of 'tiger', the copula surfaces with its own lexical tone, so that 'tiger' is analyzed as having a lexical M tone. This is the simplest case: the noun's phonological tone surfaces as such in this context, and does not modify the tone of the copula. (The same analysis is proposed for the disyllabic category exemplified by /**po˧lo˧**/ 'ram'.) By contrast, for 'horse' the copula surfaces with an H tone that must be supposed to be projected onto it by the noun. The noun 'horse', therefore, exemplifies a tone category characterized by an H tone that can only be realized on a following morpheme – a floating H tone.

This phenomenon warrants further explanation. First, the theoretical backdrop needs to be introduced (§2.3.1). On this basis, the synchronic facts about the floating tone in Yongning Na can be set out (§2.3.2). Additionally, §2.3.3 sheds comparative light on this phenomenon, showing that the floating H tone of Alawua corresponds to an overt H tone in two neighbouring dialects, and §2.3.4 discusses the possibility that floating H tones arose diachronically as a consequence of phonotactic constraints.

2.3.1 Theoretical backdrop: a quick introduction to floating tones

Floating tones, sometimes called vowelless tones (Goldsmith 2002: 84), are entities postulated to explain categorical modifications in the tonal string. They do not surface directly but affect the tones of neighbouring morphemes. For example, in Bamana (a language of the Mande subgroup of Niger-Congo), the definite article has a floating L tone as its signifier, as illustrated in (1-2). The examples are reproduced as (1a) and (2a) with tone indicated by accents; equivalents using tone-letters are provided as (1b) and (2b).

(1) a. Mùsò tɛ́ yàn
woman NEG.be here
'There is no woman/there are no women here.'

b. Mu˩so˩ tɛ˥ yan˩
woman NEG.be here
'There is no woman/there are no women here.'

(2) a. Mùsó-` tɛ́ yàn
woman-ART NEG.be here
'The woman is not here.'

b. Mu˩so˥- ˩ tɛ˥ yan˩
woman-ART NEG.be here
'The woman is not here.'

In (2), there is ample evidence of the presence of a floating L tone. First, contact between the L tone of the noun 'woman' and the floating L tone of the definite article results in the addition of an H tone at the end of the noun, in accordance with a general rule that inserts a buffer high tone between two low tones, hence /**mùsó**/ (L.H) instead of /**mùsò**/ (L.L). Second, the floating L tone triggers downstep[1] because the following morpheme, the negation marker /**tɛ́**/, carries an H tone.[2]

Comparative evidence generally indicates that floating tones originate from the reduction (complete segmental ellipsis) of a syllable. The Bamana article is believed to derive from an earlier form ***-ò**; the full form (comprising both a vowel and a tone) "is still attested in numerous varieties spoken on the geographic periphery of the Manding area: Mandinka, Xasonka, Worodugukan, Marka-Dafin, some Kagoro dialects" (Vydrin 2016).

A purely tonal morpheme arises when a morpheme that was originally expressed segmentally (as a syllable with a tone) is reduced to a mere tone, only manifesting itself through its association with another morpheme. This is not the only diachronic scenario, however: a morpheme can acquire a floating tone through the loss (complete segmental ellipsis) of one of its syllables. For instance, in earlier stages of the Igbo language, disyllabic verbs are postulated; in the present state of the language, "the syllabicity of the final syllables is lost, leaving floating tones, which (…) shift to the left and knock the first tones off in front of the verb morpheme" (Hyman & Shuh 1974: 94).

Concerning the choice of terms, Voorhoeve (1967: 424) initially used the labels "presegmental tonemes" and "postsegmental tonemes". In his search for an ade-

[1]Downstep refers to a distinctive lowering of tone. For historical background on this notion, see Rialland (1997). Downstep is not marked in the transcript here, because it is a phonological consequence of the presence of a floating L tone. Alternatively, one could provide a surface phonological representation marking downstep with the conventional exclamation mark: /**Mùsó !tɛ́ yàn**/.

[2]An alternative analysis of the Bamana data, following Dumestre (1987: 24-25), posits that the tone change is from L.H.H to L.L.H rather than from L.L.L to L.H.L. This alternative simplifies the analysis of trisyllables by assuming that the category of nouns exemplified by /**mùsó**/ 'woman' has a lexical LH pattern. Under this assumption, the underlying and surface tones of 'woman' in (2) would be identical (LH), with the change from the underlying form to the surface form occurring in (1). However, which of these analytic options is favoured has no bearing on the floating tone analysis: all authors agree that the floating tone triggers downstep of the following H in (2).

quate cover term for both sets, he seems to have been aware of the awkwardness of the pleonastic label "nonsegmental tonemes", which he grazed (but skirted) by referring to "nonsegmental H and L". He introduced the notion of *floating tones* in 1971. In the early 1970s, tonologists often enclosed this newly coined term in quotation marks at first occurrence, as illustrated by the following commentary about Fe'Fe' (Bantu, Niger-Congo):

> [In the word 'pot' /**cə̀g ´**/] an earlier high tone suffix was present. Historically, there was an accompanying vowel, but synchronically, a mere "floating" high tone is posited. (…) [T]his high tone causes the preceding L tone to raise to a raised-low tone via the process of low-raising. (Hyman & Shuh 1974: 86)

The term gradually gained acceptance and is now standard in studies of Bantu (see e.g. Franich et al. 2012: 33), of other languages of Africa (Idiatov & Van de Velde 2020: 102; Snider 2021; Green & Konoshenko 2022), and beyond.

The notion of floating tone is here applied to one of the lexical H tone categories of Yongning Na. This is motivated by the observation that the Na tone at issue is never realized on the word to which it is lexically attached: it can only be anchored to a following morpheme. In the present volume, "floating tone" is used as a synchronic concept, "the concept of *floating* having here the meaning of non-realised tones in an isolated context" (Somé 2000: 61).

An important caveat is that adopting this concept from Bantu tone studies does not imply that the diachronic origin is the same. In Bantu, all floating tones are assumed to originate from the loss of segmental material – the *segmental ellipsis* of a syllable –, whereas in Yongning Na, comparative evidence (set out in §2.3.3) suggests that the evolutionary mechanism was different.

2.3.2 The synchronic facts about the floating tone in Yongning Na

An example illustrating the floating H tone category in the Alawua dialect of Yongning Na is provided by the monosyllabic form for 'horse', realized in isolation as /**ʐwæ˧**/. The word for 'colt', realized in isolation as /**ʐwæ˧zo˧**/, offers a neat opportunity to extend the analysis to disyllables: the H tone that appears in /**ʐwæ˧zo˧ ɲi˥**/ 'is (a/the) colt' is interpreted as reflecting a floating H tone lexically attached to the noun 'colt', in a manner exactly parallel to /**ʐwæ˧ ɲi˥**/ ('is (a/the) horse').[3]

[3]To preview results to be discussed later: noun-plus-copula combinations behave tonally like object-verb combinations, of which a detailed account is presented in Chapter 6. The rules yielding the surface phonological tone pattern are syntactically conditioned: not all combinations of a #H tone and an L tone yield an /M…M.H/ sequence.

Since this floating H tone is the only type of H tone that may be lexically attached to a monosyllable, it is convenient to transcribe it as a simple H tone on monosyllabic nouns in the dictionary (Michaud et al. 2024) and in examples within this volume. For example, 'horse' is transcribed as //**ʐwæ**˥//. (The double slashes are used to distinguish underlying phonological forms from surface phonological ones.) For disyllables, however, an opposition exists between this floating H tone and a word-final H tone (as in /**hwæ**˧**ʈʂæ**˥/ 'squirrel'). This complexity of syllabic anchoring necessitates the use a nonstandard symbol: a symbol not used in the International Phonetic Alphabet. Desperate tones call for desperate measures: the pound symbol # was chosen to indicate the end of a lexical word, so that the word is notated as /**ʐwæ**˧**zo**#˥/ and the tonal category is designated as #H.

To illustrate further, consider the #H-tone word 'little brother' and the M-tone word 'little sister', which, when spoken in isolation, both carry an M tone on each syllable (/**gi**˧**zɯ**˧/ 'little brother', /**go**˧**mi**˧/ 'little sister'). In combination with the copula, the former yields /**gi**˧**zɯ**˧ **ɲi**˥/ ('is little brother'), with a M.M+H tone sequence, whereas the latter yields /**go**˧**mi**˧ **ɲi**˩/ ('is little sister'), with a M.M+L tone sequence. The analysis proposed is that 'little brother' possesses a final H tone which remains unassociated unless it can associate to a following syllable – a floating H tone.

The association of this floating H tone is conditioned by specific morphosyntactic factors. For instance, the H tone does not surface when the noun is followed by the possessive clitic /=**bv̩**˧/: thus, **gi**˧**zɯ**˧=**bv̩**˧/ ('of (a/the) little brother') is tonally identical to /**go**˧**mi**˧=**bv̩**˧/ ('of (a/the) little sister'). Whether the floating H tone surfaces or not in a given context is not solely a function of the syntactic class of the added morpheme. To preview data presented in §5.4.1, note that another clitic, the ASSOCIATIVE, can host a floating tone. Such complexities contribute to the necessity for a book-length description and analysis of tone in Yongning Na.

2.3.3 The floating H tone of Alawua corresponds to an overt H tone in two neighbouring dialects

Observations on a neighbouring dialect, that of the village of Pujjo **pʰv̩**˧**dʐo**˧ (Labai 拉伯), offers indirect supporting evidence for the analysis of the Alawua tonal category to which the word 'horse' belongs. My only contact with the Labai dialect to date has been through two work sessions with Mr. Lamu Gatusa 拉木·嘎吐萨, a researcher at the Academy of Social Sciences in Kunming. The monosyllables of Alawua that are analyzed as bearing a (floating) H tone correspond to monosyllables with an overt H tone in Labai. Examples are presented in Table 2.5. When one of the #H-tone words in Table 2.5 is spoken in isolation, its H tone does not surface (//M// and //#H// are neutralized to /M/), whereas in Labai H-tone items are realized as such. The #H::H correspondence among monosyllables is the most widely populated. The sole exception is the pair **hæ̃˩::hæ̃˥** for 'gold' (shown at the bottom of Table 2.5), for which no explanation can presently be offered. For the sake of completeness, Table 2.6 shows the correspondences for syllables carrying M tone in Labai. (A recording of some of these words is available online from the Pangloss Collection {0004489}.) Importantly, all monosyllables that belong to the M tone category in Alawua correspond to M-tone monosyllables in Labai.

Table 2.5: Tone correspondences between Alawua and Labai for monosyllabic nouns carrying H tone in Labai.

gloss	Alawua		Labai	correspondence
	/in isolation/	//lexical form//		
earth	**ʈʂe˧**	**ʈʂe#˥**	**tɕi˥**	#H::H
hail	**dzo˧**	**dzo#˥**	**dzo˥**	#H::H
sky	**mv̩˧**	**mv̩#˥**	**mv̩˥**	#H::H
fire	**mv̩˧**	**mv̩#˥**	**mi˥**	#H::H
star	**kɯ˧**	**kɯ#˥**	**kɯ˥**	#H::H
snow	**bi˧**	**bi#˥**	**mbi˥**	#H::H
pond	**ɖwæ˧**	**ɖwæ#˥**	**ɳɖwæ˥**	#H::H
canal	**qʰæ˧**	**qʰæ#˥**	**qʰæ˥**	#H::H
urine	**dʑɯ˧**	**dʑɯ#˥**	**ɳɖʐɯ˥**	#H::H
gall	**kɯ˧**	**kɯ#˥**	**kɯ˥**	#H::H
blood	**sɤ˧**	**sɤ#˥**	**sɤ˥**	#H::H
head	**ʁo˧**	**ʁo#˥**	**ʁo˥**	#H::H
Pumi	**bɤ˧**	**bɤ#˥**	**bɤ˥**	#H::H
man	**zo˧**	**zo#˥**	**zo˥**	#H::H
bronze	**æ̃˧**	**æ̃#˥**	**æ˥**	#H::H
salt	**tsʰe˧**	**tsʰe#˥**	**tsʰe˥**	#H::H
gold	**hæ̃˧**	**hæ̃˩**	**hæ̃˥**	L::H

On disyllables, the floating H tones of Alawua likewise correspond to H tones in Labai: disyllables that have a #H tone in Alawua display an H tone on their second syllable in Labai (while disyllables with M tone in Alawua carry M tone on

Table 2.6: Tone correspondences between Alawua and Labai for monosyllabic nouns carrying M tone in Labai.

gloss	Alawua	Labai	tone correspondence
small dike	**bo˧**	**mbu˧**	M::M
intestine	**bv̩˧**	**bv̩˧**	M::M
dew	**ɖʐv̩˧**	**ɳɖʐɯ˧**	M::M
wind	**hæ̃˧**	**hæ̃˧**	M::M
tobacco	**jɤ˧**	**jɤ˧**	M::M
field	**lv̩˧**	**lv̩˧**	M::M
wound	**mi˧**	**mi˧**	M::M
hole	**qʰv̩˧**	**qʰɚ˧**	M::M
serf	**wɤ˧**	**wɤ˧**	M::M
liquor/spirits	**ʐɯ˧**	**ʐɯ˧**	M::M
water	**dʑɯ˩**	**ɖʐɯ˧**	L::M
lake	**hi˩**	**hɯ˧**	L::M
thread	**kʰɯ˩**	**kʰɯ˧**	L::M
silver	**ŋv̩˩**	**ŋv̩˧**	L::M
bridge	**dzo˩**	**ndzo˧**	L::M
iron	**ʂe˩**	**ɕi˧**	L::M
bone	**ɻ̩˥**	**ɚ̃˧**	H::M
pond	**ɖwæ˥**	**ɳɖwæ˧**	H::M
rib	**ɬo˥**	**hõ˧**	H::M
earth	**di˩˥**	**di˧**	LH::M
rain	**hi˩˥**	**hɯ˧**	LH::M

both syllables in Labai). Some examples of these correspondences are provided in Table 2.7.

Another neighbouring dialect, that of the village of Wujiao 屋脚 (located just across the border with the county of Muli in Sichuan, 四川凉山州木里县屋脚乡), yields similar tonal correspondences. Field notes kindly provided in 2012 by Xu Duoduo 许多多, then a graduate student at Tsinghua University, reveal that words classified in the #H tone category in Alawua (which neutralize to M in isolation) are realized with an overt H tone in Wujiao. Xu Duoduo transcribed this H tone phonetically as ⁵³ (high-to-mid) based on commonly observed patterns in isolated production; in the absence of a ⁵⁵ (high-level) tone in her notes, this can

Table 2.7: Examples illustrating the M::M and #H::MH tone correspondences for disyllabic nouns between Alawua and Labai.

correspondence	meaning	Alawua	Labai
M::M	little sister	**go˧mi˧**	**go˧mi˧**
	ancestor	**ə˧pʰv̩˧**	**ə˧pʰɚ˧**
	Bai	**ɬi˧bv̩˧**	**li˧bv̩˧**
	mother	**ə˧mi˧**	**ə˧mi˧**
	body	**gv̩˧mi˧**	**gv̩˧mv̩˧**
	heel	**mv̩˧ʈʰɯ˧**	**mi˧ʈʰɯ˧**
	thigh	**do˧bæ˧**	**do˧bæ˧**
	buttock	**do˧bv̩˧**	**do˧bv̩˧**
	nostril	**ɲi˧qʰv̩˧**	**ɲi˧qʰɚ˧**
	back	**gv̩˧dv̩˧**	**gv̩˧dv̩˧**
	breast	**ʁɑ˧pv̩˧**	**ŋgɑ˧pv̩˧**
	belly	**bi˧mi˧**	**bi˧mi˧**
	plait	**hæ̃˧pɤ˧**	**hæ̃˧pɤ˧**
	sun	**ɲi˧mi˧**	**ɲi˧mi˧**
	moon	**ɬi˧mi˧**	**li˧mi˧**
	stone	**lv̩˧mi˧**	**lv̩˧mi˧**
	powder	**tsɑ˧bɤ˧**	**tsɑ˧mbɑ˧**
	hot spring	**ɻ̩˧qʰv̩˧**	**ɚ˧qʰɚ˧**
	paddy field	**ɕi˧lv̩˧**	**ʂɯ˧lv̩˧**
#H::MH	little brother	**gi˧zɯ#˥**	**gɯ˧zɯ˥**
	grandson	**ʐv̩˧v̩#˥**	**ʐv̩˧v̩˥**
	granddaughter	**ʐv̩˧mi#˥**	**ʐv̩˧mi˥**
	sole	**mi˧bɤ#˥**	**mi˧bɤ˥**
	nose	**ɲi˧gɤ#˥**	**ɲi˧ŋgɤ˥**
	craftsman	**po˧ɖʐɯ#˥**	**po˧ɖʐv̩˥**
	forehead	**to˧kɤ#˥**	**to˧kɤ˥**
	host	**dɑ˧pv̩#˥**	**ndɑ˧pv̩˥**

confidently be interpreted as an H tone. Examples include [**kʰv̩⁵³**] 'dog' (homophone: 'to steal'), [**kv̩⁵³**] 'garlic', and [**hṽ̩⁵³**] 'hair'. The corresponding words in Alawua are phonologically identical, except that the H tone is floating: //**kʰv̩**#˥// 'dog', //**kʰv̩**#˥// 'to steal', //**kv̩**#˥// 'garlic' and //**hṽ̩**#˥// 'hair'. No counterexample has been found.

These comparisons neatly confirm the H tone category independently postulated for the Alawua dialect. Needless to say, the tonal correspondences between Alawua, Labai and Wujiao warrant a systematic investigation based on a much larger dataset than that collected so far for Labai and Wujiao. The tone system of Labai calls for an in-depth study in its own right, with detailed comparisons to Alawua, Lataddi (Dobbs & La 2016, Fily 2022), and other dialects. While the tone system of Wujiao appears, at first glance, to be more similar to that of Alawua, it nonetheless exhibits sufficient differences to require its own comprehensive description.

2.3.4 The creation of floating H tones: a consequence of phonotactic constraints?

The correspondence between floating H tones in Alawua and overt H tones in Labai raises the question of whether an earlier overt H tone became floating in Alawua, or conversely, whether an earlier floating H tone became overt in Labai. Cross-linguistically, the more common scenario is that overt tones are set afloat by changes in metrical structure. For instance, in Manding languages (a branch of the Mande subgroup of Niger-Congo), it has been hypothesized that, at one stage of language history, heavy syllable rhymes (VV or VN) could carry up to two tonal levels, and that, at a later stage exemplified by Bambara, the distinction between heavy and light rhymes was lost, with the second tone on former heavy rhymes becoming floating (see Creissels & Grégoire 1993; also Konoshenko 2008 on developments in Liberian Kpelle concerning the delinking of H in an earlier LH category).

It seems reasonable to hypothesize that in Naish languages the diachronic change was likewise from overt tones to floating tones. A specificity of Alawua, in contrast with Labai, is the prohibition of tone-group-initial H tone. As a consequence of this exceptionless rule, it is impossible at the surface phonological level to have an H tone on a monosyllable spoken in isolation. Supposing that there used to be an overt H tone on monosyllables, the opposition between tone categories M and H for monosyllables would be threatened when the number of possible contrasts for a group-initial syllable collapsed from three (H, M, L) to two (M, L). Delinking the H tone could then be seen as a response to this threat, preserving the lexical distinctions.

The argument that H tones are set afloat *in order to preserve lexical distinctions* must be wielded with great care, since some phonological oppositions are irreversibly lost in the course of language history. "The linguistic system, with its myriad phonetic and semantic pressures effecting changes simultaneously and at times antagonistically, always emerges functionally unscathed, its semantic clarity intact" (Silverman 2015: 697). For now, I am content to record the fact that the tonal oppositions were maintained; determining *why* is a much more arduous task.

Regarding disyllables, supposing that there once existed nouns carrying a H.M pattern, their initial H tone would likewise have been affected when the prohibition against tone-group-initial H tone set in. One might imagine that this initial H tone would be set afloat as a response, similar to the case in monosyllables. However, this scenario does not take into account the state of the tone system as a whole. The representation shown in Figure 2.1 brings out the implausibility of a change whereby the initial H tone on disyllabic nouns is set afloat while the tonal category characterized by a final H tone remains unchanged. It does not appear likely that the *HM category could leap over the *MH category, as it were, to yield a floating tone (scenario 1). Instead, the change affecting word-initial H tones plausibly initiated a push chain, represented as scenario 2 in Figure 2.1: the *initial* H moves to a final position, and the *final* H is set afloat.[4] Scenario 2 is supported by the observed correspondences between Alawua and Labai shown in Table 2.7: disyllables with a floating H tone in Alawua correspond to disyllables with a final H tone in Labai.

To conclude this brief foray into diachronic territory, a caveat is in order. The scenarios presented in Figure 2.1 presuppose that, at the time of the change, the rest of the system was as it is now: specifically, that the M and H# categories were already as they are now. They also presuppose that the state of affairs in Labai is more conservative than that in Alawua; indeed, the tone patterns observed in Labai are reproduced in Figure 2.1 with the simple addition of an asterisk, as if they were proto-forms. These assumptions are not unreasonable given the close geographic proximity between the two dialects (see Map 1.1). Nonetheless, the history of these dialects is not without complexity, as evidenced by the diversity of tonal correspondences for monosyllables shown in Table 2.5.

The logical next step in diachronic analysis is to conduct a comprehensive comparison of the tone systems of Alawua and Labai. As noted in the introduc-

[4] I am grateful to Henriëtte Daudey and Denis Creissels for closely-argued discussion of this topic.

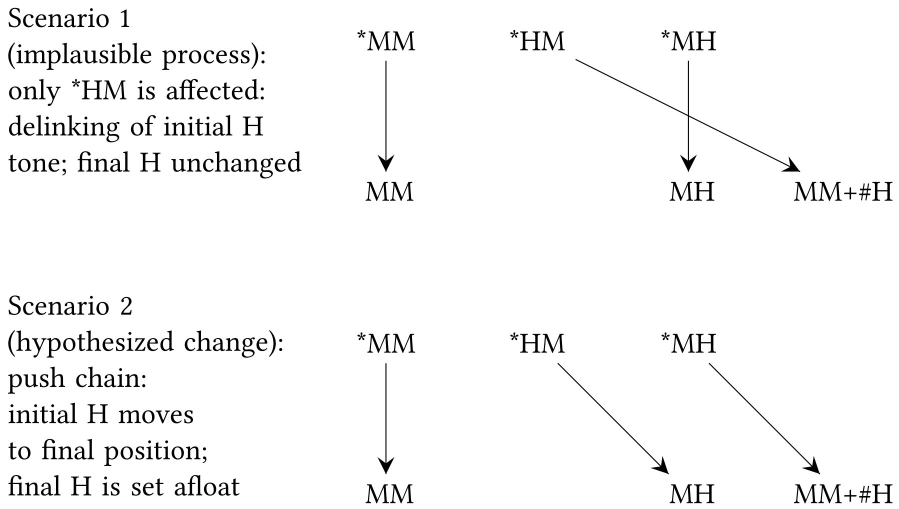


Figure 2.1: Two scenarios of evolution for *MM, *HM and *MH tone patterns over disyllabic nouns.

tion (§1.4.6), the ultimate research goal, conceived as a collective endeavour, is to document in detail the synchronic tone systems of a number of Naic-speaking villages and gradually reconstruct the evolution from a non-tonal stage to each of the present-day varieties. For the present argument, it suffices to conclude that there is solid comparative evidence for a process of delinking of H tones, which has resulted in the present-day floating H tones of the Alawua dialect.

2.3.5 Word-final H tone and the 'flea' H tone category

It was mentioned above that the words 'squirrel' and 'flea', realized with an M.H pattern in isolation (as /**hwæ˧ʈʂæ˥**/ and /**kv̩˧ʂe˥**/, respectively), belong to different lexical tone categories.

The word 'squirrel' has a straightforward tonal behaviour: its H tone attaches to the final syllable of the lexical word. This pattern is consistent across contexts. Under the present analysis, the first syllable receives an M tone by default, yielding a surface M.H pattern.

This pattern might alternatively be analyzed as consisting of an MH contour associated with the first syllable, with syllable-by-syllable association of successive levels: M on the initial syllable and H on the second. Such an analysis would contrast a word-initial MH tone with another category (MH#) in which the MH contour is associated with the word's final syllable (e.g. /**hwɤ˧li˧˥**/ 'cat'). However, there is no evidence to support the existence of a MH contour here. The analysis adopted for the category illustrated by ‘squirrel’ is therefore as a lexical-word-final H tone, transcribed as H#.

By contrast, the word ‘flea’ proves more elusive. ‘Flea’ is an apt eponym for the H tone category to which this example word belongs, serving as a mnemonic for its propensity to hop around from one host to another. When a word carrying this tone is pronounced in isolation, its H tone associates with the final syllable: for instance, /**kv̩˧ʂe˥**/ ‘flea’ has an M.H sequence on the phonological surface. However, when the copula is added, the result is /**kv̩˧ʂe˧ ɲi˥**/ (‘is (a/the) flea’), with an H tone appearing on the copula – a surface pattern identical to that observed with the floating H tone, as exemplified by /**gi˧zɯ˧**/ (‘little brother’) and /**gi˧zɯ˧ ɲi˥**/ (‘is (a/the) little brother’). Moreover, when the noun is followed by the possessive clitic /=**bv̩˧**/, no H tone reaches the surface: the form is /**kv̩˧ʂe˧=bv̩˧**/ (‘of (a/the) flea’), with M tone on both the noun and the clitic. This pattern mirrors that observed with the floating H tone in the corresponding ‘little brother’ example (/**gi˧zɯ˧=bv̩˧**/ ‘of (a/the) little brother’). Consequently, it appears that the ‘flea’ H tone is not anchored on the lexical word’s final syllable, nor does it behave in the same manner as the floating H tone in ‘little brother’ (#H), which never surfaces on the word to which it is lexically attached.

One possibility entertained in early reflections was that the association of this type of H tone might be specified relative to a higher prosodic unit. In Na, the syllable is the smallest relevant unit for tonal association, while the tone group is the highest relevant unit: successive tone groups are entirely independent from the point of view of their phonological tones. (On the notion of tone group, see Chapter 7.) Between these extremes, additional levels may be distinguished:

- the lexical word, to which tone categories are lexically associated;
- the tonal word, which is a combination of lexical words (such as a noun plus a verb in S+V or O+V combinations, or a noun plus a noun in compounds);
- the tonal phrase, comprising a tonal word along with any attached clitics and affixes.

In an article published in 2015, I proposed that the ‘flea’ H tone attaches to the right edge of the tonal phrase (Michaud 2015). Although lexically associated with a word, this type of H tone “hops” to the rightmost position within this higher prosodic domain. In cases where the final syllable of the tonal phrase is a suitable host, the H tone attaches to it; otherwise, as with the possessive suffix, the H tone remains unassociated and does not make it to the surface phonological level.

Upon further reflection, the search for a single defining characteristic of the ‘flea’ H tone relative to a specific level in the prosodic hierarchy appears unpromising. While this tone tends to be realized later than the word to which it is lexically associated – a behaviour it shares with the floating H tone –, it is not a sufficiently distinctive property to define the category. For example, the main consultant’s family name displays the ‘flea’ H tone: when spoken in isolation, it is realized as /**lɑ˧tʰɑ˧mi˥**/; when the associative clitic /=**ɻ̩˩**/ is added, it yields /**lɑ˧tʰɑ˧mi˧=ɻ̩˥**/ (‘the Latami family; the Latamis’); and with the addition of the agent adposition /**ɳɯ˧**/, it yields /**lɑ˧tʰɑ˧mi˧=ɻ̩˧-ɳɯ˥**/ (‘by the Latami family’). Impressionistically, this might suggest that the ‘flea’ tone tends to “hop” or “glide” to the right edge of the tonal phrase as the morphotonological opportunity arises. However, this is not a defining property, as the floating H tone yields the same results with these added syllables.

The extent to which a certain lexical tone tends to be realized close to the edge of a given level in the prosodic hierarchy is a statistical tendency, not to be taken as a defining property of that tone. Instead, the various lexical tones must be understood in terms of the full range of their interactions with other morphemes, as summarized in the tables presented in Chapters 2 to 6.

Thus, it does not appear feasible to pinpoint the exact phonological nature of the ‘flea’ H tone by proposing a single defining characteristic. Rather, it is more appropriate to regard it as one tone within the system, defined by the set of oppositions in which it participates across the full range of morphotonological contexts. For convenience, a nickname may be applied: the tone of ‘little brother’ (#H) is referred to as a *floating* H tone, while the tone of ‘flea’ is labelled a *hopping* H tone.[5] This is, however, merely a convenient nickname and should not be mistaken for a definition. The symbol chosen for transcribing the ‘flea’ H tone is H$, with the dollar sign added to distinguish it from the other two lexical H tones: H# (lexical-word-final H tone) and #H (floating H tone). The choice of this

[5] In the first edition of this book, the tone of ‘flea’ was nicknamed the *gliding* H tone. In retrospect, this choice of name appears unfortunate, as ‘glide’ is an established term in phonology, where it refers to semivowels, and in phonetics, where it describes the time course of pitch transitions ‘gliding’ from one target to another. For instance, Hyman & Leben (2000: 61) report the tones of the Hausa words /**káì**/ ‘head’ and /**káí**/ ‘you [masculine]’ as being realized as “an extra-H gliding down to a raised L”. In this context, the term ‘gliding’ refers to a gradual pitch movement, not a change in the syllabic anchoring of a tone. To avoid potential confusion, all five occurrences of the nickname “*gliding* H tone” have been replaced with “*hopping* H tone” in this edition.

arbitrary symbol reflects the abstract nature of this tone category as one of the distinctive tones within the Alawua system.

This category is relatively small, with only fifty example words in the dictionary (Michaud et al. 2024), compared with 209 examples for the word-final H tone category (H#) and 368 examples for the floating H tone category (#H). Nevertheless, it is firmly attested, and there are productive rules of compounding, prefixation and suffixation that feed into this category, as will be detailed in later chapters.

In summary, disyllabic (and polysyllabic) nouns with H tone must be divided into three categories, labelled H#, H$ and #H. An H tone on the final syllable of a disyllabic or polysyllabic noun may have different origins. It may be the realization of a High tone that is anchored to the final syllable of the lexical word (H#), or it may be the realization of an H$ tone. It is impossible to distinguish these in isolation. The third category – #H – denotes a noun that carries a floating H tone. To determine the lexical tone of a word, it must be observed in various contexts. For nouns, telltale environments are: when spoken in isolation (i.e. in tone-group-final position), in tone-group-internal position, and when followed by a clitic (e.g. the possessive). Only by matching the behaviour of a word in these various contexts can its lexical tone be identified with certainty.

2.3.6 An added complexity concerning L tone: A postlexical rule for all-L tone groups

'Sheep', realized in association with the copula as /**jo˩ ɲi˩˧**/ (i.e. with a low-rising contour on the verb), is a case in which the noun's phonological tone (an L tone) is hypothesized to surface as such. A slight complexity is that the copula following it surfaces with a low-rising tone. This is consistent with the exceptionless observation that an utterance cannot carry low tone on all of its syllables. In other words, sequences such as L+L (monosyllabic noun+copula) and L.L+L (disyllabic noun+copula) cannot surface as such, due to a general prohibition against all-L tone groups in the Alawua dialect of Yongning Na. The contour observed at the end of a sequence of L tones is interpreted as resulting from a postlexical addition of an extra tone. (This phonological rule is formulated in §7.1.1 as Rule 7: "If a tone group only contains L tones, a postlexical H tone is added to its last syllable".) The same analysis applies to the tonal category of disyllables exemplified by /**kʰv̩˩mi˩˧**/ 'dog'.

Concerning the transcription of low-rising contours, one might initially regard the choice between notation as LM or LH as a nonissue, since there is no surface contrast between LM and LH contours. However, in the case of /**jo˩ ɲi˩˥**/ (‘is (a/the) sheep’) or /**kʰv̩˩mi˩˥**/ (realization of ‘dog’ in isolation), there is a language-internal argument for analyzing the endpoint of the contour (i.e. the postlexical tone) as H rather than M. As will be set out in §2.4.3, the M tone in Na is phonologically inert; if the postlexical tone added to all-L sequences were M, this would be the only instance of a rule adding an M tone to a syllable that already hosts another tone. Therefore, the postlexical tone added at the end of a sequence of L tones is analyzed as H, and the rising contour found in the nouns ‘dog’ and ‘wilderness’, and in the phrase ‘is a sheep’, will henceforth be written as LH (hence /**kʰv̩˩mi˩˥**/, /**dʑɯ˩nɑ˩mi˩˥**/, and /**jo˩ ɲi˩˥**/, respectively).

The question of why monosyllabic L-tone nouns, when spoken in isolation, surface with M tone rather than with an LH contour (comprising their lexical L tone plus a postlexical H tone) is addressed in §2.4.6.

2.3.7 Tonal contours as sequences of level tones on the same syllable

As mentioned in the static overview presented earlier, no phonological falling contours occur in Alawua: no syllable carries HL, HM, or ML contours. Moreover, tone-group-initial H is never observed.

Rising contours, on the other hand, do occur. They are restricted to the final syllable of a tone group. A rising contour is not found on a non-group-final syllable, except in some special cases discussed in §7.3. The two observed rising contours are M-to-H and L-to-H (with the latter representing the neutralization of underlying LM and LH). Unlike the low-rising contour, the phonological behaviour of the mid-rising contour (MH) is straightforward. When a word is tone-group-final, the MH contour is realized as such: that is, as a rising tone beginning at a non-low pitch – for instance in /**ʈʂʰæ˧˥**/ ‘deer’ and /**hwɤ˧li˧˥**/ ‘cat’. (Note that when a word is pronounced in isolation, it forms its own tone group, so that its beginning is also the beginning of the tone group and its end is likewise the end of the tone group.) When there is a following syllable within the tone group, the MH contour unfolds, with its H component projecting onto that syllable. Unlike the floating High tone (#H), which cannot attach to a following clitic, the MH contour unfolds over any available syllable. With the copula, this produces /**ʈʂʰæ˧ ɲi˥**/ ‘is (a/the) deer’ and /**hwɤ˧li˧ ɲi˥**/ ‘is (a/the) cat’. With the possessive, it yields /**ʈʂʰæ˧=bv̩˥**/ ‘of (a/the) deer’ and /**hwɤ˧li˧=bv̩˥**/ ‘of (a/the) cat’. These observations provide strong evidence for analyzing the tonal contours of Yongning Na as sequences of level tones.

To preview the result of the analysis, low-rising contours can also be decomposed into level components. But they raise some subtle issues for description and analysis, which are addressed in the following paragraphs.

2.3.8 An alternative analysis of the //LM// and //LH// categories: Could the two terms of the opposition be //LM// and //LML//?

Concerning the two categories of tones that neutralize to a low-rising contour in isolation, illustrated by /**ʐæ˩˧**/ ‘leopard’ and /**bo˩˧**/ ‘pig’, at least two analytical options are open. Assuming (for reasons which will be explained below) that the perfective suffix normally carries an M tone unless affected by preceding tones, the L tone observed on the suffix in (3), contrasting with the M tone in (4), could be put down to a floating L tone (parallel to the floating H tone found in the tone category illustrated by ‘horse’). In this analysis, the floating L tone would remain unassociated when ‘to buy leopards’ is produced without a suffix, and would associate to the suffix when one is available.

(3) ʐæ˩ hwæ˧-ze˩
ʐæ˩˧ hwæ˧ -ze˧
leopard to_buy pfv
‘has bought leopards’

(4) bo˩ hwæ˧-ze˧
bo˩˧ hwæ˧ -ze˧
pig to_buy pfv
‘has bought pigs’

The first analytical option is thus to consider that the underlying tone pattern for ‘to buy leopards’ is //LM#L// – that is, an LM tone followed by a floating L tone. The L tone on the suffix in (3) would result from the association of the floating L tone. This //LM#L// pattern contrasts with the simpler underlying //LM// pattern for ‘to buy pigs’. (Recall that double slashes denote underlying phonological tone, whereas simple slashes indicate surface phonological tone.) In this view, the difference between the two object-plus-verb phrases would be attributed to a //LM#L// vs. //LM// tone contrast on the noun.

An alternative analytical option is suggested by the static observation that within a tone group,[6] an H tone is invariably followed by L tones. In this light, the lowering of the tone of the perfective suffix in (3) could be ascribed to the influence of a preceding H tone, which depresses the tones of all following syllables. In this scenario, the underlying form would be //**ʐæ˩ hwæ˥-ze˩**//, with H tone (and not M tone) on the verb, originating from a lexical LH tone on the noun ‘leopard’.

[6] About the key speech unit referred to here as a *tone group*, see full details in Chapter 7.

Of these two options, the second is currently favoured because there is no evidence elsewhere in the language for floating L tones. There are strong reasons to adopt the concept of a floating H tone in this dialect's morphotonology (including comparative-diachronic evidence presented in §2.3.4), whereas positing a floating L tone would be an *ad hoc* theoretical move. Nonetheless, the evidence is not overwhelming enough to entirely reject the analysis of the 'leopard' category as a sequence of three levels – LM plus a floating L tone (LM+#L). Such cases of analytical indeterminacy are important for understanding the system's evolutionary potential, a topic which will be addressed in Chapter 10.

Under the present analysis, the tone categories //LM// and //LH// contrast not only on monosyllables but also on disyllables. These two tones surface identically except when the word is followed by a clitic. For instance, //**bo˩mi˧**// 'sow, female pig' and //**bo˩ɬɑ˥**// 'boar, male pig' have the same surface phonological tone pattern both in isolation and when followed by the copula. In Alawua, L.M.L and L.H.L do not contrast at the surface phonological level, nor do L.M and L.H in final position within a tone group; accordingly, the surface forms may be transcribed interchangeably as /**bo˩mi˧**/ or /**bo˩mi˥**/ for 'sow' and as /**bo˩mi˧ ɲi˩**/ or /**bo˩mi˥ ɲi˩**/ for 'is (a/the) sow'; and as /**bo˩ɬɑ˧**/ or /**bo˩ɬɑ˥**/ for 'boar' and /**bo˩ɬɑ˧ ɲi˩**/ or /**bo˩ɬɑ˥ ɲi˩**/ for 'is (a/the) boar'. The contexts that allow these two lexical tone categories to be distinguished are exemplified by /**bo˩mi˧ = bv̩˧**/ 'of (a/the) sow' vs. /**bo˩ɬɑ˧ = bv̩˩**/ (which could also be transcribed as /**bo˩ɬɑ˥ = bv̩˩**/) 'of (a/the) boar': in the latter phrase, the possessive clitic // = **bv̩˧**// receives L tone, as shown in Table 2.4 above.

For the first category ('sow'), an analysis of the tone pattern as LH is ruled out: if it were transcribed as /**bo˩mi˥**/, then the form with the possessive would have to be /‡ **bo˩mi˥ = bv̩˥**/ ('of (a/the) sow'), since the possessive surfaces with the same tonal level as the noun's second syllable; but the tone sequence H+H is never observed elsewhere in the Alawua dialect of Yongning Na. By an exceptionless rule, a syllable following an H-tone syllable receives L tone; this will be referred to in §7.1.1 as "Rule 4". In view of the above argument, 'of (a/the) sow' is transcribed as /**bo˩mi˧ = bv̩˧**/, and the category illustrated by 'sow' is analyzed as //LM//, hence //**bo˩mi˧**//.

As for 'boar', a phonological analysis as //**bo˩ɬɑ˥**// makes good phonological sense, since all following tones are lowered to L, as expected after an H tone. When a word of this tonal category is followed by the possessive // = **bv̩˧**//, the latter carries L tone, just as after a disyllable with H# tone, as illustrated in (5a-5b), where notation as LH is adopted for 'boar'.

(5) a. hwæ˧ʈʂæ˥=bv̩˩
hwæ˧ʈʂæ˥ =bv̩˧
rat POSS
'of (a/the) rat'

b. bo˩ɬɑ˥=bv̩˩
bo˩ɬɑ˥ =bv̩˧
boar POSS
'of (a/the) boar'

Furthermore, all compounds involving the tone category of 'boar' (e.g. 'boar's head', 'boar's blood', and so on) have the same pattern, which can be described as L followed by H, then a sequence of L tones (L.H.L…). This is again parallel to the //H#// category, where compounds follow a pattern of M followed by H, then a sequence of L tones (M.H.L…).

Under the present analysis, both disyllables and monosyllables undergo neutralization of the underlying //LM// and //LH// categories when realized in isolation. By convention, the result of this neutralization is transcribed as /LH/.[7]

For disyllables of the 'sow' and 'boar' types, as for monosyllables of the 'pig' and 'leopard' types discussed at the outset of this section (§2.3.8), an alternative to the //LM//-vs.-//LH// analysis would be to posit an opposition between //LM// and //LML//. Under this approach, the tone pattern of 'sow' would be analyzed as //LM//, and that of 'boar' as //LML//. This analysis equally captures the fact that the tones of syllables following nouns of the 'boar' type are lowered to L: by Rule 5 ("All syllables following an H.L or M.L sequence receive L tone"), L.M.L must be followed by additional L tones.

This was the analysis I initially adopted, including in the Yongning Na glossary deposited in 2011 in the Sino-Tibetan Etymological Dictionary and Thesaurus (STEDT). At the time, describing this tone category as a sequence of three levels did not seem excessively complex, given the presence of other intricate categories such as //L+MH// and //LM+#H//. However, those two lexical tone categories are composed of two distinct parts, one associated to the beginning of the word and the other to its end, whereas //LML// would be the only pattern specifying three consecutive tonal levels. Moreover, it would be the only category for which an //ML// sequence is posited. For these reasons, notation as //LH// is adopted here.

[7]In the first edition of this book, /LH/ was used for monosyllables – hence /**bo˩˥**/ for 'pig' and /**ʐæ˩˥**/ for 'leopard' – but /LM/ was used for disyllables – hence /**bo˩mi˧**/ for 'sow', /**bo˩ɬɑ˧**/ for 'boar', and /**nɑ˩hĩ˧**/ for 'Naxi'. The same treatment is now applied to the LM-vs.-LH opposition across words of one or two syllables, hence /**bo˩mi˥**/ for 'sow', /**bo˩ɬɑ˥**/ for 'boar', and /**nɑ˩hĩ˥**/ for 'Naxi'.

That said, an analysis as //LML// remains a viable alternative. Such cases of analytical uncertainty are not merely abstract issues for phonological theorists: they also shed light on the system's potential for change, as language learners must navigate these competing analytical options when constructing their own phonological systems.

2.3.9 Cases of neutralization of the opposition between //LM// and //LH//: Is the product /LM/ or /LH/?

Monosyllables belonging to the //LM// and //LH// tone categories, such as //**bo˩˧**// 'pig' and //**ʐæ˩˥**// 'leopard', are realized in isolation with the same low-rising contour. Phonetically, both a low-to-mid and a low-to-high realization are acceptable. My consultants occasionally corrected my productions of this tone category when the starting point was not low enough, as this could lead to confusion with //MH//; on the other hand, they never corrected me for a mistaken endpoint (whether too high or too low).

To investigate whether there was a preference for a [Mid] or [High] endpoint in the phonetic realization of this low-rising contour, I tried fishing for corrections from Mrs. Latami (consultant F4). On several occasions, I deliberately produced two variants of words such as 'pig' and 'leopard', both with a low starting point: one with a moderate rise (approximately up to F_0 mid-range), and another with a strong, rapid rise towards a [High] final target. When asked which sounded better, the consultant consistently responded that both were correct (/**ɲi˧-bæ˧ | ho˩˥**/: *two-*CLF *correct*).

In this study, I choose to transcribe the product of the neutralization of //LM// and //LH// as /LH/ at the surface phonological level. There is no decisive phonetic argument for this notation, however. The outcome of tone neutralization tends to be phonetically less definite than that of consonantal neutralization. For instance, the opposition between coronal and retroflex stops in Yongning Na is neutralized before /**ɯ**/, with the resulting sound clearly realized as a retroflex: [**ʈʰɯ**] is a well-formed syllable, whereas [**tʰɯ**] is not. On the other hand, the product of the neutralization of //H// and //M// in isolation spans the entire phonetic space corresponding to these two tones: it is a non-low tone, but it may not be appropriate to assign it a more precise phonetic label, such as either 'high' or 'mid'. (The topic of phonetic implementation of tone sequences is addressed in Chapter 9.)

2.3.10 On the anchoring of tones to word boundaries

The M and L tone categories of the Alawua dialect of Yongning Na are hypothesized to associate with the *first* syllable of a lexical item and to spread from there across the entire word. By contrast, the three categories of H tones have their mode of syllabic association specified respective to the *last* syllable of the lexical item. There are thus tone categories that are anchored at the beginning of the word and others at its end. Two tone categories combine both anchorages: they are composed of two parts, with the first anchored at the beginning of the lexical word, and the second at its end. These are //LM+#H// and //LM+MH#//, exemplified in Table 2.9 by //**nɑ˩hĩ#˥**// 'Naxi' and //**õ˩dv̩˧˥**// 'wolf', respectively.

The '+' symbol in //LM+#H// and //LM+MH#// denotes the juncture between the first and second parts of these tones. Thus, in the phrase /**nɑ˩hĩ˧ ɲi˥**/ 'is (a/the) Naxi', the H tone is interpreted as the manifestation of a floating H tone, and the lexical tone of this category is analyzed as //LM+#H//: a //LM// contour followed by a floating H tone, //#H//. Likewise, //LM+MH#//, exemplified by //**õ˩dv̩˧˥**// 'wolf', is analyzed as a tonal category consisting of two parts: a //LM// tone followed by a final //MH// contour. In both cases, the pound symbol indicates the syllabic anchoring of the second part of these two-part tone categories: beyond the lexical word boundary in //LM+#H//, and on the last syllable of the lexical word for //LM+MH#//.

These two tone categories may seem awfully complex, as each consists of two parts that associate at opposite ends of the word. This complexity is real, and probably goes a long way towards explaining why only two such tone categories exist in Alawua, rather than the full range of theoretically possible combinations: for instance, there is no //LM+H$// tone. However, seen from within the Na tone system, the behaviour of //LM+#H// and //LM+MH#// is not all that complicated, as it follows directly from that of their constituent elements. Each part of the tone behaves as it does when occurring independently, with its own mode of association. The mode of association of the first part (//LM//) is straightforward, as is that of the second part (//#H// and //MH#//, respectively). The '+' symbol in //LM+#H// and //LM+MH#// serves as a notational convention to reflect the fact that these tones consist of two separate components, each of which associates independently. Their complexity stems from the coexistence of these two anchoring patterns, rather than from any additional structural interaction between them.

2.4 General observations about the system

Some generalizations about the Yongning Na tonal system emerge from the observations made above.

- There are three tonal levels: H(igh), M(id) and L(ow).
- The tone-bearing unit is the syllable, more specifically the syllable rhyme. There is no distinction in terms of syllable weight, and thus no need for a decomposition into moras: any syllable rhyme, including syllabic consonants, can function as a tone-bearing unit for one or two tonal levels.
- Among the six theoretically possible contours (HM, HL, MH, ML, LH, and LM), only the three rising contours (MH, LH, and LM) are attested as lexical categories, exemplified in monosyllables by //**ʈʂʰæ˧˥**// 'deer', //**ʐæ˩˥**// 'leopard', and //**bo˩˧**// 'pig'.
- At the surface phonological level, contours are restricted to tone-group-final position.
- At the surface phonological level, LM and LH are neutralized to LH.

Stated differently, each syllable within a tone group carries one of three levels, H, M, or L, while the final syllable carries one of the following: H, M, L, MH, or LH. There are no phonological falling contours (HL, HM or ML) on a single syllable, only rising contours.

The following paragraphs propose an overview of the system of lexical tones for nouns and offer some reflections on its structure.

2.4.1 Usefulness of an autosegmental approach

A first general observation that can safely be made in view of the data presented so far is that tone in Yongning Na is best analyzed within an autosegmental framework: that is, a model in which tones are *auto*nomous from *segment*s (i.e. vowels and consonants). Autosegmental models were originally developed for West African tone systems, but have been convincingly applied to certain languages of the Tibeto-Burman area (see, in particular, Hyman & VanBik 2002). The choice of this framework is motivated by language-internal evidence; it is by no means dictated by *a priori* theoretical commitments.

I have had the opportunity to learn and study two strikingly different tone systems of Asia: that of Yongning Na, which has phonetically simple yet morphophonologically complex tones, and that of Northern Vietnamese, which has phonetically complex but morphophonologically inert tones. In my view, it is clear that Yongning Na should be analyzed as a level-tone system, unlike Vietnamese, where "there are no objective reasons to decompose (…) tone contours into level tones or to reify phonetic properties like high and low pitch into phonological units such as H and L" (Brunelle 2009c; see also Brunelle et al. 2010, Kirby 2010, 2011). This topic is discussed further in §12.1.1.

2.4.2 Recapitulation of the lexical tone categories

Tables 2.8 and 2.9 set out an analysis of the six tone categories of monosyllabic nouns and the eleven tone categories of disyllabic nouns. To date, no single morphosyntactic context that brings out all the tonal contrasts of nouns has been found: each context brings out some oppositions while neutralizing others.

For instance, addition of the copula distinguishes //M// from //#H//, a contrast that is neutralized in isolation. On the other hand, addition of the copula neutralizes tonal contrasts that are present in isolation among the disyllabic tone categories //#H//, //MH#// and //H$//. All three yield /M.M+H/ in the copular frame, whereas in isolation, they are realized as /M.M/, /M.MH/ and /M.H/, respectively. Thus, to determine the lexical tone of a word, it is necessary to elicit it in several contexts.

Tables 2.8 and 2.9 provide information on the tone categories (i) in isolation, (ii) when followed by the copula //**ɲi˩**//, in frame (6), and (iii) when followed by the possessive clitic //=**bv̩˧**//.

(6) ʈʂʰɯ˧ ______ ɲi
DEM.PROX *target item* COP
'This is (a/the) ______.'

A recording of disyllabic nouns in frame (6) is available online under the identifier *NounsInFrame* {0004528}.

This set of three contexts is sufficient to bring out all tonal oppositions, except for the contrast between //LM// and //LH// in monosyllables. This opposition only surfaces in highly restricted contexts. As mentioned in §2.2.2, one such context is in combination with the verb 'to buy'. For instance, the //LM//-tone word 'pig' yields /**bo˩ hwæ˧-ze˧**/ 'bought pigs', whereas the //LH//-tone word 'leopard' yields /**z̩æ˩ hwæ˧-ze˩**/ 'bought leopards'.

Table 2.8: The lexical tone categories of monosyllabic nouns.

analysis	in isolation	+COP	+POSS	//example//	meaning
// LM //	LH	L+H	L+H	**bo˩˧**	pig
// LH //	LH	L+H	L+H	**ʐæ˩˥**	leopard
// M //	M	M+L	M+M	**lɑ˧**	tiger
// L //	M	L+LH	L+M	**jo˩**	sheep
// #H //	M	M+H	M+M	**ʐwæ˥**	horse
// MH# //	MH	M+H	M+H	**ʈʂʰæ˧˥**	deer

The proximal demonstrative //**ʈʂʰɯ˥**// always carries the same surface tone in (6), regardless of the tone category of the following noun. Consequently, only the tonal pattern of the target noun and copula is indicated in Tables 2.8 and 2.9. Conversely, no tone is indicated for the copula in frame (6), as its surface tone varies according to the tonal category of the noun.

Dots indicate syllable boundaries within the lexical word, while the '+' symbol marks the juncture between the noun and a following morpheme. For example, in Table 2.9, disyllabic L-tone nouns have the pattern L.LH in isolation and L.L+H when followed by either the copula or the possessive clitic. The word for 'dog' exemplifies this: it appears as /**kʰv̩˩mi˩˥**/ in isolation, yielding /**kʰv̩˩mi˩ ɲi˥**/ 'is (a/the) dog' and /**kʰv̩˩mi˩=bv̩˥**/ 'of (a/the) dog'.

The leftmost column in the tables ("Analysis") lists the lexical tone categories. The penultimate column contains example words, transcribed according to the phonological tone categories, following the conventions set out in §2.4.4.

In view of this picture of the noun tone system, the distributional observations made above can be flipped around. Instead of stating that "a monosyllabic noun that carries an M tone in isolation may belong to one of three distinct underlying categories", we can now say that the three non-contour lexical tones – //M//, //L//, and //#H// – all neutralize to /M/ when a monosyllable is pronounced in isolation. Among disyllables, //M// and //#H// neutralize to /M.M/; //H$// and //H#// neutralize to /M.H/; and //LM//, //LH//, and //LM+#H// neutralize to /L.M/.

When the possessive clitic //=**bv̩˧**// is added to a monosyllabic noun, as in /**ʈʂʰæ˧=bv̩˥**/ 'of the deer', tonal contours unfold over the two syllables of the resulting phrase. Specifically, //LH// yields /L+H/ (as does //LM//, following neutralization with //LH//), and //MH// yields /M+H/.

This last point provides evidence for a distinction between tonal contours (//LM//, //LH//, and //MH#//) on the one hand and the floating H tone (//#H//)

Table 2.9: The lexical tone categories of disyllabic nouns.

analysis	in isolation	+COP	+POSS	//example//	meaning
// M //	M.M	M.M+L	M.M+M	**po˧lo˧**	ram
// #H //	M.M	M.M+H	M.M+M	**ʐ̩wæ˧zo#˥**	colt
// MH# //	M.MH	M.M+H	M.M+H	**hwɤ˧li˧˥**	cat
// H$ //	M.H	M.M+H	M.M+M	**kv̩˧ʂ̩e˥$**	flea
// H# //	M.H	M.H+L	M.H+L	**hwæ˧ʈʂ̩æ˥**	squirrel
// L //	L.LH	L.L+H	L.L+H	**kʰv̩˩mi˩**	dog
// L# //	M.L	M.L+L	M.L+L	**dɑ˧ʝ̩i˩**	mule
//LM+MH#//	L.MH	L.M+H	L.M+H	**õ˩dv̩˧˥**	wolf
//LM+#H//	L.H	L.M+H	L.M+M	**nɑ˩hĩ#˥**	Naxi
// LM //	L.H	L.M+L	L.M+M	**bo˩mi˧**	sow
// LH //	L.H	L.H+L	L.H+L	**bo˩ɬɑ˥**	boar

on the other. The second component of a contour is realized on the possessive, whereas the floating H tone is not.

Why does the possessive clitic receive an H level when the noun has an MH contour but not when the noun has a floating H tone? One way to think of it – admittedly an *ad hoc* hypothesis, based on the investigator's non-native intuition – is that anchoring a //#H// tone is quite a different matter from hosting the H level of a //MH#// contour that is anchored to a preceding syllable. The //MH#// tone has a stable anchorage on the noun's last syllable, allowing it to unfold in the usual way: its M part is realized on the noun's last syllable, while its H part lands on the clitic. The MH sequence is firmly moored to the syllabic string at one of its ends (the M part), and this mooring suffices for the contour as a whole to hold firm and not lose its second part. In an intuitive sense, the H level, as it were, protrudes from the noun, jutting out onto the clitic. By contrast, the //#H// tone does not materialize because the necessary anchorage is not established in the first place. In derivational terms, a distinction must be made between *tonal anchoring* and the later stage of *contour unfolding*.

A step-by-step schematic representation is provided in Figure 2.2, taking as an example a disyllabic noun belonging to the //MH#// tone category. Stage 1 represents the input: the noun has //MH#// tone, and the copula has //L// tone. Stage 2 shows the anchoring of the MH tone pattern onto the last syllable of the lexical word, a specification indicated by the # symbol in the label MH#. Stage 3 depicts the one-to-one mapping of tonal levels to available syllables. The MH contour associates sequentially, starting from its anchorage point: the noun's last syllable receives M, and the following syllable (the copula) receives H. Since no syllable remains available for the copula's lexical L tone, it remains unassociated and does not surface. (Remember that there are no falling contours in Yongning Na: ‡ HL is not a possible sequence over a monosyllable, any more than ‡ HM or ‡ ML.) The noun's first syllable, left toneless in this process, is assigned a default M level in Stage 4. Stage 5 presents the final phonological surface form.

Figure 2.3 illustrates an analogous process for the possessive clitic //=**bv̩˧**// in this tonal category of nouns.

By contrast with MH#, the floating H tone (#H) does not anchor to any of the syllables of the word to which it is lexically attached, and the possessive clitic is unable to provide such anchorage. Since this H tone receives no syllabic anchorage, either from the word to which it is lexically attached or from the possessive clitic that follows it, it remains unassociated and does not surface at all in this context. This is represented as Figure 2.4. The figure for the 'flea' tone, H$, would be identical with Figure 2.4: the possessive clitic is not a suitable host, so the H$ remains unassociated and does not surface at all in this context.

The behaviour of the word-final H tone (H#) is shown in Figure 2.5. In this case, the L tone on the clitic results from an exceptionless phonological rule whereby all tones following H are lowered to L (see §7.1.1).

These step-by-step representations provide more detail than tonologists with experience in level tones may find necessary. It does not appear indispensable to draw similar figures for the other lexical categories of Yongning Na, though this could be offered as an exercise for an introductory phonology class.

2.4.3 M as a default tone

The analysis set out above assumes that M serves as a default tone: syllables that are not specified for tone receive M. For instance, a //#H// tone carried by a disyllabic noun can only manifest itself on a following word: the H tone, though lexically attached to the noun, never appears on the noun itself. Both syllables of the noun receive /M/ tone in the surface phonological form. Under the present analysis, this is understood as default tone assignment. Likewise, /M.L/ is observed as a surface pattern on disyllabic and polysyllabic words, such as /**dɑ˧ʝi˩**/ 'mule', but this pattern is analyzed as the manifestation of a lexical-word-final L tone (notation: //L#//): the /M/ tone on the first syllable is analyzed as a default tone, not a lexically specified one. Evidence for this analysis will be presented in the course of the discussion, drawing on the combinatorial properties of tones, such as the tone rules that apply in compounding.

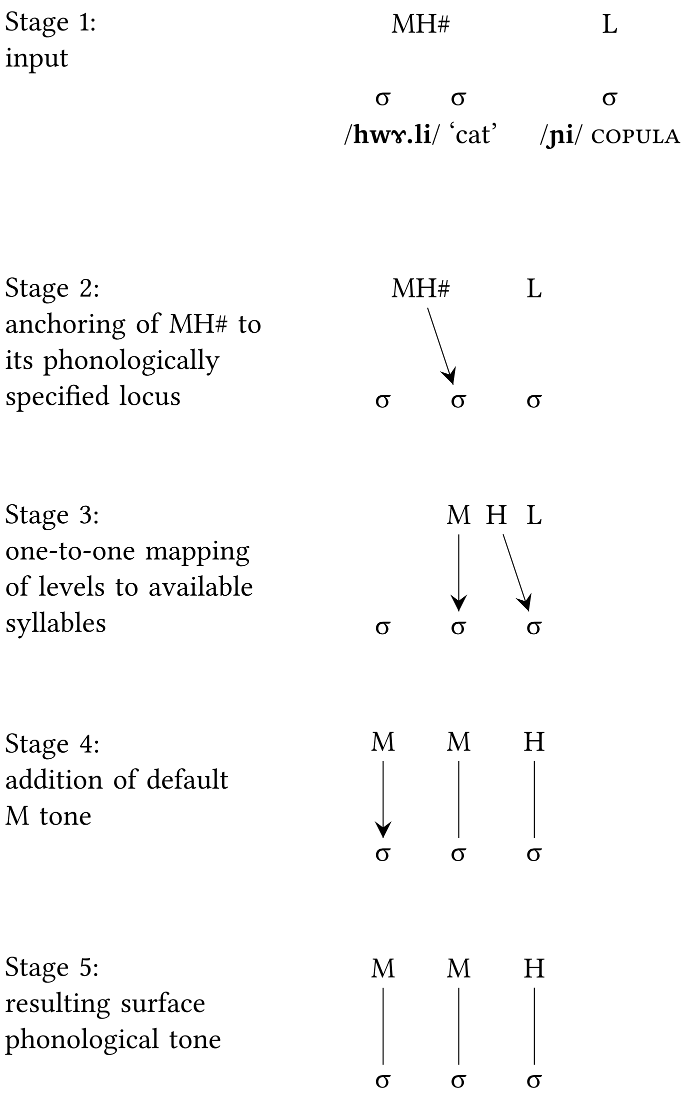


Figure 2.2: A detailed representation of tone-to-syllable association for /**hwɤ˧li˧ ɲi˥**/ 'is (a/the) cat'.

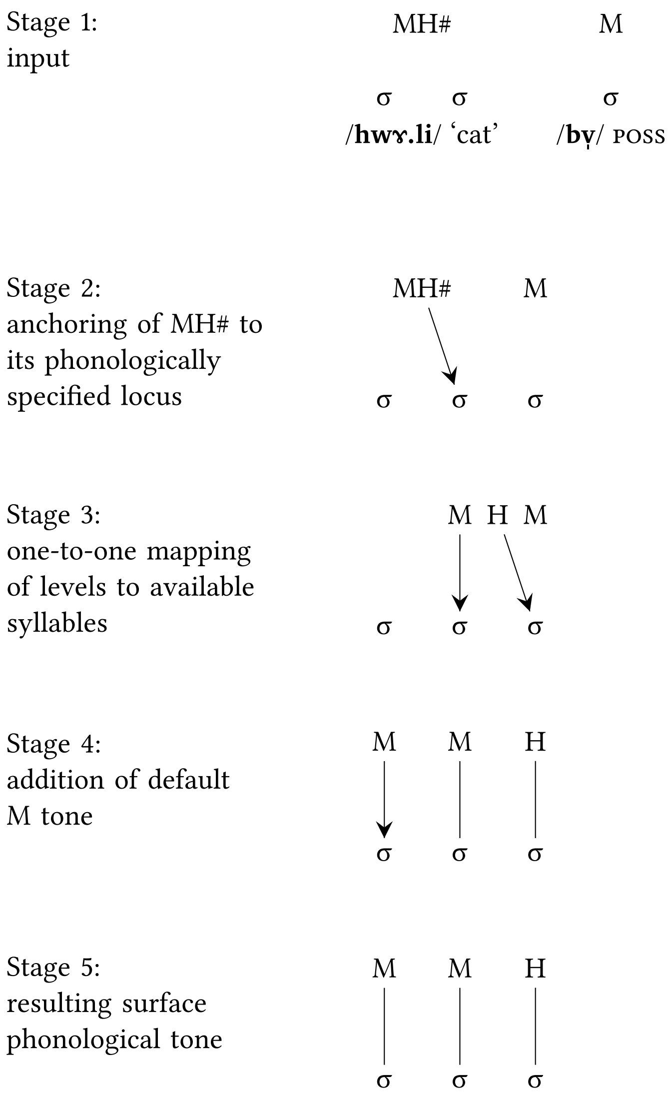


Figure 2.3: A detailed representation of tone-to-syllable association for /**hwɤ˧li˧=bv̩˥**/ ‘of (a/the) cat’.

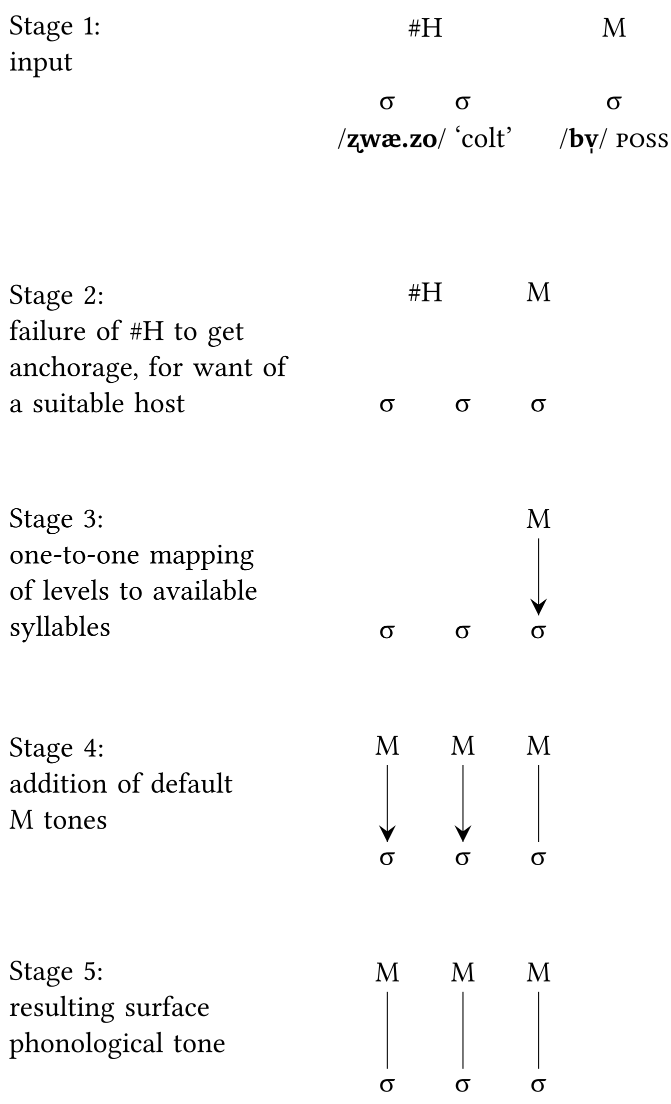


Figure 2.4: A detailed representation of tone-to-syllable association for /**ʐwæ˧zo˧=bv̩˧**/ ‘of (a/the) colt’.

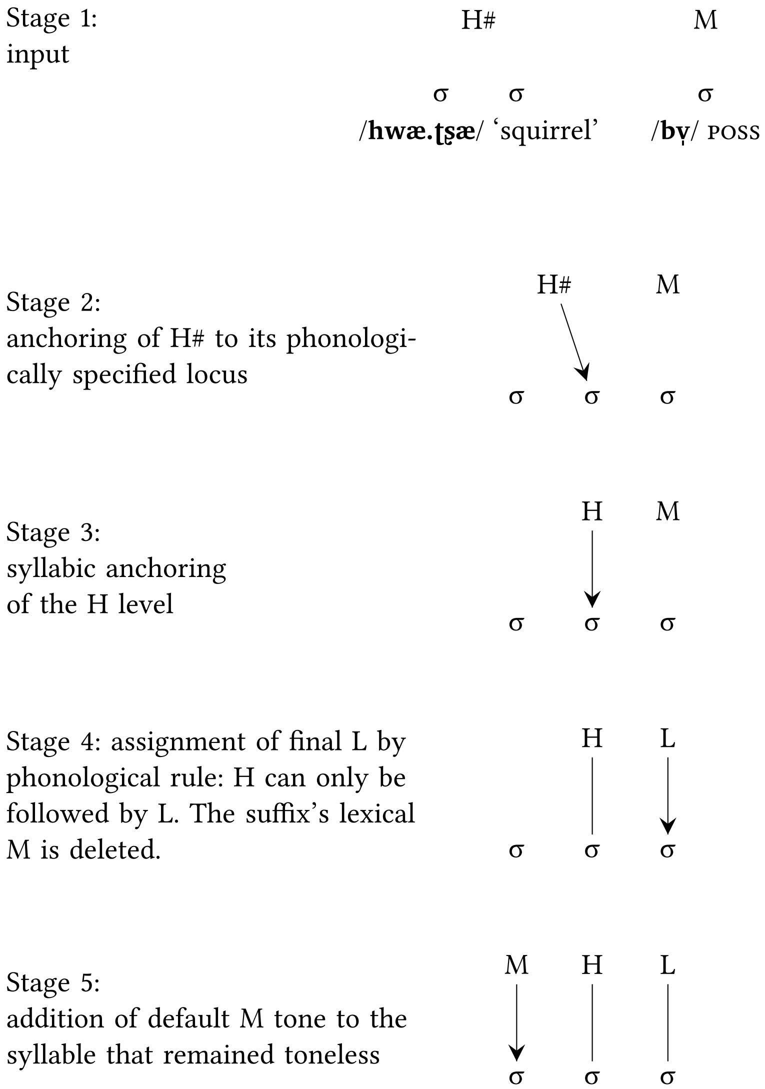


Figure 2.5: A detailed representation of tone-to-syllable association for /**hwæ˧ʈʂæ˥=bv̩˩**/ ‘of (a/the) squirrel’.

One may be tempted to push this analysis further and attempt to eliminate the specification of M tone altogether in the inventory of lexical tones, for instance dispensing with the M component in the analysis of the //LM// tone category, illustrated by disyllables such as /**bo˩mi˧**/, with surface L.M. There is, however, a motivation for considering the M tone of their second syllable as phonologically specified at the underlying lexical level: this M tone blocks L-tone spreading.

The background to this analysis is the observation that L tone spreads progressively ("left-to-right") onto syllables that are unspecified for tone (this is referred to as Rule 1; see §7.1 for further detail). In Table 2.9, disyllables that carry L tone on both of their syllables are accordingly analyzed as possessing a simple lexical L tone. In transcriptions, L tone is indicated on both syllables by convention (e.g. the word for 'dog' is written //**kʰv̩˩mi˩**//), so as to stay close to surface forms, but at a lexical level, these words are analyzed as carrying a simple //L// tone. Disyllabic nouns that have a /L.M/ pattern (L on the first syllable, M on the second) are analyzed as possessing a phonological /M/ that blocks L-tone spreading.

Under an approach that dispenses with M at the lexical level, the syllabic anchoring of all L tones would need to be specified. This would, to some extent, parallel H tones, which fall into three categories with different modes of association to the syllabic string: H#, #H, and H$. However, if the L.M pattern were reanalyzed as a word-initial L tone, it would be necessary to specify that it does not spread, unlike other L tones. Reanalyzing the //LM// category as a non-spreading L, contrasting with a spreading L, is a theoretical possibility, but one which (at present) appears to me to be less consistent with the rest of the description than positing a combination of two levels as the underlying lexical tone.

Yet another alternative would be to analyze the /L.H/, /M.L/, and /L.L/ surface patterns as the realization of initial //L//, final //L//, and //L.L// (with L tone specified on both syllables), respectively, thereby avoiding any reference to L-tone spreading. However, given that L-tone spreading is such a commonly attested process in the Alawua dialect of Yongning Na, this does not appear to be a promising line of analysis.

Moreover, any description that avoids M tones at the lexical level would also require an alternative means of accounting for MH contour tones. It would then be necessary to posit a separate type of H tones: a contour-creating H tone, in addition to the three types recognized so far. This is hardly an appealing analytical option.

For these various reasons, it seems reasonable to adopt a model in which M is included in the lexical specification of certain tonal categories.

2.4.4 The notation of tonal categories in lexical entries

This section explains the choices made for the notation of tonal categories in lexical entries (as headwords in dictionary entries and in the interlinear glossing of texts), as exemplified for nouns in Tables 2.8 and 2.9.

One typographical option would be to indicate the phonological category in superscript at the beginning or end of the word, e.g. //**ʐwæ**$^{\#\mathrm{H}}$// for ‘horse’ and //**õ.dv̩**$^{\mathrm{LM+MH}\#}$// for ‘wolf’. This notation, which separates tone from vowels and consonants, makes good sense in view of the analysis proposed here, according to which tone in Na is lexically associated with entire lexemes, not individual syllables. However, working out the tone-to-syllable mapping requires complete familiarity with the tonal rules of Yongning Na. A transcription that more closely resembles surface phonology therefore seemed preferable, marking tone at the end of each syllable. International Phonetic Alphabet tone-letters (Chao 1930) were chosen: ˥ for High, ˧ for Mid, ˩ for Low, ˩˧ for Low-to-Mid, ˩˥ for Low-to-High, and ˧˥ for Mid-to-High.

This system is strictly equivalent to the Africanist notation that uses accents. For instance, //**bo˩˧**// ‘pig’ could be written as //**bǒ̄**// in Africanist notation, and //**ʐæ˩˥**// ‘leopard’ as //**ʐæ̌**//. Tone-letters are preferred over accents for want of a satisfactory solution to the typographic issue of diacritic combinations, such as how to indicate a rising contour on a syllable like /ɻ̩̃/.

In China, numbers rather than tone-letters have become standard in phonetic notation. The strict numerical equivalents are as follows: ˥ corresponds to ^{5}, ˧ to ^{3}, ˩ to ^{1}, ˩˧ to ^{13}, ˩˥ to ^{15}, and ˧˥ to ^{35}. However, this is not how the numerical system is actually used in China. In order to provide the necessary background and clarify the complexities associated with the Chinese use of the numerical system, a detour through the history of Chao Yuen-ren’s initial proposal and its adoption in China is in order.

In his initial proposal to the International Phonetic Association, Chao Yuen-ren introduced constraints to limit the full combinatorial potential of the five-point scale. “In order not to make distinctions too fine, points 2 and 4 are used either alone or with each other, but not in combination with 1, 3, or 5” (Chao 1930: 25). This restriction reduced the number of proposed tone-letters to thirteen (rather than twenty-five) for “straight tones” – tones composed of two pitch targets, which may be identical or different. Additional restrictions applied to “circumflex tones”, which involve three pitch targets. Not only were the {1, 3, 5} and {2, 4} sets of levels never allowed to mix, but certain sequences were quietly excluded, presumably on grounds of phonetic implausibility. These include ˩˥˩

(bottom-top-bottom), ˩˥˥ *(bottom-top-top)*, and their mirror images ˥˩˥ *(top-bottom-top)*, ˥˩˩ *(top-bottom-bottom)*, ˥˥˩ *(top-top-bottom)*, and ˩˩˥ *(bottom-bottom-top)*. Table 2.10 reproduces the full set from the original publication.

Table 2.10: Full set of tone-letters proposed in Chao (1930: 25).

straight tones		circumflex tones		short tones	
tone-letter	name	tone-letter	name	tone-letter	name
˩˩	11	˩˧˩	131	˩	1
˩˧	13	˩˥˧	153	˨	2
˩˥	15	˨˦˨	242	˧	3
˨˨	22	˧˩˧	313	˦	4
˨˦	24	˧˩˥	315	˥	5
˧˩	31	˧˥˩	351		
˧˧	33	˧˥˧	353		
˧˥	35	˦˨˦	424		
˦˨	42	˥˩˧	513		
˦˦	44	˥˧˥	535		
˥˩	51				
˥˧	53				
˥˥	55				

The inventory in Table 2.10 was never implemented with full consistency, however. In *A grammar of Modern Chinese* (Chao 1968), Chao Yuen-ren himself deviated from his earlier principle of not blending the {1, 3, 5} and {2, 4} levels by proposing the notation ^{214} (˨˩˦) for the third tone of Beijing Mandarin – a convention that later became standard.[8] The lack of an explicit, principled basis for the original inventory ultimately limited its practical applicability.

Once these restrictions were relaxed, however, the use of tone-letters became impractical. A set consisting of five short tones, twenty-five “linear” tones and a hundred and twenty-five “circumflex” tones was manageable neither in handwritten form nor in the pre-digital, pre-Unicode typographical world. This led to the widespread adoption of numerical notation, particularly in large-scale dialect surveys (Simmons et al. 1957: 45-58). In the process, however, the original distinction between short, straight, and circumflex tones was lost.

[8] Lee & Zee (2003: 110) propose ^{213} (˨˩˧) instead of ^{214} (˨˩˦), similarly disregarding the earlier restriction on mixing the {1, 3, 5} and {2, 4} subsets.

The convention that emerged in China required at least two numbers for each tone: one for its beginning and one for its endpoint. This practice is implicitly based on the notation adopted for Chinese tones, whereby the four-way tonal contrast in Beijing Mandarin is transcribed using the labels ^{55}, ^{24}, ^{214}, and ^{51}. Such a convention creates a difficulty when transcribing level-tone languages such as Japanese, Pumi or Yongning Na. To reflect the insight that there is no phonologically relevant pitch movement in the course of the H, M, and L tones in Yongning Na, one might consider doubling the number indicating each tone's relative pitch level, transcribing the H tone as ^{55}, the M tone as ^{33}, and the L tone as ^{11}. But this notation would be incongruent with the conventions used in China, where the numbers are selected to reflect pitch curves, not abstract phonological entities. The systematic use of at least two numbers in the standard five-point tonal notation encourages linguists to pay attention to any differences in pitch between a tone's beginning and endpoint – differences that are crucial for systems where tones do not simply consist of sequences of level pitches (see §12.1.1), but irrelevant for level-tone systems. For instance, since an L tone is often realized phonetically (in Yongning Na, as also in other level-tone languages) as a pitch fall rather than a flat, sustained low pitch, it is typically transcribed by Chinese linguists as ^{21} or ^{31}. (Thus, the L tone of Naxi is transcribed as ^{31} by Hé Jírén & Jiāng Zhúyí 1985.) Notation as ^{11} would not accurately reflect the phonetic realization of the Low tone and would thus be counterintuitive for users accustomed to this system. On the other hand, notation as ^{31} would obscure the phonological nature of the L tone, misleadingly suggesting that the tones transcribed as ^{31} (L in the present description) and ^{13} (LM) are mirror images of each other.[9]

For tonologists accustomed to the Chinese-style numerical notation, I therefore strongly recommend the following set of equivalences: ^{5} for ˥, ^{3} for ˧, ^{1} for ˩, ^{13} for ˩˧, ^{15} for ˩˥, and ^{35} for ˧˥. Tables 2.11 and 2.12 spell out these equivalences for the lexical tone categories of nouns, taking up the same examples as in Tables 2.8 and 2.9. These equivalences may initially seem counterintuitive, but they minimize the risk of phonological misinterpretation.

[9] The present reflections on notational choices for level tones and their influence on the phonological analysis of tone in Naish languages build on a talk given at the International Conference on Phonetics of the Languages in China (Michaud 2013b). The same point is made by Jackson Sun about Va, a three-tone language of the Austroasiatic family: "Va distinguishes three tonal levels: high, mid, and low. What matters in this system is relative pitch register, as the actual pitch contours vary according to syllable type and the position of a syllable inside a word. (…) It is therefore misleading to assign steady tone letters (…) or tone numerals (…) to underlying tones in Va" (Sun 2018: 139).

Table 2.11: Equivalences between tone-letters and tone digits for transcribing the lexical tone categories of monosyllabic nouns.

analysis	with tone-letters	with tone digits (underlying form)	with tone digits (surface form)	meaning
// LM //	**bo˩˧**	**bo^{13}**	**bo^{15}**	pig
// LH //	**ʐæ˩˥**	**ʐæ^{15}**	**ʐæ^{15}**	leopard
// M //	**lɑ˧**	**lɑ^{3}**	**lɑ^{3}**	tiger
// L //	**jo˩**	**jo^{1}**	**jo^{3}**	sheep
// #H //	**ʐwæ˥**	**ʐwæ^{5}**	**ʐwæ^{3}**	horse
// MH# //	**ʈʂʰæ˧˥**	**ʈʂʰæ^{35}**	**ʈʂʰæ^{35}**	deer

Table 2.12: Equivalences between tone-letters and tone digits for transcribing the lexical tone categories of disyllabic nouns.

analysis	with tone-letters	with tone digits (underlying)	with tone digits (surface)	meaning
// M //	**po˧lo˧**	**po^{3}lo^{3}**	**po^{3}lo^{3}**	ram
// #H //	**ʐwæ˧zo#˥**	**ʐwæ^{3}zo#^{5}**	**ʐwæ^{3}zo^{3}**	colt
// MH# //	**hwɤ˧li˧˥**	**hwɤ^{3}li^{35}**	**hwɤ^{3}li^{35}**	cat
// H$ //	**kv̩˧ʂe˥$**	**kv̩^{3}ʂe^{5}$**	**kv̩^{3}ʂe^{5}**	flea
// H# //	**hwæ˧ʈʂæ˥**	**hwæ^{3}ʈʂæ^{5}**	**hwæ^{3}ʈʂæ^{5}**	squirrel
// L //	**kʰv̩˩mi˩**	**kʰv̩^{1}mi^{1}**	**kʰv̩^{1}mi^{15}**	dog
// L# //	**dɑ˧ʝi˩**	**dɑ^{3}ʝi^{1}**	**dɑ^{3}ʝi^{1}**	mule
//LM+MH#//	**õ˩dv̩˧˥**	**õ^{1}dv̩^{35}**	**õ^{1}dv̩^{35}**	wolf
//LM+#H//	**nɑ˩hĩ#˥**	**nɑ^{1}hĩ#^{5}**	**nɑ^{1}hĩ^{5}**	Naxi
// LM //	**bo˩mi˧**	**bo^{1}mi^{3}**	**bo^{1}mi^{5}**	sow
// LH //	**bo˩ɬɑ˥**	**bo^{1}ɬɑ^{5}**	**bo^{1}ɬɑ^{5}**	boar

In the process of mapping tonal categories onto syllables transcribed in the International Phonetic Alphabet, some cases are straightforward. For instance, the LM tone category can be represented by associating both levels with a monosyllable, as in //**bo˩˧**// ‘pig’, and distributing them over the two syllables of a disyllable, as in //**bo˩mi˧**// ‘sow’. However, not all cases are this simple, necessitating detailed explanations of the notational choices made here.

The first decision concerns the transcription of the floating H tone of monosyllabic nouns. Here, it is represented as a simple H tone, without an explicit indication that it is floating. Thus, 'horse' is transcribed simply as //**ʐwæ˥**// rather than //**ʐwæ#˥**//. The rationale for this choice is that, on monosyllables, there is only one type of H tone – namely, this floating H, which can only surface after the syllable to which it is lexically attached. Since no contrast exists among different types of H tones in this context, it was deemed economical to dispense with the pound symbol '#' (an explicit indicator of the tone's mode of syllabic association).

For nouns of two or more syllables, on the other hand, three types of H tones must be distinguished: H#, #H, and H$. In these cases, some indication of syllabic anchorage is indispensable in lexical transcriptions. At least two diacritic marks are required to differentiate among H#, #H and H$. The first of these three tones is transcribed with a simple H-tone mark ˥ on the last syllable, as its mode of anchoring appears as phonologically simplest: it remains fixed on the final syllable and never shifts. Hence, 'squirrel' is transcribed as //**hwæ˧ʈʂæ˥**// (rather than the fully explicit //**hwæ˧ʈʂæ˥#**//).

The decision to mark a Mid tone ˧ on the first syllable of such words follows from the principle of keeping lexical forms close to the surface phonological forms. When a disyllable with lexical H#, #H or H$ tone is produced in isolation, its first syllable carries M tone, which is analyzed here as a default tone (assigned to syllables lexically unspecified for tone; see §2.4.3). In the transcription of lexical forms, it thus appears preferable to provide a tone for each syllable. The same principle underpins the transcription of 'colt' as //**ʐwæ˧zo#˥**// rather than //**ʐwæ.zo#˥**//, 'flea' as //**kv̩˧ʂe˥$**// rather than //**kv.ʂe˥$**//, and 'cat' as //**hwɤ˧li˦**// rather than //**hwɤ.li˦**//.

In the case of L-tone words, the L tone is indicated on both syllables, introducing some redundancy. For instance, 'dog' is transcribed as //**kʰv̩˩mi˩**// rather than //**kʰv̩˩mi**//.

Conversely, in disyllabic words where the tone pattern is analyzed as LM+#H, the lexical transcription omits the M tone, by convention. This is to avoid stacking two tone marks on the same syllable, which would misleadingly suggest the presence of a contour. The lexical transcription of tone therefore consists simply of L on the first syllable and #H on the second. Thus, 'Naxi (person)' is written as //**nɑ˩hĩ#˥**// rather than //**nɑ˩˧hĩ#˥**// or //**nɑ˩hĩ˧#˥**//.

Since is not easy to master all these conventions at once, the dictionary (Michaud et al. 2024) provides several layers of tonal information to facilitate gradual familiarization with the Yongning Na tone system and enhance accessibility to the online linguistic materials. Specifically, each entry includes (i) a bespoke field indicating the lexical tone, (ii) the complete underlying (lexical) form, in which tonal specifications are encoded in the way explained above, and (iii) the surface form, which reflects the actual pronunciation in isolation. For instance, the entry for ‘dog’ clarifies that the lexical tone is Low (“Tone: L”) and provides the underlying form **kʰv̩˩mi˩**, where L tone is indicated on both syllables, alongside the surface form **kʰv̩˩mi˩˥**, which is how the word in pronounced.

2.4.5 Attested and unattested lexical tones

The static regularities identified in §2.1 can be reformulated in dynamic terms, as the outcome of a set of phonological tone rules. These rules will be set out in §7.1.1, and discussed in detail throughout Chapter 7, which also discusses the phonological unit within which they apply: the tone group. The following is a preview of the full set of rules, to which the reader will need to refer frequently in subsequent chapters.

Rule 1: L tone spreads progressively (“left-to-right”) onto syllables unspecified for tone.

Rule 2: Syllables that remain unspecified for tone after the application of Rule 1 receive M tone.

Rule 3: In tone-group-initial position, H and M are neutralized to M.

Rule 4: The syllable following an H-tone syllable receives L tone.

Rule 5: All syllables following an H.L or M.L sequence receive L tone.

Rule 6: In tone-group-final position, H and M are neutralized to H if they follow an L tone.

Rule 7: If a tone group only contains L tones, a postlexical H tone is added to its final syllable.

Three key observations are particularly relevant here: (i) L tone spreads progressively (“left-to-right”); (ii) all tones following H are lowered to L; and (iii) H and M are neutralized to M in tone-group-initial position. These generalizations, together with the fact that no falling contours occur on a single syllable, account for the absence of all the unattested lexical tone patterns in monosyllables. They also explain most of the unattested patterns for disyllables, such as ‡ H.L, ‡ H.M, ‡ M.LM, and ‡ ML.M.

However, one theoretically possible combination is unattested despite being compatible with the language’s phonotactics: there is no //LM+H$// lexical tone category for disyllables, even though the categories //LM+MH#// and //LM+#H//

are attested. (As for //LM+H#//, it is indistinguishable from //LH//, since both lead to the same tonal association in a disyllable: L on the first syllable, and H on the second.) This gap can be interpreted as evidence that //H$// is not only less widespread in the lexicon than #H and H# – as was pointed out in §2.3.5 – but is also less deeply integrated into the system.

2.4.6 Phonological regularities and morphotonological oddities

Looking back at the data in Tables 2.8 and 2.9, it is tempting to seek phonological regularities that capture all the observed patterns. However, such a search soon encounters facts that resist phonological generalizations.

For instance, there is no obvious reason why L should surface as M in isolation. This may be linked to the prohibition of all-L tone groups (see §2.3.6), and *a fortiori* of all-L utterances. However, for verbs, this state of affairs is repaired by the addition of a postlexical final H tone, so that verbs with lexical //L// tone surface with /LH/ contours in isolation (see Table 6.2). If the tone system were governed solely by phonological rules applying uniformly across all morphosyntactic contexts, one would expect lexical //L// on a noun to surface as /LH/, not as /M/.

A similarly puzzling case concerns the //L// tone on disyllabic nouns. A word such as //**kʰv̩˩mi˩**// 'dog' yields /**kʰv̩˩mi˩˥**/ in isolation, as expected. However, when followed by the copula, it surfaces as /**kʰv̩˩mi˩ ɲi˥**/ 'is (a/the) dog': the copula loses its lexical //L// tone. There is no obvious reason why this should be so: one might instead expect a /L.L.L/ sequence, //† **kʰv̩˩mi˩ ɲi˩**//, which, following the addition of a postlexical H tone to avoid an all-L tone group, would be realized as /† **kʰv̩˩mi˩ ɲi˩˥**/.

This asymmetry in the tonal behaviour of the copula after a //L//-tone noun, depending on the number of syllables in the noun, highlights a crucial aspect of Yongning Na tone: many tone rules have narrowly restricted domains of application. They operate within highly specific morphosyntactic contexts and are sensitive to both the number of syllables and the internal structure of the morphemes concerned.

These reflections on the overall outlook of the Yongning Na tone system will be taken up in §12.2, in light of the detailed morphotonological account to which we now turn (Chapters 3-6).

3 Nominal classifiers

A nominal classifier is "a type of limited noun that occurs only after numerals (…), and whose selection is determined by a preceding (overt or implicit) noun" (Matisoff 1973: 88). While classifiers are, morphologically speaking, nouns, their tonal behaviour is highly specific and exhibits remarkable complexity. This justifies dedicating a separate chapter to their analysis rather than including them within Chapter 2 on the lexical tones of nouns.[1]

The term 'classifier' is understood here in the syntactic sense of any noun that may appear directly after a numeral. This includes measure words ('inch', 'armspan', 'heap', …) and time words ('day', 'month', 'year', …), which immediately follow the numeral. In view of their great importance in the language and the richness of their tone patterns, classifiers are dealt with in a chapter of their own. The main focus is on their tonal behaviour in numeral-plus-classifier phrases (§3.1), but classifiers also appear in demonstrative-plus-classifier constructions (§3.2). Moreover, numeral-plus-classifier and demonstrative-plus-classifier phrases can interact tonally with a preceding noun (§3.3).

Devoting tens of pages to nominal classifiers, after an extensive chapter on the tones of nouns (Chapter 2), may test the patience of even the most willing reader. Will the archipelago of tables never come to an end? Is there no more concise way to describe tone in Yongning Na than by recording lump after lump of morphotonological detail?

Indeed, the tone patterns of classifiers phrases are a disappointment to the linguist, in that they cannot be accounted for by phonological sandhi, nor do they follow a single, well-defined set of morphotonological rules.[2] Instead, the tonal paradigms of the various categories of classifiers need to be learnt one by one. The tables documenting these patterns are, admittedly, the most heavy-going in this volume. Yet paradoxically, this chapter, which deals with the most irregular part of the Yongning Na tone system, carries a reassuring message: there *is* a limit to the seemingly endless complexity of the system, even in its least predictable areas. I venture to claim (taking full responsibility for this immodest stance) that this chapter comes close to an exhaustive description of its object,

[1] In the first edition, this chapter was Chapter 4. It has been repositioned to follow immediately after the chapter on nouns.

[2] For clarifications on the terms 'tone sandhi', 'morphotonology', and 'tonal morphology', see §8.4.1.

thereby showing that a comprehensive account of tone in Yongning Na is not beyond reach.

3.1 Numeral-plus-classifier phrases

Within the Naish language group, tonal alternations in numeral-plus-classifier phrases are vestigial in Naxi (as spoken in the plain of Lijiang), limited in Laze, and ubiquitous in Yongning Na. In the Alawua dialect of Yongning Na, the tone of the classifier is affected by the numeral to such an extent that arriving at the classifier's underlying tone is far from straightforward. For instance, the classifier for 'day' carries an H tone in /**ɲi˧-ɲi˥**/ 'two days', an LH tone in /**so˩-ɲi˩˥**/ 'three days', and an M tone in /**ʐv̩˧-ɲi˧**/ 'four days'; and the classifier for 'year' has an MH tone in /**ɲi˧-kʰv̩˧˥**/ 'two years', an LH tone in /**so˩-kʰv̩˩˥**/ 'three years', and an L tone in /**ʐv̩˧-kʰv̩˩**/ 'four years'. This section proposes a synchronic description and analysis.[3]

For each noun documented during fieldwork, a note was made of its typical classifier(s), and this piece of information was included in the dictionary of Yongning Na (Michaud et al. 2024), following the example of Eugénie Henderson's dictionary of Bwe Karen (Henderson 1997). Naturally, this does not capture the full range of stylistic possibilities in classifier choice, which is best studied through actual language use: "a noun can be accompanied by various classifiers depending on context, so it may not be adequate to describe one of these as the primary classifier at a lexical level" (François 2000: 167).[4] Nonetheless, this information proved valuable in assembling a comprehensive list of classifiers, yielding over one hundred monosyllabic items. (Disyllabic classifiers will be addressed separately in §3.1.8.)

3.1.1 Elicitation procedures

The logical first step in studying the tones of classifiers was to conduct systematic elicitation with numerals.[5] The first dataset, elicited in 2009, covered numerals

[3] An earlier version of this section was published as Michaud (2013c). Among other differences, subcategories of tones for classifiers are now distinguished using subscript letters, e.g. M_a and M_b instead of the earlier notation as M1 and M2.

[4] *Original text:* le même nom peut être accompagné de plusieurs classificateurs selon les contextes, sans qu'il soit légitime d'en privilégier un comme fondamental dès le lexique.

[5] Classifiers are bound forms and cannot be elicited in isolation (i.e. without an accompanying numeral or demonstrative). Numerals, too, do not usually appear on their own, but it is possible, at a push, to elicit them in isolation. The forms from one to ten are: /**ɖɯ˩˥**/ '1', /**ɲi˩˥**/ '2', /**so˧**/ '3', /**ʐv̩˧**/ '4', /**ŋwɤ˧**/ '5', /**qʰv̩˧˥**/ '6', /**ʂɯ˧**/ '7', /**hõ˧˥**/ '8', /**gv̩˧**/ '9', and /**tsʰe˧**/ '10'.

from 1 to 30. However, the results were less than fully consistent, partly because the consultant was unaccustomed to lengthy counting tasks. The same noun-plus-classifier combination was sometimes realized with different tone patterns from one elicitation session to another. When such discrepancies were pointed out, the consultant (Mrs. Latami) would identify one pattern as correct and dismiss the others as mistaken. Yet these *a posteriori* judgments also wavered: a variant that had been brushed aside as erroneous might come up again in a later session, with the consultant now insisting on its correctness.

At first, I wrongly assumed that only one tone pattern could be correct. Over time, however, it became clear that some of the phrases allow two variants. For instance, the combination of '9' with the classifier for threads can be realized as either /**gv̩˧-kʰɯ˥**/ or /**gv̩˧-kʰɯ˩**/. Taking into account both the occasional occurrence of elicitation errors and the existence of genuine variants, a comprehensive description could finally be reached, dissipating earlier perplexity and frustration.

The data collected in 2009, covering numerals from 1 to 30, reveals that some classifiers have tantalizingly similar yet not identical behaviour. For instance, the classifier for days and the classifier for steps (as in steps of a staircase) follow the same tone patterns for numerals up to thirteen but differ for 14, 15, 16, 18, 19, and 22, as shown in Table 3.1. Thus, 'fourteen steps' is //**tsʰe˩ʐv̩˩-ɖwæ˥$**// (tone: L+H$; surface phonological form: /**tsʰe˩ʐv̩˩-ɖwæ˥**/), whereas 'fourteen days' is //**tsʰe˩ʐv̩˩-ɲi˩**// (tone: L; surface phonological form: /**tsʰe˩ʐv̩˩-ɲi˩˥**/).

This finding led to the decision to include the full range of numerals from 1 to 100 in a second set of recordings. In total, 2,810 numeral-plus-classifier phrases were recorded. Within this dataset, the ranges [50..59] and [80..89] were omitted, as previous elicitation had shown that their tone patterns were identical to those of [40..49] and [60..69], respectively. Shortening the list of numerals in this manner helped mitigate consultant fatigue.

The entire set of transcribed recordings is available online, with both a surface phonological transcription and an indication of the underlying tonal pattern.

No amount of continuous speech would be sufficient to document all the numeral-plus-classifier combinations from 1 to 100, hence the choice to resort to systematic elicitation. However, some of these combinations are also attested in the transcribed narratives.

3.1.2 Results: Nine tonal categories for monosyllabic classifiers

The nine classifier tone categories (H_a, H_b, MH_a, MH_b, M_a, M_b, L_a, L_b, and L_c) are illustrated in Table 3.2.

Table 3.1: Surface phonological form of the classifiers for days and for steps, in association with numerals from 1 to 30.

NUM	CLF.days	CLF.steps	comparison
1	**ɖɯ˧-ɲi˥**	**ɖɯ˧-kʰwɤ˥**	same
2	**ɲi˧-ɲi˥**	**ɲi˧-kʰwɤ˥**	same
3	**so˩-ɲi˩˥**	**so˩-kʰwɤ˩˥**	same
4	**ʐv̩˧-ɲi˧**	**ʐv̩˧-kʰwɤ˧**	same
5	**ŋwɤ˧-ɲi˧**	**ŋwɤ˧-kʰwɤ˧**	same
6	**qʰv̩˧-ɲi˥**	**qʰv̩˧-kʰwɤ˥**	same
7	**ʂɯ˧-ɲi˧**	**ʂɯ˧-kʰwɤ˧**	same
8	**hõ˧-ɲi˥**	**hõ˧-kʰwɤ˥**	same
9	**gv̩˧-ɲi˧**	**gv̩˧-kʰwɤ˧**	same
10	**tsʰe˩-ɲi˩˥**	**tsʰe˩-kʰwɤ˩˥**	same
11	**tsʰe˩ɖɯ˩-ɲi˩˥**	**tsʰe˩ɖɯ˩-kʰwɤ˩˥**	same
12	**tsʰe˩ɲi˩-ɲi˩˥**	**tsʰe˩ɲi˩-kʰwɤ˩˥**	same
13	**tsʰe˩so˩-ɲi˩˥**	**tsʰe˩so˩-kʰwɤ˩˥**	same
14	**tsʰe˩ʐv̩˩-ɲi˩˥**	**tsʰe˩ʐv̩˩-kʰwɤ˥**	*different*
15	**tsʰe˩ŋwɤ˩-ɲi˩˥**	**tsʰe˩ŋwɤ˩-kʰwɤ˥**	*different*
16	**tsʰe˩qʰv̩˩-ɲi˩˥**	**tsʰe˩qʰv̩˩-kʰwɤ˥**	*different*
17	**tsʰe˩ʂɯ˩-ɲi˩˥**	**tsʰe˩ʂɯ˩-kʰwɤ˩˥**	same
18	**tsʰe˩hõ˩-ɲi˩˥**	**tsʰe˩hõ˩-kʰwɤ˥**	*different*
19	**tsʰe˩gv̩˩-ɲi˩˥**	**tsʰe˩gv̩˩-kʰwɤ˥**	*different*
20	**ɲi˧tsi˧-ɲi˧**	**ɲi˧tsi˧-kʰwɤ˧**	same
21	**ɲi˧tsi˧ɖɯ˧-ɲi˥**	**ɲi˧tsi˧ɖɯ˧-kʰwɤ˥**	same
22	**ɲi˧tsi˧ɲi˧-ɲi˧**	**ɲi˧tsi˧ɲi˧-kʰwɤ˥**	*different*
23	**ɲi˧tsi˧so˩-ɲi˩˥**	**ɲi˧tsi˧so˩-kʰwɤ˩˥**	same
24	**ɲi˧tsi˧ʐv̩˧-ɲi˧**	**ɲi˧tsi˧ʐv̩˧-kʰwɤ˧**	same
25	**ɲi˧tsi˧ŋwɤ˧-ɲi˧**	**ɲi˧tsi˧ŋwɤ˧-kʰwɤ˧**	same
26	**ɲi˧tsi˧qʰv̩˧-ɲi˥**	**ɲi˧tsi˧qʰv̩˧-kʰwɤ˥**	same
27	**ɲi˧tsi˧ʂɯ˧-ɲi˧**	**ɲi˧tsi˧ʂɯ˧-kʰwɤ˧**	same
28	**ɲi˧tsi˧hõ˧-ɲi˥**	**ɲi˧tsi˧hõ˧-kʰwɤ˥**	same
29	**ɲi˧tsi˧gv̩˧-ɲi˧**	**ɲi˧tsi˧gv̩˧-kʰwɤ˧**	same
30	**so˧tsʰi˧-ɲi˧**	**so˧tsʰi˧-kʰwɤ˧**	same

Table 3.2: One example of each of the nine tonal categories of monosyllabic classifiers.

classifier	tone	description: classifier for...
ɖwæ˥$_a$	H_a	steps (of stairs)
ɲi˥$_b$	H_b	days
hɑ̃˦$_a$	MH_a	nights
kv̩˦$_b$	MH_b	people, persons
nɑ˧$_a$	M_a	tools
dzi˧$_b$	M_b	pairs of non-separable objects, e.g. shoes
dze˩$_a$	L_a	pairs of separable objects, e.g. pots, bottles
dzi˩$_b$	L_b	trees, bamboos
ʐɤ˩$_c$	L_c	lines, patterns (in weaving or drawing)

These nine tonal categories were brought out by distributional analysis, grouping together monosyllabic classifiers that have the same tonal behaviour in combination with numerals. The choice of labels for these categories was guided by structural considerations. The tone patterns in association with the numerals 6 and 8 are not highly informative, as nearly all tonal oppositions are neutralized in this context (yielding only H# and H$). Likewise, numeral-plus-classifier phrases involving 3 and 10 only exhibit two distinct patterns. The numerals 4 and 5 allow for the identification of four tonal groups, but if these patterns were taken as indicative of the classifiers' lexical tones, the system would comprise only two High tones (#H and H#) and two Low tones (L# and L), with no Mid tones or contour tones. This would be strikingly different from the lexical tone system of the other monosyllabic nouns found in Yongning Na, which includes H, M, L, and two types of rising contours, analyzed as MH and LM.

On the other hand, the tone patterns associated with the numerals 1 and 2 provide meaningful labels for tonal categories of classifiers. These four patterns are H, MH, M, and L, all of which also occur as lexical tones for nouns. They are therefore adopted as category labels, with subscript letters added to distinguish subcategories (two for H, MH, and M, and three for L), ordered by decreasing frequency (e.g. category M_a contains more classifiers than M_b). The same approach is used for verbs, where two subcategories of L tones (L_a and L_b) and three subcategories of M tones (M_a, M_b, and M_c) are recognized (see Chapter 6).

However, tonal subcategories are established separately for nouns and verbs, with no basis for identifying them across parts of speech. For instance, the label 'L_a' used for verbs does not refer to the same category as 'L_a' for classifiers. To preclude potential misunderstandings, one could use distinct arrays of subscript letters, such as Greek letters for classifier subcategories and Latin letters for verb subcategories. But it seemed advisable to use Latin letters throughout to avoid an extravagant profusion of symbols.

A comprehensive summary of the tone patterns these nine categories yield in combination with numerals from 1 to 100 is presented in Tables 3.3 to 3.10. These tables encapsulate, in tightly packed form, the information required to generate the surface phonological tone patterns of all numeral-plus-classifier phrases in the Alawua dialect of Yongning Na.

The mass of information presented in Tables 3.3 to 3.10 may appear overwhelming at first glance. Were it not for the clear evidence from recorded data, one might suspect that this multiplicity of tone patterns was an artefact of the elicitation procedures.

Variants are separated by a slash (/). For typographical clarity, the information is split between two sets of tables: the first set (Tables 3.3 to 3.6) presents the categories with H and MH tones (H_a, H_b, MH_a, and MH_b), while the second (Tables 3.7 to 3.10) covers the categories with M and L tones (M_a, M_b, L_a, L_b, and L_c).

Numeral-plus-classifier phrases typically form a single tone group.[6] However, speakers can choose to split them into two groups for expressive (emphatic) purposes, as will be discussed in Chapter 7. The juncture indicated by '–' does not mark a division into separate tone groups but rather separates the phrase into two structural parts: one for the tens and another for the units and classifier. This juncture appears after the first two syllables in phrases containing numerals above twenty. Such phrases consist of two syllables referring to the tens ('two-ten' for twenty, 'three-ten' for thirty, etc.), followed by the last digit and the classifier – except in round numbers (twenty, thirty, etc.), where no zero is indicated and the classifier follows directly, as in /**ʐv̩˧tsʰi˩-kʰv̩˩**/ 'forty years'. For instance, /**so˧tsʰi˧so˩-ɲi˩**/ '33 days' can be represented as /**so˧tsʰi˧–so˩-ɲi˩**/ to show the juncture between the two parts (glosses for the four syllables: 'three - ten – three - CLF.days').

[6] The *tone group* is the unit within which tonal processes apply in Yongning Na. It could also be referred to as *phonological phrase*. Chapter 7 is devoted to this morphotonological unit, which is fundamental to Na prosody.

Table 3.3: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. H and MH tones. Numerals from 1 to 25.

	H_a	H_b	MH_a	MH_b
1	H$	H$	MH#	MH#
2	H$	H$	MH#	MH#
3	L	L	L	L
4	#H	#H	L#	L
5	#H	#H	L#	L
6	H$	H$	H#	H$
7	#H	#H	MH#	MH#
8	H$	H$	H#	H$
9	#H	#H	L#	L
10	L	L	L	L
11	L	L	L	L
12	L	L	L	L
13	L	L	L	L
14	L+H$	L	L+H#	L+H#
15	L+H$	L	L+H#	L+H#
16	L+H$	L	L+H#	L+H#
17	L	L	L	L
18	L+H$	L	L+H#	L+H#
19	L+H$	L	L+H#	L+H#
20	#H	#H	MH#	MH#
21	H$	H$	MH#	MH#
22	H$	#H	MH#	MH#
23	–L	–L	–L	–L
24	#H	#H	–L#	–L
25	#H	#H	–L#	–L

Table 3.4: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. H and MH tones. Numerals from 26 to 50.

	H_a	H_b	MH_a	MH_b
26	H$	H$	H#	H$
27	#H	#H	MH#	MH#
28	H$	H$	H#	H$
29	#H	#H	–L#	–L
30	#H	#H	MH#	MH#
31	H$	H$	MH#	MH#
32	H$	#H	MH#	MH#
33	–L	–L	–L	–L
34	#H	#H	–L#	–L
35	#H	#H	–L#	–L
36	H$	H$	H#	H$
37	#H	#H	MH#	MH#
38	H$	H$	H#	H$
39	#H/–L	#H	–L#	–L
40	L#–	L#–	L#–	L#–
41	L#–H$ / L#–	L#–H$ / L#–	L#–MH# / L#–	L#–MH#
42	L#–H$ / L#–	L#–#H / L#–	L#–MH# / L#–	L#–MH#
43	L#–L	L#–L / L#–	L#–L	L#–L
44	L#–#H / L#–	L#–#H / L#–	L#–L#	L#–L
45	L#–#H / L#–	L#–#H / L#–	L#–L#	L#–L
46	L#–H$	L#–H$	L#–H#	L#–H$
47	L#–#H	L#–#H	L#–MH#	L#–MH#
48	L#–H$	L#–H$	L#–H#	L#–H$
49	L#–#H / L#–L	L#–#H / L#–	L#–L#	L#–L
50	L#–	L#–	L#–	L#–

Table 3.5: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. H and MH tones. Numerals from 51 to 75.

	H_a	H_b	MH_a	MH_b
51	L#–H$ / L#–	L#–H$ / L#–	L#–MH / L#–	L#–MH#
52	L#–H$ / L#–	L#–#H / L#–	L#–MH / L#–	L#–MH#
53	L#–	L#–	L#–L	L#–L
54	L#–#H / L#–	L#–#H / L#–	L#–L#	L#–L
55	L#–#H / L#–	L#–#H / L#–	L#–L#	L#–L
56	L#–H$	L#–H$	L#–H#	L#–H$
57	L#–#H	L#–#H	L#–MH#	L#–MH#
58	L#–H$	L#–H$	L#–H#	L#–H$
59	L#–#H / L#–	L#–#H / L#–	L#–L#	L#–L
60	LM–H$	LM–H$	LM–H#	LM–#H
61	LM–H$	LM–H$	LM–H#	LM–#H
62	LM–H$ / LM–#H	LM–#H	LM–H#	LM–H#
63	LM–H$ / LM–L	LM–H$ / LM–L	LM–H#	LM–H#
64	LM–#H	LM–#H	LM–H#	LM–H#
65	LM–#H	LM–#H	LM–H#	LM–H#
66	LM–H$	LM–H$	LM–H#	LM–H#
67	LM–#H	LM–#H	LM–H#	LM–H#
68	LM–H$	LM–H$	LM–H#	LM–H#
69	LM–#H / LM–L	LM–L	LM–H#	LM–H#
70	L#–	L#–	L#–	L#–
71	L#–H$ / L#–	L#–H$	L#–MH# / L#–	L#–MH#
72	L#–H$ / L#–	L#–#H	L#–MH# / L#–	L#–MH#
73	L#–L	L#–L	L#–L	L#–L
74	L#–#H	L#–#H	L#–MH# / L#–	L#–L
75	L#–#H	L#–#H	L#–MH# / L#–	L#–L

Table 3.6: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. H and MH tones. Numerals from 76 to 100.

	H_a	H_b	MH_a	MH_b
76	L#–H$	L#–H$	L#–H# / L#–	L#–H#
77	L#–#H	L#–#H	L#–MH# / L#–	L#–MH#
78	L#–H$	L#–H$	L#–H# / L#–	L#–H#
79	L#–#H / L#–L	L#–#H	L#–L#	L#–L
80	LM–H$	LM–H$	LM–H#	LM–H#
81	LM–H$	LM–H$	LM–H#	LM–H#
82	LM–H$	LM–#H	LM–H#	LM–H#
83	LM–H$	LM–H$ / LM–L	LM–H#	LM–H#
84	LM–#H	LM–H$	LM–H#	LM–H#
85	LM–#H	LM–#H	LM–H#	LM–H#
86	LM–H$	LM–H$	LM–H#	LM–H#
87	LM–#H	LM–#H	LM–H#	LM–H#
88	LM–H$	LM–H$	LM–H#	LM–H#
89	LM–#H	LM–#H	LM–H#	LM–H#
90	L#–	L#–	L#–	L#–
91	L#–H$	L#–H$	L#–MH# / L#–	L#–MH# / L#–
92	L#–H$	L#–#H	L#–MH# / L#–	L#–MH# / L#–
93	L#–L	L#–L	L#–L	L#–L
94	L#–#H	L#–#H	L#–L# / L#–	L#–L
95	L#–#H	L#–#H	L#–L# / L#–	L#–L
96	L#–H$	L#–H$	L#–H# / L#–	L#–H# / L#–
97	L#–#H	L#–#H	L#–MH# / L#–	L#–MH# / L#–
98	L#–H$	L#–H$	L#–H# / L#–	L#–H# / L#–
99	L#–L	L#–#H	L#–L# / L#–	L#–L
100	#H	#H	MH#	MH#

Table 3.7: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. M and L tones. Numerals from 1 to 25.

	M_a	M_b	L_a	L_b	L_c
1	M	M	L#	L#	L#
2	M	M	L#	L#	L#
3	M	M	L	M	M
4	L	L	H#	H#	H#
5	L	L	H#	H#	H#
6	H#	H$	H#	H#	H#
7	M	M	L#	L#	L#
8	H#	H$	H#	H#	H#
9	L	L	H#	H# / L#	H#
10	M	M	L	M	L
11	M	M	L#	L#	L#
12	M	M	L#	L#	L#
13	M	M	L#	L#	L#
14	L+H#	L	L+H#	L+H#	L+H#
15	L+H#	L	L+H#	L+H#	L+H#
16	L+H#	L	L+H#	L+H#	L+H#
17	M	M	L#	L#	L#
18	L+H#	L	L+H#	L+H#	L+H#
19	L+H#	L	L+H#	L+H#	L+H#
20	M	M	–L	–L	–L
21	M	M	–L#	–L#	–L#
22	M	M	–L#	–L#	–L#
23	–L / M	M / –L	–L	–L	–L
24	–L	–L	H#	H#	H#
25	–L	–L	H#	H#	H#

Table 3.8: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. M and L tones. Numerals from 26 to 50.

	M_a	M_b	L_a	L_b	L_c
26	H#	H$	H#	H#	H#
27	M	M	–L#	–L#	–L#
28	H#	H$	H#	H#	H#
29	–L	–L	H#	–L# / H#	H#
30	M	M	–L#	–L	–L#
31	M	M	–L#	–L#	–L#
32	M	M	–L#	–L#	–L#
33	–L	–L	–L	–L	–L
34	–L	–L	H#	H#	H#
35	–L	–L	H#	H#	H#
36	H#	H$	H#	H#	H#
37	M	L	–L#	–L#	–L#
38	H#	H$	H#	H#	H#
39	–L	–L	H# / –L#	H# / –L#	H# / –L#
40	L#–	L#–	L#–	L#–	L#–
41	L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
42	L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
43	L#–	L#–M / L#–	L#–M / L#–L	L#–M / L#–	L#–M / L#–
44	L#–H# / L#–	L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
45	L#–H# / L#–	L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
46	L#–H# / L#–	L#–H$ / L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
47	L#–M / L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
48	L#–H# / L#–	L#–H$ / L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
49	L#–H# / L#–	L#–L	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
50	L#–	L#–	L#–	L#–	L#–

Table 3.9: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. M and L tones. Numerals from 51 to 75.

	M_a	M_b	L_a	L_b	L_c
51	L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
52	L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
53	L#–	L#–M / L#–	L#–M / L#–L	L#–M / L#–	L#–M / L#–
54	L#–H# / L#–	L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
55	L#–H# / L#–	L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
56	L#–H# / L#–	L#–H$ / L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
57	L#–M / L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
58	L#–H# / L#–	L#–H$ / L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
59	L#–H# / L#–	L#–L	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
60	LM–H#	LM–H$	LM–H#	LM–H#	LM–H#
61	LM–H# / L+MH#–M	LM–H$	LM–H#	LM–H#	LM–H#
62	LM–H# / L+MH#–M	LM–H$	LM–H#	LM–H#	LM–H#
63	LM–H# / L+MH#–M	LM–H$	LM–H#	LM–H#	LM–H#
64	LM–H# / L+MH#–L	LM–H$	LM–H#	LM–H#	LM–H#
65	LM–H# / L+MH#–L	LM–H$	LM–H#	LM–H#	LM–H#
66	LM–H#	LM–H$	LM–H#	LM–H#	LM–H#
67	LM–H# / L+MH#–M	LM–#H	LM–H#	LM–H#	LM–H#
68	LM–H#	LM–H$	LM–H#	LM–H#	LM–H#
69	LM–H# / L+MH#–L	LM–H$ / LM–L	LM–H#	LM–H#	LM–H#
70	L#–	L#–	L#–	L#–	L#–
71	L#– / L#–M	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
72	L#– / L#–M	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
73	L#– / L#–M	L#–M / L#–	L#–L	L#–M / L#–	L#–M / L#–
74	L#– / L#–H#	L#–L	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
75	L#– / L#–H#	L#–L	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–

Table 3.10: The underlying tone patterns of the nine categories of numeral-plus-classifier phrases. M and L tones. Numerals from 75 to 100.

	M_a	M_b	L_a	L_b	L_c
76	L#–H# / L#–	L#–H$ / L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
77	L#–M / L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L# / L#–
78	L#–H# / L#–	L#–H$ / L#–	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
79	L#–L / L#–H#	L#–L	L#–H# / L#–	L#–H# / L#–	L#–H# / L#–
80	LM–H#	LM–H$	LM–H#	LM–H#	LM–H#
81	LM–H# / L+MH#–M	LM–H$	LM–H#	LM–H#	LM–H#
82	LM–H# / L+MH#–M	LM–H$	LM–H#	LM–H#	LM–H#
83	LM–H# / L+MH#–M	LM–H$	LM–H#	LM–H#	LM–H#
84	LM–H# / L+MH#–L	LM–H$	LM–H#	LM–H#	LM–H#
85	LM–H# / L+MH#–L	LM–H$	LM–H#	LM–H#	LM–H#
86	LM–H#	LM–H$	LM–H#	LM–H#	LM–H#
87	LM–H# / L+MH#–M	LM–#H	LM–H#	LM–H#	LM–H#
88	LM–H#	LM–H$	LM–H#	LM–H#	LM–H#
89	LM–H# / L+MH#–L	LM–H$ / LM–L	LM–H#	LM–H#	LM–H#
90	L#– / L#–	L#–	L#–	L#–	L#–
91	L#–M / L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L#
92	L#–M / L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L#
93	L#–M / L#–	L#–M / L#–	L#–M / L#–	L#–M / L#–	L#–M / L#–
94	L#–L / L#–H#	L#–	L#–H# / L#–	L#–H# / L#–	L#–H#
95	L#–L / L#–H#	L#–	L#–H# / L#–	L#–H# / L#–	L#–H#
96	L#–H# / L#–	L#–H$ / L#–	L#–H#	L#–H# / L#–	L#–H#
97	L#–M / L#–	L#–M / L#–	L#–L# / L#–	L#–L# / L#–	L#–L#
98	L#–H# / L#–	L#–H$ / L#–	L#–H#	L#–H# / L#–	L#–H#
99	L#–L / L#–H#	L#–L	L#–H#	L#–H# / L#–	L#–H# / L#
100	M	M	L#	L#	L#

Entries in Tables 3.3 to 3.10 that begin with '−' do not have any specified tone on their first part; this part receives a Mid tone, by default. Using again '33 days' as an example, its tonal pattern is −L, meaning that a Low tone associates after the juncture, yielding /…**so˩-ɲi˩**/. Since Low tones do not spread regressively ('right-to-left'), the first part of the phrase receives a default M tone (/**so˧tsʰi˧**…/), resulting in the final output /**so˧tsʰi˧so˩-ɲi˩**/.

Similarly, entries whose tone pattern ends with '−' lack a specified tone on their second part, which receives its tones through the phonological tone rules governing tone groups in the Alawua dialect of Yongning Na. For example, 'forty years' has the tone pattern L#−: a final L tone associates to the first half of the phrase, yielding /**ʐv̩.tsʰi˩**…/. As previously mentioned, tones do not spread leftward; the first syllable thus receives a default Mid tone, giving /**ʐv̩˧tsʰi˩**…/. At this point, a phonological rule applies (see §7.1.1): Rule 5, which states that "All syllables following an H.L or M.L sequence receive L tone". This means that a tone cannot be flanked by higher tones within a tone group: sequences such as | MLM |, | HMH |, | HLM | or | MLMH | do not occur. Consequently, the only possible tone on the second part of the phrase is L. The final output is /**ʐv̩˧tsʰi˩-kʰv̩˩**/, as shown in Figure 3.1.

3.1.3 Why little evidence about the tones of classifiers can be gleaned from the free forms in which they originate

In the preceding sections, the tones of classifiers were analyzed on the basis of their synchronic distributional properties. Mention needs to be made of other potential sources of evidence, explaining why they have not, so far, provided decisive insights into the phonological nature of classifiers' tonal categories.

In principle, relevant evidence could come from classifiers that transparently correspond to a free form: either a noun or a verb. For example, the classifier for blows, /**dɑ˧˥**/, is a cognate object of the verb /**dɑ˧˥**/ 'to hit, to strike', as illustrated in (1).

(1) ɖɯ˧-dɑ˧ tʰi˥-dɑ˩
ɖɯ˧ dɑ˧˥ tʰi˧- dɑ˧˥
one CLF.blows DUR to_strike
'to strike a blow' *(Sister3.135)* {0004344#S135}

The tonal correspondence with the verb appears transparent: both the verb and its grammaticalized form as a classifier carry MH tone. Similarly, 'mountain, hill' is /**ʁwɤ˧**/, and as a classifier it yields /**ʁwɤ˧**/ 'heap (of something)', which is

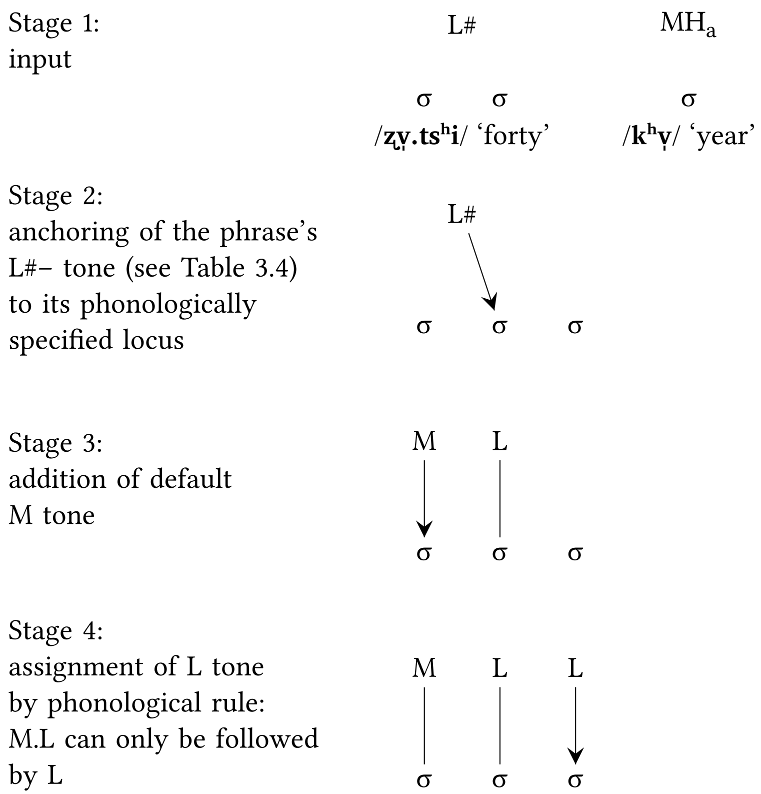


Figure 3.1: A detailed representation of tone-to-syllable association for the numeral-plus-classifier phrase /**ʐv̩˧tsʰi˩-kʰv̩˩**/ 'forty years'.

phonologically identical to the free noun, including its M tone. Likewise, 'beam', /**ɖʐo˥**/, is identical in form to its self-classifier, /**ɖʐo˥**/.

However, such tonal identity is not found in all examples. A different pattern for H-tone nouns is exemplified by /**kɯ˥**/ 'star', which yields /**kɯ˧**/ as a self-classifier, belonging to the M$_b$ category. Similarly, 'bowl' is /**qʰwɤ˩˧**/ (LM tone) as a noun but /**qʰwɤ˧˥**/ (MH$_a$ category) when used as a classifier meaning 'bowlful'.

To date, too few classifiers can be straightforwardly linked to full nouns (or verbs) for the search of tonal correspondences between lexical word and classifier to yield robust generalizations. While isolated cases may suggest a tonal relationship between free forms and classifiers, the overall picture remains too inconsistent for this to serve as a reliable means of determining classifier tonal categories.

3.1.4 Borrowed classifiers

Classifiers borrowed from Chinese constitute another potential source of evidence for the tonal categories of Yongning Na classifiers. Specifically, it can be assumed that the tone subcategories assigned to recent borrowings function synchronically as unmarked, default categories.

Table 3.11 presents the eight observed loanwords: four with L tone, three with MH tone, and one with H tone.

Table 3.11: Classifiers borrowed from Chinese.

borrowed classifier			Chinese donor form		
form	tone	CLF for:	script	*Pinyin*	note
po˩$_b$	L$_b$	pack	包	*bāo*	presumably a recent loan
tsʰe˩$_b$	L$_b$	thumb, inch	寸	*cùn*	presumably a recent loan
mæ˩$_a$	L$_a$	monetary unit	元	*yuán*	Middle Chinese *ngjwon (Baxter 2000); Old Chinese *[ŋ]o[r] (Baxter & Sagart 2014)
mæ˩$_a$	L$_a$	10,000	萬	*wàn*	Middle Chinese *mjonH (Baxter 2000); Old Chinese *C.ma[n]-s (Baxter & Sagart 2014)
tɕi˧˥$_a$	MH$_a$	pound	斤	*jīn*	presumably a recent loan
pv̩˧˥$_a$	MH$_a$	step, stride	步	*bù*	presumably a recent loan
te˧˥$_a$	MH$_a$	generations	代	*dài*	presumably a recent loan
mo˥$_a$	H$_a$	acre	亩	*mŭ*	presumably a recent loan

The number of examples is small, and a detailed study of the successive layers of Chinese loanwords in Na (following the method applied to Hani by Sagart & Shixuan 2001) remains a task for the future. Nonetheless, keeping these major limitations in mind, an interesting pattern emerges: the two subsets of L-tone classifiers in Table 3.11, each containing two items, appear to align with a distinction between recent and earlier loanwords.

The classifiers for ‘pack’ (e.g. a pack of cigarettes) and ‘thumb, inch’ (used in its standard sense as a measurement of length, ⅓ dm) are excellent candidates for recent loanwords, and both carry L$_b$ tone. By contrast, ‘monetary unit, yuan’ and ‘10,000’ are not part of the same layer of loanwords. Their history remains to be worked out, but their nasal initial consonant, in contrast with the glide-initial onset of their Mandarin counterparts, constitutes sufficient evidence that they are not recent loans. Both carry L_a tone. This pattern suggests that the current default subcategory of L tones for classifiers is L_b.

The three MH-tone classifiers in Table 3.11 are all presumably recent loans and all carry MH_a tone. Thus, the current default subcategory of MH tones for classifiers appears to be MH_a.

As for the H-tone classifier /**mo˥**$_\mathbf{a}$/, ‘Chinese acre’, corresponding to 亩 *mǔ*, it is presumably a recent loan, but given that it is the sole H-tone item in the dataset, no general conclusions can be drawn.

The way forward here would consist in a comprehensive study of Chinese loanwords in Na.

3.1.5 Indirect confirmation for the H, MH, M, and L categories of classifiers from frequency in surface forms

The distributional frequency of the various tonal levels (the three phonological primitives: H, M, and L) provides some degree of support – albeit weak and indirect – for the four tonal ‘super-categories’ of classifiers proposed here, namely H, MH, M, and L. Under the (admittedly simplistic) assumption that the contribution of the classifier’s lexical tone is statistically reflected in the tonal patterns of numeral-plus-classifier phrases, the labels H, MH, M, and L make good sense.

Averaging over the entire range of tone patterns from 1 to 100, it emerges that the classifiers that carry a High tone after 1 and 2 (labelled H_a and H_b in Tables 3.3 to 3.10) also show the highest proportion of H tones overall – simply counting the ‘H’ letters in the relevant columns, regardless of the tonal pattern in which they appear, be it H#, #H, H$, MH or MH# – and the lowest proportion of L tones. Conversely, the three tonal categories of classifiers that carry a Low tone after ‘1’ and ‘2’ (analyzed as L_a, L_b, and L_c) exhibit the lowest proportion of H tones and the highest of L tones (based, here again, on a raw count of ‘H’ and ‘L’ letters in the relevant columns).

The other two classifier categories, M and MH (with subgroups M_a and M_b, and MH_a and MH_b), lie in an intermediate position between these two extremes. As expected, M_a and M_b show a higher proportion of M tones and a lower proportion of H tones than MH_a and MH_b. These broad comparisons, while lacking demonstrative value, help convey an overall sense of the data’s structure.

Another indirect approach to this dataset consists in examining occasional mistakes made by the consultant. Errors in tonal realization may shed light on the underlying categories by revealing which tones are more susceptible to confusion or substitution.

3.1.6 Degree of tonal complexity and frequency of mistakes in the recordings

As mentioned earlier (§3.1.1), the task of producing long series of numeral-plus-classifier phrases was challenging for the consultant. Among the 2,810 recorded tokens, 7% have a mistaken tone pattern,[7] i.e. a tone pattern that Mrs. Latami (consultant F4) consistently judged to be incorrect (a tonal slip of the tongue) when we revisited the data after the recording sessions.

The overall amount of errors is comparable to that obtained in tonal elicitation on different languages: e.g. Zerbian & Kügler (2023: 369) report discarding 8% of tokens due to disfluencies or mispronunciations. In detail, these mistakes partly reflect the degree of complexity of the tone patterns in question. The notion that mistakes can provide insights about language dates back at least to Henri Frei's *Grammar of mistakes* (1929); this strand of research was also pursued by Fromkin (1973), Rossi & Peter-Defare (1998), and Nooteboom (2011). The usefulness of speech errors and word games to gain insights into tonal systems was demonstrated by Hombert (1986: 180–181). Speakers of Mandarin, Cantonese, Minnan and Thai tend to move the tones along with the syllables when changing a $C_1V_1C_2V_2$ sequence into $C_1V_2C_2V_1$ or $C_2V_2C_1V_1$, whereas speakers of Bakwiri (also known as Bakweri; Bantu branch of Niger-Congo) tend to leave tone patterns unchanged. (See also Wan & Jaeger (1998) on Mandarin.)

Figure 3.2 presents the distribution of mistakes according to the range of tens, showing the proportion of mistakes occurring in numeral-plus-classifier phrases for numerals between 1 and 9 (leftmost bar), 10 and 19 (second bar), etc. The ranges 50–59 and 80–89 are not represented, as they were excluded from the recordings to reduce consultant fatigue. As noted in the description of the elicitation procedure in §3.1.1, this decision was based on the fact that, in the morphotonological system of Yongning Na, the tone patterns of 50–59 are always identical with those for 40–49, and those of 80–89 mirror those of 60–69.

Figure 3.3 shows the distribution of mistakes according to the last digit: how many mistakes concern numbers ending in 1 (namely 1, 11, 21, 31, 41, and so on), in 2, etc.

The dataset is not perfectly symmetrical. In particular:

[7] This figure includes some items that were deleted from the sound files at an early stage of the study, before the decision was made to preserve the recordings unchanged.

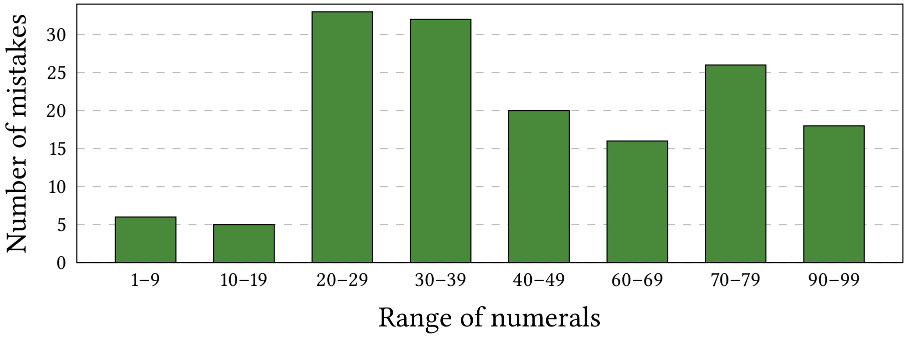


Figure 3.2: Number of mistakes in the recorded numeral-plus-classifier phrases as a factor of the range of tens.

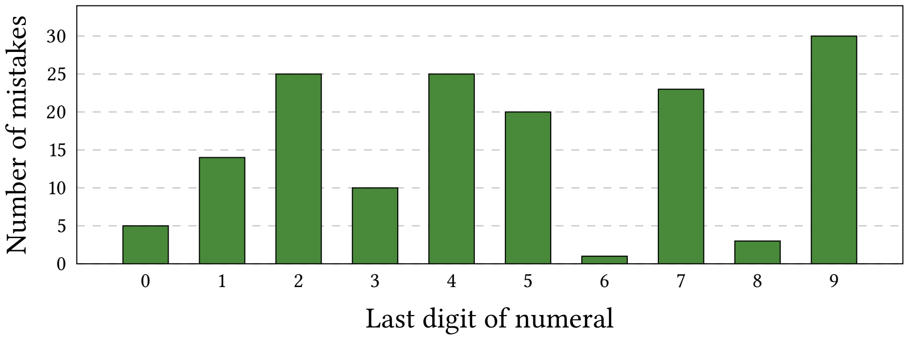


Figure 3.3: Number of mistakes in the recorded numeral-plus-classifier phrases as a function of the last digit (units).

(i) some tonal categories are represented more extensively than others;

(ii) certain combinations were repeated at recording; and

(iii) there is a slight overrepresentation of tokens in the range [1..10] relative to higher numerals.

Nevertheless, some observations can be made. Numerals ending in 6 and 8 (e.g. 6, 16, 26; 8, 18, 28) show a notably lower rate of mistakes. Likewise, numerals beginning with 6 or 8 (e.g. 60, 61, 62, 63; 80, 81, 82, 83) are slightly less prone to errors than those in adjacent ranges of ten. This is unsurprising, as numeral-plus-classifier phrases containing 6 and 8 are the least complex in tonal terms, due to the neutralization of many tonal distinctions following these numerals (as noted at the outset of §3.1.2).

By contrast, numeral-plus-classifier phrases containing numbers ending in 7 and 9, which exhibit the greatest diversity in tone patterns, are among the most frequently mistaken. The highest concentration of mistakes is found in phrases containing {2, 4, 5, 7, 9}.

3.1.7 About variants of tone patterns

As noted at the outset of this chapter, some phrases allow for variant tone patterns. This phenomenon is more frequent for higher numerals than for numerals below twenty. For instance, the combination of the numeral 47 with the classifier for round objects can be realized either as /**ʐv̩˧tsʰi˩–ʂɯ˧-ɭɯ˥**/ or as /**ʐv̩˧tsʰi˩–ʂɯ˩-ɭɯ˩**/. No observed numeral-plus-classifier phrase has more than two acceptable tone patterns.

Many of these variants can be explained in light of the existence of two possible ways to parse a numeral-plus-classifier phrase: as a single tone group or as two tone groups. If a numeral-plus-classifier phrase is treated as a single tone group, then its tone pattern is subject to the phonological rules that apply within a tone group (see §7.1.1).

For instance, the numeral 44 in association with the classifier for tools, /**nɑ˧$_{\mathbf{a}}$**/, allows for the following two variants: /**ʐv̩˧tsʰi˩–ʐv̩˧-nɑ˥**/ and /**ʐv̩˧tsʰi˩–ʐv̩˩-nɑ˩**/. The first of these, /**ʐv̩˧tsʰi˩–ʐv̩˧-nɑ˥**/, is not a well-formed tone group, as it violates Rule 5: within a tone group, all syllables following an M.L sequence receive L tone. The phrase /**ʐv̩˧tsʰi˩–ʐv̩˧-nɑ˥**/ is therefore to be analyzed as consisting of two tone groups: /**ʐv̩˧tsʰi˩ | ʐv̩˧-nɑ˥**/ (tone pattern: L# for the first tone group, H# for the second). If this phrase were treated as a single tone group, the tones of its last two syllables would be lowered to L. This is precisely what happens in the alternative attested variant: /**ʐv̩˧tsʰi˩ʐv̩˩-nɑ˩**/ (tone pattern: L#–). The two variants can therefore be described as (i) a form consisting of two tone groups and (ii) a simplified form, whose tonal pattern results straightforwardly from its treatment as a single tone group. (The same phenomenon is reported in the Lataddi dialect: see Dobbs & La 2016.)

The same applies to all tonal patterns in the ranges [40..59], [70..79], and [90..99], as the first two syllables (corresponding to 40, 50, 70 and 90, respectively) carry a Mid-plus-Low pattern. This pattern precludes any tone other than L on the following syllables within the same tone group (by Rule 5). One would therefore expect all of these combinations to have two variants. This holds true as a general rule: whenever the consultant provided a complex form and I attempted a simplified form, the latter was never rejected. On the other hand, for some combinations, only the simpler form is acceptable. For instance, for 44 with a classifier of category M_b, such as /**ɭɯ˧$_b$**/ (the classifier for round objects), the only correct (acceptable) form is /**ʐv̩˧tsʰi˩ʐv̩˩-ɭɯ˩**/ (tone pattern: L#–).

If one supposes, by analogy with category M_a, that † **ʐv̩˧tsʰi˩ | ʐv̩˧-ɭɯ˥** was acceptable at an earlier stage of the language, then it must have fallen out of use. For category M_a, where two variants are currently in common use, the consultant's intuition is that the complex variant, /**ʐv̩˧tsʰi˩ | ʐv̩˧-nɑ˥**/, sounds somewhat "slow" and "clumsy"; it conveys special emphasis and is only appropriate as part of an expressive strategy to highlight the figure in question. In summary, the integration of numeral-plus-classifier phrases into a single tone group is the general rule.

Interestingly, when a phrase ends in two Low-tone syllables, it is possible to test whether these Low tones result from the levelling down of originally non-Low tones (as in the case of /**ʐv̩˧tsʰi˩–ʐv̩˩-nɑ˩**/, mentioned above) or whether they reflect an underlying Low tone. When the phrase is divided into two tone groups, if the second group has an underlying L tone, it receives a postlexical final H tone through the application of Rule 7: "If a tone group only contains L tones, a postlexical H tone is added to its last syllable". For example, '23 years' (category MH_a) can be realized either as /**ɲi˧tsi˧–so˩-kʰv̩˩**/ or as /**ɲi˧tsi˧ | so˩-kʰv̩˩˥**/, revealing that its underlying tone pattern is M–L. By contrast, with the classifier for tools, it would be incorrect to say ‡ **ʐv̩˧tsʰi˩ | ʐv̩˩-nɑ˩˥**: the acceptable variant with a division into two groups is /**ʐv̩˧tsʰi˩ | ʐv̩˧-nɑ˥**/.

A device for forcing the division of the phrase into two tone groups consists of inserting the syllable /**lɑ˧**/ 'and' before the last digit, as in /**ʂɯ˧tsʰi˩ lɑ˩ | qʰv̩˧-ʁwɤ˥**/ '79 heaps'.

3.1.8 Disyllabic classifiers, and what they reveal about the tones of numerals

Only a few disyllabic classifiers were observed. One is a reduplicated monosyllable: /**ʈʂʰe˧~ʈʂʰe˧**/ 'the width of a room'.[8] Others are nouns, such as /**ʝi˧qʰv̩#˥**/ 'bull's horn', which can be used as a classifier, as bull horns were traditionally used for drinking or pouring liquids, for instance to pour water into a pot. This noun competes, in its classifier function, with another classifier specifically referring to hornfuls of liquids, /**qʰv̩˧tʰv̩˧**/, which is more commonly used.

'Bottle', /**to˩bi#˥**/, can be used either as a noun, as in /**to˩bi#˥ | ɖɯ˧-ɭɯ˧**/ 'a bottle', or as a classifier, as in /**ʐɯ˧ | ɖɯ˧-to˩bi˩**/ 'a bottle of liquor/spirits'. The

[8] Roselle Dobbs (p.c. 2016) points out that the classifier /**ʈʂʰe˧~ʈʂʰe˧**/ 'the width of a room' may be reduplicated from the verb /**ʈʂʰe˧$_b$**/ 'to stretch'.

Table 3.12: The tonal behaviour of disyllabic classifiers with lexical M or #H tone.

numeral	example form	output tone	meaning
1	**ɖɯ˧-qʰv̩˧tʰv̩#˥**	#H	one hornful
2	**ɲi˧-qʰv̩˧tʰv̩#˥**	#H	two hornfuls
3	**so˩-qʰv̩˩tʰv̩˩˥**	L	three hornfuls
4	**ʐv̩˧-qʰv̩˧tʰv̩#˥**	#H	four hornfuls
5	**ŋwɤ˧-qʰv̩˧tʰv̩#˥**	#H	five hornfuls
6	**qʰv̩˧-qʰv̩˧tʰv̩#˥**	#H	six hornfuls
7	**ʂɯ˧-qʰv̩˧tʰv̩#˥**	#H	seven hornfuls
8	**hõ˧-qʰv̩˧tʰv̩#˥**	#H	eight hornfuls
9	**gv̩˧-qʰv̩˧tʰv̩#˥**	#H	nine hornfuls
10	**tsʰe˩-qʰv̩˩tʰv̩˩˥**	L	ten hornfuls

classifier for ladlefuls is /**bv̩˩dze˩**/, and that for handfuls (using both hands) is /**lo˩dzi˩**/.

The disyllabic classifiers observed to date fall into one of four tonal categories: M, #H, L, and LM+#H. In terms of their behaviour in association with numerals, these four tonal categories cluster into two sets: M and #H on the one hand, L and LM+#H on the other. Tables 3.12 and 3.13 present the data. Clearly, the relevant juncture for tonal association in these phrases is the one preceding the classifier, even for numerals below twenty. Describing the tone pattern of /**ɖɯ˧-to˩bi˩**/ 'one bottle' requires recognition of a juncture preceding the classifier, i.e. /**ɖɯ˧ – to˩bi˩**/.

3.1.9 Expressing approximation with two numerals and a classifier

Two numerals can accompany a classifier, conveying an approximative number. This can be likened to the coordinative construction 'NUM or NUM' in English, as in 'one or two' and 'two or three'. This construction is less common than coordinative compounds consisting of two numeral-plus-classifier phrases, such as /**ɖɯ˧-ɲi˧ – ɲi˧-ɲi˧**/ 'one day-two days' and /**ɲi˧-ɲi˧ – so˧-ɲi˥**/ 'two days-three days', presented in §4.3. Table 3.14 sets out the results of elicitation with the noun 'day'. A dash '–' indicates that the combination is not in use. This is not to say that there is a hard-and-fast rule against the combinations marked with a dash in Table 3.14, only that they seemed less felicitous to the consultant for reasons

Table 3.13: The tonal behaviour of disyllabic classifiers with lexical L or LM+#H tone.

numeral	example form	output tone	meaning
1	**ɖɯ˧-to˩bi˩**	–L	one bottle
2	**ɲi˧-to˩bi˩**	–L	two bottles
3	**so˩-to˩bi˩˥**	L	three bottles
4	**ʐv̩˧-to˥bi˩**	#H–	four bottles
5	**ŋwɤ˧-to˥bi˩**	#H–	five bottles
6	**qʰv̩˧-to˥bi˩**	#H–	six bottles
7	**ʂɯ˧-to˩bi˩**	–L	seven bottles
8	**hõ˧-to˥bi˩**	#H–	eight bottles
9	**gv̩˧-to˥bi˩**	#H–	nine bottles
10	**tsʰe˩-to˩bi˩˥**	L	ten bottles

that may involve considerations of homophony along with frequency of use and semantics.

For instance, the expression **ɖɯ˧-ɲi˧ ɲi˧**, literally 'one-two days', actually means 'a few days', somewhat like 'a couple of days' in English. Since this expression easily covers the range from one to four, it renders the combinations 'two-three' and 'three-four' unnecessary. Among higher numbers, a reason why the combination 'seven-eight' is deemed acceptable may be that it corresponds to 'a week or so', a span of time that holds particular relevance in the consultant's current conceptualization of time, given that weeks are meaningful units due to the grandchildren's school schedule. Seen in this light, the combination 'six-seven' should be about as relevant as 'seven-eight'. Indeed, the consultant (Mrs. Latami) confirmed that it could be acceptable at a push: **qʰv̩˧-ʂɯ˧ ɲi˧** 'six or seven days'.

The fact that the expressions in Table 3.14 do not form a fully productive paradigm in the consultant's speech led to the conclusion that it would not be appropriate to attempt systematic elicitation of phrases consisting of two numerals and a classifier. There was a concern that elicitation of unusual forms – many of them new coinages – would not yield consistent results (as discussed in §1.4.4). This is one of many topics that remain for future research.

Table 3.14: Expressions conveying an approximative number: two numerals plus a classifier.

numerals	association of two numerals	meaning
1 and 2	**ɖɯ˧-ɲi˧ ɲi˧**	‘a few days’
2 and 3	–	
3 and 4	–	
4 and 5	**ʐv̩˧-ŋwɤ˧ ɲi˧**	‘four or five days’
5 and 6	**ŋwɤ˧-qʰv̩˧ ɲi˧**	‘five or six days’
6 and 7	–	
7 and 8	**ʂ̩ɯ˧-hõ˧ ɲi˧**	‘seven or eight days’
8 and 9	–	
9 and 10	**gv̩˧-tsʰe˧ ɲi˥**	‘nine or ten days’

3.1.10 Conclusions about numeral-plus-classifier phrases

From the mass of information set out above, it is clear that the tone patterns of numeral-plus-classifier phrases encapsulate information that is not derived from phonological rules. The system presented in Tables 3.3 to 3.10 is both regular and productive, in that all classifiers of a given tone category follow the same tone patterns. As this system lends itself straightforwardly to computer implementation, a simple script was written.[9] This script takes as input the classifier's tone category, its segmental composition, and a numeral (or range of numerals) from 1 to 100. The data in Tables 3.3 to 3.10 is stored within the script, allowing for tone patterns to be retrieved through table lookup. The surface phonological tone pattern is then assigned to the phrase based on the general rules governing tonal association in this dialect (rules that are encoded into the script), such as the association of simple L and M tones to all syllables within their domain, and the "left-to-right" association of tone sequences.

For instance, when the numeral 44 and the classifier for tools, /**nɑ˧**$_{\mathrm{a}}$/ (tonal category: M$_{\mathrm{a}}$), are provided as input, the script yields the following two variants for ‘44 tools’: /**ʐv̩˧tsʰi˩–ʐv̩˧-nɑ˥**/ (tone pattern: L#–H#) and /**ʐv̩˧tsʰi˩–ʐv̩˩-nɑ˩**/ (tone pattern: L#–).

In the current version of the script, all of the information set out in Tables 3.3 to 3.10 is encoded in full, specifying the tone patterns of 900 combinations (9 tone

[9]As of 2025, the script is available at https://github.com/alexis-michaud/na/blob/master/SCRIPTS/NaTone/NaTone.pl.

categories of classifiers × 100 numerals). This allows for straightforward table-lookup but is uneconomical from the perspective of linguistic modelling. The addition of some rules could significantly reduce the number of combinations that need to be explicitly specified.

In particular, the tone patterns of numerals in the range [40..49] are identical to those in [50..59]; likewise for [60..69] and [80..89]. Numerals ending in 1 also share identical tone patterns with those ending in 2, with a few exceptions in category H_b. Furthermore, the information provided for each subcategory (H_a and H_b, M_a and M_b, and so on) could be streamlined by designating one subcategory as the default and only specifying forms for the other subcategories where they diverge from this default.

However, even after such simplification, a substantial number of tonal patterns would still need to be specified individually. For instance, neither H_a nor H_b can be straightforwardly derived from the other. While the presence of an L tone in all phrases from 10 to 19 might suggest that the patterns for H_b constitute a simplified version of those for H_a, there is also a complicating factor for H_b that does not apply to H_a, namely the presence of different tones after numerals ending in 1 and 2. This observation is a striking counterexample to the pan-Naish generalization that the numerals 1 and 2 always have the same tone patterns – a generalization that holds for Naxi and Laze, as well as for all the other available Na data.

Finally, an idiosyncratic tone pattern is observed for /**to˥**/ 'armful': this classifier belongs to the H_a category, yet the combination '11 armfuls' is realized as /**tsʰe˧ɖɯ˧-to˧**/ instead of the expected † **tsʰe˩ɖɯ˩-to˧˥**.

3.2 Demonstrative-plus-classifier phrases

A demonstrative and a following classifier are always integrated into a single tone group, as illustrated in (2), where the proximal demonstrative //**ʈʂʰɯ˥**// and the classifier for pairs or sets //**dzi˧**$_b$// combine as /**ʈʂʰɯ˧-dzi˧˥**/ 'this set'.

(2) no˩bv̩˧ | ʈʂʰɯ˧-dzi˧˥ | le˧-ʑi˩, | tʰi˧-mv̩˧-kʰɯ˧˥.
no˩bv̩˧ ʈʂʰɯ˥ dzi˧$_b$ le˧- ʑi˩$_b$ tʰi˧-
given_name DEM.PROX CLF.sets/pairs ACCOMP to_bring_along DUR
mv̩˧$_a$ -kʰɯ˧˥
to_put_on CAUS
'Nobbu brought that set [of clothes], and made [her] wear [the clothes].'
(BuriedAlive3.73) {0004538#S73}

The elicitation procedure for demonstrative-plus-classifier phrases was similar to that used for numeral-plus-classifier phrases, and similar puzzles were encountered.

As explained in the previous section, the tonal categories of classifiers were established on the basis of their behaviour when combined with numerals. The nine tonal categories of monosyllabic classifiers are: $\mathrm{H_a}$ and $\mathrm{H_b}$; $\mathrm{MH_a}$ and $\mathrm{MH_b}$; $\mathrm{M_a}$ and $\mathrm{M_b}$; and $\mathrm{L_a}$, $\mathrm{L_b}$, and $\mathrm{L_c}$. As for the proximal demonstrative, /**ʈʂʰɯ#˥**/, and the distal demonstrative, /**tʰv̩#˥**/, both carry a lexical #H tone. The expectations were that, in demonstrative-plus-classifier phrases, (i) there would be no difference between proximal and distal demonstratives, since they have the same lexical tone, and (ii) all classifiers within each of the nine tonal categories would have the same tonal behaviour.

The first prediction was verified: phrases containing /**ʈʂʰɯ#˥**/ 'this' and /**tʰv̩#˥**/ 'that' always have the same tonal patterns. The second prediction, on the other hand, was not borne out: some of the tonal categories for classifiers proved to be less than fully homogeneous. The account provided below begins with the simplest cases and progresses towards the category displaying the greatest degree of divergence.

When combined with demonstratives, $\mathrm{H_a}$ and $\mathrm{H_b}$ behave identically, as do $\mathrm{MH_a}$ and $\mathrm{MH_b}$. This is the simplest part of the system: the opposition between $\mathrm{H_a}$ and $\mathrm{H_b}$ is neutralized in this context, as is the opposition between $\mathrm{MH_a}$ and $\mathrm{MH_b}$. Categories $\mathrm{M_a}$ and $\mathrm{M_b}$, by contrast, behave differently from each other, but in a consistent and straightforward manner, each displaying only one possible pattern: a demonstrative plus an $\mathrm{M_a}$-tone classifier yields L#, e.g. /**tʰv̩˧-nɑ˩**/ (classifier for tools); a demonstrative plus an $\mathrm{M_b}$-tone classifier yields #H, e.g. /**tʰv̩˧-ɭɯ#˥**/ (generic classifier).

Among Low-tone classifier categories, $\mathrm{L_a}$ and $\mathrm{L_c}$ are relatively straightforward. All $\mathrm{L_a}$-tone classifiers have the same behaviour, allowing two variants: H# and H$. Both variants are firmly attested. The speaker expresses a preference for the former, but this slight imbalance appears to be uniform across examples, suggesting that no clear (lexicalized) preference exists for one variant over the other in association with a specific classifier. $\mathrm{L_c}$-tone classifiers, on the other hand, allow no fewer than three variants: MH#, H#, and H$.

Category $\mathrm{L_b}$ is the most problematic. In the production data, three variants are attested: H#, H$, and MH#. However, the distribution of these variants is not uniform across classifiers of this category. For some classifiers (e.g. /**mi˩**$_{\mathbf{b}}$/, the classifier for animals, and /**kʰɯ˩**$_{\mathbf{b}}$/, the classifier for long objects), MH# is by far the most frequent pattern, with H$ appearing as an occasional variant and H# rarely attested. For other classifiers, by contrast, H# and H$ occur with comparable frequency, whereas MH# is seldom found.

When several variants were proposed by the investigator and the consultant was asked to evaluate their acceptability, a similar picture emerged: MH# was strongly preferred for some classifiers, with H$ judged as an acceptable variant, whereas H# was dispreferred (either rejected as incorrect or considered only marginally acceptable). For other classifiers, the preferred form was with H# tone, H$ was an acceptable variant, and MH# was not. Table 3.15 presents these two subsets, provisionally labelled as ‘type I’ and ‘type II’.

Table 3.15: The two subcategories of L$_b$-tone classifiers, based on their behaviour in association with demonstratives.

type	tone pattern in association with demonstrative	form	classifier for
L$_b$, type I	MH# most common;	**bo˩**$_b$	headdresses
	H$ attested;	**dv̩˩**$_b$	small groups
	H# dispreferred or refused	**dzi˩**$_b$	trees
		jo˩$_b$	ounces
		kʰɯ˩$_b$	long objects
		lo˩$_b$	valleys
		ɬi˩$_b$	armspans
		mi˩$_b$	animals
		pʰv̩˩$_b$	fields
		tʰv̩˩$_b$	sets of ten
		tɕʰi˩$_b$	meals
		wɤ˩$_b$	loads
		wo˩$_b$	teams of oxen
L$_b$, type II	H# most common;	**pɤ˩**$_b$	ladders, doors
	H$ attested;	**po˩**$_b$	packs
	MH# dispreferred or refused	**ʁo˩**$_b$	types, sorts
		ʂɯ˩$_b$	times
		tsʰe˩$_b$	leaves
		ʈv̩˩$_b$	large chunks

The consultant's acceptance of variants fluctuated somewhat from session to session, but the distinction between types I and II was confirmed over the course of an extensive series of elicitation sessions. Additional evidence is provided by examples from recorded texts, which reflect the same patterns.

The tone patterns for all categories of classifiers are set out in Table 3.16, using the distal demonstrative /**tʰv̩#˥**/ as an example. (As noted above, the tone patterns for the proximal demonstrative, /**tʂʰɯ#˥**/, are identical.) The corresponding online recordings are *DemClf* {0004830}, *DemClf2* {0004832}, and *DemClf3* {0004834}.

Table 3.16: The tone patterns of demonstrative-plus-classifier phrases.

tone pattern		example	
tone of CLF	tone of DEM+CLF phrase	classifier for	DEM+CLF phrase
H_a, H_b	H$ / #H	chunks	**tʰv̩˧-kʰwɤ˥$** / **tʰv̩˧-kʰwɤ#˥**
MH_a, MH_b	L#	cattle	**tʰv̩˧-pʰo˩**
M_a	L#	tools	**tʰv̩˧-nɑ˩**
M_b	#H	*generic*	**tʰv̩˧-ɭɯ#˥**
L_a	H# / H$	quantities	**tʰv̩˧-mɤ˥** / **tʰv̩˧-mɤ˥$**
L_b, type I	MH# / H$	animals	**tʰv̩˧-mi˧˥** / **tʰv̩˧-mi˥$**
L_b, type II	H# / H$	doors	**tʰv̩˧-pɤ˥** / **tʰv̩˧-pɤ˥$**
L_c	MH# / H# / H$	plains	**tʰv̩˧-di˧˥** / **tʰv̩˧-di˥** / **tʰv̩˧-di˥$**

The distinction between 'type I' and 'type II' within L_b-tone classifiers constitutes a striking additional layer of complexity within the classifier system's nine tonal categories. L_b itself is a subdivision within the broader L category of tones; its further division into types I and II thus constitutes a subdivision within a subdivision. From the perspective of phonological output, type I within the L_b category yields an outcome (MH# / H$) that does not coincide with that of any other category within the system.

3.3 Tonal interactions with a preceding noun

Occasionally, tonal interaction occurs between a numeral-plus-classifier or demonstrative-plus-classifier phrase and the preceding noun, as in (3).

(3) ə˧mi˧! | pæ˧kʰwɤ˧ so˧-ɭɯ˥ ki˩ mæ˩!
ə˧mi˧ pæ˧kʰwɤ#˥ so˩ ɭɯ˧ ki˧$_{a}$ mæ˧
INTJ silver_coin three CLF to_give DISC.PTCL
'Wow! [(S)he] is giving you three silver coins!'

According to the main consultant's recollections, (3) is the type of comment that uncles and aunts would make when a child received a significant monetary gift on the occasion of their coming-of-age ceremony.[10] Offering a single coin would be inappropriate, as gifts should come in pairs. Two coins constituted a generous gift, while three exceeded expectations, corresponding at the time to approximately half a month's salary.

The tonal derivation for the phrase 'three silver coins' is as follows:

(i) The input tones are: #H for 'silver coin', //**pæ˧kʰwɤ#˥**//; L for 'three'; and M for the classifier, //**ɭɯ˧**//.

(ii) The numeral-plus-classifier phrase 'three-CLF' carries M tone: //**so˧-ɭɯ˧**//.

(iii) The phrase resulting from the combination of (i) and (ii) carries an H# tone: //**pʰæ˧kʰwɤ˧ so˧-ɭɯ˥#**//.

Other examples of such tonal interaction include (4) and (5).

(4) mv̩˩zo˩ tʰv̩˩-ɭɯ˩˧
mv̩˩zo˩ tʰv̩˥ ɭɯ˧$_{b}$
girl DEM.DIST CLF.generic
'that girl' *(ComingOfAge2.60; Tiger2.139, 141, 147)* {0004588#S60}

(5) ə˧v̩˧ tʰv̩˧-v̩˧
ə˧v̩˧˥ tʰv̩˥ v̩˧
uncle DEM.DIST CLF.individual
'that uncle' *(Tiger2.109)* {0004545#S109}

The underlying tonal category of such expressions can be established using the same tests as for nouns, such as the addition of the copula. The expression 'three silver coins' /**pʰæ˧kʰwɤ˧ so˧-ɭɯ˥**/ in (3) yields /**pʰæ˧kʰwɤ˧ so˧-ɭɯ˥ ɲi˩**/, revealing that its tone is H#: //**pʰæ˧kʰwɤ˧ so˧-ɭɯ˥#**//. By contrast, 'that uncle', /**ə˧v̩˧ tʰv̩˧-v̩˧**/ (5), yields /**ə˧v̩˧ tʰv̩˧-v̩˧ ɲi˥**/, showing that its tone is #H: //**ə˧v̩˧ tʰv̩˧-v̩#˥**//. (As for (4), its surface form is sufficient to determine its underlying tone: it can only be //L//.)

[10]Explanations are provided in the narrative *ComingOfAge2.58ff*. {0004588#S58}.

Why is tonal change only occasional? Here are some speculations.

First, if tonal change were systematic, it would entail the neutralization of some tonal distinctions among classifiers. The number of possible tone patterns in three-syllable or longer tone groups is not as high as the combinatorial diversity in shorter tone groups. For instance, after disyllabic nouns carrying an L# tone, determiners would undergo tone lowering to L (by application of Rule 5: "All syllables following an H.L or M.L sequence receive L tone"), leading to a complete neutralization of tonal oppositions among classifiers. Tonal change would thus decrease distinctiveness, as neutralization makes surface tonal strings less rich in phonological information. It would moreover increase opacity, as more cases would arise in which the surface phonological tone of numeral-plus-classifier determiners differed from their underlying tone. Given the richness of the tone system of classifiers, increased opacity would be likely to create pressure towards simplification. The fact that tonal interaction between a noun and a following determiner occurs only occasionally may thus contribute to the preservation of the current system.

Functionally, the availability of alternative forms expands the range of stylistic possibilities.

While discourse factors are paramount in determining whether interaction occurs between the noun and the phrase that determines it, the tonal category may also play a role. All else being equal, some tones appear to resist interaction because the resulting tonal output would be non-trivial, requiring the speaker to recall a specific, morphologically conditioned tone combination rule. Other tones, by contrast, are more prone to interaction, as integrating the noun with the following determiner phrase is simply a matter of applying phonological rules.

For instance, when the noun's tone is L#, it feels easiest to integrate this noun with the following phrase, as the phonological consequence is a straightforward levelling down of the tones of the following syllables within the tone group: the noun's two syllables get M and L tone, respectively, and all syllables following this M.L sequence receive L tone by Rule 5. Conversely, division into two tone groups requires the additional effort of preserving the tones of the determiner phrase from lowering. While this is possible, it is less economical. As the path of least resistance leads to adopting the pattern in which the noun is integrated with its determiner, keeping it in a tone group of its own requires a stylistic motivation: an intention to emphasize the two constituents.

The implication for the analysis of the tone system is that some tones are more combination-prone than others, as the tone combination rules in which they are involved follow directly from phonological principles and thus tend to be exceptionless and straightforward to implement. The spreading of L# onto following syllables stands as the best example. Other tones, by contrast, are less combination-prone because the tone changes they undergo are more complex and allow several variants. The H$ tone category is a case in point.

Such asymmetries within the tone system, with some tones being more change-prone than others, would be reflected in the frequency of tonal interactions between a noun and a following determiner (N+DEM+CLF, N+NUM+CLF). Systematic verification of these hypotheses must be deferred to a later stage.

3.4 Concluding note

While the tone patterns of classifiers may appear staggeringly complex, phrases containing classifiers are frequent in discourse, a factor known to favour the preservation of irregular morphology (Wu et al. 2019, Round et al. 2024). As for the system's diachronic origin, some reflections are set out in §10.2.2.

If anyone has read this entire chapter with uninterrupted excitement and engagement, they are invited to imagine how much more elaborate the picture would have been had Yongning Na had the fortune to be used as a medium to express number-theoretic concepts such as ordinals and fractions. Tibetan, for instance, has ordinals as well as ways of expressing fractions and other mathematical concepts (Liu et al. 2010), which tend to diffuse into neighbouring languages under the cultural influence of Tibetan (e.g. Nar-Phu, spoken in the Manang district of Nepal: Noonan 2003: 342). Dzongkha, a Bodic language spoken in Bhutan, displays fascinating complexities such as the use of fractions within its counting system (Mazaudon 1985). But in Yongning Na, there are neither ordinal numbers nor fractions. Expressions that contain ordinal numbers in other languages (such as English and Chinese) correspond to turns of phrase with cardinal numbers in Yongning Na.

For instance, in (6), 'on the eighth day' is expressed as 'one day [after] seven days', i.e. 'the day after seven days had elapsed'.[11] In (7), the equivalent of 'the first person who will come' is 'whoever comes', understood by inference from context

[11]The context of example (6) is as follows. A young woman choked after swallowing boiled eggs too hastily; her in-laws, believing she had died, buried her. She regained consciousness when grave robbers set her upright to strip her of the valuable garments in which she had been interred. When she returned to her mother's home, the mother was terrified, fearing that her daughter had become a ghost. According to ghost lore, spirits are thought to return within a limited period of seven days, so the mother instructed her daughter to remain in hiding until the seventh day had passed, and only then return home.

as 'the *first* person who comes'.[12] Other examples are found in *Renaming.29, 38* {0004534#S29} and *Funeral.2* {0004571#S2}.

(6) ʂɯ˧-hɑ̃˧ gv̩˧ | ɖɯ˧-ɲi˥, | ə˧mi˧ lɑ˧ | mv̩˩˦ | tʰi˧-to˧~to˧, | ŋv̩˩ ɲi˥ tsɯ˩ | mv̩˩!
ʂɯ˧-hɑ̃˦ gv̩˧c ɖɯ˧-ɲi$˥ ə˧mi˧ lɑ˧ mv̩˩˦ tʰi˧-
seven-CLF.nights to_elapse one-CLF.days mother and daughter DUR
to˧~to˧ ŋv̩˩a -ɲi˩ tsɯ˦ mv̩˧
to_hug to_cry CERTITUDE REP AFFIRM
'On the eighth day (*literally*: the day after seven days had gone by), mother and daughter fell into each other's arms and wept!'
(BuriedAlive3.129) {0004538#S129}

(7) hĩ˧ | ɲi˩ le˧-tsʰɯ˩-ɻ̩˩-dʑo˩ ...
hĩ˥ ɲi˩ le˧- tsʰɯ˩a ɻ̩˩ -dʑo˥
person INTERROG.who ACCOMP to_come to_turn_towards TOP
'The first person who comes along... (*literally:* whoever comes along...)'
(Renaming.14) {0004534#S14}

Contact with Chinese, including training in mathematics at school, now exerts pressure towards the adoption of ordinals and fractions, but this has resulted is the wholesale borrowing of Chinese forms.

[12] This example requires some explanation of context. The parents of a sick infant are advised to build a bridge and to request the first person who crosses this bridge to bestow a new name upon their child. This act constitutes a symbolic adoption, intended to grant the child a fresh start in life. The stranger gives the infant a new name along with a small token, such as a button from their garment. Example (7) is part of the instructions given to the parents regarding what they should do for their child.

4 Compound nouns

Tonal processes within the noun phrase constitute a major part of the Yongning Na tone system. They also shed light on evolutionary processes. In Na, as in other Sino-Tibetan languages that have undergone considerable phonological erosion (such as Tujia 土家语, Bai 白语, Namuyi 纳木依语, and Shixing 史兴语), many roots that were once phonologically distinct have become homophonous. As a result, there is a strong tendency towards disyllabification. The study of synchronic tonal processes reveals which processes – such as compounding and affixation – feed into which tonal categories of disyllabic nouns. It also highlights, by contrast, those disyllabic nouns whose tones deviate from expected patterns based on currently productive rules. In turn, this draws attention to these outlier nouns, raising the question of how they acquired their tone – whether they date back to a time when different tonal rules applied, for instance.

Tonal phenomena within the noun phrase in Na will be presented by starting with compounds (this chapter), then examining combinations between nouns and grammatical morphemes (Chapter 5).

Compounding is a highly productive word formation process in Na. "Compounding is the prevalent morphological process" (Lidz 2010: 344). This holds true for many other languages of East and Southeast Asia as well (for Sinitic, see Arcodia 2012: passim). Determinative compounds (the *tatpuruṣa* compounds of Sanskrit grammar) are more common than coordinative compounds (Sanskrit *dvandva*). In Yongning Na, these two types follow distinct tonal rules. For instance, the determinative compound 'nanny goat's back' /**tsʰɯ˧mi˧-gv̩˧dv̩˥**/ carries H# tone (a final H tone), whereas the coordinative compound 'father and mother' /**ə˧dɑ˧-ə˧mi#˥**/ carries #H tone (a floating H tone), even though the input tones are the same: both 'nanny goat' and 'father' have H$ tone, and both 'mother' and 'back' have M tone. Determinative and coordinative compounds will therefore be examined separately.

4.1 Determinative compound nouns. Part I: The main facts

In Yongning Na, determinative compounds follow the order *determiner plus head*, as is generally the case in Sino-Tibetan (Michailovsky 2011).

In some tonal languages, possessive constructions (genitival syntagms) and compounds (complex lexemes) are distinguished by their tone patterns. In Kita Malinke (Mande branch of Niger-Congo), for instance, 'the meat of the cow' is /**mìsí sùbû**/, whereas 'beef, cow meat' is /**mìsì-súbú**/ (Creissels & Grégoire 1993). The latter is characterized by tonal compactness (*compacité tonale*): the tone pattern of the compound is determined by that of its first component, the determiner. Similarly, in Yongning Na, possessive constructions do not involve tonal changes, whereas compounds do – although the tone changes are more complex than in Malinke, as will be explained further below. The two constructions are conspicuously different: in possessive constructions, the possessive marker /=**bv̩˧**/ is inserted after the determiner, before the head noun (e.g. /**hwɤ˧li˦**/ 'cat', /**ɬv̩˦**/ 'brains', /**hwɤ˧li˧=bv̩˥ | ɬv̩˦**/ 'brains of the cat'). The determiner and the possessive particle form a single tone group, while the head noun retains its tone pattern as it appears in isolation. By contrast, in compounds, tones are not simply concatenated from their constituent elements. The present analysis proceeds in increasing order of abstraction, moving from the surface phonological patterns of compounds to their underlying tonal system.

Determinative compounds are sometimes classified into "free" and "fixed" combinations. The former consist of two nouns that are not habitually associated, such as /**gi˧nɑ˧mi˧-njɤ˥ɭɯ˩**/ 'bear's eye', where the two nouns are combined into a noun phrase only in the context of a specific utterance. The latter constitute lexicalized combinations, such as /**ʐ̩i˩hṽ̩#˥**/ 'body hair (of humans)', literally 'monkey's hair'. Cross-linguistically, compounds tend to stray away from regular morphophonological patterns as well as from the semantics that one would expect on the face of their constituting elements. However, the degree of divergence may vary: in some cases, the meaning is specialized while the phonological form remains indistinguishable from that of a newly coined compound; conversely, the phonological form may be irregular while the meaning remains fully transparent on a strictly synchronic basis.

> Thus in Zarma, *háw bîì* /ox/black/ is a syntagm with a perfectly regular form, which would be expected to mean 'black ox', but which refers to the buffalo – an animal that resembles the ox, and whose colour is black. In *cùrò bîì*, one easily recognizes *cúrò* 'bird' and *bîì* 'black', but the meaning is 'guinea fowl'; in this case, semantic specialization is accompanied by a tonal irregularity: a syntagm meaning 'black bird' would be expected to have the form *cúrò bîì*. (Creissels 1991: 121)
>
> *Original text:* Ainsi en zarma, *háw bîì* /bœuf/noir/ est un syntagme de formation parfaitement régulière dont on attendrait qu'il signifie « bœuf

noir », mais qui désigne le buffle (animal semblable au bœuf et de couleur noire). Dans *cùrò bíì*, nous reconnaissons facilement *cúrò* « oiseau » et *bíì* « noir », mais la signification est « pintade » ; dans ce cas, le figement sémantique s'accompagne d'une irrégularité tonale : le syntagme signifiant « oiseau noir » serait *cúrò bíì*.

The lack of a straightforward match between semantic regularity and morphophonological similarity must be taken into account when analyzing a tone system. Semantics cannot serve as the sole criterion for identifiying morphophonologically irregular compounds. The present chapter therefore does not distinguish between "fixed" and "free" combinations but rather between regular and irregular ones. From a morphotonological perspective, the relevant parameter is whether a compound's tone pattern follows productive rules.

4.1.1 The role of the number of syllables

Tonal changes in compounding are only observed when the second term – the head – has fewer than three syllables. That is, tone changes occur in compounds of the form σ+σ, σ+σσ, σσ+σ, σσ+σσ, σσσ+σ or σσσ+σσ. By contrast, no tone change takes place when the head has three or more syllables (e.g. in σ+σσσ, σσ+σσσ or σσσ+σσσ). What matters, then, is not the total number of syllables in the compound but the number of syllables in the head. This is illustrated is examples (1a–1c).

(1) a. lo˧ʂv̩˩ | -hi˩nɑ˧mi#˥
lo˧ʂv̩˩ hi˩nɑ˧mi#˥
Loshu (village name) lake
'Lugu Lake' (*literally:* 'the lake of Loshu')

b. ɬi˧di˩-di˩mi˩
ɬi˧di˩ di˧mi˧
Yongning (place name) large_plain
'Yongning plain'

c. gi˧nɑ˧mi˧-njɤ˥˩ɭɯ˩
gi˧nɑ˧mi#˥ njɤ˩ɭɯ˧
bear eye
'bear's eye'

The place names in (1a) and (1b) share the same syntactic structure: Loshu **lo˧ʂv̩˩** (Chinese: Luòshuǐ 落水) is the name of a village on the shore of Lugu

Lake, and **ɬi˧di˩** is the name of Yongning. In both cases, the relationship is one of determiner and head: 'the lake of **lo˧ʂv̩˩**' and 'the plain of **ɬi˧di˩**'. In (1a), both constituents retain their lexical tones: /**lo˧ʂv̩˩**/ 'Luoshui' and /**hi˩nɑ˧mi#˥**/ 'lake'. The compound 'Lugu Lake', /**lo˧ʂv̩˩ | -hi˩nɑ˧mi#˥**/, thus consists of two tone groups (this is indicated by the symbol '**|**', which stands for a tone group boundary). If it were a single tone group, its expected tone pattern would be † **lo˧ʂv̩˩-hi˩nɑ˩mi˩**, by application of Rule 5: "All syllables following an H.L or M.L sequence receive L tone" (for a list of the tone rules, see §7.1.1). In (1b), by contrast, the expected tone change does take place: the lexical tone of 'plain' is M (/**di˧mi˧**/), but within this compound, it is lowered to L by application of Rule 5. Example (1c), from /**gi˧nɑ˧mi#˥**/ 'bear' and /**njɤ˩ʈ͡ʂɯ˧**/ 'eye', shows that tonal change still occurs when the determiner has three syllables, provided that the head has no more than two.

It is clear from the data set out below (in Tables 4.2 to 4.6) that heads undergo more tonal changes than determiners in compounding. In particular, there are numerous cases where an H tone that is lexically associated with the determiner shifts to the final syllable of the head. When this occurs in a σσ+σ compound, the distance between the H tone's original position and its surface position in the compound is no greater than one syllable. In a σσ+σσ compound, this distance increases to two syllables. If the same process applied to a σσ+σσσ compound, the H tone would move three syllables away from its original lexical position, making the shift significantly more complex. This is by no means a cognitive impossibility: highly intricate tonal phenomena are firmly attested cross-linguistically, and by some estimates the Alawua dialect of Yongning Na qualifies as a complex system (see §12.2). Nevertheless, the observed pattern, whereby the compound is divided into two parts, appears to reflect a tendency for tonal computation to be circumvented, thereby placing a limit on generative complexity. The asymmetry, whereby σσσ+σσ compounds undergo tonal change while σσ+σσσ compounds do not, makes intuitive sense in light of the greater computational load required for the latter.

There is thus a preference (in this particular language and dialect) for tonal processes that do not involve tonal shifts of more than two syllables at a time. Long-distance tone movement is avoided. This observation will be taken up in §4.2.8.

4.1.2 How the tone patterns were collected

Some lexicalized compounds appear in narratives, such as /**ə˧mi˧-ʁæ˧ʈv̩˥**/ 'mother's neck' in *Tiger2.86* {0004545#S86}. Others were encountered during vocabulary elicitation sessions and are consigned in the dictionary of Yongning Na (Michaud et al. 2024). To obtain a comprehensive set of tonal combinations for determiners and heads, systematic elicitation was also conducted.

The main language consultant, Mrs. Latami Dashilamu (F4), was reluctant to accept semantically implausible combinations. Over time, she came to understand that the unusual pairings I proposed were intended to elicit specific tonal sequences, yet she remained firmly committed to common-sense usage. For instance, compounds such as 'flea's back' and 'flea's liver' were deemed acceptable –they did not stretch plausibility too far in the consultant's view –, and I gratefully recorded them. But she would not have accepted combinations as farfetched as 'chief's beetle's kidney basket' – an example from Hyman (2007a), later discussed by Evans (2010: 225).

Compounds that did not make sufficiently good sense in her view, such as 'woman's blood', were produced either as possessive constructions – an English equivalent would be 'blood of a/the woman', as opposed to the desired 'woman's blood' – or else as ungrammatical juxtapositions of citation forms.

Nevertheless, it was possible to elicit all tonal combinations in the end, by searching through the lexicon for the least implausible pairings and discussing possible contexts with the consultant. In the case of 'woman's blood', her argument was that there is nothing inherently distinct about the blood of women, as opposed to that of men: it is simply human blood. To work around this, I imagined a scenario in which a man-eating demon craves *woman's blood*.

Each combination was then checked multiple times by using different lexical items with the same tones and by eliciting tokens across elicitation sessions. This cautious approach made the elicitation process slower than it might have been with a consultant willing to generate arbitrary combinations among nouns. However, such conservative behaviour has its advantages: one might suspect that a consultant whose imagination roamed entirely free from the trammels of common sense could occasionally take similar liberties with the language's tonal system, too.

Transcribed recordings of over 1,500 compounds are available online from the Pangloss Collection (references: *DetermCompounds1* {0004450} through *DetermCompounds16* {0004491}). For each compound, these recordings provide the input tones and the lexical forms of both the determiner and the head, as illustrated in

Figure 4.1. In all cases where examples also appear in texts, the tone patterns obtained through elicitation are identical with those found in spontaneous speech. Table 4.1 provides an example noun for each tonal category used in the elicitation of body-part compounds.

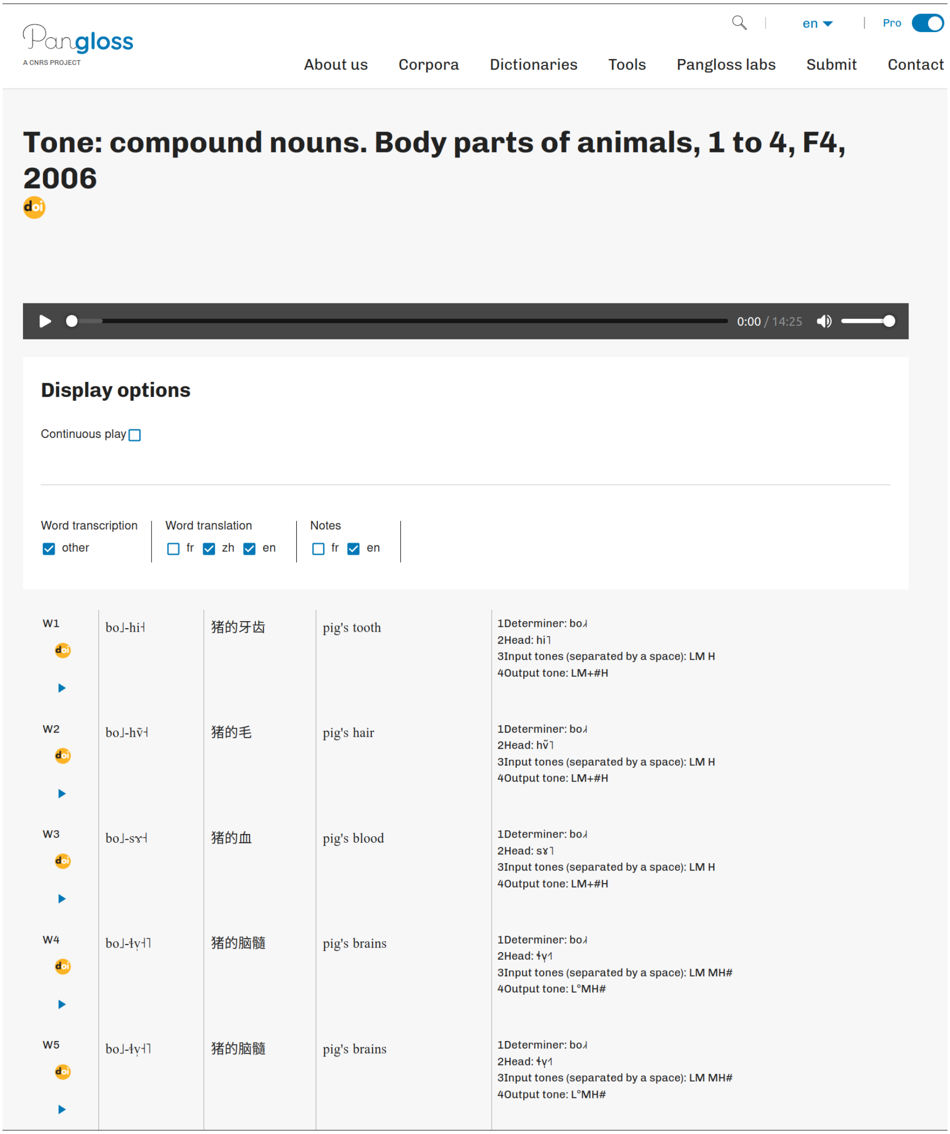


Figure 4.1: First lines of the document *DetermCompounds1* {0004450} as displayed in the online interface.

Table 4.1: Example words used to elicit body-part compound nouns.

tone	determiners	meaning	heads	meaning
LM	**bo˩˧**	pig	**ɣɯ˩˧**	skin
M	**lɑ˧**	tiger	**bv̩˧**	intestine
L	**jo˩**	sheep	**mɤ˩**	fat
#H	**ʐwæ˥**	horse	**sɤ˥**	blood
MH#	**ʈʂʰæ˧˥**	deer	**ɬv̩˧˥**	brains
M	**po˧lo˧**	ram	**gv̩˧dv̩˧**	back
#H	**ʐwæ˧zo#˥**	colt	**ɲi˧gɤ#˥**	nose, snout
MH#	**hwɤ˧li˧˥**	cat	**qv̩˧ʈʂæ˧˥**	throat
H$	**hwɤ˧mi˥$**	she-cat	**hu˧mi˥$**	stomach
L	**kʰv̩˩mi˩**	dog	**nv̩˩mi˩**	heart
L#	**dɑ˧ʝi˩**	mule	**ɬi˧pi˩**	ear
LM+MH#	**õ˩dv̩˧˥**	wolf	**ʝi˩ʈʂæ˧˥**	waist
LM+#H	**nɑ˩hĩ#˥**	Naxi person	**njæ˩qʰæ#˥**	eye sand, rheum
LM	**æ˩mi˧**	hen	**njɤ˩ɭɯ˧**	eye
LH	**bo˩ɬɑ˥**	boar	**hi˩ʐæ˥**	uvula
H#	**hwæ˧tsɯ˥**	rat	**ʁæ˧ʈv̩˥**	neck

4.1.3 The facts: Surface phonological tone patterns

The tone patterns of compound nouns in Alawua are set out in Tables 4.2 to 4.6 as a function of the tones of their constituent elements. The leftmost column indicates the tone of the determiner, while the top row indicates the tone of the head. For example, 'tiger' /**lɑ˧**/ has lexical M, and 'skin' /**ɣɯ˩˧**/ has lexical LM. The tone of the compound 'tiger's skin' can be found at the intersection of row M and column LM in Table 4.2. The cell at this intersection contains 'M.L', indicating that the surface phonological tone pattern of the compound is M.L: /**lɑ˧-ɣɯ˩**/ 'tiger's skin'. Tables 4.2 and 4.3 present data for compounds with monosyllabic heads, and Tables 4.4 and 4.5 cover disyllabic heads.

As with simple nouns, compounds must be elicited in at least two contexts to determine their lexical tone categories, since oppositions such as //M// vs. //#H// are neutralized in isolation. Compound nouns were therefore elicited both (i) in isolation and (ii) in a carrier frame: (2).

Table 4.2: Surface phonological representation of the tones of compound nouns. Monosyllabic determiner and monosyllabic head. Leftmost column: tone of determiner; top row: tone of head.

tone	LH; LM	M	L	H	MH
LM	L.M			L.M, L.M.H	L.MH
LH	L.H	L.L	L.H		
M	M.L	M.M, M.M.H	M.L	M.M, M.M.H	M.MH
L	L.LH				
H	M.H	M.M, M.M.H			M.L
MH	M.H			M.H, M.M.H	

Table 4.3: Surface phonological representation of the tones of compound nouns. Disyllabic determiner and monosyllabic head. Leftmost column: tone of determiner; top row: tone of head.

tone	LH; LM	M	L	H	MH
M	M.M.L	M.M.M, M.M.M.H	M.M.L	M.M.M, M.M.M.H	M.M.L
#H	M.M.H	M.M.M, M.M.M.H			M.M.L
MH#	M.M.H	M.M.MH			M.M.H
H$	M.M.H	M.M.M, M.M.M.H	M.M.H, M.M.M.H	M.M.M, M.M.M.H	M.H.L
L	L.L.H	L.L.LH			L.L.H
L#	M.L.L				
LM+MH#	L.M.H	L.M.MH	L.M.H, L.M.M.H		
LM+#H	L.M.H	L.M.M, L.M.M.H	L.M.H	L.M.M, L.M.M.H	L.M.H
LM	L.M.L	L.M.M	L.M.L	L.M.M, L.M.M.H	L.M.MH
LH	L.H.L				
H#	M.H.L				

This elicitation procedure is the same as that applied earlier to simple nouns (see §2.4.2). Accordingly, frame (2) is reproduced from example (6) of Chapter 2. The lexical tones of the demonstrative and the copula are indicated in the interlinear glosses, following the findings of Chapter 2. The tone-group boundary separating the demonstrative from what follows is also indicated, by means of the vertical bar '|'. No tone is given for the copula in the surface phonological representation (first line) because its surface tone varies according to the tone category of the target noun.

(2) ʈʂʰɯ˧ | ______ ɲi.
ʈʂʰɯ˥ ______ ɲi˩
DEM.PROX *target item* COP
'This is (a/the) ______.'

In cases where the copula surfaces with its lexical L tone, a single tone pattern is recorded in the corresponding table cell. For instance, 'tiger's skin' /**lɑ˧-ɣɯ˩**/ (tone: /M.L/) yields /**lɑ˧-ɣɯ˩ ɲi˩**/ 'is tiger's skin', with L tone on the copula. The table thus simply indicates 'M.L', with no additional notation, meaning that the pattern remains unchanged upon addition of the copula.

Conversely, in cases where the copula bears a tone other than its lexical L, the altered tonal string is noted after a comma. For instance, the information 'M.M, M.M.H' provided at the intersection of row M and column M in Table 4.2 indicates that the tonal string of the compound is M.M when said in isolation, e.g. /**lɑ˧-bv̩˧**/ 'tiger's intestine', but that adding the copula yields M.M.H: /**lɑ˧-bv̩˧ ɲi˥**/ 'is tiger's intestine'.

In Table 4.2, there are no examples of simple M.M: all disyllabic M.M compounds plus copula yield M.M.H, so that the 'M.M.H' part in 'M.M, M.M.H' may seem redundant. However, this is not superfluous, since in Yongning Na, a surface M.M sequence always requires disambiguation: /M.M/ constitutes the neutralization of underlying //M// and //#H//, exemplified in Chapter 2 by //**po˧lo˧**// 'ram' (M) and //**ʐwæ˧zo#˥**// 'colt' (#H).

When tonal variants exist, alternatives are separated by slashes. For example, in row #H, column H$ of Table 4.4, the entry 'M.L.L / M.M.H' indicates that compounds in this category may have either of these two patterns, e.g. /**ʐwæ˧-hu˩mi˩**/ or /**ʐwæ˧-hu˧mi˥**/ for 'horse's stomach'.

To enhance readability, sequences of four M tones (M.M.M.M) are abbreviated as 'M…M'. Additionally, adjacent cells with identical contents are grouped using dashed-line boxes. For instance, since all disyllabic compounds with an L-tone determiner follow the pattern /L.LH/, the entire 'L' row in Table 4.2 is enclosed in

Table 4.4: Surface phonological representation of the tones of compound nouns. Monosyllabic determiner and disyllabic head. Leftmost column: tone of determiner; top row: tone of head.

tone	M	#H	MH#	H$	L	L#	LM+MH#; LM+#H; LM; LH	H#
LM; LH	L.M.M	L.M.M, L.M.M.H	L.M.MH / L.H.L	L.M.H, L.M.M.H	L.M.L	L.M.L / L.L.H	L.H.L (=L.M.L)	L.M.H / L.L.H
M	M.M.M	M.M.M, M.M.M.H	M.M.MH	M.M.H, M.M.M.H	M.L.L	M.M.L	M.L.L	M.M.H
L	L.L.LH			L.L.H	L.L.LH	L.L.H	L.H.L	L.L.H
#H	M.M.H	M.M.M, M.M.M.H	M.H.L	M.L.L / M.M.H	M.H.L	M.M.H	M.H.L	M.M.H
MH#	M.M.H	M.M.M, M.M.M.H	M.M.MH	M.H.L	M.H.L	M.M.H	M.H.L	M.M.H

Table 4.5: Surface phonological tones of compound nouns. Disyllabic determiner and disyllabic head. Leftmost column: tone of determiner; top row: tone of head.

tone	M	#H	MH#	H$	L	L#	LM+MH#; LM+#H; LM; LH	H#
M	M…M	M…M, M…M.H	M…MH	M.M.M.H, M…M.H / M.M.H.L	M.M.L.L	M.M.M.L	M.M.L.L	M.M.M.H
#H	M.M.M.H	M…M, M…M.H	M.M.H.L	M.M.M.H, M…M.H / M.M.H.L / M.M.M.H	M.M.H.L	M.M.M.H	M.M.H.L	M.M.M.H
MH#		M…MH	M.M.H.L	M.M.H.L / M.M.M.H	M.M.H.L	M.M.M.H	M.M.H.L	M.M.M.H
H$		M…M, M…M.H	M.M.H.L / M.H.L.L		M.M.H.L	M.M.M.H	M.M.H.L	M.M.M.H
L	L.L.L.H	L.L.L.LH	L.L.H.L	L.L.L.H	L.L.H.L	L.L.L.H	L.L.H.L	L.L.L.H
L#	M.L.L.L	M.L.L.L						
L+MH#	L.M.M.H	L.M.M.M, L.M.M.M.H	L.M.H.L			L.M.M.H	L.M.H.L	L.M.M.H
LM+#H			L.M.H.L	L.M.M.H,	L.M.H.L			
LM	L.M.M.M		L.M.M.MH	L.M.M.M.H	L.M.L.L	L.M.M.L	L.M.L.L	
LH	L.H.L.L							
H#	M.H.L.L							

Table 4.6: Surface phonological representation of the tones of compound nouns. Compounds with a trisyllabic #H-tone determiner: 'bear'+body part.

head		compound	
form	tone	surface form	surface tone
ɣɯ˩˧ 'skin'	LM	**gi˧nɑ˧mi˧-ɣɯ˥**	M.M.M.H, M.M.M.H.L
bv̩˧ 'intestine'	M	**gi˧nɑ˧mi˧-bv̩˧**	M.M.M.M, M…M.H
mɤ˩ 'grease'	L	**gi˧nɑ˧mi˧-mɤ˧˥**	M.M.M.MH
sɤ˥ 'blood'	H	**gi˧nɑ˧mi˧-sɤ˧**	M.M.M.M, M…M.H
ɬv̩˧˥ 'brains'	MH	**gi˧nɑ˧mi˧-ɬv̩˩**	M.M.M.L
gv̩˧dv̩˧ 'back'	M	**gi˧nɑ˧mi˧-gv̩˧dv̩˧**	M.M.M.M.M, M…M.H
ɲi˧gɤ˧ 'nose'	#H	**gi˧nɑ˧mi˧-ɲi˧gɤ˧**	M.M.M.M.M, M…M.H
qv̩˧ʈʂæ˧˥ 'throat'	MH#	**gi˧nɑ˧mi˧-qv̩˥ʈʂæ˩**	M.M.M.H.L
hu˧mi˥$ 'stomach'	H$	**gi˧nɑ˧mi˧-hu˧mi˥**	M.M.M.M.H, M…M.H
nv̩˩mi˩ 'heart'	L	**gi˧nɑ˧mi˧-nv̩˥mi˩**	M.M.M.H.L
ɬi˧pi˩ 'ear'	L#	**gi˧nɑ˧mi˧-ɬi˧pi˥**	M.M.M.M.H
ʝi˩ʈʂæ˧˥ 'waist'	LM+MH#	**gi˧nɑ˧mi˧-ʝi˥ʈʂæ˩**	M.M.M.H.L
njɤ˩ɭɯ˧ 'eye'	LM	**gi˧nɑ˧mi˧-njɤ˥ɭɯ˩**	M.M.M.H.L
ʁæ˧ʈv̩˥ 'neck'	H#	**gi˧nɑ˧mi˧-ʁæ˧ʈv̩˥**	M.M.M.M.H

a single box labelled 'L.LH'. However, the process of grouping cells has not been extended indiscriminately. For example, in the bottom left corner of Table 4.2, the 'H' and 'MH' rows were not merged, as the tonal pattern /M.H/ likely arises for different reasons in these two cases (§4.2.4 and §4.2.5).

The purpose of Tables 4.2 to 4.6 is to set out the facts in a legible and unambiguous manner. This is a step towards the long-term goal of developing a full-fledged linguistic model, incorporating novel approaches to capturing both regularities and irregularities in paradigms (Sagot & Walther 2013, Bonami & Sagot 2017).

In view of the rarity of trisyllabic nouns, only one trisyllabic determiner was included in the data: /**gi˧nɑ˧mi#˥**/ 'bear' (tone: #H). The data is set out in Table 4.6.

4.1.4 Analysis of underlying tone patterns

The surface tonal sequences observed in compounds, as reported in Tables 4.2 to 4.6, can in most cases be reduced to simple tone categories, such as //L//, //LM//

or //LH//, by analyzing them in light of the phonological tone rules set out in §7.1.1. The following two examples illustrate this process.

- The sequence /L.LH/ can be interpreted as the realization of a simple //L// tone: it spreads over the two syllables of the compound, yielding //L.L//, and this sequence is further supplemented by a postlexical H tone due to Rule 7 ("If a tone group only contains L tones, a postlexical H tone is added to its last syllable").
- The sequence /M.M/ in /**lɑ˧-bv̩˧**/ 'tiger's intestine' could reflect an underlying //M// or //#H// tone pattern (recall that //#H// is a floating H tone, which is only realized after the lexical item at issue). The fact that the copula receives an H tone when following this compound (/**lɑ˧-bv̩˧ ɲi˥**/ 'is tiger's intestine') reveals that the underlying tone pattern is #H.

The results of this analysis are presented in Tables 4.7 to 4.11, which provide all the information needed to generate the surface phonological patterns of compounds, following the standard tone-to-syllable association rules.

In describing these patterns, reference must be made to a juncture internal to the tone group, which separates the determiner from the head. This juncture is indicated by the symbol '–', which was already used in the description of numeral-plus-classifier phrases in Chapter 3. Thus, –L refers to an L tone attaching to the second part of an expression.

For example, the output –L indicated in Table 4.10 for a disyllabic determiner carrying M tone (first row) and a disyllabic head carrying L tone (sixth column) means that the determinative compound carries L tone on its second part, i.e. after the juncture between the two nouns. Thus, the combination of /**po˧lo˧**/ 'ram' and /**nv̩˩mi˩**/ 'heart' yields a compound with L tone on its second part, hence /**po.lo.nv̩˩mi**/ (L tone associates to the first syllable after the juncture between the two nouns).

The association of tones to the other three syllables then follows the regular phonological tone rules of Yongning Na. First, the L tone spreads onto the following syllable (by Rule 1), yielding /**po.lo.nv̩˩mi˩**/. Then the first two syllables receive M tone (by Rule 2), producing /**po˧lo˧-nv̩˩mi˩**/.

A step-by-step representation of this process is provided in Figure 4.2. A similar representation is shown in Figure 4.3 for #H– (a floating H tone attaching to the first part of the expression). In this case, the #H tone is inserted before the juncture between the two nouns, i.e. it associates to the second syllable of the compound. From there:

- The H tone attaches to the following syllable (this is the defining property of the #H tone category, which associates *after* the word to which it is lexically attached), i.e. on the third syllable.
- The first and second syllables receive M tone by Rule 2.
- The fourth syllable receives L tone by Rule 4.

Four tonal categories of head nouns always behave the same way in compounds: the opposition among LM, LH, LM+#H, and LM+MH# is neutralized. These tone categories for heads are therefore pooled together in Tables 4.7 to 4.11. Among determiners, the opposition between LH and LM on monosyllables is also neutralized; accordingly, these two tones are pooled together in the tables.

Table 4.7: The underlying tonal categories of compound nouns. Monosyllabic determiner and monosyllabic head. The tone of the determiner is indicated in the leftmost column, and the tone of the head in the top row.

tone	LM; LH	M	L	H	MH
LM; LH	LH	LM	LH	LM+#H	LM+MH#
M	–L	#H	–L	#H	MH#
L	L				
H	#H–	#H			–L
MH	H#			H$	

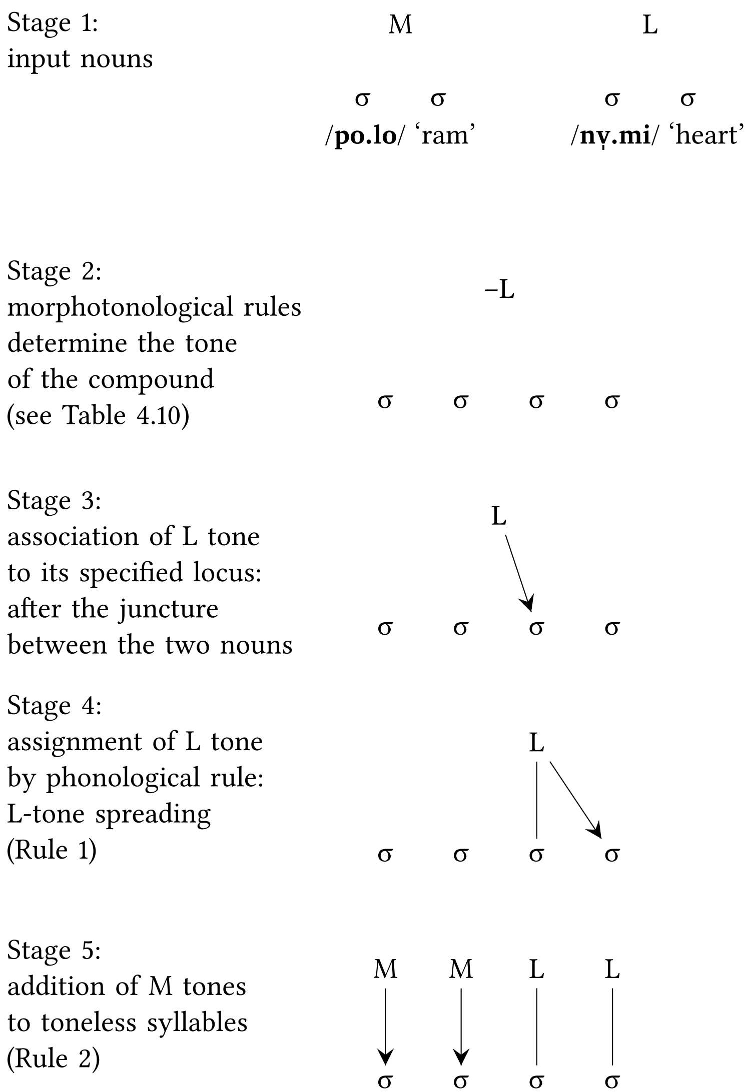


Figure 4.2: First illustration of the anchoring of tones relative to a morpheme break inside a complex expression (notation: '–'): step-by-step representation of tonal association of the –L tone of the compound /**po˧lo˧-nṿ˩mi˩**/ 'rat's stomach'.

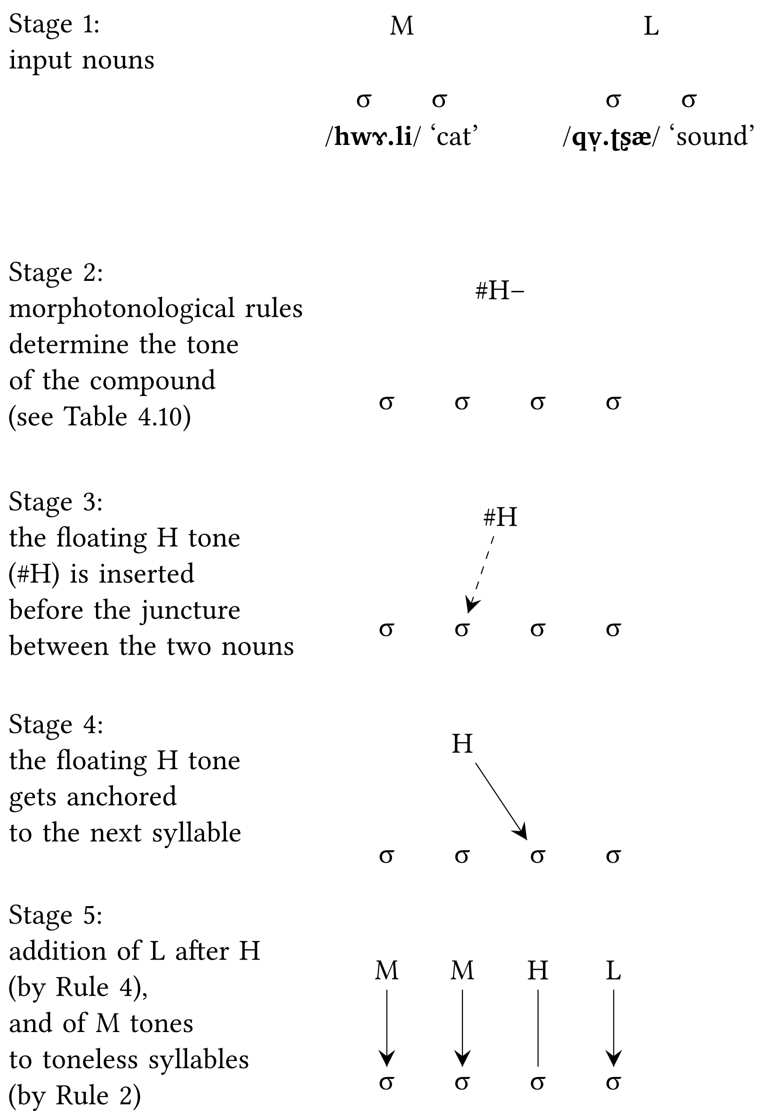


Figure 4.3: Second illustration of the anchoring of tones relative to a morpheme break inside a complex expression (notation: ‘–’): step-by-step representation of tonal association of the #H– tone of the compound /**hwɤ˧li˧-qv̩˥ʈʂæ˩**/ ‘cat’s sound / cat sounds’.

Table 4.8: The underlying tonal categories of compound nouns. Disyllabic determiner and monosyllabic head. The tone of the determiner is indicated in the leftmost column, and the tone of the head in the top row.

tone	LH; LM	M	L	H	MH
M	–L	#H	–L	#H	–L
#H	H#	#H			
MH#		MH#			H#
H$	#H–	#H	H$	#H	H#–
L	L+H#	L			L+H#
L#	L#–				
LM+MH#	LM+MH#–	LM+MH#	LM+H$		
LM+#H		LM+#H	LM+H#	LM+#H	LM+H#
LM	LM–L	LM	LM–L		LM+MH#
LH	LH				
H#	H#–				

4.2 Determinative compound nouns. Part II: Discussion

Let us now proceed to an analysis of the patterns presented above. At this stage, it may be useful to consider possible ways in which the tones of the determiner and head could combine. Some languages, such as Mandarin, have no tone change at all in compounds, demonstrating that tone change is not necessary to compounding. A first theoretical possibility would thus be the simple concatenation of the input tones. Under such a configuration, given the *determiner-head* order of Yongning Na compounds, the tone of the determiner would be expressed first. Given the constraints on tone sequences within a tone group (summarized as Rules 1–7), this would leave little room for the tone of the head to be expressed. Since an H tone precludes the realization of any other tone on subsequent syllables (by phonological rules 4 and 5), determiners carrying a lexical tone that includes an H level (i.e. one of #H, H$, MH#, LM+MH#, LM+#H, and H#) would neutralize all tonal oppositions on head nouns. Likewise, the L level in the lexical categories L and L# would spread rightward throughout the compound. The tone

Table 4.9: The underlying tonal categories of compound nouns. Monosyllabic determiner and disyllabic head. The tone of the determiner is indicated in the leftmost column, and the tone of the head in the top row.

tone	M	#H	MH#	H$	L	L#	LM+MH#; LM+#H; LM; LH	H#
LM; LH	LM	LM+#H	LM+MH# / L+#H–	LM+H$	L+#H–	L+#H– / L+H#	L+#H–	LM+H# / L+H#
M	M	#H	MH#	H$	–L	–L#	–L	H#
L	L			L+H#	L	L+H#	L+#H–	L+H#
#H	H#	#H	#H–	–L / H#	#H–	H#	#H–	H#
MH			MH#	#H–				#H

Table 4.10: The underlying tonal categories of compound nouns. Disyllabic determiner and disyllabic head. The tone of the determiner is indicated in the leftmost column, and the tone of the head in the top row.

tone	M	#H	MH#	H$	L	L#	LM+MH#; LM+#H; LM; LH	H#
M	M	#H	MH#	H$ / #H–	–L	–L#	–L	H#
#H	H#		#H–	H$ / #H– / H#	#H–	H#	#H–	
MH#		MH#		#H– / H#			MH#–	
H$		#H	#H– / H#–				#H–	
L	L+H#	L	L+H#	L+H#	L+#H–	L+H#	L+#H–	L+H#
L#	L#–							
LM+MH#	LM+H#	LM+#H	LM+MH#–	LM+MH#–/ H#	LM+MH#–	LM+H#	LM+MH#–	LM+H#
LM+#H				LM–H$	LM+#H–			
LM	LM–		LM+MH#		LM–L	LM–L#	LM–L	
LH	LH							
H#	H#–							

Table 4.11: Examples and underlying tonal categories of compound nouns with a trisyllabic, #H-tone determiner: 'bear'+body part.

head		compound	
form	tone	underlying form	underlying tone
ɣɯ˩˧ 'skin'	LM	**gi˧nɑ˧mi˧-ɣɯ˥**	H#
bv̩˧ 'intestine'	M	**gi˧nɑ˧mi˧-bv̩#˥**	#H
mɤ˩ 'grease'	L	**gi˧nɑ˧mi˧-mɤ˧˥**	MH#
sɤ˥ 'blood'	H	**gi˧nɑ˧mi˧-sɤ#˥**	#H
ɬv̩˧˥ 'brains'	MH	**gi˧nɑ˧mi˧-ɬv̩˩**	L#
gv̩˧dv̩˧ 'back'	M	**gi˧nɑ˧mi˧-gv̩˧dv̩#˥**	#H
ɲi˧gɤ˧ 'nose'	#H	**gi˧nɑ˧mi˧-ɲi˧gɤ#˥**	#H
qv̩˧ʈʂæ˧˥ 'throat'	MH#	**gi˧nɑ˧mi˧-qv̩˥ʈʂæ˩**	#H–
hu˧mi˥$ 'stomach'	H$	**gi˧nɑ˧mi˧-hu˧mi˥$**	H$
nv̩˩mi˩ 'heart'	L	**gi˧nɑ˧mi˧-nv̩˥mi˩**	#H–
ɬi˧pi˩ 'ear'	L#	**gi˧nɑ˧mi˧-ɬi˧pi˥**	H#
ʝi˩ʈʂæ˧˥ 'waist'	LM+MH#	**gi˧nɑ˧mi˧-ʝi˥ʈʂæ˩**	#H–
njɤ˩ɭɯ˧ 'eye'	LM	**gi˧nɑ˧mi˧-njɤ˥ɭɯ˩**	#H–
ʁæ˧ʈv̩˥ 'neck'	H#	**gi˧nɑ˧mi˧-ʁæ˧ʈv̩˥**	H#

of the head could be expressed only when the determiner has M tone: M plus L would yield –L, M plus #H would yield #H, and so forth.

A second possibility would be the complete neutralization of tonal oppositions on either the determiner or the head. Neutralization of oppositions on the determiner would be unexpected, given the determiner-first order of compounds and the predominantly perseverative directionality of tone spreading and tone reassociation in Yongning Na. The loss of tonal oppositions among determiners would drastically reduce the number of tonal patterns at the surface phonological level. Such neutralization of tonal oppositions on the head is attested in the neighbouring language Shixing (also known as Xumi), where only the tone of the determiner is expressed: all tonal oppositions on the head are neutralized, such that the 3×3 tonal combinations among nouns reduce to just three tone patterns on compounds (Chirkova & Michaud 2009).

A third possibility would be the assignment of a replacive tone in compounds. This is reported to be widespread in the Mande subgroup of the Niger-Congo family, e.g. in Dan-Gwɛɛtaa (Vydrin 2016) and Kpelle (Welmers 1969, 1973: 132, Konoshenko 2014b: 239). In Naish, the lower-level subgroup within Sino-Tibetan to which Yongning Na belongs, no clear instance of replacive tone has been observed to date. The closest parallel in Naish is found in the Laze language, where compounds with input tones M and H, M and L, and M and MH tend to adopt an H.H tone pattern as they lexicalize (Michaud 2008a). However, the H.H pattern is not the only one carried by compounds: if the tone of the head is L, the compound receives a different tone pattern.

The state of affairs found in Yongning Na bears some similarities to both the first and second theoretical possibilities: there is a tendency for output tone patterns to consist of the concatenation of the input tones, alongside a degree of neutralization of tonal oppositions (such that the set of output tone patterns is smaller than the set of input combinations). On the one hand, approximately half of the patterns can be straightforwardly explained in terms of successive association of the two input tones, followed by application of the general rules governing tonal adjustments within a tone group. On the other hand, the remaining patterns could not be fully accounted for using a set of rules.

The tone patterns of determinative compounds thus present a composite picture (play on words intended). Since not all patterns align neatly with any of the three theoretical possibilities listed above, it must be acknowledged that compounds have tone rules of their own, which differ subtly from the straightforward concatenation of input tones.

This conclusion may come as a slight disappointment to the linguist, whose job is to account for all observations through a model that is as simple and elegant as possible. As in many other domains of the Yongning Na tone system, the relationship between the lexical tones of words and the tones that surface in context cannot be reduced to a set of sandhi rules operating at a purely phonological level. (Clarifications concerning the terms ‘tone sandhi’, ‘morphotonology’, and ‘tonal morphology’ are provided in §8.4.1.) The behaviour of the three-syllable noun ‘bear’, /**gi˧nɑ˧mi#˥**/, when used as a determiner, illustrates this point: it patterns almost like disyllabic #H-tone nouns, such as ‘colt’, /**ʐwæ˧zo#˥**/, but not quite. When the head is an L-tone monosyllable, the output tone is MH#, rather than the expected #H.

In view of these observations, the present discussion of the combinations in Tables 4.7 to 4.11 is arranged in order of increasing complexity.

The simpler cases can be described as follows: the tone of the determiner expresses itself first, and subsequently, the tone of the head surfaces to the extent permitted by the tones already assigned. The tone patterns of compounds in which the determiner has H#, LH or L# tone are so straightforward as to appear trivial: in all three cases, only the tone of the determiner is expressed. When the determiner carries H# tone (i.e. an H tone associated with its last syllable), tonal oppositions on the head are neutralized, regardless of the number of syllables: see the final rows of Tables 4.7 to 4.11. The resulting compound carries H#–, meaning that the determiner's final syllable bears an H tone. This can be analyzed as the direct association of the H# tone with the determiner: H on its final syllable and, by default, M on its first syllable. The lowering of all following tones to L follows from Rules 4 and 5: "A syllable following an H-tone syllable receives L tone" and "All syllables following an H.L or M.L sequence receive L tone". This is represented in Figure 4.4, using the compound 'rat's stomach' (/**hwæ˧tsɯ˥-hu˩mi˩**/) as an example. In this compound, no trace remains of the H$ tone (the 'flea' tone) carried by the head.

This same analysis extends to the two other tone categories of disyllabic determiners after which all tonal oppositions are neutralized: LH and L#. The realization of either of these tone patterns on the first part of the compound (the determiner) precludes the occurrence of any tone other than L on the subsequent syllables, by Rules 4 and 5:

- LH tone, unfolding as L.H over the two syllables of the determiner, triggers the assignment of L tone to the following syllable by Rule 4 ("The syllable following a H-tone syllable receives L tone"), and further, the assignment of L tone to any subsequent syllables by Rule 5 ("The syllables following an H.L or M.L sequence receive L tone").

- L# tone, unfolding as M.L over the two syllables of the determiner (through association of an L tone to the second syllable of the determiner, and of a default M tone to its first syllable), likewise depresses the tones of following syllables to L by Rule 5 ("The syllables following a H.L or M.L sequence receive L tone").

In the remaining cases, which constitute a majority, tonal oppositions on the head are not entirely neutralized: establishing the tone of the compound requires knowledge of the head noun's tone. This observation raises doubts about the adequacy of the representation proposed in Figure 4.4, which assumes a determiner-driven tonal association process. If the process were a straightforward step-by-step association – first applying the determiner's tone, then the head's –, one would expect the complete neutralization of all tonal oppositions on the head when the determiner carries MH# tone. On the analogy of Figure 4.4, one would expect the tone pattern for 'cat's ear' to be † **hwɤ˧li˧-ɬi˥pi˩**. But the observed tone is H#: /**hwɤ˧li˧-ɬi˧pi˥**/.

The double daggers ‡ added to the labels 'Stage 2', 'Stage 3', and 'Stage 4' in Figure 4.5 highlight the hypothetical nature of that representation: it is proposed

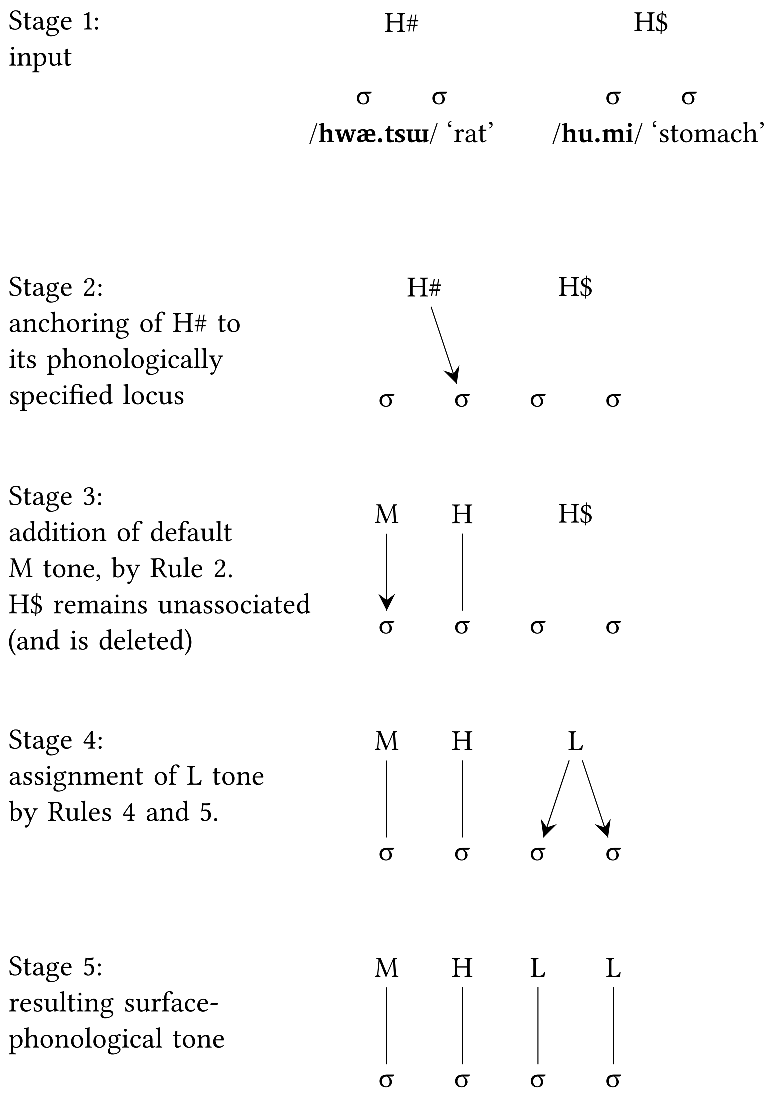


Figure 4.4: A hypothesis about how H# tone on the determiner associates to the entire compound. Example: /**hwæ˧tsɯ˥-hu˩mi˩**/ ‘rat’s stomach’.

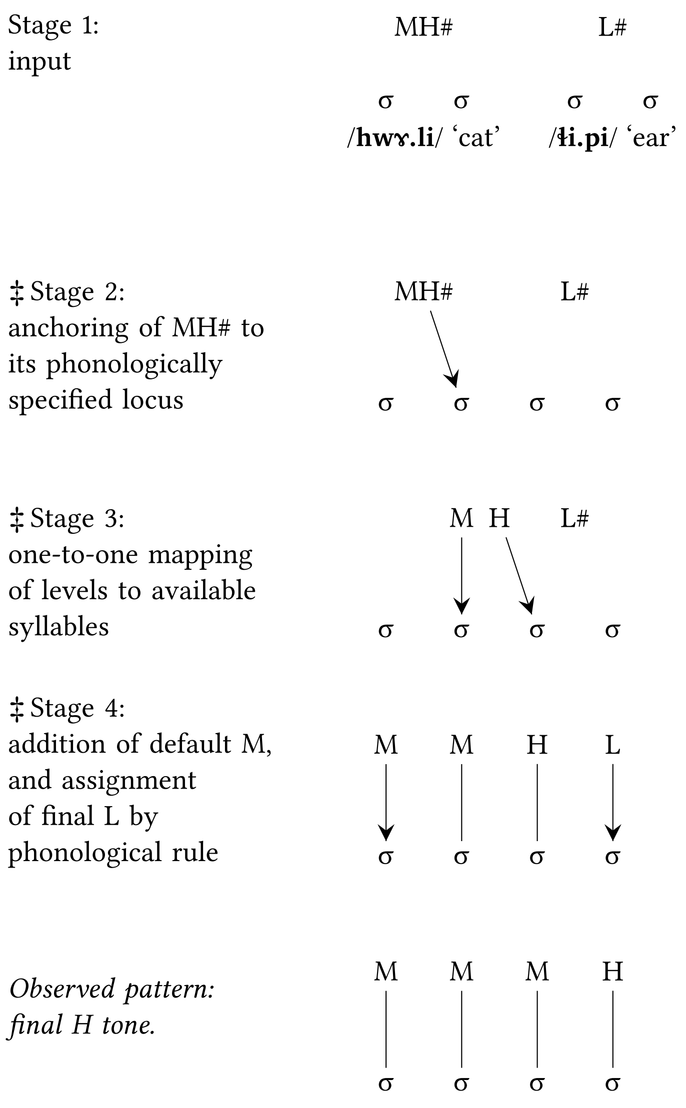


Figure 4.5: Tone-to-syllable association expected for ‘cat’s ear’ under the mistaken hypothesis of determiner-driven tone association: ‡ **hwɤ˧li˧-ɬi˥pi˩**, as contrasted with the observed pattern: /**hwɤ˧li˧-ɬi˧pi˥**/.

solely to illustrate the *ad hoc* character of the analysis in Figure 4.4. Tone association in compounds is not consistently driven by the determiner. Nonetheless, the representation in Figure 4.4 does correspond to a genuine pattern, albeit one specific to H# tone. This tone is anchored to a word's final syllable; in compounds, when the first element of the compound (the determiner) carries H# tone, this tone remains moored to the determiner's final syllable.

A few combinations in Tables 4.7 to 4.11 appear counter-intuitive in view of the input tones. For instance, when an M-tone determiner is combined with an M-tone monosyllabic head, the resulting compound carries a floating H tone, #H. This outcome is all the more unexpected as M is generally expected to be inactive (see §2.4.3). By contrast, an M-tone determiner and an M-tone *disyllabic* head yield a compound with a simple M tone. Crucially, the unexpected #H output does not result from a general phonological rule of Na that turns any sequence of two M tones into a floating H. Rather, it results from the *morpho*phonological rules that apply specifically within this syntactic construction. Since these rules cannot be straightforwardly reduced to a single generalization, they are best presented in table form, as in Tables 4.7 to 4.11.

The discussion of the more complex tone patterns that follows is arranged by the tone category of the determiner.

4.2.1 LM-tone determiners

An LM tone on the determiner invariably results in the assignment of L on the first syllable of the compound and M on the second syllable. A relatively high number of tone sequences are allowed after /L.M/, allowing some tone categories of the head to be expressed in full. Over three syllables, three patterns are observed: /L.M.L/, /L.M.M/, and /L.M.H/. Over four syllables, ‡ L.M.L.M and ‡ L.M.L.H are ruled out, as the sequence /M.L/ can only be followed by /L/, by virtue of Rule 5 ("All syllables following an H.L or M.L sequence receive L tone"). The tones that are compatible with an initial LM pattern are observed to surface as expected: LM plus #H, LM plus MH#, LM plus H$, and LM plus H# are realized straightforwardly as concatenations of the two input tones. As anticipated, an M-tone head has no effect on the final tone pattern, which is simply LM. Likewise, the combinations LM plus #L and LM plus H# in quadrisyllabic compounds surface as such.

Some of the surface patterns are analytically indeterminate. For instance, the combination of a monosyllabic LM determiner and a monosyllabic MH# head yields L.MH (e.g. /**bo˩-ɬv̩˧˥**/ 'pig's brains'). This could be analyzed in two ways: either as L–MH# (L on the first part, and MH# on the second) or as LM+MH#

(where LM contributes L to the first syllable and M to the second, and MH# assigns an MH contour to the last syllable, which in this case is also the second syllable). Since both analyses generate the same output, the simpler approach is to describe the pattern as the concatenation of the two input tones, i.e. //LM+MH#//. Under this analysis, the same applies to trisyllabic (σσ+σ) and quadrisyllabic (σσ+σσ) compounds: the surface patterns (L.M.MH for σσ+σ and L.M.M.H for σσ+σσ) follow straightforwardly from the general rules of tone-to-syllable mapping summarized in §7.1.

In a disyllabic compound with an LM-tone determiner, an L or LM tone on the head cannot express itself, as the determiner's LM contour has already projected its endpoint (M) on the second syllable. While the M tone functions as a default tone in some contexts (see §2.4.3), in this case it is specified as the endpoint of an LM contour and thus behaves as a fully specified tone. Such cases are not typologically unusual, a point to which the discussion in Chapter 12 will return. In derivational terms, this suggests that at the point when the head's tone could come into play, both syllables of the compound are already specified for tone, leading to the neutralization of LM, M, and L as the second components of σ+σ compounds with an LM determiner.

By contrast, in three- and four-syllable compounds, there is still room for the expression of an L level originating in an L or LM tone on the head. These combinations result in L.M.L in trisyllabic compounds and L.M.L.L in quadrisyllabic ones, in accordance with Rule 5 ("All syllables following an H.L or M.L sequence receive L tone").

To sum up, the tones of all compounds with an LM determiner obtain through the concatenation of the tones of their two components, subject to the phonological rules that apply at the level of tone groups (as recapitulated in Chapter 7).

4.2.2 M-tone determiners

After M-tone determiners, L and LM tones on heads are neutralized due to Rule 5 ("All syllables following an H.L or M.L sequence receive L tone"). Beyond these two categories, one would expect the tone of the second component of the compound to express itself fully, given that M behaves in some respects as a default tone (see §2.4.3) and is generally phonologically neutral (i.e. inert). This prediction is not entirely borne out, however.

As noted at the outset of this chapter, the tone pattern #H (a floating H tone) that results from the combination of an M-tone determiner with a monosyllabic M-tone head does not conform to the regularities observed for the other combinations. Similarly, the #H– variant observed for a disyllabic M-tone determiner plus an H$-tone head remains unexplained. A third unexpected outcome is the –L pattern arising from the combination of a disyllabic M-tone determiner with a monosyllabic MH# head. Given that in all other analogous cases (σσ+σσ, σ+σσ, and σ+σ), MH# surfaces as expected, one would anticipate the same here. These three cases confirm an earlier observation: not all patterns can be explained in terms of a set of rules applying throughout the morphophonological system.

Mrs. Latami (the main consultant for this study) appears to have acquired a large number of tone patterns individually during childhood, learning morphotonology in a manner comparable to how children acquire morphology in rGyalrongic or Kiranti languages, to cite two subgroups of Sino-Tibetan known for their flamboyant morphology (Michailovsky 1975, Van Driem 1990, Sun 2000, Jacques 2004). However, it may now be too late to investigate children's acquisition of Yongning Na morphotonology in the form documented in this volume: by the mid-2010s, school-age children in Alawua were exposed to a great deal of Mandarin, making it unlikely that much of the morphotonology was being transmitted. (The influence of bilingualism with Mandarin is examined in §10.5.)

4.2.3 L-tone determiners and what they reveal for the analysis of the head noun

The first of the seven phonological tone rules of Yongning Na is that L tone spreads progressively ("left-to-right") onto syllables that are unspecified for tone. The tone patterns of compounds with L-tone determiners thus provide an interesting testing-ground for determining whether lexical tones that begin with an M tone in their surface form are underlyingly specified for tone on their first syllable. If they were, that initial M tone would be expected to block L-tone spreading; conversely, if the first syllable is unspecified for tone, it should receive an L tone through spreading.

The observed patterns support the analysis that disyllables with High tone (#H, H$, and H#) are unspecified for tone on their first syllable: when the head carries one of these tones, an L-tone determiner spreads L onto the first syllable of the head.

The same reasoning can be extended to the L# tone, a type of L tone that associates in word-final position. L plus L# yields H#, as in the compound /**kʰv̩˩mi˩-ɬi˩pi˥**/ 'dog's ear', where the L tone on the first syllable of the head is analyzed as resulting from L-tone spreading (cf. /**kʰv̩˩mi˩**/ 'dog' and /**ɬi˧pi˩**/ 'ear'). (The final H in this compound's tone pattern is discussed further below.)

The weight of this argument is admittedly somewhat diminished by the fact that tone patterns after an L-tone determiner cannot be generated in full through the application of a set of general rules. It remains unclear to what extent the processes involving L tone in this particular morphosyntactic context (determinative compounds) relate to the general rule of L-tone spreading. When the determiner and the head are both monosyllabic, all tonal oppositions are neutralized: the compound carries a simple L tone. In the other three length combinations (σσ+σσ, σσ+σ, σ+σσ), the picture is more complex. A floating H tone (#H) on the head is always disregarded, yielding L. In combination with an M-tone head, the result is also L, except for σσ+σσ, which yields L+H# (surface phonological tone sequence: L.L.L.H). The presence of a final H is not due to a general rule limiting the propagation of L: for instance, the compound //**jo˩-gv̩˩dv̩˩**// 'sheep's back' (input tones: L plus #H) carries a simple L tone, which spreads over both syllables of the head. (It surfaces as /**jo˩-gv̩˩dv̩˩˥**/, with a final rise, due to postlexical H-tone addition: Rule 7.) Similarly, an L plus L combination in disyllables yields an outcome that is incompatible with the hypothesis that the tones of the determiner and head are simply concatenated: L+#H− (surface form: L.L.H.L).

As with M-tone determiners (§4.2.2), the behaviour of L-tone determiners resists phonological generalizations. In some cases it could seem as if dissimilation were at play. For example, in /**kʰv̩˩mi˩-nv̩˥mi˩**/ 'dog's heart' (from an L plus L input), the H tone on the first syllable of the head might be interpreted as a case of dissimilation, hypothesizing a process whereby the L tone on the head dissimilates to an initial H tone. But L plus L produces a simple L output in other length configurations (σσ+σ, σ+σ, and σ+σσ), which shows that there is no general (phonological) mechanism of dissimilation between successive L tones. Once again, the evidence suggests that tone combination rules are not derived from a uniform set of phonological principles. They need to be learnt individually.

4.2.4 H-tone determiners: #H, H#, and H$

As with L-tone determiners, the tone of compounds formed with determiners having an H tone (#H, H# or H$) varies depending on the number of syllables in the head noun. Attempts to generate the tones of these compounds from the input tones via a set of rules were unsuccessful. Nonetheless, a general observation can be made:

> #H and H$ (the floating H tone and the 'hopping' H tone) are never observed to reassociate more than one syllable to their right.

This constraint is specific to determinative compounds and does not apply elsewhere in the morphotonology. For example, in (3), the H$ tone shifts two syllables away from the word to which it is lexically attached (the family name /**lɑ˧tʰɑ˧mi˥$**/ ‘Latami’).

(3) lɑ˧tʰɑ˧mi˧ =ɻ̍˧ ɳɯ˥
lɑ˧tʰɑ˧mi˥$ =ɻ̍˩ ɳɯ˧
Latami (family name) ASSOCIATIVE A
‘by the Latamis’ (Field notes.)

In compounds with a monosyllabic head, the #H tone is preserved in six out of ten cases. Since this tone attaches at the end of the word that carries it, this preservation effectively corresponds to a one-syllable shift to the right from its original position. When the head is disyllabic, the #H tone is never present in the output, as if it could not move more than one syllable away from its original position without changing its nature. Instead, in eight of the sixteen cases, the compound carries a fixed, word-final H tone (H#).

Under this analysis, the H# tone observed in compounds where both the determiner and the head carry H# tone should be attributed to the head rather than the determiner.

The same generalization accounts for the fact that the H$ tone never surfaces in compounds with a disyllabic head. On the other hand, no hypothesis can be proposed as to why it surfaces when the head is an L-tone monosyllable but does not surface in combination with any other monosyllabic head. Interestingly, the seven cases that have two or three variant forms all involve an H$-tone head noun, suggesting that this tone category is relatively more unstable than the other types of H tones.

Compounds with a #H-tone determiner yield the same output tone regardless of whether the determiner is monosyllabic or disyllabic. This finding corroborates the initial hypothesis that the tone category of monosyllables illustrated by /**ʐwæ#˥**/ ‘horse’ is phonologically identical to that of disyllables such as /**gi˧zɯ#˥**/ ‘little brother’.[1]

4.2.5 MH-tone determiners

In determinative compounds, MH tone, like other tones containing an H level, is never observed to move more than one syllable to the right. When the determiner is disyllabic and the head monosyllabic (σσ+σ), an MH tone on the determiner

[1] As was explained in §2.4.4, for typographical simplicity, and given the absence of an opposition among different types of H tones on monosyllabic nouns, ‘horse’ is transcribed as /**ʐwæ˥**/ rather than /**ʐwæ#˥**/, omitting the information on the segmental anchoring of its H tone.

moves onto the last syllable of the compound. In σ+σ, on the other hand, the MH tone does not move as a whole: it appears to remain associated with the determiner, and to project its H level onto the head – except when the head has a #H or MH tone.

Once again, these observations remain fragmentary: no set of rules can be formulated to systematically generate the tones of these compounds from the input tones.

4.2.6 Determiners carrying LM+MH# tone or LM+#H tone

The behaviour of LM+MH# and LM+#H when they appear on determiners provides further evidence for their phonological analysis. These two lexical categories unfold as L.M over the first two syllables of polysyllabic compounds, which supports the analysis proposed in §2.4.2: both involve an LM tonal sequence. If LM+MH# were instead analyzed as L+MH#, dispensing with the first M in the expression 'LM+MH#', its initial L tone would be expected to spread, creating a sequence of L tones followed by a final MH contour. Likewise, reanalyzing LM+#H as L+#H would not be promising at all, despite the apparent gain in descriptive economy: if the first part of this tone were simply L, its behaviour in compounds would be incomprehensible.

Beyond this general observation, the interpretation of individual tone combinations is not straightforward. The cases that seem to make good sense in terms of the input tones do not greatly outnumber those that appear opaque. For instance, when LM+MH# is followed by a monosyllabic head carrying #H or MH, the resulting compound receives a pattern comprising a 'hopping' H tone (H$), exactly like in compounds where a monosyllabic MH-tone determiner associates with a #H-tone or MH-tone head. But the parallel ends here: when the head is an L-tone monosyllable, an MH-tone determiner yields a final H tone (H#), whereas an LM+MH# determiner produces a 'hopping' (H$).

4.2.7 About cases of neutralization of tonal oppositions on the head

Four tonal categories of disyllabic head nouns always behave identically: the distinctions among LM, LH, LM+#H, and LM+MH# are neutralized. This neutralization is not a direct consequence of the seven phonological tone rules set out in §7.1.1 and reiterated at various points in this volume. In principle, one could imagine a morphotonological rule whereby an L-tone determiner and a head carrying LM+MH# tone would yield L–LM+MH# in the compound through simple concatenation of the two tones. Under such a rule, the expected output for 'sheep's waist' (with input tones L and LM+MH#) would not be /**jo˩-ɟi˥ʈʂæ˩**/ but rather † **jo˩-ɟi˩ʈʂæ˧˥**. This form would not violate any well-formedness constraints. What, then, might explain its absence?

As observed at the outset of §4.2, in simpler cases, the tone of the determiner is realized first, followed by the tone of the head to the extent that it is compatible with the tones already assigned. The four categories LM, LH, LM+#H, and LM+MH# all begin with an L tone; in almost all cases, expression of this L tone on the head results in the creation of an /H.L/ or /M.L/ sequence at the juncture between the determiner and the head. Consider, for instance, a compound consisting of an M-tone determiner such as /**po˧lo˧**/ 'ram' and an LM+MH#-tone head such as /**ɟi˩ʈʂæ˧˥**/ 'waist'. The tone assignment process can be hypothesized as follows:

(i) The M tone of the determiner associates first, yielding M tone on the first two syllables of the compound /**po.lo-ɟi.ʈʂæ**/ 'ram's waist', hence /**po˧lo˧-ɟi.ʈʂæ**/.

(ii) Since M is a non-spreading tone, the tone of the head can surface. By left-to-right association, the first syllable of the head receives L tone, corresponding to the first tone level in the LM+MH# pattern. This results in /**po˧lo˧-ɟi˩ʈʂæ**/.

(iii) The final syllable receives L tone due to one of the phonological rules that govern tone association in the Alawua dialect of Yongning Na: "All syllables following an H.L or M.L sequence receive L tone" (Rule 5). This yields /**po˧lo˧-ɟi˩ʈʂæ˩**/.[2]

The situation exemplified by /**po˧lo˧-ɟi˩ʈʂæ˩**/ 'ram's waist' is widespread: there are many cases where the tone of the compound can be analyzed as resulting from the successive association of the determiner and head's tones, with any necessary adjustments dictated by phonological tone rules. Why, then, does the combination of input tones L and LM+MH# deviate from this pattern?

[2]An alternative approach would be to assume that the full lexical tone pattern of the head is initially assigned, yielding ‡ **po˧lo˧-ɟi˩ʈʂæ˧˥**, and that this form, which is disallowed due to the presence of a trough in the middle (the L in the M.M.L.MH sequence), is subsequently repaired by deleting the final MH and replacing it with L. Under this view, Rule 5 could be reformulated as Rule 5': "All syllables following an H.L or M.L sequence are lowered to L". Rule 5' would apply after full association of the LM+MH# tone pattern, whereas under the present account, Rule 5 applies as soon as the M.L sequence is created, i.e. as soon as an L tone associates to the syllable /...**-ɟi˩**.../. Since both approaches yield the same outcome, the choice between them depends on theoretical preferences.

A possible explanation is that, at an earlier historical stage, this input (L and LM+MH#) yielded L.L.MH (phonologically: L+MH#) for a σ+σσ string, through successive association. If the rest of the system at that stage resembled its present state, this L+MH# output would have been an outlier. Among compounds with a L-tone determiner, it would have been the only one to contain anything other than L and H tones, as it would have been the only combination in which a tone pattern beginning in L could be fully realized on the head without contravening phonological constraints. In all other cases, the opposition between LM, LH, LM+#H, and LM+MH# would have been neutralized by Rule 5: "All syllables following an H.L or M.L sequence receive L tone". Thus, L+MH# would have stood as the sole counterexample to this local pattern of neutralization. It is conceivable that speakers regularized this unique output (*L+MH#) through analogy with the more common pattern.

In other words, the proposed scenario is as follows. First, a near-complete pattern of neutralization emerged due to a phonological rule, with a single exception: *L+MH# from input L and LM+MH#. Later, this exception was eliminated through analogy, rendering the pattern fully exceptionless within this corner of Yongning Na morphotonology. This development enhanced the internal consistency of the tone system by ensuring greater uniformity in the inventory of tone patterns of compounds. It resulted in the current pattern of full neutralization of LM, LH, LM+#H, and LM+MH# tones on head nouns in compounds. On the other hand, it also disrupted the straightforward correspondence between input tones and output tones by introducing an exception to the general principle of successive association. (The simplicity of the successive-association pattern makes it appealing to the phonologist, but apparently less so to speakers of the language: rule simplicity is one thing; simplicity of output forms is another.) As a result, the new output for input L and LM+MH# became more aligned with other compound tone patterns but it had to be learnt independently.

This account is admittedly purely speculative. However, formulating hypotheses of this kind can be useful in constructing a coherent understanding of the tone combination rules, rather than viewing them as a mere collection. Successive association of input tones in compounds may never have been an exceptionless rule at any diachronic stage; indeed, it is conceivable that cases where output tones directly reflect successive association were *fewer* in number at some point in the past than they are now. Nonetheless, reasoning in terms of a prototypical pattern (in this case, successive association) remains useful: it allows for the systematic identification and analysis of deviations, providing insight into the structural factors that may have shaped these patterns. Further progress in this strand of research will require broader dialectal coverage than is currently available. The topic of the diachronic dynamics of the Yongning Na tone system is taken up in Chapter 10.

4.2.8 A tendency to avoid long-distance movement of tones

It was noted at the outset of this chapter, when discussing compounds of five syllables and more, that Yongning Na has a tendency to avoid processes leading to long-distance tone movement – specifically, reassociation of an H tone more than two syllables away from the word to which it is lexically attached. This tendency is confirmed by observations about the tone patterns that appear in compounding. A floating H tone (#H) on the determiner, combined with an M tone on the head, yields a final H tone (H#) on the compound, rather than a floating H tone (#H). Consequently, the H tone does not shift further away, as a floating tone would. It seems as if a determiner's floating H tone loses its ability to float in the process of compounding. To put it impressionistically: the H tone only floats once; at the outcome of the compounding process, it does not retain any potential for further movement. It appears highly significant that the only σσ+σσ compounds that retain a floating H tone result from an input in which the head originally carried this tone, i.e. cases where the floating H tone does not move in the process of compounding.

Likewise, when the determiner carries H$ or MH#, the H level in the lexical tone tends to remain close to the determiner's edge, even though its anchoring may be modified in compounding (as set out in §4.2.4).

The association of an L tone to long stretches of syllables does not constitute a counterexample to the tendency to avoid long-distance tone movement. Tone spreading, neutralizing all tonal oppositions on a portion of the tone group, does not involve tonal movement in the sense of reassociating a tone away from the word to which it is lexically attached.

4.2.9 Brief remarks about slips of the tongue

Hesitations, variants, and tonal slips of the tongue can offer insights into the tone system. A full-fledged study of this captivating topic would require more fine-grained tools than those used so far. The boundary between an acceptable variant and a commonly occurring mistake is not entirely clear, and the main consultant's judgments sometimes wavered between the two. While the greatest care was exercised to verify the data, the initial dichotomy between mistakes and acceptable variants would need to be followed up with targeted experiments to assess variant acceptability on a more precise scale. (On the notion of gradient acceptability, see Kirby & Yu 2007, Coetzee & Pater 2008, Goldrick 2011.) An attempt at quantifying and analyzing mistakes is made in the discussion of numeral-plus-classifier phrases (§3.1.6); for compounds, only preliminary observations can be offered here.

In studying slips of the tongue, it is useful to consider, for each surface phonological tone pattern, (i) the rules that can give rise to it, and (ii) the morphosyntactic constructions in which it occurs. For instance, the L tone category does not appear in any σσ+σ determinative compound (see Table 4.8), so that cases where a σσ+σ compound is mistakenly realized with an L tone cannot be attributed to analogy with other σσ+σ compounds. (Examples include the realization of 'dog's brains' as ‡ **kʰv̩˩mi˩-ɬv̩˩** instead of /**kʰv̩˩mi˩-ɬv̩˥**/ in the recording *DetermCompounds12* {0004463#W121}.) A possible explanation is interference between tone rules that apply to compounds of different syllabic structures, given that σ+σσ compounds with the same tonal input yield an L tone (see Table 4.9). Alternatively, the erroneous pattern (the slip of the tongue) could result from misapplying a phonological rule of tone spreading: "L tone spreads progressively onto syllables unspecified for tone" (Rule 1; see §7.1.1). In this case, the mistake would consist in grouping the two nouns together with a solely phonological adjustment instead of a morphophonological one.

Tonal slips of the tongue – perhaps better termed "slips of the larynx"? – in the Na data also suggest a special closeness between certain tonal categories. The M tone category, for instance, appears more prone to confusion with #H than with other tones. This is illustrated by errors involving /**lɑ˧**/ 'tiger', in which the expected tone pattern is replaced with that of a #H-tone noun: ‡ **lɑ˧-ɬv̩˩** 'tiger's brains' instead of /**lɑ˧-ɬv̩˦**/, and ‡ **lɑ˧-hu˩mi˩** 'tiger's stomach' instead of /**lɑ˧-hu˧mi˥$**/. (Both errors occur twice in the recordings: *DetermCompounds6* {0004454} and *DetermCompounds7* {0004456}.) There are also cases of substitution among H$, MH#, and H#, such as ‡ **hwɤ˧li˧-sɤ˥** instead of /**hwɤ˧li˧-sɤ˦**/ for 'cat's blood', and ‡ **hwɤ˧li˧-ɬv̩˦** instead of /**hwɤ˧li˧-ɬv̩˥**/ for 'cat's brains'. Finally, the speaker occasionally confuses LM+H# and LM+MH# tones.

4.2.10 Perspectives for comparison across speakers

The dataset analyzed here was provided by the primary consultant, Mrs. Latami (F4). Additional data was elicited from three other speakers: F5, who is F4's daughter-in-law; M21, a relative of F4, belonging to the same generation; and M23, who is M21's son. Information about these three speakers was provided in §1.4.3. To summarize briefly, all three are less proficient speakers than F4 – M21 due to long stays away from Yongning, and F5 and M23 due to a generation gap: both are fluent speakers of Southwestern Mandarin.

Unsurprisingly, some cross-speaker differences emerge. Analyzing these differences is crucial to understanding the dynamics of the tone system. Some reflect lexical variation, with certain nouns belonging to different categories in different consultants' speech. For instance, 'flea' is /**kv̩˧ʂe˥$**/ (LH tone) in F4's speech but /**kv̩˧ʂe#˥**/ (#H tone) in M21's. Similarly, 'boar' is /**bo˩ɬɑ˥**/ (LH tone) in F4's speech but /**bo˩ɬɑ˦** in F5's and /**bo˩ɬɑ˧**/ in M21's.[3] The tonal difference for 'boar's snout' – /**bo˩ɬɑ˥-ɲi˩gɤ˩**/ (LH) in F4's speech and /**bo˩ɬɑ˧ɲi˧gɤ#˥**/ (LM+#H) in M21's – does not appear to stem from differences in the tonal rules governing compounds, but from a difference in the lexical tone of the determiner. Both compounds follow the regularities summarized in Tables 4.7 to 4.11, but the input combination is different.

A full assessment of each consultant's tone system and lexicon is therefore a necessary first step before selecting compound nouns for elicitation. Only on this basis can cross-speaker differences in the tone rules be reliably brought out.

To begin with a comparison within the same generation, Table 4.12 presents two differences between F4 and M21.

Table 4.12: Differences between speakers F4 and M21 in the tones of compounds.

input tones	output tones, with example compounds		
	F4	M21	meaning
M+M (σ+σ)	#H	L#	
	lɑ˧-bv̩#˥	**lɑ˧-bv̩˩**	tiger's intestine
#H+H$ (σσ+σσ)	H#, H$ or #H–	–L	
	ʐwæ˧zo˧-hu˧mi˥	**ʐwæ˧zo˧-hu˩mi˩**	colt's stomach

The output obtained when combining two M tones over monosyllables differs between the two speakers, and in both cases, it is somewhat unexpected: why not simply concatenate the input tones, yielding M tone for the compound, as occurs in the other M+M compounds (σσ+σσ, σσ+σ, and σ+σσ)? For both F4 and M21, the outcome deviates from the apparent default. Likewise, the combination of #H and H$ over disyllables (σσ+σσ) differs between the two speakers, who both rejected the other's variant when I tested it on them.

[3]The dictionary records variants for the various consultants: see Michaud et al. (2024: x-xi).

An intriguing characteristic of these patterns is their closeness: despite the differences, M21's patterns bear a family resemblance to F4's. For instance, M21 assigns –L tone to the phrase 'colt's stomach' (σσ+σσ), a tone also found in F4's data, albeit in compounds where the input nouns with these tones are monosyllabic (σ+σ). Such observations suggest that subtle processes of phonological convergence may be at play, whereby individuals bound by social ties tend to accommodate to one another's tone patterns. One could speculate that speakers seeking to foster a sense of community, for instance in relaxed conversations with relatives, occasionally adopt new patterns to emphasize shared linguistic ground over divergence. Given that a single surface tone pattern is open to several phonological interpretations (e.g. /M.M/ may be the realization of underlying //M// or //#H//), this process of accommodation could lead to a proliferation of variant forms across speakers. In other words, speakers' adoption of tone patterns from their customary interlocutors could explain the extent of the observed pool of morphotonological variation.

By a related process, speakers may tend to select from within this pool of variation those patterns that they believe will be most accessible to their addressee, avoiding variants that they feel are most sharply at variance with the addressee's linguistic habits. Conversely, self-assertive speakers wishing to distance themselves from the addressee might favour patterns that they feel are most divergent from their interlocutor's norms, potentially leading to the spread and eventual stabilization of a particular variant within that speaker's idiolect.

To test these hypotheses, one could examine dialogues, comparing F4's tone patterns in conversations with different family members – whose own tone systems would need to be documented with the greatest possible precision. This line of research holds promise for identifying key mechanisms in the evolution of the tone system, while also offering insights into its synchronic state. However, carrying out such a study would be far from straightforward. As anthropologists are well aware, speech recording in the uplands of Socialist Asia is a sensitive activity (Milan 2024), posing particular challenges for the collection of spontaneous conversations from multiple speakers.

4.2.11 Exceptional items

Some compounds have lexical tones that differ from those expected under the synchronic tone rules. The irregularity may stem either from the determiner, as in the first four lines of Table 4.13, or from the head, as in the last line. These examples are examined individually below.

Table 4.13: Compounds whose tones differ from those expected under the synchronic tone rules.

observed compounds	tone pattern	expected pattern	irregular word	meaning	tone
nɑ˩hĩ˧-kʰɯ˥dʑi˩ ‘Naxi leggings’; **nɑ˩hĩ˧-bɑ˧lɑ˥** ‘Naxi clothes’	LM+#H–; LM+H#–	LM+MH#–; LM+#H–	**nɑ˩hĩ#˥**	Naxi	LM+#H
ɲi˧gɤ˧-dʑɯ˦ ‘mucus’	MH#	#H	**ɲi˧gɤ#˥**	nose	#H
mv̩˧-ʁo˥ ‘sky’	H#	#H	**mv̩˥**	sky	#H
ʝi˧bv̩˦ ‘cows’ stable’	MH#	#H	**ʝi˥**	ox	#H
lv̩˧mi˧-tsɑ˩bɤ˩ ‘fine sand’; **qʰɑ˧dze˧-tsɑ˩bɤ˩** ‘sweetcorn flour’; **dze˧ɭɯ˧-tsɑ˩bɤ˩** ‘wheat flour’	–L	M	**tsɑ˧bɤ˧**	powder	M

4.2.11.1 The noun 'Naxi'

The noun 'Naxi' (/**nɑ˩hĩ#˥**/, tone: LM+#H) yields irregular tonal patterns in two quadrisyllabic compounds (see the recording *DetermCompounds16* {0004491}):

(i) With MH# tone: /**nɑ˩hĩ˧-kʰɯ˥dʐi˩**/ 'Naxi leggings'. The expected pattern would be † **nɑ˩hĩ˧-kʰɯ˧dʐi˦** (underlying tone: LM+MH#–), but this form is not acceptable. The observed tone is LM+#H–. A compound that follows the regular tone pattern is /**nɑ˩hĩ˧-ŋwɤ˧pʰæ˦**/ 'Naxi tile'.

(ii) With L tone: /**nɑ˩hĩ˧-bɑ˧lɑ˥**/ 'Naxi clothes'. The expected pattern would be † **nɑ˩hĩ˧-bɑ˥lɑ˩** (underlying tone: LM+#H–, which can also be described as LM+MH#–), but this form is also not acceptable. The observed tone is LM+H#. A regularly patterned compound is /**nɑ˩hĩ˧-sɯ˥tʰi˩**/ 'Naxi knife'.

The noun 'Naxi' has a special status in Yongning Na. On the one hand, it refers to an ethnic group perceived as distinct from the Na: the Naxi of the Lijiang plain, some of whom settled in Yongning in the early 20th century but have retained their distinct costumes and language (on the cultural divide between the Naxi and Na, see Appendix B, in particular the end of §B.3.4). On the other hand, it consists of the endonym of the Na combined with the word for 'person, human being', which creates a potential for ambiguity: its independence from 'Na' is problematic. The exceptional treatment of this noun may reflect a deliberate effort to avoid its perception as a compound meaning 'Na person'.

4.2.11.2 The nouns 'nose' and 'hair'

'Nasal mucus' is /**ɲi˧gɤ˧-dʐɯ˦**/ (tone: MH#); based on its component nouns, /**ɲi˧gɤ#˥**/ 'nose' and /**dʐɯ˩**/ 'water', the expected tone would be #H († **ɲi˧gɤ˧-dʐɯ#˥**). Similarly, 'hair (on the head)', /**ʁo˧hṽ̩˦**/, also has MH# tone, whereas the expected tone would be #H († **ʁo˧hṽ̩#˥**), given that both input nouns carry H tone: /**ʁo˥**/ 'head; top' and /**hṽ̩˥**/ 'hair'.

The etymology of these two words is self-evident. However, mucus is not merely a type of water ('nose-water'). The difference between hair on the head and on the body may seem subtler, yet many languages distinguish between the two with distinct lexical roots, e.g. French and Lao (Enfield 2006: 187-188). In Na, 'hair (on the head)' and 'body hair' are both lexicalized compounds; the latter is /**ʐi˩hṽ#˥**/, literally 'ape hair'. The irregular tone patterns of 'nasal mucus' and 'hair (on the head)' may thus reflect early lexicalization. The discrepancy between their tones and the output of the currently productive tone rules for compounds could be due to earlier tone rules that applied at the time of their formation. Alternatively, their divergence from the regular tone pattern may have arisen as these compounds evolved into lexical units rather than transparent compositions. While this second possibility might seem less plausible at first glance, item-by-item tone change accompanying lexicalization is a salient characteristic of the tone system of Laze, a language closely related to Na.[4] This possibility should therefore not be lightly dismissed.

4.2.11.3 The noun 'flour, powder'

The word for 'powder, flour' is /**tsɑ˧bɤ˧**/, with M tone. According to the synchronically productive rules, combining this noun with the M-tone determiners /**lv̩˧mi˧**/ 'stone', /**qʰɑ˧dze˧**/ 'sweetcorn', and /**dze˧ɭɯ˧**/ 'wheat' should yield a simple M-tone output, e.g. † **lv̩˧mi˧-tsɑ˧bɤ˧** for 'fine sand'. But the attested forms, shown in Table 4.14, all follow an M.M.L.L tone pattern, corresponding to underlying –L (an L tone on the second part of the compound).

Table 4.14: Compounds with /**tsɑ˧bɤ˧**/ 'flour, powder'.

determiner	compound with /**tsɑ˧bɤ˧**/ 'flour, powder' as head noun		
	expected tone: M	attested tone: –L	meaning
/**lv̩˧mi˧**/ 'stone'	† **lv̩˧mi˧-tsɑ˧bɤ˧**	**lv̩˧mi˧-tsɑ˩bɤ˩**	fine sand
/**qʰɑ˧dze˧**/ 'sweetcorn'	† **qʰɑ˧dze˧-tsɑ˧bɤ˧**	**qʰɑ˧dze˧-tsɑ˩bɤ˩**	sweetcorn flour
/**dze˧ɭɯ˧**/ 'wheat'	† **dze˧ɭɯ˧-tsɑ˧bɤ˧**	**dze˧ɭɯ˧-tsɑ˩bɤ˩**	wheat flour

The principle applied in the present study is that two lexical items of similar phonological structure whose tones differ in at least one morphosyntactic

[4] Laze has four lexical tones for monosyllables (for predicates: H, M, L, and MH; for nouns: H, M, L, and a floating H tone). In theory, this could yield up to sixteen tone patterns for disyllables, but only seven are attested. Tone changes occur as lexicalization takes place. Numerous combinations of two input tones other than H yield H+H. For instance, 'dog' is /**kʰɯ˧**/ and 'to beat' is /**ɖɯ˩**/, a combination which should yield M.L but instead surfaces as H.H. This process plays a key role in the lexical integration of disyllables in Laze (Michaud 2008a, 2009, Michaud & Jacques 2012).

context must be assigned to different lexical tone categories. Mechanical application of this principle would lead to recognizing /**tsɑ˧bɤ˧**/ 'flour, powder' as the sole member of an umpteenth (twelfth) tonal category of disyllabic nouns. But it makes much more sense to seek an explanation for the irregular behaviour of the compounds in Table 4.14.

The word /**tsɑ˧bɤ˧**/ is a Tibetan loanword (from *rtsam pa* 'roasted flour'). The tone lowering observed in the latter part of the compounds in Table 4.14 is part of a broader pattern. It echoes observations about a larger set of words of Tibetan origin: given names. Compounds involving Tibetan loanwords will be referred to for short as 'Tibetan compounds'. This topic will be taken up in the analysis of given names in §4.4.

4.2.11.4 The noun 'sky'

The disyllabic form of the noun 'sky' is /**mv̩˧ʁo˥$**/. The monosyllabic root for 'sky' is /**mv̩˥**/. If the second syllable were /**ʁo˥**/ 'head; top', the expected tone of the compound would be #H tone, based on the regular patterns set out in Tables 4.7 to 4.11. But the second syllable could also be the postposition 'on', which is itself likely to have grammaticalized from 'head'. This postposition is no longer in common use (the common form is //**bi˩**// 'on; at': see §5.4.1), making it difficult to study its combinatorial properties and establish its lexical tone.

4.3 Coordinative compounds

4.3.1 The main facts

In the closely related language Naxi, the tones of coordinative compounds are simply the concatenation of those of their constituents. For instance, /**ɲi˧nv̩˩-jæ˧kæ˩zɯ˧**/ 'wife and husband' is built from /**ɲi˧nv̩˩**/ 'wife' and /**jæ˧kæ˩zɯ˧**/ 'husband'.[5] In Yongning Na, by contrast, coordinative compounds are tonally active to a degree comparable with determinative compounds.

However, coordinative compounds are less common than determinative compounds and more difficult to elicit systematically. Syntactically, coordinative constructions can apply to any pair of nouns. This is illustrated by the names of public houses in English: combinations like "Fox and Hounds" or "Dog and Duck"

[5] This tonally inert compound is of little phonological interest; on the other hand, it has some ethnolinguistic significance. As the Naxi of Lijiang like to point out, this compound places the wife before the husband, in contradiction of Confucian principles. About the family structure of the Na and Naxi, and its history, see §B.3.

refer to hunting traditions, while others, such as "Bear and Ragged Staff", refer to heraldry. Once the pattern is established, new coordinative compounds can be created at will, such as the humorous "Snail and Salad", where the relationship between the two terms – and their relationship to the food served in the pub – is offered to the customer's fancy.

In Yongning Na, any two nouns can be linked by means of the conjunction /**lɑ˧**/ 'and', but coordinative compound nouns are not as easy to coin: the two nouns must refer to entities that are commonly paired together.

Three main sources of coordinative compounds were identified:

- Pairs of animal names denoting the two sexes, or one sex and an offspring, such as 'ewe and ram', 'ram and ewe', 'ewe and lamb', and 'ram and lamb';
- Pairs of kinship terms, such as 'uncle and nephew' and 'mother and daughter';
- Successive numerals followed by the same classifier, as in 'two or three years', 'five or six months', or 'four or five days'.

A broad sample of the first two types can be found in the online recording *CoordCompounds* {0004561}, while the third type appears in *CoordCompounds2* {0004662} and *CoordCompounds3* {0004869}. Since the consultant (Mrs. Latami, F4) preferred to remain within the bounds of common sense, non-matching pairs such as 'mother and nephew' or 'grandmother and brother' were avoided.

The data is set out in Tables 4.15-4.23. As elsewhere, a slash separates variants. Some tone patterns proposed by the investigator but rejected by the consultant are marked with a double dagger ‡ in the output column. For example, given that the combination /**ʐv̩˩ɬi˩-ŋwɤ˩ɬi˩˥**/ ≈ /**ʐv̩˩ɬi˩-ŋwɤ˥ɬi˩**/ 'four or five months' has two possible variants, an attempt was made to extend the L-tone variant /**ʐv̩˩ɬi˩-ŋwɤ˩ɬi˩˥**/ to other combinations of two L-tone nouns, such as 'nephews and nieces' /**ze˩v̩˩-ze˥mi˩**/. The consultant rejected the L-tone variant (‡ **ze˩v̩˩-ze˩mi˩˥**). This piece of information is indicated in the output column by '(‡ L)'.

Three suffixes appear repeatedly in the table: the female/augmentative suffix /**-mi˩**/, the male suffix /**-pʰv̩˥**/, and the child/male/diminutive suffix /**-zo˥**/. These suffixes are discussed in more detail in §5.1.

Most examples are quadrisyllabic, formed from two input disyllables (σσ+σσ). The two disyllabic examples (σ+σ) at the top of Table 4.15 are written without a hyphen, reflecting the intuition that they are more strongly integrated than the others.

Table 4.15: Coordinative compounds of fewer than four syllables, arranged by input tones.

compound	meaning	input	output
mv̩˧di˦˥	universe ('sky'+'earth')	H and LM	MH#
zo˧mv̩˥	child ('son'+'daughter')	H and LH	H#
ə˧mi˧-mv̩˩	mother and daughter	M and LH	–L
ə˧mi˧-zo#˥	mother and son	M and H	#H
ə˧dɑ˧-mv̩˥	father and daughter	H$ and LH	H#
ə˧dɑ˧-zo#˥	father and son	H$ and H	#H

All four trisyllabic examples follow the structure *disyllable plus monosyllable* (σσ+σ). This pattern arises from morphological and semantic factors rather than a phonological preference. All kinship terms referring to one's elders (e.g. elder sister or brother, mother, aunts, father, uncles) include a schwa prefix (presented in §5.1.5), making them disyllabic, whereas 'daughter' and 'son' are monosyllabic. Since terms for elders conventionally precede those for younger family members, the four kinship pairs in Table 4.15 ('mother and daughter', 'mother and son', 'father and daughter', and 'father and son') naturally conform to the σσ+σ pattern. Thus, the prevalence of this structure stems from the morphological properties of kinship terms and their semantic ordering rather than from a phonological preference for an initial disyllabic term.

Hexasyllabic compounds can also be formed, such as /**ŋwɤ˩-ɬi˩mi˩-qʰv̩˥-ɬi˩mi˩**/ (*Dog2.64)* {0004660#S64}) 'the fifth and sixth months'.

Table 4.16: Quadrisyllabic compounds with M as the first input tone.

compound	meaning	input	output
ə˧pʰv̩˧-ʐv̩˧v̩#˥	great-uncle and great-nephews	M and M	#H
ə˧pʰv̩˧-ʐv̩˧mi#˥	great-uncle and great-nieces		
ə˧si˧-ə˧pʰv̩#˥	3rd-generation ancestors		
ə˧si˧-ʐv̩˧mi#˥	(great-)grandmother and granddaughters		
ə˧si˧-ʐv̩˧v̩#˥	great-grandmother and grandsons		
gv̩˧dv̩˧-gv̩˧mi˧	(human) body	M and M	M
jo˧mi˧-po˧lo˧	ewe and ram		
ɖʐwæ˧mi˧-ɖʐwæ˧pʰv̩˧	male and female sparrow		
ɖɯ˧ɬi˧-ɲi˧ɬi˧	one or two months		
ɲi˧ɬi˧-so˩ɬi˩	two or three months	M and M	–L
bæ˧mi˧-bæ˧pʰv̩#˥	female and male duck	M and #H	#H
bæ˧mi˧-bæ˧zo#˥	female duck and duckling		
bv̩˧mi˧-bv̩˧zo#˥	female yak and yak calf		
ʂɯ˧ɬi˧-hõ˧ɬi#˥	seven or eight months	M and H$	#H
ə˧mi˧-ze˩mi˩	aunt and niece	M and L	–L
ə˧mi˧-ze˩v̩˩	aunt and nephew		
bv̩˧mi˧-bv̩˩ʂwæ˩	female and male yak		
dzo˧mi˧-dzo˩pʰv̩˩	female and male lizard		
so˧ɬi˧-ʐv̩˩ɬi˩	three or four months		

Table 4.17: Quadrisyllabic compounds with #H as the first input tone.

compound	meaning	input	output
gi˧zɯ˧-go˧mi#˥	little brothers and sisters	#H and M	#H
bæ˧zo˧-bæ˧mi#˥	duckling and female duck		
bæ˧pʰv̩˧-bæ˧mi#˥	male duck and female duck		
tsʰɯ˧zo˧-to˧qɑ˥	kids and little nanny goats	#H and M	H#
ʐv̩˧v̩˥-ʐv̩˩mi˩	grandchildren	#H and #H	H#–
hwɤ˧pʰv̩˧-hwɤ˧zo#˥ / **hwɤ˧pʰv̩˧-hwɤ˥zo˩**	tom-cat and kitten	#H and #H	#H / #H–
ho˧mi˧-ho˧pʰv̩#˥ / **ho˧mi˧-ho˥pʰv̩˩**	female and male pheasant		
dʑi˧mi˧-dʑi˧zo#˥ / **dʑi˧mi˧-dʑi˥zo˩**	female and baby buffalo		
dʑi˧zo˧-dʑi˧mi#˥ / **dʑi˧zo˧-dʑi˥mi˩**	baby and female buffalo		
lɑ˧mi˧-lɑ˧pʰv̩#˥ / **lɑ˧mi˧-lɑ˥pʰv̩˩**	female and male tiger		
lɑ˧mi˧-lɑ˧zo#˥ / **lɑ˧mi˧-lɑ˥zo˩**	female and baby tiger		
ʐv̩˧ɲi˧-ŋwɤ˧ɲi#˥ / **ʐv̩˧ɲi˧-ŋwɤ˥ɲi˩**	four or five days		
ʁv̩˧pʰv̩˧-ʁv̩˧mi#˥	male and female crane	#H and MH	#H
hwɤ˧pʰv̩˧-hwɤ˧mi˥	tom-cat and she-cat	#H and H$	H#
hwɤ˧zo˧-hwɤ˧mi˥	cats: kitten and parents		
ŋwɤ˧ɲi˧-qʰv̩˩ɲi˩	five or six days	#H and H$	–L
ʂɯ˧ɲi˧-hõ˩ɲi˩	seven or eight days		
ʐwæ˧zo˧-ʐwæ˥mi˩ / **ʐwæ˧zo˧-ʐwæ˧mi˥**	colt and mare	#H and L	#H–/ H#
pʰɤ˧pʰv̩˧-pʰɤ˥mi˩ / **pʰɤ˧pʰv̩˧-pʰɤ˧mi˥**	male and female hyena		
gv̩˧ɲi˧-tsʰe˩ɲi˩ / **gv̩˩ɲi˩-tsʰe˩ɲi˥**	nine or ten days	#H and L	–L / L+H#
kʰv̩˧zo˥-kʰv̩˩mv̩˩ / **kʰv̩˧zo˧-kʰv̩˧mv̩˥**	male and female puppies	#H and H#	H#– / H#

Table 4.18: Quadrisyllabic compounds with MH# as the first input tone.

compound	meaning	input	output
ə˧ʑi˧-ə˧pʰv̩˧˥	elders, grandparents	MH# and M	MH#
æ˧mv̩˧-go˧mi˥	sisters, female siblings	MH# and M	H#
ə˧ʑi˧-zv̩˥v̩˩	grandmother and grandsons	MH# and #H	#H–
ə˧ʑi˧-zv̩˥mi˩	grandmother and granddaughter		
æ˧mv̩˧-gi˥zɯ˩	brethren, brothers		
ɖɯ˧kʰv̩˧-ɲi˥kʰv̩˩	one or two years	MH# and MH#	MH#–
ə˧v̩˧-ze˥v̩˩	uncle and nephew	MH# and L	MH#–
ə˧v̩˧-ze˥mi˩	uncle and niece		
zo˧hṽ̩˧-mv̩˥zo˩	descendants		
ɲi˧kʰv̩˧-so˧kʰv̩˥ (‡ **ɲi˧kʰv̩˧-so˥kʰv̩˩**)	two or three years	MH# and L	H#
ʂɯ˧kʰv̩˧-hõ˥kʰv̩˩ / **ʂɯ˧kʰv̩˧-hõ˧kʰv̩˥**	seven or eight years	MH# and H#	MH#– / H#

Table 4.19: Quadrisyllabic compounds with H$ as the first input tone.

compound	meaning	input	output
tsʰɯ˧mi˧-po˧lo˥	nanny goat and billy goat	H$ and M	H#
ə˧dɑ˧-ə˧mi#˥	father and mother, parents	H$ and M	#H
qʰv̩˧ɬi˥-ʂɯ˩ɬi˩	six or seven months	H$ and M	H#–
ə˧ɲi˧tsʰi˧ɲi#˥ **ə˧ʝi˧-tsʰi˧ʝi#˥** **ʈʂʰæ˧mi˧-ʈʂʰæ˧zo#˥**	these days these years doe and stag	H$ and #H	#H
hwɤ˧mi˧-hwɤ˧zo#˥ / **hwɤ˧mi˧-hwɤ˧zo˥$**	she-cat and kitten	H$ and #H	#H / H$
hwɤ˧mi˧-hwɤ˥pʰv̩˩ / **hwɤ˧mi˧-hwɤ˧pʰv̩#˥** / **hwɤ˧mi˧-hwɤ˧pʰv̩˥$**	she-cat and tom-cat	H$ and #H	#H– / #H / H$
qʰv̩˧ɲi˥-ʂɯ˩ɲi˩ / **qʰv̩˧ɲi˧-ʂɯ˥ɲi˩** / **qʰv̩˧ɲi˧-ʂɯ˧ɲi#˥** (‡ **qʰv̩˧ɲi˧-ʂɯ˧ɲi˥$**) **hõ˧ɲi˥-gv̩˩ɲi˩** / **hõ˧ɲi˧-gv̩˥ɲi˩** / **hõ˧ɲi˧-gv̩˧ɲi#˥** (‡ **hõ˧ɲi˧-gv̩˧ɲi˥$**)	six or seven days eight or nine days	H$ and #H	H#– / #H– / #H
ɲi˧ɲi˧-so˧ɲi˥ (‡ **ɲi˧ɲi˥-so˩ɲi˩**)	two or three days	H$ and MH#	H#
ɖɯ˧ɲi˧-ɖɯ˥hã˩ (‡ **ɖɯ˧ɲi˧-ɖɯ˧hã˥**)	one day and one night		#H–
ʈʂʰæ˧mi˧-ʈʂʰæ˧zo#˥ **ɖɯ˧ɲi˧-ɲi˧ɲi#˥**	doe and fawn one or two days	H$ and H$	#H
hõ˧ɬi˥-gv̩˩ɬi˩ **ɲi˧ɲi˥ | -so˩ɲi˩˧**	eight or nine months two or three days	H$ and L	H#–
ə˧ʝi˧-ʂɯ˥ʝi˩	in the past	H$ and LM+#H	#H–

Table 4.20: Quadrisyllabic compounds with L as the first input tone.

compound	meaning	input	output
bv̩˩ʂwæ˩-bv̩˥mi˩ / **bv̩˩ʂwæ˩-bv̩˩mi˩**	male yak and female yak	L and M	L / L+#H–
kɤ˩pʰv̩˩-kɤ˩mi˥ / **kɤ˩pʰv̩˩-kɤ˥mi˩**	male and female falcon		
dzo˩pʰv̩˩-dzo˩mi˩ / **dzo˩pʰv̩˩-dzo˥mi˩**	male and female lizards		
mv̩˩zo˩-ə˥mi˩ / **mv̩˩zo˩-ə˩mi˥**	young woman and (her) mother	L and M	L+#H– / L+H#
gv̩˩ɬi˩-tsʰe˥ɬi˩ / **gv̩˩ɬi˩-tsʰe˩ɬi˥**	nine or ten months		
mv̩˩zɯ˩-ni˥mi˩	brothers and sisters	L and #H	L+#H–
ʐwæ˩mi˩-ʐwæ˩zo˩	mare and colt	L and #H	L
so˩ɲi˩-ʐv̩˩ɲi˩	three or four days		
ŋwɤ˩ɬi˩-qʰv̩˥ɬi˩	five or six months	L and H$	#H–
ɟi˩mi˩-ʐɤ˥qo˩	cow and calf	L and L	L+#H– (‡ L)
ze˩v̩˩-ze˥mi˩	nephews and nieces		
ɟi˩bv̩˩-ɟi˥mi˩	bull and cow		
pɤ˩mi˩-pɤ˥pʰv̩˩	female and male frog		
pʰɤ˩mi˩-pʰɤ˥zo˩	female hyena and hyena pup		
ʐv̩˩ɬi˩-ŋwɤ˩ɬi˩˥ / **ʐv̩˩ɬi˩-ŋwɤ˥ɬi˩**	four or five months	L and L	L / L+#H–
so˩kʰv̩˩-ʐv̩˩kʰv̩˥	three or four years	L and L#	L+H#

Table 4.21: Quadrisyllabic compounds with L# as the first input tone.

compound	meaning	input	output
ʐwæ˧sɯ˩-ʐwæ˩zo˩	stallion and colt	L# and #H	L#–
ʐwæ˧sɯ˩-ʐwæ˩mi˩	stallion and mare	L# and L	L#–
gv̩˧kʰv̩˩-tsʰe˩kʰv̩˩	nine or ten years		
ʐv̩˧kʰv̩˩-ŋwɤ˩kʰv̩˩	four or five years	L# and L#	L#–
ŋwɤ˧kʰv̩˩-qʰv̩˩kʰv̩˩	five or six years	L# and H#	L#–

Table 4.22: Quadrisyllabic compounds with LM as the first input tone.

compound	meaning	input	output
ɑ˩ʁo˧-ʑi˧dv̩˧	the household	LM and M	LM–
pv̩˩tsɯ˧-pv̩˥mi˩	small and large combs	LM+MH# and L	LM+MH#–
pɤ˩tɕi˧-pɤ˥mi˩	tadpole		
æ˩mi˧-æ˧ʂwæ˥	hen and cock	LM and H#	LM+H#
æ˩mi˧-æ˧tsɯ˥	hen and chicks		
bo˩mi˧-bæ˧bv̩˥	sow and piglets		
ʐæ˩pʰv̩˧-ʐæ˩mi˩	male and female leopard	LM and LM+#H	LM–L
dv̩˩mi˧-dv̩˥pʰv̩˩	female and male weasels	LM+#H and LM	LM+#H–
ɑ˩mi˧-ɑ˥pʰv̩˩	female and male goose	LM+#H and LM+#H	LM+#H–
ɖɯ˩zo˧-ɖɯ˥mi˩	female and male mule		

Table 4.23: Compounds of four to six syllables with H# as the first input tone.

compound	meaning	input	output
qʰv̩˧kʰv̩˥-ʂɯ˩kʰv̩˩	six or seven years	H# and MH#	H#–
hõ˧kʰv̩˥-gv̩˩kʰv̩˩	eight or nine years	H# and L#	H#–
æ˧ʂwæ˥-æ˩mi˩	cock and hen	H# and LM	H#–
ŋwɤ˩ɬi˩mi˩-qʰv̩˥ɬi˩mi˩	the fifth and sixth months		

4.3.2 Discussion: Tonal variability and lexical diversity

The existence of variants was already observed for some determinative compounds (see Tables 4.7 to 4.11), but the overall proportion of combinations allowing variants was low: only 7 out of 257, all of which had H$ tone on the head noun. In the case of coordinative compounds, on the other hand, less regularity is observed. Not only are there tonal variants, but also compounds with identical input tones that yield different outputs. Among quadrisyllabic compounds, two different outputs (on different examples) are found for no fewer than six tonal combinations: those with input tones M and M, #H and M, MH# and M, H$ and M, H$ and #H, and L and #H. A seventh combination, #H and #H, even has three different outputs. Overall, for quadrisyllabic compounds, roughly one in four tone combinations has two or three distinct outputs.

The general picture is thus one of great tonal diversity. However, this does not mean that coordinative compounds are characterized by a general looseness in tone patterns, whereby two or three tonal variants would be acceptable for any given combination of input tones. In cases where the same tonal input yields different outputs in different coordinative compounds, an attempt was made to substitute one pattern for the other. For instance, input nouns with H$ and M tones yield a compound with #H tone in the case of ‘little brothers and little sisters’ (//**gi˧zɯ˧-go˧mi#˥**//) but a compound with H# tone in the case of ‘kids and little nanny goats’ (//**tsʰɯ˧zo˧-to˧qɑ˥**//). On this basis, tones #H and H# were tested as substitutes for each other in coordinative compounds. The consultant rejected these substitutions, as documented in Table 4.24. This indicates that each coordinative compound acquires a habitual tone pattern to the exclusion of others. Further comparison across speakers (initially within a single family, then

Table 4.24: Attempted tonal variants for coordinative compounds.

meaning	input tones	attested form	attempted variant
little brothers and little sisters	H$ and M	**gi˧zɯ˧-go˧mi#˥**	‡ **gi˧zɯ˧-go˧mi˥**
father and mother	H$ and M	**ə˧dɑ˧-ə˧mi#˥**	‡ **ə˧dɑ˧-ə˧mi˥**
kids and little nanny goats	H$ and M	**tsʰɯ˧zo˧-to˧qɑ˥**	‡ **tsʰɯ˧zo˧-to˧qɑ#˥**
nanny goat and billy goat	H$ and M	**tsʰɯ˧mi˧-po˧lo˥**	‡ **tsʰɯ˧mi˧-po˧lo#˥**
ancestors	MH# and M	**ə˧ʐi˧-ə˧pʰv̩˦**	‡ **ə˧ʐi˧-ə˧pʰv̩˥**
sisters	MH# and M	**æ˧mv̩˧-go˧mi˥**	‡ **æ˧mv̩˧-go˧mi˦**

extending to other micro-dialects) will be necessary to trace the paths through which these idiosyncratic preferences develop.

Tonal diversity relates in subtle ways to the semantic diversity of coordinative compounds. Importantly, the meaning of such compounds is not always predictable from their two constituents. For instance, /**hwɤ˧zo˧-hwɤ˧mi˥**/, composed of the words for ‘kitten’ and ‘she-cat’, does not mean ‘kitten and she-cat’ (i.e. a mother and her offspring) but rather refers to cats in general, as a species. Meanwhile, the terms for male and female puppies, /**kʰv̩˧zo#˥**/ and /**kʰv̩˧mv̩#˥**/ respectively, have been repurposed as names for human newborns: an unattractive name is intentionally chosen to ward off malevolent spirits believed to threaten the child’s life. This practice is known in Chinese as “milk name” (乳名 *rǔmíng*) and constitutes one of the many “names intended to avoid attracting the unwanted attention of gods, sparing the name-bearers the misfortunes wrought by the god’s wrath or jealousy” (Chen 2016: 118). The child’s real name is given only after a couple of months, or sometimes as late as one full year after birth. Over time, the terms /**kʰv̩˧zo#˥**/ and /**kʰv̩˧mv̩#˥**/, along with their compound /**kʰv̩˧zo˥-kʰv̩˩mv̩˩**/, have become culturally specialized and are no longer used to refer to actual puppies.

Coordinative compounds thus range from novel elicited combinations, which a consultant may never have conceptualized before (e.g. ‘male and female jackal’), to highly lexicalized expressions. The overall number of examples is too small to determine with confidence, within this diverse landscape, which outputs are currently productive. A logical next step in investigating this issue would be to assess the degree of lexicalization of the various compounds.

The tone patterns observed thus far are summarized in Tables 4.25 and 4.26. When different (and mutually exclusive) patterns are attested for different compounds, these patterns are separated by a semi-colon. In cases of free variation (i.e. over the same compound), the patterns are separated by a slash. To save space, tone categories for which no examples have been found are simply omitted from the table.

Table 4.25: The tones of coordinative compounds with monosyllabic second noun. A question mark indicates that no example was found.

type of 1st noun	tone of 1st noun	tone of 2nd noun		
		LM	L	#H
monosyllables	#H	MH#	H#	?
disyllables	M	?	–L	#H
	H$	?	H#	#H

Sixty percent of the combinations are identical with those found on determinative compounds. There is a considerable proportion of combinations for which no single example was found, however. In addition to the cells containing a question mark in the table, one must also take into account the empty columns, which are omitted from the table for conciseness. In total, only 35 combinations were observed out of a theoretically possible 223 (6×6 for σ+σ compounds, 11×6 for σσ+σ compounds, and 11×11 for σσ+σσ compounds). Since there is no inherent restriction on input combinations, the gaps in Tables 4.25 and 4.26 must be considered accidental. This is due in part to the limitations of available materials, but the scarcity of examples, coupled with the diversity of their tone patterns, also raises morphotonological questions: coordinative compounds are much less common than determinative compounds. It may be that there is no such thing as a full-fledged set of productive tone combination rules governing the tone patterns of coordinative compounds, and that new coinages are based on analogy with the best example at hand: an established (lexicalized) compound perceived (on grounds that may fluctuate) as falling under the same tone rules. A touch of expressivity may also be at play in the process: greater weight placed on one of the two elements in the compound, for semantic-stylistic reasons, could contribute to the selection of one tone pattern over another. This could help explain the observed diversity.

Table 4.26: The tones of coordinative compounds consisting of two disyllables. A question mark indicates that no example was found.

	2nd noun							
tone of 1st noun	M	#H	MH#	H$	L	LM+#H	LM	H#
M	M; #H	#H	?	?	–L	?	?	?
#H	H#; #H	H#–; H#/#H–	#H	H#	#H– / H#	?	?	H# / H#–
MH#	MH#; H#	MH#–	?	?	MH#–	?	?	?
H$	H#; #H	#H; #H– / H#	#H–	#H	?	#H–	?	?
L	L+H# / L+#H–	L; L+#H–	?	?	L+#H–	?	?	?
L#	?	L#–	?	?	L#–	?	?	?
LM+MH#	?	?	?	?	LM+MH#–	?	?	?
LM+#H	?	?	?	?	?	LM+MH#–	LM+MH#–	?
LM	LM–	?	?	?	?	LM–L	?	LM+H#
LH	?	?	?	?	?	?	?	?
H#	?	?	?	?	?	?	H#–	?

4.4 Compound given names and other "Tibetan compounds"

In Yongning, given names are of Tibetan origin. They consist of a combination of two disyllabic names. For instance, /**ʝi˧tɕi˧-ɖɯ˩mɑ˩**/ is composed of /**ʝi˧tɕi˧**/ and /**ɖɯ˩mɑ#˥**/; similarly, /**ɖɯ˩ɖʐɯ˧-tsʰɯ˩ɻ̍˩**/ combines /**ɖɯ˩ɖʐɯ˧**/ and /**tsʰɯ˧ɻ̍#˥**/. Table 4.27 provides a list of disyllabic names and attested combinations. The corresponding forms in Written Tibetan were proposed by Nathan Hill and Tsering Samdrup (p.c. 2016), and remain to be confirmed by eliciting the written forms from a monk in Yongning. Question marks indicate uncertain identifications. A dash '–' signals the absence of attestations in the set of recorded texts, indicating the need for additional data.

Compound given names constitute a specific area in the morphotonology of Yongning Na. Their tonal behaviour differs from that of coordinative compounds. The tones of the second name are lowered to L in all cases, even when they could theoretically be expressed without contravening any phonological rule. For instance, /**ɖɯ˩ɖʐɯ˧**/ and /**tsʰɯ˧ɻ̍#˥**/ could combine as † **ɖɯ˩ɖʐɯ˧-tsʰɯ˧ɻ̍#˥**, by successive association of the tones of the two components. This tone pattern is attested elsewhere in the language, as seen in four-syllable expressions such as /**nɑ˩bɑ˧-ʁɑ˧ɭɯ#˥**/ (Nabbahralee, the name of a mountain).

The lowering to L applies exclusively to given names and does not affect compound names consisting of a term of address followed by a two-syllable given name. For instance, a woman named *Gisso* /**ki˧zo#˥**/ may be addressed as /**ə˧mi˧-ki˧zo#˥**/ ("Mother Gisso, Aunt Gisso") by her nephews and nieces. This term of address is not realized as † **ə˧mi˧-ki˩zo˩**, as would be expected if it followed the pattern observed in compound given names.

The lowering of the final two syllables in given names, as in /**ɖɯ˩ɖʐɯ˧-tsʰɯ˩ɻ̍˩**/ and all other examples in Table 4.27, cannot be put down to the general tone rules of Yongning Na. Nor can it be explained as the result of lexicalization at an earlier stage of the language's history, since the lowering process operates as an exceptionless synchronic rule, applying to all compound given names. Speakers retain a clear awareness of the two components of these names, each of which has its own lexical tone. One of the two elements serves as the usual term of address, which may be the first or, less commonly, the second. For instance, the name that was bestowed on me by a priest of the Yongning monastery is *Yishi Diddeo* /**ʝi˧ʂɯ˥-ti˩ɖo˩**/; Mrs. Latami (F4) chose to address me by the second part, *Diddeo* /**ti˧ɖo˥**/, which she pronounced with its original lexical tone rather than pronouncing it as a fragment extracted from the compound (which would

Table 4.27: Yongning Na given names and identifications with Tibetan names proposed by Nathan Hill and Tsering Samdrup.

name	Tibetan	attested combinations
ɖɯ˩ɖʐɯ˧	Rdo rje	**ɖɯ˩ɖʐɯ˧-tsʰɯ˩ɻ̩˩**, **ɖɯ˩ɖʐɯ˧-ɬa˩-tsʰo˩**
ɖɯ˩ma#˥	Sgrol ma	**ɖɯ˩ma˧-ɬa˩tsʰo˩**, **ɖɯ˩ma˧-pv̩˩ʈʰɯ˩**
dʑɤ˩tsʰi#˥	Bde skyid?	**dʑɤ˩tsʰi˥-ɖɯ˩ma˩**, **dʑɤ˩tsʰi˥-pv̩˩ʈʰɯ˩**
gv̩˧ma˧	?	**gv̩˧ma˧-tsʰɯ˩ɻ̩˩**
ʝi˧ʂɯ˥	Ye shes	**ʝi˧ʂɯ˥-ti˩ɖo˩**
ʝi˧tɕi˧	Yid ches?	**ʝi˧tɕi˧-ɖɯ˩ma˩**, **ʝi˧tɕi˧-ɬa˩mv̩˩**
kɤ˧zo#˥	Skal bzang?	**kɤ˧zo˧-tsʰɯ˩ɻ̩˩**
ki˧zo#˥	Skal bzang?	**ki˧zo˧-ɖɯ˩ma˩**, **ki˧zo˧-ɬa˩mv̩˩**
la˩ma˩	Bla ma	–
ɬa˧mv̩˥$	Lha mo	–
ɬa˧tsʰo#˥	Lha mtsho	–
na˧dʑi#˥	Rnam rgyal	–
ɲi˩ma#˥	Nyi ma	–
no˩bv̩˧	Nor bu	**no˩bv̩˧-tsʰɯ˩ɻ̩˩**
no˧no˧	?	**no˧no˧-ɖɯ˩ma˩**
pæ˩pʰæ˧˥	Spen pa?	–
pi˧ma˧	Padma	**pi˧ma˧-ɬa˩mv̩˩**, **pi˧ma˧-ɬa˩tsʰo˩**
pʰi˧tsʰo#˥	Phun tshogs	**pʰi˧tsʰo˧-ɖɯ˩ɖʐɯ˩**
pv̩˩ʈʰɯ˧	Bu phrug? Bu khrid?	–
ɻ̩˩ʈʂʰe#˥	Rin chen	**ɻ̩˩ʈʂʰe˧-ɖɯ˩ma˩**, **ɻ̩˩ʈʂʰe˧-tsʰɯ˩ɻ̩˩**
ta˩dʑɤ#˥	Dar rgyes?	–
ʈæ˧ʂɯ˧	Bkra shis	**ʈæ˧ʂɯ˧-ɖɯ˩ma˩**, **ʈæ˧ʂɯ˧-ɬa˩mv̩˩**, **ʈæ˧ʂɯ˧-pæ˩pʰæ˩**, **ʈæ˧ʂɯ˧-ʈæ˩ʈv̩˩**, **ʈæ˧ʂɯ˧-tsʰi˩ti˩**
ʈæ˩ʈv̩#˥	Dgra ’dul?	–
tɕʰi˧ɖv̩#˥	Spyi ’dul?	–
ti˧ɖo˥	?	–
tsʰi˧ti#˥	?	–
tsʰɯ˧ɻ̩#˥	Tshe ring	**tsʰɯ˧ɻ̩˧-la˩mv̩˩**, **tsʰɯ˧ɻ̩˧-ɖɯ˩ma˩**, **tsʰɯ˧ɻ̩˧-ɬa˩mv̩˩**, **tsʰɯ˧ɻ̩˧-pʰi˩tsʰo˩**

have resulted in L tone).[6] This is one of many pieces of evidence demonstrating that compounds such as /**ʝi˧ʂɯ˥-ti˩ɖo˩**/ remain readily decomposable.

Should the tonal lowering of the latter part of compound given names be considered as another instance of a tone rule applying in a highly specific morphosyntactic context: a tone combination rule that holds in given names, and nowhere else – not even in other sets of proper names, such as place names?

Interestingly, a similar process of lowering is found in compounds involving the word for 'powder, flour': /**tsɑ˧bɤ˧**/ (as reported in §4.2.11.3). According to the synchronically productive rules, the combination of this word with /**lv̩˧mi˧**/ 'stone', /**qʰɑ˧dze˧**/ 'sweetcorn', and /**dze˧ɭɯ˧**/ 'wheat' should yield a simple M-tone output, e.g. † **lv̩˧mi˧-tsɑ˧bɤ˧** for 'fine sand'. But the observed forms, shown in Table 4.14, all carry an M.M.L.L tone pattern, corresponding to underlying –L (an L tone on the second part of the compound), just like the compound given names in Table 4.27. The label 'Tibetan compounds' is proposed for these compounds. From a synchronic point of view, this label may seem invalid, since none of the consultants has any knowledge of Tibetan (either spoken or written) and hence any clear awareness of Tibetan loanwords as such. The hypothesis here is that at some earlier point in time – one to four centuries ago? – speakers of Yongning Na who had some knowledge of Tibetan (a smattering would have been enough) applied a specific tonal treatment to compounds containing Tibetan words, by imitation of what they perceived as the Tibetan pattern. Tibetan was a prestige language in Yongning at least since the fourteenth century, and up until the mid-twentieth century (see Appendix B, §B.1), so it is not unlikely that

[6]F4's son (M18) argued that the shortened name should be *Yishi* /**ʝi˧ʂɯ˥**/, but F4 maintained her choice without further explanation other than personal preference. The decision partly reflects concerns about homonymy within the extended family: the pool of available names is small, making homonymy a genuine issue. Moreover, concerns about inappropriate use of proper names run deep in Asian cultures. Strict conventions govern the use of personal names, as evidenced, for instance, by the taboo against uttering (or writing) the names of important persons (emperors, but also one's elders), a practice known in China as *bìhuì* 避讳 (discussed in detail by Adamek 2012). In Yongning Na, such concerns are not limited to elders, as illustrated by the following anecdote. In the course of telling the story *BuriedAlive2*, consultant F4 realized that the name of a protagonist, /**ʈæ˧ʂɯ˧-no˩bv̩˩**/, coincided with that of a household member. She immediately substituted another name, /**no˩bv̩˧-tsʰɯ˩ɻ̩˩**/, to avoid any undesirable association between her family member and the character's shameful behaviour (*BuriedAlive2.114* {0004537#S114}). The consultant was so concerned by this coincidence that she would have preferred me to delete the recording entirely. But I was not overjoyed at the prospect of outright deletion. Fortunately, the consultant found comfort in the explicit disclaimer which is present within the narrative, to the effect that the character's name was /**no˩bv̩˧-tsʰɯ˩ɻ̩˩**/ and *not* /**ʈæ˧ʂɯ˧-no˩bv̩˩**/, and she agreed that we should complete the transcription and allow access to this linguistic document.

processes of imitation of prosodic patterns took place. This may have happened at the time when the Tibetan words were borrowed, or at a later stage, when speakers of Yongning Na who were aware of the status of these words as Tibetan in origin made efforts to copy (what they felt to be) Tibetan prosodic patterns. Cases of contact with a prestigious language can result in a range of unusual changes by imitation, including hypercorrections: see Meillet (1936: 99-103) on a possible case concerning Germanic and Romance, and Ferlus (2001) on cases in the Vietic subgroup of Austroasiatic; a highly speculative application to Tibetan has also been proposed (Ferlus 2003).

Supposing that imitation of Tibetan was at play in these compounds, it remains to be explained why lowering tone to L on the latter part of compounds had a Tibetan ring to Na ears, and at which point in history the adoption of this prosodic pattern took place. For want of a command of Tibetan, I am not in a position to investigate topics of historical contact with this language – one of many topics that remain for future investigations.

4.5 Compound nouns containing adjectives

The tonal categories of adjectives will be brought out in §6.1.3. As background to the present discussion, here is a preview of the results: adjectives fall into four tonal categories – L, M, H, and MH –, with L-tone items further subdivided into L_a and L_b.

Compounds containing adjectives are a difficult topic in Yongning Na, not least because they arise through lexicalization and cannot be elicited systematically. Unlike *noun plus noun* compounds, which can be explored through on-the-fly coinage by consultants to bring out a full set of synchronic rules, compounds containing adjectives do not lend themselves to such elicitation. As a preliminary to studying lexicalized compounds, it is useful to examine adjectival phrases.

4.5.1 A productive construction: N+Adj+relativizer

In Yongning Na, adjectives are associated with nouns through the construction N+Adj+relativizer/nominalizer /**-hĩ˥**/. For instance, //**ɖɯ˩$_a$**// ‘big’[7] yields //**ɖɯ˩-hĩ˩**// ‘(which is) big’, and //**ʂɯ˧˥**// ‘new’ yields //**ʂɯ˧-hĩ˥$**// ‘(which is) new’. These follow the noun as a separate tone group, as illustrated in (4)-(7).

[7]As explained on the first page of the introduction, morpheme-level transcriptions indicate lexical tone using tone symbols supplemented by subscript letters $_{a\ b\ c}$ to distinguish subcategories of lexical tones. The subcategories for verbs and adjectives are set out in Table 6.2 of §6.1.1.

(4) ɲi˧zo˧ | ɖɯ˩-hĩ˩˥
ɲi˧zo˧ ɖɯ˩a -hĩ˥
fish large NMLZ
'big fish'

(5) pʰi˧ | ʂɯ˧-hĩ˥
pʰi˧ ʂɯ˦˥ -hĩ˥
linen_cloth new NMLZ
'brand new linen cloth'

(6) tɕʰo˩˥ | ʂɯ˧-hĩ˥
tɕʰo˩˧ ʂɯ˦˥ -hĩ˥
ladle new NMLZ
'new ladle'

(7) qʰwɤ˦˥ | ʂɯ˧-hĩ˥
qʰwɤ˦˥ ʂɯ˦˥ -hĩ˥
bowl new NMLZ
'new bowl'

No tonal interaction takes place between the noun and adjective. If an intensifier is substituted for the relativizer, the construction becomes a statement, as in (8). The construction (9) likewise means 'the fish is big'.

(8) ɲi˧zo˧ | ɖɯ˧ | ʐwæ˩˥.
ɲi˧zo˧ ɖɯ˩a ʐwæ˩
fish large INTS
'The fish is very big.'

(9) ɲi˧zo˧ | ɖɯ˧.
ɲi˧zo˧ ɖɯ˩a
fish large
'The fish is big.'

In example (10), the numeral-plus-classifier phrase has the effect of nominalizing a construction that would otherwise mean 'the fish is/was really big', rather than 'a very big fish'.

(10) ni³³ zɔ³³ dɯ⁵⁵ ʐwæ¹³ dɯ³³ mi³¹
fish big INTS one CLS
鱼 大 很 一 量词
'a very big fish' (example 187 from Lidz 2010: 215; her glosses and tone

marking. CLS: classifier; [55]: High tone; [33]: Mid tone; [31]: Low-falling tone; [13]: Low-rising tone.)

The word order Noun+Adjective within a noun phrase always signals a lexicalized item. To use a textbook example from English, the phrasal, non-lexicalized expression 'black bird' contrasts with 'blackbird' (a specific species). A similar distinction exists in Yongning Na. For instance, //**ʐɯ˧nɑ˩**//, composed of //**ʐɯ˧**// 'liquor/spirits' and //**nɑ˩$_b$**// 'black', does not mean 'black liquor' in a descriptive sense (i.e. liquor of a black colour) but refers specifically to a type of strong, high-quality spirits. This is a disyllabic noun requiring its own dictionary entry; it is not a phrasal construction that simply attributes a quality to the noun's referent. Likewise, the nouns //**ə˧mi˧-ɖɯ˩**// and //**ə˧mi˧-tɕi˩**//, referring to the mother's older sisters and younger sisters respectively, are lexical units, despite their transparent internal structure: they consist of //**ə˧mi˧**// 'mother' plus the adjectives //**ɖɯ˩$_a$**// 'large' and //**tɕi˩$_a$**// 'small'.

Compounds formed with the adjectives 'big' and 'small' also exist for maternal uncles, //**ə˧v̩˦**//: //**ə˧v̩˧-tɕi˥**// for 'mother's younger brother', and //**ə˧v̩˧-ɖɯ˦**// for 'mother's elder brother'. However, constructions incorporating the relativizer //**-hĩ˥**// are more common. To specify whether one is referring to the elder or younger maternal uncle, one typically says /**ə˧v̩˦ | ɖɯ˩-hĩ˩˥**/ for the former and /**ə˧v̩˦ | tɕi˩-hĩ˩˥**/ for the latter, as shown in (11a–11b).

(11) a. ə˧v̩˦ | ɖɯ˩-hĩ˩˥
ə˧v̩˦ ɖɯ˩$_a$ -hĩ˥
uncle big NMLZ
'mother's elder brother, elder maternal uncle'

b. ə˧v̩˦ | tɕi˩-hĩ˩˥
ə˧v̩˦ tɕi˩$_a$ -hĩ˥
uncle small NMLZ
'mother's younger brother, younger maternal uncle' *(Caravans.75, 76, 78, 79, 177–179, 196, 259, Elders3.23, 31, 32)* {0004530#S75}

A study of successive occurrences within the same text confirms that the construction with the relativizer/nominalizer //**-hĩ˥**//, although seemingly cumbersome, is the standard construction to associate an adjective with a noun. Notably, this construction is not replaced by a synthetic, compact N+ADJ construction at later occurrences. For instance, example (13) appears only a few sentences after (12) within the same narrative, yet it retains the nominalizer construction.

(12) mv̩˩zo˩˥ | ɖɯ˩-hĩ˩˥, | zo˧mv̩˥ | ɖɯ˩-hĩ˩˥ | ɖɯ˧-ɭɯ˧ dʑo˩ tsɯ˩.
mv̩˩zo˩ ɖɯ˩$_a$ -hĩ˥ zo˧mv̩˥ ɖɯ˩$_a$ -hĩ˥ ɖɯ˧-ɭɯ˧ dʑo˩$_b$ tsɯ˧˥
young_lady large NMLZ child large NMLZ one-CLF EXIST REP
'It is said that [this couple] had a big girl, a big child (=a child who thought very seriously for her age).' *(Reward.59)* {0004446#S59}

(13) mv̩˩zo˩˥ | ɖɯ˩-hĩ˩-ki˥ (…)
mv̩˩zo˩ ɖɯ˩$_a$ -hĩ˥ -ki˧
young_lady large NMLZ DAT
'to his elder daughter, [the father said…]' *(Reward.65)* {0004446#S65}

4.5.2 Lexicalized compounds with a N+ADJ structure

The adjectives that appear in lexicalized combinations with nouns in the examples provided above are //**nɑ˩$_b$**// 'black, dark', //**ɖɯ˩$_a$**// 'large', and //**tɕi˩$_a$**// 'small'. Is it merely coincidental that 'black' is also the adjective used in the textbook example of English *black bird* and *blackbird*? The compound noun *blackbird* refers to *Turdus merula*, a species of thrush, whereas the noun-adjective combination *black bird* refers to any bird of black colour. The former, *blackbird*, carries stress on the first element of the compound (for short: "first-element stress"), whereas the latter, *black bird*, carries primary stress on *bird*. The meaning of *black bird* is compositional, being deducible from the meanings of its elements, whereas the meaning of *blackbird* is not. First-element stress is generally interpreted as a marker of degree of lexicalization. A study of English compounds observes that "the number of adjectives that work in the way that *black* does in our *exemple-type* seems to be very restricted" (Bauer 2004: 9). Examples are shown in Table 4.28. However, after an extensive corpus-based investigation exploring factors such as the frequency of the particular collocations and contrasting patterns of premodification, the author concludes that the gaps in English adjective-plus-noun compounds are likely to be accidental. In Na, as in English, the set of adjectives appearing in compound nouns is not closed.

In English, there is variation across speakers (and even for a single speaker) in stress assignment, both in intuition-based judgments and in actual speech. In Yongning Na, by contrast, noun-plus-adjective compounds are conspicuously different from attributive constructions, as the latter require a relativizer. This structural difference makes these compounds easy to identify. Their tonal analysis, however, is less straightforward. Examples are shown in tabular form, classified by the tone of the adjective: L$_a$ in Table 4.29, L$_b$ in Table 4.30, M in Table 4.31,

Table 4.28: Types of adjectives that appear in compound nouns in English (from Bauer 2004: 9).

type of adjectives	examples	example compounds
some colour adjectives	*black*, *blue*, *brown*, *green*, *grey*, *red*, *white*	*blackboard*, *blue-tit*, *brownstone*, *greenfly*, *greyhound*, *redfish*, *whiteboard*
grand in words of family relationships	*grand*	*grandfather*
a miscellaneous set of monosyllabic gradable adjectives	*broad*, *dry*, *free*, *hard*, *hot*, *mad*, *small*, *sweet* (among others)	*broadcloth*, *dry-cell*, *freepost*, *hardboard*, *hotbed*, *madman*, *small-arm*, *sweetcorn*
a small set of non-gradable monosyllabic adjectives	*blind*, *dumb*, *first*, *quick* (= ‘alive’), *square*, *whole*	*blindside*, *dumbcluck*, *first-day*, *quicksand*, *squaresail*, *wholestitch*
a very small number of disyllabic adjectives	*bitter*, *narrow*, *silly*	*bitter-cress*, *narrow-boat*, *sillyseason*

and H in Table 4.32. (No compounds with MH-tone adjectives have yet been observed.) All these items are lexicalized: for instance, the phrase /**ʈʂʰæ˧nɑ˥**/ refers to a legendary stag that only spirits are able to hunt, and is thus distinct from an attributive construction meaning ‘black-coloured deer’.

The compounds exhibit some tonal diversity. Among the five compounds associated with monosyllabic roots carrying H tone, three have L# tone, and two have H# tone.

Table 4.29: Examples of compounds containing the L_a-tone adjectives /**mo˩**$_a$/ 'old', /**ɖɯ˩**$_a$/ 'large', /**tɕi˩**$_a$/ 'small', and /**pʰv̩˩**$_a$/ 'white'. Note that no monosyllabic form is attested synchronically for 'stone' and 'ard'.

head noun			compound		
form	tone	meaning	form	tone	meaning
hĩ˥	H	person	**hĩ˧mo˥**	H#	elderly person
ʐwæ˥	H	horse	**ʐwæ˧mo˥**	H#	old horse
si˥	H	wood	**si˧mo˥**	H#	old wood, old tree
lv̩˧mi˧	?	stone	**lv̩˧mo˥**	H#	old stones
tsʰo˩	L	human being	**tsʰo˩mo˩**	L	old man
æ˩gv̩˩	?	ard[a]	**æ˩mo˥**	LH	used ard (out of use)
ʁo˥	H	head	**ʁo˧ɖɯ˦**	MH#	tadpole
zo˥	H	son	**zo˧ɖɯ˧**	M	eldest son
mv̩˩˥	LH	daughter	**mv̩˩ɖɯ˩**	L#	eldest daughter
ə˧mi˧	M	mother	**ə˧mi˧-ɖɯ˩**	L#	mother's elder sister
ə˧v̩˦	MH#	maternal uncle	**ə˧v̩˧-ɖɯ˦**	MH#	mother's elder brother
ə˧bo˥$	H$	paternal uncle	**ə˧bo˧-ɖɯ˦**	MH#	father's elder brother
mv̩˩˥	LH	daughter	**mv̩˩tɕi˥**	LH	youngest daughter
zo˥	H	son	**zo˧tɕi˥**	H#	youngest son
ə˧mi˧	M	mother	**ə˧mi˧-tɕi˩**	L#	mother's younger sister
ə˧v̩˦	MH#	maternal uncle	**ə˧v̩˧-tɕi˥**	H#	mother's younger brother
ə˧bo˥$	H$	paternal uncle	**ə˧bo˧-tɕi˥**	H#	father's younger brother
tɕɯ˧	M	cloud	**tɕɯ˧pʰv̩˩**	L#	white cloud

[a]The ard, also known as scratch plough, is the type of ploughing implement used in Yongning. Unlike the plough, the ard has a symmetrical share that traces a shallow furrow but does not invert the soil (Haudricourt & Jean-Brunhes Delamare 1955).

Table 4.30: Examples of compounds containing the $\mathrm{L_b}$-tone adjective /**nɑ˩**$_\mathrm{b}$/ ‘black’.

head noun			compound		
form	tone	meaning	form	tone	meaning
hṽ̩˥	H	hair	**hṽ̩˧nɑ˩**	L#	wild animal
ʂe˥	H	meat	**ʂe˧nɑ˩**	L#	lean meat
si˥	H	wood	**si˧nɑ˥**	H#	deep forest
kʰv̩˥	H	dog	**kʰv̩˧nɑ˥**	H#	dog *(in formal speech)*
tɕʰi˥	H	thorn	**tɕʰi˧nɑ˥**	H#	prinsepia
ʐɯ˧	M	liquor/spirits	**ʐɯ˧nɑ˩**	L#	high-quality spirits
njɤ˩˥	LH	eye	**njɤ˧nɑ˩**	L#	eyeball
ʈʂʰæ˧˥	MH	deer	**ʈʂʰæ˧nɑ˥**	H#	legendary black stag

Table 4.31: Examples of compounds containing the M-tone adjectives /**pv̩˧**/ ‘dry’, /**bæ˧**/ ‘stupid’, /**tʰi˧**/ ‘clever’, /**tsʰi˧**/ ‘hot’, and /**ʂæ˧**/ ‘long’.

head noun			compound		
form	tone	meaning	form	tone	meaning
hɑ˥	H	food	**hɑ˧pv̩˩**	L#	dry cooked rice (as opposed to gruel)
zo˥	H	son	**zo˧bæ˩**	L#	idiot
mv̩˩˥	LH	daughter	**mv̩˩tʰi˩**	L	clever woman
dʑɯ˩	L	water	**dʑɯ˩tsʰi˩**	L	hot water
zɯ˧	M	life	**zɯ˧ʂæ˧**	M	long life

Table 4.32: Examples of compounds containing the H-tone adjectives /**qʰæ˥**/ ‘cold’ and /**ɖæ˥**/ ‘short’.

head noun			compound		
form	tone	meaning	form	tone	meaning
dʑɯ˩	L	water	**dʑɯ˩qʰæ˩**	L	cold water
zɯ˧	M	life	**zɯ˧ɖæ#˥**	#H	short life

This difference in tone does not appear to correlate with the degree of semantic specialization of the compounds: whether they are transparent or non-transparent. For instance, the compounds ‘wild animal’ and ‘lean meat’ share the same tone pattern (L#) and derive from the same input tones (H and L_b), yet they differ in their degree of transparency. The noun /**hɣ̩̃˧nɑ˩**/, literally ‘black hair’, does not refer to a type of hair but means ‘wild animal’, denoting the *possessor* of dark hair by synecdoche.[8] (In the Indian linguistic tradition, /**hɣ̩̃˧nɑ˩**/, literally ‘black hair’, meaning ‘wild animal’, would be classified as a *bahuvrīhi* compound, denoting a referent by means of a salient characteristic.) By contrast, the compound /**ʂe˧nɑ˩**/, ‘lean meat’, is relatively transparent: it designates a specific type of meat – lean meat as opposed to fat meat, /**ʂe˧mɤ˦**/. Traditionally, pigs and cattle were only slaughtered once a year, so that fresh meat was the exception; preserved lean meat was the norm and had a dark brown colour.

The set of three H#-tone compounds from like input tones (H and L_b) is not homogeneous in this respect either. While the meaning of /**tɕʰi˧nɑ˥**/ (literally ‘black thorn’) is highly specific – it refers to prinsepia –, /**kʰv̩˧nɑ˥**/ (literally ‘dark dog, black dog’) remains fully general. The latter does not denote any specific subset within *Canis familiaris*; the only nuance distinguishing it from the more common form /**kʰv̩˩mi˩**/ is that it belongs to formal, elevated speech.

A parameter that appears more relevant than the overall degree of semantic specialization is the extent to which the adjective’s literal meaning remains present in the compound. The hypothesis here is that the two tonal subsets differ in that, in the first subset, the adjective retains its literal interpretation, whereas in the second, it is semantically bleached.

[8] As a zoological aside, the association of darker fur with wildness (i.e. lower predisposition to domestication) is supported by studies of animal domestication (Trut 1999). This phenomenon appears to stem from a link between stress levels and melanin production, which manifests in darker fur (Burchill & Thody 1986): wild animals experience higher stress levels than their domesticated counterparts. Wild yaks, for instance, have darker hair than domestic yaks (Leslie & Schaller 2009).

From this perspective, the coherence of the two subsets is as follows: in the expressions ‘wild animal’ and ‘lean meat’, the adjective /**nɑ˩$_{\mathbf{b}}$**/ retains its face value of ‘black, dark’. By contrast, as noted above, /**tɕʰi˧nɑ˥**/ refers to a distinct species of plant (prinsepia) rather than to dark thorns, and /**kʰv̩˧nɑ˥**/ designates ‘dog’ without any implication of hair colour.

The third and final item in the second set, /**si˧nɑ˥**/ (‘wood’+‘dark’, for ‘deep forest’), is the least clear-cut case. A dense forest is naturally darker than a sparse one, making it difficult to determine the extent to which the adjective retains its face value of ‘dark’. Hypothesizing that the L#-tone compounds are those in which the adjective is understood literally, whereas H#-tone compounds (with the same input tones: H plus L_b) make a semantically bleached use of the adjective, it follows that the etymological notion of darkness in /**si˧nɑ˥**/ ‘deep forest’ has become bleached. Discussions with Mrs. Latami support this hypothesis.

Analysis of this issue is complicated by the scarcity of adjectives that share the same tone as ‘black’ (//**nɑ˩$_{\mathbf{b}}$**//). The only other example identified so far is //**dʑɤ˩$_{\mathbf{b}}$**// ‘good’ (see §6.1.3).

A similar lack of one-to-one correspondence between input tones and output tones is also found for compounds containing the adjective /**ɖɯ˩$_{\mathbf{a}}$**/ ‘large, big’. The compound nouns /**zo˧ɖɯ#˥**/ ‘eldest son’ and /**ʁo˧ɖɯ˧˥**/ ‘tadpole’ (literally ‘big head’) have different tonal patterns (#H and MH#, respectively), although both are made up of a noun root that has H lexical tone and the adjective /**ɖɯ˩$_{\mathbf{a}}$**/ ‘big’.

To venture speculative hypotheses about the origin of these variegated tone patterns:

- *Variation in the adjective itself:* One possibility is that the adjective in tonally different compounds is not the same. In synchrony, there exists an adjective /**nɑ˥**/ (not homophonous with /**nɑ˩$_{\mathbf{b}}$**/ ‘black, dark’) meaning ‘important, serious (e.g. a wound)’. Some adjectival compounds may have incorporated this adjective or another, now-lost adjective pronounced [**nɑ**]. However, this hypothesis does not look promising: it is not suitable to the pair of words ‘eldest son’ and ‘tadpole’, where the adjective appears recognizably identical (‘big’). The etymology of ‘tadpole’ as ‘big head’ is super-clear: an apt designation for this small and distinctive creature (even though such terms are prone to playful deformation or replacement by expressive coinages).

- *Truncation from longer words:* Some of these compounds may not originate from the direct combination of a monosyllabic noun and adjective but instead derive from the truncation of longer nouns. For instance, the syllable /**si˧**/ in /**si˧nɑ˥**/ could result from the truncation of the disyllabic /**si˧ɕi˦**/ 'forest'.[9] Evidence for the possibility of such truncation comes from /**tʰo˧ɕi˦**/ 'pine forest', which combines a monosyllable for 'pine' with the *second* syllable of /**si˧ɕi˦**/ 'forest'. However, this explanation seems implausible for 'eldest son' and 'tadpole', which appear to derive straightforwardly from the monosyllabic nouns 'son' and 'head', respectively. If one nonetheless tries to push this hypothesis, one might speculate that 'tadpole' was originally built on the basis of a disyllabic noun, which could still survive in another dialect.

- *Reanalysis of the second morpheme:* The second part of the compound may not always function as an adjective. Instead, it could be analyzed as a suffix in some cases and as an adjective in others, analogous to the distinction proposed by Lidz (2010: 182) for the morpheme /**mɔ¹³**/, which functions as an adjective meaning 'old' but also as a suffix meaning 'dear' (indicating respect). Under this analysis, the H#-tone compounds could be morphologically parsed as /**tɕʰi˧-nɑ˥**/ 'prinsepia', /**kʰv̩˧-nɑ˥**/ 'dog', and /**si˧-nɑ˥**/ 'deep forest', contrasting with the L#-tone compounds /**hṽ̩˧nɑ˩**/ 'wild animal' and /**ʂe˧nɑ˩**/ 'lean meat'.

- *Different historical layers:* The tonal differences may reflect distinct historical strata, corresponding to different tonal rules that applied at different diachronic stages. For instance, 'deep forest', 'dog', and 'prinsepia' may predate 'wild animal' and 'lean meat', as the former compounds appear less semantically transparent. This scenario bears similarities to the previous one.

So far, consistency in the tone patterns of adjectival compounds appears to be limited to synchronically trivial patterns. For instance, it does not come as a surprise that Mid-tone /**ʐɯ˧**/ 'liquor, spirits' and Low-tone /**nɑ˩$_{\mathbf{b}}$**/ 'black, dark' yield a compound with an M+L surface tone pattern, /**ʐɯ˧nɑ˩**/: this looks like a straightforward case of concatenation. The same tone pattern is also found with another adjective that has the same lexical tone, /**dʑɤ˩$_{\mathbf{b}}$**/ 'good'. Disyllabic /**kɯ˧ dʑɤ˩**/ was readily extracted from the expression /**kɯ˧ dʑɤ˩ hɑ̃˩ dʑɤ˩**/ 'auspicious

[9]Creissels (1982: 50-52) documents synchronic variation in Mandinka, where frequently compounding nouns tend to get extracted from compounds with their compound-internal tone, which eventually replaces their original lexical tone. Such processes, operating on a word-by-word basis, contribute to the irregularity of tonal correspondences across dialects, complicating diachronic comparison and reconstruction.

day' (/**kɯ˧**/ means 'star', and /**hɑ̃˦**/ 'evening, night' is a term used to count days). The L# tone pattern of /**kɯ˧ dʐɤ˩**/, literally 'good star', results in the following two syllables receiving L tone, through Rule 5.

The tone pattern of /**kɯ˧ dʐɤ˩**/ 'good star' is the same as that of /**ʐɯ˧nɑ˩**/ 'high-quality liquor'. This shared tone could suggest that both compounds belong to the same historical layer, but more examples would be needed to investigate this possibility further.

To sum up: in view of the limited number of examples found to date and their heterogeneity, pooling them all into a summary table of tonal outcomes for N+Adj compounds does not appear particularly illuminating at this stage. Provisionally, the examples are simply listed in Table 4.29, arranged by adjective, by decreasing number of examples.

As a general observation, the tonal patterns in noun-plus-adjective compounds are not identical to those of noun-plus-verb combinations (described in Chapter 6). For instance, 'hot water', /**dʐɯ˩tsʰi˩**/, has L tone, derived from an input of L and M on the noun and adjective, respectively. In contrast, noun-plus-verb combinations (whether object plus verb or subject plus verb) with the same tonal input yield M as the output.

The diachronic trend is for monosyllabic nouns to become less frequent, being replaced by disyllables. As disyllabic nouns become lexicalized, the tonal correspondence between noun-plus-adjective combinations and the roots ceases to be transparent to speakers. This makes the tone patterns of disyllables more susceptible to replacement – whether through analogy or contact among dialects – than in cases where the root still exists as an independent monosyllabic form. In the current state of the language, 'stone' and 'ard' are attested only as disyllables, except in compounds with the adjective 'old', where they are found in monosyllabic form. This raises the question of whether it is justified to extract a monosyllable from the disyllabic compounds containing 'old'.

The root for 'stone' could be reconstructed with an H tone, as ***lv̩˥**, on the basis of its tonal behaviour in adjectival compounds: the compound /**lv̩˧mo˥**/ 'old stone' carries H# tone, like the compounds formed by adding 'old' to the H-tone monosyllables 'person', 'horse', and 'wood'. However, closer examination reveals that 'old stone' is not in common use in Yongning Na and has no clear independent meaning. It appears to be a recent coinage, attested only in a saying – example (14) – where it serves as a parallel to /**si˧mo˥**/ 'old wood'. Its tone pattern also follows that of 'old wood'. (Use of the symbol 'F' for 'Focalization' in the sentence-level transcription is explained in §9.2.2.)

(14) lv̩˧mo˥ F | dʑɯ˧ | le˧-qv̩˩; | si˧mo˥ F | le˧-dze˩ kv̩˩! | no˧ F | ə˧tse˧ | le˧-ʂɯ˧ mɤ˧-tʰɑ˦ | di˩!

lv̩˧mo˥ dʑɯ˩ le˧ qv̩˩$_a$ si˧mo˥ le˧- dze˩$_a$
old.stones water ACCOMP to_carry_away old_wood ACCOMP to_cut
-kv̩˦ no˩ ə˧tse˧ le˧- ʂɯ˧$_a$ mɤ˧- tʰɑ˦
ABILITIVE 2SG INTERROG.why ACCOMP to_die NEG PERMISSIVE
di˩$_a$
EXIST.SPATIAL

'Old stones are carried away by the stream; and old wood gets chopped down! And you, why won't you die?' *Context:* jeering an elderly person. Na tradition assigns human beings a lifespan of sixty years; people getting past seventy are considered to be well past their expected lifespan. (Field notes.)

Further analysis will require gathering more examples, sorting them into sets according to their tone patterns, identifying the historical layers that they belong to, and examining their process of formation. As a first step in this direction, the following paragraph discusses items that are currently on the verge of lexicalization.

4.5.3 N+ADJ combinations in the process of lexicalization

In between adjectival constructions such as /**ə˧v̩˦ | ɖɯ˩-hĩ˩˦**/ 'mother's elder brother' (example (11a) above) on the one hand, and lexical items such as /**ə˧v̩˧-ɖɯ˦**/ (also meaning 'mother's elder brother': see Table 4.29) on the other, there are cases that offer insights into the process of lexicalization. 'Elderly person' is /**hĩ˧mo˥**/, from /**hĩ˥**/ 'person' and /**mo˩$_a$**/ 'old'. In a set of twenty texts, this noun appears fifteen times, always in the plural, as /**hĩ˧mo˥=ɻæ˩**/; the fact that it is followed by a clitic shows that it is a full-fledged noun. But there is a higher number of occurrences (twenty-three) of /**hĩ˧ mo˥-hĩ˩**/, which also means 'elderly person', again from /**hĩ˥**/ 'person' and /**mo˩$_a$**/ 'old', but with addition of the relativizer /**-hĩ˥**/. This is not quite like the adjectival construction presented in §4.5.1: in that construction, the noun constitutes a tone group on its own, e.g. /**tɕʰo˩˧ | ʂɯ˧-hĩ#˥**/ 'new ladle' (example (6) above), whereas 'elderly person' is realized as /**hĩ˧ mo˥-hĩ˩**/, in one tone group. At a push, it would be possible to say /**hĩ˧ | mo˩-hĩ˩˦**/ 'a person that is old', but this is judged decidedly awkward in the contexts where /**hĩ˧ mo˥-hĩ˩**/ is attested. The interpretation that can be proposed is that /**hĩ˧mo˥**/ is on its way towards lexicalization – as evidenced by the tonal interaction between its two constituent morphemes – but the perception of its second syllable as an adjective remains strong enough that the relativizer is commonly added after it. The fact that the agent marker (‡ **hĩ˧mo˥ ɳɯ˩**) or the topic marker (‡ **hĩ˧mo˥ ʈʂʰɯ˩**) cannot be added shows that /**hĩ˧mo˥**/ is not yet fully lexicalized. It is compulsory to add an intervening plural or relativizer: /**hĩ˧mo˥=ɻæ˩ ɳɯ˩**/, /**hĩ˧mo˥=ɻæ˩ ʈʂʰɯ˩**/, /**hĩ˧ mo˥-hĩ˩ ɳɯ˩**/, and /**hĩ˧ mo˥-hĩ˩ ʈʂʰɯ˩**/.

For purposes of synchronic description, the notations adopted are /**hĩ˧mo˥ =ɻæ˩**/ and /**hĩ˧ mo˥-hĩ˩**/. In /**hĩ˧mo˥=ɻæ˩**/, the sequence /**hĩ˧mo˥**/ is transcribed as a lexical unit, with no hyphen or blank space between its two syllables. In /**hĩ˧ mo˥-hĩ˩**/, the first syllable is analyzed as a noun, and separated by a blank space from the adjective that follows. This notational distinction aims to draw attention to the versatility of disyllables made up of a noun and an adjective. To take another example of this phenomenon, /**zo˧bæ˩**/, from /**zo˥**/ 'son; man' and /**bæ˧**/ 'stupid; dumb (unable to speak)', has clearly nominal uses, meaning 'dumb man; stupid man'. More than twenty examples are found in the *Lake* narrative, one of whose main protagonists is a dumb person. The noun can be followed by the agent adposition: /**zo˧bæ˩ ɳɯ˩**/ (*Lake3.29* {0004348#S29}, *Lake4.24* {0004350#S24}); no intervening relativizer/nominalizer is needed for quantization purposes, as evidenced by examples (15)–(17).

(15) zo˧bæ˩ ɖɯ˩-ɣ̩˩
zo˧bæ˩ ɖɯ˧-ɣ̩˧
dumb_person one-CLF.individual
'a dumb person' *(Lake4.4)* {0004350#S4}

(16) zo˧bæ˩ ʈʂʰɯ˩-ɣ̩˩
zo˧bæ˩ ʈʂʰɯ˥ ɣ̩˧
dumb_person DEM.PROX CLF.individual
'this dumb person' *(Lake4.6)* {0004350#S6}

(17) zo˧bæ˩ tʰɣ̩˩-ɣ̩˩
zo˧bæ˩ tʰɣ̩˥ ɣ̩˧
dumb_person DEM.DIST CLF.individual
'that dumb person' *(Lake4.12-14)* {0004350#S12}

The expression /**zo˧bæ˩**/ can also directly precede a verb, as in /**zo˧bæ˩ | go˩bo˧ di˦**/ 'the dumb man drove cattle' (*Lake4.19* {0004350#S19}). In addition to such typically nominal uses, the word also functions predicatively (as an adjective): in a context of self-deprecation where someone calls himself stupid, /**zo˧bæ˩**/, another person may comfort him by saying (18).

(18) mɤ˧-zo˧bæ˩!
mɤ˧- zo˧bæ˩
NEG stupid
'[No, you are] not stupid!'

The antonym of /**zo˧bæ˩**/ in this adjectival sense is /**zo˧tʰi˧**/ 'clever; clever person', which has the same structure, from /**zo˥**/ 'son; man' and /**tʰi˧**/ 'able; capable; clever; sharp'. The two words have different tones (L# tone for /**zo˧bæ˩**/, vs. M tone for /**zo˧tʰi˧**/) despite their constituting morphemes having the same tones. This tonal difference alerts us to the possibility that the two words may belong to different historical layers. Syntactically, the two words also differ: it is not possible to say ‡ **mɤ˧-zo˧tʰi˧** 'not clever' on the analogy of /**mɤ˧-zo˧bæ˩**/ 'not stupid' (18). The first syllable of these two words is sufficiently grammaticalized for them to be used as adjectives for men and women alike, but in their nominal use they refer exclusively to men: /**zo˧bæ˩**/ means 'stupid man', and /**zo˧tʰi˧**/ 'clever man'. The corresponding words for women are /**mv̩˩-bæ˧mi˩**/ 'stupid woman' and /**mv̩˩tʰi˩**/ 'clever woman'.

4.5.4 Two lexicalized compounds with an ADJ+N structure

So far, only two lexicalized compounds with an *adjective plus noun* structure have been identified: /**pv̩˧lv̩˧**/ 'nonirrigated farmland; dry land', clearly related to /**pv̩˧**/ 'dry' and /**lv̩˧**/ 'field', and /**ɖɯ˩-hĩ˩**/ 'important people, great personages (including elders)',[10] from /**ɖɯ˩a**/ 'large' and /**hĩ˥**/ 'person'.

These words exhibiting the reverse order from the other compounds do not show phonological signs of antiquity, such as consonantal or vocalic differences relative to their etymological components. Comparative evidence from Naxi appears relevant here. Naxi has an exact equivalent to Na /**pv̩˧lv̩˧**/ 'nonirrigated farmland': the corresponding Naxi form is /**pv̩˩ɭɯ˧**/, with the same ADJ+N structure, the same meaning, and the same morphological transparency (in Naxi, 'dry' is /**pv̩˩**/, and 'field' is /**ɭɯ˧**/). Naxi also has another ADJ+N compound containing 'dry': /**pv̩˩dy˩**/ 'dry land (as opposed to water)' (Pinson 2012: 55).[11] The existence of a cognate in Naxi suggests that the compound may have some time depth, possibly even predating the split between Na and Naxi.

[10] Occurrences are found in *Sister.13* {0004342#S13}, *14*, *34*, *Sister3.31* {0004344#S31}, *36*, *38*, *41*, and *BuriedAlive3.5* {0004538#S5}.

[11] There is a typographical error in the phonetic transcription in Pinson (2012: 55), where /**pv̩˩dv̩˩**/ should read /**pv̩˩dy˩**/, as confirmed by the orthographic transcription.

As for ‘important person; great personage’, the expression found in contemporary Naxi is /**hi˧-ɖɯ˩**/, which follows the more common N+Adj ordering. Further dialectal evidence will be necessary to assess the historical depth of the Na compound and make progress in the analysis generally.

4.5.5 A lexicalized compound with a V+Adj structure

As a final observation on compounds, one single lexicalized compound with a V+Adj structure has been observed so far: /**tsʰo˧ɖɯ˩**/ ‘group dance’, a type of dance involving anywhere from ten to about a hundred participants. Its components are /**tsʰo˧**$_b$/ ‘to jump’ (in a nominalized reading) and /**ɖɯ˩**$_a$/ ‘large’. An alternative interpretation whereby /**ɖɯ˩**$_a$/ would have an adverbial reading (‘jumping a lot’) is implausible, as /**ɖɯ˩**$_a$/ does not have attested adverbial uses.

This is not a productive construction: for instance, it is not possible to create a compound meaning ‘banquet’ from the verb ‘to eat’ and the adjective ‘large’. Still, the existence of the word /**tsʰo˧ɖɯ˩**/ ‘group dance’ can be taken as evidence of occasional permeability between word classes. The distinction between nouns and verbs is more or less clear-cut from one language to another (Launey 1994, *passim*). In Naish languages, some items straddle the boundary between categories. For example, Na /**kɤ˧ʈʂɯ˩**/, like Naxi /**kɯ˧ʈʂɯ˩**/, has both verbal and nominal uses: ‘to speak’ and ‘speech; language’ in Naxi, and ‘to tell’ and ‘speech’ in Na.

4.6 Concluding note

The complexity of tone combination rules in various types of compounds in the Alawua dialect of Yongning Na, along with the existence of exceptions to these synchronic rules, arguably sheds light on the considerable diversity of patterns from one dialect to another – including differences among speakers from the same village, and even within a single idiolect. While some compounds happen to carry the same tone sequence as would the simple juxtaposition of their constituent elements, most display tone patterns that reflect their status as compounds. Among these, tonal evidence allows for distinguishing between determinative and coordinative compounds in a few cases, whereas in the others, speakers need to rely on semantic information and context.

5 Combinations of nouns with grammatical elements

This chapter brings together data on a wide range of constructions containing nouns, from morphological derivation – mostly nouns with gender suffixes – and reduplication, to combinations of nouns with particles in discourse.

5.1 Derivational affixes: Gender suffixes and kinship prefix

5.1.1 Introduction to the three main gender suffixes

The most common derivational affixes in Na are the gender suffixes /**-mi**/, /**-zo**/, and /**-pʰv̩**/,[1] which carry the meanings 'female, mother', 'son, young', and 'male', respectively. For instance, in /**ɬi˧mi˧**/ 'female roebuck', /**ɬi˩zo˩**/ (variant: /**ɬi˧zo#˥**/) 'young roebuck', and /**ɬi˩pʰv̩˩**/ (variant: /**ɬi˧pʰv̩#˥**/) 'male roebuck' (Lidz 2010: 177-179).[2]

The suffixes /**-mi**/ and /**-zo**/ also serve as augmentative and diminutive markers, respectively. The association of 'mother' with 'large' and 'son' with 'small' is common in languages of the area (see Mazaudon 2003 on Tamang and the cross-linguistic discussion by Matisoff 1992). For instance, /**kʰɤ˦**/ 'basket (carried on the back)' yields /**kʰɤ˧mi˥$**/ 'large basket' and /**kʰɤ˧zo#˥**/ 'small basket'.

In many suffixed forms, the original augmentative or diminutive meaning has faded. For instance, monosyllabic /**ljɤ˩˥**/ and suffixed /**ljɤ˩mi˥**/ both refer to the same object, namely the major (supporting) beams of a structure. There is no diminutive counterpart († **ljɤ˩-zo#˥**) to /**ljɤ˩mi˥**/. Similarly, for purlins, called /**ʐv̩˩ɭɯ˥**/, an augmentative form † **ʐv̩˩ɭɯ˥-mi˩** is not available. Instead, size distinctions are made using adjectives such as /**tɕi˩a**/ 'small' and /**ɖɯ˩a**/ 'large'. (On the association of adjectives with nouns in Yongning Na, see §4.5.1.)

[1] At this stage, no tone is indicated for these three suffixes, as establishing their tonal category is a key objective of §5.1.

[2] Liberty Lidz's notations are /**-mi33**/, /**-zɔ33**/, and /**-pʰu33**/. Consultant F4's word for 'grandmother's brother', /**ə˧pʰv̩˧**/, is glossed by L. Lidz as 'grandfather (father of mother or father)' in the Luoshui dialect and transcribed as /**ɑ33-pʰv̩33**/ (p. 766), but it also appears as /**ɑ33-pʰu33**/ in some instances, e.g. pp. 269, 503, and 763.

Etymologically, the second syllable of /**ɲi˧mi#˥**/ 'sun' and /**ɬi˧mi˧**/ 'moon' probably originates in the same morpheme /**mi**/. Since these two words have the same structure in Naxi (/**ɲi˧me˧**/, /**he˧me˧**/) and Laze (/**ɲie˧mie˧**/, /**ɬie˧mie˧**/), two closely related languages, their disyllabic status appears to have historical depth. The suffix /**zo**/ in /**ɲi˧zo#˥**/ 'fish', another disyllabic noun without a monosyllabic counterpart, also has a parallel in Laze (/**ze˧**/ 'son', /**ɲi˩ze˥**/ 'fish'), whereas Naxi retains a monosyllabic form, /**ɲi˧**/.

The discussion below examines gender suffixes in names of animals and peoples, as well as their augmentative and diminutive uses. Needless to say, words that contain a /**mi**/ syllable of different origin were excluded, such as the name of the Yongning monastery, /**ɖæ˩mi˧**/, a loanword from Tibetan *dgra med*.

The suffixes for *female*, *young* and *male* are related to the free morphemes /**mi˩˧**/, /**zo˥**/, and /**pʰv̩˧**/, which appear in contexts such as (1)-(3). Here, the evolution from the nouns to the derivational affixes is straightforward.

(1) ʈʂʰɯ˧ | mi˩ ɲi˥.
ʈʂʰɯ˥ mi˩˧ ɲi˩
DEM.PROX female COP
'This is a female.'

(2) ʈʂʰɯ˧ | zo˧ ɲi˥.
ʈʂʰɯ˥ zo˥ ɲi˩
DEM.PROX son/young/male COP
'This is a young/male.'

(3) ʈʂʰɯ˧ | pʰv̩˧ ɲi˩.
ʈʂʰɯ˥ pʰv̩˧ ɲi˩
DEM.PROX male COP
'This is a male.'

The free forms /**zo˥**/ 'young/male' and /**pʰv̩˧**/ 'male' have different tones (H and M, respectively), whereas the suffixes /**-zo**/ and /**-pʰv**/ always have the same tonal patterns, even sharing the same tonal variants, as in the example of 'roebuck'. However, neutralization of tonal oppositions on nouns in the process of grammaticalization as gender suffixes is not thoroughgoing: the tonal behaviour of /**-zo**/ and /**-pʰv**/ differs from that of the female suffix /**-mi**/.

From a tonal point of view, there is thus evidence that these three derivational elements have become distinct from the free nouns in which they originate. This is reminiscent of classifiers: the study of classifiers provided in Chapter 3 reveals that the tone system of classifiers is not identical to that of free nouns and that the tone of a classifier is not necessarily the closest equivalent of that of the noun from which it derives. To repeat an example from §3.1.2, there are two tonal correspondences among classifiers for H-tone nouns: one illustrated by 'beam', /**ɖʐo˥**/, which has /**ɖʐo˥$_a$**/ (category H_a) as its self-classifier, and the other illustrated by /**kɯ˥**/ 'star', which yields /**kɯ˧$_b$**/ (M_b tone category) as a self-classifier. Seen in this light, differences in tone between a noun as a full form and as a derivational suffix are not a particularly out-of-the-way finding in the context of Yongning Na morphotonology.

The difference in tone patterns between /**-mi**/ and the other two suffixes suffices to establish that suffixes are not toneless. But ascertaining the tones of these suffixes is not an easy matter. A simple test consists in combining them with an M-tone noun: the M tone has properties that make it suitable for use in tonal tests. M can be followed by any tone (unlike H, which can only be followed by L), and it does not spread (unlike L), so it would seem to offer the best possible context for the lexical tone of the following morpheme to manifest itself in. This works out well for verbs: a useful tonal test consists in observing a verb's tone after an M-tone morpheme such as the negation prefix, /**mɤ˧-**/. In this context, H-tone verbs surface with H tone, MH-tone verbs with MH tone, and so on (see §6.1.1). But the three gender suffixes all yield the same result after an M-tone monosyllable: for instance, /**lɑ˧**/ 'tiger' yields /**lɑ˧mi#˥**/ 'female tiger', /**lɑ˧zo#˥**/ 'baby tiger', and /**lɑ˧pʰv̩#˥**/ 'male tiger'. After disyllabic M-tone nouns, on the other hand, the results are different: thus, /**si˧gɯ˧**/ 'lion' yields /**si˧gɯ˧-mi˩**/ 'female lion', with L tone on the suffix, and /**si˧gɯ˧-zo#˥**/ 'baby lion', with a floating H tone. On this slender basis, the two sets of suffixes are provisionally transcribed as carrying lexical L and H tone, respectively. They are transcribed hereafter as //**-mi˩**//, //**-zo˥**//, and //**-pʰv̩˥**//. It must be cautioned that this tentative identification does by no means encapsulate all the information about the tonal behaviour of these two types of suffixes, which is set out in tabular form below.

5.1.2 The facts

Tables 5.1 to 5.9 present the data concerning the suffixes //**-mi˩**//, //**-zo˥**// and //**-pʰv̩˥**//. The examples are arranged by tone of the suffixed form. Tables 5.1 to 5.7 present disyllables, and Tables 5.8 and 5.9 present trisyllables. Nouns in which the suffixes are augmentative or diminutive and not gender suffixes are italicized (i.e. nouns that do not refer to animals or ethnic groups). As elsewhere, a slash separates variants. A dash '–' in a cell indicates that the form does not exist: for instance, it is not possible to use a word suffixed with /**-zo**/ for ‘piglet’ (the attested form is /**bæ˧bv̩˥**/). A double dagger ‡ preceding a noun indicates that it is a form that was proposed by the investigator and rejected by the consultant. For instance, “**ʐæ˩mi#˥** (‡ **ʐæ˩mi˩**)” for ‘female leopard’ indicates that the tonal variant ‡ **ʐæ˩mi˩** was proposed by the investigator on the analogy of the existence of an L-tone variant /**ʐ̩i˩mi˩**/ for ‘female ape’, whose root belongs in the same category as ‘leopard’, and that it was rejected by the consultant.

For two of the items in Table 5.1 that lack a monosyllabic counterpart – ‘bee’ and ‘thumb’, which are not synchronically attested as monosyllables –, the tone of the root can be inferred through internal reconstruction. Since there is a substantial number of examples (seven) of LM-tone disyllables corresponding to LM-tone monosyllables, and there is no other attested source for LM-tone disyllables, /**dze˩mi˧**/ ‘bee’ and /**lo˩mi˧**/ ‘thumb’ can be hypothesized to derive historically from LM-tone roots: ***dze˩˧** and ***lo˩˧**.

But for most of the items that lack a monosyllabic counterpart, internal reconstruction does not lead to a clear-cut conclusion, because several tone categories of roots feed into the same tone categories of disyllables. For instance, /**ɲi˧zo#˥**/ ‘fish’ may have originated in a monosyllabic root of any tone category except H, and M-tone words with the /**-mi˩**/ suffix, such as /**ɖɤ˧mi˧**/ ‘fox’, may be derived from any of the following four tone categories of monosyllables: LH, L, H or MH. The last two items in Table 5.7, ‘hwamei (a species of bird)’ and ‘(cigarette) lighter’, carry a tone that does not correspond to any of the attested correspondences.

The purpose of Tables 5.1 to 5.9 is to provide a bird’s eye view of the tonal correspondences. It does not present information about etymology and frequency of use: for instance, that /**ɻ̩̃˧mi˧**/ ‘treek trunk’ (in Table 5.5) etymologically means ‘big bone’; that /**po˧lo˧**/ is a more common form for ‘ram’ than the /**-pʰv̩˥**/ suffixed form /**jo˧pʰv̩#˥**/ ≈ /**jo˩pʰv̩˩**/ (literally ‘male sheep’) shown in Table 5.4; or that /**-zo˥**/ suffixed forms are more common than /**-pʰv̩˥**/ suffixed forms to refer to male mules and water buffalo (Table 5.7). Some such facts are adduced in the discussion below; they can be looked up in the corresponding entries in the dictionary (Michaud et al. 2024).

Table 5.1: Nouns with gender suffixes or augmentative/diminutive suffixes. Disyllabic words. LM-tone roots. Nouns in which the suffixes are augmentative or diminutive and not gender suffixes are italicized.

correspondences	root	suffixed forms		
		//**-mi˩**// ‘female’/AUG	//**-zo˥**// ‘child’/DIM	//**-p^{h}v̩˥**// ‘male’
first type	sow	**bo˩mi˧**	–	**bo˩p^{h}v̩˧**
	hen	**æ̃˩mi˧**	–	–
	thumb	**lo˩mi˧**	–	–
	bee	**dze˩mi˧**	–	–
second type	yak	**bv̩˧mi˧**	**bv̩˧zo#˥** / **bv̩˩zo˩**	**bv̩˧p^{h}v̩#˥** / **bv̩˩p^{h}v̩˩**
	sparrow	**ɖʐwæ˧mi˧**	**ɖʐwæ˧zo#˥** / **ɖʐwæ˩zo˩**	**ɖʐwæ˧p^{h}v̩#˥** / **ɖʐwæ˩p^{h}v̩˩**
	hawk	**kɤ˧mi˧**	**kɤ˧zo#˥** / **kɤ˩zo˩**	**kɤ˧p^{h}v̩#˥** / **kɤ˩p^{h}v̩˩**
	steamer	**bv̩˧mi˧**	**bv̩˩zo˩** / **bv̩˧zo#˥**	–
third type	weasel	**dv̩˩mi#˥** / **dv̩˩mi˧**	**dv̩˩zo#˥** / **dv̩˩zo˧**	**dv̩˩p^{h}v̩#˥** / **dv̩˩p^{h}v̩˧**
	goose	**ɑ˩mi#˥** / **ɑ˩mi˧**	**ɑ˩zo#˥** / **ɑ˩zo˧**	**ɑ˩p^{h}v̩#˥** / **ɑ˩p^{h}v̩˧**
	lizard	**dzo˩mi#˥** / **dzo˩mi˧**	**dzo˩zo#˥** / **dzo˩zo˧**	**dzo˩p^{h}v̩#˥** / **dzo˩p^{h}v̩˧**
	ladle	**tɕ^{h}o˩mi#˥** / **tɕ^{h}o˩mi˧**	**tɕ^{h}o˩zo#˥** / **tɕ^{h}o˩zo˧**	–
	Na (people)	**nɑ˩mi#˥** / **nɑ˩mi˧**	**nɑ˩zo#˥** / **nɑ˩zo˧**	–
fourth type	jackal	**p^{h}ɤ˩mi˩**	**p^{h}ɤ˧zo#˥** / **p^{h}ɤ˩zo˩**	**p^{h}ɤ˧p^{h}v̩#˥** / **p^{h}ɤ˩p^{h}v̩˩**
	road, path	**ʐɤ˩mi˩**	–	–

Table 5.2: Nouns with gender suffixes or augmentative/diminutive suffixes. Disyllabic words. LH-tone roots. Nouns in which the suffixes are augmentative or diminutive and not gender suffixes are italicized.

correspondences	root	suffixed forms		
		//**-mi˩**// ‘female’/AUG	//**-zo˥**// ‘child’/DIM	//**-pʰv̩˥**// ‘male’
first type	leopard	**z̧æ˩mi#˥** (‡ **z̧æ˩mi˩**)	**z̧æ˩zo#˥** (‡ **z̧æ˩zo˩**)	**z̧æ˩pʰv̩#˥** (‡ **z̧æ˩pʰv̩˩**)
	monkey	**ʑ̩i˩mi#˥** / **ʑ̩i˩mi˩**	**ʑ̩i˩zo#˥** / **ʑ̩i˩zo˩**	**ʑ̩i˩pʰv̩#˥** / **ʑ̩i˩pʰv̩˩**
	buffalo	**tʰɑ˩mi#˥** (‡ **tʰɑ˩mi˩**)	**tʰɑ˩zo#˥**	**tʰɑ˩pʰv̩#˥**
	muntjac	**tɕʰɯ˩mi#˥** / **tɕʰɯ˩mi˩**	**tɕʰɯ˩zo#˥** / **tɕʰɯ˩zo˩**	**tɕʰɯ˩pʰv̩#˥** / **tɕʰɯ˩pʰv̩˩**
	plane (tool)	**tʰi˩mi#˥** (‡ **tʰi˩mi˩**)	**tʰi˩zo#˥** (‡ **tʰi˩zo˩**)	–
second type (isolated example)	*slope*	**to˩mi˩**	**to˩zo˩**	–
third type	*plain*	**di˧mi˧**	–	–
	woman	**mv̩˧mi˧**	**mv̩˩zo˩**	–

Table 5.3: Nouns with gender suffixes or augmentative/diminutive suffixes. Disyllabic words. M-tone roots. Nouns in which the suffixes are augmentative or diminutive and not gender suffixes are italicized. Only one type of correspondence.

root	suffixed forms		
	//**-mi˩**// ‘female’/AUG	//**-zo˥**// ‘child’	//**-pʰv̩˥**// ‘male’
tiger	**lɑ˧mi#˥**	**lɑ˧zo#˥**	**lɑ˧pʰv̩#˥**
goral	**se˧mi#˥**	**se˧zo#˥**	**se˧pʰv̩#˥**
message	**qʰwæ˧mi#˥**	–	–

Table 5.4: Nouns with gender suffixes or augmentative/diminutive suffixes. Disyllabic words. L-tone roots. Nouns in which the suffixes are augmentative or diminutive and not gender suffixes are italicized. Only one type of correspondence.

root	suffixed forms		
	//**-mi˩**// 'female'/AUG	//**-zo˥**// 'child'/DIM	//**-pʰv̩˥**// 'male'
daughter	**mv̩˧mi˧**	–	–
sheep	**jo˧mi˧**	**jo˧zo#˥ / jo˩zo˩**	**jo˧pʰv̩#˥ / jo˩pʰv̩˩**
roebuck	**ɬi˧mi˧**	**ɬi˧zo#˥ / ɬi˩zo˩**	**ɬi˧pʰv̩#˥ / ɬi˩pʰv̩˩**
bottle	**kɤ˧mi˧**	**kɤ˩zo˩ (‡ kɤ˧zo#˥)**	–
river	**dʐɯ˧mi˧**	–	–

Table 5.5: Nouns with gender suffixes or augmentative/diminutive suffixes. Disyllabic words. H-tone roots. Nouns in which the suffixes are augmentative or diminutive and not gender suffixes are italicized.

correspon-dences	root	suffixed forms		
		//**-mi˩**// 'female'/AUG	//**-zo˥**// 'child'/DIM	//**-pʰv̩˥**// 'male'
first type	cow	**ʝi˩mi˩**	**ʝi˧zo#˥**	**ʝi˧pʰv̩#˥**
	horse	**ʐwæ˩mi˩**	**ʐwæ˧zo#˥**	**ʐwæ˧pʰv̩#˥**
	dog	**kʰv̩˩mi˩**	**kʰv̩˧zo#˥**	**kʰv̩˧pʰv̩#˥**
second type	Pumi (people)	**bɤ˧mi#˥**	**bɤ˧zo#˥**	–
	pheasant	**ho˧mi#˥**	**ho˧zo#˥**	**ho˧pʰv̩#˥**
	cooking pan	**v̩˧mi#˥**	**v̩˧zo#˥**	–
third type	*door*	**kʰi˧mi˧**	**kʰi˧zo#˥**	–
	canal	**qʰæ˧mi˧**	**qʰæ˧zo#˥**	–
	tree trunk	**ɻ̩̃˧mi˧**	–	–

Table 5.6: Nouns with gender suffixes or augmentative/diminutive suffixes. Disyllabic words. MH-tone roots. Nouns in which the suffixes are augmentative or diminutive and not gender suffixes are italicized.

correspondences	root	suffixed forms		
		//**-mi˩**// ‘female’/AUG	//**-zo˥**// ‘child’/DIM	//**-pʰv̩˥**// ‘male’
first type	cat	**hwɤ˧mi˥$**	**hwɤ˧zo#˥** / **hwɤ˧zo˥$**	**hwɤ˧pʰv̩#˥** / **hwɤ˧pʰv̩˥$**
	doe	**ʈʂʰæ˧mi˥$**	**ʈʂʰæ˧zo#˥** / **ʈʂʰæ˧zo˥$**	**ʈʂʰæ˧pʰv̩#˥** / **ʈʂʰæ˧pʰv̩˥$**
	goat	**tsʰɯ˧mi˥$**	**tsʰɯ˧zo#˥** / **tsʰɯ˧zo˥$**	**tsʰɯ˧pʰv̩#˥** / **tsʰɯ˧pʰv̩˥$**
	crane	**ʁv̩˧mi˥$**	**ʁv̩˧zo#˥** / **ʁv̩˧zo˥$**	**ʁv̩˧pʰv̩#˥** / **ʁv̩˧pʰv̩˥$**
	wasp	**tɕɯ˧mi˥$**	**tɕɯ˧zo#˥** / **tɕɯ˧zo˥$**	**tɕɯ˧pʰv̩#˥** / **tɕɯ˧pʰv̩˥$**
	basket	**kʰɤ˧mi˥$**	**kʰɤ˧zo˥$** (‡ **kʰɤ˧zo#˥**)	–
	needle	**ʁo˧mi˥$**	**ʁo˧zo#˥** (‡ **ʁo˧zo˥$**)	–
	scales	**tɕɯ˧mi˥$**	**tɕɯ˧zo˥$** (‡ **tɕɯ˧zo#˥**)	–
	bowl	**qʰwɤ˧mi˥$**	**qʰwɤ˧zo˥$** (‡ **qʰwɤ˧zo#˥**)	–
	stomach	**hu˧mi˥$**	–	–
second type (isolated example)	*building*	**ʐi˧mi˧**	–	–

Table 5.7: Nouns with gender suffixes or augmentative/diminutive suffixes. Disyllabic words without a corresponding monosyllable. Nouns in which the suffixes are not gender suffixes are italicized.

possible root tone	meaning	suffixed forms		
		//**-mi˩**// 'female'/AUG	//**-zo˥**// 'child'/DIM	//**-pʰv̩˥**// 'male'
M or H	water buffalo	**dʑi˧mi#˥**	**dʑi˧zo#˥** (‡ **dʑi˩zo˩**)	**dʑi˧pʰv̩#˥**
	granddaughter	**z̨v̩˧mi#˥**	–	–
	sun	**ɲi˧mi#˥**	–	–
L or LM	duck	**bæ˧mi˧**	**bæ˧zo#˥**	**bæ˧pʰv̩#˥**
	large vat	**dzo˧mi˧**	**dzo˧zo#˥**	–
	sword	**ʁæ˧mi˧**	**ʁæ˧zo#˥**	–
LH, L, H or MH	*tummy, belly*	**bi˧mi˧**	–	–
	fox	**ɖɤ˧mi˧**	–	–
	little sister	**go˧mi˧**	–	–
	moon	**ɬi˧mi˧**	–	–
	king, lord	**ʁo˧mi˧**	–	–
	louse	**ʂe˧mi˧**	–	–
	wife	**ʈʂʰv̩˧mi˧**	–	–
LM or LH	frog	**pɤ˩mi˩**	–	**pɤ˩pʰv̩˩**
LM, LH or H	*tongue*	**hi˩mi˩**	–	–
	large comb	**pv̩˩mi˩**	–	–
	axe	**bi˩mi˩**	–	–
	heart	**nv̩˩mi˩**	–	–
	niece	**ze˩mi˩**	–	–
	mule	**ɖɯ˩mi#˥**	**ɖɯ˩zo#˥**	**ɖɯ˩pʰv̩#˥**
LM or H	*bow*	**z̨v̩˩mi˩**	**z̨v̩˧zo#˥**	–
any tone except LH	fish	–	**ɲi˧zo#˥**	–
unclear	hwamei (bird)	**tɕɯ˩mi˥**	–	–
	cigarette lighter	**tse˧mi˥**	–	–

Table 5.8: Nouns with gender suffixes or augmentative/diminutive suffixes. Three-syllable words derived from M-tone disyllables.

correspondences	root	suffixed forms		
		//**-mi˩**// ‘female’	//**-zo˥**// ‘child’	//**-pʰv̩˥**// ‘male’
first type	rabbit	**tʰo˧li˧-mi˩**	**tʰo˧li˧-zo#˥**	**tʰo˧li˧-pʰv̩#˥**
	snake	**ʐv̩˧bæ˧-mi˩**	**ʐv̩˧bæ˧-zo#˥**	**ʐv̩˧bæ˧-pʰv̩#˥**
	lion	**si˧gɯ˧-mi˩**	**si˧gɯ˧-zo#˥**	**si˧gɯ˧-pʰv̩#˥**
	earthworm	**dʑɯ˧dv̩˧-mi˩**	**dʑɯ˧dv̩˧-zo#˥**	**dʑɯ˧dv̩˧-pʰv̩#˥**
second type	demon	**si˧bv̩˧-mi#˥**	**si˧bv̩˧-zo#˥**	–
	ghost	**tsʰo˧qʰwɤ˧-mi#˥**	**tsʰo˧qʰwɤ˧-zo#˥**	–
	Bai (people)	**ɬi˧bv̩˧-mi#˥**	**ɬi˧bv̩˧-zo#˥**	–

Table 5.9: Nouns with gender suffixes or augmentative/diminutive suffixes. Three-syllable words without a corresponding disyllable.

root tone	root	suffixed forms		
		//**-mi˩**// ‘female’	//**-zo˥**// ‘child’	//**-pʰv̩˥**// ‘male’
L	bird	**v̩˩dze˩-mi˩**	**v̩˩dze˩-zo˩**	**v̩˩dze˩-pʰv̩˩**
L#	bat	**dze˧bɤ˩-mi˩**	**dze˧bɤ˩-zo˩**	**dze˧bɤ˩-pʰv̩˩**
	owl	**mo˧jo˩-mi˩**	**mo˧jo˩mi˩-zo˩**	**mo˧jo˩mi˩-pʰv̩˩**
LM+MH#	wolf	**õ˩dv̩˧-mi˥**	**õ˩dv̩˧-zo#˥**	**õ˩dv̩˧-pʰv̩#˥**
H#	camel	**njɤ˧mv̩˥-mi˩**	**njɤ˧mv̩˥mi˩-zo˩**	**njɤ˧mv̩˥mi˩-pʰv̩˩**
unclear	cicada	**dʑɯ˧dze˧mi#˥**	–	–
	vulture	**se˩gwɤ˩-mi˧**	–	–

5.1.3 Discussion

The tonal correspondences between monosyllabic roots and disyllables are not one-to-one. The diversity of these correspondences suggests that suffixed disyllables have varying degrees of lexicalization and historical depth. It would thus be misleading to consider all suffixed forms as the result of a synchronic, currently productive morphological process.

Semantically, there is a continuum from disyllables with a clearly female meaning, such as ‘sow’, to those in which the semantic contribution of the suffix has become bleached, e.g. /**kʰv̩˩mi˩**/, which simply means ‘dog’ rather than specifically ‘she-dog’. After such semantic bleaching, suffixes must be added anew to specify gender. For example, the terms for ‘camel calf’ and ‘male camel’ are based on /**njɤ˧mv̩˥mi˩**/ ‘camel’ and come out as /**njɤ˧mv̩˥mi˩-zo˩**/ and /**njɤ˧mv̩˥mi˩-pʰv̩˩**/, respectively. While these forms are readily understandable, the main consultant finds them awkward. Does this imply that /**-mi˩**/ in /**njɤ˧mv̩˥mi˩**/ ‘camel’ is still perceived as carrying a female meaning? Not necessarily: the slight weirdness of /**njɤ˧mv̩˥mi˩-zo˩**/ ‘camel calf’ and /**njɤ˧mv̩˥mi˩-pʰv̩˩**/ ‘male camel’ seems to stem from the sequences /…**mi.zo**/ and /…**mi.pʰv**/, where the suffix, as it were, re-activates the gender denotation of /**-mi˩**/.

An examination of forms that allow two tonal realizations, such as /**hwɤ˧zo#˥**/ and /**hwɤ˧zo˥$**/ for ‘male kitten’, reveals that tonal variants are item-specific (lexicalized). Among the nine words suffixed with /**-zo˥**/ or /**-pʰv̩˥**/ corresponding to an MH-tone root, only four allow both #H and H$ variants. Interestingly, of the four object names in this set – ‘basket’, ‘needle’, ‘scales’, and ‘bowl’ –, none allows tonal variation: each belongs unambiguously to one category (#H for ‘needle’ and H$ for the other three). By contrast, four of the five animal names allow both variants. The distinction between gender and size suffixes is not rigid: some object names also permit tonal variation. For instance, ‘ladle’ follows the pattern of exceptionless tonal duality for LM-tone roots: /**tɕʰo˩mi#˥**/ ≈ /**tɕʰo˩mi˧**/ and /**tɕʰo˩zo#˥**/ ≈ /**tɕʰo˩zo˧**/. Nevertheless, there appears to be a statistical tendency for animal names to retain greater tonal flexibility, which may reflect speakers’ stronger perception of their internal structure.

In some cases, it is possible to identify specific factors that have influenced the current tonal outcomes. For instance, ‘little bottle’ /**kɤ˩zo˩**/ lacks the expected #H-tone variant ‡ **kɤ˧zo#˥**. To the consultant, the latter form immediately evoked the given name /**kɤ˧zo#˥**/. If ‘little bottle’ once had both L and #H tonal variants, pressure to avoid homophony may have contributed to the exclusive retention of /**kɤ˩zo˩**/.

The discussion below follows the order of root noun tone categories as presented in Tables 5.1 to 5.9. For each tone category, suffixed forms with //**-mi˩**// are examined first, followed by those with //**-zo˥**// and //**-pʰv̩˥**//.

5.1.3.1 LM-tone roots

Tables 5.1 to 5.7 reveal no fewer than four tonal correspondences between LM-tone monosyllables and their suffixed forms: LM, as in /**bo˩mi˧**/ 'sow' and /**æ˩mi˧**/ 'hen'; M, as in /**bv̩˧mi˧**/ 'female yak', /**ɖʐwæ˧mi˧**/ 'female sparrow', and /**kɤ˧mi˧**/ 'female falcon'; LM+#H, as in /**dv̩˩mi#˥**/ 'female weasel', /**ɑ˩mi#˥**/ 'goose', /**dzo˧ mi#˥**/ 'female lizard', and /**nɑ˩mi#˥**/ 'Na woman'; and finally L, as in /**pʰɤ˩mi˩**/ 'female jackal'.

Three LM-tone nouns outside the semantic domain of animal names also take the /**-mi˩**/ suffix, exhibiting three of the four tonal patterns attested above. One has M tone: /**bv̩˧mi˧**/ 'big food steamer' (from /**bv̩˩˧**/ 'food steamer'). Another has LM+#H tone: /**tɕʰo˩mi#˥**/ 'big ladle' (from /**tɕʰo˩˧**/ 'ladle'). The third has L tone: /**ʐɤ˩mi˩**/ 'road, path' – which no longer specifically means 'large road': the monosyllable /**ʐɤ˩˧**/, likewise meaning 'road, path', is falling into disuse.

Given the diversity of these patterns, it is difficult to establish the relative chronology of the tone rules that produced the four types (LM, M, LM+#H, and L). Nonetheless, a few hints may be detected. While LM-tone monosyllables correspond to no fewer than four different tonal categories in suffixed forms, LM-tone suffixed forms correspond exclusively to LM-tone monosyllables. In other words, LM-tone monosyllables are the sole source of LM-tone disyllables. The words at issue, namely 'sow', 'hen', 'bee', and 'thumb', belong to basic vocabulary, suggesting that the LM::LM correspondence between monosyllable and disyllable reflects an older pattern.

The tone patterns for the //**-zo˥**// and //**-pʰv̩˥**// suffixes stand in regular tonal correspondence to those of the suffix //**-mi˩**//. In the second type of correspondences in Tables 5.1 to 5.7, forms suffixed with //**-zo˥**// or //**-pʰv̩˥**// always have #H (with L as a variant) when the form suffixed with //**-mi˩**// has M. In the third type, words with any of these three suffixes have LM+#H, with LM as a variant.

On the other hand, these patterns differ widely from those observed in other syntactic structures, such as compound nouns and noun-verb or noun-adjective combinations. This confirms that different syntactic structures are associated with distinct tone rules – further complicated by numerous lexicalized oddities.

5.1.3.2 LH-tone roots

Monosyllables with LH tone correspond to disyllables carrying LM+#H. However, two items also have an L-tone variant, namely /**ʑi˩mi#˥**/ ≈ /**ʑi˩mi˩**/ 'female monkey' and /**tɕʰɯ˩mi#˥**/ ≈ /**tɕʰɯ˩mi˩**/ 'female muntjac'. In contrast, /**ʐæ˩mi#˥**/ 'female leopard' does not allow this variant: the form ‡ **ʐæ˩mi˩** is not acceptable.

Outside the semantic field of animal names, /**tʰi˩mi#˥**/ 'large plane' (from /**tʰi˩˥**/ 'plane [carpentry tool]') cannot be realized as ‡ **tʰi˩mi˩**. Conversely, the word for 'large slope' (from /**to˩˥**/ 'slope') is /**to˩mi˩**/, and ‡ **to˩mi#˥** is not acceptable.

At present, it remains uncertain whether two different tone rules applied at different times – in which case the existence of two variants would be a development due to analogy or dialect contact – or whether both variants used to coexist for all items, with some later losing one of the two.

Two additional patterns are attested, each represented by a single example. These words may be relatively old: M tone in /**di˧mi˧**/ 'plain' (compare /**di˩˥**/ 'earth, land'), and LH tone in /**ljɤ˩mi˥**/ 'major beam'.

5.1.3.3 M-tone roots

Monosyllables with M tone yield disyllables with #H tone: /**lɑ˧mi#˥**/ 'female tiger' and /**se˧mi#˥**/ 'female goral'. This pattern is also found outside the semantic domain of animal names, as in /**qʰwæ˧mi#˥**/ 'message; letter'. That this rule is currently productive was confirmed by adding the augmentative suffix to the M-tone noun /**qwæ˧**/ 'bed mat', yielding /**qwæ˧mi#˥**/ 'large bed mat'.

The situation is more complex for trisyllables formed with the suffix /**-mi˩**/. Two patterns are attested: –L, as in /**si˧gɯ˧-mi˩**/ 'lionness', /**ʐv̩˧bæ˧-mi˩**/ 'female snake', and /**tʰo˧li˧-mi˩**/ 'hare'; and #H, as in /**ɬi˧bv̩˧-mi#˥**/ 'woman of the Bai ethnic group'. The latter pattern matches the correspondence found for monosyllables.

To determine which of the two is currently productive, the suffix was added to a word that does not normally take it: 'earthworm' /**dʑɯ˧dv̩˧**/, as earthworms are hermaphroditic. The elicitation scenario was as follows: a child wonders whether there are such things as male and female earthworms and asks, 'Are there such things as female earthworms? / Do female earthworms exist?' The speaker formulated the question without hesitation, using the suffixed form /**dʑɯ˧dv̩˧-mi˩**/ 'female earthworm' with –L tone (4).

(4) dʑɯ˧dv̩˧-mi˩ | ə˩-dʑo˩˥?
dʑɯ˧dv̩˧-mi˩ ə˩- dʑo˩$_b$
female_earthworm INTERROG EXIST
'Are there such things as female earthworms?/Do female earthworms exist?' (Field notes.)

5.1.3.4 L-tone roots

Monosyllabic words with L tone yield disyllables with M tone: /**jo˧mi˧**/ 'ewe', /**ɬi˧mi˧**/ 'female roebuck', and /**mv̩˧mi˧**/ 'woman'. This last example is attested in a proverb shown in (5).

(5) mv̩˧mi˧ ʈʂʰwɤ˩ mɤ˩-ɖɯ˩, | kʰv̩˧nɑ˥ ʐo˩ mɤ˩-ɖɯ˩.
mv̩˧mi˧ ʈʂʰwɤ˥ mɤ˧- ɖɯ˧$_b$ kʰv̩˧nɑ˥ ʐo˩ mɤ˧- ɖɯ˧$_b$
woman dinner (evening meal) NEG to_get black_dog lunch NEG to_get
'No dinner for the [married] woman, no lunch for the black dog.'
Explanation: 'No dinner for the married woman': if a married woman visits her natal home during the day, she cannot stay for the evening meal, as she has obligations in the household she has married into. 'No lunch for the black dog': here, 'black dog' refers to dogs in general. Dogs receive only two meals a day, one in the morning and one in the evening. (*Sister.130, 131, 139, 158, 171* {0004341#S130} and *Sister3.3, 113, 117* {0004344#S3})

The word /**mv̩˧mi˧**/ 'woman' is no longer intelligible to younger speakers, such as M23. It is likely that the correspondence between L-tone roots and M-tone disyllabic forms reflects an older tone rule. The same pattern (M tone) is found in /**kɤ˧mi˧**/ 'large bottle' and /**dʑɯ˧mi˧**/ 'large river'.

The tone patterns of the suffixes /**-zo**˥/ and /**-pʰv̩**˥/ are identical for roots with L and LM tones.

5.1.3.5 H-tone roots

For words derived from H-tone monosyllables, three tonal patterns are observed: #H tone, as in /**ho˧mi#**˥/ 'female pheasant' and /**bɤ˧mi#**˥/ 'Pumi woman'; L tone, as in /**ʐwæ˩mi˩**/ 'mare', /**ʝi˩mi˩**/ 'cow', and /**kʰv̩˩mi˩**/ 'dog' (discussed further below); and M tone, as in /**kʰi˧mi˧**/ 'main door', /**qʰæ˧mi˧**/ 'canal; large ditch', and /**ɻ̩̃˧mi˧**/ 'tree trunk' (etymologically 'large bone').

The first pattern makes good synchronic sense: the tones of the monosyllable and disyllable correspond neatly with each other, and the semantic relationship is also clear – the base noun (‘pheasant’, ‘Pumi’) is gender-neutral, while the suffixed form specifies gender (‘female pheasant’, ‘Pumi woman’). The second and third patterns are less transparent phonologically. Concordantly, the semantic relationship is less clear in some cases.

The words for ‘mare’, ‘cow’ and ‘dog’ illustrate three stages in the gradual evolution of the suffix’s meaning. The monosyllable for ‘horse’, /**ʐwæ˥**/, is in common use, and the derived word for ‘mare’, /**ʐwæ˩mi˩**/, simply specifies gender. By contrast, the monosyllable /**ʝi˥**/ for ‘cow’ is not in frequent use; there are over ten different disyllables pronounced /**ʝi**/, six of them with H tone. Here, the suffix /**-mi˩**/ serves a disambiguating function. While it retains its meaning of ‘female’, it can be said to function as a general animal suffix just as much as a female suffix. The third example, /**kʰv̩˩mi˩**/ ‘dog’, represents a further stage in this evolution: it applies to dogs regardless of sex, and the monosyllable /**kʰv̩˥**/ for ‘dog’ is seldom used (but attested, witness *Dog.1, 3, 45* {0004442} and *Dog2.68, 74-77, 79* {0004660#S68}). These observations suggest that #H is the tone of more recently derived words, while L tone reflects an earlier stage and remains lexically preserved in some older words.

This conjecture is supported by examples from outside the semantic field of animal names. The disyllable /**sɑ˩mi˩**/ *‘Cannabis indica’* (the psychotropic plant) corresponds to the monosyllabic /**sɑ˥**/ *‘Cannabis sativa’* (used for fibre production) and has L tone. Here, the suffix conveys an augmentative meaning, referring to the larger size of the leaves. (As an aside: a comparable derivation exists in Mandarin, where ‘cannabis’ *dàmá* 大麻 is formed from ‘hemp’ *má* 麻 by addition of the augmentative *dà* 大 ‘large’.) Again, this suggests that L was the earlier tone. By contrast, more recent and less clearly lexicalized disyllables, such as /**v̩˧mi#˥**/ ‘large pot’ (cf. /**v̩˥**/ ‘pot’), carry #H tone, as does /**sɯ˧ɻ̩˧mi#˥**/ ‘backbone, spine’ (compare /**sɯ˧ɻ̩#˥**/ ‘tree trunk’).

The tone patterns of the suffixes /**-zo˥**/ and /**-pʰv̩˥**/ are identical for roots with H and M tones.

5.1.3.6 MH-tone roots

MH-tone monosyllables all correspond to disyllables with H$ tone. This pattern is attested in the following animal names: /**hwɤ˧mi˥$**/ ‘she-cat’, /**ʈʂʰæ˧mi˥$**/ ‘hind’, /**tsʰɯ˧mi˥$**/ ‘nanny goat’, /**ʁv̩˧mi˥$**/ ‘female crane’, and /**tɕɯ˧mi˥$**/ ‘wasp’. Beyond the semantic domain of animal names, the same correspondence is found for /**kʰɤ˧mi˥$**/ ‘large basket’, /**ʁo˧mi˥$**/ ‘big needle’, /**tɕɯ˧mi˥$**/ ‘large scales (for weighing things)’, and /**hu˧mi˥$**/ ‘stomach, bowels’.

Words suffixed with /**-zo˥**/ or /**-pʰv̩˥**/ can carry either #H, a pattern which is widely attested with these suffixes, or H$, the tone found in items suffixed with /**-mi˩**/.

5.1.3.7 Some observations on other lexical tones

As predicted by Rule 5 ("All syllables following an H.L or M.L sequence receive L tone"; see §7.1.1), L# tone spreads onto the suffix, yielding M.L.L: /**dze˧bɤ˩-mi˩**/ 'female bat', /**dze˧bɤ˩-zo˩**/ 'little bat, pup', and /**dze˧bɤ˩-pʰv̩˩**/ 'male bat'. On this basis, a disyllabic form ***mo˧jo˩** can be confidently extracted from /**mo˧jo˩-mi˩**/ 'owl'. (Recall that the asterisk indicates a reconstructed form, not an ungrammatical one.)

The tone patterns of 'cicada', 'vulture', 'hwamei (bird)', and '(cigarette) lighter' have no attested parallel elsewhere, making it impossible, in the present stage of our knowledge, to infer the roots' tones.

5.1.3.8 Concluding observations

The patterns documented in Tables 5.1 to 5.9 are summarized in Tables 5.10 and 5.11. It must be emphasized that the distinction between currently productive patterns and older patterns for the /**-mi˩**/ suffix remains speculative. As elsewhere, a slash (/) separates variants.

Table 5.10: Tonal correspondences between monosyllabic base forms and disyllables containing the suffixes /**-mi˩**/, /**-zo˥**/ and /**-pʰv̩˥**/, with tentative indications on whether the tone pattern is currently productive.

	/**-mi˩**/		/**-zo˥**/, /**-pʰv̩˥**/
	older?	productive?	*no distinctions in productiveness*
LM	LM; L	M; LM+#H / LM	LM+#H / LM; #H / L
LH	M; LH; L	LM+#H / L	LM+#H / L
M		#H	#H
L		M	#H / L
H	L; M	#H	#H
MH		H$	#H / H$

Roots with the same lexical tones correspond to suffixed forms with diverse tones, with as many as four types of correspondences for LM tone. The total

Table 5.11: Tonal correspondences between disyllabic base forms and trisyllabic nouns containing the suffixes /**-mi˩**/, /**-zo˥**/ and /**-pʰv̩˥**/, with tentative indications on whether the tone pattern is currently productive.

	/**-mi˩**/		/**-zo˥**/, /**-pʰv̩˥**/
	older?	productive?	*no distinctions in productiveness*
M	#H	–L (L#)	#H
H		#H	
L		L	L
L#		L#–	L#–
LM+MH#		LM+H#	LM+#H
H#		H#–	–

number of noun subsets in Tables 5.1 to 5.9, excluding disyllabic roots, is 14. Given that /**-mi˩**/ and /**-zo˥**/, /**-pʰv̩˥**/ fall into different tonal categories, this results in 2×14=28 potentially distinct tonal types of suffixed nouns. Considering that many types have variants, the actual number of tonal patterns attested on suffixed nouns could be considerable.

Yet the observed set of tones remains relatively constrained, limited to six primary patterns: {M, #H, H$, L, LM, LM+#H}, apart from two outliers – tones attested in only one example each. This suggests that some tone categories encompass a large number of suffixed or compound nouns, whereas others are not fed by any currently productive morphological processes.

From a phonostylistic perspective, such asymmetries contribute to giving different lexical tone categories a specific feel of their own. Depending on their frequency among recent coinages (as well as among loanwords), certain tonal patterns might carry more or less salient *overtones* (if the pun may be allowed) of novelty vs. classicality. A prevalent pattern might be perceived as straightforward, mainstream, and workaday, and conversely, a rarely-occurring one might feel quaint and archaic-sounding. The phonostylistic associations of tonal categories, along with those of the various morphotonological rules, constitute an interesting avenue for future research, as one of the various stylistic topics that could be investigated through the study of the corpus of Yongning Na narratives.

For forms suffixed with /**-zo˥**/ and /**-pʰv̩˥**/, only five patterns are attested: {#H, H$, L, LM, LM+#H}. One additional pattern, M tone, is attested for /**-mi˩**/. The relatively lower complexity of tone patterns for /**-zo˥**/ and /**-pʰv̩˥**/ may have something to do with their more restricted lexical distribution: since nouns suffixed with ‘male’ and ‘child’ markers are fewer in number, they may have undergone greater tonal simplification through the analogical extension of productive patterns.

5.1.4 Other suffixes for ‘male’

In addition to the currently productive suffix /**-pʰv̩˥**/ for ‘male’, there exist other, non-productive suffixes. These are mentioned here for completeness, though the small number of examples greatly limits possibilities for analysis of their tone patterns.

5.1.4.1 The suffix /-ʂwæ˧/

The free form /**ʂwæ˧**/ currently means ‘castrated/neutered male’.

(6) ʈʂʰɯ˧ | ʂwæ˧ ɲi˩.
ʈʂʰɯ˥ ʂwæ˧ ɲi˩
DEM.PROX castrated_male COP
‘This is a castrated male.’

The morpheme /**ʂwæ˧**/ may, however, have previously meant ‘male’, as suggested by the noun /**æ̃˧ʂwæ˥**/ meaning ‘rooster, cock’ (compare /**æ̃˩˥**/ ‘chicken’). Notably, this noun has no counterpart with the suffix /**-pʰv̩˥**/; a form such as ‡ **æ̃˩pʰv̩#˥** is not available. Roselle Dobbs (p.c. 2016) reports that in Lataddi 喇塔地, ‘grandfather’ and ‘rooster’ sound comically similar. It is conceivable that an earlier form † **æ̃˩pʰv̩#˥** ‘rooster’ fell into disuse in the Alawua dialect (studied here) due to phonetic similarity with ‘grandfather’ (even in the absence of full homophony).

The suffix /**-ʂwæ**/ also appears in three other words, in which it carries the meaning ‘castrated male’ (see Table 5.12). Interestingly, its tone pattern differs between ‘cock’ and ‘castrated yak’, despite both having roots with the same lexical tone (LM). This tonal divergence may be related to the suffix’s different meanings in these words: ‘male’ in one case and ‘castrated male’ in the other. It is reasonable to assume that /**æ̃˧ʂwæ˥**/ ‘cock’ has greater time depth.

Concerning the tone of the suffix, two of the words in which it appears (‘castrated yak’ and ‘castrated male sheep’) display tone patterns that are among possible variants for the suffixes //**-zo˥**// ‘baby, male’ and //**-pʰv̩˥**// ‘male’, which are tentatively analyzed as bearing H tone. However, the third word, /**tsʰɯ˧ʂwæ˥**/ ‘wether, castrated male goat’, follows a different tonal pattern from words suffixed in //**-zo˥**// ‘baby, male’ or //**-pʰv̩˥**// ‘male’: /**tsʰɯ˧zo#˥**/ ≈ /**tsʰɯ˧zo˥$**/ and

/**tsʰɯ˧pʰv̩#˥**/ ≈ /**tsʰɯ˧pʰv̩˥$**/ (see Table 5.6). This constitutes evidence that the suffix /**-ʂwæ**/ does not share the same tonal behaviour as the //**-zo˥**// and //**-pʰv̩˥**// suffixes. Here, the suffix /**-ʂwæ**/ is provisionally labelled as carrying M tone and transcribed as //**-ʂwæ˧**//. However, it should be emphasized that this label primarily serves to distinguish its tone from that of the 'baby, male' and 'male' suffixes, which are provisionally transcribed as //**-zo˥**// and //**-pʰv̩˥**//, respectively.

Table 5.12: Names of animals with the suffix /**ʂwæ˧**/.

tone of root	meaning of root	suffixed form	meaning
LM	chicken	**æ̃˧ʂwæ˥**	rooster (not castrated)
LM	yak	**bv̩˩ʂwæ˩**	castrated yak
L	sheep	**jo˩ʂwæ˩**	wether, castrated male sheep
MH	goat	**tsʰɯ˧ʂwæ˥**	wether, castrated male goat

5.1.4.2 The suffix /-v̩/

Lidz (2010: 179) proposes that the suffix in /**zɛ³¹-wu³³**/ 'nephew' (in the speech of Mrs. Latami, consultant F4: /**ze˩v̩˩**/) and /**ʐu³¹-wu³³**/ 'grandson' (F4: /**ʐv̩˧v̩#˥**/) derives from the root for 'uncle/senior male relative', which appears in /**ə˧v̩˦**/ 'maternal uncle'. She further hypothesizes that this root constitutes the origin of the classifier for individuals (F4: /**v̩˧**/). Progress in comparative studies of Naish languages will hopefully allow for testing these hypotheses.

5.1.4.3 The suffix /-ʁo/

The word for 'castrated horse' is /**ʐwæ˧ʁo˩**/. Horses have been the object of great care and interest in this part of the Himalayas for at least two millenia (Wāng Níngshēng 1980; Sagart et al. 2019: 10320; Jacques to appear), so it is unsurprising that words belonging to this semantic field are numerous, some of them probably very old. This isolated example, however, is clearly insufficient for linguistic analysis.

5.1.5 The kinship prefix /ə˧-/

Another non-productive but readily identifiable affix is the kinship prefix /**ə˧-**/, found in all kinship terms referring to one's elders. It is common to various languages of the area, such as Rma (Qiang) (Evans & Huang 2007: 158–159), Yi, and Mandarin. Table 5.13 presents the examples that were observed in Yongning Na, where this prefix is “the only common noun prefix” (Lidz 2010: 167).

Table 5.13: Kinship terms with the prefix /ə˧-/.

kinship term	tone	meaning
ə˧mɑ˧	M	mother (term of address)
ə˧mi˧	M	mother; aunt
ə˧pʰv̩˧	M	brother of the mother’s mother
ə˧si˧	M	mother’s mother’s mother; ancestor
ə˧ɖo˧	M	boyfriend/girlfriend, lover
ə˧ʐ̩i˧˥	MH#	mother’s mother
ə˧v̩˧˥	MH#	mother’s brother
ə˧dɑ˥$	H$	father
ə˧bo˥$	H$	father’s brother
ə˧ɕ̩jɤ˩	L#	boyfriend/girlfriend, lover
ə˧jɤ˩	L#	mother’s elder sister
ə˧mv̩˩	L#	elder sibling (brother or sister)
ə˧tɕ̩i˩	L#	mother’s younger sister
ə˧zɯ˩ ≈ ə˩zɯ˩	L# ≈ L	dual: us two
ə˧-sɯ˩kv̩˩ ≈ ə˩-sɯ˧kv̩˥	–L ≈ LMH	1st person plural, inclusive

Monosyllabic forms do not exist, and no convincing method to extract the tone of the root has been found. It is tempting to hypothesize that the prefix does not contribute a tonal specification, and that the tone of the disyllable reflects that of the root: disyllables with M, MH#, and H$ tone would originate in roots with M, MH, and H tone, respectively, and disyllables with L# tone would originate in roots with L, LM or LH tone. However, this reasoning remains highly speculative, and no evidence has been found to explore this issue further. The root /**mi**/ in /ə˧**mi**˧/ ‘mother’ is presumably related to the free form /**mi**˩˧/ ‘female’, and the root /**pʰv**/ in /ə˧**pʰv̩**˧/ ‘grandmother’s elder brother’ to the free form /**pʰv̩**˧/ ‘male’. However, unlike these free forms, the kinship terms /ə˧**mi**˧/ ‘mother’ and /ə˧**pʰv̩**˧/ ‘brother of the mother’s mother’ have the same tone, making it problematic to infer the tones of the roots from those of the prefixed kinship terms.

From a static-synchronic point of view, it is also difficult to reach firm conclusions due to the limited amount of data: one prefix and four suffixes. With this qualification, one may observe that disyllables with M, H$ or L tone can result from either suffixation or prefixation; that disyllables with #H, H#, LM, LH or LM+#H tone can result from suffixation but not from prefixation; and that disyllables with MH# or L# tone can result from prefixation but not from suffixation. The possible origins of the various tonal categories of disyllables (by suffixation, prefixation, and compounding) are examined in §10.4.2.

With regard to the tone of the kinship prefix, it seems reasonable to analyze it as M, since it always appears with M tone except for two variants shown in the last two lines of Table 5.13. This is not substantially different from an analysis in which the prefix is underlyingly toneless, given that M behaves in some respects as a default tone, as discussed in §2.4.3.

Kinship terms in the Luoshui dialect (Lidz 2010: 167) are similar to those in Alawua (the dialect studied here). For instance, Luoshui /**ɑ^{33}ʐɯ^{33}**/ 'grandmother' is cognate with Alawua /**ə˧ʑi˧˥**/. Only three terms from L. Lidz's list are not attested in Alawua. One of these is /**ɑ^{33}pɔ^{31}**/, for 'uncle: father's elder or younger brother', which could be a borrowing from Mandarin *ābó* 阿伯 'brother-in-law; father's older brother'. Borrowing is facilitated by the similar structure of kinship terms for one's elders in both languages, with a similar prefix (in Mandarin: *ā* 阿). A different term is in use in Alawua: /**ə˧bo˥$**/, whose voiced initial suggests that it is not a recent borrowing from Mandarin, since Mandarin does not retain voiced stops.[3] Ethnological data sheds light on the fact that the terms for uncles on the father's side do not correspond neatly across dialects: the social relationship with one's father (and his household) was traditionally loose (see Appendix B, §B.3); accordingly, kinship terms for paternal relatives were not as specific as those for maternal relatives.

The peculiar structure of Na families invites linguistic speculation concerning the origin and evolution of the terms currently used for relatives on the father's side. Fu Maoji (1980: 23; 1983: 38–39) hypothesizes that /**ə˧bo˥$**/ 'uncle on the father's side' formerly referred to male relatives of the father's generation on the father's side, i.e. the father and his brothers, and that the introduction of the term /**ə˧dɑ˥$**/ 'father' led to the specialization of /**ə˧bo˥$**/ to refer specifically to uncles on the father's side. This would imply that people had a term for their paternal uncles, pooled together with their father under the term /**ə˧bo˥$**/, before they had a specific term for 'father'. This seems paradoxical: since children did not live in the same household as their paternal uncles, their link to these uncles was through the father, making the existence of a term for 'father' a logical prerequisite for conceptualizing the broader category of *male relatives on the father's side, of the father's generation.*

[3] In *Pinyin* romanization, *b* stands for a voiceless bilabial stop: /**p**/, not /**b**/.

However, Christine Mathieu (p.c. 2016) quotes Lamu Gatusa 拉木·嘎吐萨 as reporting a word in the dialect of Labai 拉伯 that is cognate with /**ə˧bo˥$**/ and refers precisely to this concept: male relatives on the father's side, belonging to the father's generation – i.e. the father and his brothers. Distinctions can be made by adding the adjective 'small' to refer to the father's younger brothers and 'big' to refer to his elder brothers. Seen in this light, Fu Maoji's reasoning could be plausible in a conceptual universe that is structured around extended families and clans rather than nuclear families. If an individual is primarily identified in terms of belonging to a household, and to a generation within the family, it is conceivable that fathers were not distinguished terminologically from their brothers. The hypothesis of an undifferentiated term for the father and his brothers must be examined in light of the fact that ties with the father's family were traditionally loose and distant, both economically and socially.

Synchronically, the extension of kinship terms from the traditional household (i.e. the mother's side) to the paternal family is occasionally observed. For instance, speakers may publicly address their father as /**ə˧v̩˧˥**/ 'uncle on the mother's side' (field notes, consultant F4). The stylistic effect is to both convey closeness – through inclusion in the household – and honour, as maternal uncles are characters to whom the highest respect is due (as explained in Appendix B, §B.3.1).

In light of such flexibility in terms of address, one could venture an alternative hypothesis about the origin of the term for 'uncle on the father's side': namely, that the words /**ə˧v̩˧˥**/ 'uncle on the mother's side' and /**ə˧bo˥$**/ 'uncle on the father's side' in the Alawua dialect may have originally referred to the mother's elder and younger brothers, respectively. Under this hypothesis, /**ə˧bo˥$**/ – corresponding to a socially less important and prestigious role than /**ə˧v̩˧˥**/ – would have been reassigned to uncles on the father's side, while /**ə˧v̩˧˥**/ would have been generalized to all of the mother's brothers irrespective of age. This process would have preserved the social hierarchy between /**ə˧v̩˧˥**/ as the more important social figure and /**ə˧bo˥$**/ as the less important one, while shifting the distinction from one of age to one of lineage (maternal vs. paternal).

This hypothesis does not seem excessively far-fetched in view of the existence of distinct terms for aunts: 'mother's elder sister', /**ə˧jɤ˩**/, and 'mother's younger sister', /**ə˧tɕi˩**/. This lends some plausibility to the idea that an earlier stage of the language may also have distinguished older and younger uncles. However, this hypothesis remains highly speculative.

The second term reported by Liberty Lidz that is not found in the present research data (Alawua dialect) is /**ɑ³³mɔ¹³**/ as another term for 'grandmother'.

The third is /**ɑ**^{33}**lɑ**^{31}/, referring to great-great-grandparents: in Alawua, the term /**ə˧si˧**/ is used for all ancestors of the great-grandmother's generation and above.

5.2 Reduplication

In languages that do not have morphophonological templates specific to reduplication, it is not always easy to distinguish between reduplication and other types of repetition or copying. For instance, Moravcsik (1978: 301) considers *very very* in the English example *He is very very bright* as a case of reduplication. This is debatable, as the intensifier can be repeated more than once (either an even number of times, 2×n, or an odd number of times: *He is very, very, very bright*), with gradual rather than categorical semantic-stylistic effects.

In Yongning Na, reduplication is not too difficult to delimit on a phonological basis, as it follows specific tonal templates and carries clearly identifiable syntactic and semantic values. Reduplication is most commonly observed with verbs: see §6.2. Reduplication involving nouns is nowhere as frequent: the only well-attested case is the reduplication of numeral-plus-classifier phrases.

5.2.1 Reduplication of numeral-plus-classifier phrases

Reduplication of a phrase consisting of the numeral 'one' (analyzed as carrying LH tone: //**ɖɯ˩˥**//) plus a classifier indicates iteration. The entire reduplicated phrase is integrated into a single tone group – a crucial unit in Na morphotonology, discussed in Chapter 7. The tonal pattern of the first half of the phrase determines that of the second, by application of the phonological rules set out in §7.1.1:

- After an H-tone classifier, the second half of the phrase receives L tone by application of Rules 4 and 5, e.g. //**ɖɯ˧-ɲi˥$**// → /**ɖɯ˧-ɲi˥~ɖɯ˩-ɲi˩**/ 'day after day' (*Reward.155* {0004446#S155}, *BuriedAlive2.85* {0004536#S85}, *Caravans.259* {0004530#S259}).

- After an M-tone classifier, the second part remains unaffected, as in //**ɖɯ˧-ʁwɤ˧**// → /**ɖɯ˧-ʁwɤ˧~ɖɯ˧-ʁwɤ˧**/ 'one heap after another' (*Housebuilding.51* {0004448#S51}).

- After an L-tone classifier, the second part is lowered to L by application of Rule 5, e.g. //**ɖɯ˧-ʑi˩**// → /**ɖɯ˧-ʑi˩~ɖɯ˩-ʑi˩**/ 'one family after another' (*Healing.94* {0004540#S94}, *Caravans.237* {0004530#S237}).

- After an MH-tone classifier, the H part of the contour lands on the first syllable of the second half: //**ɖɯ˧-kɤ˦**// → /**ɖɯ˧-kɤ˧~ɖɯ˥-kɤ˩**/ 'one tree after another' (*Housebuilding.28* {0004448#S28}). The final L tone in this expression arises through the application of Rule 4.

5.2.2 A reduplicated nominal suffix, conveying abundance

Addition of the suffix /**-ʂo˧~ʂo˩**/ to nouns conveys abundance. Examples include /**mɤ˩-ʂo˩~ʂo˥**/ 'smeared with grease, covered with grease' (7), /**ʂe˧-ʂo˧~ʂo˥**/ 'rich in meat, with lots of meat' (8), and /**si˧-ʂo˧~ʂo˥**/ 'packed with wood' (9).

(7) mv̩˩kʰv̩˦ | le˧-tsʰɯ˩-dʑo˩, | ɲi˧to˧ | ʈʂʰɯ˧-qo˧ | le˧-tɑ˦, | ʈʂʰɯ˧-qo˧ | le˧-tɑ˦, | **mɤ˩-ʂo˩~ʂo˥** tsɤ˩ tsɯ˩ | mv̩˩!

mv̩˩kʰv̩˦ le˧- tsʰɯ˩ₐ -dʑo˥ ɲi˧to˧ ʈʂʰɯ˧-qo˧ le˧-tɑ˦ mɤ˩
evening ACCOMP to_come.PST TOP mouth here up_to animal_fat
-ʂo˧~ʂo˩ tsɤ˧ tsɯ˦ mv̩˧
ABUNDANCE to_become REP AFFIRM

'The story goes that in the evening, when [the dumb man] came back, [his] mouth was smeared with grease up to here, up to here!' *(Lake3.15)* {0004348#S15}

(8) kʰv̩˩mi˩-ki˥, | ə… ʂe˧! | ɲi˧zo˧ | ɖɯ˧-kʰwɤ˥, | æ̃˩-ʂe˧ | ɖɯ˧-kʰwɤ˥, | bo˩-ʂe˧ | ɖɯ˧-kʰwɤ˥, | ʝi˧-ʂe˧ | ɖɯ˧-kʰwɤ˥, | kʰv̩˩mi˩-ki˥ | tʰv̩˧-hɑ̃˩-dʑo˩, | **ʂe˧-ʂo˧ ʂo˥** | tʰi˧-ki˧-kv̩˧ tsɯ˥ | mv̩˩!

kʰv̩˩mi˩ -ki˧ ə… ʂe˥ ɲi˧zo#˥ ɖɯ˧-kʰwɤ˥$ æ̃˩-ʂe#˥
dog DAT *hesitation* meat fish one-CLF.piece chicken_meat
ɖɯ˧-kʰwɤ˥$ bo˩-ʂe#˥ ɖɯ˧-kʰwɤ˥$ ʝi˧-ʂe#˥ ɖɯ˧-kʰwɤ˥$ kʰv̩˩mi˩ -ki˧
one-CLF.piece pork one-CLF.piece beef one-CLF.piece dog DAT
tʰv̩˧-hɑ̃˩ -dʑo˥ ʂe˥ **-ʂo˧~ʂo˩** tʰi˧- ki˧ₐ -kv̩˦ tsɯ˦ mv̩˧
that_day TOP meat ABUNDANCE DUR to_give ABILITIVE REP AFFIRM

'To the dog, erm… [one would give] meat! A piece of fish, a piece of chicken, a piece of pork, a piece of beef… On that day [New Year's Eve], one would give the dog plenty of meat!' *(Dog.35)* {0004442#S35}

(9) nɑ˩-ʐi˧mi˧ ʈʂʰɯ˧ | ɖɯ˧-ʈʂɤ˥~ʈʂɤ˩-ki˩-ze˩-se˩ | **si˧-ʂo˧~ʂo˥**-ɲi˩!

nɑ˩˧ ʐi˧mi˧ ʈʂʰɯ˧ ɖɯ˧- ʈʂɤ˧ₐ ~ ki˧ₐ -ze˧
Na (endonym) house TOP DELIMITATIVE to_count ACTIVITY to_give PFV
-se˩ si˥ **-ʂo˧~ʂo˩** -ɲi˩
COMPLETION wood ABUNDANCE CERTITUDE

'The Na house, if one is going to count every part of it, [one will realize that] it is packed with wood!' (i.e. a huge deal of wood goes into its construction) *(Housebuilding.281)* {0004448#S281}

This suffix does not have a non-reduplicated counterpart and does not belong to a broader set of reduplicated nominal suffixes. One might therefore wonder whether it should be analyzed as a reduplicated form or rather as a simple suffix that happens to consist of two identical syllables. The reason for analyzing it as a reduplicated form lies in its structural parallel with reduplicated suffixes added to adjectives (presented in §6.7.1).

The reduplicated suffix /**-ʂo˧~ʂo˩**/ can be added to a wide range of nouns, including count nouns such as those denoting persons. For instance, a household with numerous young men may be described as /**pʰæ˧tɕi˥-ʂo˩~ʂo˩**/ 'teeming with youngsters'. The broad semantic applicability of this suffix allows for the elicitation of an entire set, presented in Table 5.14. As elsewhere, the '+' sign in the transcription of surface tone patterns indicates the tone of the copula when placed after the expression as a test to ascertain the type of syllabic anchoring of a final H tone.

The tone of the suffix /**-ʂo~ʂo**/ can be hypothesized to be L# (hence the notation //**ʂo˧~ʂo˩**// adopted here), based on its behaviour after M-tone and LM-tone disyllables. The tone patterns documented in Table 5.14 are not entirely straightforward, however; they differ from those of disyllabic postpositions, discussed in §5.5.

5.3 Possessive constructions containing pronouns

Possessive constructions were discussed in Chapter 2, where the behaviour of nouns in association with the possessive /=**bv̩˧**/ served as one of the tests for determining lexical tone categories. Possessive constructions containing pronouns, however, do not follow quite the same tonal patterns. This is one of several respects in which pronouns are special.

5.3.1 The 1st, 2nd and 3rd person pronouns

A pronoun's tonal category is established by matching up its tone in isolation with its tone when a copula is added (a test that is also useful for nouns, as explained in §2.2). On the basis of the tones in /**njɤ˩ ɲi˩˥**/ 'it's me', /**no˩ ɲi˩˥**/ 'it's you' and /**ʈʂʰɯ˧ ɲi˥**/ 'it's her/him', the 1sg, 2sg, and 3sg pronouns are analyzed as //**njɤ˩**//, //**no˩**//, and //**ʈʂʰɯ˥**//, respectively.

Table 5.14: The tonal behaviour of the reduplicated suffix /**-ʂo˧~ʂo˩**/ depending on the tone of the preceding noun.

example	tone	example	surface pattern	analysis
dust	LM	**ɖæ˩-ʂo˧~ʂo˩**	L.M.L	LM+L#
pimple	LH	**ʝi˩-ʂo˥~ʂo˩**	L.H.L (=L.M.L)	LH−
star	M	**kɯ˧-ʂo˧~ʂo˥**	M.M.H+L	H#
grease	L	**mɤ˩-ʂo˩~ʂo˥**	L.L.H+L	L+H#
meat	H	**ʂe˧-ʂo˧~ʂo˥**	M.M.H+L	H#
mushroom	MH	**mo˧-ʂo˧~ʂo˥**	M.M.H+L	H#
dew	M	**ɖʐv̩˧qʰɑ˧-ʂo˧~ʂo˩**	M.M.M.L	L#
fly	#H	**bv̩˧ɻ̩˧-ʂo˧~ʂo˥**	M.M.M.H+L	H#
paste	MH#	**ho˧dʑɯ˧-ʂo˧~ʂo˥**	M.M.M.H+L	H#
mud	H$	**ɖʐæ˧qʰæ˧-ʂo˧~ʂo˥**	M.M.M.H+L	H#
egg	L	**æ̃˩ʁv̩˩-ʂo˩~ʂo˥**	L.L.L.H+L	L+H#
cake/bread	L#	**dze˧dv̩˩-ʂo˩~ʂo˩**	M.L.L.L	L#−
bean chaff	LM+MH#	**nv̩˩tsɑ˧-ʂo˧~ʂo˥**	L.M.M.H+L	LM+H#
potato	LM+#H	**jɤ˩jo˧-ʂo˧~ʂo˥**	L.M.M.H+L	LM+H#
soybeans	LM	**nv̩˩ɭɯ˧-ʂo˧~ʂo˩**	L.M.M.L	LM+L#
button	LH	**pv̩˩ɭɯ˥-ʂo˩~ʂo˩**	L.H.L.L	LH−
youngster	H#	**pʰæ˧tɕi˥-ʂo˩~ʂo˩**	M.H.L.L	H#−

To build a possessive construction with a pronoun, the possessive /=**bv̩˧**/ is generally used, as illustrated in (10). The forms are /**njɤ˧=bv̩˩**/, /**no˧=bv̩˩**/, and /**ʈʂʰɯ˧=bv̩˧**/ for the 1st, 2nd, and 3rd persons.

(10) ə˧ʝi˧-ʂɯ˥ʝi˩, | njɤ˧=bv̩˩ | ʐwæ˧ ʈʂʰɯ˧, | ŋwɤ˩-kv̩˩ ʈʂæ˥ | po˧ hɯ˧-ɲi˥!
ə˧ʝi˧-ʂɯ˥ʝi˩ njɤ˩ =bv̩˧ ʐwæ˥ ʈʂʰɯ˧ ŋwɤ˧ kv̩˦ ʈʂæ˦ po˦
in_the_past 1SG POSS horse TOP five CLF to_rob to_take_away
hɯ˧c -ɲi˩
to_go.PST CERTITUDE
'Once, long ago, five of my family's horses were stolen!' *Literally:* 'Long ago, my horses (=my family's horses), five were stolen and taken away!' *(Caravans.183)* {0004530#S183}

This is unlike the pattern for nouns: L-tone nouns with the possessive yield L+M, and H-tone nouns yield M+H.

A further complication is that the 3rd-person pronoun appears in two forms: /**ʈʂʰɯ˧=bv̩˧**/ and /**ʈʂʰɯ˧=bv̩˩**/. These are not tonal variants but semantically distinct forms. The latter, /**ʈʂʰɯ˧=bv̩˩**/, is a reduced form of /**ʈʂʰɯ˧=ɻ̍˩=bv̩˩**/, where /=ɻ̍˩/ is the associative plural. Thus, /**ʈʂʰɯ˧=bv̩˩**/ means 'their', whereas /**ʈʂʰɯ˧=bv̩˧**/ simply means 'her/his'. The associative plural /=ɻ̍˩/ is fully elided: it leaves no segmental trace, only a tonal difference on the possessive particle. This example provides an insight into the expansion of morphotonology through segmental simplifications – a type of diachronic change that is especially well-attested in Bantu languages. This topic will be further explored in section §10.3 of Chapter 10, where the Yongning Na tone system is examined from a dynamic-synchronic perspective.

A pronoun may also directly precede the noun to build a possessive construction, as in (11).

(11) njɤ˧ mv̩˩… | ə˧zɯ˩ | ʂɯ˧-bi˧, | ə˧mi˧ | tʰi˧-ʂɯ˧-kʰɯ˧!
njɤ˩ mv̩˩˥ ə˧zɯ˩ ʂɯ˧ -bi˧ ə˧mi˧ tʰi˧- ʂɯ˧ -kʰɯ˥
1sg daughter 1pl.incl to_die imm_fut mother dur to_die caus
'My dear daughter… We are going to die [=we can't avoid death, now that the tiger is at our door]; let Mum die [=let me sacrifice myself, so you can survive]!' *(Tiger.16)* {0004444#S16}

Combinations of pronouns and nouns, as in (11), were systematically elicited. The results are identical for the 1sg and 2sg pronouns, //**njɤ**˩// and //**no**˩//. They are shown in Table 5.15. The corresponding recording is: *PossessPro* {0004575}.

In all these phrases, the pronoun //**njɤ**˩// carries the same tone as in isolation: an M tone. (Neutralization of //L//, //M//, and //H// to /M/ in isolation is the general rule for monosyllabic nouns and pronouns: see §2.2.1.) The patterns for monosyllabic nouns cannot be obtained through the application of a simple set of general rules. On the other hand, the patterns for disyllables are extremely simple. They consist of the succession of the pronoun, as said in isolation: /**njɤ**˧/, followed by the noun, which carries the same tone as when it appears on its own, except that some tone levels are deleted to comply with the phonological constraints on a well-formed tone group. This affects the tone categories in which an L tone is attached to the first syllable of the noun: L, LM+MH#, LM+#H, LM, and LH. For these categories, the sequence found on the first two syllables (M tone on the pronoun, and L tone on the initial syllable of the disyllabic noun) is incompatible with any tone other than L on following syllables, by Rule 5 ("All syllables following an H.L or M.L sequence receive L tone": see §7.1.1). For instance, ‡ **njɤ**˧ **bo**˩**mi**˧ (obtained through simple concatenation) would not be

Table 5.15: The tones of possessive constructions consisting of a 1sg pronoun and a noun.

tone	head	meaning	example	tone pattern
LM	**bo˩˧**	pig	**njɤ˧ bo˩**	L#
LH	**mv̩˩˥**	daughter	**njɤ˧ mv̩˩**	L#
M	**zɯ˧**	life, existence	**njɤ˧ zɯ#˥**	#H
L	**dʐɯ˩**	water	**njɤ˧ dʐɯ#˥**	#H
#H	**hĩ˥**	human being	**njɤ˧ hĩ#˥**	#H
MH#	**tsʰɯ˧˥**	goat	**njɤ˧ tsʰɯ˧˥**	MH#
M	**po˧lo˧**	ram	**njɤ˧ po˧lo˧**	M
#H	**ʐwæ˧zo#˥**	colt	**njɤ˧ ʐwæ˧zo#˥**	#H
MH#	**hwɤ˧li˧˥**	cat	**njɤ˧ hwɤ˧li˧˥**	MH#
H$	**kv̩˧ʂe˥$**	flea	**njɤ˧ kv̩˧ʂe˥$**	H$
L	**kʰv̩˩mi˩**	dog	**njɤ˧ kʰv̩˩mi˩**	–L
L#	**dɑ˧ʝi˩**	mule	**njɤ˧ dɑ˧ʝi˩**	L#
LM+MH#	**ʝi˩ʈʂæ˧˥**	waist	**njɤ˧ ʝi˩ʈʂæ˩**	–L
LM+#H	**bi˩ʈʂʰɤ#˥**	whiskers	**njɤ˧ bi˩ʈʂʰɤ˩**	–L
LM	**bo˩mi˧**	sow	**njɤ˧ bo˩mi˩**	–L
LH	**bo˩ɬɑ˥**	boar	**njɤ˧ bo˩ɬɑ˩**	–L
H#	**kʰv̩˧nɑ˥**	dog	**njɤ˧ kʰv̩˧nɑ˥**	H#

a well-formed tone group; this is repaired to /**njɤ˧ bo˩mi˩**/, where the final M is lowered to L. The representation in Figure 5.1 assumes that the M part of the LM tone is initially associated but subsequently deleted. Alternatively, one could consider that this M does not associate at all. Psycholinguistic experiments would be necessary to approach more closely the processes taking place in speakers' brains.

The construction in Table 5.15 exemplifies minimal tonal integration between two elements. It can be described as concatenation of the two parts of the expression, followed by adjustments dictated by phonological rules.

For disyllables, the tone patterns following the 3rd person pronoun //**ʈʂʰɯ˥**// are identical with those found after the 1st and 2nd persons. For monosyllables, on the other hand, a contrast arises in the case of L-tone nouns: compare //**njɤ˧ dʐɯ#˥**// 'my water' with //**ʈʂʰɯ˧ dʐɯ˧**// 'her/his water'. This asymmetry poses yet another challenge to the language learner, who must (i) distinguish the tone patterns associated with these two sets of pronouns when they combine with

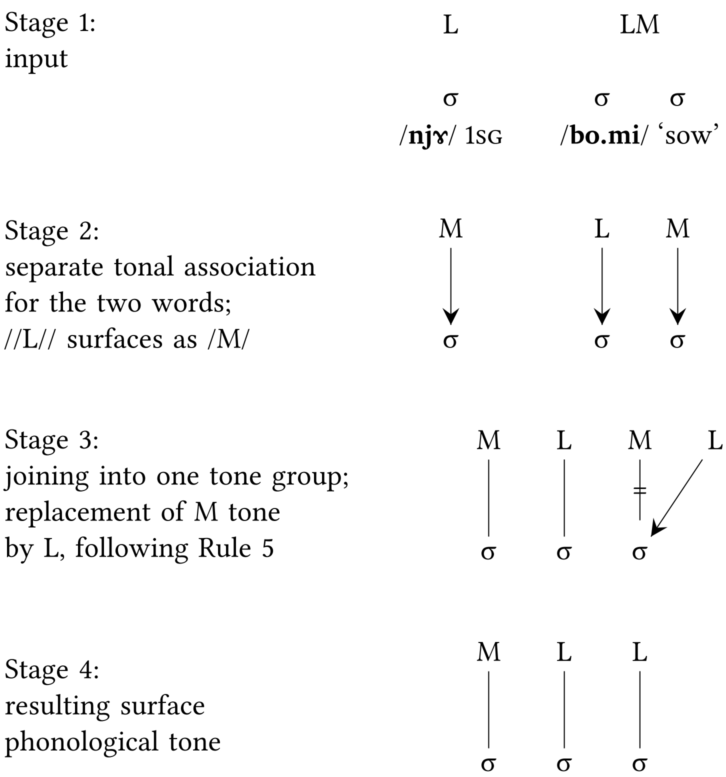


Figure 5.1: Tone-to-syllable association in /**njɤ˧ bo˩mi˩**/ 'my sow'.

a monosyllabic noun, while (ii) disregarding this difference when the following noun is disyllabic. The full set is shown in Table 5.16.

Finally, a looser construction also exists: a simple juxtaposition of the pronoun and the noun, each in its own tone group, as in (12).

(12) njɤ˧ | ɻ̩˩ʈʂʰe˧-ɖɯ˩mɑ˩ ʈʂʰɯ˩-dʑo˩, | no˧sɯ˩kv̩˩ | tʰv̩˧-ɲi˧ | lo˧ ʝi˧ hɯ˧ tsɯ˩.
njɤ˩ ɻ̩˩ʈʂʰe˧-ɖɯ˩mɑ˩ ʈʂʰɯ˧ -dʑo˥ no˧sɯ˩kv̩˩ tʰv̩˧-ɲi˧ lo˧ ʝi˧ hɯ˧$_c$
1sg given_name top top 2pl.excl that.day work to_do go.pst
tsɯ˦˥
rep
'As for my [daughter] Erchei Ddeema… that day, you (=the members of your family) had gone away to work.' *(BuriedAlive2.132)* {0004536#S132}

Table 5.16: The tones of possessive constructions consisting of a 3sg pronoun and a noun.

tone	example	meaning	with 3sg	tone pattern
LM	**bo˩˧**	pig	**ʈʂʰɯ˧ bo˩**	L#
LH	**mv̩˩˥**	daughter	**ʈʂʰɯ˧ mv̩˩**	L#
M	**zɯ˧**	life, existence	**ʈʂʰɯ˧ zɯ#˥**	#H
L	**dʐɯ˩**	water	**ʈʂʰɯ˧ dʐɯ˧**	M
#H	**hĩ˥**	human being	**ʈʂʰɯ˧ hĩ#˥**	#H
MH#	**tsʰɯ˧˥**	goat	**ʈʂʰɯ˧ tsʰɯ˧˥**	MH#
M	**po˧lo˧**	ram	**ʈʂʰɯ˧ po˧lo˧**	M
#H	**ʐwæ˧zo#˥**	colt	**ʈʂʰɯ˧ ʐwæ˧zo#˥**	#H
MH#	**hwɤ˧li˧˥**	cat	**ʈʂʰɯ˧ hwɤ˧li˧˥**	MH#
H$	**ə˧dɑ˥$**	father	**ʈʂʰɯ˧ ə˧dɑ#˥**	#H
L	**kʰv̩˩mi˩**	dog	**ʈʂʰɯ˧ kʰv̩˩mi˩**	–L
L#	**dɑ˧ʝi˩**	mule	**ʈʂʰɯ˧ dɑ˧ʝi˩**	L#
LM+MH#	**ʝi˩ʈʂæ˧˥**	waist	**ʈʂʰɯ˧ ʝi˩ʈʂæ˩**	–L
LM+#H	**bi˩ʈʂʰɤ#˥**	whiskers	**ʈʂʰɯ˧ bi˩ʈʂʰɤ˩**	–L
LM	**bo˩mi˧**	sow	**ʈʂʰɯ˧ bo˩mi˩**	–L
LH	**bo˩ɬɑ˥**	boar	**ʈʂʰɯ˧ bo˩ɬɑ˩**	–L
H#	**kʰv̩˧nɑ˥**	dog	**ʈʂʰɯ˧ kʰv̩˧nɑ˥**	H#

This is not a possessive construction in the strict sense; rather, the pronoun serves as a topic. The context for this example is as follows. A young woman is unhappy with a marriage arranged by her family and commits a small offence, which takes on huge proportions. Her mother then comes to the mother-in-law's house to make things right by talking the matter over. The young woman's mother first recounts the entire story, clarifying the actions of both parties – the members of the two families. In this situation, the first-person pronoun in the construction /**njɤ˧ | ɻ̩˩ʈʂʰe˧-ɖɯ˩mɑ˩**/ highlights that the mother is speaking on her daughter's behalf and, as head of the family, assumes some responsibility for her daughter's actions.

5.3.2 The pronoun 'oneself'

The pronoun /**õ˧˥**/ 'oneself' has a different behaviour from the 1st, 2nd, and 3rd person pronouns. It can be followed by the possessive marker /=**bv̩˧**/, as illustrated in (13).

(13) ə˧ʝi˧-tsʰi˧ʝi˧, | ɖʐɯ˧qo˩ | le˧-hwæ˧, | ɖɯ˧-tʰɑ˦ | pi˧-bi˧-bi˧, | õ˧ = bv̩˥ | ʈʂʰɯ˧-kʰwɤ˥-dʑo˩, | dʑɤ˩ ʈʂɤ˧ ɲi˥, | pi˧-zo˩, | ɑ˩ʁo˧ tʰi˧-tɕɯ˥!

ə˧ʝi˧-tsʰi˧ʝi˧	ɖʐɯ˧qo˩	le˧-	hwæ˧$_a$	ɖɯ˧$_b$	-tʰɑ˦	pi˥
nowadays	market	ACCOMP	to_buy	to_obtain	PERMISSIVE	to_say

-bi-bi	õ˦	= bv̩˧	ʈʂʰɯ˥	kʰwɤ˥$_a$	-dʑo˥	dʑɤ˩$_b$	ʈʂɤ˧$_a$
PERMISSIVE	oneself	POSS	DEM.PROX	CLF.chunks	TOP	good	to_count_as

ɲi˩	pi˥	-zo	ɑ˩ʁo˧	tʰi˧-	tɕɯ˥
CERTITUDE	to_say	ADVB	home	DUR	to_lay_up

‘These days, even if we can buy things at the market and find what we want, well, we consider the things that are our own [productions] as good (=as better than what we buy at the market), so we store them at home!’ (*Benevolence.15*) {0004582#S15}

Example (13) is the only instance found in texts where the sequence /**õ˧ = bv̩˥**/ ‘one’s own’ appears independently, rather than as part of the construction /**õ˧ = bv̩˥-õ˩**/ ‘one’s own’, ‘one’s proper’, ‘oneself’, which occurs 59 times in the available corpus (as of 2025). The latter construction is typically followed by a noun or verb. An example is shown in (14).

(14) õ˧ = bv̩˥-õ˩ hɑ˩, | õ˧ = bv̩˥-õ˩ dzɯ˩!

õ˦	= bv̩˧	õ˦	hɑ˥	õ˦	= bv̩˧	õ˦	dzɯ˥
oneself	POSS	oneself	food	oneself	POSS	oneself	to_eat

‘One’s own food, one eats it oneself! / We eat our own produce!’ *Context:* this statement summarizes the traditional self-sufficiency of the Na, who grew their own food crops. *(Agriculture.68)* {0004440#S68}

The symmetry of the construction /**õ˧ = bv̩˥-õ˩**/, which begins and ends with the morpheme /**õ˦**/ ‘self’, makes it an iconic, Narcissus-style mirror image of its meaning: namely, pointing to *self.*

The widespread use of the phrase /**õ˧ = bv̩˥-õ˩**/ (its “reproductive success”, to hijack a term from evolutionary biology) may also be linked to another matter of linguistic form, namely its tonal simplicity. Following this phrase, tonal oppositions are fully neutralized, as a following word can only carry L tones, due to a constraint formulated in the present description as Rule 5: “All syllables following an H.L or M.L sequence receive L tone”.

Outside constructions involving the possessive morpheme, the pronoun /**õ˦**/ typically combines directly with a following noun or verb, as in (15) and (16).

(15) õ˧ ə˧mv̩˥ ɲi˩-ze˩!
õ˧˥ ə˧mv̩˩ ɲi˩ -ze˧
self elder_sibling COP PFV
'This is my own brother!' *Context:* a young woman recognizes a ragged stranger attending her wedding as being her long-lost brother. (*Sister.57* {0004341#S57}; *Sister3.58* {0004344#S58})

(16) õ˧-ʂe˥, õ˩ ʈʰæ˩!
õ˧˥ ʂe˥ õ˧˥ ʈʰæ˧˥
self meat self to_bite
'Each person eats her/his own slab of meat!' *Context:* describing table manners. Each family member used to receive one slice of meat and eat it up. This differs from Han Chinese custom, where each guest picks food mouthful by mouthful, with chopsticks, from dishes placed on the table. (Field notes.)

The construction /**õ˧˥**/ plus N, '[my/one's] own N', has a tonal behaviour of its own. On the analogy of (16), new maxims can be coined, such as (17), (18), and (19).

(17) õ˧-dʐɯ˥, õ˩ ʈʰɯ˩!
õ˧˥ dʐɯ˩ õ˧˥ ʈʰɯ˩$_b$
self water self to_drink
'Each drinks from her own bottle!' *Context:* a toddler has grabbed another's bottle; parents prevent her from drinking from it. (Field notes.)

(18) õ˧-ɖæ˥, õ˩ bæ˩!
õ˧˥ ɖæ˩˧ õ˧˥ bæ˩$_a$
self dust self to_sweep
'One must sweep one's own garbage.' (Elicited example)

(19) õ˧-lv̩˥, õ˩ li˩!
õ˧˥ lv̩˧ õ˧˥ li˧$_a$
self field self to_look_after
'One must look after one's own fields.' (Elicited)

Combinations were systematically elicited. The full set of tonal combinations is presented in Table 5.17.

The behaviour of /**õ˧˥**/ in association with disyllables coincides with that of determinative compounds containing an MH-tone determiner. With monosyllables, however, the tone patterns only coincide with those of determinative compounds for nouns with LM or MH# tone.

Table 5.17: The tones of possessive constructions consisting of /**õ˧˥**/+N.

tone	example	meaning	with /**õ˧˥**/	tone pattern
LM	**ɖæ˩˧**	dirt, dust	**õ˧-ɖæ˥**	H#
LH	**mv̩˩˥**	daughter	**õ˧-mv̩˥$**	H$
M	**lv̩˧**	field	**õ˧-lv̩˥$**	H$
L	**dʐɯ˩**	water	**õ˧-dʐɯ˥$**	H$
#H	**zo˥**	son	**õ˧-zo#˥**	H#
MH#	**tsʰɯ˧˥**	goat	**õ˧-tsʰɯ˥$**	H$
M	**go˧mi˧**	younger sister	**õ˧-go˧mi˥**	H#
#H	**ʐwæ˧zo#˥**	colt	**õ˧-ʐwæ˧zo#˥**	#H
MH#	**ə˧mv̩˧˥**	elder sibling	**õ˧-ə˧mv̩˧˥ / õ˧-ə˥mv̩˩**	MH# / MH#−
H$	**ə˧dɑ˥$**	father	**õ˧-ə˧dɑ˥ / õ˧-ə˧dɑ#˥**	H# / #H
L	**kʰv̩˩mi˩**	dog	**õ˧-kʰv̩˥mi˩**	MH#−
L#	**dɑ˧ʝi˩**	mule	**õ˧-dɑ˧ʝi˥**	H#
LM+MH#	**ʝi˩ʈʂæ˧˥**	waist	**õ˧-ʝi˥ʈʂæ˩**	MH#−
LM+#H	**bi˩ʈʂʰɤ#˥**	whiskers	**õ˧-bi˥ʈʂʰɤ˩**	MH#−
LM	**ɑ˩ʁo˧**	home	**õ˧-ɑ˥ʁo˩**	MH#−
LH	**bo˩ɬɑ˥**	boar	**õ˧-bo˥ɬɑ˩**	MH#−
H#	**kʰv̩˧nɑ˥**	dog	**õ˧-kʰv̩˧nɑ˥**	H#

5.4 Monosyllabic morphemes appearing after nouns: Enclitics, suffixes, and postpositions

The analysis of morphemes as affixes, clitics, postpositions, serial verbs, "particles", and other parts of speech raises interesting issues that differ widely from one language to another, as illustrated by the diversity of proposals and viewpoints presented in a collective volume about the notion of "word" (Dixon & Aikhenvald 2002). Aikhenvald (2002: 43) proposes that there is "a multidimensional continuum, from a fully bound to a fully independent morpheme". The approach adopted here is to start out from tone patterns and move towards an analysis in light of the morphemes' tonal behaviour, rather than proceeding from predefined morphosyntactic categories. This method provides independent evidence for syntactic analysis and part-of-speech classification. For instance, the dative /**-ki**˧/ and the possessive /=**bv̩**˧/ turn out to have exactly the same tonal behaviour, distinct from that of the agentive /**ɳɯ**˧/ (see §5.4.2), suggesting that the first two belong to the same morphosyntactic category, whereas the latter constitutes a separate class. This finding can be taken as confirmation for the observation – based on syntactic behaviour – that the dative and possessive are "almost suffixal" (Lidz 2010: 155), whereas the agentive is best analyzed as a case adposition.

Enclitics, suffixes, and postpositions are grouped into four subsets on the basis of their behaviour following M-tone nouns. Those that surface with L tone in this context are considered to carry lexical L tone; likewise, those that surface with M, MH, and H tones are considered to carry M, MH, and H tones, respectively. The following sections, each of which is dedicated to one of the four subsets, show that these broad tonal categories are not fully homogeneous, but they provide a useful framework for presenting the data.

5.4.1 L-tone morphemes

This subsection examines three morphemes: the postposition meaning 'on; at' and the plural and associative clitics.

The postposition //**bi˩**// 'on; at' surfaces with L tone after an M-tone noun, e.g. in /**gv̩˧mi˧ bi˩**/ 'on the body'. Additional examples from texts and field notes include disyllabic nouns with L tone, as in /**ʐæ˩sɯ˩ bi˥**/ 'on the felt cape' (*Sister.58* {0004341#S58}). To establish a full dataset, systematic elicitation was conducted, yielding the patterns presented in Table 5.18.

The tonal patterns indicated in Table 5.18 are those observed at the surface phonological level, not the underlying tones. In the case of /**bo˩ bi˥**/ 'on (a/the) pig' and /**ʐæ˩ bi˥**/ 'on (a/the) leopard', it remains uncertain whether the underlying pattern is //L.M// or //L.H//, as both are neutralized at the surface level due to Rule 6 (see §7.1.1). Consequently, Table 5.18 does not reveal whether the tone patterns of 'on (a/the) pig' and 'on (a/the) leopard' are underlyingly identical.

Further evidence comes from the plural clitic /=**ɻæ˩**/ and the associative plural clitic /=**ɻ̩˩**/. As in Japhug, where the plural clitic /=**ra**/ conveys either plurality or collective meaning, these enclitics are not obligatory for non-singular arguments, even for human referents (Jacques 2020). Like the postposition /**bi˩**/ 'on; at', these two enclitics are analyzed as having L tone on the basis of their tonal behaviour after M-tone nouns. Their tone patterns are identical to those of /**bi˩**/.

For the plural and associative plural morphemes, an additional diagnostic is available: the possessive /=**bv̩˧**/ can be added to an *N+plural* construction to test the underlying tone category, following the method used in the study of the tones of nouns (Chapter 2). This test distinguishes //L.M//, which does not depress the tone of the following possessive, from //L.H//, which does. Table 5.19 presents the tonal outcomes for N+plural+possessive.

Table 5.18: The behaviour of the L-tone postposition //**bi˩**// 'on; at' with monosyllabic and disyllabic nouns. There is an additional 'L PRO' row because L-tone pronouns have exceptional behaviour.

example	tone	with /**bi˩**/	surface tone pattern
pig	LM	**bo˩ bi˥**	L.H
leopard	LH	**ʐæ˩ bi˥**	L.H
tiger	M	**lɑ˧ bi˩**	M.L
sheep	L	**jo˩ bi˩˥**	L.LH
2SG	L PRO	**no˧ bi˩**	M.L
horse	H	**ʐwæ˧ bi˥**	M.H
deer	MH	**ʈʂʰæ˧ bi˥**	M.H
fox	M	**ɖɤ˧mi˧ bi˩**	M.M.L
colt	#H	**ʐwæ˧zo˧ bi˥**	M.M.H
cat	MH#	**hwɤ˧li˧ bi˥**	M.M.H
she-cat	H$	**hwɤ˧mi˧ bi˥**	M.M.H
dog	L	**kʰv̩˩mi˩ bi˥**	L.L.H
mule	L#	**dɑ˧ʝi˩ bi˩**	M.L.L
wolf	LM+MH#	**õ˩dv̩˧ bi˥**	L.M.H
Naxi	LM+#H	**nɑ˩hĩ˧ bi˥**	L.M.H
sow	LM	**bo˩mi˧ bi˩**	L.M.L
boar	LH	**bo˩ɬɑ˥ bi˩**	L.H.L
rat	H#	**hwæ˧tsɯ˥ bi˩**	M.H.L

As in other contexts, expressions involving pronouns do not always follow the same tonal patterns as those containing nouns. The proximal demonstrative //**ʈʂʰɯ˥**// and the distal demonstrative //**tʰv̩˥**// yield /**ʈʂʰɯ˧=ɻæ˥$**/ and /**tʰv˧=ɻæ˥$**/, i.e. the same pattern as for monosyllabic nouns with H tone. However, the first- and second-person pronouns, //**njɤ˩**// and //**no˩**//, yield /**njɤ˧=ɻæ˩**/ and /**no˧=ɻæ˩**/ with the plural, diverging from the pattern observed for L-tone nouns such as /**jo˩=ɻæ˩˥**/ 'sheep'.

Table 5.19 brings out tonal differences between nouns and noun-plus-clitic expressions. For instance, the noun /**ə˧v̩˧˥**/ 'uncle' yields /**ə˧v̩˧=ɻæ˥**/ 'the uncles' and /**ə˧v̩˧=ɻæ˧=bv̩˥**/ 'of the uncles', i.e. a tonal alternation not found among nouns: no tonal category of disyllables ends in H tone in isolation while also yielding a final H tone on a following possessive. The H$ tone category of disyllables followed by a possessive yields M.M.M, not M.M.H, as in /**kv̩˧ʂe˧=bv̩˧**/

Table 5.19: The tonal behaviour of nouns followed by the plural clitic /=**ɻæ˩**/ plus the possessive suffix /=**bv̩˧**/.

example	tone	+PLURAL	+PLURAL+POSSESSIVE	tonal analysis
Na (ethnic group)	LM	**nɑ˩=ɻæ˥**	**nɑ˩=ɻæ˥=bv̩˩**	LH
daughter	LH	**mv̩˩=ɻæ˥**	**mv̩˩=ɻæ˥=bv̩˩**	LH
Han (ethnic group)	M	**hæ˧=ɻæ˩**	**hæ˧=ɻæ˩=bv̩˩**	L#
sheep	L	**jo˩=ɻæ˩˥**	**jo˩=ɻæ˩=bv̩˥**	L
person, human being	H	**hĩ˧=ɻæ˥**	**hĩ˧=ɻæ˥=bv̩˩** / **hĩ˧=ɻæ˧=bv̩˥**	H# / H$
deer	MH	**tʂ^{h}æ˧=ɻæ˥**	**tʂ^{h}æ˧=ɻæ˥=bv̩˩**	H#
aunt	M	**ə˧mi˧=ɻæ˩**	**ə˧mi˧=ɻæ˩=bv̩˩**	–L
younger brother	#H	**gi˧zɯ˧=ɻæ˥**	**gi˧zɯ˧=ɻæ˧=bv̩˥** / **gi˧zɯ˧=ɻæ˥=bv̩˩**	H$ / H#
maternal uncle	MH#	**ə˧v̩˧=ɻæ˥**	**ə˧v̩˧=ɻæ˧=bv̩˥** / **ə˧v̩˧=ɻæ˥=bv̩˩**	H$ / H#
she-cat	H$	**hwɤ˧mi˧=ɻæ˥**	**hwɤ˧mi˧=ɻæ˧=bv̩˥** / **hwɤ˧mi˧=ɻæ˥=bv̩˩**	H$ / H#
woman	L	**mi˩zɯ˩=ɻæ˥**	**mi˩zɯ˩=ɻæ˩=bv̩˥**	L+H#
elder sibling	L#	**ə˧mv̩˩=ɻæ˩**	**ə˧mv̩˩=ɻæ˩=bv̩˩**	L#–
wolf	LM+MH#	**õ˩dv̩˧=ɻæ˥**	**õ˩dv̩˧=ɻæ˧=bv̩˥** / **õ˩dv̩˧=ɻæ˥=bv̩˩**	LM+H# / LM+H$
Naxi (ethnic group)	LM+#H	**nɑ˩hĩ˧=ɻæ˥**	**nɑ˩hĩ˧=ɻæ˧=bv̩˥** / **nɑ˩hĩ˧=ɻæ˥=bv̩˩**	LM+H# / LM+H$
sow	LM	**bo˩mi˧=ɻæ˩**	**bo˩mi˧=ɻæ˩=bv̩˩**	LM–L
boar	LH	**bo˩ɬɑ˥=ɻæ˩**	**bo˩ɬɑ˥=ɻæ˩=bv̩˩**	LH–
young man	H#	**p^{h}æ˧tɕi˥=ɻæ˩**	**p^{h}æ˧tɕi˥=ɻæ˩=bv̩˩**	H#–

'of (a/the) flea' (see Table 2.9). Thus, the tonal pattern of /**ə˧v̩˧=ɻæ˥**/ 'the uncles' does not strictly correspond to any of the tonal categories of disyllabic nouns. Similarly, the behaviour of /**hĩ˧=ɻæ˥**/ 'the people', which yields /**hĩ˧=ɻæ˧=bv̩˥**/ 'of the people', lacks an exact counterpart among disyllabic nouns. Impressionistically, it is as if H$, the *hopping* H tone, exhibited a simplified behaviour when found in an expression containing a clitic: the H level tends to attach to the following syllable, even if that syllable is a suffix. This differs from the behaviour of H$-tone nouns, whose combinations with suffixes of different tone categories yield distinct results.

The plural clitic /=**ɻæ˩**/ is frequently used. By contrast, the associative plural /=**ɻ̍**/ has a highly specific meaning, referring to a clan or extended family, and is therefore mostly restricted to pronouns and family (clan) names, which are few in number. It cannot be used with kinship terms. However, it is not implausible that the morpheme /=**ɻ̍˩**/ that partakes in nominalization processes is in fact the associative plural. An example is shown in (20), where the reduplicated verb /**pʰæ˧~pʰæ˧**/ 'to attach' in combination with /=**ɻ̍˩**/ conveys the meaning 'a couple; a pair; a set (of things, persons…) tied together'.

(20) pʰæ˧~pʰæ˧=ɻ̍˩ ɲi˩-kv̩˩ tsɯ˩ | mv̩˩.
pʰæ˧b ~ =ɻ̍˩ ɲi˩ -kv̩˦ tsɯ˦ mv̩˧
to_tie/fasten RED ASSOCIATIVE COP ABILITIVE REP AFFIRM
'[The mountains Gemu **kɤ˧mv̩˦** and Aeshae **æ˧ʂæ˧**] make up a couple/a pair!' *(Mountains.99)* {0004573#S99}

Not unexpectedly, pronouns followed by the associative plural have a tonal behaviour of their own. The two L-tone pronouns (1SG and 2SG) follow the same tonal pattern with the associative plural as with the plural: /**njɤ˧=ɻ̍˩**/ and /**no˧=ɻ̍˩**/. However, the proximal demonstrative (also serving as 3rd-person pronoun) /**ʈʂʰɯ˥**/ and the distal demonstrative /**tʰv̩˥**/ yield /**ʈʂʰɯ˧=ɻ̍˩**/ and /**tʰv̩˧=ɻ̍˩**/ with the associative, whereas they yield /**ʈʂʰɯ˧=ɻæ˥$**/ and /**tʰv̩=ɻæ˥$**/ with the plural. This difference in tone patterns is enough to establish that the plural and associative plural do not always share identical tonal patterns, despite their identical behaviour in almost all cases.

The associative /=**ɻ̍˩**/ is presented separately in Table 5.20. Its low frequency accounts for the numerous gaps in the table.

The Alawua dialect of Yongning Na also has a dual morpheme /=**zɯ˩**/ appearing in four pronouns: first-person dual exclusive /**njæ˧=zɯ˩**/, first-person dual inclusive /**ə˧=zɯ˩**/, second-person dual /**no˧=zɯ˩**/, and third-person dual /**ʈʂʰɯ˧=zɯ˩**/. On the basis of these forms, the tone of the dual can be classified as

Table 5.20: The tonal behaviour of associative plural /=ɻ̍/.

example	meaning	tone	with /=ɻ̍/	output
–	–	LM	–	–
–	–	LH	–	–
–	–	M	–	–
njɤ˩, **no˩**	1SG, 2SG	L	**njɤ˧=ɻ̍˩**, **no=ɻ̍˩**	L#
ʈʂʰɯ˥, **tʰv̩˥**	DEM.PROX, DIST	H	**ʈʂʰɯ˧=ɻ̍˩**, **tʰv̩˧=ɻ̍˩**	L#
–	–	MH	–	–
dze˧bo˧	family name	M	**dze˧bo˧=ɻ̍˩**	L#
–	–	#H	–	–
–	–	MH#	–	–
kv̩˧tsʰɑ˥$	family name	H$	**kv̩˧tsʰɑ˧=ɻ̍˥$**	H$
lɑ˩mɑ˩	family name	L	**lɑ˩mɑ˩-ɻ̍˥$**	L+H$
ə˧ɕjo˩	family name	L#	**ə˧ɕjo˩=ɻ̍˩**	L#–
–	–	LM+MH#	–	–
–	–	LM+#H	–	–
–	–	LM	–	–
–	–	LH	–	–
–	–	H#	–	–
ɖʐɤ˩kɤ˥$	family name	LM+H$	**ɖʐɤ˩kɤ˧-ɻ̍˥$**	LM+H$

belonging to the same broad tonal category as the plural and associative, namely that of L-tone morphemes – keeping in mind that this tonal category serves as a first-pass label.

The surface phonological patterns for the postposition /**bi**˩/ 'on; at', shown in Table 5.18, are identical in every case to those for the plural clitic /=**ɻæ**˩/. The underlying forms for the postposition are more difficult to arrive at, for want of a handy test such as addition of the possessive /=**bv̩**˧/, which works for noun phrases containing the plural clitic /=**ɻæ**˩/ but not for locative phrases containing the postposition /**bi**˩/ 'on; at'. Given the full identity of surface phonological patterns between the clitic and the postposition, it is tempting to extrapolate from the surface phonological forms in Table 5.18 to the (hypothetical) underlying forms proposed in Table 5.21. These underlying forms are based on those obtained for the plural clitic /=**ɻæ**˩/ (Table 5.19).

Table 5.21: The behaviour of the L-tone postposition //**bi˩**// 'on; at' as interpreted on the analogy of the plural clitic. There is an additional 'L PRO' row because L-tone pronouns have exceptional behaviour.

example	tone	with /**bi˩**/	underlying tone
pig	LM	**bo˩ bi˧**	LM
leopard	LH	**ʐæ˩ bi˥**	LH
tiger	M	**lɑ˧ bi˩**	L#
sheep	L	**jo˩ bi˩˥**	L
2SG	L PRO	**no˧ bi˩**	L#
horse	H	**ʐwæ˧ bi˥**	H$ / #H
deer	MH	**ʈʂʰæ˧ bi˥**	H#
fox	M	**ɖɤ˧mi˧ bi˩**	–L
colt	#H	**ʐwæ˧zo˧ bi˥**	H$
cat	MH#	**hwɤ˧li˧ bi˥**	H$
she-cat	H$	**hwɤ˧mi˧ bi˥**	H$
dog	L	**kʰv̩˩mi˩ bi˥**	L+H$
mule	L#	**dɑ˧ʝi˩ bi˩**	L#–
wolf	LM+MH#	**õ˩dv̩˧ bi˥**	LM+H$
Naxi	LM+#H	**nɑ˩hĩ˧ bi˥**	LM+H$
sow	LM	**bo˩mi˧ bi˩**	LM–L
boar	LH	**bo˩ɬɑ˥ bi˩**	LH–
rat	H#	**hwæ˧tsɯ˥ bi˩**	H#–

5.4.2 M-tone morphemes: Agentive, dative and topic

This section presents three morphemes that carry M tone when following an M-tone noun and are therefore analyzed as having an M lexical tone.

Tables 5.22 and 5.23 present the tonal behaviour of the agentive /**ɳɯ˧**/, the dative /**-ki˧**/ (whose tonal behaviour is identical with that of the possessive, /=**bv̩˧**/), and the topic marker /**ʈʂʰɯ˧**/. This last morpheme can be hypothesized to be an extension of the third-person singular pronoun /**ʈʂʰɯ˥**/, which also serves as a proximal demonstrative, but in view of its tonal behaviour, the topic marker is considered to have M tone, as opposed to H tone for the third-person singular pronoun /**ʈʂʰɯ˥**/. Table 5.22 presents examples, and Table 5.23 the underlying tone patterns.

Table 5.22: Tone patterns of agentive /**ɳɯ˧**/, dative /**-ki˧**/, and topic marker /**ʈʂʰɯ˧**/ with monosyllabic and disyllabic nouns: examples in full.

example	tone	/**ɳɯ˧**/	/**-ki˧**/	/**ʈʂʰɯ˧**/
pig	LM	**bo˩ ɳɯ˧**	**bo˩-ki˧**	**bo˩ ʈʂʰɯ˧**
leopard	LH	**ʐæ˩ ɳɯ˥**	**ʐæ˩-ki˥**	**ʐæ˩ ʈʂʰɯ˥**
tiger	M	**lɑ˧ ɳɯ˧**	**lɑ˧-ki˧**	**lɑ˧ ʈʂʰɯ˧**
sheep	L	**jo˧ ɳɯ˧** / **jo˩ ɳɯ˥**	**jo˧-ki˧** / **jo˩-ki˥** / **jo˩-ki˩**[a]	**jo˧ ʈʂʰɯ˧** / **jo˩ ʈʂʰɯ˥**
horse	H	**ʐwæ˧ ɳɯ˩**	**ʐwæ˧-ki˧**	**ʐwæ˧ ʈʂʰɯ˧**
deer	MH	**ʈʂʰæ˧ ɳɯ˥**	**ʈʂʰæ˧-ki˥**	**ʈʂʰæ˧ ʈʂʰɯ˧** / **ʈʂʰæ˧ ʈʂʰɯ˥**
fox	M	**ɖɤ˧mi˧ ɳɯ˧**	**ɖɤ˧mi˧-ki˧**	**ɖɤ˧mi˧ ʈʂʰɯ˧**
colt	#H	**ʐwæ˧zo˧ ɳɯ˩**	**ʐwæ˧zo˧-ki˧**	**ʐwæ˧zo˧ ʈʂʰɯ˧**
cat	MH#	**hwɤ˧li˧ ɳɯ˥**	**hwɤ˧li˧-ki˥**	**hwɤ˧li˧ ʈʂʰɯ˥**
she-cat	H$	**hwɤ˧mi˥ ɳɯ˩** / **hwɤ˧mi˧ ɳɯ˥**	**hwɤ˧mi˥-ki˩**	**hwɤ˧mi˧ ʈʂʰɯ˧** / **hwɤ˧mi˧ ʈʂʰɯ˥**
dog	L	**kʰv̩˩mi˩ ɳɯ˥**	**kʰv̩˩mi˩-ki˥**	**kʰv̩˩mi˩ ʈʂʰɯ˥**
mule	L#	**dɑ˧ʝi˩ ɳɯ˩**	**dɑ˧ʝi˩-ki˩**	**dɑ˧ʝi˩ ʈʂʰɯ˩**
wolf	LM+MH#	**õ˩dv̩˧ ɳɯ˥**	**õ˩dv̩˧-ki˥**	**õ˩dv̩˧ ʈʂʰɯ˧** / **õ˩dv̩˧ ʈʂʰɯ˥**
Naxi	LM+#H	**nɑ˩hĩ˧ ɳɯ˩**	**nɑ˩hĩ˧-ki˧**	**nɑ˩hĩ˧ ʈʂʰɯ˧**
sow	LM	**bo˩mi˧ ɳɯ˧**	**bo˩mi˧-ki˧**	**bo˩mi˧ ʈʂʰɯ˧**
boar	LH	**bo˩ɬɑ˥ ɳɯ˩**	**bo˩ɬɑ˥-ki˩**	**bo˩ɬɑ˥ ʈʂʰɯ˩**
rat	H#	**hwæ˧tsɯ˥ ɳɯ˩**	**hwæ˧tsɯ˥-ki˩**	**hwæ˧tsɯ˥ ʈʂʰɯ˩**

[a] The third variant was not reported in the first edition of this book.

Table 5.23: Tone patterns of agentive /**ɳɯ˧**/, dative /**-ki˧**/, and topic marker /**ʈʂʰɯ˧**/ with monosyllabic and disyllabic nouns.

tone of noun	/**ɳɯ**/	/**-ki**/	/**ʈʂʰɯ**/
LM	L.M	L.M	L.M
LH	L.H	L.H	L.H
M	M.M	M.M	M.M
L	M.M / L.H	M.M / L.H / L.L[a]	M.M / L.H
H	M.L	M.M	M.M
MH	M.H	M.H	M.H
M	M.M.M	M.M.M	M.M.M
#H	M.M.L	M.M.M	M.M.M
MH#	M.M.H	M.M.H	M.M.H
H$	M.H.L / M.M.H	M.M.M	M.M.M
L	L.L.H	L.L.H	L.L.H
L#	M.L.L	M.L.L	M.L.L
LM+MH#	L.M.H	L.M.H	L.M.M / L.M.H
LM+#H	L.M.L	L.M.M	L.M.M
LM	L.M.M	L.M.M	L.M.M
LH	L.H.L	L.H.L	L.H.L
H#	M.H.L	M.H.L	M.H.L

[a] The third variant was not reported in the first edition of this book.

Note that the tones of /**bo˩ ɳɯ˧**/ and /**ʐæ˩ ɳɯ˥**/ are neutralized at the surface phonological level due to Rule 6 ("In tone-group-final position, H and M are neutralized to H if they follow an L tone": see §7.1.1). As elsewhere, variants are separated by a slash. The difference in tone patterns between these morphemes when associated with 'horse' was carefully verified: the tones are M.L in /**ʐwæ˧ ɳɯ˩**/, as opposed to M.M in /**ʐwæ˧-ki˧**/ and /**ʐwæ˧ ʈʂʰɯ˧**/.

An exceptional pattern is observed for /**di˩˥**/ 'earth': in addition to the expected /**di˩ ɳɯ˥**/, attested in (21), the form /**di˧ ɳɯ˧**/ is also acceptable (see 22). This variant is not acceptable for other LH-tone nouns; for instance, it is not possible to say ‡ **ʐæ˧ ɳɯ˧** for 'by (a/the) leopard'.

(21) di˩ ɳɯ˥ | ə˧-sɯ˩kv̩˩ li˩-dʑo˩-ɲi˩!
di˧˥ ɳɯ˧ ə˧-sɯ˩kv̩˩ li˧$_a$ -dʑo˧ -ɲi˩
earth A 1PL.INCL to_watch PROG CERTITUDE
'The Earth is watching us!' *(Reward.145)* {0004446#S145}

(22) "mv̩˧ ɳɯ˩ | ki˧! | di˧ ɳɯ˧ | ki˧!" | pi˧-ɲi˥ tsɯ˩ | mv̩˩.
mv̩˥ ɳɯ˧ ki˧$_a$ di˧˥ ɳɯ˧ ki˧$_a$ pi˥ -ɲi˩ tsɯ˦ mv̩˧
sky A to_give earth A to_give to_say CERTITUDE REP AFFIRM
'"It is a gift of the Heavens! It is a gift of the Earth!" he said.' *(Reward.121)* {0004446#S121}

Where two variants are possible, stylistic nuances can sometimes be pinpointed. One way to investigate such nuances is to combine information about the consultant's general preference with a detailed examination of examples in context.

Consultant preference is elicited without providing an indication of context. The investigator says one of the two alternatives while raising his right hand, then the second while raising his left hand. The consultant indicates whether both are acceptable and expresses a preference, often using an understatement: /**ʈʂʰɯ˧ bæ˧, | ɖɯ˧-pi˦ | ho˧˥**/, "This one is *pretty correct*", meaning "This one is better".

For instance, 'to the father' allows the two variants /**ə˧dɑ˧-ki˧**/ and /**ə˧dɑ˥-ki˩**/. Outside context, the former is considered better than the latter. A closer examination of the choice between /**ə˧dɑ˧-ki˧**/ and /**ə˧dɑ˥-ki˩**/ in discourse sheds light on subtle links between tone and pragmatic or stylistic effects. In example (23), 'to the father' was initially transcribed as /**ə˧dɑ˥-ki˩**/ but was later corrected to /**ə˧dɑ˧-ki˧**/.

(23) ə˧dɑ˧-ki˧, | ɬo˧pv̩˥ ti˩-bi˩-kv̩˩-ze˩ mæ˩!
ə˧dɑ˥$ -ki˧ ɬo˧pv̩˥ ti˩$_a$ -bi˧ -kv̩˦ -ze˧$_b$
father DAT kowtow to_hit IMM_FUT ABILITIVE PFV
mæ˧
DISC.PTCL_OBVIOUSNESS
'[The young girl] kowtows before her father, of course [just as she did before the members of the household in which she grew up: her mother's household].' *(ComingOfAge2.18)* {0004589#S18}

The context of (23) is a discussion of the respective roles of the maternal uncle and the father.

During the rite of passage to adulthood, the uncle presides over the ceremony. The father, well, we know who he is, but that doesn't really matter: it's the uncle who counts in the family, he's the one who is treated with the most consideration. When you turn thirteen, the ceremony takes place at home. The uncle is the one who takes care of everything, who keeps the whole household going. *(ComingOfAge2.4-6)* {0004589#S4}

In (23), the father is mentioned in a somewhat parenthetical manner: on the all-important occasion of a child's coming-of-age ceremony, he is not forgotten, but he is not among the central figures. The visit to the father's family is secondary to the main part of the ceremony, which takes place after all due rites have been completed at home – that is, at the mother's home. The choice of /**ə˧dɑ˧-ki˧**/ in this relatively dispassionate context highlights its linguistic nature as a form exhibiting a high degree of morphosyntactic integration, befitting carefully planned and structured speech. In /**ə˧dɑ˧-ki˧**/, 'father' (//**ə˧dɑ˥$**//) is tonally amalgamated with its suffix, forming a neatly packaged unit that can, in turn, be woven into the mesh of a larger syntactic structure. The M tone of the phrase allows for a range of diverse outcomes in morphotonological combinations.

By contrast, the form /**ə˧dɑ˥-ki˩**/ places subtle emphasis on the noun. Here, 'father' surfaces with the same tone it would have in isolation: /**ə˧dɑ˥**/ (lexical form: //**ə˧dɑ˥$**//). This makes the noun stand out, as if extracting it from the flow of speech. Moreover, from a phonostylistic point of view, the higher pitch on the syllable meaning 'father' (H tone, as opposed to M tone in /**ə˧dɑ˧-ki˧**/) has iconic value: cross-linguistically, raised pitch tends to be associated with emphasis in both tonal and non-tonal languages (see §9.2).

The High tone has culminative value in Yongning Na phonology: it can only be followed by L tones (by rules 4 and 5). This phonological constraint means that /**ə˧dɑ˥-ki˩**/ precludes any following tones other than L within the tone group. While this restriction is purely phonological in origin, its effect contributes to the phonostylistic prominence of the H-tone syllable, in this case, the main syllable of the word 'father'. For these reasons, the variant with a M.H.L tone pattern (/**ə˧dɑ˥-ki˩**/) is more suited to emotional, emphatic speech and is thus less appropriate in the context of (23), which constitutes a low-key, parenthetical statement.

These reflections are coherent with the speaker's general preference for /**ə˧dɑ˧-ki˧**/ over /**ə˧dɑ˥-ki˩**/, explaining why the former is perceived as the "better" form when assessed in isolation, outside a specific discourse context.

These observations illustrate that phonological variation can reflect subtle distinctions in communicative intent and discourse structure. Tonal variation in Na often serves pragmatic and discourse-related functions. Across the corpus, speakers' phonological choices among possible variants, as well as fine details in the phonetic realization of tones, allow them to convey nuanced effects: to highlight a referent, reinforce contrast, or mark a shift in discourse focus – or conversely, to background certain pieces of information. Such cases exemplify, in a language-specific way, broader cross-linguistic patterns in the use of intonation for prominence (information structuring) and phrasing (see §9.2 for further discussion).

In association with the agentive /**ɳɯ**˧/, the dative /**-ki**˧/, and the topic marker /**ʈʂʰɯ**˧/, the 1st- and 2nd-person pronouns behave like other L-tone items: /**njɤ**˧ **ɳɯ**˧/, /**no**˧ **ɳɯ**˧/. On the other hand, the demonstratives /**ʈʂʰɯ**˥/ (proximal, also 3rd-person singular) and /**tʰv̩**˥/ (distal) pattern differently from other H-tone items. They yield M.M in association with /**ɳɯ**˧/: /**ʈʂʰɯ**˧ **ɳɯ**˧/ and /**tʰv̩**˧ **ɳɯ**˧/ (also /**ʈʂʰɯ**˧ **lɑ**˧/, /**tʰv̩**˧ **lɑ**˧/ 'this/that one too').

5.4.3 The MH-tone morphemes /-qɑ˧˥/ (dative/comitative) and /gi˧˥/ 'behind'

The morpheme /**-qɑ**˧˥/ has both dative and comitative functions. It is analyzed as carrying MH tone based on its behaviour after M-tone words, whether monosyllabic or disyllabic, as well as after LM-tone disyllables. Table 5.24 sets out the data. The postposition /**gi**˧˥/ 'behind' follows an identical pattern.[4]

As in all other morphosyntactic contexts, the //L.H.L// pattern is neutralized with //L.M.L// at the surface phonological level. Thus, the tone pattern for 'boar' could be transcribed as /L.M.L/ in surface phonological representation. However, notation as //L.H.L// is retained here to reflect the analysis that the enclitic's tone is lowered to L due to the presence of a preceding H tone – specifically, the H component of the LH tone pattern lexically attached to the noun 'boar'.

An alternative notation as /L.M.L/ might seem preferable for remaining closer to surface phonological form and limiting abstraction. However, this could create confusion by suggesting that the lexical category //LH// is restructured as //LM// when the enclitic is added. To avoid this misleading implication, notation as L.H.L is adopted in Table 5.24.

[4] By convention, clitics are preceded by an *equals* sign, suffixes are preceded by a hyphen, prefixes are followed by a hyphen, and postpositions are written as free morphemes.

Table 5.24: The tonal behaviour of the dative/comitative marker /**-qɑ˦**/ following nouns.

example	tone	example	tone pattern
pig	LM	**bo˩-qɑ˧**	L.M (neutralized with LH on surface)
leopard	LH	**zæ˩-qɑ˥**	L.H (neutralized with LH on surface)
tiger	M	**lɑ˧-qɑ˦**	M.MH
sheep	L	**jo˩-qɑ˩˥**	L.LH
horse	H	**zwæ˧-qɑ˩**	M.L
deer	MH	**ʈʂʰæ˧-qɑ˥**	M.H
fox	M	**ɖɤ˧mi˧-qɑ˦**	M.M.MH
colt	#H	**zwæ˧zo˧-qɑ˩**	M.M.L
cat	MH#	**hwɤ˧li˧-qɑ˥**	M.M.H
she-cat	H$	**hwɤ˧mi˥-qɑ˩**	M.H.L
dog	L	**kʰv̩˩mi˩-qɑ˥**	L.L.H
mule	L#	**dɑ˧ʝi˩-qɑ˩**	M.L.L
wolf	LM+MH#	**õ˩dv̩˧-qɑ˥**	L.M.H
Naxi	LM+#H	**nɑ˩hĩ˧-qɑ˩**	L.M.L
sow	LM	**bo˩mi˧-qɑ˦**	L.M.MH
boar	LH	**bo˩ɬɑ˥-qɑ˩**	L.H.L
rat	H#	**hwæ˧tsɯ˥-qɑ˩**	M.H.L

5.4.4 The H-tone topic marker /-dʑo˥/, with observations about tonal contours in non-final position

The topic marker /**-dʑo**˥/ can appear after both nouns and verbs. The tone patterns it yields when associated with a noun are shown in Table 5.25. Interestingly, an MH-tone noun or verb preceding the topic marker is realized with an MH contour, e.g. /**ʈʂʰæ**˦**-dʑo**˩/ 'as for the deer' (from /**ʈʂʰæ**˦/ 'deer') and /**mɤ**˧**-lɑ**˦**-dʑo**˩/ 'as [he/she] did not strike' (from /**lɑ**˦/ 'to strike'). This suggests that there is a tone group boundary following the noun or verb, since tonal contours are realized only in tone-group-final position.

Such behaviour would not be unparalleled. For instance, the contrastive topic marker /**-no**˦/ and the word /**tʰi**˩˥/ 'then' systematically mark the beginning of a new tone group. But the topic marker's tonal behaviour when following a noun or verb with a tone other than MH suggests that, in these cases, it *is* integrated within the same tone group. For instance, after an M-tone noun or verb, the

Table 5.25: The tonal behaviour of the topic marker /**-dʑo˥**/ following nouns.

example	tone	example	tone pattern
pig	LM	**bo˩˧-dʑo˩**	LM.L
leopard	LH	**ʐæ˩˥-dʑo˩**	LH.L
tiger	M	**lɑ˧-dʑo˥**	M.H
daughter	L	**mv̩˩-dʑo˩˥**	LH.L
horse	H	**ʐwæ˧-dʑo˩**	M.L
deer	MH	**ʈʂʰæ˧˥-dʑo˩**	MH.L
fox	M	**ɖɤ˧mi˧-dʑo˥ / ɖɤ˧mi˧-dʑo˧**	M.M.H / M.M.M
colt	#H	**ʐwæ˧zo˧-dʑo˩**	M.M.L
cat	MH#	**hwɤ˧li˧˥-dʑo˩**	M.MH.L
she-cat	H$	**hwɤ˧mi˥-dʑo˩**	M.H.L
dog	L	**kʰv̩˩mi˩˥-dʑo˩**	L.LH.L
mule	L#	**dɑ˧ʝi˩-dʑo˩**	M.L.L
wolf	LM+MH#	**õ˩dv̩˧˥-dʑo˩**	L.MH.L
Naxi	LM+#H	**nɑ˩hĩ˧-dʑo˩**	L.M.L
sow	LM	**bo˩mi˧-dʑo˥**	L.M.H
boar	LH	**bo˩ɬɑ˧-dʑo˩**	L.M.L
rat	H#	**hwæ˧tsɯ˥-dʑo˩**	M.H.L

pattern is (M…)M.H, as in /**lɑ˧-dʑo˥**/ ‘as for the tiger’ and /**mɤ˧-hwæ˧-dʑo˥**/ ‘as [she/he] does not buy’. These expressions clearly form a single tone group. The full dataset is presented in Table 5.25.[5]

These observations are taken up in §7.3, as part of the discussion on breaches of tonal grouping: how non-final syllables can come to bear a contour while subsequent syllables become extrametrical.

[5]The data was verified by using additional nouns illustrating the tonal categories: /**zo˥**/ ‘son’ for H tone, /**õ˧˥**/ ‘oneself’ for MH tone, /**pʰɤ˧bɤ˧**/ ‘gift’ for M tone, and /**ə˧dɑ˥$**/ ‘father’ and /**mv̩˧ʁo˥$**/ ‘heavens’ for H$ tone. The results were consistent, including the occurrence of variant patterns: for ‘gift’, two variants were observed, /**pʰɤ˧bɤ˧-dʑo˧**/ and /**pʰɤ˧bɤ˧-dʑo˥**/, mirroring the variation found with ‘fox’ in Table 5.25. The only unexpected result was with ‘plain’, //**di˧qo˧**//. On the analogy of /**ɖɤ˧mi˧-dʑo˥**/, one would expect † **di˧qo˧-dʑo˥**, yet the observed pattern was M.M.M: /**di˧qo˧-dʑo˧**/. This unexpected result, confirmed across elicitation sessions, may have to do with the internal structure of this disyllable, which literally means ‘on earth’.

On the basis of its behaviour after M-tone nouns (and after M-tone verbs, which will be presented in the next chapter), the topic marker is provisionally analyzed as carrying a lexical H tone. If this tonal identification is confirmed, the morpheme constitutes an extreme case of distance between underlying form and surface forms: in texts, realizations with an L tone outnumber those with an H tone by a ratio of approximately 10 to 1. (Even more striking cases include the reported speech particle /**tsɯ˦**/ and the affirmation particle /**mv̩˧**/, whose underlying tones hardly ever surface as such: see 7.1.4.)

5.5 Disyllabic postpositions

Disyllabic locative postpositions include /**ʁo˧tʰo˩**/ 'behind', /**ʁo˧dɑ˧**/ 'in front of', /**ɬo˧tɑ˧**/ 'beside, to the side of', /**ʁwæ˧gi#˥**/ 'to the left', /**jo˩gi˩**/ 'to the right', and /**-qo˧lo˩**/ 'inside'. (The monosyllabic form /**-qo˧**/ is also attested, with the same meaning.) Table 5.26 presents the data for nouns followed by the locative postpositions /**ɬo˧tɑ˧**/ 'beside, to the side of' and /**ʁo˧tʰo˩**/ 'behind, to the back of'. The tonal behaviour of /**ʁo˧dɑ˧**/ 'in front of' is identical with that of /**ɬo˧tɑ˧**/ 'beside, to the side of' and is therefore omitted from the table. Similarly, /**ʈʰæ˧qo˩**/ 'under' behaves identically to /**ʁo˧tʰo˩**/ 'behind'. Table 5.27 provides the data for the locative postpositions /**ʁwæ˧gi#˥**/ 'to the left' and /**jo˩gi˩**/ 'to the right'. The corresponding recording is: *LocativePostp* {0004563}.

In both Table 5.26 and Table 5.27, two rows are provided for the L tone because L-tone pronouns, 1SG /**ɲjɤ˩**/ and 2SG /**no˩**/, have exceptional tonal behaviour. The difference in tonal output between pronouns and nouns is clear. For instance, with the M-tone postposition 'beside', an L-tone pronoun yields an M-tone pattern: /**no˧ ɬo˧tɑ˧**/, whereas ‡ **no˩ ɬo˩tɑ˩** is not possible. Conversely, an L-tone noun yields an L-tone pattern: /**jo˩ ɬo˩tɑ˩**/, while it is not possible to say ‡ **jo˧ ɬo˧tɑ˧**.

The system would look nice and economical if the tonal behaviour of these postpositions were identical with those for other constructions, such as determinative compounds. This is indeed the case for /**ʁo˧tʰo˩**/ 'at the back', which patterns tonally in the same way as an L#-tone head noun in determinative compounds, and for /**ɬo˧tɑ˧**/ 'beside, to the side of', which behaves like an M-tone noun does in determinative compounds. However, not all locative postpositions share this behaviour. Compare /**hwɤ˧li˧-hi˧kʰɯ˦**/ 'cat's gums (body part)' (input tones: MH# and #H; output tone: MH#) and /**hwɤ˧li˧ ʁwæ˧gi#˥**/ 'to the left of the cat' (same input; output tone: #H).

Table 5.26: The tonal behaviour of the locative postpositions /**ɬo˧tɑ˧**/ 'beside' and /**ʁo˧tʰo˩**/ 'behind'. There is an additional 'L PRO' row because L-tone pronouns have exceptional behaviour.

tone	example	meaning	beside	behind
LM	**bo˩˧**	pig	**bo˩ ɬo˧tɑ˧**	**bo˩ ʁo˥tʰo˩**
LH	**ʐæ˩˥**	leopard	**ʐæ˩ ɬo˧tɑ˧**	**ʐæ˩ ʁo˥tʰo˩**
M	**lɑ˧**	tiger	**lɑ˧ ɬo˧tɑ˧**	**lɑ˧ ʁo˧tʰo˩**
L	**jo˩**	sheep	**jo˩ ɬo˩tɑ˩**	**jo˩ ʁo˩tʰo˥**
L PRO	**no˩**	2SG	**no˧ ɬo˧tɑ˧**	**no˧ ʁo˧tʰo˩**
H	**ʐwæ˥**	horse	**ʐwæ˧ ɬo˧tɑ˥**	**ʐwæ˧ ʁo˧tʰo˥**
MH	**ʈʂʰæ˧˥**	deer	**ʈʂʰæ˧ ɬo˧tɑ˥**	**ʈʂʰæ˧ ʁo˧tʰo˥**
M	**ɖɤ˧mi˧**	fox	**ɖɤ˧mi˧ ɬo˧tɑ˧**	**ɖɤ˧mi˧ ʁo˧tʰo˩**
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ ɬo˧tɑ˥**	**ʐwæ˧zo˧ ʁo˧tʰo˥**
MH#	**hwɤ˧li˧˥**	cat	**hwɤ˧li˧ ɬo˧tɑ˥**	**hwɤ˧li˧ ʁo˧tʰo˥**
H$	**hwɤ˧mi˥$**	she-cat	**hwɤ˧mi˧ ɬo˧tɑ˥**	**hwɤ˧mi˧ ʁo˧tʰo˥**
L	**kʰv̩˩mi˩**	dog	**kʰv̩˩mi˩ ɬo˩tɑ˥**	**kʰv̩˩mi˩ ʁo˩tʰo˥**
L#	**dɑ˧ʝi˩**	mule	**dɑ˧ʝi˩ ɬo˩tɑ˩**	**dɑ˧ʝi˩ ʁo˩tʰo˩**
LM+MH#	**õ˩dv̩˧˥**	wolf	**õ˩dv̩˧ ɬo˧tɑ˥**	**õ˩dv̩˧ ʁo˧tʰo˥**
LM+#H	**nɑ˩hĩ#˥**	Naxi	**nɑ˩hĩ˧ ɬo˧tɑ˥**	**nɑ˩hĩ˧ ʁo˧tʰo˥**
LM	**bo˩mi˧**	sow	**bo˩mi˧ ɬo˧tɑ˧**	**bo˩mi˧ ʁo˧tʰo˩**
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˥ ɬo˩tɑ˩**	**bo˩ɬɑ˧ ʁo˩tʰo˩**
H#	**hwæ˧tsɯ˥**	rat	**hwæ˧tsɯ˥ ɬo˩tɑ˩**	**hwæ˧tsɯ˥ ʁo˩tʰo˩**

Table 5.27: The tonal behaviour of the locative postpositions /**ʁwæ˧gi#˥**/ 'to the left' and /**jo˩gi˩**/ 'to the right'. There is an additional 'L PRO' row because L-tone pronouns have exceptional behaviour.

tone	example	meaning	to the left of	to the right of
LM	**bo˩˧**	pig	**bo˩ ʁwæ˧gi#˥**	**bo˩ jo˧gi#˥**
LH	**ʐæ˩˥**	leopard	**ʐæ˩ ʁwæ˧gi#˥**	**ʐæ˩ jo˧gi#˥**
M	**lɑ˧**	tiger	**lɑ˧ ʁwæ˧gi#˥**	**lɑ˧ jo˩gi˩**
L	**jo˩**	sheep	**jo˩ ʁwæ˩gi˩**	**jo˧ jo˩gi˩**
L PRO	**no˩**	2SG	**no˧ ʁwæ˧gi#˥**	**no˧ jo˩gi˩**
H	**ʐwæ˥**	horse	**ʐwæ˧ ʁwæ˧gi#˥**	**ʐwæ˧ jo˥gi˩**
MH	**ʈʂʰæ˧˥**	deer	**ʈʂʰæ˧ ʁwæ˧gi#˥**	**ʈʂʰæ˧ jo˥gi˩**
M	**ɖɤ˧mi˧**	fox	**ɖɤ˧mi˧ ʁwæ˧gi#˥**	**ɖɤ˧mi˧ jo˩gi˩**
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ ʁwæ˧gi#˥**	**ʐwæ˧zo˧ jo˥gi˩**
MH#	**hwɤ˧li˧˥**	cat	**hwɤ˧li˧ ʁwæ˧gi#˥**	**hwɤ˧li˧ jo˥gi˩**
H$	**hwɤ˧mi˥$**	she-cat	**hwɤ˧mi˧ ʁwæ˧gi#˥**	**hwɤ˧mi˧ jo˥gi˩**
L	**kʰv̩˩mi˩**	dog	**kʰv̩˩mi˩ ʁwæ˩gi˩**	**kʰv̩˩mi˩ jo˥gi˩**
L#	**dɑ˧ʝi˩**	mule	**dɑ˧ʝi˩ ʁwæ˩gi˩**	**dɑ˧ʝi˩ jo˩gi˩**
LM+MH#	**õ˩dv̩˧˥**	wolf	**õ˩dv̩˧ ʁwæ˧gi#˥**	**õ˩dv̩˧ jo˥gi˩**
LM+#H	**nɑ˩hĩ#˥**	Naxi	**nɑ˩hĩ˧ ʁwæ˧gi#˥**	**nɑ˩hĩ˧ jo˥gi˩**
LM	**bo˩mi˧**	sow	**bo˩mi˧ ʁwæ˧gi#˥**	**bo˩mi˧ jo˩gi˩**
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˧ ʁwæ˩gi˩**	**bo˩ɬɑ˧ jo˩gi˩**
H#	**hwæ˧tsɯ˥**	rat	**hwæ˧tsɯ˥ ʁwæ˩gi˩**	**hwæ˧tsɯ˥ jo˩gi˩**

5.6 Adverbs

Despite their name, *adverbs* – a loosely defined class of words – can occur not only with verbs, but also with nouns and other linguistic units.

5.6.1 The homophonous adverbs /**lɑ˧**/ 'only' and 'too, and'

The tonal behaviour of the two homophonous adverbs /**lɑ˧**/ 'only' and 'too, and' is presented in Table 5.28. Eliciting expressions consisting of a noun followed by these adverbs is not easy, as such sequences do not constitute complete utterances. One illustration of the issue is the expression 'only (a/the) she-cat', which was initially recorded with an M.M.L pattern, /**hwɤ˧mi˧ lɑ˩**/, and later with an M.M.M pattern, /**hwɤ˧mi˧ lɑ˧**/. The consultant subsequently pointed out that both were incorrect and that the correct pattern was M.H.L: /**hwɤ˧mi˥ lɑ˩**/. The homophony between this phrase and 'to beat (a/the) she-cat' (likewise /**hwɤ˧mi˥ lɑ˩**/) may have contributed to the difficulty encountered at elicitation.

Table 5.28: The behaviour of /**lɑ˧**/ 'only; also' with monosyllabic and disyllabic nouns.

example	tone	example	abstract tone pattern
pig	LM	**bo˩ lɑ˧**	L.M
leopard	LH	**ʐæ˩ lɑ˥**	L.H
tiger	M	**lɑ˧ lɑ˧**	M.M
sheep	L	**jo˩ lɑ˥**	L.H
horse	H	**ʐwæ˧ lɑ˩**	M.L
deer	MH	**ʈʂʰæ˧ lɑ˥**	M.H
fox	M	**ɖɤ˧mi˧ lɑ˧**	M.M.M
colt	#H	**ʐwæ˧zo˧ lɑ˩**	M.M.L
cat	MH#	**hwɤ˧li˧ lɑ˥**	M.M.H
she-cat	H$	**hwɤ˧mi˥ lɑ˩**	M.H.L
dog	L	**kʰv̩˩mi˩ lɑ˥**	L.L.H
mule	L#	**dɑ˧ʝi˩ lɑ˩**	M.L.L
wolf	LM+MH#	**õ˩dv̩˧ lɑ˥**	L.M.H
Naxi	LM+#H	**nɑ˩hĩ˧ lɑ˩**	L.M.L
sow	LM	**bo˩mi˧ lɑ˧**	L.M.M
boar	LH	**bo˩ɬɑ˥ lɑ˩**	L.H.L
rat	H#	**hwæ˧tsɯ˥ lɑ˩**	M.H.L

Comparison with the three M-tone morphemes in Table 5.23 reveals a striking degree of similarity: the tonal behaviour of /**lɑ˧**/, /**ɳɯ˧**/, /**-ki˧**/, and /**ʈʂʰɯ˧**/ is identical for eleven of the seventeen noun categories. The only observed difference between /**lɑ˧**/ and /**ɳɯ˧**/ is in combination with L-tone monosyllables: /**lɑ˧**/ follows an L.H pattern, as in (24), whereas /**ɳɯ˧**/ follows an M.M pattern, as in (25). The data was carefully verified across work sessions. (A recording is available: *OnlyAnd* {0004569}.)

(24) jo˩ lɑ˥ | tsʰɯ˦
jo˩ lɑ˧ tsʰɯ˦
sheep and goat
'sheep and goats'

(25) jo˧ ɳɯ˧
jo˩ ɳɯ˧
sheep A
'by the sheep'

To venture a hypothesis concerning these partial similarities, it seems plausible that grammaticalization is accompanied by a tonal evolution that departs from the tone of the free root. As noted in §5.4, it may not be coincidental that the dative /**-ki˧**/ and the possessive /=**bv̩˧**/, which have identical tonal behaviour, also share the morphosyntactic property of being "almost suffixal" (Lidz 2010: 155). This distinguishes them from the agentive /**ɳɯ˧**/, analyzed as a case adposition. Viewed in this light, the differences in tone patterns may align with differences in morphosyntactic category (part of speech): one pattern is associated with suffixes, another with adpositions, a third with adverbs and conjunctions, and a fourth with discourse particles (in this case, the topic marker).

5.6.2 The adverb /pɤ˧to˩/ 'even'

The tonal behaviour of the adverb /**pɤ˧to˩**/ 'even' is presented in Table 5.29; the corresponding recording is *NounsEven* {0004565}. This data differs from that of L#-tone locative postpositions such as /**ʁo˧tʰo˩**/ 'behind' and /**ʈʰæ˧qo˩**/ 'under', discussed in 5.5, as well as from that of L#-tone heads in compound nouns.

5.7 Concluding note

The phenomena approached in this chapter are more diverse than those examined in the preceding three. Yongning Na exhibits great variety in the tonal

Table 5.29: The tonal behaviour of /**pɤ˧to˩**/ 'even'. There is an additional 'L PRO' row because L-tone pronouns have exceptional behaviour.

tone	example	meaning	N+/**pɤ˧to˩**/ 'even'
LM	**bo˩˧**	pig	**bo˩ pɤ˥to˩**
LH	**ʐæ˩˥**	leopard	**ʐæ˩ pɤ˥to˩**
M	**lɑ˧**	tiger	**lɑ˧ pɤ˧to˩**
L	**jo˩**	sheep	**jo˩ pɤ˩to˥**
L PRO	**no˩**	2SG	**no˧ pɤ˧to˩**
#H	**ʐwæ˥**	horse	**ʐwæ˧ pɤ˧to˩**
MH#	**ʈʂʰæ˧˥**	deer	**ʈʂʰæ˧ pɤ˥to˩**
M	**ɖɤ˧mi˧**	fox	**ɖɤ˧mi˧ pɤ˧to˩**
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ pɤ˧to˩**
MH#	**hwɤ˧li˧˥**	cat	**hwɤ˧li˧ pɤ˥to˩**
H$	**hwɤ˧mi˥$**	she-cat	**hwɤ˧mi˧ pɤ˥to˩**
L	**kʰv̩˩mi˩**	dog	**kʰv̩˩mi˩ pɤ˥to˩**
L#	**dɑ˧ʝi˩**	mule	**dɑ˧ʝi˩ pɤ˩to˩**
LM+MH#	**õ˩dv̩˧˥**	wolf	**õ˩dv̩˧ pɤ˥to˩**
LM+#H	**nɑ˩hĩ#˥**	Naxi	**nɑ˩hĩ˧ pɤ˧to˩**
LM	**bo˩mi˧**	sow	**bo˩mi˧ pɤ˧to˩**
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˧ pɤ˧to˩**
H#	**hwæ˧tsɯ˥**	rat	**hwæ˧tsɯ˥ pɤ˩to˩**

behaviour of grammatical elements: the dative suffix /**-ki˧**/ and the possessive clitic /**-bv̩˧**/ behave differently from the agentive adposition /**ɳɯ˧**/; a third pattern is found with the conjunction /**lɑ˧**/ 'too, and'; and yet another with the topic marker /**ʈʂʰɯ˧**/. This morphotonological complexity arises from the large number of distinct tonal paradigms rather than from the internal complexity of any single paradigm.

There remains much room for further progress in the description and analysis of this morphotonological archipelago. In particular, systematic study of the lexicon holds potential for uncovering traces of earlier morphological derivations, shedding light on historical processes. For instance, an animal suffix /**-li**/ may plausibly be reconstructed in words such as /**hwɤ˧li˧˥**/ 'cat' and /**pʰi˧li˩**/ 'butterfly', based on cognates in Naxi (/**hwɑ˥le˧**/ and /**pʰe˧le˩**/).

6 Verbs and their combinatorial properties

This chapter discusses the tones of verbs and adjectives, and their combinatorial properties.

6.1 The lexical tones of verbs

6.1.1 Overview

Among monosyllabic verbs, which constitute an overwhelming majority, seven tonal categories have been identified, as shown in Table 6.1.

Table 6.1: The seven tonal categories of monosyllabic verbs: behaviour in four different contexts.

example	in isolation	NEG	ACCOMP	V+‘a bit’
dzɯ ‘to eat’	M	M.H	M.H	M.M.M
hwæ ‘to buy’		M.M	M.M	M.H.L
tɕʰi ‘to sell’				M.M.M
bi ‘to go’			M.L	N/A
dze ‘to cut’	LH	M.L		M.M.H
ʈʰɯ ‘to drink’				M.M.MH
lɑ ‘to strike’	MH	M.MH	M.MH	M.H.L

The four contexts shown in Table 6.1 are: (i) in isolation, (ii) with the NEGATION prefix, (iii) with the ACCOMPLISHED prefix, and (iv) with **/ɖɯ˧-kʰwɤ˥$/** ‘a piece’ or **/ɖɯ˧-ʈʰɤ˥$/** ‘a drop’ as an object (yielding the meaning ‘to V a bit’).[1]

[1] The two numeral-plus-classifier phrases ‘a piece’ and ‘a drop’ share the same tone: H$. The tone patterns of numeral-plus-classifier phrases are analyzed in Chapter 3, and the H$ tone is discussed in §2.3.5.

These four contexts were chosen because their combination reveals all seven categories, whereas each individual context distinguishes only three or four. Instances of neutralization are reflected in cells that share the same tone pattern within a given column; adjacent neutralizations are marked by a dashed box, while nonadjacent ones are indicated by shading.

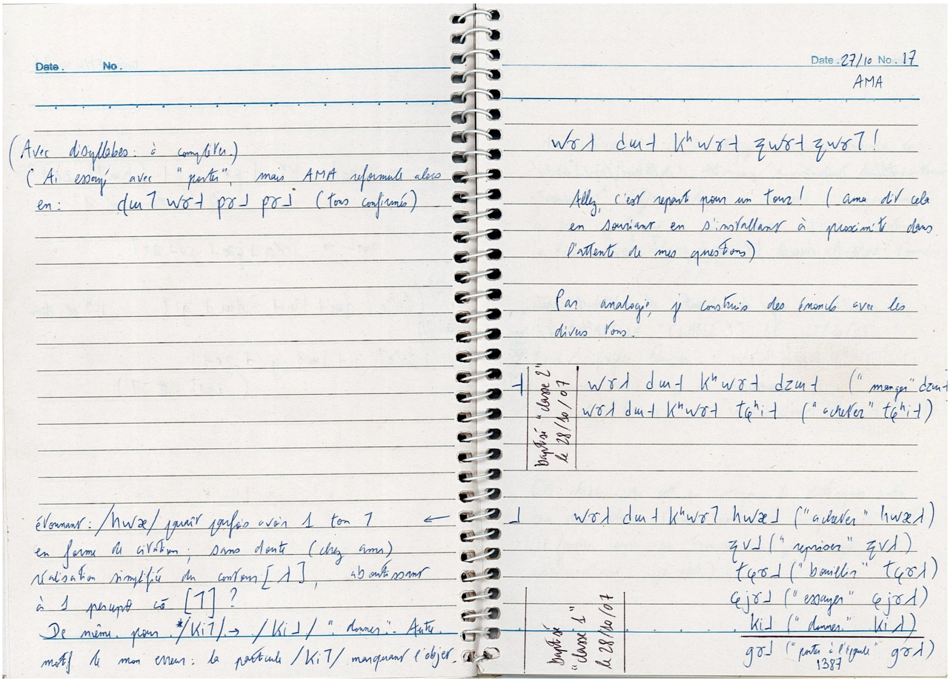


Figure 6.1: Field notes (2007). At the beginning of a work session, the consultant cheerfully said /**wɤ˧˥ | ɖɯ˧-kʰwɤ˧ ʐwɤ˧~ʐwɤ˥**/ ‘Let’s have another chat!’ This construction, ‘to V a bit’, was written down, and subsequently elicited with various verbs. It turned out to be a useful diagnostic tool for the tones of verbs.

While the data in Table 6.1 establishes the existence of seven distinct tonal categories, it allows for several analytic interpretations. Insights from the study of the tonal categories of nouns (Chapter 2) suggest that tonal diversity is greatest after an initial M tone, since this position permits the full range of H, M, L, and MH (with only LM and LH excluded by Rule 5, discussed in §7.1.1). The behaviour of the seven tonal classes of verbs following an M-tone prefix thus appeared as a promising source of evidence to establish their underlying tone. However, a complication arises: not all M-tone prefixes yield the same tonal outcomes when combined with verbs. Specifically, the verb ‘to go’ yields /**mɤ˧-bi˧**/ with the NEGATION prefix (M.M) but /**le˧-bi˩**/ with the ACCOMPLISHED prefix (M.L). Examination of a set of M-tone prefixes, such as the PROHIBITIVE /**tʰɑ˧**-/ and the DURATIVE /**tʰi˧**-/, revealed that the ACCOMPLISHED prefix is an outlier in this respect. The NEGATION was selected as the reference context for determining the phonological nature of the tones of verbs.

Observation of verbs in this context revealed four distinct surface tones: H, M, L, and MH. These were interpreted as four primary tonal categories. Within these, subcategories were identified based on their different behaviour in other contexts. Among M-tone verbs, three subcategories were distinguished: M_a, M_b, and M_c. Similarly, two subcategories were identified within the L-tone category: L_a and L_b. Thus, the seven tonal categories represented in Table 6.1 are labelled as H, M_a, M_b, M_c, L_a, L_b, and MH. In light of this additional layer of analysis, the data in Table 6.2 are reproduced in Table 6.2 with an indication of the underlying tones in its first column.

Table 6.2: The seven tonal categories of monosyllabic verbs: analysis into H, M, L, and LH tones.

tone	example	in isolation	NEG	ACCOMP	V+'a bit'
H	**dzɯ˥** 'to eat'	M	M.H	M.H	M.M.M
M_a	**hwæ˧**$_{\mathbf{a}}$ 'to buy'		M.M	M.M	M.H.L
M_b	**tɕʰi˧**$_{\mathbf{b}}$ 'to sell'				M.M.M
M_c	**bi˧**$_{\mathbf{c}}$ 'to go'			M.L	N/A
L_a	**dze˩**$_{\mathbf{a}}$ 'to cut'	LH	M.L		M.M.H
L_b	**ʈʰɯ˩**$_{\mathbf{b}}$ 'to drink'				M.M.MH
MH	**lɑ˧˥** 'to strike'	MH	M.MH	M.MH	M.H.L

Realizations in isolation, which distinguish only three subsets, make sense in light of this analysis.

- H tone is realized as M due to the neutralization of H and M in tone-group-initial position (Rule 3; see the list of phonological tone rules in §7.1.1).
- M tones (M_a, M_b, and M_c) are straightforwardly realized as M.
- L tones (L_a and L_b) are both realized as LH due to the postlexical addition of an H tone to all-L tone groups (by Rule 7).
- MH tone is straightforwardly realized as MH.

L-tone verbs thus behave unlike L-tone nouns, which surface with M tone in isolation, as explained in Chapter 2. This is one of many pieces of evidence showing that the tone system of Yongning Na is not based solely on a set of phonological rules that apply uniformly in all contexts but also includes morphotonological rules: tone rules that are specific to a given morphological context.

Tones M_a and M_b yield the same tone pattern in association with the negation prefix, as do L_a and L_b, but they are distinguished in the fourth context. Conversely, the tone pairs {M_a, MH} and {M_b, H} yield the same tonal pattern when associated with the object 'a piece'/'a drop' but are distinguished following the negation prefix. Finally, tone category M_c behaves differently from M_a and M_b after the ACCOMPLISHED prefix.

Examples of predicates belonging to the seven categories are presented in Table 6.3.

Table 6.3: Examples of the seven categories of verbs.

tone	examples
H	**dzɯ˥** 'to eat', **bv̩˥** 'to divide', **ʝi˥** 'to do', **se˥** 'to walk', **ʈʂʰæ˥** 'to wash'
M_a	**hwæ˧**$_a$ 'to buy', **hõ˧**$_a$ 'to go away.IMP', **ki˧**$_a$ 'to give', **li˧**$_a$ 'to watch', **mæ˧**$_a$ 'to catch hold of'
M_b	**tɕʰi˧**$_b$ 'to sell', **ɖɯ˧**$_b$ 'to obtain', **ɖʐæ˧**$_b$ 'to ride', **pʰæ˧**$_b$ 'to fasten', **ɲi˧**$_b$ 'to need'
M_c	**bi˧**$_c$ 'to go', **hɯ˧**$_c$ 'to go.PST', **gv̩˧**$_c$ 'to go by (of time)', **ʝi˧**$_c$ 'to come', **pv̩˧**$_c$ 'to chant', **pɻ̩˧**$_c$ 'to go out, to get out'
L_a	**dze˩**$_a$ 'to cut', **bæ˩**$_a$ 'to sweep', **ti˩**$_a$ 'to hit (gently)', **tɕi˩**$_a$ 'to write', **dzi˩**$_a$ 'to sit'
L_b	**ʈʰɯ˩**$_b$ 'to drink', **dɑ˩**$_b$ 'to weave', **do˩**$_b$ 'to see', **mɤ˩**$_b$ 'to eat food in powder form, typically tsamba (roasted flour)', **ʐwɤ˩**$_b$ 'to speak'
MH	**ɕjɤ˧˥** 'to try; to taste', **gɤ˧˥** 'to carry on one's shoulder', **lɑ˧˥** 'to strike', **tɕɤ˧˥** 'to boil', **ʐv̩˧˥** 'to sew (clothes)'

The labels used for the sets of categories {L_a, L_b} and {M_a, M_b, M_c} are deliberately abstract, for want of decisive evidence regarding the phonological nature of the categories at issue. The letters are assigned on the basis of relative frequency in the lexicon. Among L tones, the 'to cut' type is about five times as frequent as the 'to drink' type. Among M tones, the 'to buy' type is twice as frequent as the 'to sell' type, while the 'to go' type is infrequent.

The 'to cut' and 'to drink' types, labelled here as L_a and L_b, are both analyzed as containing an L tone level, since they are realized with L tone after the negation prefix. However, evidence on the phonological nature of the distinction between L_a and L_b remains limited. A theoretical possibility would be to analyze one of the two as a simple L tone, and the other as a contour (LM or LH), on the analogy of nouns, but such an analysis would be arbitrary, as there is no compelling evidence that either of these categories consists of a contour. The apparent economy gained by adopting labels similar to those used for nouns would come at a high cost in terms of descriptive adequacy. In nouns, the //LM// and //LH// contour tones surface as such in isolation (where they are neutralized to /LH/), whereas the //L// tone surfaces as /M/ in isolation. By contrast, no such difference is found between the L_a and L_b categories of verbs, both of which surface with a /LH/ contour in isolation.

6.1.2 About subsets of M-tone verbs

M-tone verbs, defined as those that are realized with M tone after the negation prefix, fall into three subsets: M_a, M_b, and M_c. From the standpoint of descriptive economy, it would be appealing to reserve the label M for one of these three and to assign the others labels drawn from the inventory of tone categories of nouns, such as #H. However, no evidence has been found to support such identifications. Hence, the choice was made to adopt noncommittal abstract labels with subscript letters.

Tone category M_c encompasses six verbs that behave like M-tone verbs of the M_a category except in a few contexts, such as when preceded by the ACCOMPLISHED prefix /**le˧-**/. These verbs are /**bi˧**$_c$/ 'to go' and its past form /**hɯ˧**$_c$/, as well as /**gɣ̩˧**$_c$/ 'to go by, to flow, to fly (of time)', /**ʝi˧**$_c$/ 'to come', /**pɣ̩˧**$_c$/ 'to chant, to perform (a sacrifice, a ritual, a festival)', and /**pɻ̩˧**$_c$/ 'to go out, to get out'. With the ACCOMPLISHED prefix and the PERFECTIVE suffix, the resulting pattern is M.L.L, as illustrated in (1)-(2); the same applies to the other four verbs: /**le˧-hɯ˩-ze˩**/, /**le˧-gɣ̩˩-ze˩**/, /**le˧-ʝi˩-ze˩**/, and /**le˧-pɻ̩˩-ze˩**/. This pattern contrasts with that of the other verbs in the M tone category (i.e. those of the M_a and M_b subtypes), which retain M tone after the ACCOMPLISHED prefix.

(1) le˧-bi˩-ze˩
le˧- bi˧$_c$ -ze˧
ACCOMP to_go PFV
'[she/he/they...] went'

(2) le˧-pv̩˩-ze˩
le˧- pv̩˧$_c$ -ze˧
ACCOMP to_chant PFV
'[she/he/they…] chanted'

The difference between M_a and M_c is not related to verb valency: all M_c-tone verbs have intransitive uses, but one of the six ('to chant') can also be used transitively; conversely, not all intransitive verbs belong to this tonal category, witness 'to die', /**ʂɯ˧$_a$**/, which yields /**le˧-ʂɯ˧**/ (*Sister3.11, 95* {0004344#S11}).

Since these six verbs have the same behaviour as M_a-tone verbs in most contexts, they could be analyzed as a subset of the M_a category. A further diacritic could be added to their tone label, yielding something such as M_a' (M_a prime). However, it appeared less awkward typographically to label them as M_c, a third subcategory within the M tone category.

The tonal behaviour of M_c verbs calls for analysis. One observation that may be relevant is that the ACCOMPLISHED prefix /**le˧**-/ can carry special semantic connotations when associated with these verbs. In examples (3) and (4), it is interpreted as 'back/to return':

(3) le˧-bi˩-dʑo˩, | ʈʂʰwɤ˧ | ɖɯ˧ mɤ˧-kv̩˧ tsɯ˥ | mv̩˩!
le˧- bi˧$_c$ -dʑo˥ ʈʂʰwɤ˥ ɖɯ˧ mɤ˧- -kv̩˦ tsɯ˦ mv̩˧
ACCOMP to_go TOP dinner to_get NEG ABILITIVE REP AFFIRM
'If [the daughter] goes back [to her mother's home after marriage, she] cannot have dinner there.' [She must not stay there for the night; she has to go back to her new home before evening.] *(Sister3.116)* {0004344#S116}

(4) lv̩˧mi˧ so˩-ʈv̩˩ pɤ˩~pɤ˩! | le˧-bi˩-ze˩!
lv̩˧mi˧ so˩-ʈv̩˩ pɤ˦ ~ le˧- bi˧$_c$ -ze˧
stone a_few to_carry ACTIVITY ACCOMP to_go PFV
'Tonight, I'll bring (back) a couple of stones! I'm going back!' *Context:* a man, compelled by his spouse to steal in order to support the family, is about to steal sweetcorn; but he decides to refrain from stealing and instead fills his basket with stones before returning home. (*Reward.77*) {0004446#S77}

On the analogy of phenomena found in Naxi, one might imagine that an additional morpheme is present after the ACCOMPLISHED prefix, its *signified* being the semantic indication '… back' and its *signifier* a floating High tone that depresses the verb's tone to L. In Naxi, 'to go back' is **le˧-wu˥ bɯ˧**; the second of these three syllables can coalesce with the first, yielding **le˦** (Michaud & He

2007, Hé & Liú 2021). It would not be particularly surprising if the Na cognate for Naxi **wu˥**, namely **wo˦** 'again, back', had undergone a similar type of reduction, leaving only a tonal trace.

However, this hypothesis does not appear particularly promising, for several reasons. First, in examples (3) and (4), it is not possible to insert **wo˦** between the ACCOMPLISHED prefix and the verb, depriving the hypothesis of direct language-internal evidence of the sort available in Naxi, where full and reduced forms can be observed side by side. Secondly, instances of syllable reduction and coalescence in Na appear to be relatively rare compared with Naxi. The only such case documented in this volume is fairly different: reduction of an L-tone morpheme (the associative plural) depresses the tone of a following possessive from M to L (see §10.3). Thirdly, the meaning 'back, again' is not always present when an M_c-tone verb is preceded by the ACCOMPLISHED prefix, and the tone pattern remains unchanged when the verb simply means 'to go, to set off', as in (5), where the motion is clearly away from a familiar place and towards an unfamiliar one.

(5) "æ.hi.hi!" pi˧, | le˧-bi˩-zo˩-kv̩˩ | tsɯ˦ | mv̩˩!

æ.hi.hi	pi˥	le˧-	bi˧$_c$	-zo˧	-kv̩˦	tsɯ˦	mv̩˧
INTJ	to_say	ACCOMP	to_go	OBLIGATIVE	ABILITIVE	REP	AFFIRM

'[The mother, uncles, aunts and other relatives of the deceased wife's family shout out a cry of defiance:] "A-hi-hi!" and they set off!' *(Sister1.81)* {0004341#S81}

To sum up, in the absence of a principled explanation for the tonal behaviour of these five verbs, it appeared best to set up a distinct synchronic tone category for them: M_c.

6.1.3 Adjectives as distinct from verbs

The question of whether adjectives constitute a distinct part of speech has been raised for various languages of Southeast Asia. For a review and discussion, with a particular focus on Tai-Kadai, see Post (2008). As DeLancey (2015: 41) observes, "[s]ome Tibeto-Burman languages have a definable adjective category, usually only marginally distinguishable from nouns or from verbs". Yongning Na is one such language: in Na, adjectives behave in most respects like stative verbs, yet the intensive /**ʐwæ˩**/ 'extremely' occurs exclusively with adjectives (whereas the intensive /**ɖwæ˦**/ can be used with verbs – 'to V a lot' – as well as with adjectives – 'very ADJ'). Also, the copula is not used with adjectives but may be added after any verb to express certainty (Lidz 2010: 354). Moreover, adjectives have some tonal specificities. These pieces of evidence warrant the recognition of adjectives as a formally distinct class of words.

Four main tonal categories have been identified for monosyllabic adjectives: L, M, H, and MH. The L tone category must be further divided into two subcategories. The L_b category only contains two examples: /**dʑɤ˩$_b$**/ 'good' and /**nɑ˩$_b$**/ 'black, dark'. Their behaviour in context is examined in detail later in this chapter.

Importantly, the MH category of adjectives and the MH category of verbs do not always have the same tonal behaviour, as illustrated in §6.4.3 below. Likewise, the L_a and L_b categories of adjectives are not fully parallel to the L_a and L_b categories of verbs in terms of tonal behaviour. As for the M category of adjectives, no evidence was found for a division into subcategories analogous to the M_a, M_b, and M_c categories set up for verbs. These discrepancies between the tone systems of adjectives and verbs exemplify the morphosyntactic ramifications of tone in Yongning Na.

Disyllabic adjectives also exist. Examples include /**pʰv̩˧ɖɯ˦**/ 'expensive', from /**pʰv̩˧**/ 'price' and /**ɖɯ˩$_a$**/ 'large', and /**lo˩ɖɯ˧**/ 'generous', from /**lo˩˥**/ 'hand' and, again, /**ɖɯ˩$_a$**/ 'large'.

In view of the range of tone categories found in nouns, verbs and adjectives in Yongning Na, it is no wonder that they yield a wealth of diverse patterns when combined among themselves and in interaction with grammatical morphemes. The structure of the Na verb phrase, as schematized by Lidz (2010: 350–351), consists of the following components: manner adverb – verb complex – causative – intensifier – tense/aspect and modal particles, and auxiliary verbs – quotative evidential. Additionally, the verb may be preceded by spatial indications such as 'forward'/'backward' and 'upward'/'downward'. The verb complex itself may be a lexical verb, an existential verb, a copula, or a serial verb construction, and it may take a verbal prefix (or two prefixes, in the case of the DURATIVE prefix followed by the negation prefix).

The following sections examine the morphotonology of the verb phrase in detail.

6.2 Reduplication

Reduplication of verbs is a highly productive process in Yongning Na. From a semantic point of view, reduplication can convey various types of divergence from the prototype of the action or activity referred to by the verb. If the verb refers to an action, the reduplicated form can warp it towards representation as an activity, as illustrated in (6).

(6) jo˧-lo˥dʐo˩ tʰi˩-kʰɯ˩~kʰɯ˩. | hæ̃˩-lo˩pv̩˩ tʰi˥-kʰɯ˩~kʰɯ˩.
jo˥ lo˩dʐo˥ tʰi˧- kʰɯ˦ ~ hæ̃˩ lo˩pv̩˦ tʰi˧- kʰɯ˦ ~
jade bracelet DUR to_put ACTIVITY gold ring DUR to_put ACTIVITY
'They adorned her with jade bracelets. They adorned her with gold rings.'
Context: a bride is being prepared for a wedding. *(BuriedAlive2.29)*
{0004536#S29}

The *simplex* form /**jo˧-lo˥dʐo˩ tʰi˩-kʰɯ˩**/ means 'to put a jade bracelet (on someone's wrist)'. It depicts a single, well-delimited event. In the absence of a distinction between singular and plural, noun phrases tend to be interpreted as singular unless a numeral-plus-classifier phrase is added, as in (7).

(7) jo˧-lo˥dʐo˩ | ɲi˧-ɭɯ˧ tʰi˧-kʰɯ˦
jo˥ lo˩dʐo˥ ɲi˧ ɭɯ˧$_b$ tʰi˧- kʰɯ˦
jade bracelet two CLF DUR to_put
'to put two jade bracelets'

By contrast, (6) means that the bride is adorned with jade bracelets and gold rings by family members. The reduplicated form leads to a plural interpretation of the nouns, presumably because successive acts of adorning are not performed with the same object: if you adorn someone more that once you would be doing this with additional ornaments, not by taking one off and putting it on again.

The use of reduplication illustrated by (6) is glossed as ACTIVITY. It represents the process denoted by the verb as an activity rather than as a neatly delimited event. In the narrative from which (6) is drawn, verb reduplication emphasizes the generosity of the bride's family as a unified social entity. They collectively go through the prescribed steps of the marriage ritual, giving generous measure of the requisite offerings; each family member's individual action partakes in a collective activity. This builds a contrast with later events, when the young woman, feeling estranged in her in-laws' home, becomes oblivious of good manners and lapses into antisocial behaviour (solitary gluttony).

The surface semantic effect of ACTIVITY is close to that of the progressive suffix /**-dʐo**˧/. For instance, the reduplicated form of /**ʐi**˦/ 'to sleep' in (8) could be replaced by the *simplex* form followed by the progressive suffix /**-dʐo**˧/, as in (9).

(8) hĩ˧ | ɖɯ˧-tɑ˦ | le˧-ʐi˩~ʐi˩.
hĩ˥ ɖɯ˧-tɑ˦ le˧- ʐi˦ ~
person all ACCOMP to_sleep ACTIVITY
'Everyone was asleep.' *(BuriedAlive2.94)* {0004536#S94}

(9) hĩ˧ | ɖɯ˧-tɑ˧˥ | le˧-ʑi˧-dʑo˥.
hĩ˥ ɖɯ˧-tɑ˧˥ le˧- ʑi˧˥ -dʑo˧
person all ACCOMP to_sleep PROG
‘Everyone was asleep.’

As noted by Lidz (2010: 373), “[w]hen stative verbs reduplicate, one gets a reading of added intensity, while reduplicating non-stative verbs gives a reading of reciprocity of action, or a semantics of back-and-forth”. An example is shown in (10), where the reduplicated adjective ‘pleased’ serves as an understatement for ‘in love’ (‘they were in love with each other’).

(10) ʁo˧dɑ˧, | no˧by̩˥-tsʰɯ˩ɻ̩˩ lɑ˩ | ʈʂʰɯ˧=zɯ˩ | fy̩˧~fy̩˩-ɲi˩!
ʁo˧dɑ˧ no˧by̩˥-tsʰɯ˩ɻ̩˩ lɑ˧ ʈʂʰɯ˧=zɯ˩ fy̩˧ ~ -ɲi˩
before proper_name and 3DU pleased RECP CERTITUDE
‘Before [my daughter joined your house]… Nobbu Ci’er and [my daughter]… they used to like each other!’ *Context:* the unhappy woman’s mother explains what the matter is with her daughter. *(BuriedAlive2.136)* {0004536#S136}

The online texts contain more than a thousand examples of reduplication. This volume about morphotonology is not the right place to explore these rich materials to delve into the study of the semantic values that arise through divergence from *simplex* meanings, and to look for semantic and stylistic common denominators to reduplication as it applies to different parts of speech.[2] Instead, the present discussion is confined to morphotonological patterns.

Table 6.4 summarizes the tone patterns of reduplicated verbs in Yongning Na. Two of these verbs differ from those in Tables 6.1-6.2: ‘to go’ does not readily lend itself to reduplication, so ‘to chant’ was substituted; likewise, ‘to drink’ is much less frequently reduplicated than ‘to speak’. The selection of example verbs in this chapter (primarily drawn from Table 6.3) aims to avoid forms that would be unnatural or jarring to native speakers for semantic, syntactic or pragmatic reasons. Crucially, the seven tonal categories of verbs are internally consistent: within each class, all verbs conform to the same tone patterns. Thus, ‘to chant’ always has the same tonal behaviour as ‘to go’, and ‘to speak’ always has the same behaviour as ‘to drink’.

The patterns in Table 6.4 hold for all the verbs for which a reduplicated form could be elicited. The only apparent exception is the disyllabic verb //**wɤ˩~wɤ˩**//

[2]For insights into these topics, see Lidz (2010: 372-373, 385, 438, 440). Additionally, reduplication in Naxi is discussed in Michaud & Vaissière (2007).

(surface form: /**wɤ˩~wɤ˩˦**/) 'to detour past, to bypass', which looks like an unmistakeable case of reduplication, but whose L tone does not correspond to any of the patterns found for other verbs. At a push, this verb can be used in monosyllabic form, as in (11).

(11) le˧-wɤ˩-ze˩
le˧- ??wɤ˩ -ze˧
ACCOMP to_bypass PFV
'[She/he/they] bypassed' (elicited example; the question marks '??' signal the problematic status of the monosyllabic verb form)

In view of the M.L.L tone pattern in (11), the monosyllabic form must be analyzed as carrying lexical L tone: //**wɤ˩**//. However, this monosyllabic form appears to result from truncation of the disyllabic form rather than the other way around. It seems reasonable to hypothesize that the disyllabic form //**wɤ˩~wɤ˩**// 'to bypass' is not synchronically derived from a monosyllabic root but belongs to the small and heterogeneous set of disyllabic verbs in Yongning Na. Consequently, this verb does not constitute a counterexample to the generalizations shown in Table 6.4.

Table 6.4: The tone patterns of reduplicated verbs.

tone	example	meaning	reduplication	surface tone	underlying tone
H	**dzɯ˥**	to eat	**dzɯ˧~dzɯ#˥**	M.M	#H
M_a	**hwæ˧$_a$**	to buy	**hwæ˥~hwæ˩**	M.L	H–
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧~tɕʰi˧**	M.M	M
M_c	**pʏ̩˧$_c$**	to chant	**pʏ̩˥~pʏ̩˩**	M.L	H–
L_a	**dze˩$_a$**	to cut	**dze˧~dze˥**	M.H	H#
L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˥~ʐwɤ˩**	M.L	H–
MH	**lɑ˧˥**	to strike	**lɑ˩~lɑ˧˥**	L.MH	LM+MH#

The tonal string that surfaces in isolation is indicated in the "surface tone" column in Table 6.4. The underlying tone (provided in the last column and also used for transcribing the reduplicated expressions in the fourth column) was arrived at by examining the behaviour of reduplicated expressions in different contexts. (The corresponding recordings are *VerbReduplObj* {0004526} and *VerbReduplObj2* {0004516}.) Table 6.5 presents the tone patterns that obtain in frames (12) and (13). Frame (13) is one of the few contexts where M_a and M_c yield different outcomes.

Table 6.5: Surface phonological representation of reduplicated verbs in two different carrier phrases.

tone	example	meaning	ACCOMP+REDUPL	'to V a little'
H	**dzɯ˥**	to eat	**le˧-dzɯ˧~dzɯ˧**	**ɖɯ˧-dzɯ˧~dzɯ˧-ɻ̩˥**
M_a	**hwæ˧$_a$**	to buy	**le˧-hwæ˥~hwæ˩**	**ɖɯ˧-hwæ˥~hwæ˩-ɻ̩˩**
M_b	**tɕʰi˧$_b$**	to sell	**le˧-tɕʰi˧~tɕʰi˧**	**ɖɯ˧-tɕʰi˧~tɕʰi˧-ɻ̩˩**
M_c	**pv̩˧$_c$**	to chant	**le˧-pv̩˥~pv̩˩**	**ɖɯ˧-pv̩˧~pv̩˥-ɻ̩˩**
L_a	**bæ˩$_a$**	to sweep	**le˧-bæ˧~bæ˥**	**ɖɯ˧-bæ˧~bæ˥-ɻ̩˩**
L_b	**ʐwɤ˩$_b$**	to speak	**le˧-ʐwɤ˥~ʐwɤ˩**	**ɖɯ˧-ʐwɤ˥~ʐwɤ˩-ɻ̩˩**
MH	**lɑ˧˥**	to strike	**le˧-lɑ˩~lɑ˩**	**ɖɯ˧-lɑ˩~lɑ˩-ɻ̩˩**

(12) le˧-V~V
le˧- V ~
ACCOMP *target verb* RED
'to V'

(13) ɖɯ˧-V~V-ɻ̩˩
ɖɯ˧- V ~ -ɻ̩˩
DELIMITATIVE *target verb* RED INCHOATIVE
'to V a little'

Finally, Table 6.6 shows the result of associating the object /**tso˧~tso˧**/ 'thing' to a reduplicated verb.

Interpretation of the tone pattern for H-tone verbs draws on information from Table 6.5. The presence of an H tone on the final syllable of /**ɖɯ˧-dzɯ˧~dzɯ˧-ɻ̩˥**/ 'to eat a little', from /**dzɯ˥**/ 'to eat', suggests that the reduplicated expression contains an H tone. Since this H tone does not surface when the expression is uttered in isolation (where the reduplicated verb appears as /**dzɯ˧~dzɯ˧**/), the tonal category cannot be //H−// (an initial H tone), nor //H#// (a final H tone), nor //H$// (a *hopping* H tone: see §2.3.5). On the other hand, its behaviour is fully compatible with interpretation as //#H// (a *floating* H tone), and this interpretation is therefore adopted in Table 6.4.

The /M.L/ surface pattern in the reduplication of M_a-tone verbs (e.g. /**hwæ˧$_a$**/ 'to buy' → /**hwæ˧~hwæ˩**/) could reflect various underlying tones, such as //L#// (a final L tone) or //H−// (an H tone associated to the first part of the reduplicated expression, i.e. its first syllable). Crucial evidence comes from the contexts documented in Table 6.5, where the first syllable of the reduplicated expression

Table 6.6: Reduplicated verbs with the #H-tone object 'things'.

tone	example	meaning	reduplication	'things'+reduplicated V
H	**dzɯ˥**	to eat	**dzɯ˧~dzɯ#˥**	**tso˧~tso˧ dzɯ˧~dzɯ˧**
M_a	**hwæ˧$_a$**	to buy	**hwæ˥~hwæ˩**	**tso˧~tso˧ hwæ˧~hwæ˥**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧~tɕʰi˧**	**tso˧~tso˧ tɕʰi˧~tɕʰi˧**
L_a	**bæ˩$_a$**	to sweep	**bæ˧~bæ˥**	**tso˧~tso˧ bæ˧~bæ˥**
L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˥~ʐwɤ˩**	**tso˧~tso˧ ʐwɤ˧~ʐwɤ˥**
MH	**lɑ˧˥**	to strike	**lɑ˩~lɑ˧˥**	**tso˧~tso˧ lɑ˥~lɑ˩**

carries an H tone, guiding towards an interpretation of the underlying pattern as //H−// and not //L#//.[3]

Reduplication of M_b-tone verbs is the simplest case: the reduplicated expression carries M tone in isolation (e.g. /**tɕʰi˧$_b$**/ 'to sell' ⟶ /**tɕʰi˧~tɕʰi˧**/) and remains unaltered in the frames shown in Table 6.5 and Table 6.6. This strongly suggests that the reduplicated verb is underlyingly M-tone.

Analysis for the remaining three tonal categories is likewise straightforward. L_a-tone verbs reduplicate into a pattern with an H tone attached to the final syllable (see Table 6.4), leading to an analysis as //H#//. The analysis for the forms issuing from L_b-tone verbs is identical to that for M_a-tone verbs set out above. Finally, the underlying representation //LM+MH#//postulated for the expression reduplicated from an MH-tone verb corresponds to its surface form, without any added complexities.

The relationship between the tones of monosyllabic verbs and those of reduplicated expressions is another matter. Why does reduplication yield a final H tone for L_a-tone verbs, for instance? Why do MH-tone verbs get an initial L in their reduplicated form? Phonology does not provide answers to these questions: the tone patterns found in reduplicated forms cannot be derived from those of the simplex forms through phonological rules. The correspondences between simple and reduplicated tones need to be learnt; they constitute one of the components of the tonal grammar of Yongning Na.[4]

[3] An initial H tone never surfaces as such, due to the neutralization of //H// and //M// in tone-group-initial position; this is referred to as Rule 3 (see §7.1.1).

[4] In Naxi, reduplication similarly lacks a clear phonological pattern: H reduplicates as H.M, M as M.M, and L as M.L (Hé Zhìwǔ 1987: 10-11). These patterns can be analyzed as originating diachronically in total reduplication: H ⟶ H.H, M ⟶ M.M, L ⟶ L.L (Michaud & Vaissière 2007). In Yongning Na, however, no such simple historical scenario can currently be proposed. This is an area where data from neighbouring dialects appears essential for advancing the diachronic analysis.

As mentioned above, disyllabic verbs are rare and constitute a heterogeneous set. Only one instance of reduplication of a disyllabic verb was observed. It follows an AABB pattern (repetition of the first syllable of the *simplex* form, then the second): /**ʂv̩˧~ʂv̩˧ɖv̩˧~ɖv̩˧˥**/ 'pensively; with a heavy heart' (*Reward.49* {0004447#S49}), from /**ʂv̩˧ɖv̩˧**/ 'to think; to miss'. This is the sole example of a correspondence between an M tone in the *simplex* form and an MH# tone in the reduplicated expression. It looks like a unique innovation rather than the outcome of a productive pattern.

6.3 Prefixes

6.3.1 M-tone prefixes

The prefixes described in this section can be analyzed either as M-tone prefixes or as toneless prefixes that receive M by default. No evidence was found to suggest that they are inherently specified for tone. For convenience, they are referred to here as M-tone prefixes.

The most common M-tone verbal prefixes are the NEGATIVE /**mɤ˧**-/, PROHIBITIVE /**tʰɑ˧**-/, DURATIVE /**tʰi˧**-/, and ACCOMPLISHED /**le˧**-/.

> The ACCOMPLISHED prefix **lə³³-** is used to give a reading of accomplishment to a verb with lexical aspect of ongoing state, process, or liminality. (…) The DURATIVE prefix **tʰɯ³³-** [in the Alawua dialect, studied in this volume: /**tʰi˧**/] is used to give a reading of ongoing action to verbs with lexical aspect of process or liminality. (Lidz 2010: 345)

These prefixes all have the same tonal behaviour, with the exception of the ACCOMPLISHED /**le˧-**/in association with M_c-tone verbs, as reported in §6.1.2. A less common prefix, /**mv̩˧-**/, conveying IMMINENCE, has a behaviour of its own, described further below.

Table 6.7 presents the tonal behaviour of the most common prefixes. To facilitate reference, redundant data is provided for three prefixes: DURATIVE, PROHIBITIVE, and NEGATION, all of which follow identical tone patterns. Some of the relevant combinations can be heard in the following online recordings: *VerbProhib* {0004522}, *VerbProhib2* {0004524}, *VerbDurative* {0004520}, and *AccompPfv* {0004518}.

Following an M-tone prefix, the lexical tones M, H, L, and MH surface in a straightforward manner. The subcategories M_a and M_b are neutralized in this context, as are L_a and L_b.

Table 6.7: The tones of verbs in association with an M-tone prefix.

tone	example	meaning	DURATIVE	PROHIBITIVE	NEGATION
H	**dzɯ˥**	to eat	**tʰi˧-dzɯ˥**	**tʰɑ˧-dzɯ˥**	**mɤ˧-dzɯ˥**
M_a	**hwæ˧$_a$**	to buy	**tʰi˧-hwæ˧**	**tʰɑ˧-hwæ˧**	**mɤ˧-hwæ˧**
M_b	**tɕʰi˧$_b$**	to sell	**tʰi˧-tɕʰi˧**	**tʰɑ˧-tɕʰi˧**	**mɤ˧-tɕʰi˧**
M_c	**pv̩˧$_c$**	to chant	**tʰi˧-pv̩˧**	**tʰɑ˧-pv̩˧**	**mɤ˧-pv̩˧**
L_a	**bæ˩$_a$**	to sweep	**tʰi˧-bæ˩**	**tʰɑ˧-bæ˩**	**mɤ˧-bæ˩**
L_b	**ʐwɤ˩$_b$**	to speak	**tʰi˧-ʐwɤ˩**	**tʰɑ˧-ʐwɤ˩**	**mɤ˧-ʐwɤ˩**
MH	**lɑ˧˥**	to strike	**tʰi˧-lɑ˧˥**	**tʰɑ˧-lɑ˧˥**	**mɤ˧-lɑ˧˥**

Table 6.8 sets out the tone patterns associated with the ACCOMPLISHED prefix /**le˧**-/ in combination with the PERFECTIVE /**-ze˧**/ and the morpheme indicating COMPLETION, /**-se˩**/. Table 6.9 lists the patterns that obtain when the verb is followed by the CAUSATIVE /**-tsæ˧**/ and CERTITUDE morpheme /**-ɲi˩**/.

Table 6.8: Tone patterns in constructions consisting of verbs with the ACCOMPLISHED M-tone prefix and a PERFECTIVE or COMPLETION morpheme.

tone	example	meaning	ACCOMP	ACCOMP+V+PFV	ACCOMP+V+ COMPLETION
H	**dzɯ˥**	to eat	**le˧-dzɯ˥**	**le˧-dzɯ˥-ze˩**	**le˧-dzɯ˧-se˥**
M_a	**hwæ˧$_a$**	to buy	**le˧-hwæ˧**	**le˧-hwæ˧-ze˧**	**le˧-hwæ˧-se˩**
M_b	**tɕʰi˧$_b$**	to sell	**le˧-tɕʰi˧**	**le˧-tɕʰi˧-ze˧**	**le˧-tɕʰi˧-se˩**
M_c	**bi˧$_c$**	to go	**le˧-bi˩**	**le˧-bi˩-ze˩**	**le˧-bi˩-se˩**
L_a	**bæ˩$_a$**	to sweep	**le˧-bæ˩**	**le˧-bæ˩-ze˩**	**le˧-bæ˩-se˩**
L_b	**ʐwɤ˩$_b$**	to speak	**le˧-ʐwɤ˩**	**le˧-ʐwɤ˩-ze˩**	**le˧-ʐwɤ˩-se˩**
MH	**lɑ˧˥**	to strike	**le˧-lɑ˧˥**	**le˧-lɑ˧-ze˥**	**le˧-lɑ˧-se˥**

The CERTITUDE morpheme is invariantly low in Table 6.9, in keeping with its lexical L tone. This confirms the absence of a floating H tone in any of the preceding expressions, as a floating tone would be expected to land on the CERTITUDE morpheme. This is important information for arriving at the underlying tone of expressions with surface M tones, such as /**le˧-hwæ˧**/ and /**le˧-tɕʰi˧**/ – hypothesizing that if they carried a floating H tone, the following morpheme, /**-ɲi˩**/, would surface with H tone.

Table 6.9: Tone patterns in constructions consisting of verbs with the ACCOMPLISHED M-tone prefix and CAUSATIVE+CERTITUDE morphemes.

tone	example	meaning	ACCOMP	ACCOMP+V+CAUS+CERTITUDE
H	**dzɯ˥**	to eat	**le˧-dzɯ˥**	**le˧-dzɯ˧-tsæ˥-ɲi˩**
M_a	**hwæ˧$_a$**	to buy	**le˧-hwæ˧**	**le˧-hwæ˧-tsæ˧-ɲi˩**
M_b	**tɕʰi˧$_b$**	to sell	**le˧-tɕʰi˧**	**le˧-tɕʰi˧-tsæ˧-ɲi˩**
M_c	**pv̩˧$_c$**	to chant	**le˧-pv̩˧**	**le˧-pv̩˧-tsæ˧-ɲi˩**
L_a	**bæ˩$_a$**	to sweep	**le˧-bæ˩**	**le˧-bæ˩-tsæ˩-ɲi˩**
L_b	**ʐwɤ˩$_b$**	to speak	**le˧-ʐwɤ˩**	**le˧-ʐwɤ˩-tsæ˩-ɲi˩**
MH	**lɑ˧˥**	to strike	**le˧-lɑ˧˥**	**le˧-lɑ˧-tsæ˥-ɲi˩**

The data in Table 6.9 also provides further illustration of the behaviour of MH and H tones, exemplified by /**le˧-lɑ˧˥**/ and /**le˧-dzɯ˥**/, respectively. For MH-tone verbs in this construction, the M part remains on the syllable to which the tone is lexically associated, and the H part is projected onto the next syllable (the causative). This is obligatorily followed by L tone on the CERTITUDE suffix, due to Rule 4: "The syllable following an H-tone syllable receives L tone". In the case of H-tone verbs, the H tone no longer appears on the morpheme to which it is lexically associated; instead, it associates to the CAUSATIVE, resulting in a surface phonological output that is identical to that for MH tone.

A related set of facts is presented in Table 6.10: it concerns the V-NEG-V construction, as illustrated in example (14).[5]

(14) ɲi˩ mɤ˥-ɲi˩, | mɤ˧-ɳv̩˥!
ɲi˩ mɤ˧- ɲi˩ mɤ˧- ɳv̩˥
COP NEG COP NEG to_know/to_get_to_know
'Whether it is actually the case… we don't know!' *Context:* two persons discuss what a third person has said.

[5]Two variants are possible: one in which the sequence forms a single tone group, as in (14), and another in which it is divided into two groups: /**ɲi˩˧ | mɤ˧-ɲi˩, | mɤ˧-ɳv̩˥!**/, with the same meaning and morphemic composition as (14). In the latter case, the first syllable has the same tone as in isolation, and the negated form follows the tonal patterns described in Table 6.7 above. These V-NEG-V constructions are typically followed by /**mɤ˧-ɳv̩˥**/ '[I/we] don't know' or /**mɤ˧-do˩**/ '[we] don't know/can't see for ourselves'. Sometimes, the V NEG-V portion undergoes focalization, as in /**hwæ˧ mɤ˧-hwæ˧ F | mɤ˧-do˩**/ '[I/we] don't know whether [they] bought [it/some] or not'. (For a discussion of intonational focalization, transcribed as 'F', see §9.2.2.) In addition to the elicited data in Table 6.10, further examples appear in texts, e.g. in *BuriedAlive3.133* {0004539#S133}, *Seeds2.85* {0004543#S85}, and *Dog.59* {0004442#S59}.

Table 6.10: Tone patterns of the V-NEG-V construction.

tone	example	meaning	V NEG V	V | NEG V
H	**dzɯ˥**	to eat	**dzɯ˧ mɤ˧-dzɯ˧**	**dzɯ˧ | mɤ˧-dzɯ˥**
M_a	**hwæ˧**$_{\mathbf{a}}$	to buy	**hwæ˧ mɤ˧-hwæ˧**	**hwæ˧ | mɤ˧-hwæ˧**
M_b	**tɕʰi˧**$_{\mathbf{b}}$	to sell	**tɕʰi˧ mɤ˧-tɕʰi˧**	**tɕʰi˧ | mɤ˧-tɕʰi˧**
M_c	**pv̩˧**$_{\mathbf{c}}$	to chant	**pv̩˧ mɤ˧-pv̩˧**	**pv̩˧ | mɤ˧-pv̩˧**
L_a	**gɯ˩**$_{\mathbf{a}}$	to be true	**gɯ˩ mɤ˩-gɯ˥**	**gɯ˩˧ | mɤ˧-gɯ˩**
L_b	**ʐwɤ˩**$_{\mathbf{b}}$	to speak	**ʐwɤ˩ mɤ˥-ʐwɤ˩**	**ʐwɤ˩˧ | mɤ˧-ʐwɤ˩**
MH	**lɑ˧˥**	to strike	**lɑ˧ mɤ˥-lɑ˩**	**lɑ˧˥ | mɤ˧-lɑ˧˥**

As noted at the beginning of this section, the prefix /**mv̩˧**-/, conveying IMMINENCE, is infrequent: only one example was found in the first twenty-five transcribed narratives. The consultant was not comfortable pairing this prefix with verbs to form a disyllabic expression and instead proposed the three constructions (15), (16), and (17). The results are shown in Table 6.11.

(15) mv̩˧-V-bi˧
mv̩˧- V -bi˧
IMMINENCE *target verb* IMM_FUT
'will V right away'

(16) tʰi˧-mv̩˧-V
tʰi˧- mv̩˧- V
DUR IMMINENCE *target verb*
'is going to V up'

(17) le˧-mɤ˧-V
le˧- mɤ˧- V
ACCOMP NEG *target verb*
'does not V / not to V'

The outcomes for H- and MH-tone verbs are far from trivial. As with other systematically elicited combinations, it seems prudent to withhold interpretation until further confirmation can be obtained, ideally from narratives.

6.3.2 The interrogative prefix

The yes/no interrogative (polar interrogative) is provisionally analyzed as carrying an L tone: /ə˩-/, though not without reservations, as discussed below. This

Table 6.11: Tone patterns of verbs with the prefix /**mv̩˧**-/, and an antonymic construction.

tone	example	meaning	**mv̩˧**-V-**bi˧**	**tʰi˧-mv̩˧**-V	**le˧-mɤ˧**-V
H	**dzɯ˥**	to eat	**mv̩˧-dzɯ˧-bi˧**	**tʰi˧-mv̩˧-dzɯ˧**	**le˧-mɤ˧-dzɯ˧**
M_a	**hwæ˧$_a$**	to buy	**mv̩˧-hwæ˧-bi˧**	**tʰi˧-mv̩˧-hwæ˧**	**le˧-mɤ˧-hwæ˧**
M_b	**tɕʰi˧$_b$**	to sell	**mv̩˧-tɕʰi˧-bi˧**	**tʰi˧-mv̩˧-tɕʰi˧**	**le˧-mɤ˧-tɕʰi˧**
M_c	**pv̩˧$_c$**	to chant	**mv̩˧-pv̩˧-bi˧**	**tʰi˧-mv̩˧-pv̩˧**	**le˧-mɤ˧-pv̩˧**
L_a	**bæ˩$_a$**	to sweep	**mv̩˧-bæ˧-bi˩**	**tʰi˧-mv̩˧-bæ˩**	**le˧-mɤ˧-bæ˩**
L_b	**ʐwɤ˩$_b$**	to speak	**mv̩˧-ʐwɤ˧-bi˩**	**tʰi˧-mv̩˧-ʐwɤ˩**	**le˧-mɤ˧-ʐwɤ˩**
MH	**lɑ˧˥**	to strike	**mv̩˧-lɑ˩-bi˩**	**tʰi˧-mv̩˧-lɑ˧˥**	**le˧-mɤ˧-lɑ˧˥**

interrogative prefix is segmentally bleached, consisting of a neutral vowel that undergoes strong regressive vowel harmony. Tonally, on the other hand, it follows distinct patterns, differing from both M-tone prefixes and the L-tone directional prefixes presented in §6.3.3. This suggests that it has a tonal specification of its own. Table 6.12 sets out the facts.

Table 6.12: The tones of verbs in association with an L-tone prefix.

tone	example	meaning	interrogative
H	**dzɯ˥**	to eat	**ə˧-dzɯ˥**
M_a	**hwæ˧$_a$**	to buy	**ə˧-hwæ˥**
M_b	**tɕʰi˧$_b$**	to sell	**ə˧-tɕʰi˥**
M_c	**pv̩˧$_c$**	to chant	**ə˧-pv̩˥**
L_a	**bæ˩$_a$**	to sweep	**ə˩-bæ˩**
L_b	**ʐwɤ˩$_b$**	to speak	**ə˩-ʐwɤ˩**
MH	**lɑ˧˥**	to strike	**ə˧-lɑ˥**

In the first edition of this book, the tone patterns for M-tone verbs were misanalyzed as L.M: ‡**ə˩-hwæ˧** 'buy?', ‡**ə˩-tɕʰi˧** 'sell?', and ‡**ə˩-pv̩˧** 'chant?'. Recognition of the M.H pattern was delayed by the hypothesis that the high pitch of the verb in this context was due to an intonational factor. In view of the cross-linguistic tendency for interrogative sentences to be associated with raised pitch, the phonetic realization (phonetic [M.H], in the upper half of the speaker's pitch range) was initially attributed to an intonational modification of an underlying /L.M/ sequence. It was eventually established, however, that the tone patterns for H-tone, MH-tone and M-tone verbs in this context were identical. The main consultant was able to match these patterns with the M.H tone pattern of disyllabic nouns (as in /**hwæ˧tsɯ˥**/ 'rat'), despite the considerable syntactic difference between an interrogative construction and a disyllabic noun.

Under the hypothesis that the interrogative prefix carries a lexical L tone, its realization before L-tone verbs is straightforward but its realization before M-, H-, and MH-tone verbs calls for an explanation: in these contexts, the prefix surfaces with an M tone (and the verb carries H). The sequences /L.M/, /L.H/, and /L.MH/ are all phonotactically permissible in Yongning Na, so there is no phonological constraint preventing an L-tone interrogative prefix from surfacing with its expected tone in these contexts. One would thus expect forms such as † **ə˩-lɑ˧˥** ('does (s)he strike?'), yet this pattern is not observed. In light of this, the hypothesis that the interrogative prefix carries a lexical L tone is not particularly compelling.

Differences in the tonal realization of a prefix depending on the tone of the following verb constitute a puzzle in view of the scarcity of cases of regressive tonal modification (modification of a tone by that of a *following* morpheme) in the Alawua dialect of Yongning Na. In this variety, tone modification is predominantly progressive (unlike vowel harmony, which operates regressively: see Appendix A, §A.2.7). That said, a schwa prefix is a prime candidate for modification by a following tone, given its phonetic and phonological lightness. The phonetic realization of schwa prefixes warrants an experimental phonetic study, focusing on parameters such as duration, fundamental frequency, and formant structure. Synchronic patterns of phonetic variation may offer clues as to how the current morphotonological patterns of these prefixes developed.

6.3.3 The marking of spatial orientation on verbs

Extensive marking of orientation on verbs is found among Na-Qiangic languages. In particular, rGyalrongic languages "have a whole array of verbal orientation prefixes, which are obligatorily present on all perfective and imperative verb forms" (Sun 2000: 180; see also Lin 2002, and, on Tangut, Jacques 2011b and Beaudouin 2023). J. Sun describes three distinct pairs of directions in rGyalrongic: eastward (i.e. in the direction of the rising sun) vs. westward; upstream vs. downstream; and uphill (upward) vs. downhill (downward). The system found in Shixing (Xumi) comprises two productive pairs of (non-obligatory) orientation prefixes, only one of which corresponds semantically with the rGyalrongic system: upward vs. downward. The other opposition is inward vs. outward. Shixing also retains traces of a third pair, hither vs. thither, which appears in a dedicated construction meaning ‘to V back and forth’ (Chirkova 2009).

This striking feature is sometimes awarded the status of criterion for phylogenetic classification, e.g. in proposals by Matisoff (2004: 105). However, cross-dialect and cross-language variation show that orientation systems are no less prone to change than other structural features of a language. If directional prefixes have great historical depth in Sino-Tibetan, Naish must be hypothesized to have lost them. In Naish, topographically-based spatial deixis is not marked by an obligatory verbal prefix. Indications of orientation, such as /**mv̩˩tɕo˧**/ ‘downward’ in (18), are better described as orientation adverbials.

(18) mv̩˩tɕo˧ mɤ˧-hɯ˧
mv̩˩tɕo˧ mɤ˧- hɯ˧$_{C}$
downward NEG to_go.PST
‘[The dog, who had come to sit on the wooden platform close to the fire pit, where dogs are not allowed] did not/would not get down!’ (*Context:* a discussion about Sister3.22. Field notes.)

The only monosyllabic indications of orientation in common use that could plausibly be considered prefixes are /**gɤ˩**-/ ‘upward’ and /**mv̩˩**-/ ‘downward’. For instance, /**ʂo˥**/ ‘to reap, to gather in’ can combine with /**gɤ˩**-/ ‘upward’ to mean ‘to reap in, to bring back to the house and into the granary’: /**gɤ˩-ʂo˥**/. These monosyllabic prefixes also occur in set constructions such as /**gɤ˩-V | mv̩˩-V**/ ‘to V in all directions’, as illustrated in (19):

(19) ʈʂʰɯ˧ne˧-ʝi˥ | gɤ˩-dɑ˦, | mv̩˩-dɑ˦, | gɤ˩-dɑ˦, | mv̩˩-dɑ˦, | (…) ɖɯ˧-so˩ ʂɯ˩ ʝi˩ tsɯ˩ | mv̩˩! |
ʈʂʰɯ˧ne˧-ʝi˥ gɤ˩- dɑ˦ mv̩˩- dɑ˦ ɖɯ˧-so˩ ʂɯ˩ ʝi˥
thus upward to_strike downward to_strike several times to_do
tsɯ˦ mv̩˧
REP AFFIRM
‘He would give blows high and low (= hither and thither), again and again! / He would strike blows in all directions, again and again!’ *Context:* an exorcist is performing a ritual. *(Healing.38)* {0004540#S38}

In this context, the prefixes retain some of their literal meaning of ‘upward’ and ‘downward’: the exorcist’s blows with his sword are aimed high, then low (close to the ground), then high again, and so on. Yet in this construction, the spatial indications take on a broader meaning, summoning up the swift, dance-like movements of the exorcist as he battles an invisible cohort of demons surrounding him. Repetition of the prefixed verb (/**gɤ˩-dɑ˥**, | **mv̩˩-dɑ˥**, | **gɤ˩-dɑ˥**, | **mv̩˩-dɑ˥**/) contributes to weakening the literal indication of spatial orientation, yielding a sense of 'in all directions' rather than 'up and down'.

When an orientation prefix is separated from the verb by other prefixes, it can form a tone group on its own, as in (20). In this example, the prefix's //L// tone surfaces as /LH/ due to the prohibition of all-L tone groups. A phonological rule referred to as Rule 7 (see §7.1) repairs all-L tone groups by addition of a postlexical H tone to their last syllable. In the first tone group of example (20), this last syllable also happens to be the *first* syllable.

(20) gɤ˩˥ | le˧-ʈʂʰo˧-se˥-dʑo˩ | tʰi˩˥ …
gɤ˩- le˧- ʈʂʰo˥ -se˩ -dʑo˥ tʰi˩˥
upward ACCOMP to_pray COMPLETION TOP then
'after one has prayed [*literally*: prayed up (to the ancestors)]' *(Dog2.54)*
{0004555#S54}

Judging from available texts, the monosyllabic form /**gɤ**˩-/ 'upward' is more frequent than /**mv̩**˩-/ 'downward'. The two prefixes /**gɤ**˩-/ 'upward' and /**mv̩**˩-/ 'downward' have the same tonal behaviour. However, the disyllabic expressions /**gɤ˩tɕo**˧/ 'upward' and /**mv̩˩tɕo**˧/ 'downward' are generally preferred. Here, the monosyllabic forms are tentatively classified as prefixes (and transcribed accordingly, with a following hyphen), while the disyllabic forms are labelled as adverbials. However, in the absence of clear language-internal criteria distinguishing the two, the divide between these categories may not be as sharp as this choice of terms suggests.

From a tonal point of view, orientation prefixes typically belong to the same tone group as the following verb, except in uncommon cases such as (20). On the other hand, orientation adverbials often constitute a separate tone group, as discussed in §7.2.1. Tables 6.13 to 6.18 present cases where the orientation prefix or adverbial is integrated into the same tone group as the verb. The PERFECTIVE suffix /-**ze**˧/ is used to determine whether the verb carries H tone (which causes the suffix to lower to L), M tone (which leaves the suffix unaffected), or MH tone (which results in the association of the H part of the contour to the suffix).

For the sake of clarity, the surface forms are provided in full, even though in many cases the tonal behaviour of the PERFECTIVE suffix /**-ze**˧/ follows straightforwardly from the tone pattern of the non-suffixed expression through the application of the seven phonological tone rules recapitulated in §7.1.1. For instance, an L.H sequence can only be followed by L, by virtue of Rule 4 (“A syllable following an H-tone syllable receives L tone”), so the L.H expression /**gɤ˩-se˥**/ ‘to walk up(ward)’ predictably yields L.H+L in (21).

(21) gɤ˩-se˥-ze˩
gɤ˩- se˥ -ze˧
upward to_walk PFV
‘walked up(ward)’

The original data is found in the online document *SpatialOrientation* {0004559}. The verb /**bi˧$_{c}$**/ ‘to go’ was inadvertently omitted in the recording. This verb can combine with the disyllabic orientation adverbials, but not with monosyllabic /**gɤ˩**-/ ‘upward’ and /**mv̩˩**-/ ‘downward’. The data is shown in (22)-(26). Note that the variant ‡ **jo˩lo˩ bi˥** for (24) was rejected.

(22) ʁo˧dɑ˧ bi˧(-ze˧)
ʁo˧dɑ˧ bi˧$_{c}$ -ze˧
forward to_go PFV
‘to go forward’

(23) ʁwæ˧gi˧ bi˧(-ze˧)
ʁwæ˧gi˧ bi˧$_{c}$ -ze˧
leftward to_go PFV
‘to go to the left’

(24) jo˩lo˩ bi˩
jo˩lo˩ bi˧$_{c}$
rightward to_go
‘to go to the right’

(25) ʁo˧tʰo˩ bi˩
ʁo˧tʰo˩ bi˧$_{c}$
backward to_go
‘to go backward’

(26) gɤ˩tɕo˧ bi˧(-ze˧)
gɤ˩tɕo˧ bi˧$_{c}$ -ze˧
upward to_go PFV
‘to go upward’

Table 6.13: The tonal behaviour of verbs after indications of spatial orientation: monosyllabic prefixes.

tone	example	meaning	'upward' prefix	'downward' prefix
H	**se˥**	to walk	**gɤ˩-se˥, gɤ˩-se˥-ze˩**	**mv̩˩-se˥, mv̩˩-se˥-ze˩**
M_a	**li˧a**	to look	**gɤ˩-li˧, gɤ˩-li˧-ze˧**	**mv̩˩-li˧, mv̩˩-li˧-ze˧**
M_b	**tsi˧b**	to set	**gɤ˩-tsi˧, gɤ˩-tsi˧-ze˧**	**mv̩˩-tsi˧, mv̩˩-tsi˧-ze˧**
L_a	**kwɤ˩a**	to throw	**gɤ˩-kwɤ˥, gɤ˩-kwɤ˥-ze˩**	**mv̩˩-kwɤ˥, mv̩˩-kwɤ˥-ze˩**
L_b	**ɻ̍˩b**	to turn	**gɤ˩-ɻ̍˥, gɤ˩-ɻ̍˥-ze˩**	**mv̩˩-ɻ̍˥, mv̩˩-ɻ̍˥-ze˩**
MH	**mi˧˥**	to push	**gɤ˩-mi˧˥, gɤ˩-mi˧-ze˥**	**mv̩˩-mi˧˥, mv̩˩-mi˧-ze˥**

Table 6.14: The tonal behaviour of verbs after indications of spatial orientation: adverbial //**ɬo˧tɑ˧**// 'to the side'.

tone	example	meaning	with adverbial 'to the side'
H	**se˥**	to walk	**ɬo˧tɑ˧ se˧, ɬo˧tɑ˧ se˧-ze˩**
M_a	**li˧a**	to look	**ɬo˧tɑ˧ li˧, ɬo˧tɑ˧ li˧-ze˧**
M_b	**tsi˧b**	to set	**ɬo˧tɑ˧ tsi˧, ɬo˧tɑ˧ tsi˧-ze˩**
L_a	**kwɤ˩a**	to throw	**ɬo˧tɑ˧ kwɤ˩, ɬo˧tɑ˧ kwɤ˩-ze˩**
L_b	**ɻ̍˩b**	to turn	**ɬo˧tɑ˧ ɻ̍˩, ɬo˧tɑ˧ ɻ̍˩-ze˩**
MH	**mi˧˥**	to push	**ɬo˧tɑ˧ mi˧˥, ɬo˧tɑ˧ mi˧-ze˥**

All the data shown in Tables 6.13 to 6.18 boils down to the surface tone patterns summarized in Tables 6.19 and 6.20. The tone indicated after a '+' sign is that carried by the PERFECTIVE /**-ze˧**/ as it appears following the directional-plus-verb combination at issue. For example, the notation L.M+H for the combination of /**gɤ˩**-/ 'upward' with a M_b-tone verb indicates the pattern /**gɤ˩-tɕi˧**/, /**gɤ˩-tɕi˧-ze˥**/ 'to set in an upward direction'.

An especially interesting aspect of this data is the lowering of the PERFECTIVE suffix (which carries a lexical M tone: //**-ze˧**//) after expressions such as /**mv̩˩tɕo˧ se˧**/ 'to go downward', yielding /**mv̩˩tɕo˧ se˧-ze˩**/ '(s)he went downward'. This contrasts with expressions such as /**mv̩˩tɕo˧ li˧**/ 'to look downward', after which the PERFECTIVE retains its lexical M tone: /**mv̩˩tɕo˧ li˧-ze˧**/ '(s)he looked downward'. The lowering influence of an overt H tone on following tones within a tone group is a hard-and-fast phonological rule, but the expression /**mv̩˩tɕo˧ se˧**/ 'to go downward' does not contain an H tone at the surface phonological level. At a deeper level, referred to in this volume as the *underlying* phonological level,

Table 6.15: The tonal behaviour of verbs after indications of spatial orientation: adverbials //**ʁo˧dɑ˧**// ‘forward’ and //**ʁo˧tʰo˩**// ‘backward’.

tone	example	meaning	‘forward’	‘backward’
H	**se˥**	to walk	**ʁo˧dɑ˧ se˧,**	**ʁo˧tʰo˩ se˩,**
			ʁo˧dɑ˧ se˧-ze˩	**ʁo˧tʰo˩ se˩-ze˩**
M_a	**li˧$_a$**	to look	**ʁo˧dɑ˧ li˧,**	**ʁo˧tʰo˩ li˩,**
			ʁo˧dɑ˧ li˧-ze˧	**ʁo˧tʰo˩ li˩-ze˩**
M_b	**tsi˧$_b$**	to set	**ʁo˧dɑ˧ tsi˧,**	**ʁo˧tʰo˩ tsi˩,**
			ʁo˧dɑ˧ tsi˧-ze˩	**ʁo˧tʰo˩ tsi˩-ze˩**
L_a	**kwɤ˩$_a$**	to throw	**ʁo˧dɑ˧ kwɤ˩,**	**ʁo˧tʰo˩ kwɤ˩,**
			ʁo˧dɑ˧ kwɤ˩-ze˩	**ʁo˧tʰo˩ kwɤ˩-ze˩**
L_b	**ɻ̩˩$_b$**	to turn	**ʁo˧dɑ˧ ɻ̩˩,**	**ʁo˧tʰo˩ ɻ̩˩,**
			ʁo˧dɑ˧ ɻ̩˩-ze˩	**ʁo˧tʰo˩ ɻ̩˩-ze˩**
MH	**mi˧˥**	to push	**ʁo˧dɑ˧ mi˧˥,**	**ʁo˧tʰo˩ mi˩,**
			ʁo˧dɑ˧ mi˧-ze˥	**ʁo˧tʰo˩ mi˩-ze˩**

Table 6.16: The tonal behaviour of verbs after indications of spatial orientation: //**ʁwæ˧lo˥**// ‘leftward’ and //**ʁwæ˧-gi#˥**// ‘to the left side’.

tone	example	meaning	‘leftward’	‘to the left side’
H	**se˥**	to walk	**ʁwæ˧lo˥ se˩,**	**ʁwæ˧-gi˧ se˧,**
			ʁwæ˧lo˥ se˩-ze˩	**ʁwæ˧-gi˧ se˧-ze˩**
M_a	**li˧$_a$**	to look	**ʁwæ˧lo˥ li˩,**	**ʁwæ˧-gi˧ li˩,**
			ʁwæ˧lo˥ li˩-ze˩	**ʁwæ˧-gi˧ li˩-ze˩**
M_b	**tsi˧$_b$**	to set	**ʁwæ˧lo˥ tsi˩,**	**ʁwæ˧-gi˧ tsi˧,**
			ʁwæ˧lo˥ tsi˩-ze˩	**ʁwæ˧-gi˧ tsi˧-ze˩**
L_a	**kwɤ˩$_a$**	to throw	**ʁwæ˧lo˥ kwɤ˩,**	**ʁwæ˧-gi˧ kwɤ˥,**
			ʁwæ˧lo˥ kwɤ˩-ze˩	**ʁwæ˧-gi˧ kwɤ˥-ze˩**
L_b	**ɻ̩˩$_b$**	to turn	**ʁwæ˧lo˥ ɻ̩˩,**	**ʁwæ˧-gi˧ ɻ̩˥,**
			ʁwæ˧lo˥ ɻ̩˩-ze˩	**ʁwæ˧-gi˧ ɻ̩˥-ze˩**
MH	**mi˧˥**	to push	**ʁwæ˧lo˥ mi˩,**	**ʁwæ˧-gi˧ mi˩,**
			ʁwæ˧lo˥ mi˩-ze˩	**ʁwæ˧-gi˧ mi˩-ze˩**

Table 6.17: The tonal behaviour of verbs after indications of spatial orientation: //**jo˩lo˩**// ‘rightward’ and //**jo˩-gi˩**// ‘to the right side’.

tone	example	meaning	‘rightward’	‘to the right side’
H	**se˥**	to walk	**jo˩lo˩ se˩˥, jo˩lo˩ se˩-ze˥**	**jo˩-gi˩ se˩˥, jo˩-gi˩ se˩-ze˥**
M_a	**li˧$_a$**	to look	**jo˩lo˩ li˥, jo˩lo˩ li˥-ze˩**	**jo˩-gi˩ li˥, jo˩-gi˩ li˥-ze˩**
M_b	**tsi˧$_b$**	to set	**jo˩lo˩ tsi˩˥, jo˩lo˩ tsi˩-ze˥ ≈ jo˩lo˩ tsi˥, jo˩lo˩ tsi˥-ze˩**	**jo˩-gi˩ tsi˥, jo˩-gi˩ tsi˥-ze˩ ≈ jo˩-gi˩ tsi˩˥, jo˩-gi˩ tsi˩-ze˥**
L_a	**kwɤ˩$_a$**	to throw	**jo˩lo˩ kwɤ˥, jo˩lo˩ kwɤ˥-ze˩**	**jo˩-gi˩ kwɤ˥, jo˩-gi˩ kwɤ˥-ze˩**
L_b	**ɻ̩˩$_b$**	to turn	**jo˩lo˩ ɻ̩˥, jo˩lo˩ ɻ̩˥-ze˩**	**jo˩-gi˩ ɻ̩˥, jo˩-gi˩ ɻ̩˥-ze˩**
MH	**mi˧˥**	to push	**jo˩lo˩ mi˥, jo˩lo˩ mi˥-ze˩**	**jo˩-gi˩ mi˥, jo˩-gi˩ mi˥-ze˩**

Table 6.18: The tonal behaviour of verbs after indications of spatial orientation: orientation adverbials //**gɤ˩tɕo˧**// ‘upward’ and //**mv̩˩tɕo˧**// ‘downward’.

tone	example	meaning	‘upward’	‘downward’
H	**se˥**	to walk	**gɤ˩tɕo˧ se˧, gɤ˩tɕo˧ se˧-ze˩**	**mv̩˩tɕo˧ se˧, mv̩˩tɕo˧ se˧-ze˩**
M_a	**li˧$_a$**	to look	**gɤ˩tɕo˧ li˧, gɤ˩tɕo˧ li˧-ze˧**	**mv̩˩tɕo˧ li˧, mv̩˩tɕo˧ li˧-ze˧**
M_b	**tsi˧$_b$**	to set	**gɤ˩tɕo˧ tsi˧, gɤ˩tɕo˧ tsi˧-ze˧**	**mv̩˩tɕo˧ tsi˧, mv̩˩tɕo˧ tsi˧-ze˧**
L_a	**kwɤ˩$_a$**	to throw	**gɤ˩tɕo˧ kwɤ˩ , gɤ˩tɕo˧ kwɤ˩-ze˩**	**mv̩˩tɕo˧ kwɤ˩, mv̩˩tɕo˧ kwɤ˩-ze˩**
L_b	**ɻ̩˩$_b$**	to turn	**gɤ˩tɕo˧ ɻ̩˩, gɤ˩tɕo˧ ɻ̩˩-ze˩**	**mv̩˩tɕo˧ ɻ̩˩, mv̩˩tɕo˧ ɻ̩˩-ze˩**
MH	**mi˧˥**	to push	**gɤ˩tɕo˧ mi˧˥, gɤ˩tɕo˧ mi˧-ze˥**	**mv̩˩tɕo˧ mi˧˥, mv̩˩tɕo˧ mi˧-ze˥**

the expressions /**mʏ̩˩tɕo˧ se˧**/ 'to go downward' and /**mʏ̩˩tɕo˧ li˧**/ 'to look downward' must be analyzed as carrying different tonal specifications, since they behave differently with the same suffix. Since tone lowering is characteristic of H tones, it seems reasonable to posit a floating H in the underlying representation of /**mʏ̩˩tɕo˧ se˧**/ 'to go downward', thus: //**mʏ̩˩tɕo˧ se#˥**//.

If the suffix //**-ze˧**// surfaced with H tone in any of the combinations in Tables 6.19–6.20, that H tone would need to be recognized as the manifestation of a floating H tone from the directional-plus-verb expression, which would contradict headlong the interpretation proposed here: that floating H tones on such expressions only manifest themselves through *lowering* of a following tone. Crucially, the suffix //**-ze˧**// does not, in fact, surface with H tone in any of the combinations in Tables 6.19–6.20. This is taken as confirmation of the present analysis, in light of which the data in Tables 6.19 and 6.20 is rewritten in Tables 6.21 and 6.22, indicating the underlying tone patterns.

In Tables 6.21 and 6.22, reference needs to be made to the juncture between the directional morpheme and the verb (similar to junctures internal to numeral-plus-classifier phrases and compound nouns, studied in previous chapters). This morpheme break is indicated by the symbol '–'. Thus, the indication 'L#–' refers to a final L tone (L#) attaching before the morpheme break. For instance, association of L#– to the syllable sequence /**ʁo.tʰo.li**/ 'to look back(ward)' requires identification of the morpheme break following the directional /**ʁo.tʰo**/ 'backward': /**ʁo.tʰo – li**/. The L tone associates to the last syllable of the first part of the expression, yielding /**ʁo.tʰo˩**/, and spreads (by Rule 1) to the following syllable, hence /**ʁo.tʰo˩ li˩**/. Finally, the expression's first syllable receives M tone (by Rule 2), yielding /**ʁo˧tʰo˩ li˩**/. This is illustrated in Figure 6.2.

Table 6.19: The surface tone patterns of verbs after indications of spatial orientation: monosyllabic prefixes.

tone of prefix	tone of verb					
	H	M_a	M_b	L_a	L_b	MH
L	L.H	L.M+M	L.M+M	L.H	L.H	L.MH

A further intricacy is that the behaviour of /**ɑ˩pʰo˩**/ 'outside' is not fully identical with that of /**jo˩gi˩**/ and /**jo˩lo˩**/ 'to the right', even though these three spatial expressions share the same lexical tone. In association with /**tʰʏ̩˧$_a$**/ 'come out', /**ɑ˩pʰo˩**/ 'outside' yields /**ɑ˩pʰo˩ tʰʏ̩˩**/ 'to go outside, to get outside', instead of the expected † **ɑ˩pʰo˩ tʰʏ̩˥**, which is not an acceptable variant.

Table 6.20: The surface tone patterns of verbs after indications of spatial orientation: disyllabic orientation adverbials.

tone of prefix	tone of verb						
	H	M_a	M_b	M_c	L_a	L_b	MH
M	M.M.M+L	M.M.M+M	M.M.M+L	M.M.M+M	M.M.L	M.M.L	M.M.MH
#H	M.M.M+L	M.M.L	M.M.M+L	M.M.M+L	M.M.H	M.M.H	M.M.L
L	L.L.L	L.L.H	L.L.H / L.L.L	L.L.L	L.L.H	L.L.H	L.L.H
L#	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L
LM	L.M.M+L	L.M.M+M	L.M.M+M	L.M.M+M	L.M.L	L.M.L	L.M.MH
H#	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L

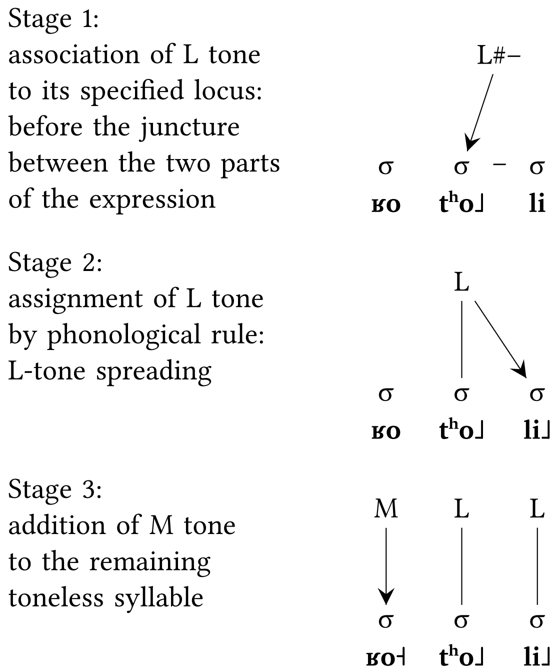


Figure 6.2: Illustration of the anchoring of tones relative to an internal juncture (notation: '–'): representation of association of L#– tone to the expression /**ʁo.tʰo – li**/ to yield /**ʁo˧tʰo˩ li˩**/ 'to look back'.

Table 6.21: The underlying tone patterns of verbs after indications of spatial orientation: monosyllabic prefixes.

tone of prefix	tone of verb					
	H	M_a	M_b	L_a	L_b	MH
L	L.H	L.M	L.M	L.H	L.H	L.MH

Closer examination of this issue reveals yet another oddity: different verbs that belong to the same tonal category, M_a, display different tone patterns when associated with /**ɑ˩pʰo˩**/ 'outside'. 'To look outside' (from /**li˧**$_{\mathrm{a}}$/ 'to look') is /**ɑ˩pʰo˩ li˥**/, and ‡ **ɑ˩pʰo˩ li˩** is not an acceptable variant. The verb /**tʰɣ̩˧**$_{\mathrm{a}}$/ 'come out' is an outlier: it is the only M_a-tone verb yielding an L.L.L tone pattern in association with /**ɑ˩pʰo˩**/ 'outside'.

Given that L.L.L and L.L.H are both acceptable variants for /**jo˩gi˩**/ and /**jo˩lo˩**/ 'to the right' when followed by an M_a-tone verb, one may speculate that the same

Table 6.22: The underlying tone patterns of verbs after indications of spatial orientation: disyllabic orientation adverbials.

tone of prefix	tone of verb					
	H	M_a	M_b	M_c	L_a	L_b	MH
M	#H	M	#H	M	L#		MH#
#H		L#		#H	H#		#H
L	L	L+H#	L+H# / L	L	L+H#		
L#	L#–						
LM	LM+#H	LM			LM+L#		LM+MH#
H#	H#–						

pattern of variation once existed for /**ɑ˩pʰo˩**/ 'outside'. Under this hypothesis, the L.L.L variant must have become dominant for 'to go outside, to get outside', to the extent that /**ɑ˩pʰo˩ tʰv̩˩**/ came to be regarded as the only correct form.

But even if one chooses to treat the combination /**ɑ˩pʰo˩ tʰv̩˩**/ 'to go outside, to get outside' as a lexicalized oddity, the behaviour of /**ɑ˩pʰo˩**/ 'outside' still differs from that of /**jo˩gi˩**/ and /**jo˩lo˩**/ 'to the right' (see Table 6.23). Generating the tone patterns requires knowledge of the morphotonological rules specific to the construction at hand. These data provide yet another compelling illustration of morphotonological complexity – one that dashes any hope of an exhaustive account relying solely on phonological rules.

Table 6.23: The tonal behaviour of verbs in association with /**ɑ˩pʰo˩**/ 'outside'.

tone of verb	example	meaning of verb	tone pattern
H	**ɑ˩pʰo˩ se˩**	to walk	L.L.L
M_a	**ɑ˩pʰo˩ li˥**	to look	L.L.H
M_a (exceptional)	**ɑ˩pʰo˩ tʰv̩˩**	to get/go	L.L.L
M_b	**ɑ˩pʰo˩ hõ˩**	to go.IMPERATIVE	L.L.L
M_c	**ɑ˩pʰo˩ bi˩**	to go	L.L.H
L_a	**ɑ˩pʰo˩ kwɤ˥**	to throw	L.L.H
L_b	**ɑ˩pʰo˩ pʰv̩˥**	to move around	L.L.H
MH	**ɑ˩pʰo˩ z̩i˥**	to sleep	L.L.H

The interrogative /**zo˩qo˧**/ 'where' has the same tonal behaviour as /**gɤ˩tɕo˧**/ 'upward' and /**mv̩˩tɕo˧**/ 'downward', as shown in Table 6.24. In combination with /**zo˩qo˧**/ 'where', the verb /**tʰv̩˧**/, whose association with /**ɑ˩pʰo˩**/ 'outside' yielded an unexpected pattern (see Table 6.23), is not any different from the other M_a-tone verbs, resulting in /**zo˩qo˧ tʰv̩˧(-ze˧)**/.

In principle, the tonal behaviour of locative constituents in Yongning Na could shed light on their morphosyntactic properties. Cross-linguistically, various configurations are attested. For instance, in Central Bantu, verbal agreement reveals a typologically uncommon pattern: "a locative noun phrase in preverbal position can be analyzed as the grammatical subject" (Creissels 2011: 34). In contrast, in Northern Sotho, where such agreement is absent, the construction is better analyzed as "an impersonal construction with a preposed locative constituent" (Zerbian 2006b). However, the Yongning Na tonal patterns presented above are not fully identical with those of any other construction. In particular, they differ from both subject-verb and object-verb constructions.

Table 6.24: The tonal behaviour of verbs in association with /**zo˩qo˧**/ 'where', with added information about a following perfective morpheme.

tone of verb	example	meaning of verb	tone pattern
H	**zo˩qo˧ se˧(-ze˩)**	to walk	L.M.M+L
M_a	**zo˩qo˧ ʂe˧(-ze˧)**	to look for	L.M.M+M
M_b	**zo˩qo˧ pʰæ˧(-ze˧)**	to attach, to fasten	L.M.M+M
M_c	**zo˩qo˧ hɯ˧(-ze˧)**	to go.pst	L.M.M+M
L_a	**zo˩qo˧ dzi˩**	to sit; to live	L.M.L
L_b	**zo˩qo˧ ɻ̍˩**	to turn towards	L.M.L
MH	**zo˩qo˧ lɑ˧˥**	to strike, to hit	L.M.MH

Let us now turn from preverbal elements to postverbal ones.

6.4 Monosyllabic postverbal morphemes

Verbs follow their objects and may be followed in turn by various morphemes, including suffixes,[6] serialized verbs, postpositions, and discourse particles, which will be referred to here under the cover term 'postverbal morphemes'.

From the point of view of tone patterns, a key observation is that not all morphemes following verbs have the same tonal behaviour, suggesting that they have lexical tones of their own. Distributional analysis constitutes a first step in investigating these tones: finding out how many tonal categories exist for a given part of speech. Once morphemes have been sorted into tonal sets, the next step consists in analyzing their underlying tones, as was done above for nouns and verbs.

[6]Lidz (2010: 349), based on data from the Luoshui dialect, proposes that "[s]uffixation is not attested on verbs in Na". However, for the dialect under description here, it appears most straightforward to recognize a few postverbal morphemes as suffixes, such as the perfective /**-ze˧**/ (analyzed by Lidz (2010: 424) as a postverbal particle). In transcriptions, a hyphen is also used in cases where a verb's grammaticalized use appears sufficiently distinct from the verb's original meaning to warrant separate recognition, for instance the immediate future /**-bi˧**/, grammaticalized from the non-imperative form of 'to go', /**bi˧**$_c$/. The decision to treat a morpheme as a serialized verb or a suffix can be difficult. My use of hyphens fluctuated over the years of preparation of this volume; as of 2025, some inconsistencies remain in the transcriptions of online primary documents in the Pangloss Collection. Consistent implementation of the usage conventions adopted in the dictionary (Michaud et al. 2024) in the glossing of the Yongning Na texts is on the to-do list for years to come.

On the one hand, in Yongning Na, all syllables carry level tones at the surface phonological level: H, M, L, and combinations into low-rising and mid-rising contours, so it makes sense to hypothesize that the lexical tone categories of postverbal morphemes are composed of these level tones and to attempt to pinpoint their phonological nature, as was done for nouns and verbs. On the other hand, one should keep in mind the structural fact that the tone systems of different word classes are not identical. In addition to differences in the number of tones (among monosyllables: seven categories for verbs, five for adjectives, six for free nouns, nine for classifiers…), tonal behaviour in context also varies across word classes, even between closely related categories such as verbs and adjectives (as pointed out in §6.1.3). Consequently, the tonal analysis of a verb cannot be assumed to apply straightforwardly to its grammaticalized counterpart.

For instance, the morpheme indicating COMPLETION, /**-se˩**/, clearly originates from the verb /**se˩**/ 'to finish, to complete'. However, this fact does not by itself justify the assumption that the grammaticalized morpheme retains the same lexical tone as the verb. Nominal classifiers, studied in Chapter 3, are a perfect illustration of the changes in lexical tone that can accompany the process of grammaticalization. Returning to postverbal morphemes, cases where a morpheme can clearly be traced back to a verb only provide indirect hints on tonal identity. The tonal categories of the grammatical morphemes examined in this chapter are therefore assembled piece by piece, and there remains room for further progress in the analysis.

The reader will be reminded at various points that a given lexical-tone label (say, L tone) assigned to different parts of speech (e.g. a verb and a suffix) does not necessarily refer to the exact same phonological entity. Rather, these represent distinct morphophonological categories that appear sufficiently similar – at the present stage of analysis – to warrant the use of the same label from among the set of phonologically distinctive tonal levels.

Monosyllabic elements will be discussed first, before turning to disyllabic postpositions and combinations of affixes.

6.4.1 L-tone postverbal morphemes

As noted above, the morpheme indicating COMPLETION, /**-se˩**/, is a grammaticalized form of the verb /**se˩**/ 'to finish, to complete'. This alone does not constitute sufficient evidence that its lexical tone is L, but its tonal behaviour after M-tone verbs (where it surfaces with L tone) points in the same direction; accordingly, it is classified as an L-tone morpheme. Since the tonal behaviour of the desiderative morpheme is the same as that of the COMPLETION morpheme, the same tonal label is applied, yielding /**-ho**˩/.

Table 6.25 presents data illustrating the tonal patterns for these two morphemes, also including the morpheme /**-sɯ**˩/ 'yet' (used in the negative construction 'not yet'). Other morphemes belonging to the same tonal class (L tone) include the inchoative, /**-ɹ̩**˩/, and the morpheme /**-dze**˩/ 'to remain; to be left over'. The latter is only observed in /**dzɯ˧-dze**˥/ 'left over after eating' and /**ʈʰɯ˩-dze**˩/ 'left over after drinking (or smoking)'. These *noncompletion resultatives*, which denote the state resulting from an action that was not carried through to completion (the incomplete consumption of an object), have handy equivalents in Mandarin Chinese: *chī shèng de* 吃剩的 for 'left over after eating' and *hē shèng de* 喝剩的 for 'left over after drinking'.

Table 6.25: The patterns of L-tone tense-aspect-modality morphemes.

tone	example	meaning	COMPLETION	DESIDERATIVE	not yet
H	**dzɯ**˥	to eat	**dzɯ˧-se**˥	**dzɯ˧-ho**˥	**mɤ˧-dzɯ˧-sɯ**˥
M_a	**hwæ˧**$_{\mathbf{a}}$	to buy	**hwæ˧-se**˩	**hwæ˧-ho**˩	**mɤ˧-hwæ˧-sɯ**˩
M_b	**tɕʰi˧**$_{\mathbf{b}}$	to sell	**tɕʰi˧-se**˩	**tɕʰi˧-ho**˩	**mɤ˧-tɕʰi˧-sɯ**˩
M_c	**pv̩˧**$_{\mathbf{c}}$	to chant	**pv̩˧-se**˩	**pv̩˧-ho**˩	**mɤ˧-pv̩˧-sɯ**˩
L_a	**bæ˩**$_{\mathbf{a}}$	to sweep	**bæ˩-se**˩	**bæ˩-ho**˩	**mɤ˧-bæ˩-sɯ**˩
L_b	**ʐwɤ˩**$_{\mathbf{b}}$	to speak	**ʐwɤ˩-se**˩	**ʐwɤ˩-ho**˩	**mɤ˧-ʐwɤ˩-sɯ**˩
MH	**lɑ**˧˥	to strike	**lɑ˧-se**˥	**lɑ˧-ho**˥	**mɤ˧-lɑ˧-sɯ**˥

The next paragraph is devoted to the nominalizing suffix /**-di**˩/, which does not fully align with the three morphemes in Table 6.25 in terms of its tonal behaviour.

The nominalizing suffix /-di˩/

In her study of the Luoshui dialect of Yongning Na, Liberty Lidz observes that "**di**[33] 'earth; place' grammaticalized into a locative nominalizer, and then further grammaticalized into a purposive nominalizer" (Lidz 2010: 184). In the Alawua dialect, the nominalizing morpheme is assigned L tone (hence /**-di**˩/) on the basis of its behaviour after M-tone verbs. However, its tone behaviour is distinct from that of the L-tone morphemes discussed in the previous paragraph. For instance, compare /**dzɯ˧-di**˧˥/ 'food, things for eating' with /**dzɯ˧-ho**˥/ 'will eat', or /**ʈʰæ˧-di**˧˥/ 'thing to bite (e.g. a toy given to teething babies)' with /**ʈʰæ˧-ho**˥/ 'will bite'. This observation supports the analysis of the nominalizing morpheme as a suffix, distinct in its morphotonological properties from serialized verbs. The data is set out in Table 6.26.

Table 6.26: The tonal behaviour of the nominalizing suffix /**-di˩**/.

tone	example	meaning	nominalizer	meaning
H	**dzɯ˥**	to eat	**dzɯ˧-di˧˥**	thing to eat, food
M_a	**hwæ˧a**	to buy	**hwæ˧-di˩**	thing to buy, product
M_b	**tɕʰi˧b**	to sell	**tɕʰi˧-di˩**	thing to sell, commodity
M_c	**pv̩˧c**	to chant	**pv̩˧-di˩**	thing to chant, ritual
L_a	**dze˩a**	to cut	**dze˩-di˩**	thing to cut, e.g. knife
L_b	**ʈʰɯ˩b**	to drink	**ʈʰɯ˩-di˩**	thing to drink, beverage
MH	**ʈʰæ˧˥**	to bite	**ʈʰæ˧-di˧˥**	thing to bite (for infant, dog…)

As an aside, consultant M21 has a different tone pattern for the L tone: L.H (e.g. /**dze˩-di˥**/ and /**ʈʰɯ˩-di˥**/) rather than L.L. This could result from analogy with other L-tone morphemes, such as the morpheme indicating COMPLETION, /**-se˩**/, which takes H tone after an L-tone verb, as discussed in §6.4.1.

6.4.2 M-tone postverbal morphemes

The IMPERATIVE, /**-hõ˧**/, is grammaticalized from the imperative form of the verb 'to go', /**hõ˧a**/, and the IMMEDIATE FUTURE, /**-bi˧**/, derives from the non-imperative form of 'to go', /**bi˧c**/. This provides an initial indication that these morphemes may belong to an M-tone category. A stronger argument comes from their behaviour in combination with the seven tonal categories of verbs, shown in Table 6.27. The table also includes data for the experiential /**-dʐɯ˧**/, which has the same tonal behaviour across all cases.

The tone patterns observed after verbs with M, L or MH tone suggest that these postverbal morphemes behave as tonally neutral elements: they allow a verb's MH tone to unfold onto them and permit a verb's L tone to spread over them. After an M-tone verb, the surface result is M, consistent with the phonologically inert nature of M tone, already noted in §2.4.3.

Other morphemes with the same tonal behaviour include /**-ɖo˧**/ 'must, have to', the VOLITIVE /**-tso˧**/, the OBLIGATIVE /**-zo˧**/, and the CAUSATIVE /**-tsæ˧**/.

In addition to the surface phonological forms, Table 6.27 proposes an analysis of the underlying tone in its final column. This analysis is based on the tone patterns when the CAUSATIVE /**-tsæ˧**/ is added after the IMMEDIATE FUTURE /**-bi˧**/,

Table 6.27: The tonal behaviour of M-tone morphemes.

tone	example	meaning	IMM_FUT	EXPERIENTIAL	IMPERATIVE	analysis
H	**dzɯ˥**	to eat	**dzɯ˧-bi˧**	**dzɯ˧-dʑɯ˧**	**dzɯ˧-hõ˧**	M
M_a	**hwæ˧$_a$**	to buy	**hwæ˧-bi˧**	**hwæ˧-dʑɯ˧**	**hwæ˧-hõ˧**	M
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧-bi˧**	**tɕʰi˧-dʑɯ˧**	**tɕʰi˧-hõ˧**	M
M_c	**pv̩˧$_c$**	to chant	**pv̩˧-bi˧**	**pv̩˧-dʑɯ˧**	**pv̩˧-hõ˧**	M
L_a	**bæ˩$_a$**	to sweep	**bæ˩-bi˩**	**bæ˩-dʑɯ˩**	**bæ˩-hõ˩**	L
L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩-bi˩**	**ʐwɤ˩-dʑɯ˩**	**ʐwɤ˩-hõ˩**	L
MH	**lɑ˧˥**	to strike	**lɑ˧-bi˥**	**lɑ˧-dʑɯ˥**	**lɑ˧-hõ˥**	H#

Table 6.28: Tone patterns of V+IMMEDIATE FUTURE+CAUSATIVE-+COPULA.

tone	example	meaning	**-bi˧-tsæ˧-ɲi˩**
H	**dzɯ˥**	to eat	**dzɯ˧-bi˧-tsæ˧-ɲi˩**
M_a	**hwæ˧$_a$**	to buy	**hwæ˧-bi˧-tsæ˧-ɲi˩**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧-bi˧-tsæ˧-ɲi˩**
M_c	**pv̩˧$_c$**	to chant	**pv̩˧-bi˧-tsæ˧-ɲi˩**
L_a	**bæ˩$_a$**	to sweep	**bæ˩-bi˩-tsæ˥-ɲi˩ (‡ bæ˩-bi˩-tsæ˩˥-ɲi˩)**
L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩-bi˩-tsæ˥-ɲi˩ (‡ ʐwɤ˩-bi˩-tsæ˩˥-ɲi˩)**
MH	**lɑ˧˥**	to strike	**lɑ˧-bi˧-tsæ˥-ɲi˩ (‡ lɑ˧-bi˧-tsæ˧˥-ɲi˩)**

as shown in Table 6.28. The copula, in its use to express certainty, was also added, as a further test to reveal the underlying tonal categories. In every case, the copula carries L, i.e. its lexical tone. This shows the absence of a floating H tone in any of the verb phrases. The first three expressions are therefore analyzed as having M tone. The underlying tonal category leading to the realizations /**lɑ˧-bi˥**/ 'will strike' and /**lɑ˧-bi˧-tsæ˥**/ (not ‡ **lɑ˧-bi˥-tsæ˩**) 'cause to strike' is interpreted as H#.

It is reassuring to note that these M-tone morphemes have the same behaviour when preceded by the negation prefix, /**mɤ˧**-/, e.g. in /**V mɤ˧-bi˧**/ 'is not going to V', /**V mɤ˧-ɖo˧**/ 'ought not to V', and /**V mɤ˧-zo˧**/ 'must not V'. Table 6.29 provides examples with /**V mɤ˧-bi˧**/, also including, in its last column, the same phrases with an added PERFECTIVE, /**-ze˧**/.

Nonetheless, M-tone morphemes are not without complexities. The verb /**mæ˧**/ 'to achieve' carries M tone, but in serial verb constructions, where it

Table 6.29: Tone patterns of V+NEGATION prefix+IMMEDIATE FUTURE.

tone	example	meaning	V NEG-IMM_FUT	V NEG-IMM_FUT-PFV
H	**dzɯ˥**	to eat	**dzɯ˧ mɤ˧-bi˧**	**dzɯ˧ mɤ˧-bi˧-ze˩**
M$_a$	**hwæ˧$_a$**	to buy	**hwæ˧ mɤ˧-bi˧**	**hwæ˧ mɤ˧-bi˧-ze˧**
M$_b$	**tɕʰi˧$_b$**	to sell	**tɕʰi˧ mɤ˧-bi˧**	**tɕʰi˧ mɤ˧-bi˧-ze˧**
M$_c$	**pv̩˧$_c$**	to chant	**pv̩˧ mɤ˧-bi˧**	**pv̩˧ mɤ˧-bi˧-ze˧**
L$_a$	**bæ˩$_a$**	to sweep	**bæ˩ mɤ˩-bi˩**	**bæ˩ mɤ˩-bi˩-ze˥**
L$_b$	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩ mɤ˩-bi˩**	**ʐwɤ˩ mɤ˩-bi˩-ze˥**
MH	**lɑ˧˥**	to strike	**lɑ˧ mɤ˧-bi˧**	**lɑ˧ mɤ˧-bi˧-ze˩**

indicates that an action has achieved its goal, its tonal behaviour is not fully identical to that of the M-tone morphemes in Table 6.27. The PERFECTIVE /**-ze˧**/ has the same tonal behaviour as /**-mæ˧**/. Their tone patterns are shown in Table 6.30. While their behaviour is in most respects similar to that of the M-tone morphemes in Table 6.27, the pattern following an H-tone verb is M.L and not M.M. This warrants distinguishing a separate tonal category, leading to the establishment of two M-tone subclasses for postverbal morphemes (M$_a$ and M$_b$).

For classifiers (discussed in Chapter 3), a total of nine tonal categories has been identified. It would not be particularly surprising if a comparable number of categories were required for affixes. In view of the greater morphosyntactic diversity of affixes in comparison with classifiers, the number could even be higher. Clearly, the added subscript letters serve merely as classificatory labels: they do not constitute an analysis of the tonal categories. A more thorough analysis

Table 6.30: The behaviour of postverbal morphemes belonging to a second subcategory of M tones: M$_b$.

tone	example	meaning	PERFECTIVE	to achieve
H	**dzɯ˥**	to eat	**dzɯ˧-ze˩**	**dzɯ˧-mæ˩**
M$_a$	**hwæ˧$_a$**	to buy	**hwæ˧-ze˧**	**hwæ˧-mæ˧**
M$_b$	**tɕʰi˧$_b$**	to sell	**tɕʰi˧-ze˧**	**tɕʰi˧-mæ˧**
M$_c$	**pv̩˧$_c$**	to chant	**pv̩˧-ze˧**	**pv̩˧-mæ˧**
L$_a$	**bæ˩$_a$**	to sweep	**bæ˩-ze˩**	**bæ˩-mæ˩**
L$_b$	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩-ze˩**	**ʐwɤ˩-mæ˩**
MH	**lɑ˧˥**	to strike	**lɑ˧-ze˥**	**lɑ˧-mæ˥**

would require a fuller inventory of morphemes. It should be borne in mind that the current size of the dictionary (Michaud et al. 2024) is approximately 3,000 words, i.e. only a small part of the full richness of the language's lexicon.

In this chapter, postverbal morphemes are simply arranged into broad tonal classes (L, M, H, and MH), while also reporting the internal diversity observed within these classes and setting up provisional subclasses, such as M_a and M_b. Subclasses M_a and M_b of M-tone postverbal morphemes include the following items:

- M_a tone: IMPERATIVE /**-hõ˧$_a$**/, IMMEDIATE FUTURE /**-bi˧$_a$**/, EXPERIENTIAL /**-dʑɯ˧$_a$**/, /**-ɖo˧$_a$**/ 'must, have to', VOLITIVE /**-tso˧$_a$**/, OBLIGATIVE /**-zo˧$_a$**/, and CAUSATIVE /**-tsæ˧$_a$**/
- M_b tone: PERFECTIVE /**-ze˧$_b$**/, and /**mæ˧$_b$**/ 'to achieve'

In the data presented in Table 6.29, the H component found in the lexical tone categories H and MH (illustrated by /**dzɯ˥**/ 'to eat' and /**lɑ˧˥**/ 'to strike', respectively) does not surface as such. Instead, it manifests itself through the lowering of the PERFECTIVE morpheme /**-ze˧$_b$**/, even though this morpheme is no fewer than three syllables away from the verb. The phrase /**dzɯ˧ mɤ˧-bi˧-ze˩**/ 'will not eat anymore' constitutes a single tone group, and the underlying presence of an H tone within this group becomes apparent through its effect on the PERFECTIVE morpheme, despite the intervening syllables. The same phenomenon is also observed in the absence of an intervening negation prefix, as shown in Table 6.31.

Table 6.31: Same data as in Table 6.29, without intervening negation prefix.

tone	example	meaning	V+IMMEDIATE FUTURE+PERFECTIVE
H	**dzɯ˥**	to eat	**dzɯ˧ bi˧-ze˩**
M_a	**hwæ˧$_a$**	to buy	**hwæ˧ bi˧-ze˧**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧ bi˧-ze˧**
M_c	**pv̩˧$_c$**	to chant	**pv̩˧ bi˧-ze˧**
L_a	**bæ˩$_a$**	to sweep	**bæ˩ bi˩-ze˥**
L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩ bi˩-ze˥**
MH	**lɑ˧˥**	to strike	**lɑ˧ bi˥-ze˩**

In these contexts, the PERFECTIVE morpheme //**-ze˧$_b$**// carries different tones depending on the lexical tone of the verb – including in the case of H-tone verbs, where the lexical H tone surfaces neither on the verb itself nor on the immediately following morpheme (the IMMEDIATE FUTURE). For H-tone verbs, the PERFECTIVE morpheme //**-ze˧$_b$**// is consistently lowered to L in each of (27), (28), and (29).

(27) dzɯ˧-ze˩
dzɯ˥ -ze˧$_b$
to_eat PFV
'have eaten'

(28) dzɯ˧-bi˧-ze˩
dzɯ˥ -bi˧$_a$ -ze˧$_b$
to_eat IMM.FUT PFV
'will eat'

(29) dzɯ˧ mɤ˧-bi˧-ze˩
dzɯ˥ mɤ˧- -bi˧$_a$ -ze˧$_b$
to_eat NEG IMM.FUT PFV
'will not eat'

Tones that are present underlyingly but only manifest themselves in a restricted set of contexts constitute a salient characteristic of the Yongning Na tone system. The surface phonological representations of 'going to buy' as /**hwæ˧-bi˧**/ and 'going to eat' as /**dzɯ˧-bi˧**/, with the same tone pattern, do not reveal the underlying presence of an H tone in the former phrase. The manifestation of the H tone of the verb is not straightforward: it does not lower an M-tone postverbal morpheme (as seen in /**dzɯ˧-bi˧**/, 'going to eat'), yet it does induce lowering on the last syllable in (28).

The lowering effect of overt H tones is an exceptionless phonological regularity, formalized as Rule 4: "The syllable following an H-tone syllable receives L tone" (§7.1.1). In certain morphosyntactic contexts, exemplified in Table 6.29, H tones exert a similar influence even though they remain floating, i.e. unassociated with a syllable. Such complexities cannot be captured by a small set of rules. This explains the abundance of tables in this chapter and, more broadly, throughout this volume.

Further data on M-tone postverbal morphemes is shown in Table 6.32, illustrating the behaviour of the PROGRESSIVE /**-dʑo˧**/. The affirmative particle /**mv̩˧**/ has the same behaviour. An intriguing peculiarity is that an MH contour on the verb does not unfold onto these morphemes. A MH-tone verb preceding these morphemes retains its MH contour, e.g. /**mɤ˧-lɑ˧˥ | -dʑo˩**/ 'as [they] did not strike…'.

Following the same descriptive approach as before, this difference in tonal behaviour requires setting up a third subcategory of M tones among postverbal morphemes: M_c.

Table 6.32: Tonal behaviour of the progressive, illustrating a third subcategory of M tones among postverbal morphemes: M_c.

tone	example	meaning	DUR+V+PROG: 'is currently V-ing'
H	**dzɯ˥**	to eat	**tʰi˧-dzɯ˥-dʑo˩**
M_a	**hwæ˧a**	to buy	**tʰi˧-hwæ˧-dʑo˧**
M_b	**tɕʰi˧b**	to sell	**tʰi˧-tɕʰi˧-dʑo˧**
M_c	**py̩˧c**	to chant	**tʰi˧-py̩˧-dʑo˧**
L_a	**bæ˩a**	to sweep	**tʰi˧-bæ˩-dʑo˩**
L_b	**ʐwɤ˩b**	to speak	**tʰi˧-ʐwɤ˩-dʑo˩**
MH	**lɑ˧˥**	to strike	**tʰi˧-lɑ˧˥-dʑo˩** (‡ **tʰi˧-lɑ˥-dʑo˩**)

6.4.3 H-tone postverbal morphemes

Two postverbal morphemes are tentatively classified under the heading of "H-tone postverbal morphemes": the relativizer/nominalizer //**-hĩ˥**// and the topic marker //**-dʑo˥**//. The rationale for this categorization is as follows.

The relativizer/nominalizer //**-hĩ˥**// behaves in many tonal contexts like the M-tone postverbal morphemes described in §6.4.2, such as the IMMEDIATE FUTURE /**-bi˧**/. However, unlike M-tone morphemes, the relativizer/nominalizer does not host the H component of an MH contour from the preceding verb; instead, the outcome is M.M, as exemplified in (30), contrasting with the M.H pattern found for M-tone morphemes, as in (31). This difference indicates that the relativizer belongs to a tonal category distinct from those previously identified as L and M.

(30) lɑ˧-hĩ˧
lɑ˧˥ -hĩ˥
to_strike RELATIVIZER/NOMINALIZER
'who strikes'

(31) lɑ˧-bi˥
lɑ˧˥ -bi˧
to_strike IMM.FUT
'is going to strike'

In assigning labels to tonal categories, I adopt the (debatable) assumption that the tone systems of different parts of speech rest on the same primitives (H, M, and L levels) and on combinations among these primitives (MH, LM, and LH). One argument for classifying the relativizer/nominalizer as an H-tone morpheme stems from the hypothesis that it derives from the noun meaning ‘person, human being’, which bears H tone: /**hĩ˥**/ (on the grammaticalization of this noun, see Lidz 2010: 164, 183). This argument is not decisive, however. As noted earlier, the process of grammaticalization often leads a morpheme to enter a tonal subsystem distinct from that of its original morphosyntactic class, sometimes resulting in a tone that is widely different from that of the source word.

Another argument for the H-tone classification is that, when combined with the relativizer/nominalizer //**-hĩ˥**//, adjectives in tone category L_b (such as /**dʑɤ˩$_b$**/ ‘good’ and /**nɑ˩$_b$**/ ‘black, dark’) yield an L.H pattern, as shown in Table 6.33. But this argument is not decisive either: since the rules are morphotonological and not simply phonological, an H tone in the output cannot be taken as direct evidence of an H tone in the input. The existence of an LH tonal category for disyllabic nouns may act as an attractor, funnelling diverse morphological combinations towards an LH output.

Adjectives and verbs behave quite differently when combined with the relativizer/nominalizer //**-hĩ˥**//. Unexpectedly, the tone patterns of verbs and those of adjectives are also very different: they coincide in only one of five categories (namely L_a). This comes as a surprise in view of the fact that, in many other contexts, the tonal behaviour of verbs and adjectives is pretty much the same. This discrepancy may have to do with semantic-syntactic differences in the function of the morpheme: when suffixed to an adjective, //**-hĩ˥**// functions as a nominalizer, whereas when suffixed to a verb, it functions as a relativizer.

The TOPIC marker //**-dʑo˥**// commonly occurs after verb phrases, as well as after noun phrases. On the basis of its tonal behaviour following M-tone verbs (and, similarly, following M-tone nouns, as described in the preceding chapter), this TOPIC marker is (provisionally) analyzed as carrying a lexical H tone. Data on its behaviour in context is presented in Table 6.34. As in Table 6.32, the MH tone does not unfold onto the following morpheme: it is not correct to say ‡ **mɤ˧-lɑ˧-dʑo˥**.

Importantly, the tone patterns for this TOPIC marker, shown in Tables 6.34 and 6.35, differ from those of the RELATIVIZER, shown in Table 6.33. Instead, the TOPIC marker exhibits stronger surface similarity to the PROGRESSIVE morpheme (documented in Table 6.32), which is analyzed as carrying lexical M tone. The sole difference between the TOPIC and PROGRESSIVE morphemes lies in their realiza-

Table 6.33: The tonal behaviour of the relativizer/nominalizer, analyzed as having a lexical H tone.

word class	tone	example	meaning	with RELATIVIZER
verbs	H	**dzɯ˥**	to eat	**dzɯ˧-hĩ˧**
	M_a	**hwæ˧$_a$**	to buy	**hwæ˧-hĩ˧**
	M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧-hĩ˧**
	M_c	**bi˧$_c$**	to go	**bi˧-hĩ˧**
	L_a	**bæ˩$_a$**	to sweep	**bæ˩-hĩ˩**
	L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩-hĩ˩**
	MH	**lɑ˧˥**	to strike	**lɑ˧-hĩ˧**
adjectives	H	**bi˥**	shallow	**bi˧-hĩ#˥**
	M	**tɕi˧**	sour	**tɕi˧-hĩ#˥**
	L_a	**hv̩˩$_a$**	red	**hv̩˩-hĩ˩**
	L_b	**dʑɤ˩$_b$**	good	**dʑɤ˩-hĩ˥**
	MH	**tʰɑ˧˥**	sharp	**tʰɑ˧-hĩ˥$**

tion following M-tone verbs: the TOPIC marker surfaces with H tone, whereas the PROGRESSIVE retains M tone.

The logical thing to do would be to recognize two subcategories of H-tone postverbal morphemes: H_a, exemplified by the RELATIVIZER, and H_b, exemplified by the TOPIC marker. But categories containing only a single member are not immensely useful. The classification of postverbal morphemes into L, M, H or MH, as proposed in §6.4.1-§6.4.4, will likely require in-depth revision as a more comprehensive picture of the tonal behaviour of postverbal elements emerges. At present, it remains premature to rigidify (hypostatize) these newly established tonal categories into a seemingly neat and tidy system. Yongning Na is replete with morphotonological anfractuosities; the present chapter, which employs "L tone", "M tone", "H tone", and "MH tone" as convenient first-pass labels, only constitutes an initial step towards an orderly inventory and analysis.

6.4.4 MH-tone postverbal morphemes

Following the same exploratory method as outlined above in the discussion of the "L", "M", and "H" sets of postverbal morphemes, an "MH" set is postulated, likewise based on fragmentary evidence. The ABILITIVE /**-kv̩˧˥**/ derives from the verb /**kv̩˧˥**/ 'to be able to'; this link, together with similarities in tonal behaviour,

Table 6.34: The tonal behaviour of the TOPIC marker with verbs.

tone	example	meaning	V+TOP	NEG+V+TOP
H	**dzɯ˥**	to eat	**dzɯ˧-dʑo˩**	**mɤ˧-dzɯ˥-dʑo˩**
M_a	**hwæ˧$_a$**	to buy	**hwæ˧-dʑo˥**	**mɤ˧-hwæ˧-dʑo˥**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧-dʑo˥**	**mɤ˧-tɕʰi˧-dʑo˥**
M_c	**bi˧$_c$**	to go	**bi˧-dʑo˥**	**mɤ˧-bi˧-dʑo˥**
L_a	**bæ˩$_a$**	to sweep	**bæ˩˧-dʑo˩**	**mɤ˧-bæ˩-dʑo˩**
L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩˧-dʑo˩**	**mɤ˧-ʐwɤ˩-dʑo˩**
MH	**lɑ˧˥**	to strike	**lɑ˧˥-dʑo˩**	**mɤ˧-lɑ˧˥-dʑo˩**

Table 6.35: The tonal behaviour of the TOPIC marker with adjectives.

tone	example	meaning	ADJ+TOP	NEG+ADJ+TOP
H	**bi˥**	shallow	**bi˧-dʑo˩**	**mɤ˧-bi˥-dʑo˩**
M	**tɕi˧**	sour	**tɕi˧-dʑo˥**	**mɤ˧-tɕi˧-dʑo˥**
L_a	**hṽ̩˩$_a$**	red	**hṽ̩˩˧-dʑo˩**	**mɤ˧-hṽ̩˩-dʑo˩**
L_b	**dʑɤ˩$_b$**	good	**dʑɤ˩˧-dʑo˩**	**mɤ˧-dʑɤ˩-dʑo˩**
MH	**tʰɑ˧˥**	sharp	**tʰɑ˧˥-dʑo˩**	**mɤ˧-tʰɑ˧˥-dʑo˩**

leads to the adoption of the label "MH" for the tone category of the ABILITIVE and other morphemes that share its tonal behaviour. Three examples are provided in Table 6.36: in addition to the ABILITIVE /**-kv̩˧˥**/, they are the PERMISSIVE /**-tʰɑ˧˥**/ and the CAUSATIVE /**kʰɯ˧˥**/. Other postverbal elements in this MH tonal class include the reported-speech particle /**tsɯ˧˥**/.

The MH tone surfaces as such after M, in keeping with the general phonological tendency whereby the M tone does not interfere with following tones. The MH tone is lowered to L after H; the interpretation proposed is that the H tone does not surface as such due to the neutralization of M and H in tone-group-initial position (this is formulated in §7.1.1 as Rule 3: "In tone-group-initial position, H and M are neutralized to M"), but that this H tone is present underlyingly and lowers all following tones to L (through Rules 4 and 5). an L tone on the verb spreads over an MH-tone postverbal morpheme (e.g. //**bæ˩$_a$**// ⟶ //**bæ˩-kv̩˩**// 'is apt to sweep'), as does the H part of an MH contour (//**lɑ˧˥**// ⟶ //**lɑ˧-kv̩˥**// 'is apt to strike'), delinking the MH tone on the postverbal morpheme.

Table 6.36: The tonal behaviour of MH-tone morphemes after verbs and adjectives.

word class	tone	example	meaning	ABILITIVE	PERMISSIVE	CAUSATIVE
verbs	H	**dzɯ˥**	to eat	**dzɯ˧-kv̩˩**	**dzɯ˧-tʰɑ˩**	**dzɯ˧ kʰɯ˩**
	$\mathrm{M_a}$	**hwæ˧**$_{\mathbf{a}}$	to buy	**hwæ˧-kv̩˧˥**	**hwæ˧-tʰɑ˧˥**	**hwæ˧ kʰɯ˧˥**
	$\mathrm{M_b}$	**tɕʰi˧**$_{\mathbf{b}}$	to sell	**tɕʰi˧-kv̩˧˥**	**tɕʰi˧-tʰɑ˧˥**	**tɕʰi˧ kʰɯ˧˥**
	$\mathrm{M_c}$	**pv̩˧**$_{\mathbf{c}}$	to chant	**pv̩˧-kv̩˧˥**	**pv̩˧-tʰɑ˧˥**	**pv̩˧ kʰɯ˧˥**
	$\mathrm{L_a}$	**bæ˩**$_{\mathbf{a}}$	to sweep	**bæ˩-kv̩˩**	**bæ˩-tʰɑ˩**	**bæ˩ kʰɯ˩**
	$\mathrm{L_b}$	**ʐwɤ˩**$_{\mathbf{b}}$	to speak	**ʐwɤ˩-kv̩˩**	**ʐwɤ˩-tʰɑ˩**	**ʐwɤ˩ kʰɯ˩**
	MH	**lɑ˧˥**	to strike	**lɑ˧-kv̩˥**	**lɑ˧-tʰɑ˥**	**lɑ˧ kʰɯ˥**
adjectives	H	**bi˥**	shallow	**bi˧-kv̩˩**	**bi˧-tʰɑ˩**	**bi˧ kʰɯ˩**
	M	**tsʰi˧**	hot	**tsʰi˧-kv̩˧˥**	**tsʰi˧-tʰɑ˧˥**	**tsʰi˧ kʰɯ˧˥**
	$\mathrm{L_a}$	**ɖɯ˩**$_{\mathbf{a}}$	large	**ɖɯ˩-kv̩˩**	**ɖɯ˩-tʰɑ˩**	**ɖɯ˩ kʰɯ˩**
	$\mathrm{L_b}$	**dʑɤ˩**$_{\mathbf{b}}$	good	**dʑɤ˩-kv̩˧˥**	**dʑɤ˩-tʰɑ˥**	**dʑɤ˩ kʰɯ˥**
	MH	**tʰɑ˧˥**	sharp	**tʰɑ˧-kv̩˥**	**tʰɑ˧-tʰɑ˥**	**tʰɑ˧ kʰɯ˥**

The phrase /**dʑɤ˩ kʰɯ˥**/ ('good'+CAUSATIVE) is in common use as a blessing on special occasions such as the New Year. It could be translated as 'Best wishes!' or 'Let there be good/happiness!' Its L.H tone pattern is also observed on combinations that do not constitute set phrases, showing that it is not an exception. The other phrases in Table 6.36 – /**bi˧ kʰɯ˩**/ ('shallow'+CAUSATIVE), /**ɖɯ˩ kʰɯ˩˥**/ ('large'+CAUSATIVE), /**tsʰi˧ kʰɯ˧˥**/ ('hot'+CAUSATIVE), and /**tʰɑ˧ kʰɯ˥**/ ('sharp' +CAUSATIVE) – all have straightforward causative meanings: 'to make shallow', e.g. to reduce the level of water in a field by decreasing the flow; 'to enlarge', e.g. to increase the size of a farm by adding another building; 'to heat up'; and 'to sharpen'.

Table 6.36 shows that not all postverbal elements grouped under the provisional heading "MH" have exactly the same tonal behaviour. A difference in tone pattern between the causative and abilitive constructions is found in association with $\mathrm{L_b}$-tone adjectives. (This difference has been verified through elicitation on several occasions; examples are found in *Caravans.203* {0004531#S203} and *ComingOfAge2.61* {0004588#S61}, *72*, *73*, *91*, *92*.) The pattern is L.MH in /**dʑɤ˩-kv̩˧˥**/ ('good'+ABILITIVE), and L.H in /**dʑɤ˩ kʰɯ˥**/ ('good'+CAUSATIVE). This requires the recognition of at least two subcategories of MH-tone postverbal morphemes. As explained above, it appears too early at present to propose a definitive inventory. The labels "L tone", "M tone", "H tone", and "MH tone" are used here as a first-pass classification.

6.5 Disyllabic postverbal morphemes

6.5.1 M.H tone

A first tonal category of disyllabic postverbal elements is illustrated by /**-kwɤ˧tɕɯ˥**/ 'after; as; because'. This postposition most often appears as part of a trisyllabic expression: /**-kwɤ˧tɕɯ˥-lɑ˩**/. A hyphen is used before the syllable /**-lɑ˩**/ to reflect the fact that this last syllable can be detached from the other two. In texts, out of 140 examples, ten are without /**-lɑ˩**/, as in (32). (The other examples are *Lake3.54*, *59*, *67*, *Sister.34*, *Sister3.133*, *Caravans.80*, *137*, *Renaming.18*, and *BuriedAlive2.48*.)

(32) bo˩-gv̩˥, | bo˩-hɑ˥ ki˩-hĩ˩=bv̩˩, | ʈʂʰwæ˩-ne˧˥ | ɖɯ˧-ɭɯ˧ | dʑo˧-kwɤ˧tɕɯ˥, | ʈʂʰɯ˧-qo˧ | tʰi˧-dzi˩-kwɤ˩tɕɯ˩, | tɕʰo˧˥ | ɖɯ˧-nɑ˧ | tʰi˧-po˧ tsɯ˥ | mv̩˩.
bo˩-gv̩˥ bo˩-hɑ#˥ ki˧a -hĩ˥ =bv̩˧ ʈʂʰwæ˩ -ne
pig_manger pig_feed to_give NMLZ POSS boat (*loan: Chinese* 船) like
ɖɯ˧-ɭɯ˧ dʑo˧b -kwɤ˧tɕɯ˥ ʈʂʰɯ˧-qo˧ tʰi˧- dzi˩a -kwɤ˧tɕɯ˥ tɕʰo˩˧
one-CLF EXIST as here DUR to_sit as ladle
ɖɯ˧-nɑ˧ tʰi˧- po˧˥ -tsɯ˧˥ mv̩˧
one-CLF.tools DUR to_bring REP AFFIRM
'As there was a pig manger, [you know,] the thing for giving swill, that was like a boat (=that had the shape of a boat), as [they] sat [in this manger]… it is said that they brought a ladle [with them].' *(Lake3.53-54)*
{0004348#S53}

Thus, while addition of /**-lɑ˩**/ (presumably the morpheme //**lɑ˧**//, meaning 'and, also') is a well-established habit, the expression can also be used without it. No special nuance of meaning was identified, except that the formulation without /**-lɑ˩**/ is perceived as more pithy and economical. Example (32) clarifies that the presence or absence of /**-lɑ˩**/ is not conditioned by the semantic interpretation of the postposition as meaning either 'when; after' or 'because, since': among the two occurrences in (32), both without an accompanying /**-lɑ˩**/, the postposition /**-kwɤ˧tɕɯ˥**/ has a causal reading ('since, because') in the first instance and a temporal reading ('as, when') in the second.

The monosyllabic form † **-kwɤ˧** is not attested in this dialect, even though it is reported in another hamlet of the Yongning plain, Walabie (**ʁwɤ˧lɑ˩-bi˩**; Chinese: 瓦拉片) (Roselle Dobbs, p.c. 2016).

The lexical tone of /**-kwɤ˧tɕɯ˥**/ is inferred from its behaviour in combination with the adjective /**dʑɤ˩b**/ 'good'. The observed pattern is /**dʑɤ˧˥-kwɤ˧tɕɯ˥-lɑ˩**/, which does not constitute a well-formed tone group (since it contains two H

Table 6.37: The tonal behaviour of /**-kwɤ˧tɕɯ˥**(**-lɑ˩**)/, 'after; because'.

word class	tone	example	meaning	V+'after; because'
verbs	H	**dzɯ˥**	to eat	**dzɯ˧-kwɤ˩tɕɯ˩(-lɑ˩)**
	M_a	**hwæ˧**$_\mathbf{a}$	to buy	**hwæ˧-kwɤ˧tɕɯ˥(-lɑ˩)**
	M_b	**tɕʰi˧**$_\mathbf{b}$	to sell	**tɕʰi˧-kwɤ˧tɕɯ˥(-lɑ˩)**
	M_c	**bi˧**$_\mathbf{c}$	to go	**bi˧-kwɤ˧tɕɯ˥(-lɑ˩)**
	L_a	**bæ˩**$_\mathbf{a}$	to sweep	**bæ˩-kwɤ˩tɕɯ˥(-lɑ˩)**
	L_b	**ʐwɤ˩**$_\mathbf{b}$	to speak	**ʐwɤ˩-kwɤ˩tɕɯ˥(-lɑ˩)**
	MH	**lɑ˦**	to strike	**lɑ˦-kwɤ˩tɕɯ˩-(lɑ˩)** / **lɑ˧-kwɤ˥tɕɯ˩-(lɑ˩)**
adjectives	H	**bi˥**	shallow	**bi˧-kwɤ˩tɕɯ˩(-lɑ˩)**
	M	**tɕi˧**	sour	**tɕi˧-kwɤ˧tɕɯ˥(-lɑ˩)**
	L_a	**hɣ̃˩**$_\mathbf{a}$	red	**hɣ̃˩-kwɤ˩tɕɯ˥(-lɑ˩)**
	L_b	**dʑɤ˩**$_\mathbf{b}$	good	**dʑɤ˩˧-kwɤ˧tɕɯ˥(-lɑ˩)**
	MH	**tʰɑ˦**	sharp	**tʰɑ˦-kwɤ˩tɕɯ˩(-lɑ˩)**

tones) and must therefore be analyzed as a sequence of two tone groups: /**dʑɤ˩˧ | -kwɤ˧tɕɯ˥-lɑ˩**/. In this context, the tones carried by /**-kwɤ˧tɕɯ˥**/, namely /M.H/, must be supposed to reflect the underlying tone of the expression: a final H tone, i.e. //H#//. As for the added morpheme, interpreted as //**lɑ˧**//, meaning 'and, also', it receives L tone through Rule 4 ("A syllable following an H-tone syllable receives L tone").

The sequence /**dʑɤ˩˧-kwɤ˧tɕɯ˥-lɑ˩**/ 'because/since [it is] good' provides evidence for cases in which a rising contour is realized on the verb and does not unfold over the postverbal expression. Another example is with MH-tone verbs and adjectives, e.g. /**lɑ˦-kwɤ˩tɕɯ˩(-lɑ˩)**/ 'because/since [someone] beat [something]'. Variants in which the MH contour unfolds over the first syllable of the following morpheme (/**lɑ˧-kwɤ˥tɕɯ˩(-lɑ˩)**/) are considered acceptable, and one example occurs in a text (*BuriedAlive2.48* {0004536#S48}), but the predominant pattern is one where the contour remains on the verb.

Since contours are only realized tone-group-finally, this suggests that there is a tone-group boundary before the postposition //**-kwɤ˧tɕɯ˥**// 'as, because'. Such behaviour is not unparalleled in this dialect. For instance, the contrastive topic marker //**-no˦**// and the word /**tʰi˩˧**/ 'then' always mark the beginning of a new tone group. This interpretation is consistent with the observation that the syllable preceding //**-kwɤ˧tɕɯ˥**// 'as, because' tends to be lengthened – a cue to the presence of an intonational boundary.[7] In early transcriptions, a comma was used to reflect a perceived pause, transcribing e.g. /**le˧-tsɑ˦, | -kwɤ˩tɕɯ˩**/ 'because [they] rowed' (*Lake3.59* {0004348#S59}).

But a difficulty with this analysis is that after certain tonal categories of verbs, the tone patterns that are observed on //**-kwɤ˧tɕɯ˥**// 'because' suggest that the verb and the postverbal element belong to the same tone group. Spreading of an L tone from the verb onto the following syllable demonstrates that the verb and its postverbal element are integrated into the same tone group.[8]

One possible way of handling this would be to postulate that the division into tone groups is determined by the lexical tone of the words at issue. There would be a single tone group (e.g. /| **hwæ˧-kwɤ˧tɕɯ˥-lɑ˩** |/ 'when [she/he] buys') except with an MH-tone verb (or an MH-tone or L_b-tone adjective), where the division would be as follows: /| **lɑ˦ | -kwɤ˩tɕɯ˩-lɑ˩** |/ 'when [she/he] strikes'. However, an insuperable difficulty arises: the MH tone permits two variants, one where the two morphemes belong to separate tone groups (no tonal interaction: /**lɑ˦ | -kwɤ˧tɕɯ˥**/) and another where they are integrated into the same group (the H part of the MH contour associates with the following syllable, triggering a lowering of the following tone to L, hence /**lɑ˧-kwɤ˥tɕɯ˩**/).

A further puzzle is that the behaviour of the MH tone (for verbs and adjectives) differs from that of the L_b tone (for adjectives). The tones of the postverbal element are all lowered to L when following an MH-tone verb, whereas they surface unscathed after the rising contour on an L_b-tone adjective. Lowering to L after a preceding H tone level is a phonological rule in the Alawua dialect of Yongning Na (Rule 4: "The syllable following an H-tone syllable receives L tone"; see §7.1.1); the tone rules operate within the tone group, never across a tone-group juncture. The fact that /**kwɤ.tɕɯ-lɑ**/ is lowered to L after an MH-tone verb (e.g. /**lɑ˦-kwɤ˩tɕɯ˩-lɑ˩**/ for 'to strike') strongly suggests that the expression /**kwɤ.tɕɯ-lɑ**/ does not constitute an independent tone group. The sequence /**-kwɤ˩tɕɯ˩-lɑ˩**/ in /**lɑ˦-kwɤ˩tɕɯ˩-lɑ˩**/ 'when (she/he) strikes' is not a well-formed tone group, since it only contains L tones.

This special situation is analyzed as the result of a stylistic process of emphasis that has become habitually associated with certain morphemes. The analysis is

[7] On articulatory, acoustic and perceptual cues to intonational boundaries, see Byrd & Saltzman (2003).

[8] In the case of M-tone verbs, it is impossible to determine the underlying division of the expression into tone groups based on surface tones, because the observed pattern is /M.M.H/ in both cases and could correspond to either of the following interpretations: M.M.H and M | M.H, e.g. /**hwæ˧-kwɤ˧tɕɯ˥-lɑ˩**/ or /| **hwæ˧ | -kwɤ˧tɕɯ˥-lɑ˩**/ for 'because (she/he) buys'. Since the M tone does not affect the following tone, the surface phonological output is the same in both cases.

set out as part of the discussion of the tone group as a key phonological unit in Yongning Na (§7.2).

6.5.2 M.L tone and H.L tone

Two categories of disyllabic postverbal elements have a tantalizingly similar behaviour. They are illustrated by the postposition /**-ʁo˧to˩**/ 'on top of; during' and the adverb /**pʰæ˥di˩**/ 'as if/it seems that', as shown in Table 6.38.[9]

Examples from texts are provided below, illustrating cases where /**pʰæ˥di˩**/ is preceded by a verb with H tone (33), L_a tone (34), and L_b tone (35), as well as by a postverbal element carrying MH tone (36).

(33) tʰi˩˥ | hĩ˧ =ɻæ˥-dʑo˩ | wɤ˩˥ | ʝi˧kʰv̩˥-dʑo˩ | mv̩˧ pʰæ˧di˥!
tʰi˩˥ hĩ˥ =ɻæ˩ -dʑo˥ wɤ˩˥ ʝi˧kʰv̩˥ -dʑo˥ mv̩˥ pʰæ˥di˩
then people PL TOP again some TOP to_understand it_seems_that
'It seems that some people understood in the end!' *(FoodShortage2.72)* {0004657#S72}

(34) mv̩˩ ʈʂʰɯ˩-v̩˩˥ | qʰwɤ˩ pʰæ˩di˥-dʑo˩ | tʰi˩˥ | kʰi˧ | tʰi˧-tv̩˧ tsɯ˥ | mv̩˩.
mv̩˩˥ ʈʂʰɯ˥ v̩˧ qʰwɤ˩a pʰæ˥di˩ -dʑo˥ tʰi˩˥ kʰi˥ tʰi˧-
daughter DEM CLF.individual clever it_seems_that TOP then door DUR
tv̩˦ tsɯ˦ mv̩˧
to_support REP AFFIRM
'That girl was clever, as it turned out: she propped herself against the door. / That girl reacted smartly: she immediately propped herself against the door [so as to keep the tiger out].' *(Tiger.14)* {0004444#S14}

(35) le˧-ʂɯ˧ le˧-nv̩˥-dʑo˩ | tʰi˩˥ | hĩ˧ | ɬo˧tɑ˧ | wɤ˩˥ | ɖɯ˧-v̩˧ ɳɯ˧ | do˩ pʰæ˩di˥ tsɯ˩ | mv̩˩!
le˧- ʂɯ˧a le˧- nv̩˥ -dʑo˥ tʰi˩˥ hĩ˥ ɬo˧tɑ˧ wɤ˩˥ ɖɯ˧
ACCOMP to_die ACCOMP to_bury TOP then person to_the_side again one
v̩˧ ɳɯ˧ do˩b pʰæ˥di˩ tsɯ˦ -mv̩˧
CLF.individual A to_see it_seems_that REP AFFIRM
'When she died and was buried… apparently, someone close by had seen [what had really happened to her]!' *(BuriedAlive2.24)* {0004536#S24}

[9] The adverb 'as if/it seems that' was transcribed with an M.L pattern in the first edition of this book: /**pʰæ˧di˩**/. Since then, it has been recategorized as H.L on the basis of its behaviour after the negation: the outcome is M.H.L.

(36) kʰv̩˩mi˩ lɑ˥ | ɖʐv̩˧nɑ˥mi˩ ʈʂʰɯ˩-dʑo˩ | ɖɯ˧-pi˧˥ | tʰi˧-kv̩˧ pʰæ˥di˩ mæ˩!
kʰv̩˩mi˩ lɑ˧ ɖʐv̩˧nɑ˥mi˩ ʈʂʰɯ˧ -dʑo˥ ɖɯ˧-pi˧˥ tʰi˧ -kv̩˧˥
dog also heron TOP TOP a_little skilful ABILITIVE
pʰæ˥di˩ mæ˧
it_seems_that OBVIOUSNESS
'The Dog and the Heron were rather talented, it seems!' *(Dog2.108)*
{0004555#S108}

Table 6.38: The behaviour of a disyllabic postposition analyzed as carrying M.L tone and a disyllabic adverb carrying H.L tone.

tone	example	meaning	V+/**pʰæ˥di˩**/ 'as if'	V+/**-ʁo˧to˩**/ 'during'
H	**dzɯ˥**	to eat	**dzɯ˧ pʰæ˧di˥**	**dzɯ˧-ʁo˧to˩**
M_a	**hwæ˧$_a$**	to buy	**hwæ˧ pʰæ˧di˩**	**hwæ˧-ʁo˧to˩**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧ pʰæ˧di˩**	**tɕʰi˧-ʁo˧to˩**
M_c	**pv̩˧$_c$**	to chant	**pv̩˧ pʰæ˧di˩**	**pv̩˧-ʁo˧to˩**
L_a	**bæ˩$_a$**	to sweep	**bæ˩ pʰæ˩di˥**	**bæ˩-ʁo˩to˥**
L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩ pʰæ˩di˥**	**ʐwɤ˩-ʁo˩to˥**
MH	**lɑ˧˥**	to strike	**lɑ˧ pʰæ˥di˩ ≈ lɑ˧ pʰæ˧di˥**	**lɑ˧-ʁo˥to˩**

The only difference in tone patterns between the M.L postposition and the H.L adverb is found in their behaviour when following an H-tone verb. This difference, however, is sufficient to recognize distinct morphotonological categories for the adverb /**pʰæ˥di˩**/ 'as if' and the postposition /**-ʁo˧to˩**/ 'during'.

At present, there is no telling whether the different tonal behaviour of these two morphemes stems from their different parts of speech (postpositions vs. adverbs), from their internal structure, or from a purely tonal difference, as two categories contrasting with each other within the same paradigm – like the various distinctions transcribed in this volume by means of subscript letters added to the tone (e.g. in the M_a, M_b, and M_c subcategories of M-tone verbs). This is one of many issues requiring further investigation.

6.6 Combinations of postverbal morphemes

6.6.1 Postverbal morphemes preceded by the negation prefix

It is common for a verb to be separated from a following morpheme by the negation, as in example (37).

(37) … pʰv̩˧ mɤ˥-tʰɑ˩-ɲi˩ ho˩ mæ˩!
pʰv̩˧˥ mɤ˧ tʰɑ˧˥ -ɲi˩ ho˩ mæ˧
to_take_off NEG PERMISSIVE CERTITUDE DESIDERATIVE OBVIOUSNESS
'[I] was not able to take off [the bracelets]!' *(BuriedAlive2.88)*
{0004536#S88}

One might expect the tonal behaviour of such expressions to be computed progressively ("left-to-right") in some simple way. But as is often the case in Yongning Na morphotonology, no simple algorithm accounts for all the data. A telling example is that of MH-tone verbs, such as /**pʰv̩˧˥**/ 'to take off'. In example (37), the H part of the MH contour reassociates to the negation prefix, yielding /**pʰv̩˧-mɤ˥**…/, which in turn triggers the lowering of all following tones to L (through Rules 4 and 5). When the postverbal morpheme carries M tone, on the other hand, the H part of the MH contour is absent from the surface phonological form, e.g. in /**pʰv̩˧ mɤ˧-bi˧**/ 'will not take off'. In such cases, the MH contour does not unfold (‡ **pʰv̩˧-mɤ˥-bi˩** is not even an acceptable variant); the contour is truncated, as it were, deleting its H component.

The data is therefore presented here in static tabular form rather than as a set of rules.

M-tone postverbal morphemes preceded by the negation

Table 6.39 shows combinations of a verb, a negation prefix, and a morpheme carrying M_a or M_b tone. The example morphemes are the IMMEDIATE FUTURE, /**bi˧$_a$**/, and /**mæ˧$_b$**/ 'to succeed, to achieve'. The verb used to illustrate the M_c tone category in Table 6.39 is /**ʝi˧$_c$**/ 'to come' rather than /**bi˧$_c$**/ 'to go' (standardly used as an example of the M_c tonal category of verbs), in order to avoid the form /**bi˧ mɤ˧-bi˧**/ 'am not going to go'. This expression is well-formed but seriously misleading outside context, as the homophony between the verb and the postverbal morpheme makes the entire expression homophonous with a V NEG-V construction meaning 'whether [she/he…] goes or not'.

MH-tone postverbal morphemes preceded by NEG or PROHIB

This paragraph examines three constructions that share the same tone patterns: (i) /**V mɤ˧-tʰɑ˧˥**/, V-NEG-PERMISSIVE: '[one] must not V'; (ii) /**V mɤ˧-kʰɯ˧˥**/, V-NEG-CAUS: 'not to let [someone] V'; and (iii) /**V tʰɑ˧-kʰɯ˧˥**/, V-PROHIB-CAUS: 'do not cause to V', 'do not let [someone] V'.

Examples from texts include (38)-(40).

Table 6.39: Combinations of verb, negation prefix and M-tone morpheme.

tone	example	meaning	'am not going to V' (tone: M_a)	'cannot manage to V' (tone: M_b)
H	**dzɯ˥**	to eat	**dzɯ˧ mɤ˧-bi˧**	**dzɯ˧ mɤ˧-mæ˩**
M_a	**hwæ˧$_a$**	to buy	**hwæ˧ mɤ˧-bi˧**	**hwæ˧ mɤ˧-mæ˧**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧ mɤ˧-bi˧**	**tɕʰi˧ mɤ˧-mæ˧**
M_c	**ʝi˧$_c$**	to come	**ʝi˧ mɤ˧-bi˧**	**ʝi˧ mɤ˧-mæ˧**
L_a	**dze˩$_a$**	to cut	**dze˩ mɤ˩-bi˩˧**	**dze˩ mɤ˩-mæ˥**
L_b	**tʰɯ˩$_b$**	to drink	**tʰɯ˩ mɤ˩-bi˩˧**	**tʰɯ˩ mɤ˩-mæ˥**
MH	**tʰæ˧˥**	to bite	**tʰæ˧ mɤ˧-bi˧**	**tʰæ˧ mɤ˥-mæ˩**

(38) le˧-ʂɯ˧-dʑo˧, | dʑi˧hỹ˥ | my̩˧ mɤ˧-kʰɯ˧˥! |
le˧- ʂɯ˧ -dʑo˧ dʑi˧hỹ˥$ my̩˧ mɤ˧- -kʰɯ˧˥
ACCOMP to_die PROG clothes to_put_on NEG CAUS
'When someone dies, [we] do not clothe [the corpse]!' *(BuriedAlive3.58)* {0004538#S58}

(39) so˩ɲi˩-so˩hɑ̃˥ | qæ˧˥ | -dʑo˩ | tʰi˩˧, | le˧-qæ˧˥, | my̩˩-mɤ˩-tʰɑ˥!
so˩ɲi˩-so˩hɑ̃˥ qæ˧˥ -dʑo˥ tʰi˩˧ le˧- qæ˧˥
3_days_and_nights to_burn TOP then ACCOMP to_burn
my̩˩ mɤ˧- -tʰɑ˧˥
to_consume/to_burn_up NEG possible
'The corpse was burnt [on the pyre] for three days and three nights, but it was not possible to burn it up!' *(Sister3.93)* {0004344#S93}

(40) ʂæ˧ɖæ˧ | di˩-tʰɑ˩-kʰɯ˥!
ʂæ˧ɖæ˧ di˩$_a$ -tʰɑ˧˥ -kʰɯ˧˥
differences_in_length EXIST PROHIB CAUS
'[The tree trunks] must not be different lengths! / There must not be differences in length!' *Context:* selecting trees that will be felled as lumber for building a house. *(Housebuilding.19)* {0004448#S19}

The data for all tone categories of verbs is shown in Tables 6.40 and 6.41, demonstrating the full identity of tone patterns across these three constructions.

Table 6.40: The tone patterns of /**V mɤ˧-tʰɑ˦˥**/ '[one] must not V' and /**V tʰɑ˧-kʰɯ˦˥**/ 'do not let [someone] V/do not cause to V'.

tone	example	meaning	'[one] must not V'	'do not cause to V'
H	**dzɯ˥**	to eat	**dzɯ˧ mɤ˧-tʰɑ˩**	**dzɯ˧ tʰɑ˧-kʰɯ˩**
M_a	**hwæ˧$_a$**	to buy	**hwæ˧ mɤ˧-tʰɑ˦˥**	**hwæ˧ tʰɑ˧-kʰɯ˦˥**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧ mɤ˧-tʰɑ˦˥**	**tɕʰi˧ tʰɑ˧-kʰɯ˦˥**
M_c	**bi˧$_c$**	to go	**bi˧ mɤ˧-tʰɑ˦˥**	**bi˧ tʰɑ˧-kʰɯ˦˥**
L_a	**dze˩$_a$**	to cut	**dze˩ mɤ˩-tʰɑ˥**	**dze˩ tʰɑ˩-kʰɯ˥**
L_b	**ʈʰɯ˩$_b$**	to drink	**ʈʰɯ˩ mɤ˩-tʰɑ˥**	**ʈʰɯ˩ tʰɑ˩-kʰɯ˥**
MH	**ʈʰæ˦˥**	to bite	**ʈʰæ˧ mɤ˥-tʰɑ˩**	**ʈʰæ˧ tʰɑ˥-kʰɯ˩**

Table 6.41: The tone patterns of V/Adj+neg-abilitive.

word class	tone	example	meaning	V/Adj+neg-abilitive
verbs	H	**dzɯ˥**	to eat	**dzɯ˧ mɤ˧-kv̩˩**
	M_a	**hwæ˧$_a$**	to buy	**hwæ˧ mɤ˧-kv̩˦˥**
	M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧ mɤ˧-kv̩˦˥**
	M_c	**bi˧$_c$**	to go	**bi˧ mɤ˧-kv̩˦˥**
	L_a	**bæ˩$_a$**	to sweep	**bæ˩ mɤ˩-kv̩˥**
	L_b	**ʐwɤ˩$_b$**	to speak	**ʐwɤ˩ mɤ˩-kv̩˥**
	MH	**lɑ˦˥**	to strike	**lɑ˧ mɤ˥-kv̩˩**
adjectives	H	**bi˥**	shallow	**bi˧ mɤ˧-kv̩˩**
	M	**tsʰi˧**	hot	**tsʰi˧ mɤ˧-kv̩˦˥**
	L_a	**ɖɯ˩$_a$**	large	**ɖɯ˩ mɤ˩-kv̩˥**
	L_b	**dʑɤ˩$_b$**	good	**dʑɤ˩ mɤ˧-kv̩˦˥**
	MH	**tʰɑ˦˥**	sharp	**tʰɑ˧ mɤ˥-kv̩˩**

Table 6.42: The tone patterns of the construction /V **mɤ˧-ho˩**/ 'will not V'.

tone	example	meaning	'will not V'
H	**dzɯ˥**	to eat	**dzɯ˧ mɤ˧-ho˥**
M_a	**hwæ˧$_a$**	to buy	**hwæ˧ mɤ˧-ho˩**
M_b	**tɕʰi˧$_b$**	to sell	**tɕʰi˧ mɤ˧-ho˩**
M_c	**bi˧$_c$**	to go	**bi˧ mɤ˧-ho˩**
L_a	**dze˩$_a$**	to cut	**dze˩ mɤ˩-ho˥**
L_b	**tʰɯ˩$_b$**	to drink	**tʰɯ˩ mɤ˩-ho˥**
MH	**tʰæ˧˥**	to bite	**tʰæ˧ mɤ˧-ho˥**

L-tone postverbal morphemes preceded by negation prefix

The tonal behaviour of L-tone postverbal morphemes preceded by the negation prefix is detailed in Table 6.42. In these expressions, the verb appears in initial position. Following an exceptionless phonological rule (Rule 3, stated in §7.1.1: "In tone-group-initial position, H and M are neutralized to M"), the verb can only receive one of two tones: L or M. Thus, L-tone verbs appear with their lexical L tone, while all others (M, H, and MH) surface with M tone.

Tone assignment on the second and third syllables does not follow a straightforward set of phonological rules. Some regularities can be observed, however. When the verb (the first syllable) has M tone, the third morpheme (following the negation prefix) surfaces with its lexical tone, meaning that the final morpheme's tone remains unchanged. For instance, the MH-tone abilitive morpheme /**-kv̩˧˥**/ retains its MH tone in /**hwæ˧ mɤ˧-kv̩˧˥**/ 'cannot buy', and the L-tone desiderative morpheme /**-ho˩**/ retains its L tone in /**hwæ˧ mɤ˧-ho˩**/ 'will not buy'. This reflects the tendency for M tone to behave as phonologically neutral: M does not impose any tonal restriction on the following syllable.

By contrast, when the initial morpheme (the verb) has H tone, the resulting patterns are remarkably diverse: the MH-tone abilitive morpheme /**-kv̩˧˥**/ is depressed to L, whereas the L-tone desiderative morpheme /**-ho˩**/ surfaces with H tone. A fully satisfactory synchronic account of these patterns remains elusive; indeed, it may be illusory to hypothesize that there are clear and cogent synchronic motivations for all patterns. It is nonetheless possible to formulate some tentative remarks about synchronic factors that might be at play.

One such factor is the culminative status of H tone in Yongning Na, reflected in the phonological rule that a tone group can contain at most one H level. For overt H tones, culminativity is manifested through the exceptionless levelling of all following tones to L. This massive effect spares no tonal category: not only does it affect tonal categories containing an H component (namely H and MH), but it also extends to M tones, despite their neutral and inert phonological status. However, it seems that in specific contexts, an H tone that does not surface overtly may still exert an effect, preventing the expression of a following H tone. If one wishes to look for a synchronic rationale for the M.M.L pattern of /**dzɯ˧ mɤ˧-kv̩˩**/ 'cannot eat', one might therefore propose that the verb's H tone, although neutralized in tone-group-initial position (in accordance with Rule 3), remains morphophonologically active.

Under this hypothesis, the effect of the verb's H tone in this context is to induce the levelling of a subsequent morpheme's tone to L if that tone contains an H component. Thus, a 'covert' H tone would preclude any H tone from appearing inside the tone group, even though it does not itself surface. The observed pattern /**dzɯ˧ mɤ˧-kv̩˩**/ 'cannot eat' (instead of ‡ **dzɯ˧ mɤ˧-kv̩˧˥**) would result from a deletion process affecting the MH tone of the third morpheme. This subtle process would specifically target morphemes with H or MH tone, reducing their tone to L, whereas M tones would remain unaffected.

Pushing the narrative further, one might attempt to account for the M.M.H pattern of /**dzɯ˧ mɤ˧-ho˥**/ 'will not eat' and /**ʈʰæ˧ mɤ˧-ho˥**/ 'will not bite' by positing that the final H tone derives from the verb's lexical H or MH tone. The idea would be that the H level is, as it were, set afloat and subsequently docks onto the final syllable of the 'will not V' construction. From a purely theoretical perspective, this hypothesis is appealing, as such a process would provide additional evidence for the analysis of the mid-rising contour as a sequence of M and H levels (set out in §2.3.7). Furthermore, it would demonstrate that these two levels are independently accessible to morphotonological processes.

However, this hypothesis does not appear promising upon closer scrutiny. First, an adequate analysis of the data in Table 6.42 must account for the entire paradigm rather than offering *ad hoc* explanations ('just-so stories') for isolated patterns. In total, three cases yield a final H tone: those in which the verb has H, L or MH tone. At best, the idea that this final H originates in the verb's lexical tone could apply to the H- and MH-tone categories of verbs. Secondly, H- and MH-tone verbs do not behave identically across the contexts presented in Tables 6.40 and 6.41. Their similar behaviour in Table 6.42 is therefore likely to be coincidental – an identity of surface forms that does not reflect identical underlying mechanisms.

A final argument against this hypothesis concerns the expected behaviour of MH tones under such a process. If Yongning Na indeed permitted the selective deletion of a specific tonal level within a lexical contour, one would expect the H portion of the MH contour of the third morpheme to be erased, yielding † **dzɯ˧ mɤ˧-kv̩˧**. However, the data suggests that lexical tones in Yongning Na are treated as unitary in this respect: either an MH tone is compatible with the preceding context, in which case it surfaces intact, or it is not, in which case it is lowered to L. There is no such thing as a morphotonological process targeting a specific tonal level within a lexical contour.

For cases where the initial morpheme has L tone, an open question remains as to why the L tone does not spread all the way to the third syllable, which would yield //† **dze˩ mɤ˩-tʰɑ˩**// 'one must not cut' (an underlying form which, through Rule 7 – "If a tone group only contains L tones, a postlexical H tone is added to its last syllable" –, would surface as /† **dze˩ mɤ˩-tʰɑ˩˥**/). The H tone observed in /**dze˩ mɤ˩-tʰɑ˥**/ does not appear to be a truncated remnant of the morpheme's MH lexical tone, as truncation of the initial portion of an MH tone is unattested elsewhere in the language. Rather, it seems that the MH tone of the third syllable is deleted by a morphotonological rule, leaving the syllable toneless. Phonology then takes over: Rule 7 assigns an H tone to the toneless syllable.

6.6.2 Postverbal morphemes preceded by the interrogative particle: Tonal oppositions are neutralized

The interrogative particle is analyzed as carrying L tone (see §6.3.2). The data in Table 6.43 reveals a complete neutralization of tonal oppositions on the morphemes following the interrogative particle. The three morphemes shown in the table have different lexical tones: the IMMEDIATE FUTURE /**-bi˧**$_\mathbf{a}$/ has $\mathrm{M_a}$ tone; the DESIDERATIVE /**-ho˩**/ has L tone; and the PERMISSIVE /**-tʰɑ˧˥**/ has MH tone. The tone patterns for /**mæ˧**$_\mathbf{b}$/ 'to manage to' are entirely identical to those in the table and are therefore not shown.

6.7 Morphemes surrounding adjectives

6.7.1 Addition of reduplicated suffixes to adjectives

Although adjectives behave in many respects like verbs (i.e. as stative verbs), as explained in §6.1.3, they do not reduplicate in the same way. Adjectives are intensified through suffixation of a reduplicated syllable that does not carry any meaning of its own. In some cases, the monosyllabic form of the adjective has fallen into disuse. For instance, /**bæ˩-lɑ˩~lɑ˥**/ 'limp, flabby (e.g. of meat without bones)' and /**bæ˩-ʁwæ˩~ʁwæ˥**/ 'loose (of knot)' point to a monosyllabic adjective

Table 6.43: The tone patterns for V+INTERROGATIVE+postverbal morpheme.

tone	example	meaning	'going to V?'	'will V?'	'can V?'
H	**dzɯ˥**	to eat	**dzɯ˧ ə˩-bi˩**	**dzɯ˧ ə˩-ho˩**	**dzɯ˧ ə˩-tʰɑ˩**
M_a	**hwæ˧ₐ**	to buy	**hwæ˧ ə˧-bi˥**	**hwæ˧ ə˧-ho˥**	**hwæ˧ ə˧-tʰɑ˥**
M_b	**tɕʰi˧_b**	to sell	**tɕʰi˧ ə˧-bi˥**	**tɕʰi˧ ə˧-ho˥**	**tɕʰi˧ ə˧-tʰɑ˥**
M_c	**bi˧_c**	to go	**bi˧ ə˧-bi˥**	**bi˧ ə˧-ho˥**	**bi˧ ə˧-tʰɑ˥**
L_a	**dze˩ₐ**	to cut	**dze˩ ə˩-bi˥**	**dze˩ ə˩-ho˥**	**dze˩ ə˩-tʰɑ˥**
L_b	**ʈʰɯ˩_b**	to drink	**ʈʰɯ˩ ə˩-bi˥**	**ʈʰɯ˩ ə˩-ho˥**	**ʈʰɯ˩ ə˩-tʰɑ˥**
MH	**ʈʰæ˧˥**	to bite	**ʈʰæ˧ ə˥-bi˩**	**ʈʰæ˧ ə˥-ho˩**	**ʈʰæ˧ ə˥-tʰɑ˩**

***bæ˩** 'loose, limp', but in the present state of the language, this root is not attested on its own.

Another lexical peculiarity is that some such expressions have shorter variants. For instance, 'short (of persons)', /**to˩ʈɯ˩~ʈɯ˥**/, has a variant /**to˩ʈɯ˩**/. Comparison with closely related dialects will be necessary to determine whether the disyllable is a shortened version of the trisyllable, or whether a disyllabic adjective /**to˩ʈɯ˩**/ underwent expressive reduplication of its second syllable. Evidence from other dialects would also be required to ascertain the lexical tone of the monosyllabic root from which the longer expressions are derived.

All instances of adjective-plus-reduplicated-suffix observed so far carry the same tone pattern (L.L.H). However, the current dataset is too limited to determine whether this is a morphotonological hallmark of *ADJ+reduplicated suffix* expressions or a mere coincidence.

6.7.2 Demonstrative and intensive constructions

In Yongning Na, in addition to the construction involving the relativizer /**-hĩ˥**/, adjectives frequently appear in demonstrative or intensive constructions, as illustrated in (41).

(41) ʈʂʰɯ˧-______-gv̩˧
ʈʂʰɯ˥ ______ gv̩˧
DEM.PROX *target adjective* to_be/to_become
'thus ADJ' (e.g. 'thus big', 'thus thick')

The morpheme /**gv̩˧**/, grammaticalized from a verb meaning 'to be; to become', serves to indicate the degree of a quality, as exemplified in (42).

(42) ʈʂʰɯ˧ | no˧ | ɲi˧ɡɤ˧ | ʂwæ˧ mɤ˧-ɡv̩˧.
ʈʂʰɯ˥ no˩ ɲi˧ɡɤ#˥ ʂwæ˧ mɤ˧- ɡv̩˧
3sg 2sg nose tall neg to_be/to_become
'Her nose is not as prominent as yours.' (Field notes.)

In the demonstrative construction 'thus Adj', all tonal oppositions on the adjective are neutralized, as shown in Table 6.44. Examples are found in narratives, for instance in *Sister.28* {0004341#S28} and *BuriedAlive3.144* {0004538#S144}.

Table 6.44: Demonstrative construction /**ʈʂʰɯ**˧-Adj-**ɡv̩**˧/ 'thus Adj'.

tone	example	meaning	'thus Adj'	meaning	tone pattern
H	**ɖæ˥**	short	**ʈʂʰɯ˧-ɖæ˧-ɡv̩˧**	thus short	M.M.M
M	**hwɤ˧**	broad	**ʈʂʰɯ˧-hwɤ˧-ɡv̩˧**	thus broad	M.M.M
L_a	**tɕi˩$_a$**	short	**ʈʂʰɯ˧-tɕi˧-ɡv̩˧**	thus short	M.M.M
MH	**ɬo˧˥**	deep	**ʈʂʰɯ˧-ɬo˧-ɡv̩˧**	thus deep	M.M.M

The same M…M tone pattern is found in cases where the adjective is reduplicated, e.g. /**ʈʂʰɯ˧-ɖɯ˧~ɖɯ˧-ɡv̩˧**/ 'thus big' (*Agriculture.55* {0004440#S55}), from /**ɖɯ˩$_a$**/ 'big' (see also *FoodShortage.43, 85* {0004557#S43}).[10]

Tonal neutralization also occurs in the construction 'as Adj (as)', /**tʰɑ˧-**reduplicated Adj**-ɡv̩˩**/, shown in Table 6.45.

Table 6.45: Tone patterns in the construction 'as Adj as'.

tone	example	meaning	'as Adj as'	meaning	tone pattern
H	**ɖæ˥**	short	**tʰɑ˧-ɖæ˥~ɖæ˩-ɡv̩˩**	as short as	M.H.L.L
M	**hwɤ˧**	broad	**tʰɑ˧-hwɤ˥~hwɤ˩-ɡv̩˩**	as broad as	M.H.L.L
L_a	**tɕi˩$_a$**	short	**tʰɑ˧-tɕi˥~tɕi˩-ɡv̩˩**	as short as	M.H.L.L
MH	**ɬo˧˥**	deep	**tʰɑ˧-ɬo˥~ɬo˩-ɡv̩˩**	as deep as	M.H.L.L

In contrast with the patterns in Tables 6.44 and 6.45, tonal oppositions are not fully neutralized in the construction shown in (43), which expresses 'thus

[10]Related constructions include dem+Adj+augmentative /**-mi˩**/, e.g. /**ʈʂʰɯ˧-ʂwæ˧-mi˧-zo˥**/ 'thus tall', where /**-zo**/ is an adverbializer (*Dog.12* {0004442#S12}), and the intensive construction /**qʰɑ˧**- Adj **-mi˧**/ 'so Adj', e.g. /**qʰɑ˧-ɖɯ˧-mi˧-hĩ˧**/ 'thus huge, extremely big' (*Lake3.28* {0004348#S28}), where /**-hĩ˥**/ is the relativizer/nominalizer.

ADJ', much like (41).[11] As shown in Table 6.46, adjectives with MH lexical tone stand out in this construction by their distinct tone pattern: M.M.H, as opposed to M.M.M for all other tonal categories.

(43) ʈʂʰɯ˧-______-ɻ̩˧
ʈʂʰɯ˥ ______ ɻ̩˧
DEM.PROX *target adjective* INCEPTIVE?
'thus ADJ' (e.g. 'thus big', 'thus thick')

Table 6.46: Tone patterns in the construction /**ʈʂʰɯ**˧-ADJ-**ɻ̩**˧/ 'thus ADJ'.

tone	example	meaning	'thus ADJ'	meaning	tone pattern
H	**bv̩**˥	thick	**ʈʂʰɯ˧-bv̩˧-ɻ̩˧**	thus thick	M.M.M
M	**fv̩**˧	happy	**ʈʂʰɯ˧-fv̩˧-ɻ̩˧**	thus happy	M.M.M
L_a	**ɖɯ˩ₐ**	large	**ʈʂʰɯ˧-ɖɯ˧-ɻ̩˧**	thus large	M.M.M
MH	**hæ**˧˥	supple	**ʈʂʰɯ˧-hæ˧-ɻ̩˥**	thus supple	M.M.H

6.8 Object followed by non-prefixed verb

Cross-linguistically, the object is the nominal argument that has strongest syntactic association to the verb. This bond is so strong that it has even been used as a defining criterion for the notion of object (Creissels 1991: 38). From a morphotonological perspective, it is reasonable to expect at least as much tonal interaction in object-verb combinations as in subject-verb combinations (which are examined below, in §6.10).

In principle, object-verb and subject-verb sequences in Yongning Na could be ambiguous. The language does not have verbal indexing for subjects or objects. Neither is there case marking, or a fixed syntactic position for subjects and objects relative to the verb. The verb follows all noun phrases, except in cases where constituents are tacked on at the end of an utterance as afterthoughts. The postposition /**ɳɯ**˧/, which marks a noun as an agent, is not obligatory. Differences in tonal behaviour between S+V and O+V sequences could, in theory, contribute to disambiguation; however, in practice, S+V without intervening morphemes is relatively uncommon, and disambiguation is generally effected through context.

[11]The identification of the morpheme /-**ɻ̩**˩/ in (43) remains uncertain; it may be the INCEPTIVE/INCHOATIVE morpheme //-**ɻ̩**˧//.

To prevent reinterpretation as subject-verb sequences, object-verb combinations were elicited in contexts designed to clarify their intended structure. For instance, pairing ‘wolf’ with ‘to eat’ naturally suggests agent role for ‘wolf’. To ensure the desired object-verb interpretation, an explicit subject noun phrase was introduced, as shown in (44). This additional noun phrase does not influence the tone of the object-verb combination, as it is followed by a tone-group boundary.

(44) lɑ˧ ɳɯ˧ | õ˩dv̩˧ dzɯ˧-ze˩.
lɑ˧ ɳɯ˧ õ˩dv̩˦ dzɯ˥ -ze˧$_b$
tiger A wolf to_eat PFV
‘The tiger ate the wolf.’

A sample of the elicited data is shown in Table 6.47, with nouns illustrating the various tone categories as objects of the L$_b$-tone verb /**do˩$_b$**/ ‘to see’. The same results obtain when numeral-plus-classifier phrases are used instead of disyllabic nouns. For instance, /**ɖɯ˧-bæ˧**/ ‘something’, made up of the numeral ‘one’ and a classifier for sorts of things, behaves tonally in the same way as disyllabic lexical units carrying the same tone (M), such as /**po˧lo˧**/ ‘ram’ and /**qæ˧do˧**/ ‘timber’. Likewise, combinations with /**ɖɯ˧-kʰwɤ˥$**/ ‘a piece’ yield the same tone patterns as those with /**dʑi˧hṽ̩˥$**/ ‘clothes’.

6.8.1 The facts

Table 6.48 presents the tone rules that apply in object-plus-verb phrases. Recordings available online include (i) a selection of combinations among monosyllables: *ObjectVerb* {0004472}, (ii) a full set, except for L$_b$-tone verbs: *ObjectVerb2* {0004474}, and (iii) L$_b$-tone verbs: *ObjectVerb3* {0004567}.

The notation adopted in Table 6.48 requires disambiguation for sequences ending in an M tone, namely M.M and L.M: object-plus-verb phrases realized with one of these patterns in isolation may have different underlying tones. Consider (45) and (46):

(45) li˩ hwæ˧
li˩˥ hwæ˧$_a$
tea to_buy
‘to buy tea’ {0004473#W18}

(46) æ̃˩ hwæ˧
æ̃˩˧ hwæ˧$_a$
chicken to_buy
‘to buy chicken’

Table 6.47: A sample of object-plus-verb combinations: the verb /**do˩**$_{\text{b}}$/ 'to see'.

tone of N	example	meaning	resulting phrase	tone pattern
LM	**bo˩˧**	pig	**bo˩ do˧**	L.M+L
LH	**ʐæ˩˥**	leopard	**ʐæ˩ do˥**	L.H
M	**lɑ˧**	tiger	**lɑ˧ do˩**	M.L
L	**jo˩**	sheep	**jo˩ do˩˥**	L.L
H	**ʐwæ˥**	horse	**ʐwæ˧ do˧˥**	M.MH
MH	**ʈʂʰæ˧˥**	deer	**ʈʂʰæ˧ do˧˥**	M.MH
M	**po˧lo˧**	ram	**po˧lo˧ do˩**	M.M.L
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ do˧˥**	M.M.MH
MH#	**hwɤ˧li˧˥**	cat	**hwɤ˧li˧ do˧˥**	M.M.MH
H$	**kv̩˧ʂe˥$**	flea	**kv̩˧ʂe˧ do˧˥**	M.M.MH
L	**kʰv̩˩mi˩**	dog	**kʰv̩˩mi˩ do˩˥**	L.L.L
L#	**dɑ˧ʝi˩**	mule	**dɑ˧ʝi˩ do˩**	M.L.L
LM+MH#	**õ˩dv̩˧˥**	wolf	**õ˩dv̩˧ do˧˥**	L.M.MH
LM+#H	**nv̩˩tɕʰi#˥**	fine chaff	**nv̩˩tɕʰi˧ do˧˥**	L.M.MH
LM	**bo˩mi˧**	sow	**bo˩mi˧ do˩**	L.M.L
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˧ do˩**	L.H.L
H#	**hwæ˧ʈʂæ˥**	squirrel	**hwæ˧ʈʂæ˥ do˩**	M.H.L

The phrases /**li˩ hwæ˧**/ 'to buy tea' and /**æ̃˩ hwæ˧**/ 'to buy chicken' both surface with an /L.M/ pattern. But they yield different results when followed by the PERFECTIVE morpheme /**-ze˧**$_{\text{b}}$/, as shown in (47)-(48).

(47) li˩ hwæ˧-ze˩
li˩˥ hwæ˧$_{\text{a}}$ -ze˧$_{\text{b}}$
tea to_buy PFV
'bought tea' {0004473#W17}

(48) æ̃˩ hwæ˧-ze˧
æ̃˩˧ hwæ˧$_{\text{a}}$ -ze˧$_{\text{b}}$
chicken to_buy PFV
'bought chicken'

Table 6.48 therefore includes information on the tonal realization of the PERFECTIVE morpheme /**-ze˧**$_{\text{b}}$/ where relevant. The tone pattern of 'to buy tea', which

Table 6.48: The tone patterns of object-plus-verb combinations.

	tone of verb					
tone of noun	H	M_a	M_b	L_a	L_b	MH
LM	L.M+L	L.M+M	L.M+M	L.M+L	L.M+L	L.MH
LH	L.L / L.H	L.H	L.L / L.H	L.H	L.H / L.L	L.MH
M	M.M+L	M.M+M	M.M+M	M.L	M.L	M.MH
L	L.L	M.M+M	M.M+M / L.L	L.L	L.L / M.L	L.L
H	M.M+L	M.L	M.M+L	M.H	M.MH	M.L
MH	M.H	M.H	M.H	M.H	M.MH	M.H
M	M.M.M+L	M.M.M+M	M.M.M+M	M.M.L	M.M.L	M.M.MH
#H	M.M.M+L	M.M.L	M.M.M+L	M.M.H	M.M.MH	M.M.L
MH#	M.M.MH	M.M.H	M.M.MH	M.M.H	M.M.MH	M.M.H
H$	M.M.M+L	M.H.L	M.M.M+L	M.M.H	M.M.MH	M.H.L
L	L.L.L	L.L.H	L.L.L	L.L.H	L.L.L	L.L.H
L#	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L
LM+MH#	L.M.M+L	L.M.H	L.M.M+L	L.M.H	L.M.MH	L.M.H
LM+#H	L.M.M+L	L.M.L	L.M.M+L	L.M.H	L.M.L / L.M.MH	L.M.L
LM	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.L	L.M.MH
LH	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L
H#	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L

exemplifies the combination of input tones //LH// (on the noun) and //M// (on the verb), is transcribed in the table as /L.M+L/, whereas that of 'to buy chicken' is simply /L.M/. Similarly, the pattern /M.M+L/ is distinguished from /M.M+M/. The tone on the ACCOMPLISHED prefix is preceded by a '+' sign.

In cases where the tone of the postverbal element follows straightforwardly from the phonological rules of tone association in Yongning Na (as summarized in Chapter 7), no further specification is indicated in the table. All L.H, M.H, L.L.H and M.M.H patterns end in H#, a final, non-floating, non-'hopping' H tone, and are followed by L tone on the PERFECTIVE suffix /**-ze˧$_b$**/. Likewise, when the final tone of the object-plus-verb phrase is L, /**-ze˧$_b$**/ always receives L tone. If the pattern ends in /MH/, then /**-ze˧$_b$**/ receives the H part of the contour.

As in the other tables, a slash separates variants. For instance, for 'has sold leopards' (input: //LH// and //M$_b$//), two realizations are acceptable, as shown in (49).

(49) /ʐæ˩ tɕʰi˥-ze˩/ ≈ /ʐæ˩ tɕʰi˩-ze˥/
ʐæ˩˥ tɕʰi˧$_b$ -ze˧
leopard to_sell PFV
'sold leopards'

For //L// and //M$_b$//, two patterns are also possible: L.L and M.M+M, as shown in (50).

(50) /jo˩ tɕʰi˩-ze˥/ ≈ /jo˧ tɕʰi˧-ze˧/
jo˩ tɕʰi˧$_b$ -ze˧$_b$
sheep to_sell PFV
'has sold sheep' {0004472#W97}

If the input is //LH// and //H//, two patterns are also possible: L.L and L.H, as illustrated by (51) and (52).

(51) /di˩ dv̩˩-ze˥/ ≈ /di˩ dv̩˥-ze˩/
di˩˥ dv̩˥ -ze˧$_b$
earth to_dig PFV
'has dug earth' {0004473#W1}

(52) /(lɑ˧ ɳɯ˧ |) tʰɑ˩ dzɯ˩-ze˥/ ≈ /(lɑ˧ ɳɯ˧ |) tʰɑ˩ dzɯ˥-ze˩/
lɑ˧ ɳɯ˧ tʰɑ˩˥ dzɯ˥ -ze˧$_b$
tiger A buffalo to_eat PFV
'(the tiger) has eaten (a) buffalo'

Finally, a note concerning the L.M.L / L.M.MH output (from input //LM+#H// and //L_b//): examples include (53), (54) and (55). The consultant expressed a preference for the L.M.MH realization.

(53) /nv̩˩tɕʰi˧ do˦˥/ ≈ /nv̩˩tɕʰi˧ do˩/
nv̩˩tɕʰi#˥ do˩$_b$
fine_chaff to_see
'to see fine chaff' {0004922#W18}

(54) /pi˩ti˧ do˦˥/ ≈ /pi˩ti˧ do˩/
pi˩ti#˥ do˩$_b$
nugget to_see
'to see nuggets (of silver)'

(55) /nɑ˩hĩ˧ do˦˥/ ≈ /nɑ˩hĩ˧ do˩/
nɑ˩hĩ#˥ do˩$_b$
Naxi to_see
'to see (the) Naxi'

The following sections offer observations about the tone patterns of object-plus-verb combinations and their relation to other aspects of the morphotonological system.

6.8.2 Object-plus-verb combinations reveal the tonal opposition between //LM// and //LH// monosyllabic nouns

The opposition between the //LM// and //LH// categories of nouns (illustrated by //**bo˩˧**// 'pig' and //**ʐæ˩˥**// 'leopard') is neutralized in most contexts. However, among object-plus-verb combinations, two contexts distinguish them: their association with an M-tone verb, e.g. /**bo˩ hwæ˧-ze˧**/ 'bought pigs' vs. /**ʐæ˩ hwæ˧-ze˩**/ 'bought leopards', and with an H-tone verb, e.g. /**bo˩ dzɯ˧**/ 'to eat pigs' vs. /**ɣɯ˩ dzɯ˩˥**/ 'to eat skin'. ('Skin' /**ɣɯ˩˥**/ was chosen as a semantically more suitable noun than 'leopard' as the object of the verb 'to eat'; nonetheless, /**ʐæ˩ dzɯ˩˥**/ 'to eat leopards' is syntactically well-formed.)

6.8.3 About tonal variants

There are interesting fine details in the distribution of tonal variants among object-plus-verb combinations. In some cases, it is possible to identify one variant that is more common in frequently occurring word combinations. For instance, with an L-tone object and an LM-tone verb, both M.M+M and L.L patterns are possible, but the former predominates for the expression ‘to drink water’: it is customary to say /**dʐɯ˧ ʈʰɯ˧**/, while the latter variant, /**dʐɯ˩ ʈʰɯ˩˥**/, is understandable but sounds unusual. Conversely, in elicitation, ‘to grab sheep’ yielded /**jo˩ ʐi˩˥**/ rather than the alternative /**jo˧ ʐi˧**/. This suggests that object-plus-verb combinations with this pair of input tones (L and LM) tend to acquire an M.M+M pattern as a result of high frequency of occurrence (which amounts to lexicalization). If so, the reason why the consultant did not produce ‘to grab sheep’ as /**jo˧ ʐi˧**/ may be because of this tone pattern’s connotation of habitual activity. There are no sheep on the consultant’s farm, so sheep-grabbing is indeed not a part of the consultant’s farming routine. The M.M+M variant /**jo˧ ʐi˧**/ could be appropriate in a context such as sheep shearing, where grabbing sheep is a routine action.

6.8.4 Exceptional combinations

Two //LH// nouns, ‘leopard’ and ‘monkey’, yield an unexpected L.M+L pattern in combination with an //H//-tone verb, instead of the expected L.L pattern, as shown in (56) and (57).

(56) ʐæ˩ dzɯ˧-ze˩ († ʐæ˩ dzɯ˩-ze˥)
ʐæ˩˥ dzɯ˥ -ze˧$_b$
leopard to_eat PFV
‘has eaten leopards’

(57) ʐi˩ dzɯ˧-ze˩ († ʐi˩ dzɯ˩-ze˥)
ʐi˩˥ dzɯ˥ -ze˧$_b$
monkey to_eat PFV
‘has eaten monkeys’

While ‘to eat leopards’ is semantically odd, as leopards are predators rather than prey, the phrase ‘to eat monkeys’ is unproblematic. The same pattern recurred in several elicitation sessions, suggesting a genuine irregularity. Pending further investigation, this is provisionally treated as an exceptional pattern requiring individual memorization.

In Table 6.48, two variants were indicated for //LH// nouns in combination with //M$_b$// verbs: L.L and L.H. Some word combinations allow both: ‘has sold leopards’ may be expressed as either /**ʐæ˩ tɕʰi˥-ze˩**/, as well as /**ʐæ˩ tɕʰi˩-ze˥**/. However, others appear to have lexicalized with one tone pattern or the other. For instance, ‘brought in the harvest’ is always /**bæ˩ ʂo˥-ze˩**/, never † **bæ˩ ʂo˩-ze˥**, whereas ‘to eat skin’ (with the same input tones) is consistently /**ɣɯ˩ dzɯ˩-ze˥**/, never † **ɣɯ˩ dzɯ˥-ze˩**. This may indicate a tendency for lexicalized combinations to receive an L.H pattern. Alternatively, L.H may be older (which would explain its higher frequency in lexicalized combinations), whereas L.L is innovative.

Two further exceptions, both on highly lexicalized items, are /**kʰv̩˧ ʂæ˦**/ ‘to hunt’ (literally ‘to lead a dog’, from //**kʰv̩˥**// ‘dog’ and //**ʂæ˦**// ‘to lead along’) and /**mv̩˧ tsʰi˦**/ ‘to light a fire’ (from //**mv̩˥**// ‘fire’ and //**tsʰi˦**// ‘to light’). The regular pattern would yield an M.L sequence – † **kʰv̩˧ ʂæ˩** and † **mv̩˧ tsʰi˩**. The observed M.MH tonal string suggests retention of a tone pattern that was once productive but is no longer so.

6.8.5 Noun plus copula behaves tonally like object plus verb

Data on the tonal behaviour of the copula //**ɲi˩$_a$**// following nouns of each tonal category were set out in Chapter 2. The pattern observed is identical to that of L$_a$-tone verbs in object-plus-verb combinations, rather than in subject-plus-verb constructions. This perfect match in tone patterns provides strong evidence that a noun preceding the copula is syntactically its object, not its subject.

6.8.6 Interrogative pronoun and verb

The combination of an interrogative pronoun and a verb is a special case of the object-plus-verb structure. However, the tonal patterns are not entirely identical, as shown in Table 6.49. Surprisingly, the three interrogative pronouns in Table 6.49, which are hypothesized to belong to the same lexical tone category, namely //LM//, show distinct tonal behaviours. These differences were carefully verified: for instance, it was confirmed that ‡ **ze˩bæ˧ lɑ˦** is not an acceptable variant for ‘strike which sort [of things]?’, any more than ‡ **ze˩gɤ˧ lɑ˥** is for ‘strike which place?’.

6.9 Object and prefixed verb

When an object combines with a prefixed verb, the two may form a single tone group, as in (58), or they may be separated by a tone group boundary, as in (59).

(58) bɑ˩lɑ˩ tʰi˥-mv̩˩
bɑ˩lɑ˩ tʰi˧- mv̩˧$_a$
clothes DUR to_put_on
‘to put on clothes’ *(ComingOfAge2.37)* {0004588#S37}

Table 6.49: The tonal behaviour of three interrogative pronouns preceding a verb: /**ze˩bæ˧**/ 'which sort', /**ze˩gɤ˧**/ 'which place', and /**zo˩qo˧**/ 'where'.

tone	example	meaning	which	which place	where
H	**dzɯ˥**	to eat	**ze˩bæ˧ dzɯ˧**	**ze˩gɤ˧ dzɯ˧**	**zo˩qo˧ dzɯ˧**
M_a	**hwæ˧$_a$**	to buy	**ze˩bæ˧ hwæ˩**	**ze˩gɤ˧ hwæ˧**	**zo˩qo˧ hwæ˧**
M_b	**tɕʰi˧$_b$**	to sell	**ze˩bæ˧ tɕʰi˧**	**ze˩gɤ˧ tɕʰi˧**	**zo˩qo˧ tɕʰi˧**
M_c	**pv̩˧$_c$**	to chant	**ze˩bæ˧ pv̩˩**	**ze˩gɤ˧ pv̩˧**	**zo˩qo˧ pv̩˧**
L_a	**bæ˩$_a$**	to sweep	**ze˩bæ˧ bæ˥**	**ze˩gɤ˧ bæ˩**	**zo˩qo˧ bæ˩**
L_b	**ʈʰɯ˩$_b$**	to drink;	**ze˩bæ˧ ʈʰɯ˦**	**ze˩gɤ˧ ʈʰɯ˩**	**zo˩qo˧ ʈʰɯ˩**
	ʐwɤ˩$_b$	to speak	**ze˩bæ˧ ʐwɤ˦**	**ze˩gɤ˧ ʐwɤ˩**	**zo˩qo˧ ʐwɤ˩**
MH	**lɑ˦**	to strike	**ze˩bæ˧ lɑ˥**	**ze˩gɤ˧ lɑ˦**	**zo˩qo˧ lɑ˦**

(59) ə˧mv̩˧ ɳɯ˥, | ʐæ˩sɯ˩˥ | tʰi˧-mv̩˧
ə˧mv̩˦ ɳɯ˧ ʐæ˩sɯ˩ tʰi˧ mv̩˧$_a$
elder_sibling A coarse_felt DUR to_put_on
'the elder brother put on [his] coarse felt cloak' *(Sister3.57)* {0004344#S57}

The same stylistic choice is open for numeral-plus-classifier phrases in object position. Example (60) illustrates a case where the object ('a bowl') and the verb ('to pour') are integrated into a single tone group. The decision to separate the prefixed verb from the object is not dictated by the length of the tone group: in (60), the tone group comprises six syllables, whereas in (59), which has similar structure, a sequence of just four syllables is split into two tone groups. The placement of tone-group boundaries is discussed in detail in §7.2.

(60) ʐɯ˧ | ɖɯ˧-qʰwɤ˧ tʰi˥-pʰv̩˩ tsɯ˩ | mv̩˩.
ʐɯ˧ ɖɯ˧ qʰwɤ˦ tʰi˧- pʰv̩˦ tsɯ˦ mv̩˧
liquor/spirits one CLF.bowls DUR to_pour REP AFFIRM
'it is said that [she] poured a bowl of liquor [for her brother].' *(Sister3.41)* {0004344#S41}

When the object and prefixed verb are integrated into a single tone group, it seems at first glance as if the tonal adjustments were purely phonological, proceeding from the beginning (in a "left-to-right" manner). In example (60), for instance, it seems that the MH contour on the numeral-plus-classifier unfolds over the prefixed verb: /**ɖɯ˧-qʰwɤ˦**/ 'one bowl(ful)' plus /**tʰi˧-pʰv̩˦**/ 'to pour' would yield /**ɖɯ˧-qʰwɤ˧ tʰi˥**.../ through unfolding of the MH contour, the H component of the MH contour reassociating to the DURATIVE prefix //**tʰi˧**-// (hence /**tʰi˥**-/). The final output, /**ɖɯ˧-qʰwɤ˧ tʰi˥-pʰv̩˩**/, would then follow from the application of Rules 4 and 5: “A syllable following an H-tone syllable receives L tone” and “All syllables following an H.L or M.L sequence receive L tone”.

A similar analysis can be applied for almost all cases, but not for quite all of them. Therefore, the process cannot be considered as purely phonological: the association of an object and a prefixed verb still belongs within morphophonology. However, the system is only a small step away from becoming a purely phonological one: if the handful of non-predictable forms were regularized on the basis of current phonological rules, tonal adjustment in this domain could be fully described in phonological terms.

The following sections set out the relevant data and discuss these phenomena in detail.

6.9.1 The facts

The data is arranged by tone. The behaviour of tones M_a, M_b, and M_c is identical, as is that of tones L_a and L_b. Consequently, only one M-tone verb is used as an example in Table 6.50, and only one L-tone verb in Table 6.52. Patterns that do not conform with the regularities discussed below are shaded in gray.

Attestations in texts are abundant: the DURATIVE prefix /**tʰi˧**-/ occurs over 700 times in twenty texts, illustrating a broad range of combinations. In example (61), this prefix appears after an LH-tone noun, a context in which it gets an H tone. Other interesting examples are found in *Funeral.69* {0004571#S69}, *108, 190, 238, 253, Healing.103* {0004540#S103}, *Housebuilding.40* {0004448#S40}, *110, 121, 217, 239, Mountains.7* {0004573#S7}, *88, 161, Reward.40* {0004446#S40}, *73, Seeds2.51* {0004542#S51}, *62* and *Sister3.41* {0004344#S41}, *95.*

(61) lwɤ˩ tʰi˥-kʰɯ˩ | tɕɤ˧-kv̩˥ mæ˩, | ə˧ʝi˧-ʂɯ˥ʝi˩!
lwɤ˩˥ tʰi˧- kʰɯ˥ tɕɤ˥ -kv̩˥ mæ˧ ə˧ʝi˧-ʂɯ˥ʝi˩
ashes DUR to_put to_boil ABILITIVE OBVIOUSNESS in_the_old_times
‘One would add ashes and boil [linen thread], in the old times!’
(FoodShortage.71) {0004557#S71}

Discussion of the above data will proceed from the most straightforward cases to the more complex ones.

Table 6.50: Objects plus the M-tone verb /**hwæ˧a**/ 'to buy' prefixed by the DURATIVE /**tʰi˧**-/.

tone	head	meaning	**hwæ˧a** 'to buy'	tone pattern
LM	**bo˩˧**	pig	**bo˩ tʰi˧-hwæ˧**	L.M.M
LH	**mṿ˩˥**	daughter	**mṿ˩ tʰi˥-hwæ˩**	L.H.L
M	**lɑ˧**	tiger	**lɑ˧ tʰi˧-hwæ˧**	M.M.M
L	**jo˩**	sheep	**jo˧ tʰi˧-hwæ˧**	M.M.M
#H	**hĩ˥**	human being	**hĩ˧ tʰi˩-hwæ˩**	M.L.L
MH#	**tsʰɯ˧˥**	goat	**tsʰɯ˧ tʰi˥-hwæ˩**	M.H.L
M	**po˧lo˧**	ram	**po˧lo˧ tʰi˧-hwæ˧**	M.M.M.M
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ tʰi˩-hwæ˩**	M.M.L.L
MH#	**hwɤ˧li˧˥**	cat	**hwɤ˧li˧ tʰi˥-hwæ˩**	M.M.H.L
H$	**kṿ˧ʂe˥$**	flea	**kṿ˧ʂe˥ tʰi˩-hwæ˩**	M.H.L.L
L	**kʰṿ˩mi˩**	dog	**kʰṿ˩mi˩ tʰi˥-hwæ˩**	L.L.H.L
L#	**dɑ˧ʝi˩**	mule	**dɑ˧ʝi˩ tʰi˩-hwæ˩**	M.L.L.L
LM+MH#	**ṿ˩tsʰɤ˧˥**	vegetables	**ṿ˩tsʰɤ˧ tʰi˥-hwæ˩**	M.M.H.L
LM+#H	**ɑ˩mi#˥**	goose	**ɑ˩mi˧ tʰi˩-hwæ˩**	L.M.L.L
LM	**bo˩mi˧**	sow	**bo˩mi˧ tʰi˧-hwæ˧**	L.M.M.M
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˥ tʰi˩-hwæ˩**	L.H.L.L
H#	**kʰṿ˧nɑ˥**	dog	**kʰṿ˧nɑ˥ tʰi˩-hwæ˩**	M.H.L.L

Table 6.51: Objects plus the H-tone verb /**dzɯ˥**/ 'to eat' prefixed by the DURATIVE /**tʰi˧**-/.

tone	head	meaning	**dzɯ˥** 'to eat'	tone pattern
LM	**bo˩˧**	pig	**bo˩ tʰi˧-dzɯ˥**	L.M.H
LH	**mv̩˩˥**	daughter	**mv̩˩ tʰi˥-dzɯ˩**	L.H.L
M	**lɑ˧**	tiger	**lɑ˧ tʰi˧-dzɯ˥**	M.M.H
L	**jo˩**	sheep	**jo˩ tʰi˩-dzɯ˩˥**	L.L.LH
#H	**hĩ˥**	human being	**hĩ˧ tʰi˩-dzɯ˩**	M.L.L
MH#	**tsʰɯ˧˥**	goat	**tsʰɯ˧ tʰi˥-dzɯ˩**	M.H.L
M	**po˧lo˧**	ram	**po˧lo˧ tʰi˧-dzɯ˥**	M.M.M.H
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ tʰi˩-dzɯ˩**	M.M.L.L
MH#	**hwɤ˧li˧˥**	cat	**hwɤ˧li˧ tʰi˥-dzɯ˩**	M.M.H.L
H$	**kv̩˧ʂe˥$**	flea	**kv̩˧ʂe˥ tʰi˩-dzɯ˩**	M.H.L.L
L	**kʰv̩˩mi˩**	dog	**kʰv̩˩mi˩ tʰi˥-dzɯ˩**	L.L.H.L
L#	**dɑ˧ʝi˩**	mule	**dɑ˧ʝi˩ tʰi˩-dzɯ˩**	M.L.L.L
LM+MH#	**v̩˩tsʰɤ˧˥**	vegetables	**v̩˩tsʰɤ˧ tʰi˥-dzɯ˩**	L.M.H.L
LM+#H	**ɑ˩mi#˥**	goose	**ɑ˩mi˧ tʰi˥-dzɯ˩**	L.M.H.L
LM	**bo˩mi˧**	sow	**bo˩mi˧ tʰi˧-dzɯ˥**	L.M.M.H
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˥ tʰi˩-dzɯ˩**	L.H.L.L
H#	**kʰv̩˧nɑ˥**	dog	**kʰv̩˧nɑ˥ tʰi˩-dzɯ˩**	M.H.L.L

Table 6.52: Objects plus the L-tone verb /**di˩ₐ**/ ‘to have’ prefixed by the DURATIVE /**tʰi˧**-/.

tone	head	meaning	**di˩ₐ** ‘to have’	tone pattern
LM	**bo˩˧**	pig	**bo˩ tʰi˧-di˩**	L.M.L
LH	**mv̩˩˥**	daughter	**mv̩˩ tʰi˩-di˥**	L.L.H
M	**lɑ˧**	tiger	**lɑ˧ tʰi˧-di˩**	M.M.L
L	**jo˩**	sheep	**jo˧ tʰi˧-di˩**	M.M.L
#H	**hĩ˥**	human being	**hĩ˧ tʰi˧-di˥**	M.M.H
MH#	**tsʰɯ˧˥**	goat	**tsʰɯ˧ tʰi˧-di˥**	M.M.H
M	**po˧lo˧**	ram	**po˧lo˧ tʰi˧-di˩**	M.M.M.L
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ tʰi˧-di˥**	M.M.M.H
MH#	**hwɤ˧li˧˥**	cat	**hwɤ˧li˧ tʰi˧-di˥**	M.M.M.H
H$	**kv̩˧ʂe˥$**	flea	**kv̩˧ʂe˧ tʰi˧-di˥**	M.M.M.H
L	**kʰv̩˩mi˩**	dog	**kʰv̩˩mi˩ tʰi˩-di˥**	L.L.L.H
L#	**dɑ˧ʝi˩**	mule	**dɑ˧ʝi˩tʰi˩-di˩**	M.L.L.L
LM+MH#	**v̩˩tsʰɤ˧˥**	vegetables	**v̩˩tsʰɤ˧ tʰi˧-di˥**	L.M.M.H
LM+#H	**ɑ˩mi#˥**	goose	**ɑ˩mi˧ tʰi˧-di˥**	L.M.M.H
LM	**bo˩mi˧**	sow	**bo˩mi˧ tʰi˧-di˩**	L.M.M.L
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˥ tʰi˩-di˩**	L.H.L.L
H#	**kʰv̩˧nɑ˥**	dog	**kʰv̩˧nɑ˥ tʰi˩-di˩**	M.H.L.L

Table 6.53: Objects plus the MH-tone verb /**ʈʰæ˦**/ ‘to bite’ prefixed by the DURATIVE /**tʰi˧**-/.

tone	head	meaning	ʈʰæ˦ ‘to bite’	tone pattern
LM	**bo˩˧**	pig	**bo˩ tʰi˧-ʈʰæ˦**	L.M.MH
LH	**mv̩˩˥**	daughter	**mv̩˩ tʰi˥-ʈʰæ˩**	L.H.L
M	**lɑ˧**	tiger	**lɑ˧ tʰi˧-ʈʰæ˦**	M.M.MH
L	**jo˩**	sheep	**jo˩ tʰi˩-ʈʰæ˩˥**	L.L.LH
#H	**hĩ˥**	human being	**hĩ˧ tʰi˩-ʈʰæ˩**	M.L.L
MH#	**tsʰɯ˦**	goat	**tsʰɯ˧ tʰi˥-ʈʰæ˩**	M.H.L
M	**po˧lo˧**	ram	**po˧lo˧ tʰi˧-ʈʰæ˦**	M.M.M.MH
#H	**ʐwæ˧zo#˥**	colt	**ʐwæ˧zo˧ tʰi˩-ʈʰæ˩**	M.M.L.L
MH#	**hwɤ˧li˦**	cat	**hwɤ˧li˧ tʰi˥-ʈʰæ˩**	M.M.H.L
H$	**kv̩˧ʂe˥$**	flea	**kv̩˧ʂe˥ tʰi˩-ʈʰæ˩**	M.H.L.L
L	**kʰv̩˩mi˩**	dog	**kʰv̩˩mi˩ tʰi˥-ʈʰæ˩**	L.L.H.L
L#	**dɑ˧ɟi˩**	mule	**dɑ˧ɟi˩ tʰi˩-ʈʰæ˩**	M.L.L.L
LM+MH#	**v̩˩tsʰɤ˦**	vegetables	**v̩˩tsʰɤ˧ tʰi˥-ʈʰæ˩**	L.M.H.L
LM+#H	**ɑ˩mi#˥**	goose	**ɑ˩mi˧ tʰi˩-ʈʰæ˩**	L.M.L.L
LM	**bo˩mi˧**	sow	**bo˩mi˧ tʰi˧-ʈʰæ˦**	L.M.M.MH
LH	**bo˩ɬɑ˥**	boar	**bo˩ɬɑ˥ tʰi˩-ʈʰæ˩**	L.H.L.L
H#	**kʰv̩˧nɑ˥**	dog	**kʰv̩˧nɑ˥ tʰi˩-ʈʰæ˩**	M.H.L.L

6.9.2 The tone of the verb surfaces when the noun phrase has LM or M tone

As observed in §2.4.3 of Chapter 2, M tends to behave as an inert tone in Yongning Na: it does not spread or otherwise affect following tones. This observation extends to the behaviour of M-tone nouns when followed by a prefixed verb: when the noun phrase has M tone, the prefixed verb surfaces with the same tonal pattern as in isolation. The same applies to LM-tone noun phrases.

6.9.3 Tonal oppositions on verbs are neutralized after a disyllabic noun phrase with L#, LH or H# tone

When a disyllabic noun phrase bears L#, LH or H# tone, it precludes the realization of any tone other than L on following syllables within the tone group. This follows from the application of Rules 4 and 5: “A syllable following an H-tone syllable receives L tone”, and “All syllables following an H.L or M.L sequence receive L tone". Consequently, all tonal oppositions on verbs are neutralized in this context.

6.9.4 Patterns that cannot be fully explained by phonological regularities

The patterns discussed in the two preceding paragraphs can be fully explained by phonological regularities that apply throughout the system. In total, roughly three of four patterns conform to these regularities, a proportion that appears high enough to support the tentative hypothesis that these tendencies represent the default case, while the remaining cases are likely to be learnt individually.

The phonological tendencies are as follows:

(i) The //LH// tone on a monosyllabic noun projects its H portion onto the next syllable (in this case the DURATIVE prefix /**tʰi˧**/).

(ii) The //#H// and //LM+#H// tones do not overtly express their H tone, which remains floating but is not deleted; this floating H tone lowers the tones of the following syllables to L.

(iii) The //H$// tone gets docked on the last syllable of the noun phrase, causing the following syllables to receive L tone, in accordance with Rules 4 and 5: "A syllable following an H-tone syllable receives L tone", and "All syllables following an H.L or M.L sequence receive L tone".

(iv) The //MH#// and //LM+MH#// tones project their final H level onto the following syllable.

From this perspective, combinations that follow these tendencies can be considered regular, while those that do not are irregular. The patterns for //L//-tone verbs are irregular because they contravene tendencies (i), (ii), and (iv). (The behaviour of L-tone verbs is analyzed further in §6.9.5.) The pattern involving a //LM+#H// noun and an //H//-tone verb is also irregular: in view of tendency (ii), one would expect L.M.L.L rather than the observed L.M.M.H, as illustrated in (62).

(62) ɑ˩mi˧ tʰi˧-dzɯ˥ († ɑ˩mi˧ tʰi˩-dzɯ˩)
ɑ˩mi#˥ tʰi˧- dzɯ˥
goose DUR to_eat
'eating a goose'

The issue of regularity and irregularity in morphotonological paradigms will be taken up in the typological discussion in Chapter 12.

6.9.5 The behaviour of L-tone verbs: Attempting a generalization

L-tone verbs exhibit the highest proportion of irregular patterns, i.e. patterns that do not follow the four phonological tendencies outlined above. Nonetheless, a generalization may still be possible. One tentative way to describe what happens in this particular morphosyntactic context is as follows: when the noun phrase contains an H tone that is not unmovably fixed, this H tone shifts to the last syllable of the resulting verb phrase.

This somewhat roundabout generalization takes into account the identical tonal treatment of prefixed verbs following disyllabic nouns with //H#// or //LH// tone. In a disyllabic noun, the H part of a //H#// or //LH// tone is firmly anchored to the second syllable. By contrast, the tones //#H//, //LM+#H//, //H$//, //MH#//, and //LM+MH#// share the property of containing an H tone whose syllabic anchoring is context-sensitive.

An apparent counterexample to this generalization is the //LH// tone: while disyllabic //LH//-tone nouns yield L.H.L.L, monosyllabic //LH//-tone nouns yield L.L.H, as shown in (63).

(63) mv̩˩ tʰi˩-do˥
mv̩˩˧ tʰi˧- do˩$_b$
daughter DUR to_see
'to see (a/the) daughter' (elicited example)

At this point, one would be justified in concluding that the morphotonology at play is fundamentally irregular and that attempts to impose systematic explanations would be unwarranted. The above generalization would thus have to be abandoned. However, experience in learning and speaking the language suggests otherwise, encouraging further speculation on how the observed patterns relate to one another.

The different treatment of //LH// tone on monosyllabic versus disyllabic nouns could be attributed to differences in syllabic association. In disyllabic nouns, the H part of the //LH// tone is unmovably anchored to the second syllable, making it a straightforward category for language learners. In monosyllabic nouns, on the other hand, the absence of a second syllable creates pressure for the H part to reassociate to a later syllable. In various morphosyntactic contexts, the H part of //LH// does not surface on the noun to which it is lexically associated. To sum up, //LH// tone has fixed, hard-and-fast syllabic anchoring in disyllabic nouns but looser anchoring in monosyllables. Seen in this light, it does not come as a great surprise that //LH// on a monosyllabic noun should behave as one of the movable

H tones (alongside //#H//, //LM+#H//, //H$//, //MH#//, and //LM+MH#//) rather than as one of the unmovable H tones.

This generalization is *ad hoc* in that it only concerns a single morphosyntactic context. However, it does not appear absurd to consider that it has psychological reality, allowing learners to acquire the tonal patterns of //#H//, //LM+#H//, //H$//, //MH#//, and //LM+MH#// (as well as //LH// in monosyllables) as a unified set.

Still on a speculative note, this analysis predicts high cross-dialect variation in //L//-tone verbs. Cases that deviate from regularities (i-iv) in §6.9.4 are especially numerous for //L//-tone verbs, and the force of analogy would tend to simplify the system by eliminating these irregularities. This is an empirical question to be investigated using data from other dialects.

6.10 Subject and verb

Tonal interaction between subject and verb is illustrated by (64–65). The verb 'to fall' carries different tones in (64a) and (64b), as does the verb 'to come' in (65a) and (65b).

(64) a. bi˧ gi˧-ze˩.
bi˥ gi˥ -ze˧$_b$
snow to_fall PFV
'It has snowed.'

b. hi˩ gi˩-ze˥.
hi˩˥ gi˥ -ze˧$_b$
rain to_fall PFV
'It has rained.'

(65) a. hĩ˧bæ˧ tsʰɯ˧-ze˥.
hĩ˧bæ#˥ tsʰɯ˩$_a$ -ze˧$_b$
guest to_come.PST PFV
'The guest has come. / The guests have come.'

b. dɑ˧pɤ˧ tsʰɯ˩-ze˩.
dɑ˧pɤ˧ tsʰɯ˩$_a$ -ze˧$_b$
priest to_come.PST PFV
'The *Ddabe* priest has come.'

Subject-plus-verb phrases are relatively infrequent in Yongning Na. This is partly due to the high frequency of postnominal morphemes and verbal prefixes, which separate the subject from the verb in the linear ordering of the sentence, and, for transitive verbs, to the subject-object-verb word order. When the verb is preceded by a particle, such as the ACCOMPLISHED prefix /**le**˧-/ or the DURATIVE prefix /**tʰi**˧-/, the noun and the verb do not interact directly; they often even belong to two different tone groups. Compare, for instance, the realizations of the MH-tone verb /**qæ**˧˥/ 'to burn' in /**mv̩**˧ **| le**˧**-qæ**˧**-ze**˥/ 'the fire burned' and /**mv̩**˧ **qæ**˩/ 'the fire burns'.

The elicited data analyzed in this section is based on intransitive verbs, so as to avoid possible confusions between S+V and O+V constructions. Following the same procedure as for nouns, two contexts were used to arrive at underlying tone categories: S+V and S+V+PERFECTIVE. For instance, (65a) without the PERFECTIVE yields /**hĩ**˧**-bæ**˧ **tsʰɯ**˧˥/ 'the guests arrive'. The tone pattern for this combination of subject and predicate can therefore be described as /M.M.MH/ and further analyzed as //MH#//: an MH contour associating with the last syllable.

6.10.1 The facts

For systematic elicitation, the following verbs were used: /**se**˥/ 'to walk', /**ʂɯ**˧$_{\mathbf{a}}$/ 'to die', /**tsʰo**˧$_{\mathbf{b}}$/ 'to jump', /**tsʰɯ**˩$_{\mathbf{a}}$/ 'to come.PST', /**ʐwɤ**˩$_{\mathbf{b}}$/ 'to speak', and /**bæ**˧˥/ 'to run'. The nouns used were kinship terms and animal names. Table 6.54 presents the results. The //LM// and //LH// tone categories of monosyllables always yield the same output, so they are pooled together in the table.

When the subject-plus-verb combination ends in an /H/ or /L/ tone, the PERFECTIVE morpheme carries /L/ tone. When it ends in an /MH/ tone, the postverbal morpheme receives the /H/ part of the contour. When it ends on an /M/ tone, the tone of the postverbal morpheme cannot be predicted; in these cases, it is indicated in the table, preceded by a '+' sign.

The recording *SubjectVerb* {0004476} contains all of the combinations in Table 6.54.

About one fourth of the tone patterns for subject-plus-verb phrases differ from the corresponding object-plus-verb phrases. Among the identical combinations are those where the noun has an //L#// or //H#// tone, since these fixed-position tones lower the tones of all following syllables to L (by application of Rules 4 and 5; see §7.1.1).

As with other types of morphosyntactic combinations, such as numeral-plus-classifier and object-plus-verb, the tone patterns in Table 6.54 cannot be derived from a set of phonological tone rules. Among the more surprising patterns is //LH// plus //LM//, yielding L.M.M+H, as in /**bo**˩**ɬɑ**˧ **ʈʰɯ**˧**-ze**˥/ 'the boar drank'.

Table 6.54: The tone patterns of subject-plus-verb combinations, in surface phonological transcription.

	tone of verb					
tone of noun	H	M_a	M_b	L_a	L_b	MH
LM, LH	L.H	L.M+M	L.M+M	L.H	L.H	L.MH
M	M.M+L	M.M+M	M.M+M	M.L	M.L	M.MH
L	M.M+L	L.L	M.M+M	L.L	L.L / M.L	L.L
H	M.M+L	M.M+L	M.M+L	M.MH	M.MH	M.L
MH	M.H	M.H	M.H	M.MH	M.MH	M.H
M	M.M.M+L	M.M.M+M	M.M.M+M	M.M.L	M.M.L	M.M.MH
#H	M.M.M+L	M.M.M+L	M.M.M+L	M.M.MH	M.M.MH	M.M.L
MH#	M.M.MH	M.M.MH	M.M.MH	M.M.MH	M.M.MH	M.M.H
H$	M.M.M+L	M.M.M+L	M.M.M+L / M.M.M+H	M.M.MH	M.M.MH	M.H.L
L	L.L.L	L.L.L	L.L.L	L.L.L	L.L.L	L.L.H
L#	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L
LM+MH#	L.M.M+L	L.M.M+L	L.M.M+L	L.M.MH	L.M.MH	L.M.H
LM+#H	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.MH	L.M.MH
LM	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.L	L.M.MH
LH	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L
H#	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L

The noun has an //LH// lexical tone, which would be expected to cause neutralization of all tonal contrasts on the following morpheme (here, on the verb). The five other combinations involving an //LH//-tone noun do indeed yield L.H.L, but that with a //L_b//-tone verb yields L.M.MH. This pattern is conspicuously unrelated to the phonological tendencies observed in the language.

In subject-verb combinations as in many other morphosyntactic contexts, the //L.M// and //L.H// sequences are neutralized on the surface. Notation as //L.H// is interchangeable with //L.M+L//. The former was chosen for the sake of descriptive simplicity: it only requires two tone symbols, and it corresponds to one of the tones attested on disyllabic nouns. (The argument for using //L.H// rather than //L.M.L// as a label for this category of disyllabic nouns was set out in §2.3.8.)

An especially interesting issue is how the PERFECTIVE acquires its tone after an M-tone verb. The interpretation proposed here is that structural gap-filling in the tonal paradigm has taken place, with far-reaching consequences for the system. This topic will be examined in detail in Chapter 10, §10.1, which explores Yongning Na tones from a dynamic perspective.

6.10.2 Variants resulting from a division into two tone groups

Some deviant patterns are observed in recorded data when the morphotonological integration of subject and verb is, as it were, incomplete: both the subject and the verb retain the tones they would carry in isolation. For example, 'the tiger jumped' can be realized as /**lɑ˧ | tsʰo˧-ze˩**/ instead of the expected, regular /**lɑ˧ tsʰo˧-ze˧**/. This corresponds to a division into two tone groups, indicated by the vertical bar '|' in the transcription.

There are, however, borderline cases where the division into two tone groups is not complete. If the subject and the verb phrase (consisting of the verb and its suffix) were truly treated as separate tone groups, a postlexical H tone would be expected in all-L sequences, such as the subject /**bo˩**/ 'pig' in (66) and the verb phrase /**tsʰɯ˩-ze˩**/ 'came' in (67). One is therefore led to conclude that these examples contain a single tone group, hence **bo˩ tsʰo˧-ze˩** and **jo˧ tsʰɯ˩-ze˩** rather than **bo˩ | tsʰo˧-ze˩** and **jo˧ | tsʰɯ˩-ze˩**.

(66) bo˩ tsʰo˧-ze˩.
bo˩˥ tsʰo˩_b -ze˧_b
pig to_jump PFV
'The pig jumped.'

(67) jo˧ tsʰɯ˩-ze˩.
jo˩ tsʰɯ˩a -ze˧b
sheep to_come.PST PFV
'The sheep have come.'

Such variants exist for all subject-verb combinations. When the subject and verb are separated in this way, a pause can be inserted before the verb, with the stylistic effect of highlighting the subject. The role of stylistic choices in the division of the utterance into tone groups, which is of great importance in Yongning Na prosody, is discussed in Chapter 7; however, the deviant tonal patterns in examples (66) and (67) have a significance that extends beyond considerations of stylistic choice. Analysis of these two examples will be taken up in §10.5.2 as part of the discussion of ongoing structural change in the Na morphotonological system.

When the demonstratives //**ʈʂʰɯ˥**// and //**tʰv̩˥**// appear in subject position, they are always separated from the verb by a tone group boundary, as illustrated by example (68).

(68) ʈʂʰɯ˧ | tsʰɯ˩-ze˩˥!
ʈʂʰɯ˥ tsʰɯ˩a -ze˧b
3SG to_come.PST PFV
'(S)he has come!'

This is unlike H-tone nouns, which tend to be integrated in the same tone group as the verb. For instance, /**ʐwæ˧ tsʰɯ˦**/ 'the horse came' – from //**ʐwæ˥**// 'horse' combined with the same verb 'to come' as in (68) – forms a single tone group. The distinct behaviour of demonstratives in this regard is consistent with other patterns observed in various nooks and crannies of the Yongning Na morphotonological system (see §5.4.2, in particular).

6.10.3 Nouns plus the existential verb /dʑo˩b/ behave tonally like subject-verb combinations

The existential verb /**dʑo˩b**/ follows the same tonal pattern as other L_b-tone verbs. Tonally, a noun followed by this existential verb behaves just like a subject-verb combination (documented in Table 6.54 above). This parallel is reflected in the data presented in Table 6.55.

Yet a subtle difference sets the existential verb /**dʑo˩b**/ apart from other L_b-tone verbs. With other verbs, an L-tone subject noun yields an L.L sequence, with

Table 6.55: The tone of the existential verb /**dʑo˩**$_b$/ in association with a noun.

example	meaning	tone	with existential	tone pattern
bo˩˧	pig	LM	**bo˩ dʑo˥(-ze˩)**	L.H
ʐæ˩˥	leopard	LH	**ʐæ˩ dʑo˥(-ze˩)**	L.H
lɑ˧	tiger	M	**lɑ˧ dʑo˩**	L.M
jo˩	sheep	L	**ɬi˧ dʑo˩**	M.L
ʐwæ˥	horse	H	**ʐwæ˧ dʑo˧˥**	M.MH
ʈʂʰæ˧˥	deer	MH	**ʈʂʰæ˧ dʑo˧˥**	M.MH
ɖɤ˧mi˧	fox	M	**ɖɤ˧mi˧ dʑo˩**	M.M.L
ʐwæ˧zo#˥	colt	#H	**ʐwæ˧zo˧ dʑo˧˥**	M.M.MH
hwɤ˧li˧˥	cat	MH#	**hwɤ˧li˧ dʑo˧˥**	M.M.MH
hwɤ˧mi˥$	she-cat	H$	**hwɤ˧mi˧ dʑo˧˥**	M.M.MH
kʰv̩˩mi˩	dog	L	**kʰv̩˩mi˩ dʑo˩˥**	L.L.L
dɑ˧ʝi˩	mule	L#	**dɑ˧ʝi˩ dʑo˩**	M.L.L
õ˩dv̩˧˥	wolf	LM+MH#	**õ˩dv̩˧ dʑo˧˥**	L.M.MH
nɑ˩hĩ#˥	Naxi	LM+#H	**nɑ˩hĩ˧ dʑo˧˥**	L.M.MH
bo˩mi˧	sow	LM	**bo˩mi˧ dʑo˩**	L.M.L
bo˩ɬɑ˥	boar	LH	**bo˩ɬɑ˥ dʑo˩**	L.H.L
hwæ˧tsɯ˥	rat	H#	**hwæ˧tsɯ˥ dʑo˩**	M.H.L

an M.L variant for some nouns but not others. In contrast, the existential verb only allows the M.L pattern for all nouns carrying lexical L tone. This systematic preference is one of the many fine-grained asymmetries that surface upon close examination.

7 Tonal association rules and the division of utterances into tone groups

This chapter sets out and discusses (i) the rules of tonal association, whereby surface phonological tones are derived from the underlying tones, and (ii) the principles underlying the division of the utterance into tone groups, a key unit for tonal processes. Several readers of this book remarked that, for them, the really engaging part of the study lay in this chapter. Here, the archipelago of tables from the earlier chapters gives way to linguistic analysis in the full sense: conveying a feel for the stylistic choices available to speakers (§7.2-7.3) and thereby shedding light on Yongning Na morphotonology in use.

The unit within which tonal processes apply in Yongning Na is referred to here as a *tone group*. In transcriptions, its boundaries are indicated by the International Phonetic Alphabet symbol '|'. Successive tone groups are entirely independent in tonological terms: tones never spread or otherwise influence one another across tone-group boundaries. In other words, tonal computation takes place separately for each tone group.

From a phonetic point of view, successive tone groups are linked to one another through a variety of phenomena, ranging from local effects such as tonal coarticulation to more global patterns, in particular *declination* at the level of utterances and higher-level discourse units. However, these intonational phenomena, discussed in Chapter 9, do not affect the phonological tonal string of the utterance. Conceptually, intonation must be distinguished from the processes by which the surface phonological string of a tone group obtains.

In prosodic hierarchy models such as the universal framework proposed by Selkirk (1986) and Nespor & Vogel (1986), made up of utterance phrase ⊃ intonational phrase ⊃ phonological phrase ⊃ phonological word ⊃ foot ⊃ syllable ⊃ mora, tone groups may be considered as corresponding to *phonological phrases*. Nonetheless, the term "tone group" is preferred here over "phonological phrase" for several reasons.

First and foremost, the defining characteristic of this phonological unit is its role as the domain of tonal processes. The constituency test for tone groups is whether tonal rules and processes apply: a tone-group boundary is where tonal computation starts anew. The emphasis laid on constituency tests aligns with the perspective set out by Tallman (2024: 2): carrying out language description and language comparison "in terms of constituency test results themselves, rather than only abstract constituency structures proposed in the linguistics literature".

Na does not have segmental rules such as the lenition of word-medial consonants found in other languages of the area, such as Qiang (Rma) (LaPolla & Huang 2003: 31–32; Sims 2014: 35–42) and Shixing (Chirkova 2009: 12–13), which provide evidence for the phonological word as a prosodic domain.

Another reason for favouring "tone group" over "phonological phrase" is that, in Yongning Na, another level is also a plausible candidate for identification as a "phonological phrase", namely the *tonal phrase*, discussed further below.

In Yongning Na, the tone group is the highest unit for tonal computation, while the syllable is the smallest: the tone-bearing unit at the surface phonological level. Between these two extremes, I propose the following additional levels:

- *the lexical word*, to which tone categories are lexically associated;
- *the tonal word*, a combination of lexical words, such as noun plus verb in S+V or O+V combinations, and noun plus noun in compounds;
- *the tonal phrase*, a tonal word plus any added clitics and affixes.

For illustration, consider the lexical word //**kʰv̩˩mi˩**// 'dog'. A tonal word is exemplified in (1): the object-plus-verb combination 'to hit a dog'. A tonal phrase is shown in (2): it consists of the tonal word (1) augmented by the PERFECTIVE suffix //**-ze˧$_b$**//.

(1) kʰv̩˩mi˩ ti˥
kʰv̩˩mi˩ ti˩$_a$
dog to_tap/to_hit_gently
'to hit a dog (gently)'

(2) kʰv̩˩mi˩ ti˥-ze˩
kʰv̩˩mi˩ ti˩$_a$ -ze˧$_b$
dog to_tap/to_hit_gently PFV
'has hit a dog (gently)'

An alternative approach would be to use "prosodic stem" instead of *tonal word* and "prosodic word" instead of *tonal phrase*, thereby allowing the use of "phonological phrase" in place of *tone group*. However, a difficulty with this alternative is that the tonal word, as defined here, tends to encompass more material than the “prosodic stem”, which “usually coincides with the morphological stem” (Van de Velde 2008a: 17; see also Downing & Kadenge 2015). For this reason, the terms “prosodic stem”, “prosodic word”, and “phonological phrase” are not adopted in the present volume. The hope is that the present description, structured in terms of *lexical words*, *tonal words*, *tonal phrases*, and *tone groups*, is explicit enough to be fully intelligible and transposable into various theoretical frameworks.

This chapter begins by addressing the phonological aspects of the system, starting with tone-to-syllable association rules (§7.1). It then moves on to the division of the utterance into tone groups, a topic which links up with stylistic considerations (§7.2).

7.1 A summary of tone-to-syllable association rules

This section provides a summary of the tone-to-syllable association rules in Yongning Na.

The description is framed in terms of derivation from an underlying level to a surface phonological level. The notion of derivation from underlying to surface phonological representations is a great help in phonological analysis, to the point that it may not seem to require special justification. However, it has come under criticism from various quarters. Some scholars argue that it is more satisfactory to conceptualize phonology in terms of sets of related surface forms rather than derivational processes. Hyman (2015), reviewing this issue, ultimately advocates for retaining underlying representations as a central analytical tool. Of course, this does not imply that underlying representations are suited to all questions of linguistic inquiry or that they constitute the final word in phonological description. The ultimate goal is to approach the actual processes at work in the speaker’s brain. In formulating generalizations – such as the tone rules proposed below – efforts were made to keep in mind psychological (cognitive) plausibility. Mid- and long-term perspectives for modelling with more elaborate tools are briefly addressed in this volume’s conclusion (Chapter 13).

One illustration of how the distinction between underlying and surface levels proves useful is provided by the LM and LH lexical tone categories: these are distinct at the underlying level, but this opposition is neutralized at the surface phonological level when words are spoken in isolation. This is due to a contextual neutralization of the M and H levels: these tones do not contrast in tone-group-final position when preceded by an L tone. To avoid ambiguity in contexts where the underlying and surface phonological levels might be confused, double slashes are used for transcriptions at the //underlying phonological level//, as against simple slashes for the /surface phonological level/. This may not be visually elegant, but desperate tones call for desperate measures.

7.1.1 The phonological tone rules

The tone-to-syllable association rules determine the surface phonological tones of a given tone group based on its underlying tones. Seven phonological tone rules have been identified in Yongning Na. For ease of reference, these are first listed below before their analysis and discussion.

Rule 1: L tone spreads progressively ("left-to-right") onto syllables unspecified for tone.

Rule 2: Syllables that remain unspecified for tone after the application of Rule 1 receive M tone.

Rule 3: In tone-group-initial position, H and M are neutralized to M.

Rule 4: The syllable following an H-tone syllable receives L tone.

Rule 5: All syllables following an H.L or M.L sequence receive L tone.

Rule 6: In tone-group-final position, H and M are neutralized to H if they follow an L tone.

Rule 7: If a tone group only contains L tones, a postlexical H tone is added to its last syllable.

The following paragraphs explain the motivation for positing these seven phonological rules.

The tone association rules for each of the lexical tone categories have been set out in the course of Chapter 2, including discussions in §2.3 and §2.3.5 on the association of tone to syllables for the three types of High tones: H#, #H, and H$. The same rules apply to the tones of more complex entities, referred to here as *tonal words*, such as compound nouns or numeral-plus-classifier phrases. Unless otherwise specified, the tone pattern associates to the tonal word syllable by syllable ("left-to-right"), one tone level at a time. When there are fewer syllables than tone levels, two levels associate with the last syllable.

Thus, L tone associates with the first syllable of the tonal word. Likewise for M tone. For LM tone, the first syllable receives L, and the second receives M. For LH, the first syllable likewise receives L, and the second receives H. These four tonal categories (L, M, LM, and LH) are the simplest in terms of tone-to-syllable association. The other tone categories (#H, MH#, H$, L#, LM+MH#, LM+#H, and

H#) each have a specific syllabic anchoring, described in §2.3, and reflected in the special symbols used in the present transcription.

In the case of the mixed tone categories LM+MH# and LM+#H, the first part (LM) associates with syllables following the usual rules (like for a simple LM tone), while the second part (MH# or #H, respectively) associates as indicated by the added symbol (#). Thus, for LM+MH# tone, the first syllable receives L, the second receives M, and the last carries an MH contour. If only two syllables are available, the MH contour associates with the second syllable, overriding its M tone. For LM+#H, the first syllable likewise receives L, and the second M, while the H level associates with the first syllable following the tonal phrase, provided that a suitable carrier is available. (These two tone categories never associate with a monosyllable.)

At this stage of tone-to-syllable mapping, some syllables remain toneless. For instance, the lexical disyllable //**dɑ.ɻi˩**// 'mule', the compound //**po.lo-ɬi.pi˩**// 'ram's ear', the numeral-plus-classifier phrase //**gv̩-ʂɯ˩**// 'nine times', and the object-plus-verb combination //**ɖɤ.mi ʑi˩**// 'to grab a fox', all of which have L# tone, are specified for tone only on their last syllable. Tonal nuclei carrying L tone constitute a mirror image of this situation: they are specified for tone only on their *first* syllable. For instance, the noun //**v̩˩dze**// 'bird', the compound //**kʰv̩˩mi-hṽ̩**// 'dog's hair', the numeral-plus-classifier phrase //**so˩-dze**// 'three pairs', and the object-plus-verb combination //**li˩ tɕʰi**// 'to sell tea' all carry L tone, which associates with their first syllable. Their remaining syllables receive a surface phonological tone through phonological rules.

First, L tone spreads, and toneless syllables receive M tone by default.

Rule 1: L tone spreads progressively ("left-to-right") onto syllables unspecified for tone.

Rule 2: Syllables that remain unspecified for tone after the application of Rule 1 receive M tone.

The phrasing of Rule 2 makes explicit that these rules must be applied in order: if Rule 2 were to apply before Rule 1, no tonally unspecified syllables would remain for L tone to spread over.

Applying Rule 1 to the examples given above yields the following results:

//**v̩˩dze**// → /**v̩˩dze˩**/ 'bird'
//**kʰv̩˩mi-hṽ̩**// → /**kʰv̩˩mi˩-hṽ̩˩**/ 'dog's hair'
//**so˩-dze**// → /**so˩-dze˩**/ 'three pairs'
//**li˩ tɕʰi**// → /**li˩ tɕʰi˩**/ 'to sell tea'

Subsequently applying Rule 2 yields:

> //**dɑ.ʝi˩**// → /**dɑ˧ʝi˩**/ ‘mule’
> //**po.lo-ɬi.pi˩**// → /**po˧lo˧-ɬi˧pi˩**/ ‘ram’s ear’
> //**gv̩-ʂɯ˩**// → /**gv̩˧-ʂɯ˩**/ ‘nine times’
> //**ɖɤ.mi ʐi˩**// → /**ɖɤ˧mi˧ ʐi˩**/ ‘to grab a fox’

After the application of Rules 1 and 2, no toneless syllables remain. (In Yongning Na, it is an exceptionless observation that every syllable carries tone at the surface phonological level.) The rules that follow pertain to tone-group boundaries: tonal oppositions are neutralized in certain positions within the tone group (Rules 3-6), and a repair rule adds an H tone on the last syllable if the whole group only contains L tones (Rule 7).

Rule 3: In tone-group-initial position, H and M are neutralized to M.

Rule 4: The syllable following an H-tone syllable receives L tone.

Rule 5: All syllables following an H.L or M.L sequence receive L tone.

Rule 6: In tone-group-final position, H and M are neutralized to H if they follow an L tone.

Rule 7: If a tone group only contains L tones, a postlexical H tone is added to its last syllable.

Rule 6 is posited on the basis of the observation that there is no opposition between H and M on a tone-group-final syllable when the preceding syllables carry L tone. This also applies to tonal contours: LH and LM contours are neutralized to LH in tone-group-final position. Thus, a tone-group-final syllable following an L-tone syllable can only bear one of the following tones at the surface phonological level: L, H, LH or MH. There is no opposition between L…L.LH and L…L.LM, any more than between L…L.H and L…L.M. In transcriptions, it appeared advisable to adhere to the principle of providing a surface transcription of tone, with no more tonal oppositions than those that are actually present at the phonological surface. This required choosing between two alternatives: transcribing the result of neutralization of M and H as M or as H. A phonological reason for adopting the latter choice was that it appeared more appropriate to represent a two-term opposition by means of the two extreme values of the tone scale (L and H).[1]

[1]At one stage, the added tone was analyzed as M (Michaud 2008b). The choice between M and H may seem to be a non-issue, since LM and LH are neutralized in this position. However, the

An unfortunate consequence is that, in transcriptions, the same word may appear with an L.M pattern in some positions and with L.H in others, even though from a phonetic point of view the realizations may be indistinguishable. This paradox is taken to an extreme in (3).

(3) mo˩kv̩˧ tʰv̩˧-kv̩˩-ze˩. | mo˩kv̩˥!
mo˩kv̩#˥ tʰv̩˧a -kv̩˦ -ze˧b mo˩kv̩#˥
meadow_mushroom to_grow ABILITIVE PFV meadow_mushroom
'[Starting in the third month,] meadow mushrooms grow. Meadow mushrooms!' *(Mushrooms.149)* {0004616#S149}

In example (3), the second occurrence of the word //**mo˩kv̩#˥**// is transcribed as /**mo˩kv̩˥**/, with an H tone on the last syllable due to its tone-group-final position, whereas the first occurrence is transcribed as /**mo˩kv̩˧**/. Phonetically, on the other hand, the second occurrence is realized *lower* than the first, due to utterance-final lowering.[2] I can't say that I am perfectly happy with this transcription, but it is principled, systematic and unambiguous. Readers are invited to think of more elegant solutions.

Rule 7 ("If a tone group only contains L tones, a postlexical H tone is added to its last syllable") is based on the observation that no tone groups contain solely L tones. It is through Rule 7 that L-tone expressions acquire a final rising contour when spoken in isolation: when the postlexical H tone is added to a syllable that carries an L tone, this results in an LH contour on that syllable. Applying Rule 7 to the same examples as above yields the following:

/**v̩˩dze˩**/ → /**v̩˩dze˩˥**/ 'bird'
/**kʰv̩˩mi˩-hṽ̩˩**/ → /**kʰv̩˩mi˩-hṽ̩˩˥**/ 'dog's hair'
/**so˩-dze˩**/ → /**so˩-dze˩˥**/ 'three pairs'
/**li˩ tɕʰi˩**/ → /**li˩ tɕʰi˩˥**/ 'to sell tea'

M tone in Yongning Na is not phonologically active: it does not spread, float, or reassociate (see §2.4.3). A tone rule affecting all-L tone groups is therefore far more likely to involve H tone than M tone. In two-tone systems, addition of a final H tone in domains having only L tone is common: it is attested in Lhasa Tibetan (Sun 1997: 498-499), Japanese (Haraguchi 1999: 19), the Bantu languages Matengo and Kimatuumbi (Odden 2005: 415), and Shixing (Chirkova & Michaud 2009). In a three-tone system, describing the added tone as M could create confusion in cross-linguistic comparisons. For these reasons, the postlexical tone of Yongning Na is analyzed here as H.

[2] For a discussion of intonational factors that come into play in the phonetic realization of tones, see Chapter 9.

7.1.2 About the ordering of rules

As mentioned above, the rules must be applied in a specific order. If Rule 7, which adds an H tone to all-L sequences, were to apply before Rule 2, which assigns M tone to toneless syllables, a sequence such as 'has not come', made up of the negation prefix /**mɤ˧-**/ and the verb /**tsʰɯ˩ₐ**/ 'to come.PST', would only have L tones (/**mɤ-tsʰɯ˩**/) at the point when Rule 7 applied. It would thus receive a final H tone (‡ **mɤ-tsʰɯ˩˥**) before undergoing the assignment of an M tone to its first syllable (‡ **mɤ˧-tsʰɯ˩˥**). However, its actual realization is /**mɤ˧-tsʰɯ˩**/, demonstrating that Rule 7 applies only after all the other rules have taken effect.

Rules 3 and 4 ("H and M are neutralized to M in tone-group-initial position" and "A syllable following an H-tone syllable receives L tone") likewise require a specific ordering. If Rule 4 applied first, an underlying sequence such as //**dzɯ˥-bi˧**// 'will eat' would have its second syllable lowered to L, yielding ‡ **dzɯ˥-bi˩**, and would then undergo Rule 3, resulting in ‡ **dzɯ˧-bi˩**. The observed surface phonological pattern, however, is /**dzɯ˧-bi˧**/, with M tone on the second syllable rather than L tone, confirming that Rule 3 must precede Rule 4 in the sequence of application.

7.1.3 A discussion of alternative formulations

The generalizations formulated here in the form of Rules 1-7 could also be captured through other formulations. For instance, it might seem simpler to collapse Rules 4 and 5 into a single rule stating that "H tone can only be followed by L tones". However, a separate rule would then be needed specifically for the M.L sequence: "All syllables following an M.L sequence receive L tone". The decision to adopt the present formulation for Rule 5 ("All syllables following an H.L or M.L sequence receive L tone") is based on the intuition that a common mechanism underlies both cases: H.L and M.L are both stepping-down sequences, moving from a higher tone level to a lower one. The generalization is that stepping-down sequences can only be followed by L tones.

This observation could also be phrased as a static constraint: "There can be no trough within a tone group" or "A tone cannot be surrounded by higher tones within a tone group". Such a constraint would rule out sequences such as ‡ MLH, ‡ MLM, or ‡ MHLM. However, while this formulation captures an important restriction, it does not specify how the offending sequences are avoided or repaired in Yongning Na. In contrast, Rule 5 provides an explicit mechanism: the tones that would otherwise result in such sequences are systematically lowered to L.

Rule 4 precludes …H.H… sequences, meaning that the H tone is culminative. In this light, it might be possible to analyze H as an HL contour. During the early stages of analysis of the Yongning Na tone system, I attempted an account in which the underlying phonological entities were not tones but rather stepwise movements along the three-level tonal scale: an upward step (from L to M or from M to H) or a downward step (from H or M to L). Proposals in this vein have been made for Japanese, where the fall from H to L can be treated as a single phonological entity – termed a *tonal accent* – rather than as a succession of two distinct tones (Kubozono 2012: 1399). Under a strictly tonal account, the tonal accents of Japanese dialects must be represented as sequences of two tones, which is less economical. A "dynamic treatment of tone" was also attempted for Igbo (Niger-Congo family): such is the title of Mary Clark's dissertation (Clark 1976).

For Yongning Na, an HL sequence might be reinterpreted as a "high fall", ML as a "low fall", LM as a "low rise", and MH as a "high rise". However, the presence of contour tones on individual syllables strongly supports a strictly tonal analysis of the system. The dynamic approach, which was ultimately abandoned for Igbo, does not appear promising for Yongning Na either. (This is of course not to say that it may not prove useful for other languages.)

Rule 4 also precludes …H.M… sequences. This can be described as a neutralization (to H.L) of the contrast between H.M and H.L.

7.1.4 Implications for the tones of sentence-final particles

In Yongning Na, as in many East Asian languages, sentence-final particles play a major role in conveying evidentiality and speaker attitude. In tonal languages, these particles may carry lexical tone (e.g. in Vietnamese), but there is a cross-linguistic tendency for an evolution towards reduced tonal distinctions or tonelessness (e.g. in Mandarin). Their position at the end of a sentence subjects them to strong intonational effects. Moreover, in Yongning Na, sentence-final position implies tone-group-final position, which exerts a strong influence on phonological tone patterns: the tone of the last syllable in a tone group is often conditioned by preceding tones. If the tone group contains an H tone or an M.L sequence, all following tones are lowered to L by Rules 4 and 5. For instance, the lexical M tone of final particle //**mæ˧**// ('obviousness') is lowered to L in (4) because the preceding M.L sequence /**le˧-ʁæ˩**/ imposes L on all subsequent tones via Rule 5.

(4) tʂe˧ tʂʰɯ˧ | le˧-ʁæ˩-ɲi˩ mæ˩.

tʂe˥	tʂʰɯ˧	le˧-	ʁæ˩$_a$	-ɲi˩	mæ˧
earth	TOP	ACCOMP	to_melt/to_fall_apart	CERTITUDE	OBVIOUSNESS

'The clods of earth fall apart / the clods of earth melt [into the water].'
(Agriculture.54) {0004440#S54}

Example (4) illustrates the majority case. The next tonal pattern in descending order of frequency is that the final particle receives an H tone projected by a contour tone lexically attached to the preceding syllable, as in (5). Here, the lexical M tone of the affirmative particle //**mæ˧**// is replaced by an H tone (hence /**mæ˥**/) through reassociation of the H component of the MH contour of the ABILITIVE //**-kv̩˧˥**//.

(5) ʐæ˩sɯ˧˦ | -dʑo˩ | ʈʂʰɯ˧ne˧-ɹ̩i˥ | tʰi˧-mv̩˧-kv̩˧ mæ˥.
ʐæ˩sɯ˩ dʑo˥ ʈʂʰɯ˧ne˧-ɹ̩i˥ tʰi˧- mv̩˧a -kv̩˧˥ mæ˧
felt TOP thus DUR to_put_on ABILITIVE OBVIOUSNESS
'This is how we used to wear felt.' *(Sister3.74)* {0004344#S74}

Only after transcribing ten texts, containing over 150 instances of //**mæ˧**//, was this particle finally observed in a context where preceding syllables did not impose a tone on it:

(6) hu˧mi˧-ʈʂʰæ˧ɣɯ˧ ʈʰɯ˧˥ | le˧-qʰwɤ˧-ze˧-mæ˧! |
hu˧mi˥$ ʈʂʰæ˧ɣɯ#˥ ʈʰɯ˩b le˧- qʰwɤ˧b -ze˧b mæ˧
stomach medicine to_drink ACCOMP to_heal PFV OBVIOUSNESS
'[nowadays, the diseased person] drinks medicines for the stomach, and [they] are healed, aren't they!' *(Healing.66)* {0004540#S66}

The phrase /**le˧-qʰwɤ˧-ze˧**/ (ACCOMP-to_heal-PFV) contains neither a contour nor a floating tone that could associate to a following syllable. Nor does it contain an H tone or an M.L sequence that would impose an L tone on subsequent syllables. In this context, which allows expression of the particle's lexical tone, it surfaces with an M. This establishes that the particle does not carry a lexical //L//, //H// or //MH// tone, and should be analyzed as carrying underlying //M// tone: //**mæ˧**//.

Similarly, the tone of the reported-speech particle //**tsɯ˧˥**// is affected by the preceding tonal environment in the vast majority of cases. Across ten narratives, it surfaces with MH tone in only eight instances out of more than three hundred.

To determine the lexical tones of sentence-final particles, elicitation proved a valuable complement to text-based observations. One such case concerns the final particle /**mo˩**/, which conveys invitation. This particle cannot appear immediately after a verb, in a ‡ V+**mo˩** construction. Instead, invitation is expressed as in (7), illustrated with the verb 'to eat' in (8).

(7) ɖɯ˧-V-ɻ̩˥ mo˩
ɖɯ˧- V -ɻ̩˥ mo˩
DELIMITATIVE *target verb* INCHOATIVE DISC.PTCL:INVITATION
'Please go ahead and V a little!'

(8) ɖɯ˧-dzɯ˧-ɻ̩˥ mo˩!
ɖɯ˧- dzɯ˥ -ɻ̩˥ mo˩
DELIMITATIVE to_eat INCHOATIVE DISC.PTCL:INVITATION
'Please have some/please eat some [of it]!'

Table 7.1 presents a set of elicited data, showing that the particle /**mo**˩/ carries L tone in all six cases. With verbs carrying H, L, and MH tones, this L tone derives phonologically from the preceding tonal sequence. Within a tone group, a syllable following an H tone can only have L tone (as observed after both H and MH tones); likewise, a syllable following an M.L sequence may only carry L tone. In the case of M tones, on the other hand, the preceding tonal sequence (M.M.M) does not impose such a constraint: it does not preclude any of L, M, H or MH on the final syllable. The L tone observed on the surface must therefore be attributed to the lexical specification of the particle, hence its analysis as //**mo**˩//, with lexical L tone.

Table 7.1: The tone patterns of the /**ɖɯ˧-V-ɻ̩˥ mo**˩/ construction.

tone	example	meaning	/ɖɯ˧-V-ɻ̩˥ mo˩/
H	**dzɯ˥**	to eat	**ɖɯ˧-dzɯ˧-ɻ̩˥ mo˩**
M_a	**hwæ˧**$_{\mathbf{a}}$	to buy	**ɖɯ˧-hwæ˧-ɻ̩˧ mo˩**
M_b	**tɕʰi˧**$_{\mathbf{b}}$	to sell	**ɖɯ˧-tɕʰi˧-ɻ̩˧ mo˩**
L_a	**bæ˩**$_{\mathbf{a}}$	to sweep	**ɖɯ˧-bæ˩-ɻ̩˩ mo˩**
L_b	**ʐwɤ˩**$_{\mathbf{b}}$	to speak	**ɖɯ˧-ʐwɤ˩-ɻ̩˩ mo˩**
MH	**lɑ˧˥**	to strike	**ɖɯ˧-lɑ˧-ɻ̩˥ mo˩**

7.2 The division of utterances into tone groups

The division of an utterance into tone groups is a central part of Na prosody. While there are some general tendencies and a few rigid rules in the division of utterances into tone groups, speakers often have several options, with different groupings affecting the relative informational prominence of the various components. Prominence (linked to information structure) and phrasing (reflecting syntactic structure) interact in the division of an utterance into tone groups. There is therefore no one-to-one correspondence between syntactic structure and tone-group division.

Tone groups can have highly diverse syntactic compositions. A tone group may consist of a single syllable: monosyllabic nouns and verbs spoken in isolation constitute a tone group on their own. Personal pronouns can associate with other words but often appear in an independent tone group, as in (9).

(9) njɤ˧ | tso˩-bi˩-zo˩-ʝi˩˦.
njɤ˩ tso˩$_a$ bi˧$_c$ -zo˧-ʝi˧
1SG to_build to_go to_have_to
'I shall go and build [a bridge].' *(Renaming.13)* {0004534#S13}

More commonly, tone groups are longer, encompassing, for example:

- a compound noun and a numeral-plus-classifier phrase, as in (10);
- a noun phrase and a verb with its affixes, as in (11);
- a numeral-plus-classifier phrase and a verb with its affixes and particles, as in (12).

(10) ə˧mi˧-mv̩˩ ɲi˩-kv̩˩
ə˧mi˧ mv̩˩ ɲi˧-kv̩˦
mother daughter two-CLF.persons
'the mother and the daughter, the two of them' *(Lake4.93)* {0004350#S93}

(11) dzɯ˧-di˧ mɤ˧-dʐo˦
dzɯ˥ -di mɤ˧ dʐo˧
to_eat NMLZ NEG EXIST
'there was no food' *(Seeds2.69)* {0004542#S69}

(12) dzo˧ | ɖɯ˧-pɤ˩ tso˩ ə˩-bi˩?
dzo˩ ɖɯ˧-pɤ˩ tso˩$_a$ ə-˩ -bi˧
bridge one-CLF to_build INTERROG IMM_FUT
'will [you] build a bridge?' *(Renaming.10)* {0004534#S10}

This section begins with the simplest case: morphemes that consistently constitute a tone group on their own.

7.2.1 Morphemes that always constitute a tone group on their own

Some morphemes always constitute a tone group on their own. They could be referred to as *tonal standalones*. These include the gap-filler /**tʰi˧˥**/ '(and) so, (and) then'; the contrastive topic marker /**-no˧˥**/; /**wɤ˧˥**/ 'again; also', which, in quite a few cases, does not have its full lexical meaning and functions as a mere gap-filler; and the intensifier /**ɖwæ˧˥**/ 'very'. The first three happen to appear in succession in (13): /**tʰi˧˥ | -no˧˥ | wɤ˧˥**/.

(13) tʰi˧-gv̩˩-se˩-dʑo˩ | tʰi˧˥ | -no˧˥ | wɤ˧˥ | qwɤ˧ tʰi˧-gv̩˩.
tʰi˧- gv̩˩ₐ -se˩ -dʑo˥ tʰi˧˥ -no˧˥ wɤ˧˥
DUR to_make/to_build COMPLETION TOP then CNTR.TOP again/also
qwɤ˧ tʰi˧- gv̩˩ₐ
fire_pit DUR to_make/to_build
'After one has finished to build [the cupboard], well, one builds the fire pit!' *(Housebuilding.144)* {0004448#S144}

The gap-fillers /**tʰi˧˥**/ '(and) so, (and) then' and /**wɤ˧˥**/ 'again' have high textual frequency. The former appears in most sentences in the narratives told by Mrs. Latami (consultant F4): over 1,500 occurrences among twenty narratives. The latter appears more than 120 times in the same twenty narratives. One might speculate that these two items owe part of their conspicuous success to their properties with regard to tone-group divisions: since they always constitute a tone group on their own, they demarcate tone groups clearly. They place a final boundary on the preceding tonal group; their own tonal realization is straightforward; and morphotonological computation subsequently starts fresh in a new tone group.

But one may just as well hypothesize an inverse causal link: that these words initially tended to be set apart due to their function as gap-fillers, and ultimately acquired the property of constituting independent tone groups. Support for this hypothesis comes from items that appear to be in the process of becoming *tonal standalones*. The adverb //**ɖɯ˧ njɤ˧**// 'continuously, ceaselessly' is a case in point. It was elicited in association with verbs exemplifying the six tone categories of verbs, yielding the results shown in Table 7.2. In narratives, however, the adverb is always followed by a tone-group boundary, as illustrated in (14), where //**ɖɯ˧ njɤ˧**// 'continuously, ceaselessly' and //**ʐwɤ˩ᵦ**// 'to say' remain in separate tone groups. (There are more than thirty examples in the first twenty texts recorded.)

(14) hĩ˧mo˥=ɻæ˩ ɳɯ˩ | ɖɯ˧-njɤ˧ | ʐwɤ˩-kv̩˧˥ | mæ˩ (…)
hĩ˧mo˥=ɻæ˩ ɳɯ˧ ɖɯ˧-njɤ˧ ʐwɤ˩ₐ -kv̩˧˥ mæ˧
elders=PL A constantly to_say ABILITIVE OBVIOUSNESS
'The elders would always say…' *(Dog2.32)* {0004555#S32}

Table 7.2: The tone patterns of phrases made up of the adverb //**ɖɯ˧ njɤ˧**// ‘continuously, ceaselessly’ followed by a verb.

tone	example	meaning	result	tone pattern
H	**dzɯ˥**	to eat	**ɖɯ˧-njɤ˧ dzɯ˧**	M.M.M
M$_a$	**hwæ˧**$_a$	to buy	**ɖɯ˧-njɤ˧ hwæ˩**	M.M.L
M$_b$	**tɕʰi˧**$_b$	to sell	**ɖɯ˧-njɤ˧ tɕʰi˧**	M.M.M
L$_a$	**dze˩**$_a$	to cut	**ɖɯ˧-njɤ˧ dze˧˥**	M.M.MH
L$_b$	**ʐwɤ˩**$_b$	to speak	**ɖɯ˧-njɤ˧ ʐwɤ˧˥**	M.M.MH
MH	**lɑ˧˥**	to strike	**ɖɯ˧-njɤ˧ lɑ˧˥**	M.M.MH

Using the context of this narrative, an attempt was made to combine the adverb with the verb, but the consultant judged this incorrect, even when the sentence was truncated after the main verb: ‡ **ɖɯ˧ njɤ˧ ʐwɤ˧˥**. This judgment underscores the fact that the data in Table 7.2 was elicited at a push: in the present state of the language, such constructions verge on the unacceptable, and the adverb is well advanced on its path towards becoming a *tonal standalone*. This example illustrates how easily different data collection methods can lead to divergent conclusions. The combination of several types of data, gathered with suitable precautions, is indispensable for cumulative progress in research.[3]

A discourse factor that arguably plays a leading role in the evolution of the adverb //**ɖɯ˧ njɤ˧**// ‘continuously, ceaselessly’ is the emphasis associated with it from a semantic-pragmatic point of view. In narratives, this adverb sometimes carries emphatic stress. The scenario would thus be one of generalization (lexicalization) of intonational emphasis, a phenomenon analyzed in §9.3.2.

7.2.2 Topicalized constituents always end a tone group

A tone group boundary invariably follows topicalized constituents. More specifically:

- The topic marker /**-dʑo˥**/ always terminates a tone group. No exception has been found among 2,000 examples from narratives.

[3] See the set of themed articles *How to study a tone language*, edited by Steven Bird and Larry Hyman, in volume 8 of the journal *Language Documentation and Conservation* (2014). A case of diverging notations in a level-tone language is analyzed by Roux (2003), leading to similar recommendations for methodological precautions. Additional perspectives on this issue are presented in Niebuhr & Michaud (2015).

- The topic marker /**ʈʂʰɯ˧**/ likewise marks the end of a tone group, except in cases where it is immediately followed by the topic marker just mentioned: /**-dʑo˥**/.
- The contrastive topic marker /**-no˧˥**/ invariably constitutes a tone group on its own, as noted above and illustrated by (15).

(15) hwɤ˧li˧˥ | -no˧˥, | ʐv̩˧kʰv̩˩-ŋwɤ˩kʰv̩˩. |
hwɤ˧li˧˥ -no˧˥ ʐv̩˧kʰv̩˩ ŋwɤ˧kʰv̩˩
cat CNTR.TOP four.years five.years
'As for the cat, [it has a lifespan of] four or five years.' *Context:* the previous discussion concerns the dog's lifespan, and the speaker now shifts the focus to cats. *(Dog2.84)* {0004555#S84}

7.2.3 Flexibility in the division into tone groups

Apart from the cases presented in §7.2.1–7.2.2, speakers generally have several options. They may choose to integrate large stretches of speech into a single tone group, or they may divide the utterance into multiple tone groups, with the stylistic effect of highlighting these individual components one after the other. This parallels observations about the intonation of numerous languages. For instance, Karcevskij (1931: 204) remarks on Russian and German: "Within certain limits, it is possible to change the position of the rhythmic breaks that separate a sentence into parts".[4]

An interesting characteristic of Yongning Na is that this division exerts a strong influence on tone, as tonal processes never apply across tone-group junctures.

For instance, /**dzɯ˧-di˧˥**/ 'things to eat; food', from /**dzɯ˥**/ 'to eat' and the nominalizer /**-di˩**/, can combine with /**mɤ˧-dʑo˧**/ 'there isn't' to express 'there isn't any food, there is nothing to eat'. The noun and the negated verb may either be integrated into a single tone group, yielding /**dzɯ˧-di˧ mɤ˧-dʑo˧˥**/, or separated: /**dzɯ˧-di˧˥ | mɤ˧-dʑo˧**/. The latter option is illustrated in (16):

(16) dzɯ˧-di˧˥ | mɤ˧-dʑo˧, | ʈʰɯ˩-di˩˧ | mɤ˧-dʑo˧!
dzɯ˥ -di˩ mɤ˧- dʑo˧$_b$ ʈʰɯ˩$_b$ -di˩ mɤ˧- dʑo˧$_b$
to_eat NMLZ NEG EXIST to_drink NMLZ NEG EXIST
'[Before mankind had learnt to grow crops], there was nothing to eat and nothing to drink!' *(Seeds2.67)* {0004542#S67}

[4] *Original text:* Dans certaines limites, nous pouvons déplacer les anti-cadences séparant les membres de la phrase.

In (16), separating the noun phrase /**dzɯ˧-di˦**/ 'food' from the negated existential verb /**mɤ˧-dʑo˧**/ 'there isn't', creating two tone groups, has the effect of emphasizing the two noun phrases, /**dzɯ˧-di˦**/ 'food; things to eat' and /**ʈʰɯ˩-di˩**/ 'drink; beverage'.

The following sentence in the story reiterates the statement 'There was no food', continuing the same strategy of highlighting the noun phrase 'food', this time adding the topic marker /**-dʑo˥**/ (17). The effect is to emphasize how dire the situation was getting.

(17) dzɯ˧-di˦ | -dʑo˩, | mɤ˧-dʑo˧-ɲi˥ tsɯ˩ | mv̩˩!
dzɯ˥ -di˩ -dʑo˥ mɤ˧- dʑo˧$_b$ -ɲi˩ tsɯ˦ mv̩˧
to_eat NMLZ TOP NEG EXIST CERTITUDE REP AFFIRM
'As for food, it's said that there was none!' *(Seeds2.68)* {0004542#S68}

The narrator then recapitulates in (18):

(18) dzɯ˧-di˧ mɤ˧-dʑo˦ | -dʑo˩ | tʰi˩˥ …
dzɯ˥ -di˩ mɤ˧- dʑo˧$_b$ -dʑo˥ tʰi˩˥
to_eat NMLZ NEG EXIST TOP so/then
'As there was nothing to eat, …' (the narrative moves on to: 'there were some exceptional, smart people, who stood up and did something about it') *(Seeds2.69)* {0004542#S69}

At this juncture, 'there was no food' is integrated into a single tone group and followed by the topic marker /**-dʑo˥**/. This provides an exemplary illustration of how larger chunks of information become incorporated into one tone group as they shift from new to old and backgrounded information.

Long tone groups, within which phonological and morphosyntactic tone rules operate freely, yield a stylistic effect of carefully constructed, poised, stately speech. Conversely, in lively discourse, pragmatic phenomena of emphasis take centre stage, and tone-group boundaries are inserted at various points to draw attention to the preceding word or phrase. Even function words can be highlighted in this manner, as in (19).

(19) ɬo˧pv̩˥ ti˩-kv̩˩ | tsɯ˦ | mv̩˩!
ɬo˧pv̩˥ ti˩$_a$ -kv̩˦ tsɯ˦ mv̩˧
kow-tow to_hit ABILITIVE REP AFFIRM
'It is said that [on that occasion, the whole family] will kow-tow!' *(Sister3.138)* {0004344#S138}

A simpler formulation would be /**ɬo˧pv̩˥ ti˩-kv̩˩ tsɯ˩ | -mv̩˩**/. The formulation in (19) emphasizes the reported-speech particle /**tsɯ˧˥**/. This evidential particle is used whenever the speaker only has indirect knowledge of an event, and hence appears repeatedly in narratives. But in the context of (19), the particle takes on its full significance: the narrator never witnessed the ritual she describes. The emphasis placed on the evidential particle in this sentence exemplifies the speaker's commitment to truthfulness and precision.

When a directional adverb is inserted between a noun and a verb, it can be integrated into the tone group, as in (20), where the noun, the adverb //**mv̩˩-tɕo˧**// 'downward', and the verb form a single tone group, just like the simple object-plus-verb combination /**kʰɯ˧tsʰɤ˧ ʈʂʰe˧**/ 'to stretch out [one's] legs' (from /**kʰɯ˧tsʰɤ˧˥**/ 'leg' and /**ʈʂʰe˧b**/ 'to stretch out').

(20) kʰɯ˧tsʰɤ˧ mv̩˥-tɕo˩ ʈʂʰe˩
kʰɯ˧tsʰɤ˧˥ mv̩˩-tɕo˧ ʈʂʰe˧b
leg downward to_stretch
'to stretch (one's) leg downward'

However, the adverb often marks the beginning of a new tone group, as in (21), where the noun is in a separate tone group from the adverb and verb.

(21) kʰɯ˧tsʰɤ˧˥ | mv̩˩tɕo˧ ʈʂʰe˧
kʰɯ˧tsʰɤ˧˥ mv̩˩tɕo˧ ʈʂʰe˧b
leg downward to_stretch
'to stretch (one's) leg downward'

Like directional adverbs, numeral-plus-classifier phrases often introduce a new tone group, as in (22). But they can also be integrated into a single tone group with a preceding noun, as in (23).

(22) ʂv̩˧~ʂv̩˧˥ | ɖɯ˧-pʰæ˧˥
ʂv̩˧~ʂv̩˧˥ ɖɯ˧-pʰæ˧˥
paper one-CLF.flat_objects
'a sheet of paper'

(23) ə˧mi˧-mv̩˩ ɲi˩-kv̩˩
ə˧mi˧ mv̩˩˧ ɲi˧-kv̩˧˥
mother daughter two-CLF.persons
'the mother and her daughter' *Literally:* 'mother and daughter, the two'
(*Tiger.11* {0004444#S11}, *51*, and *Lake4.93* {0004350#S93}, *96-98*, *125*)

The expression in (23) is a special case: /**ə˧mi˧-mv̩˩**/ is a coordinative compound meaning 'mother and daughter'. The numeral-plus-classifier phrase is clearly not intended to count mother-and-daughter pairs – otherwise, the classifier would be that for pairs rather than persons. Instead, the expression can be paraphrased as 'these two: the mother and the daughter'.

Demonstrative-plus-classifier phrases are commonly integrated with a preceding noun. For instance, in the first version of the *Lake* story, the same two characters, a mother and her daughter, are referred to as /**ə˧mi˧ ʈʂʰɯ˧-v̩˧ lɑ˩ | mv̩˩ ʈʂʰɯ˩-v̩˩˥**/, 'that mother and that daughter': see (24).

(24) ə˧mi˧ ʈʂʰɯ˧-v̩˧ lɑ˩ | mv̩˩ ʈʂʰɯ˩-v̩˩˥
ə˧mi˧ ʈʂʰɯ˧-v̩˧ lɑ˧ mv̩˩˥ ʈʂʰɯ˧-v̩˧
mother DEM-CLF.individual and daughter DEM-CLF
'that mother and that daughter' *(Lake3.52)* {0004348#S52}

The choices made by a speaker in placing tone-groupe boundaries have to do with considerations of informational prominence, as outlined below.

7.2.4 The role of information structure: Considerations of prominence

As a first illustration of how information structure influences the division into tone groups, consider (25):

(25) kʰv̩˩mi˩-ʂe˩˥, | dzɯ˧ mɤ˧-ɖo˧ pi˧-zo˥!
kʰv̩˩mi˩-ʂe˩ dzɯ˥ mɤ˧- ɖo˧ₐ pi˥ -zo
dog-meat to_eat NEG ought_to to_say ADVB
'It's said that one mustn't eat dog meat! / It's said that dog meat is something one must not eat!' *(Dog2.37)* {0004555#S37}

In (25), the noun phrase 'dog meat' is set into relief by forming a tone group on its own. Despite the absence of a morphemic indication of topicalization, it clearly functions as the topic of the utterance. In this context, tonal integration with the following verb would not be stylistically appropriate.

Likewise, in (26), the adverbial 'outside', /**ə˩pʰo˩**/, constitutes a tone group on its own. An alternative option would be to integrate it tonally with the following verb. (The tonal paradigms for *spatial adverbial+verb* combinations are set out in Chapter 6.) In this context, integration into a single tone group would be stylistically acceptable. Separation into two tone groups, however, has the effect of providing information incrementally, giving the impression that the speaker is constructing the utterance as she speaks, rather than delivering preplanned chunks of speech.

(26) ə˧ʝi˧-ʂɯ˥ʝi˩-dʑo˩, | kʰv̩˧ ʈʂʰɯ˧-dʑo˩, | dʑɤ˩˥ | ə˩pʰo˩˥ | kʰɯ˧ mɤ˥-kv̩˩!
ə˧ʝi˧-ʂɯ˥ʝi˩ dʑo˥ kʰv̩˥ ʈʂʰɯ˧ -dʑo˥ dʑɤ˩˥ ə˩pʰo˩ kʰɯ˧˥ mɤ˧- -kv̩˧˥
in_the_past TOP dog TOP TOP INTS outside to_let NEG ABILITIVE
'In the old times, one wouldn't usually let dogs go outside! / In the old times, dogs weren't usually allowed to leave the house!' *(Dog2.75)* {0004555#S75}

It is uncommon for a verb preceded by the accomplished prefix /**le˧-**/ to interact tonally with a preceding noun phrase. In *Caravans.191* {0004530#S191}, for instance, 'the uncle comes back' is realized as /**ə˧v̩˧˥ | le˧-tsʰɯ˩**/ rather than /**ə˧v̩˧ le˥-tsʰɯ˩**/, although the latter form is also acceptable. Cases where tonal interaction does take place are characterized by a high degree of semantic givenness, as illustrated in (27):

(27) ɖɯ˧-v̩˧ le˧-tsʰɯ˩, | ɲi˧-kv̩˧ le˧-tsʰɯ˧˥, | so˩-kv̩˩ le˩-tsʰɯ˩˥.
ɖɯ˧-v̩˧ le˧- tsʰɯ˩$_{a}$ ɲi˧-kv̩˧˥ so˩-kv̩˩
one-CLF.individual ACCOMP to_come.PST two-CLF three-CLF
'One person came, then two, then three.' *Context:* explanation proposed by consultant F4 during a discussion of *Lake4.126* {0004350#S126}. (Field notes.)

It would not be incorrect to say /**ɖɯ˧-v̩˧ | le˧-tsʰɯ˩, | ɲi˧-kv̩˧˥ | le˧-tsʰɯ˩, | so˩-kv̩˩˥ | le˧-tsʰɯ˩**/, but this would be inappropriate in a context where the emphasis is on the count, not on the verb.

When an explanation is added as an afterthought, a relatively long sequence of syllables can be integrated into a single tone group, as in (28), where the final tone group contains ten syllables.

(28) jɤ˧ŋɤ˧-dʑo˧, | ə˧ʝi˧-ʂɯ˥ʝi˩, | hæ̃˩-bɑ˥lɑ˩! | hæ̃˩-bɑ˥lɑ˩-bɑ˩lɑ˩ le˩-po˩ jo˩-kv̩˩ mæ˩!
jɤ˧ŋɤ˧ -dʑo˥ ə˧ʝi˧-ʂɯ˥ʝi˩ hæ̃˩-bɑ˥lɑ˩ hæ̃˩-bɑ˥lɑ˩-bɑ˩lɑ˩ le˧-
Chengdu TOP in_the_past silk silk_clothes ACCOMP
po˧˥ jo˩ -kv̩˧˥ mæ˧
to_bring to_come ABILITIVE OBVIOUSNESS
'From Chengdu, in the past... Silk!! [The people who went on caravans] would bring back silk clothing [from their journeys to Chengdu]!' *(Caravans.104-105)* {0004530#S104}

Here, the word 'silk' conveys the essential information and stands as a tone group of its own. The final tone group serves as a backgrounded explanation, a status reflected in the levelling down of all the tones in /**le˩-po˩-jo˩-kv̩˩-mæ˩**/ '[they] would bring back'.

As many as twelve syllables are bunched together in (29):

(29) ə˧ʐi˧˥, | ɖɯ˩mɑ˧-ɬɑ˩tsʰo˩ pi˩-hĩ˩ ɖɯ˩-v̩˩ dʑo˩-ɲi˩ tsɯ˩ | mv̩˩.
ə˧ʐi˧˥ ɖɯ˩mɑ˧-ɬɑ˩tsʰo˩ pi˥ -hĩ˥ ɖɯ˧ v̩˧ dʑo˧$_{a}$
grandmother GIVEN_NAME to_say REL/NMLZ one CLF.individual EXIST
-ɲi˩ tsɯ˧˥ mv̩˧
CERTITUDE REP AFFIRM
'[Among] women elders, it is said that there was one by the name of Ddeema Lhaco.' *(Elders3.11)* {0004532#S11}

The speaker places considerable emphasis on the person's name, *Ddeema Lhaco*, while the rest of the sentence is strongly backgrounded. Phonologically, the name and all that follows are integrated into a single tone group, causing all the syllables from the third to the twelfth and last to be lowered to L.

As a last example, consider (30).

(30) dzo˧ | le˧-gv̩˩ | tʰi˧-tɕɯ˥ | tʰi˩˥ | no˧ | le˧-tsʰɯ˩-ɲi˩-ze˩-mæ˩, | ə˩-gi˩! | hĩ˧ ɖɯ˧-v̩˧ mɤ˧-tsʰɯ˩! | no˩ le˩-tsʰɯ˩-ɲi˥-ze˩ mæ˩!
dzo˩ le˧- gv̩˩$_{b}$ tʰi˧- tɕɯ˥ tʰi˩˥ no˩ le˧- tsʰɯ˩$_{a}$
bridge ACCOMP to_build DUR to_put then 2SG ACCOMP to_come
-ɲi˩ -ze˧$_{b}$ mæ˧ ə˩-gi˩ hĩ˥ ɖɯ˧ v̩˧ mɤ˧- tsʰɯ˩$_{a}$
CERTITUDE PFV AFFIRM isn't_it person one CLF.individual NEG to_come
no˩ le˧- tsʰɯ˩$_{a}$ -ɲi˩ -ze˧$_{b}$ mæ˧
2SG ACCOMP to_come CERTITUDE PFV AFFIRM
'After the bridge is built, and left there [=and the person who built it waits for someone to cross], you come along! [=someone comes along: you, for instance!] [For a long time] nobody comes, [but at last] you come along!' *(Renaming.17)* {0004534#S17}

The same syntactic structure, 'you come along', is realized as two tone groups: /**no˧ | le˧-tsʰɯ˩ ɲi˩-ze˩ mæ˩**/, then repeated as a single tone group: /**no˩ le˩-tsʰɯ˩ ɲi˥-ze˩ mæ˩**/. This provides an exemplary illustration of how tone groups tend to be longer when the speaker assumes that the semantic content is already familiar to the listener.

7.2.5 Extreme cases of tonal integration: Set phrases and proverbs

Set phrases constitute a typical case of integration. For instance, formulaic expressions recapitulate affinities among animals symbolizing the twelve Terrestrial Branches. These subsets play a role in fortune-telling, where the year of birth serves as a basis for predicting whether or not an individual will be able to relate harmoniously with another. Within the twelve-year cycle, the twelve animals are grouped into four sets of three, shown in (31)-(34).

(31) by̩˧ʐv̩˧ ʝi˧ | æ̃˩ so˥-kʰv̩˩
by̩˧ʐv̩˧ ʝi˥ æ̃˩˧ so˩ kʰv̩˦a
snake ox chicken three CLF.years
'the three years of the Snake, the Ox, and the Rooster'

(32) mv̩˧gv̩˧ ʑi˦ | hwɤ˧ so˧-kʰv̩˥
mv̩˧gv̩˧ ʑi˦˧ hwɤ˧ so˩ kʰv̩˦a
dragon ape rat three CLF.years
'the three years of the Dragon, the Ape, and the Rat'

(33) tʰo˧li˧ bo˩ | jo˩ so˩-kʰv̩˦˧
tʰo˧li˧ bo˩˧ jo˩ so˩ kʰv̩˦a
rabbit pig sheep three CLF.years
'the three years of the Rabbit, the Pig, and the Sheep'

(34) lɑ˧ | ʐwæ˧ | kʰv̩˧ | so˩-kʰv̩˦˧
lɑ˧ ʐwæ˥ kʰv̩˥ so˩ kʰv̩˦a
tiger horse dog three CLF.years
'the three years of the Tiger, the Horse, and the Dog'

The tonal grouping in these four phrases is not entirely uniform. In (31), (32), and (33), the phrase is divided into two tone groups, instead of the four tone groups that would obtain if each animal name were said separately. This results in a stronger degree of integration than in a list where each item forms an independent tone group. The first tone group comprises the names of two animals, while the second contains the third animal name along with the phrase 'three years'. Each of these two tone groups consists of three syllables, creating a rhythmic balance. The tone that arises from the association of two animal names conforms to the patterns observed in coordinative compounds (as detailed in §4.3): a disyllabic M-tone noun plus a monosyllabic H-tone noun yields //#H// (e.g. //**by̩˧ʐv̩˧ ʝi#˥**//), which surfaces as M tone: /**by̩˧ʐv̩˧ ʝi˧**/.

For ease of comparison, the forms that would obtain if each noun stood in a tone group of its own are shown as (35), (36), and (37).

(35) by̩˧ʐv̩˧ | ʝi˧ | æ̃˦˧ | so˩-kʰv̩˦˧
by̩˧ʐv̩˧ ʝi˥ æ̃˩˧ so˩ kʰv̩˦a
snake ox chicken three CLF.years
modified from (31) by placing each noun in a separate tone group

(36) mv̩˧gv̩˧ | ʑi˧˥ | hwɤ˧ | so˩-kʰv̩˧˥
mv̩˧gv̩˧ ʑi˧˥ hwɤ˧ so˩ kʰv̩˦ₐ
dragon ape rat three CLF.years
modified from (32) by placing each noun in a separate tone group

(37) tʰo˧li˧ | bo˧˥ | jo˧˥ | so˩-kʰv̩˧˥
tʰo˧li˧ bo˩˧ jo˩ so˩ kʰv̩˦ₐ
rabbit pig sheep three CLF.years
modified from (33) by placing each noun in a separate tone group

By contrast, in (34), the presence of an odd number of syllables precludes the formation of two tone groups of equal length. As a result, the symmetrical structure observed in (31), (32), and (33) cannot be achieved. Instead, each animal name is assigned to a separate tone group, yielding a total of four tone groups. This illustrates the influence of rhythmic factors in tonal organization. (For an introduction to the notoriously complex domain of linguistic rhythm, see Niebuhr 2009a, Cummins 2012, House 2012.)

Proverbs provide another example of tightly-knit tonal integration. An example is shown in (38).

(38) hĩ˧dzɑ˧ | ɖʐe˧ | tʰɑ˧-ʝi˥, | ɻ̩̃˧ko˩ mi˩ tʰɑ˩-tʰv̩˩.
hĩ˥ dzɑ˥ ɖʐe˧ tʰɑ˧- ʝi˥ ɻ̩̃˧ko˩ mi˧ tʰɑ˧- tʰv̩˧ₐ
person poor money PROH to_borrow shinbone wound PROH to_get
'The poor must not borrow money; the shinbone must not receive wounds.' (Field notes.)

The proverb warns about vulnerability: one must beware of avoiding hits to fragile spots. The listener is expected to know that a blow to the shin is especially painful and, by analogy, to imagine how hard it is for a poor person to repay a loan with added interest. The sequence /**ɻ̩̃˧ko˩ mi˩ tʰɑ˩-tʰv̩˩**/ 'the shinbone must not receive wounds' forms a single tone group, with the stylistic effect of presenting it as a self-evident truth – an established fact rather than a statement coined on the fly by the speaker. If produced as an on-the-spot utterance, the division into tone groups would instead be /**ɻ̩̃˧ko˩ | mi˧ tʰɑ˧-tʰv̩˧**/ or /**ɻ̩̃˧ko˩ | mi˧ | tʰɑ˧-tʰv̩˧**/.

Importantly, even in proverbs and set phrases, speakers retain a latitude of choice in their division of the utterance into tone groups. A comparison of different versions of the same story by the same speaker reveals numerous instances of such variation. For instance, the saying 'If you see a tiger, it means your father is going to die; if you see a panther, it means your mother is going to die', which is at the heart of the narrative *Tiger*, is divided into four tone groups in (39), three in (40), and just two in (41).

(39) ʐæ˩ do˥ | ə˧mi˧ ʂɯ˧; | lɑ˧ do˩, | ə˧dɑ˧ ʂɯ˧.
ʐæ˩˥ do˩$_{b}$ ə˧mi˧ ʂɯ˧$_{a}$ lɑ˧ do˩$_{b}$ ə˧dɑ˥＄ ʂɯ˧$_{a}$
panther to_see mother to_die tiger to_see father to_die
'If you see a panther, it means your mother is going to die; if you see a tiger, it means your father is going to die.' (*Tiger.10* {0004444#S10}; *Tiger2.5* {0004545#S5}, *13*)

(40) ʐæ˩ do˥ | ə˧mi˧ ʂɯ˩, | lɑ˧ do˩ ə˩dɑ˩ ʂɯ˩.
ʐæ˩˥ do˩$_{b}$ ə˧mi˧ ʂɯ˧$_{a}$ lɑ˧ do˩$_{b}$ ə˧dɑ˥＄ ʂɯ˧$_{a}$
panther to_see mother to_die tiger to_see father to_die
'If you see a panther, it means your mother is going to die; if you see a tiger, it means your father is going to die.' *(Tiger.50)* {0004444#S50}

(41) lɑ˧ do˩ ə˩dɑ˩ ʂɯ˩, | ʐæ˩ do˥ ə˩mi˩ ʂɯ˩.
lɑ˧ do˩$_{b}$ ə˧dɑ˥＄ ʂɯ˧$_{a}$ ʐæ˩˥ do˩$_{b}$ ə˧mi˧ ʂɯ˧$_{a}$
tiger to_see father to_die panther to_see mother to_die
'If you see a tiger, it means your father is going to die; if you see a panther, it means your mother is going to die.' (*Tiger.31* {0004444#S31}; *Tiger2.111* {0004545#S111})

As in examples (16)–(19) and (27)–(30), the stylistic nuance is that a greater number of tone groups directs attention to the individual components of the sentence. In *Tiger*, the phrasing in (39) is found at first occurrence, at the beginning of the story, while later occurrences tend to have fewer tone groups, as in (40) and (41). Example (40) is an interesting intermediate case in which the 'panther/mother' and 'tiger/father' pairs are not treated symmetrically. The context is a journey through the mountains, where a mother and daughter encounter a tiger; the father is not present. This asymmetry serves to foreground the omen concerning the mother. The reference to the father is retained to ensure the proverb remains recognizable but is backgrounded through its integration into a single tone group.[5]

As a final example, let us examine (42).

[5]The order of the two clauses – 'panther/mother + tiger/father' versus 'tiger/father + panther/mother' – does not directly affect the division into tone groups. Both orders are acceptable: the ordering in (39)-(40) is not necessarily a departure from a canonical form represented by (41), or vice versa.

(42) hĩ˧ ɳɯ˩ mɤ˩-do˩, | mv̩˧ ɳɯ˩ | do˩˥!
hĩ˥ ɳɯ˧ mɤ˧- do˩$_b$ mv̩˥ ɳɯ˧ do˩$_b$
person/human A NEG to_see sky/heavens A to_see
'People may not see, but the *heavens* see!'

This saying serves as a reminder that other people's gaze is not the touchstone of good conduct, and that one's actions should be guided by the same principles whether seen or unseen. The most common realization of this saying is (42), in which the first part ('What people do not see') is integrated into a single tone group, whereas the second part is divided into two. This structure places particular emphasis on the verb /**do˩$_b$**/ 'to see, to observe', which, occurring alone in its tone group, receives a final H tone and is realized with a rising contour (/LH/), following Rule 7: "If a tone group only contains L tones, a postlexical H tone is added to its last syllable". (Variants are found in the narrative *Reward*, at sentences *28* {0004446#S28}, *36*, *62*, and *114*.)

7.2.6 Two cases of resistance to tonal integration

Some expressions resist integration into a single tone group. As noted in §4.1.1, tonal changes in compounding occur only when the second element (the head) has fewer than three syllables. A typical example is the proper name "Lugu Lake", shown in example (1a) of Chapter 4.

Similarly, the phrase /**sɑ˧ | zo˩bv̩˥l̩ɯ˩**/, meaning 'the universe, the whole world' (*Mountains.69* {0004574#S69}), is also perceived as composed of two distinct parts, /**sɑ˧**/ and /**zo˩bv̩˥l̩ɯ˩**/, even though the first syllable, /**sɑ˧**/, is no longer independently intelligible and does not appear on its own. The trisyllable /**zo˩bv̩˥l̩ɯ˩**/ can be used alone as a synonym for the four-syllable expression. The existence of this trisyllabic form may contribute to the resistance of /**sɑ˧ | zo˩bv̩˥l̩ɯ˩**/ to tonal integration. If such integration were to occur, the expected output would be † **sɑ˧-zo˩bv̩˩l̩ɯ˩** (M.L.L.L), following Rule 5: "All syllables following an H.L or M.L sequence receive L tone".

By contrast, the same argument cannot be invoked to explain the absence of tonal integration in the Na term for 'field penny-cress' (*Thlaspi arvense*), a foetid plant with round, flat pods. The plant's name, /**ʁv̩˧=bv̩˥ | v̩˩tsʰɤ˩˥**/, literally 'the crane's vegetable', would be expected to form a single tone group, as it constitutes a lexical item. However, its transparent structure as a possessive construction probably contributes to slowing down its phonological integration. Moreover, this plant is not commonly used, and the word's low frequency in discourse reduces the pressure towards phonetic-phonological simplification.

7.2.7 Illustration: Sample derivations

This section recapitulates some of the mechanisms described above by presenting sample derivations for two sentences from transcribed texts.

(43) dʐɯ˧ | ɖɯ˧-qʰv̩˧tʰv̩˧ tʰi˩-kʰɯ˩.
dʐɯ˩ ɖɯ˧-qʰv̩˧tʰv̩#˥ tʰi˧- kʰɯ˧˥
water one-CLF.hornful DUR to_put
'[People who travel all day] put a hornful of water [in their bag, so as to have something to drink].' *(Tiger2.51)* {0004545#S51}

In example (43), the noun 'water' forms a tone group on its own and is realized with M tone, following the regular pattern shown in Table 2.8. The second tone group comprises five syllables: /**ɖɯ˧-qʰv̩˧tʰv̩˧ tʰi˩-kʰɯ˩**/.

First, the tone of the numeral-plus-classifier phrase is determined via the table-lookup rules set out in Chapter 3 (Table 3.12), yielding //**ɖɯ˧-qʰv̩˧tʰv̩#˥**// 'one hornful'. The combination of this noun phrase with the prefixed verb //**tʰi˧-kʰɯ˧˥**// 'to put (into)' also follows a table-lookup process (the tone patterns for object-plus-verb phrases are set out in Chapter 6). The resulting surface tone sequence is M.M.M.L.L. Thus, knowledge of a tone group's internal morphosyntactic structure, along with the lexical tones of its components, suffices to arrive at the surface tone sequence.

As a second sample derivation, let us consider a slightly more complex example, shown in (44).

(44) ə˧mv̩˧˥ | le˧-tsʰɯ˩ | tʰi˩˥, | go˧mi˧ ɳɯ˧ | ə˧mv̩˧˥ | mɤ˧-sɯ˥ tsɯ˩ | mv̩˩!
ə˧mv̩˧˥ le˧- tsʰɯ˩ₐ tʰi˩˥ go˧mi˧ ɳɯ˧
elder_sibling ACCOMP to_come.PST gap_filler:well younger_sister A/TOP
ə˧mv̩˧˥ mɤ˧- sɯ˥ tsɯ˧˥ mv̩˧
elder_sibling NEG to_know REP AFFIRM
'[When] the big brother came back, the younger sister didn't recognize him!' *(Sister.49)* {0004341#S49}

In (44), the group | **ə˧mv̩˧˥** | 'elder sibling' simply consists of a noun, so that the tonal word and tone group coincide. Since there are no suffixes or final particles, the noun's lexical MH tone is realized on the last syllable of the tonal word, which is also the last syllable of the tone group.

The tone pattern of | **le˧-tsʰɯ˩** | 'came back' obtains as described in Table 6.8. The word /**tʰi˩˥**/ '(and) then, (and) so' always forms an independent tone group, as discussed in §7.2.1. The tone pattern of **| go˧mi˧ ɳɯ˧ |** 'by the sister' follows the pattern set out in Table 5.22.

In **| mɤ˧-sɯ˥ tsɯ˩ |**, the verb 'to know, to recognize', //**sɯ˥**//, is flanked by the negation prefix //**mɤ˧-**// and the sentence-final particle //**tsɯ˧˥**// (reported speech). The negation prefix surfaces with its lexical M tone and the verb retains its H tone (in keeping with the general pattern documented in Table 6.1), and the sentence-final particle, being preceded by an H tone, receives L through Rule 4.

The final particle //**mv̩˧**// (affirmative) was already encountered above in various examples in which it appears after a tone-group boundary and carries L tone: /**| mv̩˩**/. It does not constitute a well-formed tone group, since a tone group cannot contain L tones only. The special status of this particle, analyzed here as *extrametrical*, brings us to the topic of breaches of tonal grouping, which introduce extrametrical syllables into the system.

7.3 Cases of breach of tonal grouping and their consequences for the system

A breach of tonal grouping occurs when a non-final syllable comes to carry a contour. This causes the following syllables to become extrametrical, with repercussions for the tonal system as a whole. Considerations of analytical consistency lead to positing extrametrical syllables even in some contexts where they are not preceded by a tonal contour. This is the case, for instance, of the affirmative final particle /**| mv̩˩**/ in example (44).

7.3.1 The stylistic option of realizing a contour on a word in non-final position

Syllables that are not in final position within a tone group cannot carry a contour (//MH//, //LM// or //LH//). This restriction is a core property of the tone group as a phonological unit. But this rule is at odds with a stylistic device whereby a word is emphasized by ending the tone group immediately after that word. This device interrupts tonal computation, allowing the realization of a contour on the emphasized word, as illustrated in (45).

(45) le˧-tsɑ˧˥, | le˧-tsɑ˧˥, | le˧-tsɑ˧˥ | -kwɤ˩tɕɯ˩, | le˧-lv̩˧˥!
le˧- tsɑ˧˥ -kwɤ˧tɕɯ˥ le˧- lv̩˧˥
ACCOMP to_row because ACCOMP to_escape
'By rowing, rowing, rowing, they managed to escape!' *(Lake3.59)*
{0004348#S59}

This example occurs in a highly emotional context: a mother and her daughter are rowing for their lives, struggling against the flood that has suddenly transformed their homeland into a vast lake. The verb 'to row' is repeated, and the sentence is chopped into short tone groups, as if mimicking the oars chopping into the water in rapid, successive strokes. Phonetically, the verb is articulated strongly, each time bearing its lexical rising tone. The conjunction /**-kwɤ˩tɕɯ˩**/ is tacked on at the end as if it were an afterthought. A more structured alternative would be to integrate the conjunction with the preceding tone group, as in (46). This would yield the standard tonal pattern, with the MH contour unfolding over the verb and the first syllable of the conjunction. But this deliberate, neatly structured variant would be stylistically inappropriate in this context.

(46) le˧-tsɑ˧˥, | le˧-tsɑ˧˥, | le˧-tsɑ˧-kwɤ˥tɕɯ˩, | le˧-lv̩˧˥!
le˧- tsɑ˧˥ -kwɤ˧tɕɯ˥ le˧- lv̩˧˥
ACCOMP to_row because ACCOMP to_escape
'By rowing, rowing, rowing, they managed to escape!' *Modified from* (45)

An example using the same conjunction as in (45), but where the expected division into tone groups is respected and the expected process of unfolding of a contour tone takes place, is found in (47).

(47) lo˩dʑo˥ | ʈʂʰɯ˧ne˧-ʝi˥ | mv̩˩tɕo˧ pʰv̩˧-kwɤ˥tɕɯ˩-ɳɯ˩, | "qʰʰʰ…ə!" | pi˧ tsɯ˩ | mv̩˩. |
lo˩dʑo˥ ʈʂʰɯ˧ne˧-ʝi˥ mv̩˩tɕo˧ pʰv̩˧˥ -kwɤ˧tɕɯ˥ -ɳɯ
bracelet thus downward to_take_off because TOP
qʰʰʰ…ə! pi˥ tsɯ˧˥ mv̩˧
onomatopoeia:burp! to_say REP AFFIRM
'When [the man] took off [the buried woman's bracelets], like this, [the corpse made a gurgling sound]: Buuurp!' *(BuriedAlive2.48)* {0004536#S48}

This is the only example found so far where the H part of a verb's MH contour reassociates to the conjunction //**-kwɤ˧tɕɯ˥**//, in contrast to numerous examples where this contour surfaces as such on the verb preceding this conjunction (for instance in *Dog.49* {0004442#S49}, *Tiger.46* {0004444#S46}, *BuriedAlive3.65* {0004538#S65}, *Caravans.80* {0004530#S80}, *Sister.50* {0004341#S50}, *Sister3.133* {0004344#S133}, *Seeds2.34* {0004542#S34}, *Renaming.18* {0004534#S18}, and *Funeral.51* {0004571#S51}). Example (47) provides just enough evidence to show that realization with contour spreading is possible. Contour unfolding may once have been the norm, while realization with a contour on the verb may originally have constituted a conspicuous stylistic effect. Be that as it may, the latter is currently much more common than the former, to the point that realization with contour unfolding is now a stylistically marked option.

Realizations of contours in non-final position are not uncommon, each instance carrying a distinct stylistic twist. For instance, in (48), avoidance of contour unfolding has an emphatic effect, making the statement more eloquent by reinforcing the sense of certitude conveyed by the particle /**-ɲi˩**/. Contour unfolding (/**le˧-ʈʰæ˧-ɲi˥**/) would be less forceful in this context.

(48) “mɤ˧-dʐɤ˩ ɲi˩, | ə˧-sɯ˥! | ə˧ʝi˧-ʂɯ˥ʝi˩, | ə˧mi˧-mv̩˩ ɲi˩-kv̩˩, | zo˩no˥, | ɬi˧di˩-di˩mi˩ qo˩ dzi˩, | le˧-wo˥ | le˧-hɯ˩-zo˩, | ə˧ʑi˧-ə˧pʰv̩˧-ki˥ | le˧-hɯ˩-dʑo˩, | ʈʂʰɯ˧ne˧-ʝi˥, | ʐæ˩ ɳɯ˥ | le˧-ʈʰæ˦ | lɑ˧ ɳɯ˧ | **le˧-ʈʰæ˦** | **-ɲi˩**” | pi˧-zo˩!

mɤ˧- dʐɤ˩$_b$ ɲi˩ ə˩- sɯ˥ ə˧ʝi˧-ʂɯ˥ʝi˩ ə˧mi˧ mv̩˩˥
NEG good COP INTERROG to_know once_upon_a_time mother daughter
ɲi˧-kv̩˦ zo˩no˥ ɬi˧di˩-di˩mi˩ qo˧ dzi˩$_a$ le˧-wo˥le˧-hɯ˩
two-CLF.persons well Yongning_plain inside to_dwell went_back
-zo ə˧ʑi˦ ə˧pʰv̩˧ -ki˧ le˧- hɯ˧$_c$ -dʑo˥
ADVB grandmother grandmother’s_brother ALL ACCOMP to_go.PST TOP
ʈʂʰɯ˧ne˧-ʝi˥ ʐæ˩˥ ɳɯ˧ le˧- ʈʰæ˦ lɑ˧ ɳɯ˧ **le˧-** **ʈʰæ˦**
thus panther A ACCOMP to_bite tiger A ACCOMP **to_bite**
-ɲi˩ pi˥ -zo
CERTITUDE to_say ADVB

‘[Elders would tell stories about the tiger eating people,] saying: “That is very bad, you know! Once upon a time, a mother and her daughter who lived in the Yongning plain went to see the grandmother and her brothers; and they were actually attacked by the big cats!”’ *Literally:* ‘they were bitten by the panther, by the tiger’ *(Tiger.51)* {0004444#S51}

A second stylistic consequence of the absence of contour unfolding in (48) is the symmetry between the clauses /**ʐæ˩ ɳɯ˥ | le˧-ʈʰæ˦**/ ‘(a/the) panther bit’ and /**lɑ˧ ɳɯ˧ | le˧-ʈʰæ˦**/ ‘(a/the) tiger bit’, which would be diminished if the tone patterns differed (/**le˧-ʈʰæ˦**/ in the first case, and /**le˧-ʈʰæ˧-ɲi˥**/ in the second). This symmetry is crucial in linking the story to the Na saying ‘If you see a tiger, it means your father is going to die; if you see a panther, it means your mother is going to die’ (example (41) above). The story is about a tiger killing a young woman’s mother, so there is an apparent mismatch with the saying (which associates *panther* with *mother* and *tiger* with *father*). Mention of both a panther and a tiger in (48) is not to be taken literally, as there is no panther in the story. Instead, this parallel mention clarifies that the seeming contrast between *panther* and *tiger* is irrelevant, guiding towards a broader interpretation: that encountering a big cat is an ill omen, foreboding the death of a parent. Example (48) thus provides a clear hint that the panther and tiger in the saying are no more distinct from each other than *cats* and *dogs* in the English expression *it's raining cats and dogs*.

Another construction for which the set of narratives contains examples without the unfolding of a rising contour is /**ʈʂʰɯ˧ne˧ gv̩˧˥**/, which combines the adverb 'thus, in this way' with the verb 'to take place, to occur', as illustrated in (49).

(49) ʈʂʰɯ˧-ʑi˧˥ | -dʑo˩, | ʈʂʰɯ˧ne˧ gv̩˧˥ | -ɲi˩! | tʰv̩˧-ʑi˧˥ | -dʑo˩, | ʈʂʰɯ˧ne˧ gv̩˧˥ | -ɲi˩!

ʈʂʰɯ˧-ʑi˧˥ -dʑo˥ ʈʂʰɯ˧ne˧ gv̩˧$_{c}$ -ɲi˩

DEM.PROX-CLF.households TOP thus to_take_place CERTITUDE

tʰv̩˧-ʑi˧˥

DEM.DIST-CLF.households

'This is what happened to this household! And this is what happened to that household!' *Context:* the narrator reports how her grandmother used to teach children the proper way to behave by drawing on real-life examples of events that had occurred in the neighbourhood. *(Elders3.44)* {0004532#S44}

In this example, a tone-group boundary follows both occurrences of the phrase /**ʈʂʰɯ˧ne˧ gv̩˧˥**/, setting it into relief. The following morpheme also stands out by not being incorporated within the same tone group. This morpheme, /**-ɲi**˩/, is a grammaticalized form of the copula, used to convey "an epistemic strategy that marks a high degree of certitude" (Lidz 2010: 497). The context clarifies what happens here: this passage is in reported speech, where the narrator adopts the tone of voice of her mother's mother, whom she regards as the highest authority on traditional lore.

> My grandmother knew about everything! In the old days, she would also tell us stories about people in the village and what we must learn from them: "This household, this is what happened to them! That household, this is what happened to them! One must develop habits of doing good! One mustn't do wrong!" (*Elders3.44-45* {0004533#S44})

The stylistic choice of inserting a tone-group boundary before the morpheme indicating certitude conveys the assertiveness of the character to whom this passage in reported speech is assigned. Realization as /**ʈʂʰɯ˧ne˧ gv̩˧˥ | -ɲi**˩/ rather than /**ʈʂʰɯ˧ne˧ gv̩˧-ɲi˥**/ conveys an authoritative stance. On the other hand, when the phrase 'This is how it happened' is used more casually, as an introduction to a narrative, it forms a single tone group, as seen in (50). A related phrasing, which likewise constitutes a set phrase and is therefore integrated into one tone group, is presented in (51).

(50) ʈʂʰɯ˧ne˧ gv̩˧-ɲi˥ tsɯ˩ mv̩˩.
ʈʂʰɯ˧ne˧ gv̩˧$_{c}$ -ɲi˩ tsɯ˧˥ mv̩˧
thus to_take_place CERTITUDE REP AFFIRM
'This is how it happened.' *(BuriedAlive2.1)* {0004536#S1}

(51) ʈʂʰɯ˧ne˧ gv̩˧-kv̩˥.
ʈʂʰɯ˧ne˧ gv̩˧$_{c}$ -kv̩˧˥
thus to_take_place ABILITIVE
'This is how it would happen.' (*Sister3.149* {0004344#S149}, *Tiger.2* {0004444#S2})

Example (52) illustrates that the stylistic device whereby a tone group is cut short after a certain word can be applied as early as the first syllable of a sentence. A more strongly integrated formulation would be /**pʰo˩-hɯ˩-kwɤ˩tɕɯ˥-lɑ˩**/, without any special emphasis on the verb 'to flee/to rush'.

(52) pʰo˩˥ | hɯ˧-kwɤ˧tɕɯ˥-lɑ˩ | tʰi˩˥, | go˧mi˧ ʈʂʰɯ˧-v̩˧-dʑo˩, | le˧-ŋv̩˩, | le˧-ŋv̩˩, | le˧-ŋv̩˩, | le˧-ŋv̩˩, | le˧-ŋv̩˩-zo˩!
pʰo˩$_{a}$ hɯ˧$_{c}$ -kwɤ˧tɕɯ˥-lɑ˩ tʰi˩˥ go˧mi˧ ʈʂʰɯ˥
to_flee/to_rush to_go.PST after then younger_sister DEM.PROX
v̩˧ -dʑo˥ le˧- ŋv̩˩ -zo
CLF TOP ACCOMP to_cry ADVB
'After he rushed away, [his] younger daughter cried her eyes out!' *(Sister3.68)* {0004344#S68}

7.3.2 Consequences for the tone system: The emergence of extrametrical syllables

The phenomenon whereby a tone group is cut short after a certain word (noun or verb) has implications for the overall architecture of the tone system. In cases where the portion of the tone group that is cut off can stand on its own as a tone group, the tenets of the system remain unaffected, as illustrated in example (53).

(53) hæ˧, | kʰv̩˩mi˩-ʂe˩˧ | dzɯ˧-kv̩˩!
hæ˧ kʰv̩˩mi˩-ʂe˩ dzɯ˥ -kv̩˧˥
Chinese dog_meat to_eat ABILITIVE
'The Chinese (Han) eat dog meat!' (Field notes, 2012.)

Example (53) lays emphasis on 'dog meat'. In the Na world view, dogs and men are close friends: according to myth, the dog agreed to exchange its original sixty-year lifespan for the thirteen-year lifespan initially granted to man (see the narrative *Dog*). Eating dog meat is therefore taboo among the Na, and the realization that some other ethnic groups do consume it comes as a shock. An unmarked phrasing of (53) would be /**hæ˧ | kʰv̩˩mi˩-ʂe˩ dzɯ˩-kv̩˥**/, in which a single tone group spans both the object and the verb, allowing for regular tonal computation.

By contrast, when particles or conjunctions are left stranded, as in (45), they do not constitute a well-formed tone group on their own. The rules recapitulated in §7.1.1, such as the addition of a final H tone to all-L sequences, do not apply to them – otherwise, one would expect a final rising contour in (45): † **le˧-tsɑ˧˥ | -kwɤ˩tɕɯ˩˧**. These stranded syllables are neither assigned a final H tone nor integrated into the following tone group.

Several analytical options are open here. One possibility is to consider that, at some phonological level, the division into tone groups remains unchanged. This would imply that a contour can be realized in a non-final position within a tone group – an implication which contradicts headlong the definition of the tone group adopted in the present analysis. A preferred option is to consider that the emphasis laid on a word, and the consequent realization of a contour on that word, modifies the division of the utterance into tone groups, leaving certain syllables stranded. These syllables acquire extrametrical status. The notion of extrametricality rescues the general rule that serves as one of the key criteria for defining the tone group as a phonological unit, namely that tonal contours appear only in tone-group-final position.

Consider example (54):

(54) pv̩˩ɭɯ˥ | ɖʐɤ˧˥ | ki˩ tsɯ˩ | mv̩˩.
pv̩˩ɭɯ˥ ɖʐɤ˧˥ ki˧ tsɯ˧˥ mv̩˧
button to_pluck to_give REP AFFIRM
'It is said that [he] plucked a [button from his jacket] and gave it [to the child]. / He plucked a button and gave it [to the child].' *(Renaming.23)* {0004534#S23}

At least three stylistic options are open here. The most tightly-knit configuration involves a single tone group: /**ɖʐ̩ɤ˧ ki˥ tsɯ˩**/.[6] The most analytic alternative consists of two full-fledged tone groups: /**ɖʐ̩ɤ˧˥ | ki˧ tsɯ˧˥**/. The third possibility, found in (54), is intermediate between the previous two: the verb /**ɖʐ̩ɤ˧˥**/ 'to pluck' is realized with its lexical MH contour, as if it were tone-group-final, while the following syllables are all lowered to L, as if they still belonged to the preceding tone group. The syllables /**ki˩ tsɯ˩**/ are extrametrical: they do not constitute a full-fledged tone group on their own.

This range of stylistic variation is a salient characteristic of Yongning Na. Among other potential consequences for the evolution of the tone system, extrametrical syllables at the end of a tone group may, in some cases, come to be reanalyzed as part of the following tone group – provided that the sequence of (surface) tones allows for such reinterpretation. A case in point is the highly frequent sequence consisting of the topic marker //**-dʑo˥**// and the discourse marker //**tʰi˩˥**// 'so, then'. The latter systematically forms an independent tone group, as noted in §7.2.1. However, in the narratives recorded by Mrs. Latami (consultant F4), no pause precedes it, whereas there tends to be a pause before the topic marker. As a result, the two morphemes are pronounced in rapid succession.

This situation creates a discrepancy between two levels: the division into tone groups, on the one hand, and linguistic rhythm, on the other. Now, the topic marker //**-dʑo˥**// most often surfaces as /**-dʑo˩**/, due to the presence of an H tone earlier in the tone group, and the sequence /**-dʑo˩**/ + /**tʰi˩˥**/ would constitute a well-formed tone group. The tone sequence L.LH can be the surface realization of either underlying //L.LH// or underlying //L//. One may speculate that the high discourse frequency of the sequence /**-dʑo˩ tʰi˩˥**/, which on the phonological surface resembles a tightly-integrated tone group, favours its reinterpretation as a single tone group. Such a reinterpretation appears particularly likely in contexts such as (55).

(55) … gɯ˩-ʝi˥ | -dʑo˩ | tʰi˩˥ …
gɯ˩-ʝi˥ -dʑo˥ tʰi˩˥
really TOP then
'… in actual fact, …' *(Mountains.58)* {0004573#S58}

As explained in Appendix A (§A.5), the expression /**gɯ˩-ʝi˥**/ 'really, truly' (from /**gɯ˩**/ 'authentic, true') is well on its way towards reduction to a monosyllable: to my ears, it sounds like [**gi˩˥**] except when hyperarticulated. In (55),

[6] For the sake of simplicity, the affirmative final particle is omitted from the present analysis; it will be discussed in §7.3.3.

for instance, /**gɯ˩-ʝi˥**/ sounds very much like a monosyllable carrying a rising contour: [**gi**˩˥]. Now, a rising contour signals the end of a tone group. While the topic marker that follows, /**-dʑo**˩/, does not constitute a well-formed tone group, the sequence /**-dʑo**˩ **tʰi**˩˥/ does. To labour the point: although in data from speaker F4 the division into tone groups is clearly /**…-dʑo**˩ | **tʰi**˩˥ |/, this sequence could easily be interpreted by a language learner as an L-tone group: /| **-dʑo**˩ **tʰi**˩˥ |/.

7.3.3 Further examples of extrametrical elements

Additional language-internal evidence for resorting to the concept of extrametricality in describing the Na tone system comes from the affirmative particle //**-mv̩**˧// and the expression /**ə**˩**-gi**˩/ 'isn't it!', 'right!'

The affirmative particle //**-mv̩**˧// cannot host an H tone from the preceding reported-speech particle //**tsɯ**˦//: the sequence is realized as /**tsɯ**˦ **mv̩**˩/, not ‡ **tsɯ**˧ **mv̩**˥ or ‡ **tsɯ**˦ | **mv̩**˧. This case appears to be best analyzed in terms of extrametricality.

The expression /**ə**˩**-gi**˩/ 'isn't it!', 'right!' is commonly appended at the end of an utterance. Two observations suggest that it constitutes a tone group on its own. First, a preceding LH or MH contour does not unfold over it, as would be expected inside a single tone group (*Caravans.257* {0004530#S257}, *287*; *Housebuilding.113* {0004448#S113}; *Mountains.159* {0004573#S159}; *Sister3.86* {0004344#S86}). Secondly, the expression /**ə**˩**-gi**˩/ is often preceded by a short (perceived) pause. On the other hand, the fact that it only contains L tones implies that it does *not* constitute a tone group on its own, otherwise it would be realized as /**ə**˩**-gi**˩˥/ (following Rule 7). The latter form, /**ə**˩**-gi**˩˥/, is perfectly acceptable and attested in the narratives, but it is a full-fledged question ('Is it true?'), whereas /**ə**˩**-gi**˩/ is more phatic, almost a gap-filler. For these reasons, /**ə**˩**-gi**˩/ is here analyzed as extrametrical.

To sum up, the tone group may be interrupted after the last syllable of a word (generally, though not exclusively, a verb or noun), leaving some syllables stranded. They are described as having extrametrical status. Such syllables do not constitute a full-fledged tone group: specifically, Rule 7 does not apply to them. Yet these extrametrical syllables are still subject to the lowering influence of the preceding tones: Rules 4 and 5 apply to them as if they were inside the tone group.

In the transcriptions, extrametrical syllables will henceforth be preceded by a diamond symbol '◊', introduced in 2018 (after the publication of the first edition of this book) as a juncture marker distinct from the pipe symbol '|', which is used for a standard tone-group boundary. The diamond signals that the syllables following this mark, up to the next tone-group boundary, are extrametrical. Thus, example (54) is rewritten as (56).

(56) pv̩˩ʈʐɯ˥ | ɖʐɤ˧˥ ◊ ki˩ tsɯ˩ mv̩˩.
pv̩˩ʈʐɯ˥ ɖʐɤ˧˥ ki˧ tsɯ˧˥ mv̩˧
button to_pluck to_give REP AFFIRM
'It is said that [he] plucked a [button from his jacket] and gave it [to the child]. / He plucked a button and gave it [to the child].' *(Renaming.23)* {0004534#S23}

7.3.4 Deviant tone patterns and Mandarin loanwords: Does the existence of extrametrical syllables facilitate the introduction of loanwords with a non-final rising tone?

The phenomena described above in terms of *extrametricality* are clearly marginal. Yet they pave the way for increasingly significant changes to the tone system as a whole. They introduce unusual tone patterns that may become consolidated through loanwords: once a pattern exists in a language, however peripheral it may be, it becomes available for accommodating foreign sound combinations.

For instance, the gap-filler *jiùshi* 就是, 'quite right; exactly, precisely, just' is borrowed as /**tɕo˧˥ʂɯ˩**/, with a word-internal MH contour.[7] At first blush, this appears to contravene a basic phonotactic rule of Na: the restriction of contours to tone-group-final position. However, the process of emphasis described in §7.3.1 introduces tone-group-internal contours, which have now become habitually associated with certain morphemes. The existence of these rising contours arguably facilitated the adoption of Mandarin loanwords with similar tone patterns. In turn, loanwords contribute to the gradual spread of what was previously a deviant phonotactic configuration.

Pushing the argument one step further, if the emphatic function of tone-group-internal contours predates the borrowing of the gap-filler *jiùshi* 就是 'exactly' as /**tɕo˧˥ʂɯ˩**/, this expressive dimension may well have favoured the retention of the rising tone at borrowing. Emphasis is particularly well-suited to this item: a hint of insistence is not out of place for a gap-filler, as it can help signal to the addressee that the speaker wishes to keep their speech turn open. The word /**tɕo˧˥ʂɯ˩**/ also serves as a rejoinder ('Exactly!'), in Yongning Na as in Mandarin. In this function too, a touch of emphasis is welcome, reinforcing the intended

[7] Note that the borrowing is from Southwestern Mandarin, where the syllable /**tɕo**/ 就 carries a rising tone, unlike the falling tone it has in Standard Mandarin (on Southwestern Mandarin: Gui 2001, Pinson & Pinson 2008).

message of convergence of viewpoints between interlocutors. In other words, the item's lexical tone in Mandarin is congruent with its expressive interpretation within the Na prosodic system.

A cross-linguistic analogue to this situation can be found in the success of the Vietnamese loanword *nhà quê* in French. The original Vietnamese expression is a derogatory term meaning 'yokel, hayseed, country bumpkin, backwoods person'. It was borrowed into French as *niakoué* (also spelt *niacoué*), initially as a pejorative designation for the Vietnamese, and later for the Chinese as well. Lexical tones were lost in the process of borrowing, but the vowels and consonants match exactly: Vietnamese /**ɲa.kwe**/ was borrowed as /**ɲa.kwe**/. Initial **ɲ** carries expressive connotations in French. It is almost only found in slang words: *gnan-gnan* 'mawkish, mushy', *gniaf* 'cobbler', *gn(i)ard* 'child', *gn(i)ouf* 'prison', *gnognot(t)e* 'worthless stuff', *gnolle* 'futile person', *gnôle* 'alcohol, hard stuff', *gnon* 'blow', *niaque (gnaque)* 'combativeness'. The villain in Lyon's puppet theatre is tellingly named *Gnafron*. The only exception is an Italian loanword, *gnocchi*, which apparently managed to gain integration despite the slangy overtones of its initial **ɲ**. Seen in this light, the adoption of Vietnamese *nhà quê* /**ɲa.kwe**/ into French was presumably facilitated by the expressive undertones of its initial consonant to French ears.

Returning to Yongning Na, word-initial contours in Chinese words used by Mrs. Latami (consultant F4) are not restricted to expressively loaded items. The word for 'television' is a case in point.[8] Realizations in isolation fluctuate between /**tjɤ˩˧ʂɯ˧**/ (with a tone pattern unattested in Yongning Na, apart from Chinese borrowings), /**tjɤ˧ʂɯ˧**/ (suggesting an underlying M or #H tone), and /**tjɤ˩ʂɯ˧˥**/ (pointing unambiguously to an underlying LM+MH# tone). The noun's tonal behaviour in context is similarly variable. The tones of /**tjɤ˩ʂɯ˧ li˥**/ 'to watch television', /**tjɤ˩ʂɯ˧-qo˥**/ 'on TV', and /**tjɤ˩ʂɯ˧ ɲi˥**/ 'is (a/the) TV' suggest that the noun carries either LM+#H tone or LM+MH# tone (in light of the data presented in §5.4.2, §6.8, and §2.4.2, respectively), but a contour is sometimes realized on the first syllable, as in the variant /**tjɤ˩˧ʂɯ˧ ɲi˥**/ for 'is (a/the) TV'.

This variability reflects a broader state of flux affecting recent Chinese loanwords, also evident in the fluctuating realization of the rhyme of 'television' – alternating between /**jɤ**/, which conforms to Na phonotactics, and /**je**/, which does not but is phonetically closer to the Chinese model. These words remain perceived by consultant F4 as Chinese rather than fully integrated Na items and

[8] Remember that these borrowings are from Southwestern Mandarin, where tone values are nearly the reverse of those in Standard (Beijing) Mandarin, so that the syllable *diàn* 电 in *diànshì* 电视 'television' carries a rising tone, not a falling tone as in Standard Mandarin.

have yet to acquire a stable Na form. Nonetheless, their presence paves the way for gradual changes in the morphotonological system.

7.4 Concluding note

The division of an utterance into tone groups plays a central role in conveying phrasing and prominence. In this respect, the Na facts appear closely parallel to the division of sentences into intonational groups in English (or French) – extensively studied languages for which a wealth of references is available (on French, see for instance Vaissière 1975, Di Cristo 1998, Rossi 1999, Martin 2015). A striking characteristic of Na is the constant interaction between these intonational choices and the language's tonal processes. The notion of tone groups in Na is deeply intertwined with morphotonology. Boundaries between tone groups not only demarcate intonational units but also impose a strict constraint on tonal interactions, shaping the realization of tones at several levels.

The existence of extrametrical syllables further complicates this picture, loosening constraints at the margins of the system and thereby providing a path for the introduction of atypical tonal configurations. Although these syllables remain peripheral to the core system, they nonetheless create a space in which new tonal patterns can emerge and stabilize, particularly in the context of loanword adaptation. The interaction of extrametricality with the incorporation of present-day Mandarin borrowings is especially revealing. While the emergence of tone-group-internal rising contours seems to have been initially tied to emphasis marking, their consolidation through loanwords constitutes an ongoing process, whereby peripheral phonotactic combinations are progressively lexicalized and integrated into the language's tonal system. This reinforces the idea that intonational structures, even when seemingly marginal, can play a role in shaping a language's phonological evolution.

The following chapter further explores the complex interplay between morphotonology and intonational organization in Yongning Na, a key issue in tonal studies.

8 Intonation: Theory and typology

> … a theory of tone must provide some means for describing intonational processes independently of tonal patterns, as well as a procedure for integrating the two structures.
>
> (Clements 1979: 547)

> If, as seems to be the case, the complexity of intonation is typical of human complexity, then there is still a long way to go before (…) intonation yields all of its secrets.
>
> (Vaissière 2004: 256)

The field of intonation studies remains characterized by a diversity of seemingly irreconcilable approaches – an inexhaustible source of misunderstandings. The detailed discussion of various frameworks in Barnes & Shattuck-Hufnagel (2022b) suggests that little progress has been made in resolving these differences since Di Cristo (1998) examined them nearly three decades ago.

> Researchers, we have noticed, often use shared terminology in subtly different ways. These differences, frequently stemming from underlying differences in the goals researchers have set for themselves and their models, are also often either unacknowledged or unexplored by researchers, whose subsequent disagreements are animated precisely by these unspoken divergences. (Barnes & Shattuck-Hufnagel 2022a: 1)

In view of this persistent lack of consensus, it is necessary to address some conceptual issues. Are intonational phenomena best analyzed using the same theoretical constructs as lexical tone, or do they require an entirely separate framework? Should intonation be conceived primarily in terms of phonological categories, or does it have ties to other areas of language structure – such as morphology, syntax, and pragmatics – that call for a more pluridisciplinary approach? Divergent answers to these questions have led to competing models that, rather than converging towards a unified account, have continued to evolve along largely parallel trajectories.

Against this backdrop, careful reflection on the choice of framework and the definition of key terms – beginning with the notion of intonation itself – appears indispensable. This chapter engages with the theoretical landscape of intonation studies, establishing theoretical and typological foundations through an open discussion of the key issues at stake. The next chapter (Chapter 9) turns to the specifics of Yongning Na, presenting the detailed empirical work delving into its intonation system as well as issues of phonetic implementation.

This chapter does not seek to provide a comprehensive survey of existing models – an endeavour that would go beyond the scope of this work – but instead aims to clarify the theoretical choices adopted in this study. §8.1 examines the limitations of phonetic-phonological definitions of intonation, which tend to be too narrow in scope. In §8.2, I critically assess approaches that have influenced recent research, particularly the autosegmental-metrical framework. A central argument of this chapter is that this widely used model is not ideally suited for tonal languages. Section §8.3 explores the typological distinction between tonal and non-tonal intonation, emphasizing its crucial importance to intonational typology.

Building on this critical discussion, §8.4 proposes a constructive approach, outlining what I consider a sound foundation for advancing the study of intonation.

8.1 Issues with a phonetic-phonological definition of intonation

In his book simply titled *Intonation*, Alan Cruttenden begins by defining a 'suprasegmental' domain (also referred to as the 'prosodic' domain), which he contrasts – following a long-standing tradition – with the segmental domain, that of phonemes. For instance, the adjective *nice* consists of three phonemes: a nasal consonant /n/, a diphthong /aɪ/, and a fricative /s/. This constitutes its segmental composition: /naɪs/.

> But there are clearly other features involved in the way a word is said which are not indicated in a segmental transcription. The word *nice* might be said softly or loudly; it might be said with a pitch pattern which starts high and ends low, or with one which begins low and ends high; it might be

> said with a voice quality which is especially creaky or especially breathy. Such features generally extend over stretches of utterances longer than just one sound and are hence often referred to as suprasegmentals (…). Alternatively, the shorter term PROSODIC is sometimes used and I shall generally prefer this term in this book. (Cruttenden 1986: 1)

So far, intonation has not been explicitly mentioned. It makes a discreet entrance in the following sentence, parenthetically.

> Prosodic features may extend over varying domains: sometimes over relatively short stretches of utterances, like one syllable or one morpheme or one word (…); sometimes over relatively longer stretches of utterances, like one phrase, or one clause, or one sentence (intonation is generally relatable to such longer domains). (Cruttenden 1986: 1)

These two extended quotations highlight a paradoxical observation: Alan Cruttenden does not provide a definition of intonation, as if the meaning of the term were self-evident. He contrasts the *segmental* dimension of an utterance (its consonants and vowels) with its *suprasegmental* features, suggesting that the latter have a structure of their own. The phrase 'prosodic features' implies that prosody is organized into discrete units – features, a fundamental concept in phonology (Martinet 1955: 182). Within the domain of prosodic features, one is left to infer that only some are intonational in nature. Readers of Cruttenden's textbook are left to determine for themselves, as they progress through the text, what is actually meant by 'intonation'. Are we to understand that intonation is an abstract linguistic structure pertaining to non-segmental units?

In an encyclopaedia article carrying the same title, 'Intonation', Francis Nolan provides a more explicit characterization.

> The term intonation refers to a means for conveying information in speech which is independent of the words and their sounds. Central to intonation is the modulation of pitch, and intonation is often thought of as the use of pitch over the domain of the utterance. However, the patterning of pitch in speech is so closely bound to patterns of timing and loudness, and sometimes voice quality, that we cannot consider pitch in isolation from these other dimensions. (…) For those who prefer to reserve 'intonation' for pitch effects in speech, the word 'prosody' is convenient as a more general term to include patterns of pitch, timing, loudness, and (sometimes) voice quality. (Nolan 2006: 433)

In this passage, the negative characterization of intonation as a domain independent of phonemes is enriched by an important observation concerning the level of the word. Stating that intonation is "independent of the words and their sounds" clarifies that intonation operates above the level of the word, and therefore above lexical phenomena such as accent (in a language like English) and tone (in languages such as Bambara, Mandarin or Yongning Na). However, Francis Nolan does not explicitly describe intonation as an abstract structure; rather, his characterization remains grounded in the idea that intonation is essentially a matter of how spoken communication employs the three acoustic parameters of fundamental frequency (and its perceptual counterpart, pitch), intensity, and duration.

Confident in the effectiveness of a method that had already proved its worth in the 'segmental' domain, many linguists pursued the study of intonation as a problem of phonology. Their aim has been to discern, beneath the seemingly infinite variation of phonetic substance, a structure based on discrete (categorical) oppositions. Beginners seeking introductory reading on intonation are typically directed to works on the *phonology of intonation*, also referred to as *intonational phonology* (discussed further in the next section, §8.2). The phonetician-phonologist tells us "what intonation consists of, and how we can visualize it and analyze it phonologically" (Nolan 2006: 434).

At first glance, this approach aligns with the principles of the scientific method: it is rooted in empirical observation of phonetic phenomena (in particular, the analysis of fundamental frequency curves, which, for English, are regarded as the primary correlate of intonation) and guided by a commitment to theoretical generalizations. Phonological analysis is seen as the culmination of a process that begins with experimental phonetics and ultimately leads to phonological modelling.

A careful reader, however, will detect indications that phonetic-phonological approaches to intonation encounter certain limitations – limitations that scholars acknowledge without fully drawing out their implications. David Crystal notes that "intonation is not a single system of contours and levels, but the product of the interaction of features from different prosodic systems – tone, pitch-range, loudness, rhythmicality and tempo in particular" (Crystal 1975: 11). This remark suggests the need to move beyond a view that reduces intonation to melodic properties alone – namely, contours (rises and falls) and levels (on a pitch scale). The reference to "different prosodic systems" opens the door to recognizing multiple dimensions of intonation, each of which could, in principle, be approached with its own distinct methods. However, this opening is immediately narrowed to the familiar phonetic dimensions commonly described as "suprasegmental" (Lehiste

1970), much as they were in Nolan's characterization of intonation quoted above: F_0, intensity, and duration. While this list is formally open-ended (as signalled by the phrase 'in particular…'), in practice it highlights three specific phonetic dimensions – ones that naturally lend themselves to phonetic-phonological study.

Once one has come to believe that intonation can be held under one's gaze and brought under phonological scrutiny (to paraphrase Francis Nolan's formulation cited above: "how we can visualize it and analyze it phonologically"), it is difficult to relinquish the hope of such quasi-immediate access. This expectation is all the more deeply ingrained given that the intonation of English – the most widely taught language, and consequently the most extensively studied by linguists – can be described more readily than that of many other languages in terms of patterns of fundamental frequency, known as 'tones' or 'tunes' in the British tradition of intonation studies (O'Connor & Arnold 1973).

Practitioners of language description, however, point out that studying intonation is far from straightforward.

> This phenomenon [=intonation] plays a crucial role in spoken communication, yet its specific properties pose considerable challenges for the linguist, since the methods that have proved their worth in other areas do not appear to be fully suited to the analysis of intonation. (Creissels 1994: 173)
>
> *Original text:* L'importance de ce phénomène [=l'intonation] dans la communication orale est considérable, mais sa spécificité gêne beaucoup le linguiste, car les méthodes d'analyse qui ont fait leurs preuves dans d'autres domaines ne semblent pas convenir vraiment pour l'analyse de l'intonation.

Is the phonetician-phonologist well-equipped to tackle the complex domain of intonation? The question may seem surprising, given the seemingly evident connection between intonation (which is inherently linked to spoken language) and phonetics-phonology. However, to the extent that, as Pierre Cotte writes, behind the proclaimed ideal of objectivity, "we do linguistics in a thoroughly subjective way, and our temperament dictates our approach" (Cotte 1993), it may be worth asking why one chooses to become a phonetician-phonologist. To devote oneself to the study of the sound structure of language is to embrace an area of linguistic inquiry that appears particularly amenable to rigorous analysis, whether in diachronic phonology, synchronic phonology or experimental phonetics. The best generalizations of historical phonetics achieve remarkable levels of precision, even if they allow for important exceptions in detail; it is difficult to match this degree of demonstrable rigour in pragmatics, semantics or even syntax. As for synchronic phonology, it is a favoured domain for those drawn to mathematical models – even if the rigour of such models is sometimes more apparent than real (Karttunen 2006). Experimental phonetics, the other facet of the *Janus bifrons* that is phonetics-phonology, deploys an impressive experimental apparatus (see, for example, Vaissière et al. 2010, Gick et al. 2020), which, by positioning the researcher as an external observer, may lessen the necessity of engaging with languages as instruments of communication. It is entirely possible to describe a phonological opposition without speaking the language in which it occurs – an approach widely practised by Peter Ladefoged, for example (Ladefoged & Maddieson 1996).

The purpose of these remarks is neither to question the considerable value of such work nor to dispute the fact that a phonetician-phonologist may experience languages in the way many linguists do (in his autobiography, Martinet uses the expression 'living out languages': Martinet 1993). Rather, the point is to highlight a general tendency: phoneticians-phonologists are not typically among the most generalist of linguists; they are not necessarily the best informed about the full range of linguistic diversity. This makes it paradoxical to entrust phonologists with the task of elucidating how intonation functions, since the study of this domain requires attention to communicative dimensions that are not exclusively (or even primarily) phonetic-phonological. This observation may explain an important part of the considerable discrepancies between different authors' views on intonation.

8.2 Issues with tonal models of intonation

In addition to what appears in retrospect as a somewhat too exclusive focus on phonetic and phonological aspects, an additional hurdle in the study of intonation – particularly in tonal languages – has been the widespread adoption of tonal models for describing intonation. This tendency has shaped the field in ways that warrant careful reconsideration. Given this context, the following arguments need to be made to lay a solid theoretical foundation for the present descriptive work:

- A clear distinction needs to be maintained between *tonal intonation* and *non-tonal intonation*.
- The presence or absence of intonational tones is a typological parameter that varies across languages.

While these arguments are conceptually straightforward, and their conclusions appear unambiguous, the current state of intonation studies – where fundamental distinctions are not always upheld consistently – calls for a careful, step-by-step exposition of the premises.

8.2.1 Intonation and tone: functionally distinct, despite phonetic similarities

Scholars have long been aware of the phonetic similarities between intonation and tone. In the mid-17th century, the European authors who devised a Latin-based writing system for Vietnamese (de Rhodes 1651) had to develop a notation for a six-way tonal contrast. One of the tones was left unmarked, grave and acute accents were used for two others, and the tilde for a fourth. For the remaining two tones, symbols from sentence-level punctuation were used: the full stop was placed below the vowel to indicate tone 4 (orthographic *nặng*), based on the perceived similarity between its final glottal constriction and the intonational expression of *finality*; and the question mark, in a reduced form placed above the vowel, was used for tone 5 (orthographic *hỏi*), reflecting its final rise (Haudricourt 2010). To the authors of this system, the newly devised tone marks functioned as mnemonic cues to pronunciation, via a phonetic similarity with intonation in Romance languages.

When Chao Yuen-ren devised a system of "tone-letters" three centuries later (Chao 1930), he proposed it as a tool not only for the transcription of lexical tones but also for intonation. He illustrated its application to English intonation, distinguishing different ways of saying *Yes* and *Where does he live*. The original article is entirely composed in International Phonetic Alphabet, as was the standard for the journal *Le Maître phonétique*; for convenience, this excerpt from Chao (1930: 26) is given in English orthography.

42	**jes˦˨**	Ordinary affirmation.
51	**jes˥˩**	Of course.
24	**jes˨˦**	Go on, I'm anxious to hear the rest of it.
13	**jes˩˧**	I'm listening.
15	**jes˩˥**	But, —.
11	**ɦjes˩**	I understand of course.
44	**jẽˑs˦**	It's all right, although you made a mess of it.
55	**jẽˑs˥**	I heard all about that sort of thing.
351	**jes˧˥˩**	I should be most delighted.
3513	**jes˧˥˩˧**	So far as that's concerned, only —.

ʍɛə꜒	**dəz꜓**	**i:꜔**	**liv꜔꜖**	Ordinary interrogation.
ʍɛə꜖	**dəz꜔**	**i:꜓**	**liv꜒**	Where did you say he lived?
ʍɛə꜖	**dəz꜕**	**i:꜓**	**liv꜒꜖**	No matter where he eats.
ʍɛə꜔꜒	**dəz꜔**	**i:꜖**	**liv꜔꜒**	I didn't ask…, I asked *how* he lived.
ʍɛə꜖	**dəz꜓**	**i:꜒**	**liv꜖꜒**	Don't you know where he lives?

Under Chao's proposal, there is no ambiguity as to whether tone-letters are used to transcribe tones (which he calls "tonemes", emphasizing their analogy with phonemes, with which they share a distinctive function) or for intonation (which he refers to as "tone values"). "Each tone-letter consists of a vertical reference line (…), to which a simplified time-pitch curve of the tone represented is attached, for tonemes to the left of the line, and for tone-values to its right" (Chao 1930: 24-25).[1] Chao's approach, which applies similar (but not identical) transcriptional tools to intonational differences in English and tonal oppositions in Cantonese, underlines the phonetic similarities between these two domains while maintaining a clear functional distinction.

This distinction is also maintained in early auditory observations on the phonetic realization of tone in Lingala, a two-tone language of the Bantu branch:

> … the only possible variations in the intonation of a word or sentence are these:
>
> (a) A widening or narrowing of the interval between the high and the normal tones.
> (b) A raising or lowering of the pitch of voice, i.e. a change of key.
> (c) A gradual rise or fall of the pitch of voice, i.e. a continuous change of key.
>
> In Lingala the only two variations that seem to exist are (a) and (b). The gradual fall of the pitch of the voice during a sentence is so slight as to be almost imperceptible. There is, however, another modification which affects the last syllable of a phrase or sentence only. This may be called the final cadence, and means that a high tone becomes a high-falling, while a tone that is normal becomes normal falling to low. (Guthrie 1940: 472–473)

While Guthrie describes the intonation of Lingala in terms of five different levels, there is no confusion between these five intonational levels and the two tones:

[1]The symbols devised for transcribing intonation, with the stylized pitch curve placed to the right of the reference bar, were not incorporated into the International Phonetic Alphabet.

> Although there are actually five different levels used the language remains essentially two-tone, as in learning forms the only thing to be noticed is whether any syllable has a high or a normal tone. It is, moreover, interesting to notice how regular is the system of tone ranges. Emphasis shifts the intonation to the next higher range. Interrogation moves the tones two ranges higher, while the use of the subjunctive reduces the pitch to the next lower range. (Guthrie 1940: 475–476)

Description of intonation in terms of a finite number of levels was a prevailing trend at the time in American structuralist approaches to intonation. Analyses of English intonation published shortly after Guthrie's study assume the existence of four relevant pitch levels: extra high, high, mid, and low (Pike 1945, Trager & Smith 1951: 42). In Trager & Smith's system, these four levels combine with four degrees of stress (primary, secondary, tertiary, and weak), yielding a symmetrical system of no fewer than sixteen "pitch allophones". This system is rather contrived, and the links between its sixteen units and linguistic functions appear tenuous at best. Fortunately, Trager & Smith's proposal concerns English, a language for which informed native speakers have been able to provide articulate critical feedback:

> … this reviewer, at least, simply does not hear the neatly symmetrical distribution of pitch allophones with phonemes of stress as Trager and Smith describe it, he often hears nothing to justify the writing of *plus* junctures where his colleagues write them, he is sometimes in serious doubt whether to write primary or secondary stress, and he is openly astonished at the apparent claim by Trager and Smith that in final position they can distinguish four allophones of each of four pitch phonemes before each of three terminal junctures. (…) Readers dislike being told that they can 'easily supply other examples' when the most patient effort leaves them utterly baffled. (Sledd 1955)

Writing about intonation in Yongning Na is a bigger scientific responsibility, as few native speakers are likely to scrutinize the linguist's claims in such detail.

A general issue with Trager & Smith's framework for studying intonation is that it suffers from the same immoderate ambition as Hall & Trager's framework for the *analysis of culture*, which sought to provide "a hypothesis and methodology for the analysis of culture as a whole and specific cultural systems… a general analytic scheme into which all cultural activities, at all levels of integration and complexity, can be fitted" (Hall & Trager 1953: 57). By contrast, Guthrie's proposals have much to commend them. He clearly distinguishes the two level tones from the intonational factors that influence their realization. Moreover, although the four proposed phonetic ranges are presented in an order based on form, from narrowest to widest, the analysis hinges on the linguistic functions associated with variations in tone range. This is a fruitful approach, yielding a wealth of interesting observations.

In the development known as the "autosegmental-metrical" approach to intonation, on the other hand, the distinction between tone and intonation becomes threatened by the choice of a tonal treatment of intonation.

8.2.2 Autosegmental-metrical models of intonation

The autosegmental-metrical framework of intonation studies continues the American approach to intonation as a sequence of pitch phonemes or significant levels – exemplified by Pike (1945), Wells (1945), Trager & Smith (1951), and Hockett (1955) –, but it borrows from models of tone in African languages and actually treats intonation as if it were tonal. In this framework, pitch accents – arranged in a linear sequence – are considered the fundamental building blocks of an intonation contour (see Ladd 1996 and Gussenhoven 2004 for overviews). The core principles of autosegmental-metrical models draw on concepts from autosegmental tonology, including level tones, downstep, and tone spreading.

The autosegmental-metrical framework has dominated discussions of intonation since the 1980s. Its appeal lies partly in its perceived potential for speech synthesis (Pierrehumbert 1981) and prosodic annotation, notably through the ToBI system ("Tones and Break Indices") introduced by Silverman et al. (1992).

A broader perspective on autosegmental-metrical models of intonation reveals them to be somewhat hybrid and, in some respects, perplexing systems (as noted by Martin 2001 and Wightman 2002). 'Intonational tones' are abstract entities, yet their labels also serve as a phonetic transcription for linguistically significant aspects of F_0 curves. Tonal labels are typically assigned by eye – based on F_0 curves superimposed on spectrographic displays – more than by ear, whereas 'boundary tones' are meant to reflect the perceived cohesion between successive words and are thus grounded both in auditory impressions and in observed F_0 curves.

Interestingly, before becoming one of the main proponents of autosegmental-metrical models, Bob Ladd was convinced that he had "clearly demonstrate[d] the inadequacy of any approach to English intonation which treats contours as sequences of significant pitch levels" (Ladd 1978: 517).

> In short, linguistic systems force users to identify certain signals as discretely different from one another; and linguists' analyses should reflect these discrete differences. But an analysis of intonation in terms of pitch levels forces us to distinguish points along a gradient as also being discretely different – even though they are not – because the theory provides no principled way of knowing when changing a certain feature in a sequence is going to produce a 'modulation', and when it is going to produce a 'very different tune'. No amount of tinkering with theoretical mechanisms can remedy this defect; the best that any pitch-level theory can do is ignore it. To continue to ignore the difference between the gradient and the all-or-none by forcing it into a pre-ordained system of distinctions is only to put off reaching an understanding of how intonation really functions in language. (Ladd 1978: 539)

This line of thought (which makes excellent sense to me) continued to be pursued by some scholars even at the height of the autosegmental-metrical paradigm. Alternatives to tonal models of intonation include the Kiel Intonation Model and its developments (Niebuhr & Kohler 2004, Kohler 2005, Niebuhr 2007, 2010, 2022), superpositional approaches (Lindau 1986, Vaissière 2002, 2004, Grønnum 1991, 1998a,b, 2022), and various others (Delattre 1966, Martin 1977, 2015, Fónagy 1989, Rossi 1999, Hirst & Di Cristo 1998). But these approaches remain outside the mainstream of intonation studies (Zerbian 2010, Michaud & Vaissière 2015), while "Tones and Break Indices" systems are developed for an increasing range of languages. In addition to the original system, tailored for Mainstream American English, there exist ToBI systems for German (GToBI), Tokyo Japanese (J-ToBI), Seoul Korean (K-ToBI), Athens Greek (GrToBI), European and Brazilian Portuguese (P-ToBI), Mandarin, Cantonese, Dutch, French, Italian, Spanish, Catalan, Hindi, etc.

As Vaissière (2002) notes, "To be fair to the original spirit of Janet Pierrehumbert, who intended to describe American English and carefully avoided generalization in her thesis, applying ToBI symbols to a new language requires prior re-evaluation of the underlying principles". The question can be formulated in typological terms: does the language under description possess *tonal intonation*, that is, intonation encoded by tones that are treated on a par with lexical and morphological tones?

The question of whether a language has tonal intonation may appear as a non-issue to researchers working within autosegmental-metrical models, which, by definition, operate with the same conceptual tools – including tonal representations – across all languages. To some extent, this issue is terminological: there often exist straightforward equivalences between descriptions framed in tonal vs. non-tonal terms. For instance, in their analysis of phrasing in French, Fougeron & Jun (1998: 49–51) use the labels H* and H% (or L%) to correspond to Delattre's (1966) notions of *minor continuation* and *major continuation*, respectively. Such equivalences allow for conversions between different frameworks applied to the same language. However, when examined from a typological perspective, the uniform use of tonal labels across structurally different prosodic systems creates considerable difficulties. The motivation for using tonal labels in descriptions of intonational phenomena is analogous to the use of the /±ATR/ (Advanced Tongue Root) feature, borrowed from West African studies (e.g. Snider 2023 on Chumburung), to analyze four-way vowel height oppositions in systems with sets such as /**i-e-ɛ-a**/. This approach circumvents the need for a multi-valued /open/ feature, seen as theoretically cumbersome (Calabrese 2000). Yet typological considerations expose the limits of this strategy. Should French and other Romance languages be included in cross-linguistic studies of ATR? A sensible approach is to begin with a core set of languages that unambiguously have ATR systems (a crucial test being ATR vowel harmony) and to proceed with caution when extending the concept.

A similar methodological prudence is warranted when evaluating claims about tonal intonation across languages. A closer look reveals that H% and L% are used with widely different meanings across languages such as Kinande, French, Vietnamese, and Bemba.

In Kinande, the phrase-final H% is a *bona fide* tone, which interacts with lexical tones – for instance, triggering neutralization of certain lexical tone oppositions when nouns are said in isolation (Hyman 2014: 558).

In French, where neither lexical nor morphological tone is present, no language-internal evidence supports an interpretation of H% as tonal. Instead, it can be viewed as equivalent to Delattre's *major continuation*, with an added specification of phonetic realization.

In Vietnamese, Hạ & Grice (2010) label phrase-final rising pitch movements as H%, but this analysis is motivated by theory-internal considerations rather than by structural similarities between these movements and the language's lexical tones.

Finally, in Bemba, which like Kinande belongs to the vast group of Bantu languages, Kula & Hamann (2016) describe boundary tones but remain noncommittal as to whether these entities are genuinely tonal. Their analysis posits two left-edge boundary tones (H- and L-) and two right-edge boundary tones (H% and L%). The left-edge tones reflect global pitch range expansion and compression, and from the perspective adopted in the present volume, they are clearly non-tonal. The right-edge boundary tones (H% and L%) resemble the intonational tones reported for Kinande, but their precise status remains unclear. The authors explicitly acknowledge this uncertainty: "[i]t remains to be investigated whether this boundary tone replaces the lexical tone" (Kula & Hamann 2016: 331).

To sum up, while the generalized use of boundary tone labels such as H% and L% may seem economical from a theoretical perspective, it obscures typological differences (Ladd 2008). This practice often leaves readers struggling to figure out whether intonation in a given language is genuinely encoded by tones or whether tone and intonation constitute separate levels of description.

Paradoxically, the widespread application of tonal models of intonation to typologically diverse languages has an unintended yet valuable consequence for intonational typology: it can prompt linguists describing tonal languages to address explicitly the degree of similarity between intonational tones and other tonal elements in the language (typically, lexical and grammatical tones). The following section takes up the typological challenge of examining specific cases of reported intonational tones to clarify the key distinction between tonal and non-tonal intonation.

8.3 Intonational typology: Tonal vs. non-tonal intonation

To investigate the typology of tonal vs. non-tonal intonation, a natural starting point is to examine cases where a language's tones have been reported to serve intonational functions, as in the Kinande example mentioned above (where phrase-final H% is a *bona fide* tone, which interacts with lexical tones). Such instances provide crucial insight into the extent to which intonation can be integrated into a language's tonal system.

8.3.1 Instances of intonational tones in the world's languages

In some tonal languages, specific components of intonation are encoded by tones that are treated on a par with lexical and morphological tones. Tones that thus serve as markers for functions at the phrasal level will be referred to as *intonational tones.*

This extension of the notion of tone beyond its primary domains (lexical and morphological tone) is justified by the structural similarities between lexical / morphological tone and certain intonational phenomena. However, it does not imply an unrestricted expansion of the concept of tone to encompass all aspects of intonation, as seen in some versions of autosegmental-metrical models of intonation. Instead, the distinction between intonational tones and broader intonational phenomena must be carefully maintained.

Firstly, tone may indicate sentence mode.

> The most commonly encountered cases involve a tonal means to distinguish interrogatives from declaratives. In Hausa, an L is added after the rightmost lexical H in a yes/no question, fusing with any pre-existing lexical L that may have followed the rightmost H (…). As a result, lexical tonal contrasts are neutralized. In statements, [**káì**] 'head' is tonally distinct from [**káí**] 'you [masculine]'. But at the end of a yes/no question, they are identical, consisting of an extra-H gliding down to a raised L. (Hyman & Leben 2000: 61)

The Hausa example is described as a case of intonational tone, rather than the superimposition of an intonational contour onto an underlyingly unchanged tone sequence.

Secondly, tone may serve a phrasing function. In some languages, certain junctures of the utterance are characterized by the addition of boundary tones, which, though introduced by postlexical rules, are integrated into the tone sequence on a par with lexical tones. L. Hyman (p.c. 2012) notes that such phenomena are "rampant in African tone systems": for instance, in Luganda, the phrase-final boundary tone behaves like any other H tone, except that it is inserted into the tonal string later than the lexical tones. Any sequence of preceding toneless moras is raised to the H level (provided that at least one L remains before it). For example, /**omulimi**/ 'farmer' is pronounced with an all-L pattern (/**òmùlìmì**/), but at the end of an utterance marked by this H% boundary tone it is pronounced L.H.H.H: /**òmúlímí**/. The phrase-final boundary tone of Luganda is transcribed as H%, where the '%' sign, representing a boundary, serves as a functional indication of the tone's origin.

A third intonational function that may be served by tone is the conveyance of prominence. A clear example of such intonational tone is encountered in Naxi, where a word that carries lexical L or M tone on its last syllable can be focused through the addition of an H tone that aligns at the right edge of the word, causing the tone of the last syllable to become rising (Michaud 2006b: 72).

In order to understand how intonational tones emerge and evolve, it is useful to examine not only clear-cut cases such as those reviewed above, but also borderline instances, doubtful cases, and spurious examples of intonational tone. Such is the object of the following sections.

8.3.2 Spurious cases of tonal intonation. Part I: Utterance types in Tanacross Athabaskan

A study of Tanacross Athabaskan, a two-tone language of the Dene family, uses the tonal notation advocated within autosegmental-metrical models (intonational phonology), positing four intonation contours for four utterance types: H* L% for declaratives, H* H% for polar questions, L* L% for imperatives, and H+L* L% for open (*wh-*) questions (Holton 2005: 263). The author espouses the underlying assumption of this notation – that tone and intonation are of the same nature at a certain phonological level: "Both tone and intonation can be viewed as strings of binary tone values with certain associations between the tone values and the segmental tone bearing units" (Holton 2005: 267).

This, in turn, raises the question of whether intonational tones partake in the language's tone rules. In Tanacross Athabaskan, tone spreading is conditional on the lexical tone of the stem: the tone of the stem syllable conditions the tones assigned to preceding syllables. The author examines whether the final L tone found in declaratives influences tone spreading, given that an additional tone in the sequence could, in principle, alter its application. However, the analysis shows that the pre-stem tone spread constraint is sensitive only to underlying lexical tone and remains unaffected by the hypothesized intonational "tone" (Holton 2005: 270). The search for categorical effects of intonational "tones" on the tonal string thus yields a clear negative result.

In my view, this settles the matter: Tanacross Athabaskan does not have tonal intonation. In retrospect, this conclusion also clarifies the author's caveat that "[t]he notation consists of two types of 'pitch-accents', or tones (not to be confused with lexical tones discussed in the previous section)" (Holton 2005: 263). Thanks to such analyses, the facts can be clearly understood. However, the confusion introduced by autosegmental-metrical notation is not easily dispelled. To call two different entities by the same name, assert their identity at one level, and yet insist on keeping them distinct is to place a considerable burden on the reader.

8.3.3 Spurious cases of tonal intonation. Part II: Mandarin interjections

The treatment of the interjection /**a**/ (transcribed as 啊 in Chinese writing) in a learner's dictionary of Standard Mandarin provides a clear example of spurious tonal identification. This dictionary treats the interjection as if it bore lexical tone, establishing four distinct entries corresponding to the four lexical tones of Mandarin: with tone 1, the interjection purportedly expresses that "the speaker gets to know something pleasant"; with tone 2, it signals a "call for repetition"; with tone 3, it conveys "surprise or disbelief"; and with tone 4, it indicates "sudden realization of something" (Huángfǔ Qìnglián 1994, entry "a"). This categorization is based on phonetic similarities between the pitch patterns of the four tones and the intonational variants of the interjection, as summarized in Table 8.1.

In reality, there is a considerable phonetic difference between the four-way division of the Mandarin tonal space and the intonational gradations observed in the realization of interjections. Interestingly, the dictionary glosses the "tone-4" variant of /**a**/ as the "*sudden* realization of something" (emphasis added). However, this interjection can just as well express realization without any hint of suddenness (Lín Yǔtáng 1972, entry "啊"). The fundamental frequency (F_0) of the interjection decreases gradually, in a manner that does not resemble tone 4, which is an abruptly falling tone. The mention of suddenness was probably added because the intonational signalling of this nuance tends to shorten the interjection, thereby increasing its surface similarity with tone 4.

From a functional point of view, there should be no confusion: the phonetic realization of interjections in Mandarin is purely intonational, "with varying, indeterminate accent, like English *Oh! ah! aha!*" (Lín Yǔtáng 1972, entry "啊"). Mandarin interjections are not subject to tonal coding; the interjection /**a**/ has a wide range of possible realizations and expressive effects. The four dictionary entries merely isolate four such realizations and grant them distinct status based on their phonetic resemblance to the four lexical tones of Mandarin.

This example illustrates the risk of misinterpreting intonational phenomena as tonal, but it also hints at the potential for intonational patterns to be reanalyzed as tonal in the course of language evolution. The possibility of such reanalysis invites closer scrutiny of contexts where intonational patterns display a privileged association with specific grammatical morphemes. In particular, sentence-final particles in Vietnamese and Muong illustrate how intonation can become closely tied to certain morphemes, to the extent that it raises questions about the boundary between intonation and tone.

8.3.4 Sentence-final particles in Vietnamese and in Muong

Several studies have examined the interaction between lexical tone and intonation over final particles in Vietnamese (Hạ & Grice 2010, Brunelle et al. 2012, Mac et al. 2015). There is broad consensus that these particles bear a lexical tone, which remains phonologically intact despite phonetic modification by intonation. Such modification occurs through variations in the amplitude of the lexical

Table 8.1: Phonetic basis for the four-way categorization of the nuances expressed by the interjection /**a**/ in Mandarin, as proposed in some dictionaries.

tone	characterization in dictionary	example	translation of example	F_0 on interjection	canonical realization of tone
1	"speaker gets to know something pleasant"	啊！我考过了！ *ā! wǒ kǎo guò-le!*	Wow! I passed the exam!	overall high F_0	level, in the upper part of the speaker's range
2	"call for repetition"	啊，是吗？ *á, shì ma?*	Oh, is that right?	rising	rising
3	"surprise or disbelief"	啊？你在这儿干吗？ *ǎ? nǐ zài zhèr gànmá?*	Huh? What are you doing here?	falling-rising	falling from mid-low to lowest, with final rise in isolation
4	"sudden realization of something"	啊，现在我知道了。 *à, xiànzài wǒ zhīdào-le.*	Aha! Now I understand.	falling	sharply falling, from high starting point

tone's F_0 curve – either expansion or compression – and through shifts in register (upward or downward) (Brunelle et al. 2012: 4). These phenomena are well accounted for within a superpositional model of prosody, in which tone and intonation are conceived as superimposed: local tonal modulations are grafted onto a modulation that extends over larger domains, such as the intonational phrase or the entire sentence.

In Muong, which is closely related to Vietnamese, a study in which I participated (Michaud et al. 2021) reaches different conclusions: final particles do not bear a lexical tone but instead serve as hosts for intonational phenomena. However, certain intonational patterns appear to have become entrenched: specific sentence-final particles developed privileged associations with specific tunes. For instance, interrogative intonation in Muong is typically realized as a clear rise over the last syllables of the sentence, most clearly over final particles. Yet other intonational patterns also occur, and they are consistently associated with specific particles.

For example, the final particle /**hɔ**/, used to seek confirmation or agreement, is pronounced with a descending F_0 contour, ending in a glottal stop. The same intonational pattern – which is distinct from any of Muong's five lexical tones – is also observed over the final particle /**kwa**/, which conveys agreement. Intuitively, this suggests the emergence of a tonal niche, in which a specific communicative function (here, expressing explicit assent) becomes linked to an available phonetic pattern. Speakers of Muong seem to exploit the expressive potential of a glottal stop or constriction, incorporating it into the language's intonational repertoire.

Functionally, this intonational device is all the more effective because its use is currently restricted to just a few morphemes. Without venturing to predict its future fate – particularly in the context of Muong's gradual replacement by Vietnamese –, it is theoretically conceivable that this intonational tune might come to be used on a larger set of morphemes. Over time, it might even stabilize as a distinct lexical tone category.

The case of Muong highlights a scenario in which an intonational pattern becomes associated with specific morphemes, potentially opening a pathway toward *tonalization*. Similar questions about the interplay between lexical tone and sentence-level intonation are also relevant in languages where intonation appears to override or modify lexical tone. A particularly interesting case is found in Phake, a Tai-Kadai language spoken in Assam, where tone changes are reported in the expression of negation and sentence mode.

8.3.5 The expression of negation and sentence mode in Phake

Phake, a Tai-Kadai language of Assam (India), has six lexical tones, as well as cases of "changed tones" (Morey 2008: 234–240). Three distinct processes have been reported:

(i) If a verb has the second tone (High falling), it changes to a rising tone when negated. This rising tone is identical in form to the language's sixth lexical tone; speakers perceive the change as categorical in nature (tonal replacement). This process also appears to be extending to verbs carrying other tones (S. Morey, p.c. 2013).

(ii) Observations from the 1960s and 1970s indicate that changing the lexical tone of the last syllable in a sentence to the sixth tone (a rising tone) yields interrogative sentence mode (Banchob 1987). More recent fieldwork confirms the same phenomenon, but instead of identifying the "changed tone" with one of the six lexical tones, it is described as "a special questioning tone (…). This questioning tone first rises and then falls, and here is arbitrarily notated as 7" (Morey 2008: 234).

(iii) Finally, an eighth tone is reported: an *imperative tone* "that exhibited glottal constriction and creaky voice" (Morey 2008: 239).

Observation (ii) can be reinterpreted as involving neutralization of tonal oppositions: it does not appear implausible that question intonation in Phake overrides the lexical tone of the sentence's final syllable. Likewise, imperative intonation has a salient influence on certain tense-aspect-modality markers in Phake, which may go so far as to override their lexical tone. As observed by Martinet (1957), "the fluctuating needs of communication and expression are reflected more directly and immediately in intonation than in any other section of the phonic system". The phonation type associated to imperative mode in Phake has a clear iconic motivation: contraction of the laryngeal sphincter conveys an attitude of authority, as a general ethological principle (see Fónagy 1983: 113–126). The "imperative tone" of Phake may thus reflect a cross-linguistic tendency towards a *command intonation* that is short, sharp, and high.[2]

[2] Stephen Morey (p.c. 2016) reports that a similar "imperative tone" occurs in Tai Khamti, a nearby language with five citation tones. When tones in Khamti were first marked in the 1990s, special tone marks were used for features such as the imperative, replacing the citation tone marks in texts. Whether these cases involve full neutralization of lexical distinctions remains to be investigated.

It is perhaps significant that "changed tones" are reported in an area where the dominant languages are non-tonal. Phake speakers are also fluent in Assamese, a non-tonal language, which may exert pressure towards simplification of the Phake tone system. Neutralization of tonal contrasts in some contexts could be a result of such pressure. Overall, however, it would seem that intonation does not easily win the day over lexical tone. Some experimental evidence on this topic comes from a study of the Austroasiatic language Kammu, whose two dialects differ in the presence or absence of lexical tones. A comparison of the two dialects finds that their intonational systems are basically identical, except that, in the tonal dialect, intonation patterns adapt to the lexical tones where intonation would otherwise jeopardize the identity of lexical tones (Karlsson et al. 2012).

The findings from Phake suggest that intonation tends to be constrained by considerations of the contrastive integrity of the tonal system. Another window onto tensions between intonational and tonal phenomena is offered by languages of the Dene family, where negation-related prosodic patterns provide evidence for an interplay between grammatical function, tone, and intonation.

8.3.6 The expression of negation in Alaskan Dene

Variegated prosodic developments related to negation are found in five Alaskan Dene languages – Koyukon, Lower Tanana, Middle Tanana, Upper Tanana, and Tanacross. These developments include:

- a floating tone that associates with the negative verb stem in Upper Tanana;
- a high tone associated with the negative suffix in Lower Tanana;
- an utterance-level intonational pattern in Tanacross and Middle Tanana.

Lovick & Tuttle (2024) argue that these tonal and intonational phenomena originate in an intonational source: an emphatic form of negative verb suffixation that remains partially productive in Koyukon. "In all languages where this tonal pattern exists, it involves a high, extra-high, or rising pitch either on the negative suffix or on the stem itself. An example illustrating this pattern for Koyukon is given in [the example below], which contrasts an affirmative form (1) with its corresponding standard negated form (2)."

(1) k^hiːl lɛhæn
k^hiːl lɛ–hæn
boy 3sg.s:cnj:pfv:vv–stand:ipfv
'the boy is standing'

(2) iːhæʔǽ
iː–hæʔ–ǽ
3SG.S:NEG.PFV:VV–stand:IPFV–NEG
'he is not standing'

The pattern "plays a different role in the linguistic systems of these varieties: in Koyukon, it is a pragmatically motivated intonational pattern broadly related to emphasis; in Lower Tanana and Upper Tanana, the negative high can be viewed as a tonal distinction in addition to constriction tone, with primarily grammatical functions; and in Tanacross and Middle Tanana, it has been reinterpreted as an utterance-level intonational pattern" (Lovick & Tuttle 2024: 399).

Alaskan Dene thus provides an example where tonal changes associated with negation appear to have originated in expressive strengthening, eventually developing into functionally distinct prosodic patterns across related languages.

8.3.7 Non-tonal intonation as a widespread typological possibility

The cases of tonal intonation reviewed in the preceding section illustrate a highly interesting but relatively uncommon phenomenon. Languages in which intonation is integrated into the tonal system are a minority. Non-tonal intonation – where intonational functions are not carried by tones – is a far more widespread pattern.

In most East and Southeast Asian languages, available descriptions suggest that intonation is not implemented by the addition of tones in the manner observed for Kinande, Hausa, Luganda, and Naxi (§8.3.1). A well-documented case is Standard Mandarin. Intonation in Mandarin is particularly salient, exerting a strong influence on the phonetic realization of lexical tones. However, intonation does not alter the phonological identity of lexical tones. Tone and intonation are phonetically superimposed on each other, while remaining at structurally distinct levels. Even in cases where intonation profoundly affects phonetic realization, it does not modify the underlying sequence of tones – a fact recognized since the pioneering work of Chao (1929). The informational prominence of a syllable is reflected in local phenomena such as curve expansion and lengthening on the target syllable, along with some modifications in supraglottal articulation. Conversely, phonetic reduction is observed elsewhere in the utterance, notably in the form of post-focus compression of the F_0 range (Xu 1999).

It seems intuitively clear that multi-level tone systems (e.g. Ngamambo, Wobe) cannot allow the degree of intonational flexibility in tonal realization that is pervasive in Mandarin or Vietnamese, as such flexibility would jeopardize the identification of the tonal sequence. The need to distinguish a wide range of categorically distinct tone sequences makes it less economical to encode phrasing and prominence via F_0 modulations superimposed on the tonal string. As a result, languages with multi-level tone systems tend to favour other means of conveying phrasing and prominence: either by integration these functions into the tonal string itself (i.e. through *intonational tones* as defined here) or by relying on non-intonational mechanisms, such as word order and topicalization and focalization morphemes.

Experimental verification of these hypotheses is greatly complicated by the multifarious differences among the languages to be compared. It is hoped that the growing availability of detailed monographs – such as the present one – can contribute to a gradual clarification of these typological perspectives.

The preceding discussion, which included two spurious examples of intonational tone (§8.3.2-8.3.6) and three instances of borderline cases (§8.3.4-8.3.6), suggests that cases of intonational tones are relatively rare in the world's languages. (Yongning Na follows the majority pattern, in that it exhibits no tonal intonation; tone in Na is restricted to lexical and morphological differentiation.)

In light of the critical reflections and typological perspectives set out in the above sections, let us now take a more constructive perspective and outline what I consider a sound basis for advancing our understanding of intonation. It is preferable to adopt a descriptive vocabulary suited to the data, even if it is not mainstream, rather than force the data into an inadequate model.

8.4 Choice of framework and definition of terms

8.4.1 Intonation and its components: Definition of terms

The present study adopts a functional perspective that allows for distinctions that are essential in prosody studies: in particular, distinguishing tone from intonation, and also distinguishing an abstract level of description from the level of phonetic realizations.[3]

Mario Rossi's definition of intonation is particularly promising, as it is grounded in linguistic function, not on phonetic properties. "Intonation, which

[3]The latter distinction is blurred in frameworks where 'tone' is treated as synonymous with F_0. For instance, Hyman & Monaka (2008) define 'tonal' in phonetic terms, as 'realized by F_0', and 'non-tonal' as 'realized by parameters other than F_0' (such as phonation types). The equation of 'tone' with 'F_0' may seem self-evident, to the point that defining tone differently might appear unnatural. Yet, from a classical linguistic perspective, the distinction is crucial: F_0 is an acoustic parameter, whereas linguistic tone is a functional concept.

has long been conflated with one of its privileged parameters – pitch –, is a linguistic system designed to order and organize the information that the speaker intends to communicate to the addressee(s) in their message, and to linearize the hierarchy of syntactic structures" (Rossi 2001).[4] In the study of intonation, it seems methodologically sound to begin with a characterization framed at this level of abstraction: that of the linguistic system.

Rossi distinguishes several schools of thought regarding the concept of *intonation* (Rossi 1999: 32-33). His historical perspective on linguistic theories is particularly useful for clarifying the issues at stake. Adopting a concrete understanding of *intonation as melody* – historically the earliest acception of the term – fosters a monoparametric approach, centred primarily on the acoustic parameter of fundamental frequency, and an interpretation in terms of phonological units – the H and L tones of the autosegmental-metrical framework. However, it is also possible to conceptualize intonation in a manner that Rossi terms *morphological*, in which he identifies influences dating back at least to the Prague School. One key insight from this perspective is the recognition that intonation comprises several components: "[i]ntonation is a part of prosody, which is a whole made up of stress, intonation, and rhythm" (Rossi 2001: 103).[5]

Here, *stress* refers to "lexical stress, a feature of the morpheme, and the properties attached to it" (*ibid.*).[6] This definition is adequate in the context of Romance languages (Mario Rossi's primary area of study). However, from the perspective of prosodic typology, it must be broadened to encompass other prosodic phenomena that share the distinctive function of stress and that likewise operate at the level of an entire morpheme. These include: (i) tone, as in Na, Mandarin, Yorùbá, and so on; (ii) tonal accent, as in Japanese[7] and Swedish; and (iii) phonation-type

[4] *Original text:* L'intonation, qui a longtemps été confondue avec l'un de ses paramètres privilégiés, la mélodie, est un système linguistique destiné à organiser et à hiérarchiser l'information que le locuteur entend communiquer à l'allocutaire ou aux allocutaires dans son message, et à linéariser la hiérarchie des structures syntaxiques.

[5] *Original text:* L'intonation est une partie de la prosodie qui est un ensemble constitué de l'accentuation, de l'intonation et du rythme.

[6] *Original text:* Nous entendons *accentuation* au sens de Garde (1965) : l'accentuation est constituée exclusivement de l'accent lexical, trait du morphème et des propriétés qui lui sont attachées.

[7] As mentioned in §7.1.3, several competing analyses of the prosodic systems of Japanese dialects have been proposed (for a review, see Kawahara 2015). The concept of tonal accent, also called pitch accent, has been presented by Hyman (2009) as a typical example of how *not* to do prosodic typology: the argument is that "alleged pitch-accent systems freely pick-and-choose properties from the tone and stress prototypes, producing mixed, ambiguous, and sometimes analytically indeterminate systems which appear to be intermediate". However, describing pitch-accent systems as "intermediate" between stress and tone does little to clarify the issue.

registers, as in Mon-Khmer languages (Ferlus 1979, Thongkum 1988, Brunelle & Kirby 2016, Tạ et al. 2022, Brunelle et al. 2022), as well as nasality when its domain extends to an entire morpheme (e.g. in Émérillon (Teko), a Tupi-Guarani language; Rose 2002). Since these features serve to encode lexical contrasts, they meet the criterion for phonemic status. This is why some authors refer to lexical tones as *tonemes* (Pike 1948), a term that links them to the established analytical tools of phonemic theory, such as the concept of *allotones* to designate tonal variants, like the allophones of segmental phonemes (on the widespread phenomenon of allotonic variation, see, in particular, Ternes 2006, Jordheim & Ophaug 2015) and the concept of *tonotactics* as an extension of *phonotactics* (Zerbian & Kügler 2023: 365).

Since Rossi's concept of *stress* has been broadened in this way, its designation should also be changed. To refer to morpheme-level phonological entities that carry lexical contrasts, we propose the term *lexical prosodic properties*. This more inclusive designation extends the typological reach of the framework to languages in which lexical prosodic properties are not accentual.

This set of definitions allows for a typological contextualization of Pierre Delattre's characterization of intonation:

> Intonation is one of the prosodic phenomena. These phenomena are also known as suprasegmental features because they do not relate separately to 'segments', i.e. vowels and consonants, but to words and sense groups. The other prosodic phenomena are stress ([in French:] final stress, emphatic stress), rhythm, syllabification, and pauses. (Delattre 1966: 1-2)
>
> *Original text:* L'intonation est un des phénomènes prosodiques. Ces phénomènes sont aussi connus sous le nom de traits suprasegmentaux parce qu'ils ne portent pas séparément sur les « segments », les voyelles et les consonnes, mais sur les mots et les groupes de sens. Les autres phénomènes prosodiques sont l'accent (accent final, accent d'insistance), le rythme, la syllabation, et la pause.

Delattre includes under the broad label of "stress" the phenomenon of group-final stress in French – which is not actually lexical stress but a boundary marker signalling junctures in the utterance. In the diagram in Figure 8.1, by contrast,

Hyman's questioning of pitch accent as a typological category could be extended to tone, in view of the considerable heterogeneity of prosodic systems described as "tone systems" (see Brunelle & Kirby 2016 and discussion in §12.1.1). In the current state of prosodic typology, it is not obvious that there is much to gain from prohibiting tonal accent as a descriptive label.

this phenomenon (which belongs to *phrasing*: constituent segmentation) is considered part of *intonation*, not of *lexical prosodic properties*. Delattre also includes emphatic stress under this broad "stress" category, yet emphatic stress is, likewise, not lexical, and is treated here as part of the pragmatic dimension of intonation: prominence (see §9.2.1). Placing lexical stress, final stress (group stress), and emphatic stress in three separate sub-divisions of prosody, as proposed in Figure 8.1, may seem counter-intuitive to scholars accustomed to viewing them as different types of a single phenomenon. The reluctance is likely to be especially strong among French-speaking linguists, since in French, the term *accent* appears in the designation of all three (*accent lexical, accent de groupe, accent d'insistance*). One common objection is that such a separation "would run the risk of rendering opaque the strong entanglement that exists between intonation and stress in French" (Lacheret-Dujour & Beaugendre 1999: 13).[8] However, while these phenomena are indeed closely interwoven, their interactions do not undermine the functional distinction between elements with a lexical contrastive function and those without. It would thus seem unwise to abandon this distinction.

These adjustments lead to the representation of the components of prosody shown in Figure 8.1, further refining the model proposed in Vaissière & Michaud (2006) (a study on prosodic constituents that focused on French). Rossi's notion of *rhythm* is replaced here with the more general term *performance factors* (Grosjean et al. 1979, Kohler 2010), which subsumes rhythm. The representation in Figure 8.1 builds on earlier proposals that, in my view, are fundamentally similar in their essentials (Coustenoble & Armstrong 1937, Delattre 1965, Martin 1977, Rossi 1999). In this framework, prosody is conceived as comprising three major components: (i) lexical prosodic properties, (ii) intonation, and (iii) performance factors, including rhythm.

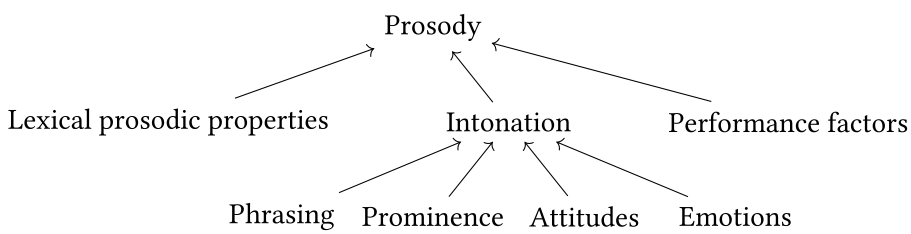


Figure 8.1: A schematic representation of the components of prosody.

[8] *Original text:* … risquerait de rendre opaque l'intrication forte qui existe entre l'intonation et l'accentuation en français.

It should be emphasized that the tree-like diagram in Figure 8.1 is a highly simplified representation. It does not attempt to capture the subtle interconnections among the various components of prosody, e.g. between rhythm and lexically distinctive suprasegmentals. Its purpose is simply to provide a visual summary of the core distinctions made here. The representation in Figure 8.1 highlights the fact that intonation is a complex, abstract structure. Its four components can be grouped into two sets:

- Two sub-systems of linguistic structure: syntactic intonation *(phrasing)*, which primarily reflects syntax in the broad sense, and pragmatic intonation *(prominence)*, which encodes information structure;

- Attitudinal and emotional dimensions, which convey speaker attitudes and emotions.

Speech phrasing and pragmatics are so central to intonation that it may be tempting to define intonation exclusively in terms of these functions. As Rossi puts it:

> Thus, the role of a grammar of intonation would be to determine the structure of intonational categories and units linked in one way or another to syntax and semantics-pragmatics, in other words to identify the intonational forms governed by higher cognitive devices. (Rossi 2001: 105)
>
> *Original text:* Ainsi une grammaire de l'intonation aurait pour rôle de déterminer la structure des catégories et des unités intonatives liées d'une façon ou d'une autre à la syntaxe et à la sémantique-pragmatique, en d'autres termes d'identifier les formes intonatives gouvernées par les dispositifs cognitifs supérieurs.

Indeed, one might argue that removing the expression of attitudes and emotions from Figure 8.1 would not fundamentally alter the conceptual framework used here. Yet these last two components are essential to a comprehensive understanding of intonation. Intonation, as Bolinger famously described it, is a "half-tamed savage" (Bolinger 1978: 475). *Phrasing* belongs to the tamer, more intellectual, more structured domain; it is most clearly manifested in deliberate oral renderings of carefully composed texts. *Prominence*, though still describable within a linguistic system and exhibiting clear cross-linguistic differences, is a less domesticated dimension of intonation: its stronger manifestations can at times interfere with phrasing as dictated by syntactic structure. Finally, the expression of attitudes and emotions transcends linguistic structure to some extent and can partly be accounted for in ethological terms, for instance through the Frequency Code:

> … an innately specified 'Frequency Code' (…) associates high acoustic frequency with the primary meaning of 'small vocalizer' and thus secondary meanings as 'subordinate, submissive, non threatening, desirous of the receiver's goodwill, etc.' and associates with low acoustic frequency the primary meaning of 'large vocalizer' and such secondary meanings as 'dominant, aggressive, threatening', etc. (Ohala 1984: 1)

Still on issues of terminological clarification, the phrase "syntactic intonation" may appear to be somewhat of a misnomer, insofar as intonational phrasing does not stand in a strict, one-to-one relationship with syntactic units. This has been recognized since the early classics of phonetics (Grammont 1933) and has been further confirmed by later research (Selkirk 1972; 2000: 231; Martin 1981). Nonetheless, the phrase "syntactic intonation" is retained here, given that knowledge of a sentence's syntax provides a sufficient basis for the synthesis of an acceptable fundamental frequency contour (Vaissière 1971).

Rossi is led to consider that "the primitives of intonation are two-sided signs: intonational morphemes" (Rossi 2001). This notion of *intoneme* is reminiscent of Delattre's reflections, which distinguish between form and function: it is necessary to analyze "on the one hand the significant oppositions that rely on intonation, and on the other the shape of the intonation curves" (Delattre 1966: 1).[9] In terms of form, the acoustic correlates of prosody are manifold. They include variations in fundamental frequency, duration, and intensity, as well as phonation type, but also allophonic variation in the realization of phonemes. Thus, prosody has correlates at the respiratory (i.e. subglottal) level, the glottal level, and the supraglottal level (see Erickson 1998). All these parameters contribute to prosody simultaneously, to varying degrees.

8.4.2 Tone sandhi, morphotonology, and tonal morphology

The concepts of 'tone sandhi', 'morphotonology', and 'tonal morphology' also require clarification. *Tone sandhi* refers to phonologically specified tone changes that apply automatically within a given phonological domain. In other words, *tone sandhi* refers to categorical tone change conditioned by phonological context. The seven tonal rules of Yongning Na presented in §7.1.1 (reproduced below) are tone sandhi rules in this sense, as they govern adjustments among neighbouring syllables within a given phonological domain (the tone group).

[9] *Original text:* « … d'une part les oppositions significatives qui reposent sur l'intonation, de l'autre la forme des courbes d'intonation ». The phrase 'intonation curves' might suggest that Delattre confines the phonetic dimension of intonation to fundamental frequency, which can be visualized as a curve. However, a close reading his article – which is highly recommended – makes it clear that this is not the case.

Rule 1: L tone spreads progressively ("left-to-right") onto syllables unspecified for tone.

Rule 2: Syllables that remain unspecified for tone after the application of Rule 1 receive M tone.

Rule 3: In tone-group-initial position, H and M are neutralized to M.

Rule 4: The syllable following an H-tone syllable receives L tone.

Rule 5: All syllables following an H.L or M.L sequence receive L tone.

Rule 6: In tone-group-final position, H and M are neutralized to H if they follow an L tone.

Rule 7: If a tone group only contains L tones, a postlexical H tone is added to its last syllable.

Morphotonology refers to categorical change in tone governed by rules that are specific to particular syntactic phrase types, such as object plus verb, subject plus verb, numeral plus classifier, and so on.

Finally, *tonal morphology* refers to grammatically specified tone changes marking certain morphosyntactic categories. The phrase "tone cases" is used in descriptions of some Bantu languages, such as UMbundu (Schadeberg 1986), where a noun's tone varies according to its case. The addition of a specific tone to a verb can express a given tense-aspect-modality category.

In Yongning Na, there is no tonal morphology in the sense of tonal morphemes. The floating High tone of Yongning Na is not a morpheme (i.e. a form-meaning pairing): it constitutes one of the lexical tone categories of nouns. Consequently, only the first two of the three concepts outlined above – sandhi, morphotonology, and tonal morphology – are relevant to the present study.

Some authors adopt "a broad and loose usage of the term" *sandhi*, encompassing both sandhi and morphotonology as defined here (Chen 2000: x). While such an approach may be suitable for large-scale language surveys (such as the survey of Chinese dialects in Chen 2000), distinguishing between sandhi and morphotonology is crucial in the present study. The reason for undertaking a book-length description is that tonal alternations in Yongning Na are not purely phonological: alongside sandhi (as defined here, as a phonological phenomenon), the language exhibits a wide range of syntactically restricted tone rules – i.e. morphotonology.

Based on these terminological clarifications, the next chapter (Chapter 9) finally turns to intonation in Yongning Na.

9 Intonation in Yongning Na: Description and analysis

This chapter presents observations on intonation in Yongning Na, with particular attention to its interaction with tone.

The approach taken in this volume has been to set out the phonological and morphophonological facts first, before turning to intonation. In practice, tones and intonation need to be addressed together as a matter of course, since they share an all-important phonetic correlate: voice fundamental frequency (F_0). This connection is particularly evident in Na, where tones are specified solely in terms of pitch. In Na, unlike in various other East and Southeast Asian languages, tones do not have length or phonation-type characteristics as part of their phonological definition (as discussed in §12.1.1).

The structure of this chapter is as follows: §9.1 examines *phrasing*, also referred to as *syntactic intonation*, while §9.2 focuses on intonational *prominence* (the conveyance of information structure), following a major division within intonation – as shown in the diagram in Figure 9.1, reproduced from the preceding chapter.

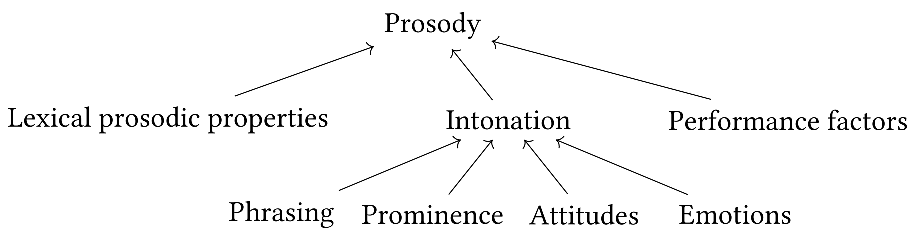


Figure 9.1: A schematic representation of the components of prosody.

The interaction between tone and intonation is then discussed in §9.3. On this basis, §9.4 addresses the phonetic implementation of tone. Next, §9.5 recapitulates the various observations about tone implementation by illustrating them through examples of mistaken notations from early field notes. Lastly, by way of conclusion, §9.6 sets out prospects for future experimental phonetic modelling.

9.1 Syntactic intonation: Phrasing and junctures

The most important unit in the prosodic organization of Na speech is the tone group, examined in Chapter 7, where much information relevant to phrasing has already been set out in detail. From a phonological point of view, tonal processes operate independently within each tone group. From the point of view of intonation, however, the picture is more nuanced. On the one hand, there is a degree of independence between tone groups; in particular, reference values of F_0 do not match across tone-group boundaries, as detailed in the next section. On the other hand, tone groups are integrated into higher-level prosodic units, characterized by intonational phenomena extending over several successive tone groups (§9.1.2).

9.1.1 Gaps in reference F_0 values for tones at junctures between tone groups

Full resetting of F_0 does not take place at each boundary between two tone groups, but there can be a sizeable *discrepancy in reference values for tones* across such boundaries, such that an M tone in final position within a tone group and another M tone in initial position within the following tone group can have widely different fundamental frequency values.

An illustration is found in the elicited sentence (1), where the M tones of /**hu˧mi˧**/ 'stomach' are realized distinctly lower than those of /**ʐwæ˧zo˧=bv̩˧**/ 'of (a/the) horse'.

(1) ʈʂʰɯ˧ | ʐwæ˧zo˧=bv̩˧ | hu˧mi˧ ɲi˥.
ʈʂʰɯ˥ ʐwæ˧zo#˥ =bv̩˧ hu˧mi˥$ ɲi˩
DEM.PROX colt POSS stomach COP
'This is the stomach of a colt.' *(DetermCompounds7.140)* {0004456#W140}

In (1), the M tones in the final tone group precede an H tone, and are coarticulated accordingly: with a lowered pitch (see §9.4.3).

However, successive tone groups are not entirely independent from the point of view of phonetic implementation. Rather, they are part of higher-level prosodic units characterized by intonational phenomena spanning several successive tone groups.

9.1.2 Above the tone group: Sentence and prosodic paragraph

A first useful cross-linguistic descriptive concept in this space is the *sentence* – also referred to as the *utterance*, a term that aims to bring out its grounding in a communicative context, as emphasized by Culioli (1995). No attempt will be made here to define it in any elaborate way. Let us simply note that, at sentence level, the neutral affirmative sentence is the basic, archetypal pattern from which other sentence modes depart (Thorsen 1980). Abstracting away from other dimensions of prosody (such as lexical prosodic properties, prominence, and the expression of attitudes and emotions), the F_0 curve of a sentence typically rises to a peak located on one of its first syllables. The gradual, noncategorical decrease in fundamental frequency known as 'declination' takes place over the course of the sentence. Fundamental frequency thus fluctuates within a gradually narrowing range, until final lowering, which signals the end of the sentence. (This corresponds to Tune 1 as described for English by Armstrong & Ward 1926.)

Final lowering is a more localized phenomenon, typically affecting the last syllable of declarative utterances, but this is subject to cross-language variation.

> Final lowering can be total, leading to the neutralisation of High and Low tones, as in Akan. It can be realized as a low register plateau, as in Chichewa or in Bemba (for the long stretched one), or as a gradual fall, as in Embosi, or as a sharp fall, as in Bemba (for the one syllable one). Its domain can be short (one syllable) or rather long (a stretch of tone bearing units). (Downing & Rialland 2016: 4-5)

Declination and final lowering are common across languages, as is their suspension to convey non-assertiveness (in questions, or to convey nuances of doubt and uncertainty).

Studies of read speech in languages such as English or French note that the peak F_0 value in a sentence tends to decrease from the first to the last sentence in a paragraph (Lehiste 1975). Paragraph-final position is typically characterized by an extra-low F_0, often accompanied by a change in phonation type and reduced intensity (Vaissière & Michaud 2006: 50–52). On the analogy of written text, it has been proposed to refer to *prosodic paragraphs* as a unit above sentence level in spontaneous speech (Grobet & Simon 2001). The term 'paragraph' has been criticized by linguists who object to applying concepts from written text analysis to spoken language. Nevertheless, it remains a convenient label, in view of the similarities between the division of a written text into paragraphs and that of speech into prosodic paragraphs – both of which allow for a high degree of stylistic freedom.

From a phonetic point of view, prosodic phenomena at the levels of the sentence and the prosodic paragraph interact with tones. The phonetic realization of a phonological H, M or L differs considerably depending on the position of the syllable within the sentence and the prosodic paragraph. For instance, in Yongning Na, the 1st, 2nd, and 3rd-person pronouns //**njɤ˩**//, //**no˩**//, and //**ʈʂʰɯ˥**// all have M tone in isolation due to the neutralization – in this position – of the lexical categories L, M, and H. Their surface phonological representation in isolation is therefore /**njɤ˧**/, /**no˧**/, and /**ʈʂʰɯ˧**/, respectively. But in my first notes I transcribed them with H tone, as [**njɤ˥**], [**no˥**], and [**ʈʂʰɯ˥**]. This was due to their realization on a high pitch in isolation. In that context, their pitch is noticeably higher than that of an M tone occurring in tone-group-initial position later within a prosodic paragraph.

Thus, in my first field notes, I transcribed (2) as ‡ **njɤ˥ mɤ˧-dzɯ˥**; I was led to this phonologically inappropriate notation by the considerable phonetic difference in pitch between the first and second syllables, as well as the similarity in pitch between the first and last syllables of this short sentence. The effects of phrasing and coarticulation had to be brought to light before a correct transcription of the surface phonological form of the pronouns could be arrived at.

(2) njɤ˧ | mɤ˧-dzɯ˥.
njɤ˩ mɤ˧- dzɯ˥
1SG NEG to_eat
'I don't eat.'

9.2 Pragmatic intonation

"Information structure is a vast topic of research that has been pursued within different theoretical frameworks" (Krifka 2008: 244), and with different objectives in view. A few general observations about information structure in Yongning Na will be followed by a discussion of two phenomena that belong to *pragmatic intonation*: emphatic stress (§9.2.1) and focalization through local intonational modification of tone (§9.2.2).[1]

The following observations about Qiang (Rma) also apply to Yongning Na (and other Sino-Tibetan languages):

> The structure of the clause is to some extent affected by pragmatic factors, but this only applies to the order of noun phrases in the clause. The utterance-initial position is the unmarked topic position (though secondary topics can follow the primary topic), while the position immediately before the verb is the unmarked focus position, and so the focused

[1]This section contains passages adapted from Michaud & Brunelle (2016).

> element will generally appear there. The verb always appears in final position; there is no possibility for the actor of a clause to appear in postverbal position, even if it is focal. The only exception to this is the occasional afterthought clarification of a noun phrase that was omitted or expressed as a pronoun in the clause. (LaPolla & Huang 2003: 221)

Information structure in Na has accordingly been described as 'topic-comment', extending an observation made by Chao Yuen-ren about Sinitic: "the grammatical meaning of subject and predicate in a Chinese sentence is topic and comment, rather than actor and action" (Chao 1968: 69; see also Shi 2000; LaPolla 2009).

> The primary information structure in Na is topic/comment rather than subject/predicate. (…) [A] topic can be a nominal argument, which the rest of the sentence will comment upon, but the topic can also be an adverbial, an independent clause, or a dependent clause. (Lidz 2010: 296)

Example (3) provides an illustration.

(3) le˧-dzɯ˥ | bi˧mi˧ go˩.
le˧- dzɯ˥ bi˧mi˧ go˩$_a$
ACCOMP to_eat stomach/belly to_ache
'[If] [you] eat [of it], [your] stomach [will] hurt!' (Field notes. Context: on the mountain, pointing out a berry that is not edible.)

9.2.1 Emphatic stress and its toned-down avatars

An 'up' arrow ↑ placed to the left of a syllable indicates emphatic stress, as in (4).

(4) tʰi˧˥, | ə˧mv̩˧ ʝi˥-hĩ˩ | -dʑo˩, | ↑zo˧ ɲi˥ tsɯ˩ ◊ mv̩˩.
tʰi˧˥ ə˧mv̩˧˥ ʝi˥ -hĩ˥ dʑo˥ zo˥ ɲi˩ tsɯ˧˥ mv̩˧
then elder_sibling to_do REL TOP boy/son COP REP AFFIRM
'The elder [of the two siblings] was a boy.' *(Sister.5)* {0004341#S5}

In many contexts, emphatic stress appears on a constituent that can be predicted to receive focus prosody. For example, in sentence (4), one would expect the focus to be in the immediate preverbal position – the usual unmarked focus position in verb-final languages. Emphatic stress can thus be viewed as an extreme form of focus prosody – an extreme along a continuum: there is no hard-and-fast boundary between emphatic stress and milder realizations of focus prosody.

Emphatic stress presents an interpretative ambiguity: it may signal either "broad focus" or "narrow focus", to use terms proposed by Lambrecht (1994). This phenomenon is extensively studied in the literature on focus projection (e.g. Selkirk 1995).

The phonetic realization of emphatic stress includes effects on the articulation of vowels and consonants. For instance, in (5), the second syllable of /**dʑɤ˩↑bv̩˥**/ 'to play' is realized with stronger trilling of the /**b**/ than is found in non-emphatic contexts. This can be considered an example of 'articulatory prosodies' in the sense of Kohler & Niebuhr (2011) and Niebuhr (2013).

(5) mv̩˩zo˩=ɻæ˩ lɑ˥ | ə˧ʝi˧-ʂɯ˥ʝi˩ | tʰi˦˥, | dʑɤ˩↑bv̩˥-ɲi˩ tsɯ˩ ◊ mv̩˩!
mv̩˩zo˩ =ɻæ˩ lɑ˧ ə˧ʝi˧-ʂɯ˥ʝi˩ tʰi˦˥ dʑɤ˩bv̩˥ -ɲi˩ tsɯ˧˥
girl ASSOCIATIVE with long_ago so/then to_play CERTITUDE REP
mv̩˧
AFFIRM
'The story goes that at that time, long ago, he would have fun [i.e. flirt] with girls!' *(Caravans.231)* {0004530#S231}

Emphatic stress in Na appears to be essentially the same phenomenon as in English and French, motivating the adoption of this label (proposed by Coustenoble & Armstrong 1937). Prototypical realizations of emphatic stress have been shown to involve supplementary activity of the expiratory muscles, resulting in a sudden rise in subglottal pressure during the articulation of a consonant (Benguerel 1973, Carton et al. 1976, Ohala 1978b, Fant et al. 1996), hence the term "force-accent" used by Kohler (2003).

This category has been somewhat neglected in intonation studies, as the dominance of tonal models of intonation led researchers to focus their attention mostly on the acoustic parameter of fundamental frequency. But it is a good candidate for the status of universal of human language. It is the most extreme manifestation of intonational emphasis, with linguistic functions ranging from attitudinal and emotional to pragmatic.

In speech, toned-down realizations of emphatic stress are more common than its prototypical realization: the physiological effort at the subglottal level is often mimicked through such strategies as F_0 excursions and consonant lengthening. Needless to say, the description of emphatic stress as a language universal by no means implies that it is uniformly present in all languages and all oral genres. Like other linguistic phenomena, it comprises important language-specific and speech-style-specific dimensions. Its frequency of use varies greatly from language to language, from speaker to speaker, and from style to style; its stylistic effect is inversely proportional to its frequency of use.

9.2.2 Focalization through local intonational modification of tone

In Yongning Na, there can be focalization through local intonational modification of tone on the last syllable of the word in focus. The syllable gets lengthened. An H or M level tone changes to a dipping contour, while the phonetic range of an MH or LH rising contour gets expanded. As for L tone, which is canonically realized with a decrease in fundamental frequency, its falling contour becomes more noticeable under focalization. This phenomenon will be referred to, for short, as *intonational focalization.*

The notation adopted is '**F**' (for 'Focalization'), placed after the syllable at issue. This may seem inconsistent with the choice to place the upward arrow for emphatic stress (↑) *before* the stressed syllable (§9.2.1). There is a phonetic basis for this different treatment, however: while emphatic stress is strongest at the *onset* of the syllable, intonational focalization primarily affects the syllable *rhyme.*

This phenomenon was first identified in examples where the syllable receiving this intonational modification carried H tone. The modification of H tone is more conspicuous than that of M, L, MH or LH: its realization becomes a rapid dipping contour, for instance in examples (6) and (7).

(6) hĩ˧-ki˧ | ɖɯ˧-kʰwɤ˥ F | mɤ˧-pi˥!
hĩ˥ -ki˧ ɖɯ˧-kʰwɤ˥\$ mɤ˧- pi˥
person DAT 1-CLF.pieces NEG to_say
'(S)he did not say anything to the people present! / (S)he did not greet anyone!' (Field notes, 2009)

(7) no˧ | njɤ˧-ki˧ | ɖɯ˧-sɑ˥ F | hwæ˧-mɤ˧-zo˧!
no˩ njɤ˩ -ki˧ ɖɯ˧-sɑ˥\$ hwæ˧$_{a}$ mɤ˧- -zo˧$_{a}$
2SG 1ST DAT 1-CLF.things to_buy NEG OBLIG
'You don't need to give me anything! / There is no need to buy any presents for me!' *(TraderAndHisSon.34)* {0004632#S34}

In (6), the numeral-plus-classifier phrase /**ɖɯ˧-kʰwɤ˥**/ 'one piece' is given prominence through the phonetic realization of the classifier /**kʰwɤ˥**/ with a noticeable fall plus lengthening. Likewise for /**ɖɯ˧-sɑ˥**/ 'anything' (literally 'one thing') in (7). A spectrogram illustrating this phenomenon is shown in Figure 9.2.

This phenomenon is found in rapid speech as well as in slow repetitions.

It was later realized that the same type of prominence-lending local intonational modification also applied to the two rising contours: high-rising (MH) and low-rising (LH). For these contours, the effect consists of lengthening and F_0

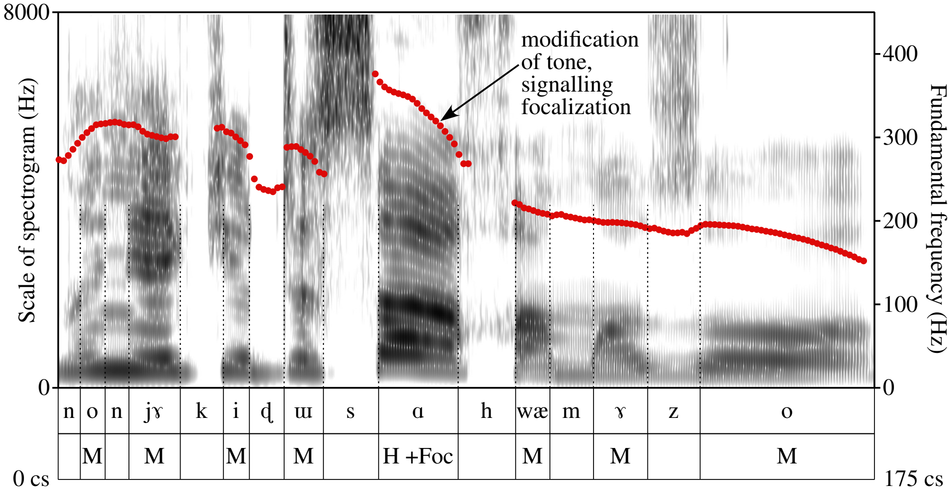


Figure 9.2: Spectrogram and F_0 curve for example (7), showing a noticeable fall plus lengthening on the syllable /**sɑ˥**/ in /**ɖɯ˧-sɑ˥**/ 'anything'. Top line of annotation: segments; bottom line: tones, with the added mention '+Foc' for the syllable receiving intonational focalization.

range expansion. Since these are already phonological contours, which have inherently greater duration than simple levels, the modification is less salient perceptually than for M and H. Recognition of the existence of this phenomenon for the L level came last.

On the analogy of Naxi, where a reduced M- or L-tone syllable can give rise to contours (Michaud & He 2007) that bear some phonetic similarity to those of Yongning Na, it was first hypothesized that a reduced syllable must be present in examples such as (6): ‡ **ɖɯ˧-kʰwɤ˥-ə˩ mɤ˧-pi˥**. Likewise, example (8) was initially transcribed as ‡ **ə˧-dzɤ˧~dzɤ˥-ə˩ dzɯ˧**. Subsequent investigation showed that these notations were inappropriate, however. The observed dip in fundamental frequency, accompanied by lengthening, is a purely intonational device, not an indication of an additional syllable.

(8) ə˧-dzɤ˧~dzɤ˥ F | dzɯ˧!
ə˧-dzɤ˧~dzɤ˥ F dzɯ˥
slowly FOCALIZATION to_eat
'Eat slowly!/Take your time!' (Polite invitation to eat.)

The realization of focalization comprises a movement in F_0, a slight lengthening, and perhaps a slight change in the vowel, as if a final *schwa* target were added.

This process is specific enough for tonal distinctions to be preserved: headlong conflict with the underlying tone system is avoided. To put it differently, intonational focalization does not lead to neutralization of lexical tone. Similarly, emphatic stress remains perceptually distinct from lexical tone, as it is cued by various phonetic dimensions beyond fundamental frequency. The possibility of a misperception of lexical tone caused by intonational phenomena is therefore limited.

9.2.3 Borderline cases

About two hundred instances of intonational focalization are indicated in the first twenty transcribed texts. Some cases are clearer than others. For one thing, cases where this focalization is superimposed on rising contours, and on L tones, tend to be less salient perceptually for phonetic reasons, as mentioned above. For another, there are borderline cases, where it is not obvious whether to add a ‘**F**’ in the transcription or not. Borderline cases do not by themselves cast doubt on the categorical nature of the phenomenon. Intonational focalization can be toned down, in the same way as emphatic stress has toned-down avatars shading into non-emphatic realizations; this is a common situation in the field of intonation. On the other hand, it should be borne in mind, when consulting the texts, that the ‘**F**’ and ‘↑’ symbols (for intonational focalization and emphatic stress, respectively) cannot be assigned with the same degree of certainty as consonants, vowels, and tones. For instance, the syllable /**li˩**/ ‘to see’ is transcribed in (9)[2] with an indication of intonational focalization, on the basis of the auditory impression that it stands out in the flow of speech. The proposed translation for /**hĩ˧ li˩ F mɤ˩-ʁo˩**/ in (9) is ‘… could not *even* see people’, chosen to reflect the pragmatic implications of focalization on the verb ‘to see’. The sequence /**hĩ˧ li˩ mɤ˩-ʁo˩**/, without intonational focalization, would simply mean ‘… could not see people’.

(9) le˧-mo˩, | le˧-mo˩, | njɤ˩ɭɯ˧ | ʈʂʰɯ˧ne˧ gv̩˧˥, | hĩ˧ li˩ F mɤ˩-ʁo˩!

le˧-	mo˩$_{a}$	njɤ˩ɭɯ˧	ʈʂʰɯ˧ne˧-ʝi˥	gv̩˧$_{c}$	hĩ˥	li˧$_{a}$
ACCOMP	old	eye	thus	to_occur	person	to_look_at
F	mɤ˧-	ʁo˧				
FOCALIZATION	NEG	to_be_able_to/to_manage				

‘[The dog] got older and older; and so, its eyes could not even look at people anymore / it could not even see people anymore!’ *(Dog2.80)* {0004555#S80}

[2] In (9), the second and third syllables of /**ʈʂʰɯ˧ne˧-ʝi˥**/ ‘thus’ are coalescent, and fully undistinguishable on the spectrogram (Figure 9.3). For convenience, a simplified transcription as [**ʈʂʰɯ˧ne˧ gv̩˧˥**] is adopted in the surface phonological transcription of the sentence, and in the annotation to the spectrogram, in preference to /**ʈʂʰɯ˧ne˧-ʝi˧ gv̩˧˥**/. This case of coalescence is analyzed in Appendix A, §A.3.1.

But looking at the spectrogram in Figure 9.3, it appears that the syllable /**li**˩/ 'to see' is not strikingly different from its neighbours in terms of either duration or fundamental frequency. The average fundamental frequency of vowel /**i**/ in /**li**˩/ is two semitones lower than that of /**ĩ**/ in the preceding syllable, /**hĩ**˧/ (210 Hz vs. 237 Hz), a phonetic difference that is in keeping with the phonological tones (M vs. L). The decrease in fundamental frequency between /**li**˩/ and the two L-tone syllables that follow can be interpreted as a straightforward case of *declination*, a phenomenon mentioned in §9.1.

The data is nonetheless compatible with the hypothesis (based on auditory impression) that the syllable /**li**˩/ 'to see' is highlighted by intonational means. In terms of duration, the syllable /**li**˩/ is 20 centiseconds long, which is slightly above the average for the passage shown in Figure 9.3 (100 centiseconds for seven syllables, i.e. about one syllable every 15 centiseconds). This difference in length

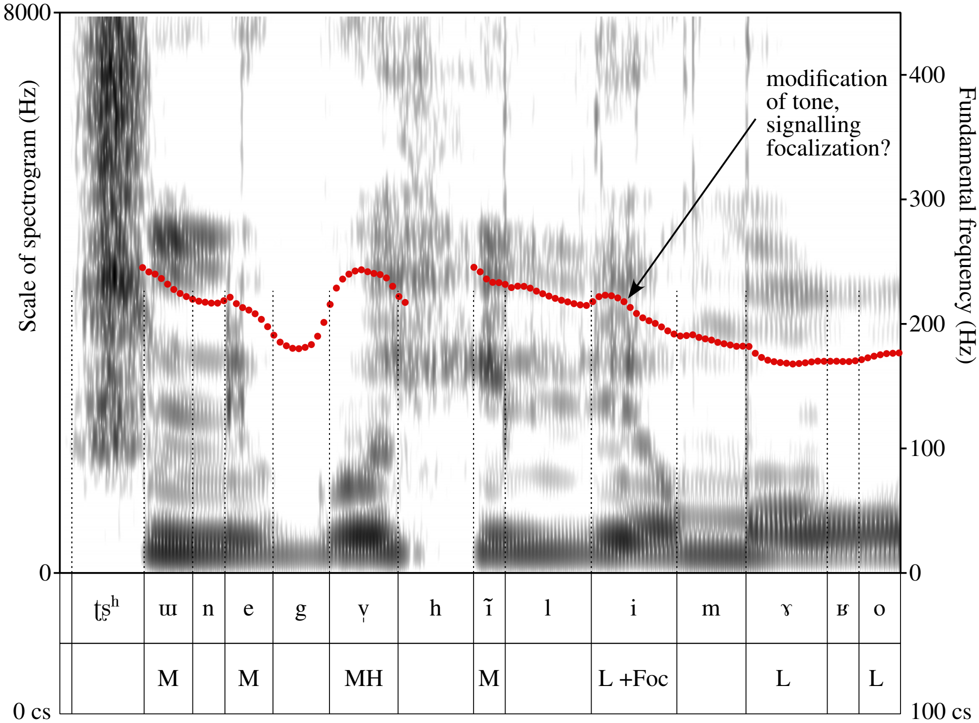


Figure 9.3: Spectrogram and F_0 curve for example (9), showing a slight fall plus hints of lengthening on the syllable /**li**˩/ 'to see'. Top line of annotation: segments; bottom line: tones, with the added mention '+Foc' for the syllable presumed to receive intonational focalization.

cannot be dismissed as linked to the syllable's phonemes: if anything, the vowel /**i**/ would be expected to be intrinsically *shorter* than other vowels (Di Cristo & Hirst 1986, Whalen et al. 1998). As for fundamental frequency, the display in Figure 9.3, covering the range from 0 Hz to 450 Hz, gives the impression of a relatively smooth curve over the last four syllables, but if the scale is changed to a narrower range, as in Figure 9.4, the hump at the beginning of the vowel /**i**/ in /**li**˩/ becomes more salient visually. This hump, which results in a dip of 2.5 semitones in the brief course of this vowel, could be significant: there is no obvious contextual reason why the hump should be present, therefore it makes sense to interpret it as one of the phonetic correlates of a local intonational modification signalling some sort of emphasis, such as the phenomenon of intonational focalization transcribed here as '**F**'.

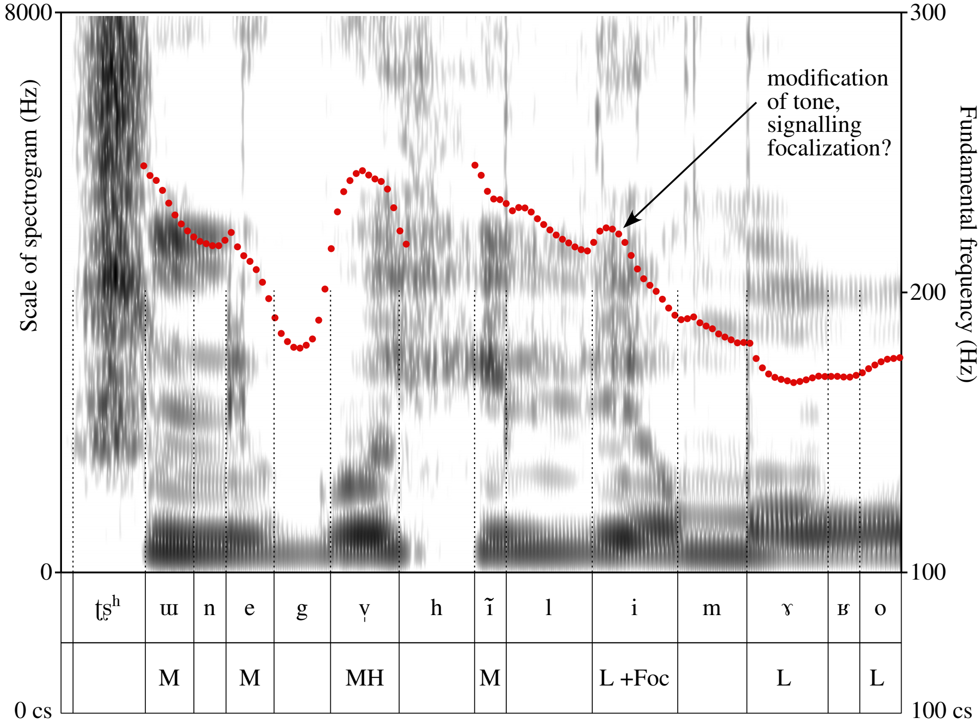


Figure 9.4: Same data as in Figure 9.3, adopting a narrower range for F_0 display.

When transcribing narratives, one way to check with the speaker whether to classify a given case as having intonational focalization or not consists in playing the passage at issue, then repeating it with and without the telltale fall in pitch and lengthened rhyme, and asking the consultant to indicate which of the two realizations fits better. But this entails no guarantee: even supposing that the investigator is successful in producing the intended distinction, the consultant may not base her decision on the realization heard on the recording. She may go for the realization which she considers (in retrospect) would have been more suitable in that context.

Pending experimental verification through perceptual tests, it seems likely that different speakers have different degrees of sensitivity to intonational detail. Toned-down versions of intonational focalization may go unnoticed by some speakers. It is a fact of life that hearers can fail to pick up intended clues. A linguist's aphorism has it that, in human communication, misunderstanding is the general case, and mutual understanding is a special case ("la compréhension est un cas particulier du malentendu": Culioli 1990: 39). The world is a cemetery of cultures, and every text a tomb for lost allusions.[3]

9.3 Tone and intonation: Superposition and interaction

The observations presented separately about two major components of intonation – phrasing in §9.1 and prominence in §9.2 – now call for an account of how they combine with each other, as well as with tone. It is all very well to distinguish different functional levels: in particular, tone on the one hand, and intonational modifications on the other. But since tone and intonation share a key phonetic parameter – fundamental frequency –, a phonetic issue is bound to arise: how is information from the different functional levels conveyed through the same acoustic channel?

The general approach adopted here is superpositional, in keeping with Chao Yuen-ren's intuition of tones as ripples riding over a larger wave (sentence intonation). The various components are conceived as superimposed: local modulations (reflecting tones, intonational focalization and emphatic stress, as well as phrasing) are grafted onto an overall sentence-level pattern that depends on

[3]Lost allusions are often staged in literary works, among them Proust's *In Search of Lost Time*. The grandmother's sisters design allusions that are so carefully veiled that they are unintelligible to the intended addressees: "… they, in their horror of vulgarity, had brought to such a fine art the concealment of a personal allusion in a wealth of ingenious circumlocution, that it would often pass unnoticed even by the person to whom it was addressed." Scott Moncrieff translation revised by Terence Kilmartin. *Original text:* « Celles-ci par horreur de la vulgarité poussaient si loin l'art de dissimuler sous des périphrases ingénieuses une allusion personnelle qu'elle passait souvent inaperçue de celui même à qui elle s'adressait. »

sentence mode and speaker attitude (and emotion) – with some phenomena operating over wider domains such as the *prosodic paragraph*. Thus, the analytical baseline consists in assuming that the functional distinction between lexical tones and intonation is preserved: that listeners process F_0 curves in such a way as to retrieve both tone and intonation, teasing them apart in perception, as it were.

Against this general background, a range of different situations are observed in practice. A compromise has to be found, in each speech act, between the competing demands of clarity, on the one hand, and expressivity, on the other. Depending on context, the relative weight of these two components can be very different. For instance, when picking up the phone, speakers of Mandarin say *wèi* 喂, lexically a tone-4 syllable, i.e. with sharply falling pitch; but in this context, the lexical tone can be overridden by interrogative intonation, and the pitch is often rising. One theoretically possible interpretation would be that the lexical tone, tone 4, is changed to another, say, tone 2, the rising tone. But rather than treating this case as an instance of tone change, it makes better sense to consider it as an extreme case where the lexical tone has so little communicational relevance, and the expression of speaker attitude such importance, that the end result bears little trace (if any) of the lexical tone.

As a general rule, it seems clear that Na speakers are careful to avoid too great a distortion of the tonal string due to intonational emphasis. While no specific phonetic study has so far been conducted on Yongning Na data, it seems reasonable to assume that the situation is comparable to Naxi, where a study of the three basic tones (H, M and L) under emphasis reveals a proportionally milder effect of emphasis on F_0 than on intensity, as compared to English data (Michaud 2005: 107–167; Michaud et al. 2015).

There are nonetheless cases in which pragmatic intonation interacts with tones to the point of tampering with the tonal string. Such cases are analyzed below. Rather than dismissing them as marginal, it seems more forward-looking to heed the suggestion of Marsault et al. (2024) and Dingemanse (2024: 258) to pay special attention to elements traditionally considered as peripheral – an approach that is seemingly iconoclastic but turns out to be highly heuristic.

9.3.1 Influence of emphatic stress on the tonal string

In some rare cases, intonational modifications go so far as to affect the string of tones for the utterance. In its most vehement manifestations, emphatic stress intrudes into a sentence's prosody, wreaking havoc on tonal contrasts. Example (10) is a case in point.

(10) pʰv̩˩-tɕæ˥ɻæ˩ gv̩˩-kv̩˩-ze˩ ◊ mæ˩!
pʰv̩˩-tɕæ˩ɻæ˥ gv̩˧ -kv̩˦ -ze˧b mæ˧
very_white to_become ABILITIVE PFV AFFIRM
'[after boiling, linen thread] can become really white!' *(FoodShortage.73)* {0004557#S73}

The usual pronunciation is /**pʰv̩˩-tɕæ˩ɻæ˥**/ 'very white'. In (10), the second syllable is realized phonetically with extremely high fundamental frequency on the syllable /**tɕæ˩**/, which is also considerably lengthened. From a phonetic point of view, its phonetic L tone is conspicuously disregarded. One way of looking at this modification would be to describe it as due to an intonational overlay: functionally, one could consider transcribing as /**pʰv̩˩-↑tɕæ˩ɻæ˥-gv̩˩**/, where the arrow ↑ indicates emphatic stress, and the phonological tonal string is unchanged.

This forcible intonational modification does interact with the phonological tone string of the tone group, however. If the modification of the second syllable in /**pʰv̩˩-tɕæ˩ɻæ˥-gv̩˩**/ only took place on an intonational level, one would expect the phonological tonal string to remain unchanged, in which case the third syllable would retain its phonological H tone. But what is observed is that the third and fourth syllables in (10) are lowered to L: /**ɻæ˩-gv̩˩**/. This is precisely what is expected if the second syllable carries H tone. This phenomenon is therefore analyzed as involving a categorical tone change, from an L.L.H sequence, /**pʰv̩˩-tɕæ˩ɻæ˥**/, to an L.H.L sequence, /**pʰv̩˩-tɕæ˥ɻæ˩**/.

This example illustrates how an expressive coinage possessing a lilt of its own undergoes continuous attraction from the more central structures that make up the language's phonological system, tending towards its integration into the language's phonological categories, and interacting with these categories in the process.

Another example where emphatic stress interacts with tone – in this case obscuring a syllable's lexical tone – consists of the set of eight extra-distal locative expressions, which share a specific intonation. The argument proposed here is that this habitual intonation constitutes a path towards the loss of lexical tone on the grammatical morphemes at issue. These locative expressions are listed in Table 9.1, where their first syllable is transcribed with an exclamation mark '!' instead of a tone mark, as a provisional device to represent their telltale intonation.

The two sets of extra-distal locatives presented in Table 9.1 are structurally parallel, and they share the same intonation. The realization of their first syllable is highly expressive and allows variants. Either it starts on an extra-high pitch and glides downward, at a slope left to the speaker's discretion: a sharp fall

Table 9.1: Extra-distal locative expressions carrying specific intonation.

locative	meaning	1^{st} σ	meaning of 1^{st} σ
gɤ!-qo˧	way up	**gɤ˩-**	upward
gɤ!-ʈʂʰɯ˧qo˧	way up there (PROX)	**gɤ˩-**	upward
gɤ!-tʰv̩˧qo˧	way up there (DIST)	**gɤ˩-**	upward
gɤ!-tʰv̩˧-gi#˥	way up in that direction	**gɤ˩-**	upward
dɤ!-qo˧	way over there	?	?
dɤ!-ʈʂʰɯ˧qo˧	way over there (PROX)	?	?
dɤ!-tʰv̩˧qo˧	way over there (DIST)	?	?
dɤ!-tʰv̩˧-gi#˥	way over in that direction	?	?

or a prolonged descent. Or it is rising, the details of the rise (duration, slope, and peak height) being again left to the speaker's discretion. As far as could be ascertained, the sharply falling variant emphasizes the great distance to the place at issue (paraphrase: 'in a place far, far away'), whereas the rising variant (exemplified in Figure 9.5) is used to direct the listener's attention to that place, against a background of shared knowledge (paraphrase: 'that faraway place, you know'). Consider (11), for instance: the context to this example is that a mother thinks that her daughter has died; the young woman's lover tells the mother that her daughter is still alive, and that she is in hiding. The young man knows where the young woman is, but does not wish to disclose it to the mother. The extra-distal locative in (11) does not emphasize the great distance: paraphrase as 'Your daughter is in a faraway place' would be inappropriate. In this context, the syllable /**dɤ**/ in **dɤ!-tʰv̩˧qo˧**/ 'way over there' is realized as rising: see the fundamental frequency tracing in Figure 9.5.

(11) no˧ mv̩˩ | dɤ!-tʰv̩˧qo˧ dʑo˩.
no˩ mv̩˩˧˥ dɤ!-tʰv̩˧qo˧ dʑo˩$_b$
2SG daughter DEM.EXTRA_DISTAL EXIST.animated_beings
'Your daughter is [in a place that I know,] way over there.'
(BuriedAlive3.105) {0004538#S105}

The perceived pitch of the syllable /**dɤ**/ in these locative expressions could be stylized by means of Chao Yuen-ren's tone letters for intonational notation, placed on the right hand-side of the vertical line that serves as a reference, hence [**dɤ꜒꜖-qo˧**] for a fall from the top to the bottom of the pitch range, [**dɤ꜖꜒-qo˧**]

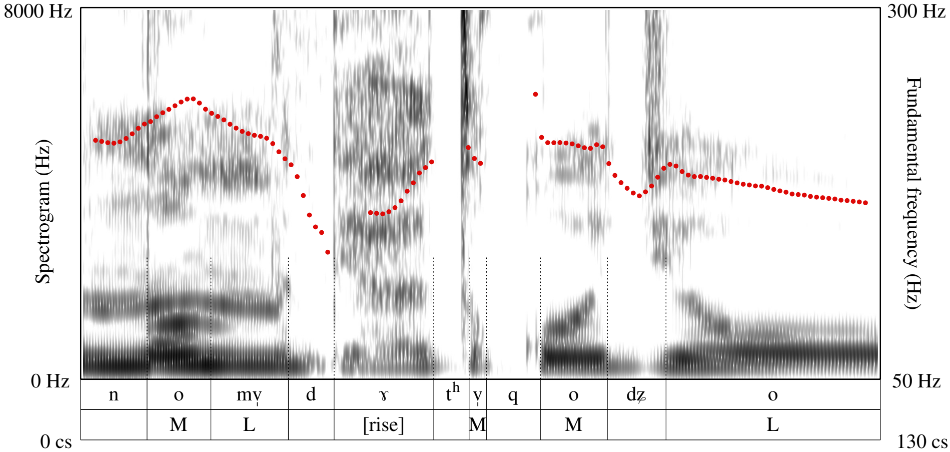


Figure 9.5: Spectrogram and F_0 curve for example (11), showing the rising realization of the syllable /**dɤ**/ in /**dɤ!-tʰv̩˧qo˧**/ ‘way over there’.

for a rise followed by a plateau, and [**dɤ˧˥-qo˧**] for a rise from mid-range to top of pitch range. However, a consistent choice of notations would require a full-fledged study of expressive phenomena in Yongning Na; this falls outside the scope of this exploratory chapter.

Identification of the prefix /**gɤ˩-**/ ‘upward’ in the locative expressions /**gɤ˩-qo˧**/, /**gɤ˩-ʈʂʰɯ˧qo˧**/, /**gɤ˩-tʰv̩˧qo˧**/ and /**gɤ˩-tʰv̩˧-gi#˥**/ of Table 9.1 is not problematic, as this prefix is attested elsewhere in the language, in a number of productive constructions, with the same meaning that it has in the extra-distal locatives (see §6.3.3). On the other hand, it is an issue how to analyze the extra-distal locatives /**dɤ!-qo˧**/, /**dɤ!-ʈʂʰɯ˧qo˧**/, /**dɤ!-tʰv̩˧qo˧**/ and /**dɤ!-tʰv̩˧-gi#˥**/, because the syllable /**dɤ**/ is synchronically orphaned. It is peculiar not only in its intonational realization, but also in its segmental composition. Apart from extra-distal locative expressions, this combination of initial and rhyme is only attested in Yongning Na in /**hṽ̩˧dɤ˧ɻ̩#˥**/ ‘clumsy’ and /**õ˧dɤ˧ɻ̩˧**/ ‘fundamental(ly)’. In both of these words, the syllable /**dɤ**/ is followed by /**ɻ̩**/; since rhotic sounds are known (cross-linguistically) to have especially strong coarticulatory effects (see for instance West 1999), it is far from implausible that the combination /**dɤ**/ in these two words results from the phonologization of phonetic coarticulation.

My best guess at the time of writing is that the syllable /**dɤ**/ in extra-distal locatives originates in an expressive deformation of the extra-distal morpheme //**dv̩˩**//. This morpheme appears in /**dv̩˩-tɕo˧**/ ‘that way, that direction’, an expression now almost fallen into disuse which is structurally parallel to /**ʈʂʰɯ˧-tɕo˧**/ ‘this way’ and /**tʰv̩˧-tɕo˧**/ ‘that way’. The expression /**dv̩˩-tɕo˧**/ alone is not enough to determine the tone of the extra-distal morpheme //**dv̩**// with certainty: it is necessary to observe a morpheme in several contexts of occurrence in order to arrive at its lexical tone, as explained step by step in the discussion of the lexical tones of nouns in Chapter 2, and verified in the discussion of grammatical words in Chapters 5–6. But the presence of an L tone on the first syllable in /**dv̩˩-tɕo˧**/ 'that way, that direction' suggests that the tone of the extra-distal morpheme /**dv̩**/ is likely to be L or LM, hence the provisional adoption of an internal reconstruction as *****dv̩˩**.

From a synchronic point of view, however, there is no evidence for positing an initial lexical L tone in the extra-distal locatives /**dɤ!-qo˧**/, /**dɤ!-ʈʂʰɯ˧qo˧**/, /**dɤ!-tʰv̩˧qo˧**/, and /**dɤ!-tʰv̩˧-gi#˥**/. This set has the same prosodic realization as the set consisting of /**gɤ!-qo˧**/, /**gɤ!-ʈʂʰɯ˧qo˧**/, /**gɤ!-tʰv̩˧qo˧**/, and /**gɤ!-tʰv̩˧-gi#˥**/ (shown in Table 9.1), but this realization is so distant from what one would expect on the basis of the lexical L tone of /**gɤ˩-**/ 'upward' that it looks like a case of neutralization of all tonal oppositions on the first syllable of these locatives.

While the synchronic data provides no evidence for one tonal notation rather than another, it also provides no evidence against adopting a notation of /**dɤ-**/ with L tone, as suggested by language-internal evidence. Therefore, notation as /**dɤ˩-**/ is provisionally adopted, rewriting Table 9.1 as Table 9.2.

Table 9.2: Extra-distal locative expressions carrying specific intonation.

locative	meaning	1st σ	meaning of 1st σ
gɤ˩-qo˧	way up	**gɤ˩-**	upward
gɤ˩-ʈʂʰɯ˧qo˧	way up there (PROX)	**gɤ˩-**	upward
gɤ˩-tʰv̩˧qo˧	way up there (DIST)	**gɤ˩-**	upward
gɤ˩-tʰv̩˧-gi#˥	way up in that direction	**gɤ˩-**	upward
dɤ˩-qo˧	way over there	***dv̩˩-**	*DEM.DIST
dɤ˩-ʈʂʰɯ˧qo˧	way over there (PROX)	***dv̩˩-**	*DEM.DIST
dɤ˩-tʰv̩˧qo˧	way over there (DIST)	***dv̩˩-**	*DEM.DIST
dɤ˩-tʰv̩˧-gi#˥	way over in that direction	***dv̩˩-**	*DEM.DIST

The considerable intonational modification of the initial syllable in these locative expressions is transcribed in narratives through the addition of the mark for emphatic stress '↑' (see §9.2.1). This mark does not tell the full story, but has the advantage of bringing attention to the presence of a strong intonational modification resulting in a gap between the hypothesized lexical tone and the phonetic realization. Such unstable situations hold special potential for reinterpretation and change.

Like emphatic stress, intonational focalization can become habitually associated with a word, again blazing a path towards the lexicalization of new tone patterns.

9.3.2 Influence of intonational focalization on the tonal string

This paragraph presents a case in which intonational focalization has become habitually associated with a phrase. The classifier /**sɑ˥**/ ‘thing’ only appears in the phrase /**ɖɯ˧-sɑ˥**/ ‘anything’, itself restricted to negative contexts: typical examples are shown in (12) and (13).

(12) ɖɯ˧-sɑ˥ F | mɤ˧-dʑo˧.

ɖɯ˧	sɑ˥	F	mɤ˧-	dʑo˧$_b$
one	CLF.things	FOCALIZATION	NEG	to_possess

‘[I/they] don’t have anything at all / [I/they] have nothing at all.’

(13) ɖɯ˧-sɑ˥ F | mɤ˧-ʝi˥

ɖɯ˧	sɑ˥	F	mɤ˧-	ʝi˥
one	CLF.things	FOCALIZATION	NEG	to_do

‘to do nothing, to slob around the place, to do bugger all’

Among thirty examples of /**ɖɯ˧-sɑ˥**/ found in twenty-five transcribed narratives, all but one are accompanied by intonational focalization. The existence of this solitary example without intonational focalization (*TraderAndHisSon.24* {0004632#S24}) is enough to demonstrate that the association of this focalization with /**ɖɯ˧-sɑ˥**/ is habitual, and should not be considered a lexicalized characteristic of this expression in F4’s speech.

Moreover, intonational focalization can affect the string of tones in an utterance by influencing the division into tone groups.

As discussed in Chapter 7, the division of the utterance into tone groups is a fundamental dimension of Yongning Na prosody. The sifting of examples reveals that intonational focalization does not necessarily entail the presence of a following tone-group boundary, as shown by (14).

(14) ə˧ʝi˧-ʂɯ˥ʝi˩ | ɖɯ˩ɖʐɯ˧-ɬɑ˩ts^{h}o˩ ɳɯ˩ | ɖɯ˧-ʑi˩-ki˩ F ki˩-zo˩!

ə˧ʝi˧-ʂɯ˥ʝi˩	ɖɯ˩ɖʐɯ˧-ɬɑ˩ts^{h}o˩	ɳɯ˧	ɖɯ˧-ʑi˩	-ki˧	F
long_ago	*proper name*	A	one-CLF.households	DAT	FOCALIZATION

kiꜜ$_{a}$ -zoꜜ$_{a}$
to_give OBLIG
'Long ago, Ddeezzhi Lhaco had to marry into someone's family!'
(BuriedAlive3.143) {0004538#S143}

In the vast majority of examples, such a boundary is present, however. Moreover, there are examples where the intonational focalization is associated to the insertion of a tone-group boundary at a place where it would otherwise be highly unexpected, as in (15).

(15) zo˧ F | ə˧mi˧ ɳɯ˧ | mv̩˩-ki˥ ʐwɤ˩.
zo˥ F ə˧mi˧ ɳɯ˧ mv̩˩˧ -ki˧ ʐwɤ˩$_{b}$
son/man FOCALIZATION mother A daughter/girl DAT to_speak
'The young man's mother talked to the young woman.' *(BuriedAlive3.161)* {0004538#S161}

The emphasis placed on /**zo˧**/ 'son, man' causes the insertion of a tone-group boundary inside the phrase 'the young man's mother'. This results in a different sequence of tones than is found in the phrase 'the boy's mother', which, outside this context, is /**zo˧-ə˧mi˥**/. The division of this phrase into two tone groups results in the non-application of the tone rules which hold in determinative compounds, since tone rules never apply across tone-group boundaries.

On the basis of these findings, we are now in a position to turn to the topic of the phonetic implementation of tone in relation to intonation.

9.4 The phonetic implementation of tone

This section aims to convey a feel for the tonal and intonational space of Yongning Na through a sample of what appear as salient factors in the phonetic implementation of tone sequences. Four domains will be mentioned: phonotactics, coarticulation, morphology, and pragmatics. But first, an introductory note will highlight the distance between surface phonological tone and phonetic realization.

9.4.1 On the distance between surface phonological tone and phonetic realization

A large gap exists between phonological representations (framed in terms of sequences of level tones) and the observed fundamental frequency contours: from a phonetic perspective, "both F_0 height and F_0 velocity are relevant parameters (…) even for the simplest level tone languages" (Yu 2010: 1). To begin with an anecdote: back in the 1970s, at a time when producing F_0 tracings required expert technical skill, a specialist in Bantu tone asked a phonetician to generate an F_0 tracing from a recording illustrating a specific phonological phenomenon. Upon receiving the desired tracing, this renowned tonologist said that there must be an error, as the F_0 curve did not match the tone pattern that his trained ear had clearly perceived in the recording. In reality, the F_0 detection was entirely accurate; the issue lay in his expectation of a neatly binary F_0 tracing that would directly reflect the phonological tone sequence (J. Vaissière, p.c. 2001). This episode highlights a fundamental point: even in languages with relatively straightforward prosodic systems, such as Tokyo Japanese, F_0 curves are shaped by multiple factors and do not transparently reflect phonological tone. Without the guidance of a language consultant, it is simply impossible to determine, for instance, whether the lowering of F_0 on the final syllable of an utterance is due to an L tone on that syllable or to an intonational final lowering of an M-tone syllable. Arriving at tonal contrasts requires factoring out intonation, and vice versa.

As an illustration from Yongning Na, Figure 9.6 shows spectrograms of two phrases: /**bo˩-ɬv̩˧˥**/ 'pig's brains' (*DetermCompounds1to4.4-5* {0004451#W4}) and /**bo˩-ɬv̩˧ ɲi˥**/ 'is (a/the) pig's brains' (*DetermCompounds5.3-4* {0004452#W3}, *DetermCompounds7.13* {0004456#W13}, *DetermCompounds12.5* {0004462#W5}).

The rise in F_0 on the rhyme /**v̩**/ in the upper panel of the figure is consistent with its phonological description as an MH tone, while the flatter shape on the same rhyme in the lower panel, followed by a higher F_0 on the copula, is uncontroversially compatible with phonological description as a sequence of M on one syllable and H on the next. On the other hand, as emphasized by Cruz & Woodbury (2014) and Morey (2014), there is no way to read phonological tones off F_0 tracings. Figure 9.6 illustrates a phenomenon that becomes immediately apparent when examining any piece of experimental evidence: variability in the realization of tone.

Thus, glottalization is found at the end of both tokens in Figure 9.6, exerting a lowering influence on F_0 towards the offset of voicing. Glottalization is common in utterance-final position in Yongning Na, i.e. serving as a cue to intonational junctures between utterances. In these materials, the phrases /**bo˩-ɬv̩˧˥**/ 'pig's brains' and /**bo˩-ɬv̩˧ ɲi˥**/ 'is (a/the) pig's brains' both occur in absolute final position.

Another source of variability is the fact that level tones allow for some degree of variation within tonal space (F_0), much as vowels have a certain range of variation within the acoustic space (essentially defined by the first three formants,

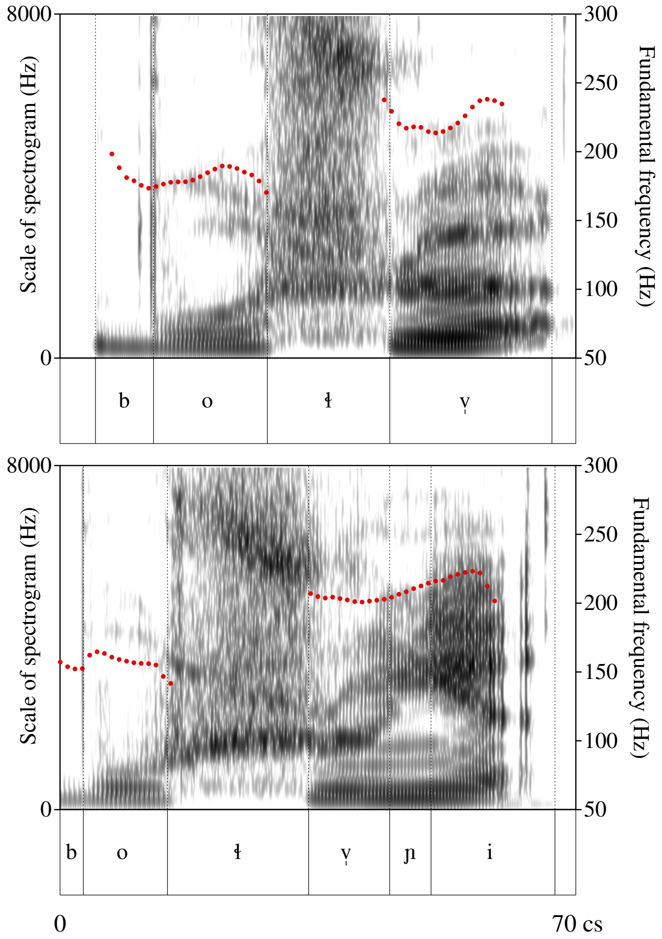


Figure 9.6: Spectrogram and F_0 tracing of ‘pig’s brains’ and ‘is (a/the) pig’s brains’.

F1-F2-F3). In Yongning Na, phonological rising tones never occur in initial position within a tone group. As a result, the identification of an initial L tone, as in /**bo**˩/ 'pig', is not jeopardized by its realization with a slight rise, as seen in the upper panel of Figure 9.6. Seen in this light, it does not come as a surprise that /**bo**˩/ 'pig' is realized with noticeably different F_0 in the two panels of the figure.

9.4.2 Phonotactic factors

A first set of salient factors in the phonetic implementation of tone concerns phonotactics.

Yongning Na has phonological constraints on possible tonal contrasts in initial, medial and final position inside a tone group. An M tone in initial position only contrasts with L, so that it has greater freedom of phonetic variation than an M tone in final position, where the range of possible contrasts is broader. (Recall that, following a penultimate M tone, a tone group can end in any of L, M, H, MH or LH; this places precise constraints on a final M tone's position within the tonal space.) Since the opposition between M and H is neutralized in tone-group-initial position, a phonetically high realization carries no risk of misinterpretation by native listeners. As a consequence, M tones in this position have the entire upper part of the tonal phonetic space as their range of intonational variation.

Two other key phonological facts about Yongning Na tone conspire towards framing the phonetic realizations of tone sequences: (i) there is no contrast between /H.M/ and /H.L/ sequences (only /H.L/ is observed), and (ii) the contrast between //LM// and //LH//, which is postulated at the underlying phonological level, is neutralized in surface phonological forms. Consequently, for any pair of successive syllables within a tone group, it is sufficient to classify the tone of the second as (i) higher than the preceding tone, (ii) on the same level as the preceding tone, or (iii) lower than the preceding tone; said differently, a two-step shift on the tone scale (i.e. from /L/ to /H/, or from /H/ to /L/) never contrasts with a one-step shift (i.e. from /L/ to /M/, or from /H/ to /M/). This is unlike the closely related language Naxi, where all combinations of tones in disyllables are attested, including /L.H/, /L.M/, /H.M/, and /H.L/.

The realization of rising contours is likewise subject to strong phonological constraints. As noted in Chapter 2, only two types of contours exist on monosyllables at the surface phonological level: low-rising and mid-rising. There is no phonological opposition between LM and LH contours at the surface phonological level. The product of this neutralization is labelled LH for the structural reasons set out in §2.3.9. Phonetically, it is a low-rising contour whose endpoint is not as high as that of the MH contour. To approximate phonetic realizations in terms of three levels, they are closer to [LM] than to [LH].

This pattern makes sense both in terms of production and in terms of perception. In production, reaching as high a phonetic target for the low-rising contour as for the mid-rising contour would require greater effort, as it would require a larger rise in F_0. In perception, a rise reaching a final target similar to that of the MH contour might make it more difficult to distinguish the (phonologically contrastive) low-rising and mid-rising contours. There are thus phonetic reasons why the outcome of the neutralization of LM and LH should be phonetically closer to [LM] than to [LH].[4]

9.4.3 Tonal coarticulation: Language-specific dimensions in tonal implementation

Based on expert listening, Creissels proposes stylized representations of the phonetic realizations of certain tone sequences (in a two-tone system), which clarify that tone implementation is language-specific. It is all the more important to describe such tonal coarticulation patterns as they hold potential for diachronic changes in the tone system.

For instance, Figure 9.7 illustrates an observation made in some West African tonal languages: the first in a sequence of L tones following an H tone carries a pitch level that is intermediate between that of the preceding H tone and that of the following L tones.

> Such realizations can be seen as the inception of a phenomenon of propagation: if the raising of the first in a sequence of L tones following an H tone becomes more noticeable, it can result in misperception as an H tone. (Creissels 1994: 215–216)
>
> *Original text:* On observe par exemple dans certaines langues que, sans que son identification comme ton bas soit remise en cause, le premier d'une séquence de tons bas succédant à un ton haut est réalisé à un niveau intermédiaire entre celui du ton haut qui le précède et celui des tons bas suivants. (…) On peut voir dans de telles réalisations l'amorce d'un phénomène de propagation : en effet, si le réhaussement du premier d'une séquence de

[4]From the point of view of the tone system, this creates a tension between phonological categories and phonetic realizations. Such discrepancies hold potential for reanalysis of the system by language learners, especially in the present social context, where exposure to Na is increasingly limited due to the pervasive presence of Mandarin. As a result, children may not acquire the tonal system in its full richness. This issue will be taken up in Chapter 10.

> tons bas succédant à un haut s'accentue, on peut aboutir à la confusion avec un ton haut (…).

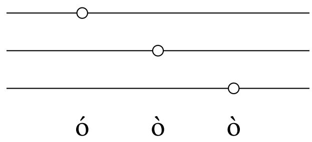


Figure 9.7: Stylized representation of the realization of an H.L.L sequence in some West African languages: as a gradual decrease in pitch from the first syllable to the third (Creissels 1994: 217).

Another pattern reported in some West African languages is phonetic anticipation of following tones: in an L.L.H sequence, for instance, the second L tone is realized with a higher fundamental frequency than the first, in a gradual progression towards the H tone (Creissels 1994: 216). Auditory impression suggests that the opposite pattern is found in Yongning Na. If an M tone is followed by L, this M tone is realized with a higher fundamental frequency than when followed by H. This has the effect of enhancing the contrast in pitch between M and the following tone. The M tone can thus be said to have three allophonic variants (*allotones*): the highest occurs before L, a mid-range variant appears before M or MH, and the lowest is found before H.

Similarly, in Yongning Na, an L tone is realized lower before M or H than when followed by another L. Using the vertical position of letters to indicate relative pitch, the phonetic realization of an M.M.L sequence can be approximated as [M.$^{\text{M}}$.$_{\text{L}}$], while an L.L.H sequence may be represented as [L.$_{\text{L}}$.$^{\text{H}}$].

These coarticulatory phenomena link up with observations about the realization of like-tone sequences, L.L.L… and M.M.M… (H.H.H… sequences never occur, since an H tone is always followed by L; this is formalized in §7.1.1 as Rule 4: "The syllable following an H-tone syllable receives L tone"). Sequences of L tones decline towards a clearly low final L target. The first L tone in such a sequence may be realized in a relatively high range of fundamental frequency and, when extracted from context, may sound like [M]. However, this does not threaten its correct identification within the tone sequence: a sequence such as L.L.L cannot be mistaken for M.M.L, because the latter requires a significant phonetic drop from M to L (approximated as [M.$^{\text{M}}$.$_{\text{L}}$]).

The perceptual distinction between these sequences is further reinforced by their characteristic phonetic trajectories. In contrast to an M.M.L sequence, realized with a slight upward shift in fundamental frequency from the first to the second M level followed by a pronounced drop to L ([M.$^{\mathrm{M}}$.$_{\mathrm{L}}$]), a sequence of like tones (L.L.L or M.M.M) is realized with a gradual decrease in fundamental frequency but without any abrupt transitions, making it perceptually distinct. Thus, an M.M.M sequence cannot be confused with M.M.L, even if its last tone is depressed by intonational factors such as final lowering. While final lowering affects the last syllable, it does not raise the preceding one (schematic representation: [M.M.$_{\mathrm{M}}$]). By contrast, an M.M.L sequence involves a raising of the penultimate syllable: [M.$^{\mathrm{M}}$.$_{\mathrm{L}}$].

These auditory impressions are summarized in Table 9.3. (As before, the symbol σ is used to stand for syllables.) Acoustic measurements reported in a phonetic pilot study conducted by a speaker of Na (Ā Huì 2016) provide some empirical support for these observations. The logical next step would be to conduct a state-of-the-art experimental study. Readers with an interest in experimental phonetics may seek more detail than the schematized representations in Figure 9.7 and Table 9.3 can encapsulate.

Experimental-phonetic studies of level tones that propose models of tonal implementation are available for several two-tone systems, such as Dinka (Remijsen & Ladd 2008), Kinyarwanda (Myers 2003), Sotho (Zerbian & Barnard 2010a,b), and Tswana (Zerbian & Kügler 2023); three-tone systems, including Sumi (Teo 2014: 48–65), Ao (Coupe 2003: 100–106), and Yorùbá (Laniran & Clements 2003); and four-tone systems, e.g. Mambila (Connell 2003, 2016). These studies confirm that issues of tone realization are conditioned by a host of language-specific parameters.

No experimental phonetic tools are deployed to identify such parameters in the Yongning Na prosodic system, in line with Fischer-Jørgensen's view that "the linguistic analysis, which may perfectly well be made on auditory basis, must come first" (Fischer-Jørgensen 1949: 4). Nonetheless, the prospect of future experimental verification has been kept in mind throughout, and the observations proposed here are intended as a foundation for subsequent phonetic experiments. A study of how tone sequences are implemented in Yongning Na would be a useful addition to the literature on contextual tonal variation and segmental effects on tone in Asian languages, as examined by Abramson (1979), Gandour & Potisuk (1992) and Luksaneeyanawin (1998) for Thai, Brunelle (2009c) for Vietnamese, and Sun & Shih (2021), Kuang et al. (2023) for Mandarin.

Grasping the phenomenon of anticipatory phonetic dissimilation in M.L and M.H sequences is crucial to the identification of tones in Yongning Na. The system allows for a great range of phonetic variation in sequences of L tones or M tones, as long as the pitch trajectory remains consistent: L-tone sequences may start from a relatively high pitch, provided that they descend clearly towards

Table 9.3: A schematic representation of the realization of some tone sequences in Yongning Na.

phonetic outline	interpretation	cues to tonal identification
[σ.σ.σ.σ.σ]	[M.M.M.M.M]	gentle decline in F_0; overall mid range of F_0; final lowering is not considerable
[σ.σ.σ.σ.σ]	[L.L.L.L.L]	slightly steeper decline in F_0 than for all-M sequences; slightly lower starting point and endpoint
[σ.σ.σ.σ]	[M.M.M.L]	raising of the syllable before last, and clear difference in pitch between the last two syllables
[σ.σ.σ.σ]	[L.L.L.H]	lowering of the syllable before last, and clear difference in pitch between the last two syllables

a phonetic low target without intervening upward shifts. Likewise, M-tone sequences may be strongly affected by declination (a gradual, noncategorical decrease in fundamental frequency in the course of the utterance: see §9.1) without putting their identification at risk. Listeners confronted with a sequence of syllables with decreasing fundamental frequency but no noticeable upward jumps can safely interpret it as carrying either M.M.M... or L.L.L... tones. The distinction between these two rests on the overall slope of the F_0 decline and the F_0 range reached at the end of the sequence.

While anticipatory phonetic dissimilation in M.L and M.H sequences is especially salient, preplanning in tone production is also noticeable at the level of the entire tone group. For instance, an M.M.H sequence begins on a lower pitch than M.M.M, which itself begins lower than M.M.L. A striking illustration of this effect was unintentionally captured during elicitation using the frame in (16).

(16) no˧ ɳɯ˧ | V-zo˧-ho˩.
no˩ ɳɯ˧ V -zo˧$_{a}$ -ho˩
2SG A V OBLIG FUT
'you must now V / you will now have to V.'

In the sequence 'you will now have to eat', /**no˧ ɳɯ˧ | dzɯ˧-zo˧-ho˥**/, the two M-tone syllables of the utterance-initial tone group, /**no˧ ɳɯ˧**/ (2SG plus A), were realized with high pitch. By contrast, the two subsequent M-tone syllables, /**dzɯ˧-zo˧**/, were realized lower than average, thereby maximizing the contrast with the following H tone. The result is a considerable drop in F_0 from /**ɳɯ˧**/ to /**dzɯ˧**/, despite the identical phonological tone of these syllables.

A third set of factors influencing the phonetic implementation of tones has to do with morphology.

9.4.4 Morphological factors: The weaker realization of function words

It is a well-established cross-linguistic observation that grammatical words tend to be less strongly articulated than lexical words: their lighter semantic weight is reflected in a weaker phonetic realization. In languages with distinctive stress, many grammatical words do not carry lexical stress, and those that do are sometimes destressed in discourse. In some languages that have tone, there are cases of toneless grammatical words, for instance in Mandarin (Lín Màocàn & Yán Jǐngzhù 1980, Chen & Xu 2006). Even in languages where there is no phonological difference between lexical and grammatical words (e.g. in French, which does not have lexical stress, and in Vietnamese, which has tone on all syllables including grammatical morphemes), a phonetic difference between these two categories is noticeable. The extent of this difference varies across languages: grammatical words undergo less hypo-articulation in Vietnamese than in English or French (Brunelle et al. 2015).

Colleagues report that in some West African languages, the phonetic realization of tone sequences is unaffected (or almost unaffected) by the nature of the tone-bearing words, so that an M tone on a grammatical morpheme is realized in the same way as on a verb or noun (Jacqueline Leroy, p.c. 2007; Larry Hyman, p.c. 2011). In Yongning Na, on the other hand, the intonational backgrounding of grammatical words, such as sentence-final particles, is easily perceptible.

The prohibitive prefix /**tʰɑ˧-**/ is a case in point. When followed by a reduplicated verb of tone category M_a, the resulting tone pattern is M.H.L, e.g. /**tʰɑ˧-ki˥~ki˩**/ 'do not give away, do not distribute'. Phonetically, the syllable /**tʰɑ˧**/ is realized with rising fundamental frequency. Averaging across the eight tokens in the recording *VerbProhib* {0004522}, F_0 rises by 6.7% (standard deviation: 3.6). This rise is probably too small to be perceived as such: glissandos require a minimum slope of approximately 14% over a long tone-bearing unit (over 300 ms) or 18% over a short one (≈100 ms) to be perceptible (Rossi 1971, Gsell 1979). The rise during /**tʰɑ˧-**/ can be compared to the 16% jump in F_0 (standard deviation: 5.0) from the prefix to the following H-tone syllable.

To use a concept from a model developed for Mandarin speech synthesis, grammatical words could be said to have a lower *strength coefficient*. In this model (called Stem-ML, for *Soft Template Mark-up Language*), the tones of syllables with a high strength coefficient are realized close to their lexical templates, whereas syllables with a lower coefficient undergo stronger coarticulation with adjacent tones (Kochanski & Shih 2003). The anticipatory rise in the course of the prohibitive prefix /**tʰɑ˧-**/ can be interpreted as a typical case of coarticulation between an M-tone syllable with low strength coefficient and a following H-tone syllable with high strength coefficient.

Future studies of this topic could undertake direct phonetic comparisons between segmentally and tonally homophonous pairs, such as the noun /**bv̩˧**/ 'intestine' and the possessive /=**bv̩˧**/, or the noun for 'person, human being', /**hĩ˥**/, and its grammaticalized avatar as a relativizer, /**-hĩ˥**/.

The final factor examined here in relation to the phonetic implementation of tones concerns pragmatic modulation.

9.4.5 Pragmatic factors: Intonational foregrounding and intonational backgrounding

The intonational foregrounding of specific words for pragmatic reasons (as detailed in §9.2) goes hand in hand with the intonational backgrounding of words that are less prominent in terms of information structure. Thus, when a consultant pronounces a sequence of numeral-plus-classifier phrases, such as /**ɖɯ˧-kʰwɤ˥ | ɲi˧-kʰwɤ˥ | so˩-kʰwɤ˩˥ | ʐv̩˧-kʰwɤ˧**/ ('one piece, two pieces, three pieces, four pieces…'), the tone of the classifier changes (e.g. High in /**ɖɯ˧-kʰwɤ˥**/ and /**ɲi˧-kʰwɤ˥**/, Low-to-High in /**so˩-kʰwɤ˩˥**/, and Mid in /**ʐv̩˧-kʰwɤ˧**/), but its consonants and vowels remain unchanged. Consequently, the speaker's attention is directed towards the new information – essentially: the correct tonal sequence for the phrase. In terms of the continuum from hyper-articulation to hypo-articulation (Lindblom 1990), the classifier's consonant and vowel are hypo-articulated. Specifically, the unvoiced lateral /**ɬ**/ in classifiers such as /**ɬi˧**/ 'month' and /**ɬi˩**/ 'armspan' is occasionally realized as voiced, despite the presence of a phonemic distinction between /**ɬ**/ and /**l**/ in Na. Pursuing such observations would provide insight into the field of allophonic dispersion of Na phonemes. Given the nature of the corpus, numerous tokens of each numeral and classifier are available, offering a solid empirical basis for statistical analysis.

9.5 By way of recapitulation: Examples of mistaken tonal identification

This section recapitulates observations made above about intonation and tone implementation by illustrating them through examples of mistaken notations from early field notes. These cases of erroneous tone identification arose because I was unaware of the language-specific factors shaping the final prosodic form of utterances. By shedding light on the process of categorical interpretation of the pitch of successive syllables, these examples illustrate common pitfalls in tone analysis. They are arranged in thematic order.

9.5.1 Phonotactics and tone-group boundaries

The tone sequence M.L.L.L in (17) was mistakenly transcribed as ‡ MH.M.L.L (‡ **le˦-mv̩˧pʰæ˩-ze˩**) on my first field trip.

(17) le˧-mv̩˩pʰæ˩-ze˩
le˧- mv̩˩pʰæ˦ -ze˧ b
ACCOMP to_forget PFV
'[I/you...] have forgotten' (Field notes. Also found in *Dog.4* {0004442#S4})

Transcription as ‡ MH.M.L.L is relatively close to the phonetic realization, with a slight rise on the first syllable followed by a gradual decline in fundamental frequency over the remaining three syllables, reaching a low target on the final syllable. However, identifying the correct tone sequence requires recognizing that the crucial phonological cue is the overall higher fundamental frequency on the first syllable. Details in its phonetic contour – whether flat, rising or falling – are phonologically irrelevant in this context, since phonological contours only occur in tone-group-final position. What matters is that the first syllable has a higher mean F_0 than those that follow. Since the first tone in a tone group can only be M or L at the surface phonological level, it can safely be identified as M. The subsequent lower tone can then safely be identified as L. By application of Rule 5, all following tones must also be L.

It could be that, for the listener, the identification of the sequence in (17) as a single syntactic unit (a verb phrase) facilitates its recognition as a single tone group, which in turn provides guidance in tone identification. But the interplay between morphosyntactic and phonological information in speech comprehension falls within the domain of psycholinguistic research, which lies beyond the scope of this volume.

The next example further illustrates the importance of tone-group boundaries. At a stage when I had not yet worked out clearly the existence of tone groups, sentence (18) was transcribed as ‡ **njɤ˥ ʈɯ˥ hwæ˧-bi˧-ze˧**. The sequence of three M tones over the last three syllables was identified correctly, but it seemed to me that the tone of the first two syllables was higher, which led me to transcribe them as H.

(18) njɤ˧ ʈɯ˧ | hwæ˧-bi˧-ze˧!
njɤ˩ ʈɯ˧ hwæ˧$_a$ -bi˧$_a$ -ze˧$_b$
1SG A to_buy IMM.FUT PFV
'I'm paying for it! / I'm the one who's buying [it]!'

Once it was recognized that there were two tone groups in (18), the considerable phonetic difference in pitch between them could be interpreted as a difference in their overall F_0 register. As explained in §9.1.1, resetting of reference values for tones at junctures between tone groups means that phonologically identical tones may have different fundamental frequencies across tone-group boundaries.

A proficient speaker of Yongning Na would not interpret the high pitch on /**njɤ˧ ʈɯ˧**/ as an H.H sequence, since such sequences do not occur in the language (due to Rules 4 and 5: within a tone group, an H tone can only be followed by L tones). Further examples are shown in Table 9.4. In all of these cases, my early transcriptions were based on perceived dissimilarities in pitch between successive syllables, overlooking the fact that the overall F_0 register can differ from one tone group to the next.

Table 9.4: Some examples of mistaken tonal transcriptions due to oversight of resetting at junctures between tone groups.

meaning	tone sequence	early, mistaken transcription	current transcription
you go and buy [it]	M | M.M	**no˥ hwæ˧hõ˧**	**no˧ | hwæ˧-hõ˧**
one mountain	M | M.M	**ʁwɤ˥ ɖɯ˧-ɭɯ˧**	**ʁwɤ˧ | ɖɯ˧-ɭɯ˧**
a drop of water	M | M.M	**dʐɯ˥ ɖɯ˧-ʈʰɤ˧**	**dʐɯ˧ | ɖɯ˧-ʈʰɤ˧**
this is a mountain	M | M.L	**ʈʂʰɯ˥ ʁwɤ˧ ɲi˩**	**ʈʂʰɯ˧ | ʁwɤ˧ ɲi˩**
to light a fire	M | M.MH	**mv̩˥ tʰi˧-tsʰi˧˥**	**mv̩˧ | tʰi˧-tsʰi˧˥**
I cut a piece	M.M | M.M.H	**njɤ˥ ʈɯ˥ ɖɯ˧-kʰwɤ˧ dze˥**	**njɤ˧ ʈɯ˧ | ɖɯ˧-kʰwɤ˧ dze˥**

9.5.2 Tonal coarticulation: Anticipatory dissimilation before an L tone

The tone sequence M.M.L was often misidentified as M.H.L, for instance in the determinative compound ‘tiger’s ear’, shown in (19).

(19) lɑ˧-ɬi˧pi˩
lɑ˧ ɬi˧pi˩
tiger ear
‘tiger’s ear’ *(DetermCompounds6.17)* {0004456#W17}

This compound was initially transcribed as ‡ **lɑ˧-ɬi˥pi˩**, owing to the fact that the pitch of the second syllable, /**ɬi**˧/, is higher than that of the first, /**lɑ**˧/. This effect arises due to anticipatory dissimilation before an L tone, as described in §9.4.3: the sequence can be stylized phonetically as [M.^{M}.$_{L}$]. The perceptible difference in pitch between the first two syllables serves as a cue to the presence of an L tone on the third syllable, distinguishing it from an M tone affected by *declination* or *final lowering*. Obviously, this difference must be factored out when identifying the phonological tone of the second syllable, /**ɬi**˧/.

In what can be described as a form of “chain shift”, the M.L.L sequence tends to occupy some of the phonetic space left unfilled (or less densely populated) due to the phonetic raising of M before L. Just as M.M.L was misperceived as M.H.L, a misidentification arose whereby M.L.L was transcribed as M.M.L – at a stage when the foreign learner’s ear was not yet attuned to the Alawua tonal space.

In (20), the phrase /**ɖɯ˧-ʁæ˩bæ˩**/ ‘one plateful’ was initially transcribed as ‡ **ɖɯ˧-ʁæ˧bæ˩**. Due to final lowering, there is a phonetic drop in F$_0$ from the second syllable (/**ʁæ**˩/) to the last syllable (/**bæ**˩/), even though both carry the same phonological tone (L). This drop may be as salient perceptually (or even more salient) than the one from the first syllable (/**ɖɯ**˧/) to the second (/**ʁæ**˩/), even though the latter transition reflects a genuine phonological contrast (M vs. L).

(20) ʁæ˧bæ˧-qo˧ ɳɯ˧ | ʁæ˧bæ˧ | ɖɯ˧-ɭɯ˧-qo˧ ɳɯ˧, | ɖɯ˧-ʁæ˩bæ˩!
ʁæ˧bæ˧ -qo˧ ɳɯ˧ ʁæ˧bæ˧ ɖɯ˧ ɭɯ˧$_{b}$ -qo˧ ɳɯ˧
wooden_plate inside ABL/TOP wooden_plate one CLF inside ABL/TOP
ɖɯ˧ ʁæ˩bæ˩
one CLF.platefuls
‘[The sweets used to be presented] on a plate! [They were placed] on a plate; [one used to prepare] a plateful!’ *(Funeral.216)* {0004571#S216}

9.5.3 Effects of pragmatic intonation

In my first field notes, example (21) was transcribed as ‡ **kɯ˩ɻ̩˧ ʈɤ˥** due to an intonational strengthening of the verb. The phrase was provided in response to a question concerning the verb used with ‘erhu, Chinese violin’: I had initiated a sentence, /**ʈʂʰɯ˧ ɳɯ˧ | kɯ˩ɻ̩˧** .../, ‘(S)he ... the violin’, while making the gesture of playing. The consultant responded by placing emphasis on the verb in her answer, producing a form of emphatic stress as described in §9.2.1.

(21) kɯ˩ɻ̩˧ ʈɤ˧
kɯ˩ɻ̩˧ ʈɤ˧a
Chinese_violin to_pull
‘to play erhu (Chinese violin)’

Likewise, the sentence (22) was initially transcribed as ‡ **ɖɯ˧-bæ˧ | mɤ˧-tsɤ˥**, with H tone on the verb. I believe that this was because the verb /**tsɤ˧**/ is realized phonetically with higher fundamental frequency than the preceding negation prefix. As a grammatical morpheme, the negation prefix is intonationally weaker, as discussed in §9.4.4.

(22) ɖɯ˧-bæ˧ | mɤ˧-tsɤ˧!
ɖɯ˧ bæ˧a mɤ˧- tsɤ˧
one CLF.sorts NEG to_become/to_be
‘It’s not the same!’

The phrase in example (23) was transcribed as ‡ **mv̩˧ʁo˥ tɕɯ˥** instead of /**mv̩˧ʁo˥ | tɕɯ˧**/. The final syllable has a high informational load, and accordingly receives some intonational emphasis. Furthermore, since there is no opposition between M and H tones in tone-group-initial position, phonetic realizations in this context can extend into the speaker’s higher tonal range without risk of phonological confusion.

(23) mv̩˧ʁo˥ | tɕɯ˧.
mv̩˧ʁo˥ tɕɯ˧
sky/heavens cloud
‘The sky is cloudy.’

As a final example, let us consider (24).

(24) zo˧ ɳɯ˧ | ʝi˧!
zo˥ ɳɯ˧ ʝi˥
man A to_do
'This is a man's job! / This type of work [viz. ploughing] is men's part!'
(Field notes.)

In (24), the first two syllables form a tone group and carry the same surface phonological tone (M tone). In my early notes, however, I jotted down this example as ‡ **zo˥ ɳɯ˧ ʝi˧**, as I was influenced by the intonational emphasis carried by the first syllable. This emphasis results in a higher pitch on /**zo˧**/ than on the following syllable, which, as a grammatical element, is prone to intonational backgrounding, a phenomenon studied in §9.4.4.

The final section in this chapter on intonation in Yongning Na is devoted to prospects for in-depth experimental phonetic study of intonation in Yongning Na.

9.6 By way of conclusion: Prospects for experimental phonetic modelling

The present book follows the approved order of business, whereby the system – the morphotonology and intonation – needs to be worked out before launching into experimental investigations into acoustic correlates and fine phonetic details (Mazaudon 2014, Rice 2014). A fine-grained study of the phonetic implementation of tone in Yongning Na has yet to begin in earnest, but this mid- to long-term perspective has always been kept in view from the outset of fieldwork. It has motivated continuous efforts to collect data that will be suitable for such research, including high-fidelity audio recordings and, for some of the documents, an accompanying electroglottographic signal.

The ultimate research objectives of the exploration of the phonetic implementation of tone bear a strong link to the analytic framework adopted here. Every deviation from an abstract, ideal phonetic template for a given phonological entity (whether a tone or a phoneme) is, as a matter of principle, to be accounted for as resulting from an intonational factor, unless it can be straightforwardly explained by coarticulation processes. Thus, the ultimate goal is to assess the contribution of various factors to the final phonetic realization of each syllable, disentangling the different components of the speech signal and clarifying their linguistic interpretation. Computer modelling constitutes a promising tool for this endeavour, as it allows intonational phenomena to emerge as components that are not predictable from the utterance's contrastive units: the sequence of phonemes and the tonal string parsed into tone groups.

This perspective opens into an ambitious research agenda that extends beyond the glottal level (the phonetic implementation of tonal sequences in terms of patterns of fundamental frequency and phonation types) to encompass the subglottal and supraglottal levels as well. It includes a vast range of questions, such as those associated with "segmental intonation" in the sense of Niebuhr (2009b): for instance, the extent to which tone-group boundaries are cued by fine phonetic details in the realization of segments (vowels and consonants). Such investigations could adopt a *data science* approach, integrating data-driven contour clustering, as proposed by Kaland (2023), with high-level modelling of supraglottal gestures (Stavness et al. 2017, Gick et al. 2020).

This concluding note brings to a close the pair of chapters devoted to intonation. The next two chapters present further analyses of the specific dynamics of Yongning Na. Chapter 10 examines its tonal system from a dynamic-synchronic perspective, and Chapter 11 places it in its dialectal and areal context.

10 Yongning Na tones in dynamic-synchronic perspective

> The past century of phonetic research has illuminated our understanding of the production of sound, the properties of the acoustic signals, and to a certain extent, the perception of speech sounds. But the search for the originating causes of sound change itself remains one of the most recalcitrant problems of phonetic science.
>
> Labov (1979: 1)

> It is often claimed that the causes of phonetic changes remain unknown. However, this assertion is incorrect. Crucially, there are multiple factors at play, not just one single cause. A common mistake among those who have addressed this question is that, upon identifying a cause of phonetic change, they have assumed it to be the sole cause and attempted to attribute all changes to it.
>
> Grammont (1933: 175)[1]

The synchronic description proposed in the present volume provides a foundation for examining the historical dynamics of tone in Na. As mentioned in §1.3.2, a dynamic approach to synchrony brings out patterns of synchronic variation which, in turn, offer glimpses into diachronic evolution.

[1] *Original text:* On enseigne partout qu'elles [les causes des changements phonétiques] sont encore inconnues et mystérieuses. C'est inexact. Mais il n'y a pas une cause, il y en a un grand nombre, et l'erreur de la plupart de ceux qui se sont occupés de la question a été précisément, lorsqu'ils ont reconnu une cause de changements phonétiques, de croire qu'elle était la seule cause et de vouloir tout y ramener.

Variability in tone patterns tends to be high in level-tone systems with rich morphotonology, and diachronic change tends to occur at a faster pace than in other areas of the linguistic system (such as syntax). The view that tonological models should be designed in such a way as to accommodate patterns of variation is found e.g. in a discussion of Bambara, a Mande language:

> Clearly, any hypothesis about the system that underlies the tonal productions of Bambara speakers should be able to account, with minimal adjustments, for observed patterns of variation. (Creissels 1992: 8)
>
> *Original text:* il est clair que toute hypothèse sur l'organisation du système sous-jacent à un corpus de productions tonales de bambarophones doit pouvoir au prix d'un minimum d'aménagements rendre compte de possibilités éventuelles de variation.

As more data becomes available on the dialectal diversity of Na, it will be possible to investigate patterns of contact and variation with increasing precision. This chapter discusses five topics, in no particular order, and making no attempt at exhaustiveness: structural gap-filling, analogy, syllable reduction, disyllabification, and the influence of bilingualism with Mandarin.

10.1 Gap-filling in tonal paradigms: The example of subject-plus-verb phrases

When a phonological unit drifts far enough from its original pronunciation for a new combination to occupy the abandoned slot, structural gap-filling can occur, leading to changes in the phonological system. For instance, in Yongning Na it is likely that the sound [**ʁ**] was originally an empty-onset filler (see Appendix A, §A.3.3). The syllable /**ʁo**/ in /**ɑ˩ʁo˧**/ 'house' is reconstructed as a simple ***o** at the proto-Naish stage (Jacques & Michaud 2011). It remains phonemically onsetless in Laze ([**ɑ˥wu˥**], phonemically /**ɑ˥u˥**/) and in Naxi ([**mi˧wu˩**], phonemically /**mi˧u˩**/). Once ***o** syllables came to be realized as [**ʁo**] at the surface phonological level in Yongning Na, no [**o**] or [**wo**] syllables remained. But this phonetic slot was subsequently filled by syllables with other origins: the syllable /**wo**/ is now firmly attested in words such as /**wo˥**/ 'hard', /**wo˩**$_{\mathbf{b}}$/ 'classifier for teams of oxen', /**wo˩kɤ#˥**/ 'swing', and /**wo˩˥**/ 'turnip leaves'. The emergence of [**wo**] syllables precipitated the phonemicization of what was originally an empty-onset-filler: the syllable /**ʁo**/ in /**ɑ.ʁo**/ 'house' must now be analyzed as composed of two phonemes, an initial /**ʁ**/ and the vowel /**o**/.

Table 10.1: The tone patterns of subject-plus-verb combinations, in surface phonological transcription.

	tone of verb					
tone of noun	H	M_a	M_b	L_a	L_b	MH
LM, LH	L.H	L.M+M	L.M+M	L.H	L.H	L.MH
M	M.M+L	M.M+M	M.M+M	M.L	M.L	M.MH
L	M.M+L	L.L	M.M+M	L.L	L.L / M.L	L.L
H	M.M+L	M.M+L	M.M+L	M.MH	M.MH	M.L
MH	M.H	M.H	M.H	M.MH	M.MH	M.H
M	M.M.M+L	M.M.M+M	M.M.M+M	M.M.L	M.M.L	M.M.MH
#H	M.M.M+L	M.M.M+L	M.M.M+L	M.M.MH	M.M.MH	M.M.L
MH#	M.M.MH	M.M.MH	M.M.MH	M.M.MH	M.M.MH	M.M.H
H$	M.M.M+L	M.M.M+L	M.M.M+L / M.M.M+H	M.M.MH	M.M.MH	M.H.L
L	L.L.L	L.L.L	L.L.L	L.L.L	L.L.L	L.L.H
L#	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L	M.L.L
LM+MH#	L.M.M+L	L.M.M+L	L.M.M+L	L.M.MH	L.M.MH	L.M.H
LM+#H	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.MH	L.M.MH
LM	L.M.M+L	L.M.M+M	L.M.M+M	L.M.L	L.M.L	L.M.MH
LH	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L	L.H.L
H#	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L	M.H.L

Gap-filling can also take place in tonal paradigms. This section is devoted to a plausible case in subject-plus-verb constructions.

Table 6.54, repeated here as Table 10.1, presents the tonal behaviour of subject-plus-verb combinations, covering both monosyllabic and disyllabic subjects. Two contexts were used to arrive at underlying tone categories: S+V and S+V+perfective. For instance, 'the guests arrive' is /**hĩ˧-bæ˧ tsʰɯ˧˥**/, and addition of the perfective yields /**hĩ˧-bæ˧ tsʰɯ˧-ze˥**/ 'the guests have arrived'. The tone pattern of this subject-predicate combination, /M.M.MH/, is analyzed as //MH#//: an MH contour associating to the last syllable.

A key analytical challenge raised by the dataset in Table 10.1 concerns the surface phonological tone sequences ending in M+L, M+M, and M+H. In these notations, the tone following the '+' sign is that carried by the perfective /**-ze˧**/ when suffixed to the subject-plus-verb combination. The question is how the perfective suffix acquires its /L/, /M/ or /H/ tone in these combinations.

The full set of relevant expressions is comprised of M.M+L, M.M.M+L, M.M+M, M.M.M+M, M.M.M+H, L.M+M, L.M.M+M, and L.M.M+L. Among these, those ending in /M+L/, as in (1), and those ending in /M+M/, as in (2), are commonplace. On the other hand, only one expression ends in /M+H/: it results from the combination of a //H\$//-tone subject and a //M_b//-tone verb, as in (3).

(1) lɑ˧ se˧-ze˩
lɑ˧ se˥ -ze˧$_b$
tiger to_walk pfv
'the tiger walked' (input tones: M on noun and H on verb) *(SubjectVerb.31)* {0004477#W31}

(2) lɑ˧ ʂɯ˧-ze˧
lɑ˧ ʂɯ˧$_a$ -ze˧$_b$
tiger to_die pfv
'the tiger died' (input tones: M on noun and M on verb) *(SubjectVerb.22)* {0004477#W22}

(3) hwɤ˧mi˧ tsʰo˧-ze˥
hwɤ˧mi˥\$ tsʰo˧$_b$ -ze˧$_b$
she_cat to_jump pfv
'the she-cat jumped' (input tones: H\$ on noun and M_b on verb) *(SubjectVerb.85-87)* {0004477#W85}

The hypothesis proposed here is that the pattern ending in /M+H/ is an innovation.

The PERFECTIVE can receive one of three tones in subject-plus-verb plus PERFECTIVE constructions: /M/, /H/ or /L/. Cases in which the PERFECTIVE carries /M/ tone are the simplest: the morpheme does not receive any tonal specification from what precedes, and surfaces with its lexical M tone. The surface strings /M.M+M/ (for monosyllabic nouns) and /M.M.M+M/ (for disyllabic nouns) can therefore be analyzed as //M//. Likewise, /L.M+M/ (for monosyllabic nouns) and /L.M.M+M/ (for disyllabic nouns) can be analyzed as //LM//.

Cases where the PERFECTIVE receives /H/ tone look like typical instances of the floating H tone, //#H//. This tone, which does not surface in isolation but can be manifested on a following syllable, is frequently observed in Yongning Na. It is the lexical tone of a class of nouns, exemplified by 'little brother', which surfaces in isolation as /**gi˧zɯ˧**/ 'little brother' but yields /**gi˧zɯ˧ ɲi˥**/ when followed by the copula, as explained in §2.3.

As noted in §2.3.5, the //H$// tone shows signs of variability: it is the lexical tone with the greatest number of morphotonological variants across different morphosyntactic contexts. In subject-plus-verb constructions, its combination with a //$\mathrm{M_b}$//-tone verb allows for two possibilities: M.M.M+L and M.M.M+H. The latter tonal string, M.M.M+H, is not attested in any of the other subject-plus-verb combinations. This distribution suggests the hypothesis that this variant is an innovation that filled an empty slot in the surface phonological patterns.

Under the hypothesis that the tone pattern M.M.M+H in subject-plus-verb plus PERFECTIVE combinations represents an innovation, the pre-existing M.M.M+L surface pattern could have been analyzed as //#H//. The floating H tone was not manifested directly but triggered a lowering of following tones – in this instance, a lowering of the tone of the PERFECTIVE morpheme.[2]

This analysis is reflected in Table 10.2, which leaves out the problematic variant M.M.M+H for the combination of a subject carrying H$ tone with a verb carrying $\mathrm{M_b}$ tone.

Table 10.2 is a reconstruction of the set of tone rules that applied in subject-plus-verb constructions prior to the appearance of the M.M.M+H variant. If this reconstruction represents a historical reality, it is likely to have shallow time

[2]The floating H tone of Yongning Na, transcribed as //#H//, exists not only as a lexical category on nouns (as discussed in §2.3), but also as the output of some morphotonological rules, such as those that apply in compound nouns. Cases where an H tone does not surface but lowers the following tones (to L) are found in several areas of Yongning Na morphotonology. For instance, the phrase /**mv̩˩tɕo˧ se˧**/ 'to walk downward' depresses a following PERFECTIVE suffix (//**-ze˧$_\mathrm{b}$**//, which has lexical M tone) to L: /**mv̩˩tɕo˧ se˧-ze˩**/ '(s)he walked downward' (*SpatialOrientation.59-60* {0004559#W59}). This is interpreted as evidence of the presence of a floating H tone in the expression 'to walk downward': //**mv̩˩tɕo˧ se#˥**// (see §6.3.3).

Table 10.2: A phonological analysis of the tones of subject-plus-verb combinations, leaving aside the M.M.M+H variant of the combination of an H$-tone subject and an M_b-tone verb.

	tone of verb					
tone of noun	H	M_a	M_b	L_a	L_b	MH
LM, LH	LM	LM	LH	LH	LH	LM+MH#
M	M	M	#H	M.L	M.L	M.MH
L	L	M	#H	L	L	L
H	#H	#H	#H	MH#	MH#	L#
MH	H#	H#	H#	MH#	MH#	H#
M	M	M	#H	#H	#H	MH#
#H	#H	#H	#H	MH#	MH#	#H
MH#	MH#	MH#	MH#	MH#	MH#	#H
H$	#H	#H	#H	MH#	MH#	H#
L	L	L	L	L	L	L+H#
L#	L#–	L#–	L#–	L#–	L#–	L#–
LM+MH#	LM–+#H	LM–+#H	LM–+#H	LM+MH#	LM+MH#	LM+H$
LM+#H	LM–	LM–	LM–+#H	LH–	LM+MH#	LM+MH#
LM	LM–	LM–	LM–+#H	LH–	LH–	LM+MH#
LH	LH–	LH–	LH–	LH–	LM+MH#	LH–
H#	H#–	H#–	H#–	H#–	H#–	H#–

depth: the amount of observed idiolectal and dialectal variation suggests that such a change could have taken place within a couple of generations. At this reconstructed stage, a tone rule must be posited whereby the //#H// tone in subject-plus-verb constructions does not surface, but depresses following tones to L. In view of the general architecture of the Na tone system, this rule is not an *ad hoc* device to explain away an unaccountable observation: a rule to the same effect operates in other morphosyntactic contexts.

In the present state of the language, on the other hand, the M.M.M+H variant has settled in, and the simplest phonological interpretation of this pattern is as the manifestation of a floating H tone – an interpretation that conflicts with the earlier system. Interpretation of the M.M.M+H pattern as underlying //#H// by language learners entails an in-depth modification of the tonal system: as the //#H// slot in the system comes to be occupied by the new, innovative form, the M.M.M+L surface pattern – previously analyzable as reflecting an underlying //#H// – requires a new phonological interpretation, as do the other surface patterns ending in /M+L/ in Table 10.1.

One possible phonological reanalysis would be to interpret these cases as involving a floating L tone, //#L//, which would thus enter the language's tonal system. The surface phonological patterns in subject-plus-verb plus PERFECTIVE constructions would then be interpreted as shown in Table 10.3, where the newly introduced //#L// category is highlighted, bringing out its relatively pervasive presence in the table.

Devoting an entire section to the discussion of a single tonal variant may seem dreadfully disproportionate. This variant deserves special attention, however, because it exemplifies the constant tension between underlying forms and surface phonological forms, shedding light on types of evolution taking place in level-tone systems. From the point of view of surface phonological forms, the innovative pattern discussed here can be viewed in the light of a simplification: at the (hypothetical) conservative stage presented in Table 10.2, the combination of //H$// and //$M_b$// involves an H tone in the input but none in the output; in the innovative form, by contrast, an H tone is present in both input and output, creating a closer match between the two. From the perspective of underlying forms, on the other hand, this development introduces an analytical puzzle for linguists – and quite possibly for language learners as well. Such cases allow for several analytical options and hence hold potential for diachronic change.

Table 10.3: A phonological analysis of the tones of subject-plus-verb combinations positing floating L tones.

	tone of verb					
tone of noun	H	M_a	M_b	L_a	L_b	MH
LM, LH	LH	LM	LM	LH	LH	LM+MH#
M	#L	M	M	M.L	M.L	M.MH
L	#L	L	M	L	L	L
H	#L	#L	#L	MH#	MH#	L#
MH	H#	H#	H#	MH#	MH#	H#
M	#L	M	M	#L	#L	MH#
#H	#L	#L	#L	MH#	MH#	#L
MH#	MH#	MH#	MH#	MH#	MH#	#L
H$	#L	#L	#H / #L	MH#	MH#	H#
L	L	L	L	L	L	L+H#
L#	L#–	L#–	L#–	L#–	L#–	L#–
LM+MH#	LM–+#H	LM–+#H	LM–+#H	LM+MH#	LM+MH#	LM+H$
LM+#H	LM–+#H	LM–	LM–	LH–	LM+MH#	LM+MH#
LM	LM–+#H	LM–	LM–	LH–	LH–	LM+MH#
LH	LH–	LH–	LH–	LH–	LM+MH#	LH–
H#	H#–	H#–	H#–	H#–	H#–	H#–

10.2 Analogy

10.2.1 General principles

Analogy is a process of language change whereby a linguistic form is altered to resemble another form or pattern within the language system. The typical result is that less common word forms are reshaped to conform with more common patterns, levelling down irregularities. But the driving force behind analogy is not likely to be a preference for regularity: irregular forms can endure over centuries, so long as they are frequent enough in discourse that they do not go out of memory.

The real driving force is likely to be that a speaker is in doubt about a certain form, and coins a plausible form on-the-fly by imitating other forms that exist in the language.[3] For instance, at some point, a speaker of English who was in doubt about the past tense of *to dive* reasoned that *dove* is to *dive* as *drove* is to *drive*, and introduced an innovative form, *dove*, which has now become standard in North America, replacing the earlier form *dived*. From a morphological point of view, analogy can be viewed as a process of regularization. From the point of view of phonetic change, on the other hand, the piecemeal modifications introduced by analogy tend to obfuscate regular correspondences.

Case studies of analogical reanalysis reveal the complexity of individual situations. For instance, in the Bantu language Eton, the stem of the possessive 'my' ends in /**ɔ**/ in association with nouns of classes 1 and 3: /**-amɔ**/, and in /**a**/ elsewhere: /**-ama**/. This irregularity is due to a mechanism of analogical morphophonological reanalysis that changed the original /**a**/ of the class 1/3 forms to /**ɔ**/. In Eton, there is a |**ɔ**| morphoneme whose morphologically conditioned realizations include /**wa**/. Commonly occurring sequences of /**w**/+/**a**/, although separated by a morpheme boundary, were reinterpreted as realizations of this morphoneme (Van de Velde 2008b). This example illustrates how morphophonological analogy can disregard morphological boundaries.

Analogy is, by definition, irregular and unpredictable. One may nonetheless believe that evidence from case studies gradually adds up, allowing for an increasingly refined understanding of this phenomenon.

> [I]t is possible to some extent to constrain the space of hypotheses involving analogy, and research on the general principles of analogy is of utmost importance for historical linguistics. (Jacques 2016: 239)

[3] This perspective was proposed by Nathan Hill (p.c. 2015).

To date, studies about the principles of analogy (e.g. Kuryłowicz 1944, Lahiri 2000, Hill 2007, Blevins & Garrett 2009, Juge 2013, Hill 2014) contain little discussion of tone, while studies on tone (e.g. Pike 1948, Fromkin 1978, Pulleyblank 1986, Gussenhoven 2004: 229-231) contain little on analogy. Yet it seems intuitively clear that morphotonology, like other aspects of morphophonology, can be subject to analogical levelling.

10.2.2 Analogy in Yongning Na morphotonology

Traces of analogy are found among the tones of compounds and affixed forms, as was pointed out in several sections (§4.4, 4.5.2, 5.1.3.2, 5.1.3.8, and 6.4.1). It appears highly plausible that the tantalizingly similar, but not identical tonal paradigms of classifiers – H_a and H_b, M_a and M_b, MH_a and MH_b, L_a, L_b and L_c (see Chapter 3) – have undergone a degree of analogical levelling without becoming fully identical. The existence of variants for some combinations, along with consultants' occasional hesitations and confusions (errors) during elicitation sessions, all point to the presence of contradictory pressures: on the one hand, a tendency towards analogical simplification, and on the other, a counteracting tendency to maintain the distinct identity of the different classes.

This is an area where the description of a single language variety reaches its limits, and a variationist approach would be called for. Such a study could begin with the closest language varieties, studying phenomena of accommodation among speakers within the hamlet under study, before extending the investigation to dialects spoken in the Yongning basin and its immediate vicinity.

10.3 Syllable reduction

Syllable reduction is a salient topic in the phonology of Naxi, a language closely related to Na. The topic was encountered in Naxi narratives and investigated in a phonetic study.

> The Western dialect of Naxi has four lexical tones: High, Mid, Low and Rising; the latter is rare in the lexicon. Rising contours on monosyllables are frequent in connected speech, however, as a result of a process of syllable reduction: reduction of a morpheme carrying the High tone results in re-association of its tone to the syllable that precedes it in the sentence, creating a rising contour. An experiment (with one speaker and five listeners) establishes that there is not only one rising contour that originates

> in tonal reassociation, as reported in earlier descriptions, but two: Low-to-High and Mid-to-High – as could be expected by analogy with phenomena observed in Niger-Congo languages and elsewhere. A second set of experiments (…) investigates the reduction of Mid- and Low-tone syllables: they reduce to [**ə̄**] and [**ə̀**], respectively, and coalesce with the preceding syllable (in Naxi, syllabic structure is simply *consonant+glide+vowel*). Unlike High-tone syllable reduction, this process stops short of complete tonal de-linking. (Michaud & He 2007: 237)

The topic was taken up by a native speaker, He Likun 和丽昆, who carried out a highly detailed study. Not only did he bring out further patterns, but he also advanced the discussion about the applicability of the notions of floating tones and tonal reassociation, confirming that complete tonal de-linking (producing floating tones) is not the majority case.

> Previous studies report that some syllables carrying high tone in Naxi get deleted (in function words), leaving only a "floating tone" which reassociates to the previous syllable in the sentence. Re-examination of this phenomenon reveals that syllable reduction is extremely common in present-day Naxi, and that it is not limited to high-tone syllables: there are as many as twelve possible tonal combinations in syllable reduction. These reduced syllables behave in ways that are tantalizingly similar to floating tones, but do not entirely match this concept. The study of reduction and coarticulation phenomena in Naxi opens a window onto the phonetic complexity of tonal variation and change; it provides insights into evolutionary paths in tone systems, and their ties to processes of grammaticalization. (Hé & Liú 2021)

In Na, syllable reduction leading to tonal change is nowhere as salient as in Naxi. Nonetheless, it is attested in at least one case, described in §5.3.1. The 3rd-person pronoun, when combined with the possessive, appears in two forms: /**ʈ͜ʂʰɯ˧=bv̩˧**/ and /**ʈ͜ʂʰɯ˧=bv̩˩**/. These are not tonal variants but semantically distinct forms. The latter, /**ʈ͜ʂʰɯ˧=bv̩˩**/, is a reduced form of /**ʈ͜ʂʰɯ˧=ɻ̩˩=bv̩˩**/, where /=**ɻ̩˩**/ is the associative plural. Thus, /**ʈ͜ʂʰɯ˧=bv̩˩**/ means 'their', whereas /**ʈ͜ʂʰɯ˧=bv̩˧**/ simply means 'her/his'. In the expression /**ʈ͜ʂʰɯ˧=bv̩˩**/, the associative plural /=**ɻ̩˩**/ is fully elided: it leaves no segmental trace, only a tonal difference on the possessive particle. This example illustrates how segmental simplifications can contribute to the expansion of morphotonology, a type of diachronic change that is especially well-attested in Bantu languages.

10.4 Disyllabification

As mentioned at the outset of Chapter 4, many roots that were once phonologically distinct have become homophonous in Na, as in other Sino-Tibetan languages that have undergone considerable phonological erosion (such as Tujia, Bai, Namuyi, and Shixing). As a consequence, there is a strong tendency towards disyllabification.

If each tonal combination of two monosyllables were to create a distinct tonal category for the resulting disyllable, this would multiply the number of tones by squaring: six tones on monosyllables could theoretically yield 6×6=36 tonal categories on disyllables. The actual number is much smaller: eleven tone categories for disyllabic nouns. Investigating the relationship between the tones of monosyllables and those of disyllables holds promise for an understanding of the dynamics of the tone system.

10.4.1 A dynamic analysis of compound nouns

The analysis of compound nouns in Chapter 4 aimed to elucidate the relationship between input nouns and the resulting compound. The notations chosen for the tones of compounds were designed to bring out their internal makeup. For instance, the combination of a #H-tone determiner and an LM-tone head yields an M.H surface phonological tone pattern, as in (4).

(4) ʐwæ˧-ɣɯ˥
ʐwæ˥ ɣɯ˩˧
horse skin
'horse's skin' (*DetermCompounds6.24* {0004454#W24}, *7.67-68* {0004456#W67}, *12.44* {0004462#S44})

The process leading from the input tones to the tone of the compound can be interpreted as follows: the lexical tone of the determiner, being a floating H tone (which is never realized on the lexical item itself but only on a following syllable), associates to the second syllable of the compound. The assignment of surface tones then takes place according to the general rules governing the association of #H tone in Yongning Na. Since the tone group contains a following syllable that can host the floating tone (namely, the second syllable of the compound), the H tone attaches there, while the first syllable of the compound receives M by default (through Rule 2). In Chapter 4, the notation used for this combination was #H−, using the symbol '−' to stand for the last syllable of the first component of the compound. The complexity of this notation is justified by the fact that it reflects a hypothesis about how the compound's tone pattern obtains. Such notations are referred to below as *source-oriented*.

In terms of end result, on the other hand, the compound in (4) belongs to the tone category H#: it carries a final H tone that does not shift. Disyllabic compounds made up of the combination of a #H-tone determiner and an LM-tone head thus feed into the lexical category of H# disyllables. This equally valid notation, H#, is referred to below as *result-oriented*.

Likewise, the source-oriented notation –L corresponds to the result-oriented notation L#: assigning an L tone after the juncture between the two parts of the disyllabic compound yields the same outcome as assigning a final L tone to the entire expression. Table 10.4 provides a summary.

Table 10.4: Source-oriented and result-oriented notations of the tones of compounds: three examples.

		phonological analysis	
input tones	surface phonological tone	source-oriented	result-oriented
#H and LM	M.H	#H–	H#
M and LM	M.L	–L	L#
M and L	M.L	–L	L#

The table presenting the tone patterns of disyllabic compounds (Table 4.2 in Chapter 4) is rewritten below as Table 10.5, adopting a result-oriented notation that eliminates all references to the juncture between the two components of the compound (transcribed by means of the symbol '–' in Table 10.4). Each row corresponds to a tonal category of determiners, and each column to a tonal category of heads.[4]

All of the tone categories observed on disyllabic nouns in Yongning Na appear in Table 10.5, with the sole exception of M. This reveals that the synchronically productive tone rules that apply in compounds feed into all of the tone categories of disyllables except M.

10.4.2 Possible origins of disyllables in view of their tone

Table 10.6 flips around the morphotonological rules set out in Chapters 4 and 5 to provide an overview of possible origins of disyllabic items in view of currently

[4]The same treatment cannot be extended to compounds of more than two syllables: describing their tone patterns without reference to the juncture between the input nouns would require a complete change of notation, for instance specifying the tone of each syllable individually.

Table 10.5: The tones of disyllabic compounds, adopting a result-oriented notation. The four combinations transcribed differently from Table 10.4 are set in italics.

tone	LH; LM	M	L	#H	MH#
LM	LM	LM	LM	LM+#H	LM+MH#
LH	LH	L	LH		
M	*L#*	#H	*L#*	#H	MH#
L	L				
#H	*H#*	#H			*L#*
MH	H#			H$	

productive tone rules. A '–' indicates that no example was found. H# tone as an outcome of suffixation is labelled as *dubious* because no firmly established pattern of correspondence has been identified between monosyllables and suffixed forms carrying //H#// tone; only isolated tokens exist, whose analysis remains problematic. For instance, /**tse˧mi˥**/ 'cigarette lighter' has //H#// tone and looks like it results from suffixation, but no corresponding monosyllable has been found, so that there is no possibility to establish a tone correspondence between root and suffixed form.

The bird's-eye view in Table 10.6 can provide a clue for the analysis of disyllabic words whose etymology is uncertain.

10.4.3 Recovering the tones of nouns on the basis of compounds

It is tempting to try to work backwards from the tones of compounds to those of their constituting elements. For instance, 'elder sibling (brother or sister)' is most commonly realized as /**ə˧mv̩˩**/ (tone: L#), but it has a variant with MH# tone: /**ə˧mv̩˧˥**/. The tone of the coordinative compound /**ə˧mv̩˧-gi˥zɯ˩**/ 'brothers' (made up of 'elder sibling' + 'younger brother') is the one expected for an input MH# tone, not an input L# tone. This could suggest that the MH# variant of 'elder sibling', /**ə˧mv̩˧˥**/, was the form that entered into the formation of the compound. Seen in this light, the relative rarity of the MH# variant in present-day speech, where /**ə˧mv̩˩**/ is far more common, suggests that the MH# variant /**ə˧mv̩˧˥**/ is not a recent innovation but a form which is currently losing ground to /**ə˧mv̩˩**/.

However, the greatest care must be exercised when attempting to recover earlier tones through such inferences, as different tone rules may have applied at

Table 10.6: Possible origins of disyllabic items, in view of currently productive tone rules.

tone	compounding	suffixation	prefixation
M	–	yes	yes
#H	yes	yes	–
MH#	yes	–	yes
H$	yes	yes	yes
L	yes	yes	yes
L#	yes	–	yes
LM+MH#	yes	–	–
LM+#H	yes	yes	–
LM	yes	yes	–
LH	yes	yes	–
H#	yes	*dubious*	–

different stages of the language's history. As pointed out by Nathan Hill (p.c. 2016), it remains entirely possible that the MH# variant /**ə˧mɣ̩˥**/ for 'elder sibling' was not inherited but instead inferred from the compound – whatever the origin of the compound's tone may have been –, making it an innovation.

10.5 The influence of bilingualism with Mandarin

Language contact is well known to be a key factor in linguistic change. an exemplary illustration of how the study of present-day contact dynamics can shed light on prosodic systems is provided by Kubozono (2007) in his analysis of Kagoshima Japanese. The Kagoshima dialect has two prosodic patterns at word level: one (Tone A) with a high tone on the penultimate syllable (i.e. a fall in pitch from the penultimate to the last syllable) and the other (Tone B) with a high tone on the final syllable (i.e. no pitch fall within the word). At the time of study, this dialect was undergoing tonal change due to influence from Tokyo Japanese, the national standard: words that involve an abrupt pitch fall in Tokyo tended to be reinterpreted as carrying Tone B in Kagoshima, and those without one as Tone A. This sheds light on the issue whether the Kagoshima prosodic system is best analyzed as tonal or accentual: "the tonal changes in question can best be understood if an accentual analysis of Kagoshima Japanese prosody is adopted in preference to the traditional tonal analysis" (Kubozono 2007: 348; further supporting evidence from a follow-up study of twenty speakers is reported by Ota et al. 2016).

Since the present volume adopts a synchronic perspective, past contact between Na, Tibetan, Chinese, Pumi, Lisu, Naxi and other languages is not investigated (apart from brief remarks in §11.3.1). Instead, this section focuses on the current landscape of language contact, in which Mandarin has, by far, the leading role. To take the example of the main consultant, Mandarin is the only language other than her mother tongue of which she has any knowledge.[5] The guiding principle in focusing on the influence of Mandarin is that "extracting as much historical information from clear contact phenomena as possible before attempting greater time depths may be the order of investigation most likely to be fruitful" (Souag 2010: 485).

Mandarin is a relatively recent arrival in this area. The feudal chieftain of Yongning spoke Na, and up until the mid-20th century the Na language held a dominant position in the Yongning basin. Few Han Chinese had settled in Yongning, and those who did learnt Na, which was the locally dominant language and served as the lingua franca in the Yongning marketplace for speakers of other languages, such as Pumi, Lisu, and Nosu (Nuosu). While there can be no doubt that the Na language received various influences in the course of its history, bilingualism was not widespread among native speakers of Yongning Na: locally, the expectation was that speakers of other languages would learn some Yongning Na, rather than the other way round. It is reported by Mrs. Latami that numerous Na speakers had very little command of other languages, or none at all. This situation is somewhat uncommon in this region, at the Sichuan-Yunnan border. For instance, the small Na-speaking community in the neighbouring county of Muli 木里 (Shuiluo 水落 township) is bilingual in Shixing (Xumi) and has some command of Tibetan and Pumi, while the variety of Na spoken in Guabie 瓜别 has long been subject to influence from other languages, in particular Pumi and Nosu.

[5] Since moving to Lijiang (2010), she has had relatively frequent contacts with Naxi speakers, however, leading to at least one amendment to her Na vocabulary. The Na refer to the Naxi as //**nɑ˩hĩ#˥**//, a calque modelled on the structure of the Naxi word /**nɑ˩çi˧**/, made up of an endonym which is segmentally identical in both languages (/**nɑ**/) combined with the word for 'person, human being': Naxi /**çi˧**/, Na //**hĩ˥**//. Initially, the main consultant used the Na pronunciation //**nɑ˩hĩ#˥**// when speaking with Naxi people in Lijiang. But to a Naxi listener, this pronunciation does not sound right, as Naxi does not have nasalization in the syllable for 'person, human being'. Whether at the suggestion of Naxi speakers or through spontaneous accommodation, she shifted to referring to the Naxi as /**nɑ˩ɕi˥**/, denasalizing the second syllable. This amounts to borrowing the word from Naxi, instead of calquing it using Na morphemes.

While Yongning still preserves the role of a meeting place and market centre for inhabitants of surrounding mountain villages, for instance for the Nosu and Pumi (Wellens 2006: 85), language shift from Na to Mandarin is now under way in Yongning. All Na speakers today have at least some command of Mandarin. While some members of the community express regret that their language is falling into gradual disuse, proficiency in Mandarin – one of the keys to success in present-day society – is widely valued far above proficiency in Na. The blending of Na with Mandarin is not stigmatized but instead accepted with tolerance, an attitude that facilitates linguistic change.

Within China, all level-tone systems are currently subject to the same pressure towards alignment with the tone system of Mandarin. The sociolinguistic landscape of Yongning Na offers interesting opportunities for studying the effects of bilingualism – specifically, *non-egalitarian bilingualism*, to adopt a notion from Haudricourt (1961) – in languages with widely different tone systems. The account of Na tone presented in Chapters 2 to 7 of this volume makes it clear how much this system differs from those of the Chinese dialects to which Na speakers are currently exposed, namely Southwestern Mandarin and Standard Mandarin. As discussed in Chapter 12, tones in Yongning Na are phonetically simple, consisting of three level tones and their combinations, whereas Mandarin tones are phonetically complex. The Yongning Na tone system also includes certain oppositions that are neutralized when a word is said in isolation.

When they learn Mandarin, Na speakers come to terms with a tonal system that is structured in a fundamentally different way. In Mandarin, leaving aside marginal cases of toneless syllables and tone sandhi, each syllable has its own tone, which surfaces as such in isolation. In contrast, the discrepancy between underlying forms and surface forms in Yongning Na makes its tones difficult to handle for bilingual speakers who receive greater exposure to Mandarin than to Na.[6]

10.5.1 Loss of tone categories that do not surface in isolation

Among younger speakers, especially those who went to boarding school and predominantly use Southwestern Mandarin, there is a tendency to overlook the tonal oppositions that are neutralized in isolation, such as that between the //H#//

[6]On the effects of bilingualism with Mandarin on the tone systems of other minority languages in rural Southwest China, see the two case studies presented by Stanford & Evans (2012). These concern (i) Sui (Tai-Kadai family), which has a system of six phonetically complex tones in unchecked syllables and two in checked syllables, and (ii) Southern Qiang / Rma (Sino-Tibetan family), whose tone system is based on a binary opposition between H and L levels.

and //H$// tones. The surface tone pattern of a word is reinterpreted as its underlying pattern, causing an upheaval in the architecture of the tonal system.

For instance, the family name of the main language consultant is realized in isolation as /**lɑ˧tʰɑ˧mi˥**/, while its associative form ('the Latami clan, the Latami family') is /**lɑ˧tʰɑ˧mi˧=ɻ̍˥**/. This reveals that the family name belongs to tone category H$, indicating that the underlying form is //**lɑ˧tʰɑ˧mi˥$**//. But if these alternations are ignored and the M.M.H surface tone pattern in /**lɑ˧tʰɑ˧mi˥**/ is taken at face value, the word will instead be interpreted as carrying an H tone on its final syllable, leading to its classification within phonological category //H#//, i.e. //**lɑ˧tʰɑ˧mi˥**#//. As a consequence, when building phrases, e.g. forming an associative plural ('the Latami clan') by adding the suffix /**=ɻ̍˩**/, the H tone will be left sitting on the final syllable of the noun, yielding ‡ **lɑ˧tʰɑ˧mi˥=ɻ̍˩**. This pattern is observed in the speech of consultant F5, a speaker proficient in both Na and Mandarin, who was 35 years old at the time of fieldwork.

An especially difficult opposition to acquire is that between the LM and LH tone patterns over disyllables, as it only surfaces when the noun is followed by a clitic. For instance, //**bo˩mi˧**// 'sow, female pig' and //**bo˩ɬɑ˥**// 'boar, male pig' carry the same tones not only in isolation (/**bo˩mi˥**/ and /**bo˩ɬɑ˥**/) but also in most other contexts. The few environments in which this tonal opposition becomes apparent include constructions such as /**bo˩mi˧=bv̩˧**/ 'of (a) sow' vs. /**bo˩ɬɑ˥=bv̩˩**/ 'of (a) boar'. Even there, the H component of the lexical LH pattern is not detectable as such at a phonetic level: given tonotactic restrictions, the sequences L.H.L and L.M.L are neutralized on the phonological surface, so that there is no phonetic cue to the presence of an H tone in /**bo˩ɬɑ˥=bv̩˩**/. The H tone in //**bo˩ɬɑ˥**// is only manifested indirectly: it induces lowering of the following possessive clitic to L. In view of the complexity of this area of the morphotonological system, it does not come as a surprise that the lexical opposition between LM and LH has been lost by some speakers, such as F5 and F6, for whom the tones of 'sow' and 'boar' have fully merged.

10.5.2 Simplification of morphosyntactic tone rules

Examining the tables in Chapters 3 to 6, the complexity of Na tonal morphosyntax may look mind-boggling. But the rules are productive, and the syntactic structures at issue (object-plus-verb, compound nouns…) are so frequent that they were not particularly challenging for first-language learners steeped in a Na-speaking environment. For speakers with limited exposure and practice, however, these tonal patterns become problematic: they are difficult to recall and apply if one does not practise the language regularly. This holds true for the visiting linguist as well as for many Na speakers below the age of sixty, many of whom use Mandarin more than Na in daily life.

Deviant tonal patterns serve as harbingers of language change. To the linguist, such patterns serve as early indicators of ongoing structural reorganization within the tonal system. These range from occasional slips of the tongue to entrenched habits.

Instances of hesitation and tonal simplification can be observed in the speech of M23, a bilingual language consultant. In subject-plus-verb and object-plus-verb constructions, as well as in compounds, the //L// tone, which surfaces as /M/ in isolation, tends to be neutralized with //M//. For instance, consultant M23 consistently realizes 'the sheep came' as /**jo˧ | tsʰɯ˩-ze˩**/ (example (67) in Chapter 6), instead of the expected form in (5). In the view of the main consultant (Mrs. Latami, F4), the M.L.L pattern for 'the sheep came' is a mere variant, and is moreover condemned as a slip of the tongue; in M23's speech, it has become the usual form.

(5) jo˩ tsʰɯ˩-ze˩˥
    jo˩ tsʰɯ˩$_a$ -ze˧$_b$
    sheep to_come.PST PFV
    'the sheep have come'

Similarly, in M23's speech the phrase 'sheep's muzzle' is //**jo˧-ɲi˧gɤ˧**//, i.e. a simple concatenation of the surface forms of the two nouns, diverging from the conservative pattern found in F4's speech, shown in (6).

(6) jo˩-ɲi˩gɤ˩˥
    jo˩ ɲi˧gɤ˧
    sheep nose/muzzle
    'sheep's muzzle'

In the conservative form, (6), the tonal pattern of the compound is phonologically identical with the lexical L tone of the determiner, //**jo˩**// 'sheep'. This yields //**jo˩-ɲi˩gɤ˩**// 'sheep's muzzle', which surfaces as /**jo˩-ɲi˩gɤ˩˥**/ following the postlexical addition of a final H tone, in accordance with Rule 7: all-L tone groups are not allowed in Yongning Na, and if a tone group only contains L tones, a postlexical H tone is added to its final syllable. A speaker needs a good command of Na morphotonology to implement this conservative tonal pattern. The L tone of 'sheep', which does not even surface in isolation, extends over no fewer than three syllables in succession in the compound, whereas the lexical tone of the head noun does not get to express itself. The underlying form of ‘nose, muzzle’ is //**ɲi˧gɤ˧**//, which surfaces as /**ɲi˧gɤ˧**/ in isolation but is changed to /…**ɲi˩gɤ˩**/ in (6). Such tonal processes are alien to Mandarin. In light of this typological discrepancy, it is unsurprising that less proficient speakers of Yongning Na who are exposed to Mandarin on a day-to-day basis should have hesitations and (occasionally or regularly) opt for a direct concatenation of the tones as they surface in isolation, as is the case in their second language (Mandarin), instead of applying the morphotonological rules of Na.

These examples illustrate the complexity of phenomena of language variation and change. As noted earlier, a given change may constitute a simplification from one point of view and a complexification from other points of view. Adopting an M.L.L tone pattern (/**jo˧ tsʰɯ˩-ze˩**/) for ‘the sheep arrived’, instead of the conservative L.L.L pattern in (5), can be seen as a simplification insofar as the subject bears the same tone as in isolation. However, it simultaneously increases the frequency of contexts in which the lexical L tone of ‘sheep’ does not surface, making it more complex for language learners to arrive at the identity of this word’s underlying lexical tone.

10.5.3 Straightening out irregular tone patterns

In addition to losing some tone categories, less proficient speakers tend to regularize irregular patterns, for want of the detailed knowledge of exceptions that fluent speakers possess. For instance, the word for ‘powder, flour’ is /**tsɑ˧bɤ˧**/, which bears M tone. According to the synchronic rules that govern the tone patterns of compound nouns, its combination with /**lv̩˧mi˧**/ ‘stone’ should yield a simple M-tone output, † **lv̩˧mi˧-tsɑ˧bɤ˧**. (This compound means ‘fine sand’.) However, in the speech of the older generation of speakers, compounds involving /**tsɑ˧bɤ˧**/ ‘powder, flour’ have an irregular tone pattern: the second part (the head noun ‘powder, flour’) systematically carries L tone. These compounds belong in a set of expressions referred to as ‘Tibetan compounds’ (see §4.2.11.3 and §4.4): they contain Tibetan loanwords and consistently have L tone on their second part. Examples were provided in Table 4.14 and Table 4.27.

This irregularity is levelled out in the speech of less proficient speakers, who produce these compounds with an M.M.M.M tonal string (phonological analysis: /M/). Thus, instead of the conservative realizations, they use forms such as ‡ **lv̩˧mi˧-tsɑ˧bɤ˧** ‘fine sand’, ‡ **qʰɑ˧dze˧-tsɑ˧bɤ˧** ‘sweetcorn flour’, and ‡ **dze˧ɭɯ˧-tsɑ˧bɤ˧** ‘wheat flour’.

10.5.4 Cases where MH tone fails to unfold: Towards a syllable-tone system?

In Yongning Na, plentiful synchronic evidence supports the analysis of contours as sequences of levels. There are nonetheless some weak hints of a tendency for MH contours to be treated as units associated with a syllable. When saying MH-tone monosyllabic nouns in the frame 'This is …', Mrs. Latami (consultant F4) occasionally realized the target noun with a rising contour, as illustrated in (7). The standard realization is provided in (8).

(7) ‡ ʈʂʰɯ˧ | tsʰɯ˧˥ ɲi˩. ≈ ‡ ʈʂʰɯ˧ | tsʰɯ˧˥ ɲi˥.
ʈʂʰɯ˥ tsʰɯ˧˥ ɲi˩
DEM.PROX goat COP
'This is a sheep.'

(8) ʈʂʰɯ˧ | tsʰɯ˧ ɲi˥.
ʈʂʰɯ˥ tsʰɯ˧˥ ɲi˩
DEM.PROX goat COP
'This is a sheep.'

When her attention was drawn to these discrepancies, the consultant said: /**ɖɯ˧-bæ˧ lɑ˧ ɲi˥**/, "It's just the same". Intonationally, realization of the MH contour on the syllable to which it is lexically attached tends to occur when the word is emphasized.

This tendency surfaces here and there in the recordings. An example is the phonetic realization of /**ʈʂʰæ˧-pɤ˥to˩**/ 'even a deer' as [**ʈʂʰæ˧˥-pɤ˥to˩**] (in the recording *NounsEven.7* {0004566#W7}), where:

- [**pɤ˥**] carries an H tone due to reassociation of the H part of the MH contour of /**ʈʂʰæ˧˥**/ 'deer', as expected; and
- [**ʈʂʰæ˧˥**] exhibits an MH contour, reflecting incomplete dissociation of the H component of its phonological MH contour.

This is only a weak tendency, which by no means warrants the conclusion that Yongning Na is on its way towards adopting a syllable-tone system. A detailed cross-linguistic phonetic study would be required to assess how widespread such tendencies are among the world's level-tone systems. Such a study might reveal that Yongning Na is not exceptional in this respect. The computation of tone sequences is not a purely mechanical process, and slips of the tongue whereby an

H tone does not fully dissociate from the syllable to which it is lexically attached may not come as a surprise to linguists familiar with level-tone systems. Even so, this synchronic tendency appeared well worth mentioning in relation to the influence currently exerted by Mandarin.

Naxi, a close relative of Yongning Na, can be hypothesized to constitute an example of a language which the influence of Mandarin is nudging towards reinterpretation of its tones as units attached to the syllables, rather than as High, Mid and Low levels that can combine among themselves. Of the four tones of Naxi, High, Mid, Low, and Rising, the last is especially revealing in this respect. It is clearly an innovation that emerged in a system previously containing only three levels: High, Mid and Low. A plausible historical scenario is that rising contours developed through processes of syllable reduction and subsequently became lexicalized in certain words. Once such rising tones had become part of the Naxi lexicon, the way was paved for the assignment of a /LH/ tonal sequence to Mandarin words with rising tone. Borrowings from Southwestern Mandarin then consolidated this marginal tonal category by giving it substantial lexical expansion (Michaud 2003, 2006b, Michaud & He 2007). In recent decades, bilingualism with Mandarin gradually tilted the perception of Naxi tones towards a syllable-tone system – to the point that it is now an open question whether the rising tone of Naxi should be analyzed as a sequence of levels (L+H or M+H) or as an indecomposable phonological unit (akin to contours in Mandarin). Given that most Naxi speakers are highly proficient in Southwestern Mandarin, it is unclear to what extent they maintain a cognitive distinction between the tone systems of the two languages. To venture a hypothesis, the lexical rising tone of Naxi currently appears to function primarily as an indecomposable contour.

10.5.5 A topic for future research: Influence of language contact on intonation

Intonation is particularly susceptible to carry-over from one language to another in the speech of bilinguals (see e.g. Lai & Gooden 2022). A striking example involving Vietnamese speakers in a French-speaking environment is reported by Dô et al. (1998: 401). In the case of Yongning Na, a full-fledged study of intonation should include a description and analysis of the intonation patterns used by Na speakers when speaking Mandarin, comparing them with their intonation in Na and investigating cases of interaction between the two. This is not an easy topic, however. The main consultant uses Mandarin reluctantly and hesitantly, lacking confidence and feeling awkward in the language. Her own assessment is that she will never be able to speak "proper Chinese" and will make do with "pig Chinese" until her last breath.

A further complication is that the variety of Chinese to which she was occasionally exposed until 2000 was Southwestern Mandarin, whereas since 2000 she has had regular exposure to Standard Mandarin through television. While the tone systems of these two dialects are almost identical phonologically, the four tones' phonetic templates differ sufficiently to complicate accommodation to their respective intonation patterns. Consultant F4 does her part in dialogues with Mandarin-speaking relatives and acquaintances, but recording these interactions and transcribing them with the speakers would have run counter to the implicit focus of our collaboration on her mother tongue, Yongning Na. The consultant's comfort zone was respected, and no pieces in Mandarin were recorded.

As a (lame) consolation for not being able to offer data on this topic, here is an example from another language illustrating how language contact can exert a strong influence on intonation. Wolof, one of the nontonal languages of the Niger-Congo family, has been described as typologically unusual in that it does not have any intonational marking of focus (Rialland & Robert 2001). Information structure is conveyed through verbal morphology: there are three "nonfocusing conjugations" and three "focusing conjugations". The latter "vary according to the syntactic status of the focused constituent: subject, verb or complement (in the wide sense of any constituent that is neither subject nor main verb)" (Rialland & Robert 2001: 895). As Wolof ascends "from its origins in the heartland of Senegal to the status of urban vernacular and national *lingua franca*" (McLaughlin 2008: 142), it is increasingly acquired as a second language by speakers of diverse linguistic backgrounds.

Informal discussion with a speaker of Wolof who also speaks some Bambara suggests that, in her speech, focus is clearly marked intonationally: the focused constituent 'Peer' (a proper name) in (9) is pronounced with a raised pitch, unlike in (10), where it is not under focus.

(9) … Peer moo ko lekk
Peer SUBJEMPH.3SG OPR to_eat
'It was Peer who ate it.' (SUBJEMPH stands for *subject-emphatic*, and OPR for *object pronoun*.)

(10) … Peer lekk na
Peer to_eat PFT.3SG
'Peer has eaten.' (No focused constituent.)

This attests to the possibility of carry-over of intonation patterns from Bambara to Wolof in bilingual speakers. Such contact-induced influences can have far-reaching consequences for the evolution of a language's intonation system.

11 Yongning Na in dialectal and areal context

As a preliminary to the typological reflections set out in Chapter 12, this brief chapter offers a few tentative observations on Yongning Na in its dialectal and areal context.

11.1 Na dialectology: A promising field of study

Since the publication of the first edition of this volume, significant progress has been made in Na dialectology, expanding both the breadth and depth of coverage. In terms of scope, the dialect of Luggu (**lo˧gv̩˩**; in Chinese: Běiqúbà 北渠坝), which is not mutually intelligible with that of Yongning, has been documented (Lǐ Zǐhè 2021; the fieldwork location is Yanba village, Daxing, Ninglang county: 宁蒗县大兴镇堰坝村). Closer to Yongning, a paper is dedicated to tone in the dialect of Lomae (**lo˩mæ˩**; in Chinese: Qiánsuǒ 前所) (Xu 2015). In terms of depth, the dialect of Lataddi, previously described by Dobbs & La (2016), has been the subject of a Ph.D. dissertation (Fily 2022). These additions contribute to a growing empirical basis for the comparative study of tone across Na dialects, ultimately aiming at understanding tonogenesis in Naic languages – a topic that is recognized as highly complicated (Li 2024: 170).

Beyond Na proper, it appears interesting to touch upon a few other Sino-Tibetan languages of the area with which Na may have been in contact in the past. The observations presented here are admittedly preliminary and tentative.

11.2 Naxi and Laze: Close relatives, but not part of a convergence area

It has been observed that "the same phonological processes such as tone sandhi and lengthening obviously make reference to different prosodic domains within the same language family (Bantu)" (Zerbian 2006a: 132). High diversity is also found within Naish, even though this lower-level grouping is incomparably smaller than Bantu.

Comparison of Na with Naxi and Laze is fundamental for diachronic research, as these two languages are Yongning Na's closest known relatives (see §1.1.3). But from an areal point of view, I have not been able to find evidence of diffusional convergence between Na and its sister languages. It seems that these languages have not formed a convergence area. On the contrary, their histories have been marked by strong social, political and cultural *divergence* since the 14th century. The Naxi chiefdom of Lijiang underwent increasing Sinicization (and finally came under direct Chinese administration in the 18th century), whereas in Yongning and Muli, the feudal chieftain system persisted until the mid-20th century due to the failure of military campaigns to subdue the Liangshan Yi area, which constitutes the gateway to these peripheral regions (see Appendix B, §B.1).

The Laze are a small group of some four hundred people who migrated from Yanbian to their current location in Muli towards the end of the 19th century. They are reported to be among the speakers of Naish languages who left Yanbian as it became predominantly Yi-speaking, but little is known about sociolinguistic configurations prior to the influx of Yi clans. Consequently, I am not in a position to assess how areal convergence may have contributed to shaping the prosody of Laze in past centuries. The present-day areal situation is not well-understood either, as the other Naish dialects spoken in the vicinity of the Laze villages remain, to my knowledge, undocumented.

From a prosodic point of view, Naxi and Laze currently exemplify the *nonrestricted tone* type as defined by Voorhoeve (1973): tones are assigned to individual syllables without regard to the tonal pattern of the entire word or phrase. In particular, in these two languages there is no restriction on the number of successive H tones, unlike in Yongning Na, where H tone is culminative (that is, there is at most one H tone per tone group). Interestingly, culminativity has been proposed as an areal characteristic of the languages that have long coexisted in the county of Muli (Chirkova 2012: 160).[1]

In terms of this important property of the prosodic system, Yongning Na does not pattern with Naxi or Laze, but with languages that have long been spoken

[1]Muli was a semi-independent kingdom ruled by Pumi hereditary lama kings until the mid-20th century. It is important to clarify that the claim is not that Muli county as a whole constituted a single convergence area, but rather that convergence took place in certain parts of Muli that were historically multilingual over extended periods. The high mountains and deep valleys of Muli create formidable obstacles to communication, and different parts of the county constitute strikingly different sociolinguistic environments, with different languages in contact and different relationships of prestige among ethnic groups and their languages.

in Muli, such as Pumi, Namuyi, Shixing (Xumi), Lizu,[2] and the local dialect of Tibetan. Among these, Pumi deserves particular attention, as it is also spoken in Yongning.

11.3 Comparison with Pumi

The prosodic system of Na is remarkably close to that of Pumi (also known as Prinmi), a neighbouring Sino-Tibetan language.

11.3.1 The tone group and its role in conveying information structure

In Wadu 瓦都 Pumi, as in Yongning Na, utterances are structured into *tone groups*.

> Within a tone group, the underlying tone of one lexical element (usually the left-most element) spreads (usually rightwards) to the adjacent morphemes in the same tone group (…). The remaining elements in a tone group are assigned default low surface tone. Tone does not spread across tone group boundaries. (Daudey 2014: 66)

As in Yongning Na, the tone group plays a key role in conveying information structure.

> [S]ome elements always combine with others into a single tone group, some elements always form a tone group by themselves, and for some elements, speakers can decide to combine or not combine them into tone groups. The latter elements are the most interesting, in that they allow the speaker to express pragmatic differences through the choice of combining them or not. (Daudey 2014: 68)

The parallel with the observations on Na set out in Chapter 7 is striking. Such similarities raise the question of whether language contact has played a role. The variety of Pumi studied by Henriëtte Daudey (Wadu Pumi) is spoken in the plain of Yongning, where the Na and the Pumi have coexisted on good terms for at least eight centuries. The similarities could thus stem from language contact. The two

[2] *Lizu* is not to be confused with *Lisu* 傈僳语. The former is an Ersuish language, spoken by approximately 7,000 people who reside along the banks of the Yalong 雅砻 River (Tibetan: Nyag chu) (Chirkova & Chen 2013); the latter is a Yi (Loloish) language spoken by about 900,000 people in a wide area that straddles boundaries between China (Yunnan and Sichuan), Thailand, Myanmar, and India.

groups "frequently intermarry and so a fair amount of Pumi speak or understand Yongning Na to some degree. The reverse is not necessarily true" (Daudey 2014: 5). However, similar characteristics are also observed in another dialect of Pumi, that of Niuwozi 牛窝子, which is not in contact with Yongning Na. This dialect, spoken near the Ninglang county seat, is in contact with another variety of the Na language (Luggu **lo˧gv̩˩**; in Chinese: Běiqúbà 北渠坝), not mutually intelligible with the Yongning variety. (As a piece of anecdotal evidence illustrating the degree of mutual comprehension: one of the daughters of consultant F4 married a Na speaker from that area; the differences in dialect led the couple to adopt Mandarin as their language of communication.)

While the linguistic terminology used in the description of Niuwozi Pumi differs slightly from that used for Wadu Pumi and Yongning Na, the observed prosodic patterns appear highly similar:

> Under the influence of intonation, the underlying H tone of a phonological word is readily removable when it is situated in the final unit of the clause (…). When this happens, the phonological word is merged with the other prosodic domain (removal of the original boundary of phonological word due to a loss of H tone in a following word […]). Sometimes, a series of low tones may appear in the ending syllables of an utterance after the boundary of phonological word is eliminated (…). (Ding 2014: 69)

Thus, the similarities in prosodic organization between Pumi and Na are not necessarily the outcome of areal convergence. As pointed out in §7.4, the division of an utterance into intonation phrases plays a central role in conveying phrasing and prominence in the most diverse languages. Within the Sino-Tibetan family, a distant parallel to the Na facts is found in Zhuokeji rGyalrong, where "clauses that are juxtaposed without any overt linkage marker that denotes coordination or consecutivization, or any morphosyntactic marking that signals dependency of one clause on the other" can be integrated into one group (called "intonation unit") to highlight "strong rhetorical links" between these clauses (Lin 2009: 208).

11.3.2 Other similarities

Further similarities emerge in phonological rules. In both Na and Pumi, each prosodic domain requires at least one non-L tone; when none is present, an H tone is added to the final syllable, yielding a rising tone, LH (Ding 2014: 60). Concerning numeral-plus-classifier phrases, Ding (2014: 69) observes that tone patterns "are not utterly predictable from the tones of the two formatives, as other factors beyond phonology are at work". This situation is closely parallel to that found in Na (studied in Chapter 3), although judging from Picus Ding's description, the degree of complexity would seem to be somewhat lower in that particular variety of Pumi than in the Na variety under study.

The high number of points of similarity suggests that further comparison of Na and Pumi could be highly revealing.

11.4 Comparison with Yi

Naish languages have remarkable typological similarities with languages of the Yi (Loloish) branch of Burmese-Yi (Burmese-Lolo), particularly in their morphosyntactic profiles. Tonal alternations arise in comparable contexts, such as compounds and numeral-plus-classifier phrases. In Nosu, for instance, tone ^{33} shifts to a sandhi tone ^{44} (rendered orthographically with a final *x*) when followed by another ^{33} tone. This dissimilatory change is morphosyntactically conditioned, as illustrated by the tonal contrast between *nga gu* 'I called (someone)' and *ngax gu* '(Someone) called me', which reflects a grammatical distinction between agent and patient (Gerner 2013: 28). However, the extent of such morphotonological alternations remains limited: tone sandhi is reported in only eight contexts, described over three pages of a grammar totalling more than five hundred (Gerner 2013: 28–30).

At present, the working hypothesis is that such tonal parallels between Yi and Naish are instances of homoplasy: independent developments arising under similar structural pressures, most notably the drive towards monosyllabicization. This tendency, widely attested across East and Southeast Asia, is generally considered to originate from the influence of Old Chinese, which exerted direct or indirect pressure on languages from several families, including Sino-Tibetan, Tai-Kadai, Hmông-Miên, and Austroasiatic. Yet this process has been more radical in Yi than in Chinese itself, with segmental reduction reaching a more extreme degree (Michaud 2012: 125, drawing on Haudricourt 1991).

11.5 A hypothesis about contact between two-level and three-level tone systems

Contact between two-level and three-level tone systems appears as an especially interesting topic for areal studies of tone. Pumi, with which Na has been in at least occasional contact for centuries, has a two-level system. Among Na dialects, two-level systems are found in Lataddi, on the edge of the swamp area known as the Grass Sea, which forms the eastern end of Lugu Lake (Dobbs & La 2016, Fily 2022), and in Shuiluo 水落, in the county of Muli 木里 (source: unpublished field notes, 2009). The dialect of Luoshui 落水, geographically close to the Yongning plain, has three levels, although the highest level has a relatively marginal status (Lidz 2010). Alawua (the dialect studied in this volume) likewise has three levels, but with a strongly restricted distribution. By contrast, dialects spoken further to the west and northwest, such as Labai 拉柏, clearly have three levels (L, M, and H).

Past contact between two-level and three-level tone systems may account for some synchronic phenomena found in Alawua, such as the exceptionless phonological rule prohibiting H tone in tone-group-initial position, effectively reducing the number of tonal contrasts to two in that position. This rule has far-reaching consequences: at the surface phonological level, H tone (as distinct from M) cannot occur on a monosyllable pronounced in isolation, nor can the tonal patterns H.M, H.MH, H.L or H.H occur on disyllables in isolation, since an isolated form (often referred to as a *citation form*) constitutes a tone group in itself. As discussed in §2.3.4, correspondences between overt H tones in the Labai dialect and floating H tones in Alawua suggest that word-initial H tones in Alawua may have been displaced from their original position on the first syllable of the word as a response to the enforcement of the rule prohibiting initial H. If this scenario is correct, the next question is why word-initial H tones became phonotactically illicit in the first place. At this stage, one can entertain the hypothesis that contact with a two-level tone system played a role, as speakers of such a system might have had special difficulty maintaining a three-way tonal opposition in initial position.

Synchronic case studies of contact between two-level and three-level tone systems provide valuable insights into the types of processes that such contact can induce, as well as their possible consequences for the linguistic systems involved. Ongoing investigations into contact between Mano and Kpelle, two Mande languages that have different tonal systems – Kpelle having a binary tonal contrast (Low vs. High) and Mano possessing three contrastive level tones (Low, Mid, and High) –, are exemplary (Konoshenko & Khachaturyan 2024). Findings from such studies, combined with additional data on present-day dialectal diversity within Na, hold potential for progress in the study of the historical role that contact may have played in shaping the tonal systems observed today in Na dialects.

Mention of typological parallels between Naish tone systems and those of Mande provides a natural transition to the discussion of typological topics in the next chapter.

12 Typological perspectives

The following statement sets a possible stage for typological inquiry:

> [L]anguages may differ at virtually all levels in their process of categorisation – not only in how they group sounds into emic categories (phonemes) but also in the way their particular constraints group these phonemes into meta-categories (classes of phonemes). These constraints, in turn, have to be defined system-internally, even when they derive from such supposedly universal parameters as sonority. Haspelmath (2007: 129) reminds us that "structural categories of language are language-particular, and we cannot take pre-established, *a priori* categories for granted". Such a stance does not rule out the possibility of universal generalisations, but entails that they can only be based on the empirical study of language-internal structures, and the acknowledgment of cross-linguistic diversity. (François 2010: 429)

In this passage, François emphasizes that every language has its own emic categories, which can only be discovered through an in-depth, language-internal analysis. Universals should not be assumed *a priori*: they are to be investigated empirically and explored through cross-linguistic comparison. This stance does not amount to a relativistic view in which each language requires an entirely distinct conceptual framework. Instead, it opens up a comparative research programme that highlights functional similarities across languages.

12.1 Tonal typology

This section clarifies the typological distinction between "level tones" and "complex tones", which constitutes the background to the classification of Yongning Na tones as "level tones" (§12.1.1). Some reflections are then set out (in §12.1.3) concerning the typological profile of Na prosody, shaped by the tone rules described in the preceding chapters.

12.1.1 Level tones and phonetically complex tones

In what can broadly be termed as "Africanist" usage, the term "level tone" refers to *a tone defined solely by a discrete level of relative pitch*. Level tones (a phrase used interchangeably with "tonal levels") are monodimensional: they are defined along a single parameter, F_0. This differs from segmental phonology, as vowels and consonants are defined along intersecting parameters, such as voicing, nasality, and place of articulation. Since level tones do not freely combine multiple properties, they are not analyzed further in terms of features: level tones constitute phonological primitives (Clements et al. 2011: 20; Hyman 2011).

Level-tone systems have two to five levels of relative pitch: L vs. H; L vs. M vs. H; L vs. M vs. H vs T(op); or B(ottom) vs. L vs. M vs. H vs. T(op). Systems with more than three levels are relatively uncommon (e.g. Bariba: Welmers 1952, and Bench, also known as Gimira: Wedekind 1983, 1985). One single case of a six-level system has been reported: Chori (Dihoff 1977), for which a reanalysis is possible (Odden 1995). Bariba, Bench, and Chori are spoken in Sub-Saharan Africa, an area where level tones are especially common. However, level-tone representations have proved useful beyond the Sub-Saharan area, for which they were initially developed (on languages of the Americas: Gomez-Imbert 2001; Hargus & Rice 2005; Girón Higuita & Wetzels 2007; Michael 2010; on languages of Asia: Ding 2001; Hyman & VanBik 2002, 2004; Donohue 2003, 2005; Evans 2008; Jacques 2011a).

In Yongning Na, the morphotonological alternations studied in the preceding chapters provide overwhelming evidence for a level-tone analysis. In level-tone systems, a phonetic contour is *the realization of two or more level tones on a single syllable*. These contours are phonologically decomposable: the observed F_0 movement results from interpolation between successive tonal levels.

There are some languages for which attempts to decompose contours into level tones have been less successful, however, to the point of casting doubt on the relevance of such an approach for these languages. In the Austroasiatic and Tai-Kadai language families, no convincing evidence has been found for the decomposition of contours into simpler units (e.g. Morey 2014: 639).

In the field of Sinitic languages (Chinese dialects), Chao Yuen-ren's early 20th-century work on Mandarin (Chao 1929, 1933) brought to light the complexities of its tone system. Following sustained exchanges with Chao, Kenneth Pike proposed a typological divide between (i) register-tones, defined solely in terms of discrete pitch levels, and (ii) contour-tones, about which he concluded: "the glides of a contour system must be treated as unitary tonemes and cannot be broken down into end points which constitute lexically significant contrastive pitches" (Pike 1948: 10).

Later studies have emphasized the role of phonation-type characteristics. In some prosodic systems, phonation types are merely low-level phonetic characteristics that occasionally accompany tone (see, for example, the investigation of the effect of creaky voice on Cantonese tonal perception by Yu & Lam 2014). In others, phonation type constitutes a distinctive feature which is phonologically independent of tone (*orthogonal* to tone, to use a term that has long been popular in the speech sciences: see e.g. Fant 1967: 636; Ohala 1978a), as in the Oto-Manguean languages Mazatec (Garellek & Keating 2011) and Trique (DiCanio 2012). Finally, in a third type of system, phonation-type characteristics are part and parcel of the definition of tones. Experimental studies of this third type of tone system include Rose (1982, 1989, 1990) for the Wu branch of Sinitic, Edmondson et al. (2001) for Yi and Bai, Mazaudon & Michaud (2008) for Tamang, and Andruski & Ratliff (2000), Andruski & Costello (2004) and Kuang (2013) for Hmong.

> Languages such as Black Miao and Vietnamese highlight the difficulty of drawing a line in the sand separating ‘tone’ languages from ‘register’ languages. This problem is even more strikingly illustrated by Burmese (…), which has been described both as a register system (e.g. Bradley 1982, Jones 1986) and as a tone language (e.g. Watkins 2001, Gruber 2011) (…). Gruber (2011) has shown that glottalisation, creakiness and the presence of a high pitch target are all important perceptual cues, thus demonstrating that, much like Vietnamese or Black Miao, Burmese should not be analyzed in terms of pitch or phonation type alone, nor is it straightforward to decide which property is the primary acoustic cue to the contrast. (Brunelle & Kirby 2016: 194)

Pike’s two-way typology of tone, while it emphasizes typologically relevant properties of the languages which he was able to take into consideration, has some limitations when applied to cases such as Vietnamese. In the Vietnamese system, the tones contrast with one another through a set of characteristics that include specific phonation types in addition to the time course of F_0 (Alves 1995, Mixdorff et al. 2003, Brunelle et al. 2010, Nguyen et al. 2013, Mac et al. 2015, Tạ 2023). For such tones, the label “contour tones” appears underspecific. For this reason, the term “complex tones” is used here in preference to Pike’s “contour tones”.

To recapitulate the terminology used in the present discussion: *complex-tone systems* are distinguished from *level-tone systems* (which are based on discrete levels of relative pitch). Complex tones include Pike’s category of “contour tones”, with the explicit addition of tones that comprise phonation-type characteristics.[1]

[1]This definition differs from that used in the *World Atlas of Language Structures*, where “complex” refers to the number of oppositions, not to the nature of the tones: “[t]he languages with

Under this set of definitions, “contour” refers to a unitary contour: a tone that is phonologically defined in terms of an overall template specifying the time course of F_0 over the tone-bearing unit. Phonologically unitary contour tones are encoded as overall shapes: “there are no objective reasons to decompose Vietnamese tone contours into level tones or to reify phonetic properties like high and low pitch into phonological units such as H and L” (Brunelle 2009a: 94; see also Brunelle et al. 2010, Kirby 2010). In this type of system, the term “level tone” is used to refer to *a tone that does not exhibit any salient fluctuations in* F_0. For instance, Mandarin tone 1 and Vietnamese tone 1 (orthographic *ngang*) can be referred to as “level tones” because, unlike the other tones in these languages, their F_0 curve remains relatively stable over the course of the syllable. This does not entail that they are phonologically defined by a discrete level of relative pitch (on Mandarin, see Xu & Wang 2001).

The two types of phonological contour tones – sequences of levels, on the one hand, and unitary contours, on the other – can be phonetically indistinguishable, making it necessary to relate phonetic observation to functional-structural levels of description. The evidence for distinguishing these two types of contours is morphotonological. A rising contour in a Bantu language will typically exhibit phonological behaviour showing that it consists of a low level tone followed by a high level tone (Clements & Goldsmith 1984, Clements & Rialland 2007). In Yongning Na, too, there is a wealth of evidence supporting the analysis of contour tones as sequences of level tones. From a typological point of view, rather than assuming that all tones can be decomposed into level components, it is at least as reasonable to adopt the opposite standpoint: viewing contours as nondecomposable units unless there is positive evidence to the contrary (Nick Clements, p.c. 2008).

Tonal systems based on levels (tone heights) are not unheard of in Sino-Tibetan. Examples include Pumi (Matisoff 1997, Ding 2006, Jacques 2011a, Daudey 2014), Cone Tibetan (Sun 2003b, Jacques 2014), Mianchi Qiang (Rma) (Evans 2008), Shixing (Chirkova & Michaud 2009), Hakha Lai (Hyman & VanBik 2002), Kuki-Thaadow (Hyman 2010), and the Lataddi dialect of Na (Dobbs & La 2016).

The distinction between level tones and complex tones is proposed here as a rule-of-thumb distinction; it aims to bring attention to a substantial body of Asian data that may not be readily visible to some prosodic typologists and which deserves to be more widely appreciated. Needless to say, the two-way distinction

tones are divided into those with a simple tone system – essentially those with only a two-way basic contrast, usually between high and low levels – and those with a more complex set of contrasts” (Maddieson 2011).

between level tones and complex tones is by no means water-tight: there are borderline situations.

Among other research perspectives opened up by the typological distinction between level tones and phonetically complex tones, comparison between the intonation systems found alongside these two types of tonal organization appear well worth carrying out, hypothesizing that there may be different trends between these two broad types.

12.1.2 Intonation in level-tone systems and complex-tone systems

In his study of tone and intonation in Lingala, mentioned earlier (§8.2), Guthrie suggests that the tonal range is set at the sentence level. This arguably reflects a characteristic of Lingala, where successive tones are much more cohesive than in complex-tone systems. In complex-tone systems, attention is drawn to *local* phenomena of F_0 range expansion or compression.

To venture an impressionistic description of the contrast between the two types of systems: level tones exist as part of a sequence, whereas phonetically complex tones have a stronger individual identity. Level tones are subject to a range of categorical processes that modify the tonal string, such as tone spreading (where an H or L tone is copied onto successive syllables). By contrast, complex tones are less prone to categorical change and more prone to noncategorical, local intonational modifications conveying emphasis or phrasing.

This does not imply that successive complex tones are fully independent. For instance, in Mandarin, focus on one syllable affects neighbouring syllables, particularly those that follow: "focus is usually related to F_0-range-expansion of focused words that are not in the final position of an utterance and F_0-range-suppression of post-focus words" (Zhang & Hirose 2004: 449). Moreover, tonal coarticulation effects in Sinitic, Vietnamese, and Thai are strong and tend to solidify diachronically into sandhi patterns (Abramson 1979, Gandour & Potisuk 1992, Brunelle 2003, 2009b, Zhang & Liu 2011). Thus, it would be misguided to view the noncategorical intonational modification of complex tones as purely local, or conversely, to assume that level tones are entirely exempt from local, noncategorical intonational variation. Nevertheless, the following generalization seems to hold: in the study of intonation, Bantuists' attention is regularly drawn to sentence-level phenomena rather than to local pragmatic emphasis. This suggests that local variations of the kind observed in complex-tone systems – as exemplified by Vietnamese, Thai, and Sinitic –, where they do not alter the phonological nature of the tones, are less salient in level-tone systems such as those found in

Bantu languages. In Bantu prosody, scholarly attention is drawn instead to *categorical* changes (modifications that alter the phonological tonal sequence) and, secondarily, to sentence-level phenomena. Echoing Guthrie's study of Lingala, a study of Chichewa intonation likewise focuses on sentence mode, specifically on differences in F_0 between questions and statements (Myers 1996).

In light of the above typological perspectives, the following subsection attempts to convey a feel for the organization of the Na prosodic system by pointing out some of the consequences of its tone rules for the outlook of this level-tone system.

12.1.3 Typological profile of Na prosody as shaped by the tone rules

One of the salient characteristics of Yongning Na is the partial neutralization of lexical oppositions when words are said in isolation. However, this does not appear to have far-reaching implications for the organization of the tonal system as a whole. Such neutralization is found in prosodic systems that differ widely from one another, such as Japanese (Kubozono 1993), San Juan Quiahije Chatino (Oto-Manguean family) (Cruz 2011: 91), and Sotho and Tswana (Bantu) (Creissels et al. 1997, Zerbian & Barnard 2010b, Zerbian 2016).

On the other hand, the levelling rules of Yongning Na (Rules 4 and 5), which lower all tones following an H tone or an M.L sequence to L, have far-reaching consequences for surface phonological tone sequences. These two rules lead to the neutralization of tonal oppositions over large portions of tone groups – a massive levelling effect reminiscent of stress systems in which all syllables following a major stress are de-stressed. The culminative nature of its H tone makes Yongning Na strikingly different from the extensive set of systems known as 'terraced-level tone languages' (Armstrong 1968).

The concept of 'terracing' refers to categorical shifts in register affecting all subsequent tones: *downstep*, a distinctive lowering, and *upstep*, a distinctive raising. An important consequence of terracing is that tones within the same 'terrace' hang together more closely than successive tones in a language like Na, without downstep or upstep. To use an image from weaving, tones in the same terrace are tightly knit together; to use an image from woodwork, they are pegged together. This intuition is reflected in Nick Clements's proposed treatment of 'terracing' in terms of a "tone level frame".

"Within this framework, terracing is seen to be the result of (…) processes applying to the tone level frame itself, rather than directly to individual tones" (Clements 1979: 538). 'Terracing' places constraints on the range of fundamental frequency within which the tonal levels are realized, since shifts in register are distinctive. It plays a major role in shaping surface phonological tone sequences and their phonetic realization. A key property of terracing, as Clements notes, is that categorical register shifts can take place more than one time within a tone group.

> [An] important feature of tone terracing, at least in the case of downstep, is that there is no limit on the number of register lowerings that may occur within a tone group; the only limit is the external one imposed by the lexical, grammatical, or phonological factors that govern the occurrence of downstep. (Clements 1979: 540)

This type of tonal organization requires preplanning strategies, which can get highly elaborate. Brilliant speakers anticipate the amount of downsteps required in a long utterance, raise the initial pitch accordingly, and manage successive lowerings throughout. Less talented orators need to reset their F_0 when successive downsteps bring them to the lower limit before they reach the end of a tone group (Rialland 2001).

In Yongning Na, by contrast, there is no downstep, and hence no need for such long-distance preplanning. Whenever a tone group contains an H tone, this tone serves as the tone group's climax. From an information-processing perspective, once an H tone is identified, the remainder of the tone group contain no further tonal contrasts: there is nothing to expect but a sequence of phonological L tones. Phonetically, the pitch gradually descends towards its floor value; the realization of the F_0 curve on this portion of the tone group is free from the trammels of phonological distinctiveness.

To summarize, the 'tone level frame' (Nick Clements's term for the tone space at a given point in an utterance) is subject to far fewer constraints in Yongning Na than in 'terracing' tone languages. The absence of downstep or upstep, combined with the culminative nature of H tones, go a long way towards explaining the different feel of its prosody compared to that of 'terracing' tone languages. There are simply fewer possible tone sequences: for instance, while both Yongning Na and Yala (Ikom) have H, M, and L tones, Yala also has downstepped counterparts (!H, !M, and !L) and allows their full range of combinations (Armstrong 1968). Moreover, in Na, there is no contrast between a fall from M to L and one from H to L, allowing for greater phonetic freedom than in systems where the tonal space is more crowded. (This phonetic flexibility is exploited for intonational purposes, as discussed in §9.2.)

Like downstep, *downdrift* – the gradual phonetic lowering of phonologically identical tones separated by a lower tone – is absent in Yongning Na, for the same reason: the two possible sequences of a higher tone and a lower one are H.L and M.L, both of which constitute a descent to the lowest phonological level. By Rule 5, these sequences can only be followed by L tones, prohibiting configurations such as ‡ M.L.M, ‡ H.L.H or ‡ H.M.H. This is another salient trait of the typological profile of Yongning Na.

12.2 Assessing the complexity of the Na tone system

"Language complexity is a notion that has hovered in linguistics for more than a century" (Oh & Pellegrino 2023: 789), as a scale in language description (with one feature being considered as more complex than another) and as a general backdrop against which to evaluate differences among languages. An article discussing methods for measuring the degree of complexity of tone systems (Konoshenko 2014a) proposes the number of tonal contrasts and the number of tonal rules as the two main dimensions of tonal complexity. In terms of these two variables, the tone system of Yongning Na is more complex than that of Naxi and Laze, its immediate siblings in the Naish subgroup of Sino-Tibetan (§1.1.3): Na has more lexical tone categories, and richer morphotonology. A task for the future will be to assess this complexity by modelling regularities and irregularities in the paradigms that make up Na morphotonology. As a stepping-stone towards this mid- and long-term goal, some dimensions of this complexity are recapitulated below and compared with other languages. The comparisons are based not on phylogenetic or areal closeness but on synchronic typological similarities.

12.2.1 Partially regular morphotonology

Partially regular morphological paradigms are cross-linguistically widespread. Examples include the inflection of transitive verbs in Dinka (Andersen 1993: 8), where certain inflections are marked by a particular toneme for all verbs alike, while others are specific to particular classes of verbs, and the inflection of interrogative pronouns in the Australian language Anguthimri (Crowley 1981: 172).

In the Na tone system, numeral-plus-classifier phrases constitute an area where tone patterns have proliferated. As shown in Chapter 3, no fewer than nine tonal categories exist for monosyllabic classifiers, compared to five for monosyllabic nouns. Furthermore, the tonal patterns of these nine classifier categories in combination with numerals need to be learnt individually, as they do not follow from synchronically regular rules. While this complexity is not as spectacular as that of Ahmao (Hmong-Mien family), where classifiers have

"12 basic forms, each displaying a complex cluster of meanings" (Gerner & Bisang 2009), the Na data nonetheless contributes to typological generalizations, demonstrating that the tones of classifiers can be more complex than those of nouns.

12.2.2 Nouns and verbs: A comparable degree of complexity?

Keeping in mind that certain types of nouns (especially classifiers) are more complex than others in terms of their tone categories, it seems that there is no conspicuous imbalance between Na nouns and verbs in terms of tonal complexity. This differs from tonal systems in Bantu and, more broadly, in the Niger-Congo family, where verbs generally display less tonal diversity than nouns. Many Bantu languages only have two tonal types of verbs (irrespective of their number of syllables), versus three or more tonal categories for monosyllabic nouns and even more for disyllabic and longer nouns (Creissels 1994: 183). In some languages, such as Gbeya, Kissi, Baoulé, and Urhobo, there are no tonal oppositions among verbs at all (Creissels 1994: 184).

12.2.3 More progressive spreading than regressive spreading: A typologically common pattern

Under the present analysis, Yongning Na has a phonological tone rule (Rule 1) whereby L tone spreads progressively onto syllables that are unspecified for tone. In contrast, the H tone does not spread in the sense of associating to several syllables in a row; there can only be one syllable carrying H tone in a tone group. Despite this, an H tone influences the following tones within the group, causing them all to lower to L.

While the morphotonological rules are context-specific and cannot be reduced to a set of purely phonological rules, they also reveal an overall tendency towards progressive tone spreading, rather than the other way round. For instance, averaging over the entire set of tone rules applying in determinative compounds (§4.2), the determiner (which appears first) makes a larger contribution than the head to the tone pattern of the entire compound. Seen in this light, Na exhibits a clear predominance of progressive over regressive spreading.

This pattern is typologically unsurprising. Regressive spreading is firmly attested, for instance in Tswana, Odienné Dioula, and Kikwere (as analyzed by Odden 1998: 177-178), but progressive spreading is more common cross-linguistically (Creissels 1994: 206-207).

A separate but not wholly unrelated observation is that H tone in Yongning Na has a tendency to occur late within a tone group. an H tone never associates to the first syllable within a tone group. The //H#// tone associates to the last syllable of the root; so does //H$//, but it typically 'hops' from this position to a later syllable (§2.3.5); and //#H// never associates to the word to which it is lexically attached, only to a later syllable, if available (§2.3). This overall tendency is relatively common cross-linguistically. Late realization of H tone targets is more common than early realization: "perseverative tone spreading phonologizes the tendency of tone targets to be realized late" (Hyman 2007b: 19). The diachronic developments that have led to the present diversity of final H tones in Yongning Na appear to align with well-attested cross-linguistic tendencies. This is a case where the synchronic complexity is not particularly surprising when considered from the perspective of the typology of language change, that is, from a *panchronic* perspective (§1.3.2).

12.2.4 The dual status of the M tone is not a typological rarity

In the present analysis, the M tone in Yongning Na has two facets. On the one hand, it functions as a fully specified phonological tone: the M component in categories such as LM and MH cannot be omitted, as LM contrasts with L, and MH# with H#. On the other hand, M tone serves as a default tone: by Rule 2, M tone is assigned to syllables that remain toneless after the application of Rule 1 (whereby L tone spreads onto toneless syllables).

Na is not unique in this respect. In Yorùbá, too, the M tone has a dual status. It is not lexically specified – the only two lexical tones are L and H –, but after being introduced through default-tone assignment rules, M behaves as a specified tone in tonal derivations (Akinlabi 1985).

13 Conclusion

> When one aims to please others, one may fail, whereas things that we do to please ourselves always have a chance to interest someone or other.
>
> Marcel Proust[1]

The Yongning Na tone system comprises (i) a set of phonological rules governing tone-to-syllable association, set out in §7.1, and (ii) a host of rules specific to certain morphosyntactic contexts, detailed in Chapters 3-6. Different rules apply, for instance, to the association of a verb with a subject or an object, the formation of a compound noun, the combination of a numeral and classifier, or the attachment of affixes to a word. These tonal paradigms constitute the core of Yongning Na morphotonology and represent the bulk of what language learners must acquire to master this tone system.

As a conclusion, let us return to the initial puzzle presented in the introduction. It is now possible to set out the mechanisms whereby the surface phonological tone sequences of examples (1a–1b) obtain.

[1] This sentence is followed by reflections that are close to linguists' concerns: "No one is unique: our individualities are made out of a universal fabric; this is what allows for sympathy and understanding, which are such great pleasures in life. If we could analyze the soul as we analyze matter, it would become apparent that below the surface diversity of minds, as under that of material objects, there are but a few simple substances and irreducible elements; and that what we think of as our personality is made up from elements which are quite common, and which are met again pretty much everywhere in the universe." *Original text:* Quand on travaille pour plaire aux autres on peut ne pas réussir, mais les choses qu'on a faites pour se contenter soi-même ont toujours chance d'intéresser quelqu'un. Il est impossible qu'il n'existe pas de gens qui prennent quelque plaisir à ce qui m'en a tant donné. Car personne n'est original et fort heureusement pour la sympathie et la compréhension qui sont de si grands plaisirs dans la vie, c'est dans une trame universelle que nos individualités sont taillées. Si l'on savait analyser l'âme comme la matière, on verrait que, sous l'apparente diversité des esprits aussi bien que sous celle des choses, il n'y a que peu de corps simples et d'éléments irréductibles et qu'il entre dans la composition de ce que nous croyons être notre personnalité, des substances fort communes et qui se retrouvent un peu partout dans l'Univers. (*Pastiches et mélanges*, Paris: Gallimard, 1919, pp. 108–109)

(1) a. njɤ˧ ʐi˩ bi˩ -zo˩ -ho˥.
1SG to_take to_go OBLIGATIVE DESIDERATIVE
'I have to go and take [my luggage] now.'

b. njɤ˧ bi˧ -zo˧ -ho˩.
1SG to_go OBLIGATIVE DESIDERATIVE
'I have to go. / I'm afraid I have to leave.'

With morpheme-level transcriptions indicating lexical tone in terms of the Yongning Na tone categories, these sentences can be represented as (2a–2b):

(2) a. njɤ˧ | ʐi˩ bi˩-zo˩-ho˥.
njɤ˩ ʐi˩$_a$ bi˧$_c$ -zo˧$_a$ -ho˩
1SG to_take to_go OBLIGATIVE DESIDERATIVE
'I have to go and take [my luggage] now.'

b. njɤ˧ | bi˧-zo˧-ho˩.
njɤ˩ bi˧$_c$ -zo˧$_a$ -ho˩
1SG to_go OBLIGATIVE DESIDERATIVE
'I have to go. / I'm afraid I have to leave.'

A crucial point is the tone-group boundary after the 1SG subject: these utterances consist of two tone groups, and tonal processes apply independently within each group, as explained in Chapter 7.

The first tone group consists of a single syllable whose lexical tone is L. As documented in Chapter 2, an isolated L-tone syllable is realized as a level, non-low tone. This accounts for the surface phonological form /**njɤ˧**/ in (2a) and (2b).

In (2a), the second tone group consists of two serialized verbs followed by two suffixes. Following the morphotonological rules brought out in Chapter 6, //**ʐi˩$_a$**// in combination with //**bi˧$_c$**// 'to go' yields //**ʐi˩-bi˩**//. Addition of the obligative //**-zo˧$_a$**// results in //**ʐi˩-bi˩-zo˩**//. The final suffix, the desiderative //**ho˩**//, carries L tone. The tonal behaviour of an L-tone suffix depends on the length of the preceding expression:

- When attached to an L-tone verb, it retains L tone (e.g. //**ʐi˩-ho˩**// 'will take', L.L).
- When attached to an L-tone expression of two or more syllables, it carries H tone (e.g. //**ʐi˩-zo˩-ho˥**// 'will need to take', L.L.H).

Since //**ʐi˩-bi˩-zo˩**// consists of three syllables, the suffix carries H tone, yielding //**ʐi˩-bi˩-zo˩-ho˥**//.

Both tone groups, //**njɤ˧**// and //**ʐi˩-bi˩-zo˩-ho˥**//, contain at least one tone other than L. As a result, the repair rule for all-L tone groups (Rule 7 in §7.1.1) does not apply. These tone groups thus proceed unaltered to the surface phonological level as /**njɤ˧ | ʐi˩-bi˩-zo˩-ho˥**/.

In (2b), the second tone group consists of a main verb, 'to go', and the same two suffixes as in (2a). As per the morphotonological rules brought out in Chapter 6, the M tone of the main verb does not affect the tones of the suffixes. Similarly, the M tone of the obligative suffix //**-zo˧$_{\mathbf{a}}$**// leaves the following morpheme unchanged, so that all three syllables in the tone group surface with their lexical tones.

While it is satisfying to confirm that the morphotonological patterns described in this volume shed light on these and other examples, it must be acknowledged that this book represents only one step towards an advanced linguistic model of tone in Yongning Na. A mid- to long-term objective is the computer-aided analysis of individual utterances based on a computational model of the grammar (finite-state modelling), following the methodological suggestion of Karttunen (2006). Achieving this will require (i) implementing the entire tonal grammar of Yongning Na in computational scripts, (ii) glossing Yongning Na texts at the morpheme level, establishing a link from each morpheme to the lexicon, and (iii) encoding the morphosyntactic structure of each utterance. The goal will be to generate surface phonological tone patterns for utterances, specifying the full range of possible variants in the division of utterances into tone groups.

Such a model would allow for quantitative verification, across the full set of available data, of the generalizations proposed in this volume. More ambitiously, it would enable the researcher to compare the tonal patterns observed in a textual utterance (a real sentence from a text) with a set of possible alternatives. This, in turn, opens new perspectives for appraising the speaker's stylistic choices in a narrative, such as the division of the sentence into tone groups and the choice of one tonal variant over another in cases where two or more patterns are acceptable. This approach holds promise for identifying the factors underlying so-called nonconditioned variation. By modelling the various components of the language's prosody and morphosyntax, it may become possible to explore the range of stylistic possibilities offered by the linguistic system – e.g. through the choice of congruence vs. dissonance between message and form, or between syntax and prosodic phrasing. A relevant source of inspiration in this regard is Delattre's work on French intonation (1966, 1970).

A second perspective will consist in examining the phonetic implementation of surface phonological tone sequences in relation to intonation. The ultimate aim here is to assess the contribution of various factors to the final phonetic realization of each syllable, disentangling and spelling out the different components of the speech signal and their linguistic interpretation. In this approach, intonational phenomena are defined as encompassing all those aspects of the speech signal that are not predictable from the utterance's contrastive units – namely the sequence of phonemes and the tonal string as parsed into tone groups. Computational modelling may serve as a tool for identifying intonational phenomena by contrast, bringing them to the fore against the backdrop of predictable elements.

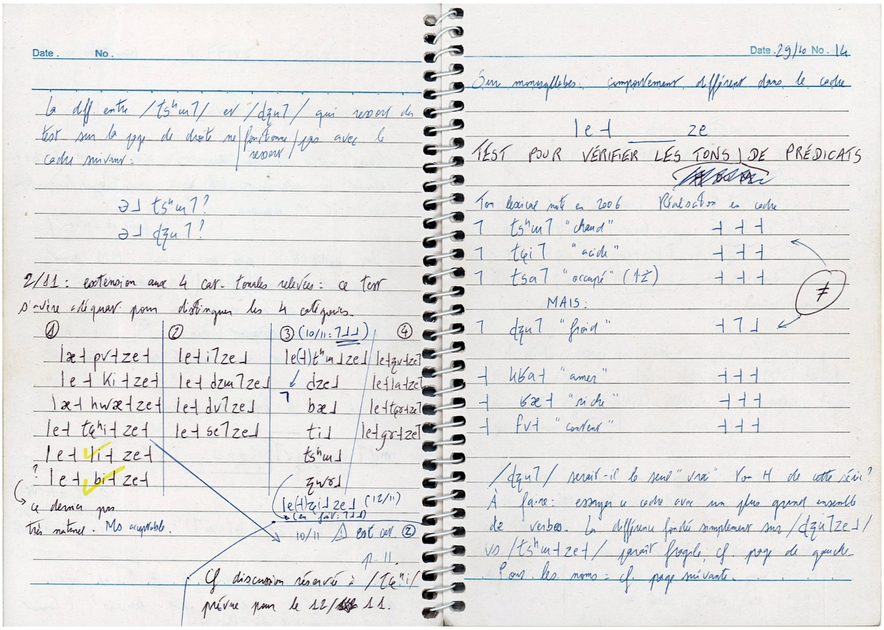


Figure 13.1: Working out tone in Yongning Na: field notes, 2007.

Appendix A: Vowels and consonants

> … without a good sense of how languages vary, not only in terms of the symbolic units such as phonemes and allophones but also in the details of their phonetic implementation, we have little hope of understanding the possible range of language. Endangered languages in particular represent an important but often-ignored source of information about what is possible in Language.
>
> Richard Wright, University of Washington
> Phonetics Laboratory website, 2014

In Yongning Na, there are no restrictions on co-occurrence of tones and segments, nor are there special classes of segments that trigger synchronic tonal processes (as is the case with depressor consonants in Bantu, e.g. in the Ikalanga language: see Hyman & Mathangwane 1998). So it did not appear appropriate to interpose a presentation of the language's segmental phonetics and phonology between the reader and the book's central topic – tone.

This Appendix provides an opportunity to smuggle a free-standing overview of the vowels and consonants of Yongning Na into this tonal study. It discusses the choices made in phonemicization, with particular emphasis on areas where phonemic analysis is cracking at the seams. Several of these observations could serve as starting points for experimental phonetic/phonological studies, building on the availability of several hours of transcribed and annotated recordings.

The vowels and consonants of Yongning Na are phonologically inert: they are not involved in synchronic phonological rules and processes. In this respect, Yongning Na stands at the opposite end of the typological continuum from a language such as Kifuliiru (Bantu), which has (i) a range of phonological rules, such as the strengthening of /**h**/, /**l**/, and /**r**/ to a plosive when preceded by a nasal, and (ii) morphological rules, such as the deletion of final consonants in resultative verb forms (Van Otterloo 2011: 37–96).

When examining vowels and consonants in Yongning Na, attention is drawn instead to their patterns of coarticulation. Due to dramatic phonological erosion since proto-Sino-Tibetan (Jacques & Michaud 2011), syllabic structure in Yongning Na has collapsed down to (C)(G)V+T, where C is a consonant, G a glide – with a severely restricted distribution –, V a syllable nucleus, and T the tone. The brackets indicate that C and G are optional.

Coarticulation constitutes a salient part of a language's sound system (Keating 1990, Kühnert & Nolan 1999). Structural approaches to phonological systems predict cross-linguistic differences in coarticulation and articulatory reduction. For example, the extent to which high, front vowels exert a palatalizing influence depends in part on the number, nature, and functional yield of existing phonemic oppositions. In Na, which contrasts /**ki**/ and /**tɕi**/, the range of allophonic variation of /**ki**/ can safely be predicted to be narrower than in Naxi, which does not have this contrast.

Phonetic studies confirm that coarticulation is language-specific. In the description that follows, special attention is paid to coarticulation, allophonic variation, and phenomena of articulatory reduction.

A.1 Consonant and vowel charts

Table A.1 presents the consonant inventory, and Figure A.1 displays the set of rhymes. The chart of rhymes includes the syllabic consonants /**ɻ̩**/ and /**v̩**/, which are discussed in §A.2.1. Their placement in the lower right-hand and upper right-hand areas of the chart serves as a rough approximation of their articulatory characteristics.

To prevent overcrowding, the following rhymes are not shown in the chart:

- Rhymes containing a glide (discussed in §A.3.1): /**wæ**/, /**wɑ**/, /**wɤ**/, /**jæ**/, /**jɤ**/, and /**jo**/.
- Nasal vowels: /**ĩ**/, /**ṽ̩**/, /**w̃ɤ**/,[1] /**æ̃**/, and /**ɑ̃**/, all of which occur exclusively after /**h**/.
- The rhymes /**ɻ̩̃**/ and /**w̃æ**/, which always constitute syllables on their own (i.e. they do not combine with an initial consonant).
- The nasal vowel /**õ**/, which occurs either after /**h**/ or on its own, i.e. in the syllables /**hõ**/ and /**õ**/.

[1]In the rhymes /**w̃ɤ**/ and /**w̃æ**/, the diacritic for nasality is placed on the glide rather than on the final vowel because the degree of phonetic nasalization decreases over the course of the rhyme. The nasal rhymes of Yongning Na are discussed in §A.2.4.

Table A.1: The initials of Yongning Na.

	bilabial / labio-dental	dental	alveolo-palatal	retroflex	velar	uvular	glottal
plosive	**p^h p b**	**t^h t d**		**ʈ^h ʈ ɖ**	**k^h k g**	**q^h q**	
affricate		**ts^h ts dz**	**tɕ^h tɕ dʑ**	**ʈʂ^h ʈʂ ɖʐ**			
nasal	**m**	**n**	**ɲ**	**ɳ**	**ŋ**		
fricative	**f**	**s z**	**ɕ ʑ**	**ʂ ʐ**		**ʁ**	**h**
lateral		**ɬ l**					
approximant				**ɻ**			

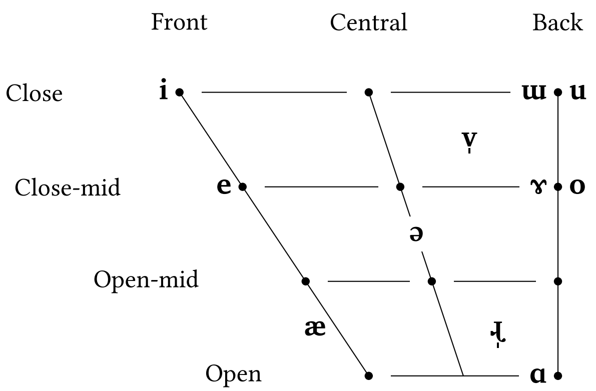


Figure A.1: The rhymes of Yongning Na.

The notation used throughout this book and in the transcription of texts is phonemic in orientation. However, a concern for phonetic transparency led to indicate the empty-onset fillers [**j**], [**ɣ**], and [**w**] in transcriptions, even though they are not phonemically contrastive (as explained in §A.3.1) and are therefore not included in Table A.1.

Tables A.2 and A.3 list all possible vowel nuclei following each initial consonant. In the interest of space, syllables with nasal rhymes are omitted. These tables only offer a first-pass view of Na phonotactics: they provide no information on lexical frequency, only a binary indication of whether a given combination occurs. A syllable is considered attested if it is firmly established in the Na lexicon, excluding combinations introduced through lexical borrowing, vowel harmony, or expressive processes, no matter how widely attested these are in the present-day state of the language.

For instance, combinations of dental stops with the vowel /**æ**/ are marked as absent because their only attestations occur in words such as /**tʰæ˩tsɯ˧**/ 'jar', borrowed from Mandarin *tánzi* 坛子, as well as /**tæ˧ɻæ˩**/ 'Adam's apple' and /**læ˧dæ˧qæ˥**/ 'armpit', where the presence of /**æ**/ in the first syllable may result from vowel harmony with the second syllable. Similarly, combinations of retroflex stops with /**ɑ**/ are indicated as nonexistent because their only occurrences are in /**ʈʂʰɑ˧lɑ˧**/ 'to chat', /**qv̩˧ɻ̩˧-ʈʂʰɑ˧nɑ˥#**/ (the name of a mountain), and /**ʈʂɑ˧tɑ˥**/ 'written sign, tracing', where the /**ɑ**/ of the first syllable may likewise result from vowel harmony.

By abstracting away from phenomena that can be identified as structural gap-filling, the tables highlight empty slots in the phonotactic inventory and cases of complementary distribution. Comments about the inventory of syllables are provided in §A.4.

Table A.2: Inventory of attested combinations of initials and rhymes, leaving aside marginal words (loanwords, expressive words…). First part.

	i	**ɯ**	**u**	**v̩**	**ɻ̩**	**e**	**ɤ**	**o**	**æ**	**ɑ**	**wæ**	**wɑ**	**wɤ**	**jæ**	**jɤ**	**jo**
Ø	√	√	√	√	√	✗	✗	√	√	√	✗	✗	√	✗	√	√
pʰ	√	✗	✗	√	√	✗	√	√	√	✗	✗	✗	✗	✗	✗	✗
p	√	✗	✗	√	√	✗	√	√	√	✗	✗	✗	✗	✗	✗	✗
b	√	✗	✗	√	√	✗	√	√	√	✗	✗	✗	✗	✗	✗	✗
m	√	✗	✗	√	✗	✗	√	√	√	✗	✗	✗	✗	✗	✗	✗
tʰ	√	✗	✗	√	✗	✗	✗	√	✗	√	✗	✗	✗	✗	✗	✗
t	√	✗	✗	√	✗	✗	√	√	✗	√	✗	✗	✗	✗	✗	✗
d	√	✗	✗	√	✗	✗	√	√	✗	√	✗	✗	✗	✗	✗	✗
tsʰ	√	√	✗	✗	✗	√	√	√	✗	√	✗	✗	✗	✗	✗	✗
ts	√	√	✗	✗	✗	√	√	√	✗	√	✗	✗	✗	✗	✗	✗
dz	√	√	✗	✗	✗	√	√	√	✗	√	✗	✗	✗	✗	✗	✗
n	√	✗	✗	√	✗	√	✗	√	✗	√	✗	✗	✗	✗	✗	✗
s	√	√	✗	✗	✗	√	√	√	✗	√	✗	✗	✗	✗	✗	✗
z	√	√	✗	✗	✗	√	✗	√	✗	√	✗	✗	✗	✗	✗	✗
ɬ	√	✗	✗	√	✗	✗	✗	√	✗	√	✗	✗	✗	✗	✗	✗
l	√	✗	✗	√	✗	√	✗	√	✗	√	✗	✗	✗	✗	✗	✗

Table A.3: Inventory of attested combinations of initials and rhymes, leaving aside marginal words (loanwords, expressive words…). Second part.

	i	**ɯ**	**u**	**v̩**	**ɻ̩**	**e**	**ɤ**	**o**	**æ**	**ɑ**	**wæ**	**wɑ**	**wɤ**	**jæ**	**jɤ**	**jo**
tɕʰ	√	√	✗	✗	✗	✗	√	√	✗	✗	✗	✗	✗	✗	✗	✗
tɕ	√	√	✗	✗	✗	✗	√	√	✗	✗	✗	✗	✗	✗	✗	✗
dʑ	√	√	✗	✗	✗	✗	√	√	✗	✗	✗	✗	✗	✗	✗	✗
ɲ	√	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
ɕ	√	√	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	√	√
ʑ	√	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
ʈʰ	√	√	✗	✗	✗	✗	√	✗	√	✗	✗	✗	✗	✗	✗	✗
ʈ	√	√	✗	√	✗	✗	√	✗	√	✗	✗	✗	✗	✗	✗	✗
ɖ	√	√	✗	√	✗	✗	√	√	√	✗	✗	✗	✗	✗	✗	✗
ʈʂʰ	√	√	✗	√	√	√	√	√	√	✗	✗	✗	✗	✗	✗	✗
ʈʂ	√	√	✗	√	√	√	√	√	√	✗	✗	✗	✗	✗	✗	✗
ɖʐ	√	√	✗	√	√	√	√	√	√	✗	✗	✗	✗	✗	✗	✗
ɳ	✗	√	✗	√	✗	✗	✗	✗	√	✗	✗	✗	✗	✗	✗	✗
ʂ	✗	√	✗	√	√	√	√	√	√	✗	√	✗	√	✗	✗	✗
ʐ	✗	√	✗	√	√	√	√	√	√	✗	√	✗	√	✗	✗	✗
ɻ	✗	✗	✗	✗	✗	✗	✗	✗	√	✗	√	✗	✗	✗	✗	✗
kʰ	√	√	✗	√	✗	✗	√	√	✗	✗	✗	✗	√	✗	✗	✗
k	√	√	✗	√	✗	✗	√	√	✗	✗	✗	✗	√	✗	✗	✗
g	√	√	✗	√	✗	✗	√	√	✗	✗	✗	✗	√	✗	✗	✗
ŋ	✗	✗	✗	√	✗	✗	√	✗	✗	✗	✗	✗	√	✗	✗	✗
qʰ	✗	✗	✗	√	✗	✗	✗	√	√	√	√	✗	√	✗	✗	✗
q	√	✗	✗	√	✗	✗	✗	√	√	√	√	✗	√	✗	✗	✗
ʁ	✗	✗	✗	√	✗	✗	✗	√	√	√	√	✗	√	✗	✗	✗
h	√	√	√	✗	✗	✗	√	√	√	√	√	✗	√	✗	✗	✗
f	✗	✗	✗	√	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗

A.2 Syllable nuclei: Vowels and syllabic consonants

A.2.1 Consonantal nuclei

In Na, certain consonantal sounds function as syllable nuclei. In the International Phonetic Alphabet, such sounds are termed 'syllabic'. Syllabic fricatives are an areal characteristic, common in neighbouring Yi languages, which belong to the Nasoid subgroup of Loloish (also known as Ngwi; see Bradley 1979: 70), as well as in Lizu (Chirkova & Chen 2013), Ersu (Chirkova et al. 2015), and Pumi (Daudey 2014: 52). Local varieties of Mandarin also have a fricative syllable nucleus, /**fv̩**/, as in Southwestern Mandarin [**fv̩**] for 'lake' (Standard Mandarin *hú* 湖; see Gui 2001, Pinson & Pinson 2008).

In the absence of coda consonants in Na, 'consonantal nuclei' are also referred to in this volume as 'consonantal rhymes'. The consonantal rhymes of Yongning Na are: [v̩], [z̩], [ʐ̩], [ɻ̩], and [ɻ̩̃].

A.2.1.1 The voiced fricative /v̩/

The voiced fricative /**v̩**/ occurs exclusively as a rhyme, never as an initial. Its friction is weaker than in Naxi, and there is some degree of formant movement towards a central articulation, tending towards [**və**]. Since friction noise is slight, even at the beginning of the rhyme, this /**v̩**/ could be described as approaching an approximant, [**ʋ**].

Martinet distinguishes between fricatives proper, which have a firm articulation and detectable fricative noise, and spirants, which have a more relaxed articulation, tending towards a vowel-like aperture (Martinet 1956: 24; 1981). In these terms, the articulation of /**v̩**/ in Yongning Na is spirant rather than fricative. After bilabial initials, /**v̩**/ tends towards trilling (though less markedly than in Naxi): /**bv̩**/ is realized close to [**ʙ̩**], /**pv̩**/ to [**pʙ̥**], and /**pʰv̩**/ to [**pʰʙ̥**].

In the absence of strong friction, the rhyme /**v̩**/ can be difficult to distinguish from the high back vowel /**o**/, especially after consonants that exert similar coarticulatory effects on both of these rhymes. In particular, uvular stops induce retraction (backing) of both /**v̩**/ and /**o**/, making the opposition between syllables such as /**qv̩**/ and /**qo**/ difficult to perceive at first. Minimal pairs such as /**mæ˧qv̩˩**/ 'tail' vs. /**-mæ˧qo˩**/ 'below; behind' constitute handy materials for learners to practise this opposition.

From a diachronic point of view, the rhyme /**v̩**/ in Yongning Na (as in Naxi) is hypothesized to originate in ***u** (Jacques & Michaud 2011). The process of fricativization is less advanced in Yongning Na than in Lijiang Naxi, suggesting a continuum among Naish languages from [**u**]-like to [**v̩**]-like realizations, with approximant [**ʋ̩**] as an intermediate stage. Such cases illustrate that International Phonetic Alphabet symbols do not tell the full story about phonological systems, as Martinet emphasized.

> One wonders whether the habit of constantly operating with graphic notations does not make some linguist[s] deaf to the gradual shifts which any painstaking observation can reveal. If one has been taught, not only that phonological systems are made up of discrete units, but also that these units are basically the same in all languages, (...) one can hardly avoid concluding that no change can take place except by means of jumps from one unit or allophone to another. Only those who know that linguistic identity does not imply physical sameness, can accept the notion that discreteness does not rule out infinite variety and be thus prepared to perceive the gradualness of phonological shifts. (Martinet 1988: 25)

A.2.1.2 Apicalized vowels

Apicalized vowels are found "on the tip of many tongues" across Sino-Tibetan (Baron 1974), and Na is no exception. The vowel /**ɯ**/ has fricative allophones after dental and retroflex fricatives and affricates, as in Naxi. For instance, /**tsʰɯ˦**/ 'goat' is realized [**tsʰz̩˦**], and /**ɖʐɯ˧**/ 'market, city' as [**ɖʐʐ̩˧**]. In phonetic transcription, it is customary in Asian linguistics to use the symbols coined by Chao Yuen-ren: [**ɿ**] and [**ʅ**], respectively. Table A.4 presents Chao's complete set of symbols for apicalized vowels, along with their closest equivalents in the International Phonetic Alphabet; symbol descriptions are taken from Pullum & Ladusaw (1986: 80, 89-90).

Table A.4: Chao Yuen-ren's symbols for apicalized vowels and their closest equivalents in the International Phonetic Alphabet.

sound	symbol	symbol description	IPA
plain apical vowel	[**ɿ**]	long-leg turned iota	**ż**
retroflex apical vowel	[**ʅ**]	right-tail turned iota	**ʐ̇**
rounded plain apical vowel	[**ʮ**]	curvy turned h	**ż̹**
rounded retroflex apical vowel	[**ʯ**]	right-tail curvy turned h	**ʐ̹̇**

Rounded apical vowels are absent in Na for two structural-diachronic reasons: (i) there is no rounding opposition among front vowels, and hence no source for a process of apicalization such as ***y** > [**ʮ**]; and (ii) among back vowels, the close vowel ***u** underwent fricativization to /**v̩**/, as described in §A.2.1.1.

In addition to distinguishing between plain and retroflex apical vowels, Yongning Na exhibits an opposition between /**i**/ and another high front vowel after the alveolopalatal initials /**dʑ**/, /**tɕ**/, and /**tɕʰ**/. The first set of syllables – syllables /**dʑi**/, /**tɕi**/, and /**tɕʰi**/ – has moderate friction during the initial, and the rhyme is not strongly apicalized.

The second set consists of syllables with an apicalized rhyme and a slightly more central vowel; these syllables can be approximated as [**dʑ̍**], [**tɕʑ̍**], and [**tɕʰʑ̍**], respectively. If one were to abstract away from the friction on the rhymes, these syllables could be transcribed phonetically as [**dʑɪ**], [**tɕɪ**], and [**tɕʰɪ**]. The phonemic analysis adopted here treats them as allophones of /**ɯ**/, yielding phonemic representation as /**dʑɯ**/, /**tɕɯ**/, and /**tɕʰɯ**/. This decision is based on structural considerations (complementary distribution) rather than phonetic similarity with the canonical [**ɯ**] vowel, and its degree of validity remains to be evaluated experimentally. One possible approach would be to investigate whether priming[2] by canonical realizations of the vowel /**ɯ**/ (as in /**ʈʰɯ˩**/ 'to drink') affects reaction times in perceptual tests. Phonetically, the syllables /**dʑɯ**/, /**tɕɯ**/, and /**tɕʰɯ**/ are not easy to distinguish from /**dzɯ**/, /**tsɯ**/, and /**tsʰɯ**/, which are shown in Table A.5 with their realizations in Chao notation and in International Phonetic Alphabet.

Table A.5: Syllables with initial dental affricate and high, unrounded vowel /**ɯ**/, and their phonetic realizations.

phonemic analysis	Chao notation	International Phonetic Alphabet
dzɯ	[**dzɿ**]	[**dz̩**]
tsɯ	[**tsɿ**]	[**tsz̩**]
tsʰɯ	[**tsʰɿ**]	[**tsʰz̩**]

An added complexity is related to Voice Onset Time (VOT) oppositions among alveolopalatal initials. Syllables with an unvoiced or aspirated initial, /**tɕ**/ or /**tɕʰ**/,

[2]Priming is a memory effect in which exposure to one stimulus (the *prime*) influences the response to another stimulus.

have stronger affrication than those with initial /**dʑ**/. This difference has coarticulatory effects on the following vowel, such that both /**i**/ and /**ɯ**/ are somewhat less open after /**tɕ**/ and /**tɕʰ**/ than after /**dʑ**/.[3]

A.2.1.3 Rhotic rhymes

In addition to the labiodental fricative and to apicalized vowels, consonantal rhymes include /**ɻ̩**/ and its nasalized counterpart, /**ɻ̩̃**/. From a phonetic point of view, the rhyme /**ɻ̩**/ does not display the considerable lowering of the third formant which is the tell-tale characteristic of rhotic vowels (Ladefoged & Maddieson 1996: 313), such as Naxi /**ɚ**/. The syllabic diacritic serves to distinguish this rhyme from the consonant phoneme /**ɻ**/, described in §A.3.6. As a rhyme, /**ɻ̩**/ can occur either on its own or following a retroflex fricative or affricate or a bilabial stop.[4] Table A.6 presents examples of syllables containing a /**ɻ̩**/ rhyme, alongside syllables with the same initial followed by the rhyme /**v̩**/.

Table A.6: Examples illustrating the phonemic contrast between /**ɻ̩**/ and /**v̩**/ after retroflex fricatives and affricates and bilabial stops.

	v̩	**ɻ̩**
ʈʂʰ	/**ʈʂʰv̩˧**/ ‘breakfast’	/**ʈʂʰɻ̩˦**/ ‘lungs’
ʈʂ	/**mv̩˧ʈʂv̩˥**/ ‘wrinkles’	/**ʈʂɻ̩˥**/ ‘to cough’
ɖʐ	/**ɖʐv̩˥**/ ‘large vein, artery’	/**ɖʐɻ̩˥**/ ‘wet, moist’
ʂ	/**ʂv̩˧ɖv̩˧**/ ‘to think’	/**ʂɻ̩˦**/ ‘full’
ʐ	/**ʐv̩˩ₐ**/ ‘delicious’	/**ʐɻ̩˩ₐ**/ ‘to knead’
pʰ	/**pʰv̩˩ₐ**/ CLF for fields	/**pʰɻ̩˩ₐ**/ ‘white’
p	/**pv̩˧**/ ‘to perform (a ritual)’	/**pɻ̩˧**/ ‘dry’
b	/**bv̩˥**/ ‘thick, coarse’	/**bɻ̩˥**/ ‘insect’

Under the present analysis, the rhyme /**ɻ̩**/ has a nasalized counterpart, /**ɻ̩̃**/. Only three items have been found so far: ‘bone’, /**ɻ̩̃˥**/, which also has the mean-

[3]A dedicated recording session (*ApicalizedVowels* {0004470}) was conducted for words containing one of the following syllables: /**dʑi**/, /**dʑɯ**/, /**tɕi**/, /**tɕɯ**/, /**tɕʰi**/, and /**tɕʰɯ**/.

[4]The opposition between /**ɻ̩**/ and /**v̩**/ following bilabial stops was not reported in the first edition of this book: it was only recognized in 2024. The lexical distribution of this opposition is documented in the dictionary (Michaud et al. 2024) starting at version 2.0 (December 2024).

ing of ‘stem’; ‘helpless, impoverished, troubled’, /**ɻ̩̃˥**/; and /**ɻ̩̃˧ʈʂwæ˩** ‘sambucus, *Toricellia angulata Oliv.*’. Alternative notations as a nasal rhotic vowel, /**ɚ̃**/ or /**œ̃˞**/, could also be considered. However, the choice of /**ɻ̩̃**/ reflects both phonetic similarity and a likely structural opposition between /**ɻ̩**/ (oral) and /**ɻ̩̃**/ (nasal).

The word for ‘bone’, transcribed here as /**ɻ̩̃˥**/, already puzzled earlier investigators. Fù Màojì (1983: 25) transcribes it phonetically as [**ʔɱɹ**] and analyzes it phonemically as /**ŋv̩ɹ**/. In this transcription system, the two-symbol sequence is not to be understood as indicating a succession of two sounds; rather, Fu Maoji employs the symbol /**ɹ**/ to denote rhoticity in the vowel. Phonemic analysis as /**ŋv̩ɹ**/ is an appealing possibility: nasality would be attributed to the initial onset, and rhoticity to a rhotic fricative rhyme, /**v̩ɹ**/. However, in the attested syllable /**ŋv̩**/, coarticulation between the initial and the rhyme does not result in full coalescence: the syllable retains two distinct parts – an initial nasal consonant [**ŋ**], with complete oral closure, and a rhyme [**ṽ̩**], which is partially nasalized phonetically but without full oral closure. A hypothetical rhotic counterpart to /**ŋv̩**/ would likewise be expected to begin with a nasal consonant. Since this is not observed, I do not analyze this syllable as a rhotic counterpart to /**ŋv̩**/. Instead, I interpret it as a monophonemic syllable consisting of a nasalized rhotic rhyme.

As for whether it should be analyzed as /**ɻ̩̃**/ or /**ṽ̩**/, both options remain open, since there is no opposition between these in onsetless syllables. On the basis of phonetic considerations, I find /**ɻ̩̃**/ the more adequate notation from a synchronic perspective.

A.2.1.4 Potential for the creation of new syllabic consonants and monophonemic syllables

In syllables with a simple CV (consonant+vowel) segmental structure, the consonant and vowel are subject to strong coarticulation. Their features tend to be realized over the syllable as a whole. In Naish languages, coarticulation in CV monosyllables tends to create compact units that become less and less tractable to straightforward segmentation into distinct phonemes. Over time, such syllables may become monophonemic.

This process is relevant not only to the emergence of apicalized vowels (described in §A.2.1.2) but also to the development of syllabic nasals. “In various Loloish languages some or all of the nasals occur as syllabics. In most such cases the diachronic source is syllables with a nasal initial and a high vowel; sometimes one dialect has nasal syllabics where others have nasals plus a high vowel. This could be called rhyme-gobbling” (Bradley 1989: 150; see also Björverud 1998: 8).

Yongning Na exhibits an intermediate stage in this development. The syllable /**mv̩**/ is phonetically realized as [**m̩**] except in careful (hyperarticulated) speech. On the other hand, /**nv̩**/ and /**ŋv̩**/ retain an oral portion following the initial nasal: these two syllables are pronounced as [**nʋ̩**] and [**ŋʋ̩**], respectively.

In the present state of the language, these phenomena remain instances of allophonic variation, but they hold potential for further evolution into more advanced stages of phonological erosion.

A.2.2 Close vowels

Table A.7 shows that the close vowels /**i**/ and /**e**/ contrast after dental fricatives and affricates, e.g. /**dzi˩**/ 'to sit' vs. /**dze˩**/ 'to fly', /**tsi˩**/ 'to boil' vs. /**tse˩**/ 'to lock', /**tsʰi˥**/ 'classifier for animal skins' vs. /**tsʰe˥**/ 'salt', and /**si˥**/ 'wood' vs. /**se˥**/ 'to walk'. The syllable /**zi**/ has only been found in /**tsʰi˧zi#˥**/ 'highland barley' and /**lɑ˧zi˥**/ 'painter'; instances of /**ze**/ are more numerous.

After the dental nasal /**n**/ and the dental lateral /**l**/, the close vowels /**i**/ and /**e**/ are marginally contrastive. The syllable /**ne**/ is only found in a grammatical morpheme, /**-ne**/ 'as, like', which appears in the interrogative /**qʰɑ˩ne˩**/ 'how', the manner demonstratives /**ʈʂʰɯ˧ne˧-ʝi˥**/ (proximal) and /**tʰv̩˧ne˧-ʝi˥**/ (distal), and constructions such as /**tɕʰɤ˧ɲi˧-ne˧-ʝi˥**/ 'every day; repeatedly, all the time'. Recognition of the morpheme /**ne**/ was delayed by the fact that the highly frequent expression /**ʈʂʰɯ˧ne˧-ʝi˥**/ 'thus, in this way' (occurring over 500 times in 25 texts) is pronounced very close to [**ʈʂʰɯ˧ni˧˥**], and hence was initially transcribed as ‡ **ʈʂʰɯ˧ni˧˥**. (This adverb is analyzed further in the section devoted to on-glides, §A.3.1.) Another marginal case where it appears reasonable to posit an /**e**/ vowel distinct from /**i**/ is in combination with initial /**l**/. The syllable /**li**/ is common, appearing in about thirty words. A [**le**] syllable appears in the ACCOMPLISHED prefix /**le˧-**/, and in derived items, such as [**njɤ˧le˧gv̩#˥**] 'daytime', which is perceived by consultant F4 as meaning literally 'the day is flowing/going by', as shown in (1).[5]

(1) njɤ˧le˧gv̩#˥
ɲi˥ le˧- gv̩˧$_{c}$
day ACCOMP to_flow/to_go_by
'daytime', interpreted by consultant F4 as having the literal meaning 'the day is flowing/going by'

[5]Assuming that the etymological glossing in (1) is correct, the change of the first syllable from /**ɲi**/ to /**njɤ**/ remains to be accounted for.

Table A.7: Distribution of the close vowels /**i**/ and /**e**/.

initial	i	e
Ø	**ʝi˥** ‘ox’	–
b p pʰ	**bi˥** ‘snow’	–
m	**mi˧** ‘wound’	–
d t tʰ	**di˧˥** ‘to hunt’	–
dz	**dzi˩** ‘to sit’	**dze˩** ‘to fly’
ts	**tsi˩** ‘to boil’	**tse˩** ‘to lock’
tsʰ	**tsʰi˥** ‘CLF.animal skins’	**tsʰe˥** ‘salt’
s	**si˥** ‘wood’	**se˥** ‘to walk’
z	**tsʰi˧zi#˥** ‘highland barley’	**ze˩mi˩** ‘niece’
n	**ni˥** ‘amaranth’	**-ne** ‘as, like’
l	**li˧ₐ** ‘to look’	**le˧-** ACCOMPLISHED
ɬ	**ɬi˥** ‘to rest’	–
dʑ tɕ tɕʰ	**tɕi˥** ‘to shake’	–
ɕ ʑ	**ɕi˧** ‘rice’	–
ɲ	**ɲi˥** ‘to listen’	–
ɖ ʈ ʈʰ	**ʈi˩ₐ** ‘to get up’	–
ɖʐ ʈʂ ʈʂʰ	–	**ʈʂe˥** ‘earth’
ʐ ʂ	–	**ʐe˥** ‘arrow’
ʁ ɻ	–	–
g k kʰ	**gi˥** ‘to owe’	–
q qʰ	**qi˧qi˧** ‘originally’	–
h	**hi˩˥** ‘rain’	–

In F4’s idiolect, the second syllable of ‘daytime’ is perceived as the ACCOMPLISHED prefix, and therefore does not constitute evidence for /**le**/ as an attested syllable in nouns. Moreover, the form /**njɤ˧le˧gv̩#˥**/ is only one of many avatars of the word for ‘daytime’ among Na dialects, some of which have /**ɬi**/ and not /**le**/ as a middle syllable (**ɲi33-ɬi31 ku33** in Lidz 2010: 297). It appeared interesting to dwell on this example nonetheless, as it provides an insight into processes whereby marginal combinations of initials and rhymes can enter the lexicon.

To sum up, the opposition of /**e**/ and /**i**/ is restricted to syllables that have a dental initial (most examples have a fricative or affricate). This is one of the many cases where a phonemic opposition is found in highly restricted contexts; in Praguean terms, these constitute extreme cases of neutralization. This issue is taken up in the discussion of the inventory of syllables, in §A.4.

Recognition of the phonemic opposition between /**i**/ and /**e**/ after dental fricatives and affricates was delayed because I initially failed to notice the existence of /**i**/ as distinct from /**ɯ**/ in this context, where both /**i**/ and /**ɯ**/ are apicalized, and their phonetic difference is a very fine one. Apicalization is stronger for /**ɯ**/ than for /**i**/. Another difference is that the lips are stretched for /**si**/, /**dzi**/, /**tsi**/, and /**tsʰi**/. In the initial transcriptions, the contrast between the two apicalized vowels was overlooked, leading to erroneous transcriptions of these syllables as ‡ **sɯ** ('wood'), ‡ **zɯ** ('barley'), ‡ **dzɯ** ('to sit'), ‡ **tsɯ** ('to boil'), and ‡ **tsʰɯ** (classifier for animal skins).

In the many contexts where the opposition between /**i**/ and /**e**/ is neutralized, the transcription follows their phonetic realization, which is closer to [**e**] after retroflex fricatives and affricates, and to [**i**] in all other contexts (after bilabials, velars, uvulars, retroflex stops and fricatives, laterals, nasals, alveolopalatals, and glottals).

As for close back vowels, [**o**] and [**u**] are only contrastive after initial /**h**/. Examples of /**ho**/ include /**tɑ˧ho˧**/ 'together', /**ho#˥**/ 'porridge, gruel', /**ho#˥**/ 'partridge', /**qo˩ho˦**/ 'wicker box', /**ho˦**/ 'to sip', /**ho˩ₐ**/ 'correct', and /**-ho˩**/ DESIDERATIVE. Examples of /**hu**/ are fewer; they include /**hu˧mi˥$**/ 'stomach', /**hu˥**/ 'to wait', and /**hu˦**/ 'to miss, to long for'. Phonetically, there is stronger friction in the initial for /**ho**/ than for /**hu**/, which could be approximated phonetically as [**hu**] vs. [**χo**]. This suggests an alternative phonemic analysis, dispensing with the opposition between /**u**/ and /**o**/, positing instead an opposition between /**h**/ and /**χ**/, and rewriting /**hu**/ and /**ho**/ as /**ho**/ and /**χo**/. The phonetic difference in vowel quality appears more salient than the difference in the initial, however, hence the choice to interpret the vowel difference as phonemic.

Another way to reinterpret the syllable [**hu**] and eliminate the /**u**/ phoneme altogether would be to grant phonemic status to initial [**f**], analyzing the syllables [**fv̩**], [**hu**] and [**ho**] as /**fv̩**/, /**hv̩**/ and /**ho**/, respectively (Roselle Dobbs, p.c. 2016): see §A.3.7.

After all initials other than /**h**/, there is no opposition between close and close-mid rounded back vowels, [**o**] and [**u**]. Phonetic realizations are close to [**o**] after dentals, velars and uvulars, and more often close to [**u**] after the other consonants. In all the contexts where the opposition is neutralized, the notation chosen is /**o**/: this appeared less cumbersome than notation as an archiphoneme /**O**/ (or /**U**/).

The two sounds [**o**] and [**u**] may have stronger phonemic status in the speech of younger speakers, whose increasing proficiency in Standard Mandarin makes them familiar with a phonemic /**u**/ (contrasting with /**oʷ**/ and /**ʷo**/; the *Pinyin* transcription of these three vowels is: *u*, *ou*, *uo*). Southwestern Mandarin does not exert direct pressure in this direction, however, since /**u**/ is fricativized in this dialect of Mandarin. For instance, the word /**tsʰu**/ 'vinegar' is pronounced [**tsʰv̩**]. This word, which is in common use in Yongning, is accordingly pronounced as [**tsʰv̩˦˩**] in the speech of the older speaker F4.

A.2.3 A neutral vowel: /ə/

A clarification is needed concerning the use of the phonetic symbol /**ə**/. In their 1985 book on Naxi, which includes a word list for Yongning Na, He Jiren and Jiang Zhuyi use /**ə**/ to represent two different vowels: a back unrounded vowel, /**ɤ**/, realized as [**ɣɤ**] in onsetless syllables, and a neutral vowel, /**ə**/, which always forms an independent syllable, harmonizes with the vowel of the following syllable, and is realized with an initial glottal stop (Michaud 2013a: 130). This confusion likely stems from the use of /**ə**/ as the official phonetic equivalent of the letter *e* in the *Pinyin* romanization of Standard Mandarin, where it was employed to represent a back unrounded vowel. In view of its phonetic realizations, which are "**ɤ**-like or **ʌ**-like" (International Phonetic Association 1949: 42), the 'ram's horn' symbol, /**ɤ**/, or the turned v, /**ʌ**/, would have been more felicitous choices than the neutral vowel /**ə**/. Linguists trained in mainland China in the first decades of the People's Republic of China tended to follow this system, which was reproduced in dictionaries and textbooks until the turn of the twenty-first century. More recent descriptions, such as Lee & Zee (2003: 110), have corrected this by distinguishing /**ɤ**/ from /**ə**/.

This misleading notation was adopted within the official phonetic transcription for Naxi and remains influential.[6]

In the present system, /**ə**/ is used exclusively for the interrogative particle and the initial syllable of certain lexical words, where it can plausibly be analyzed as a prefix. In particular, it appears in kinship terms referring to elders, such as /**ə˧mi˧**/ 'mother', realized as [**e˧mi˧**]; /**ə˧mɑ˧**/ 'mother (VOCATIVE)', realized as [**ɑ˧mɑ˧**]; and /**ə˧v̩˥**/ 'uncle', realized as [**ɤ˧v̩˥**]. The prefix's realization varies according to the quality of the vowel in the following syllable (the root morpheme): it tends towards [**æ**] before /**æ**/ and apicalized allophones of /**ɯ**/, [**ɛ**] before /**i**/, /**ĩ**/, and /**e**/, [**ɑ**] before /**ɑ**/ and /**wɤ**/, and [**ɤ**] before /**ɤ**/, /**o**/, /**ɯ**/, /**jɤ**/, /**jo**/, and /**ɻ̩**/. Before /**v̩**/, its realization depends on whether an intervening consonant is present: it is close to [**ɤ**] when immediately followed by /**v̩**/, as in /**ə˧v̩˥**/ 'uncle' (approximately [**ɤ˧v̩˥**]), but shifts towards [**æ**] when /**v̩**/ is preceded by a consonant, as

[6] In his manuscript lexicographic notes, He Jiren consistently used the turned v, /**ʌ**/, and not the neutral vowel, /**ə**/. However, the symbol /**ə**/ was reintroduced when the data was edited for publication as a dictionary (Hé Jírén et al. 2011).

in /**ə˧pʰv̩˧**/ 'mother's mother's brother' and /**ə˧mv̩˩**/ 'elder sibling' (phonetic approximation: [**æ˧pʰv̩˧**] and [**æ˧mv̩˩**]).[7]

Note, however, that phonetic (incomplete) vowel harmony is not restricted to the vowel transcribed as /**ə**/. This topic is taken up in §A.2.7.

No phonetic difference could be detected between the [**ɑ**] realization of /**ə**/ before /**ɑ**/ and the realization of the phoneme /**ɑ**/ itself. For instance, the clan name [**ɑ˧lɑ˧**] is phonemicized as /**ə˧lɑ˧**/, but an alternative interpretation as /**ɑ˧lɑ˧**/ cannot be ruled out.

A.2.4 Nasal rhymes

A.2.4.1 Nasal rhymes after the glottal /h/

Yongning Na has a relatively large inventory of nasal rhymes: it comprises /**ĩ**/, /**ṽ̩**/, /**õ**/, /**w̃ɤ**/, /**æ̃**/ and /**ɑ̃**/, as well as the syllables /**ɻ̩̃**/ and /**w̃æ**/. The first six occur after /**h**/, where they contrast neatly with their non-nasal counterparts: /**hi**/–/**hĩ**/, /**hv̩**/–/**hṽ̩**/, /**ho**/–/**hõ**/, /**hwɤ**/–/**hw̃ɤ**/, /**hæ**/–/**hæ̃**/, and /**hɑ**/–/**hɑ̃**/. Examples are provided in Table A.8.

Table A.8: Examples of /**h**/-initial syllables that are part of a correlation of nasality.

oral rhyme	nasal rhyme
/**hi˥**/ 'tooth'	/**hĩ˥**/ 'man'; /**hĩ˦**/ 'to stand'
/**hv̩˧**/ ([**fv̩˧**]) 'to like'	/**hṽ̩˩**/ 'red'; /**hṽ̩˥**/ 'hair'; /**nv̩˧hṽ̩˩**/ 'kidney bean'; /**dʑi˧hṽ̩˥$**/ 'clothes'; /**hṽ̩˧~hṽ̩˧**/ 'to stir-fry'
/**ho˦**/ 'to sip'; /**ho˥**/ 'to wait'	/**hõ˦**/ 'eight'; /**hõ˧**/ 'to go (IMPERATIVE)'
/**hwɤ˩**/ 'to pass over, to hand over'	/**hw̃ɤ˩**/ 'late'
/**hæ˧**/ 'Chinese'; /**hæ˦**/ 'lime'; /**hæ˧pɤ˧**/ 'plait'	/**hæ̃˧**/ 'wind'; /**hæ̃˩**/ 'gold'
/**hɑ˥**/ 'food'	/**hɑ̃˦**/ 'night'

[7]This phonemic analysis of the kinship prefix was suggested by Roselle Dobbs.

Diachronically, these syllables illustrate a process whereby nasality is transferred from a syllable-initial consonant cluster to the following vowel. This phenomenon is attested in several languages of Asia. In Kam-Sui (Tai-Kadai family), Sandong Sui lost the stop component of the original cluster: the stop+nasal clusters ***km-**, ***kn-**, ***tn-**, and ***kɲ-** merged with the preglottalized initials ***ˀm-**, ***ˀn-**, and ***ˀɲ-**, which remain preserved in Sui, e.g. /**ˀma¹**/ 'vegetables' and /**ˀma³**/ 'flexible', both corresponding to a proto-Kam-Sui ***ˀm** initial (Ferlus 1996: 251–252). Lakkia, by contrast, retained the initial stop, while the nasal underwent lenition, nasalizing the following vowel in the process, e.g. /**kũːi**/ 'bear', from an earlier stop+nasal cluster ***km-**.

Northern Sui dialects (Pandong 潘洞 and Yang'an 阳安) illustrate a possibility for the later evolution of glottal+nasal onsets. Here, distinctive nasality is transferred to the following vowel, and all that remains in onset position is a glottal sound, yielding [**ʔṼ**] or [**h̰Ṽ**], with the entire syllable becoming nasalized, including the initial glottal sound (Haudricourt 1967: 176). This pattern is exactly parallel to what is observed in Yongning Na and other Naish languages, as brought out in Table A.9.

Japhug, a conservative Sino-Tibetan language that preserves a broad range of initial clusters, provides a valuable point of comparison. Once correspondences among Naish languages have been established, comparison with such conservative languages suggests hypotheses for fleshing out reconstructions (Jacques & Michaud 2011: 470-471).

Table A.9: Comparative data pointing to the development of nasality in Naish from earlier */**rN-**/ onsets.

	Japhug	Fengke Naxi	Yongning Na	Laze
red	**ɣɯrni**	**hṽ̩˩**	**hṽ̩˩**	–
to stand	**rma**	**hṽ̩˩˧**	**hĩ˧˥**	**hĩẽ˥**
person	**tɯ-rme**	**hĩ˧**	**hĩ˥**	**hĩ˧**
body hair	**tɤ-rme**	**hṽ̩˥**	**hṽ̩˥**	**hṽ̩˩**
to stir-fry	**rŋu**	–	**hṽ̩˧~hṽ̩˧**	–
two	**ʁnɯs**	**ɲi˩˧**	**ɲi˥**	**ɲi˧**

Table A.9 brings out a correspondence between the /**h̰Ṽ**/ syllables of Yongning Na and etyma with initial /**rm-**/ or /**rn-**/ in Japhug. This leads to the hypothesis that the /**h̰Ṽ**/ syllables found in Naish languages originate in earlier *CNV syllables.

The hypothesis that nasal vowels in some Sino-Tibetan languages could result from the influence of syllable-initial nasals was already expressed by Huáng Bùfán (1991). On the other hand, no hypotheses had been advanced regarding the precise phonemic sequences involved in the change until Michaud et al. (2012).

The final example in Table A.9, 'two', illustrates the retention of nasals in Naxi, Na, and Laze that originate in onsets other than */**rN-**/. This suggests that the *CN- onsets that gave rise to vowel nasalization may have all gone through an intermediate */**sN-**/ stage. (For general phonetic reflections on this topic, see Ohala & Ohala 1993: 233, who note that "children learning English sometimes pronounce target *sm* and *sn* clusters as voiceless nasals".)

Before nasal rhymes, /**h**/ is itself nasalized; the lowered velum prevents the buildup of intra-oral pressure required for strong friction noise. Since the entire syllable is nasalized, an alternative phonemic analysis would be to posit a nasalized glottal fricative, /**h̰**/, contrasting with plain /**h**/. A review on "possible and impossible segments" (Walker & Pullum 1999) classifies /**h̰**/ among the set of "possible segments", as it is firmly attested as a phoneme contrasting with /**h**/ in at least two languages: Kwangali (Bantu; Ladefoged & Maddieson 1996: 132-133) and Seimat (Austronesian; Blust 1998). In the case of Yongning Na, the decision to interpret nasality as a feature of the vowel rather than the consonant is based on the observation that nasal vowels also occur in a small set of syllables that do not have an initial glottal fricative. These cases, examined in the following section, further illustrate the role of nasality as a distinctive feature in the vowel system of the language.

A.2.4.2 Onsetless nasal syllables: /õ/, /æ̃/ and /ɻ̩̃/

Among onsetless nasal syllables, /**õ**/ contrasts with /**o**/, /**æ̃**/ with /**æ**/, and /**ɻ̩̃**/ with /**ɻ̩**/. Examples of /**o**/ (realized with a glide onset, [**wo**]) include /**wo˥**/ 'hard'; examples of /**õ**/ include /**õ˦**/ '(one)self', /**õ˧ʈʂwɤ˧**/ 'mosquito', and /**õ˩dv̩˦**/ 'wolf'. Synchronically, initial glottalization contributes to a clear phonetic contrast between /**o**/, realized [**wo**], and /**õ**/, realized [**ʔõ**].

Examples of /**ɻ̩̃**/ and /**ɻ̩**/ were presented earlier (§A.2.1.3). The syllable /**æ̃**/ also has a glottalized onset, [**ʔæ̃**]. A non-nasalized counterpart, [**ʔæ**], is found in the system and is analyzed as /**æ**/, i.e. recognizing the phonemic status of both /**æ**/ and /**æ̃**/. Examples include /**æ̃˩zɯ˩**/ 'agate' vs. /**æ˩gv̩˩**/ 'ard'. The words /**æ̃˩-mi˧**/ 'hen' and /**æ˩mi˧-ʁwɤ#˥**/ 'Aemiwua' ('the village of Aemi') form a quasi-minimal pair.

From a diachronic perspective, the glottal onset of [**ʔõ**] and [**ʔæ̃**] may result from the same phenomenon of empty-onset hardening that led to the presence of an initial [**ʁ**] in words such as /**ʁwɤ˥**/ 'village', corresponding to Naxi /**wɤ˧**/. This phenomenon is discussed in greater detail in §A.3.1.

A.2.4.3 Phonemic analysis of the onsetless nasal syllable [w̃æ]

The syllable realized as [**ʔw̃æ**] occurs in a single lexical item: 'to swell, to inflate (e.g. the belly is swollen)', [**ʔw̃æ˧**]. It does not have a non-nasal counterpart. In the absence of an opposition with an oral syllable [**wæ**], nasality and initial glottalization might be considered the product of implementation rules rather than as distinctive features. From this perspective, the underlying form could be analyzed as simply /**wæ**/. However, additional evidence suggests that this analysis is insufficient. Consultant F4 pronounces the Mandarin syllables *wa*, *wan*, and *wang* (as in *wáng* 王, a common surname) as [**ʁwæ**], rather than as [**ʔw̃æ**]. If the underlying form of 'to swell' were merely /**wæ**/, one would expect these Mandarin forms to pattern with /**wæ**/ (phonetically [**ʔw̃æ**]) rather than with /**ʁwæ**/. This suggests that nasality, which plays a distinctive role in the vowel system of Yongning Na, should be granted phonemic status, whereas initial glottalization, which is not contrastive, should not. In summary, the syllable /**w̃æ**/, realized as [**ʔw̃æ**], contrasts with /**wæ**/, realized as [**ʁwæ**] (see §A.3.3).

A.2.4.4 The syllable /ĩ/

The syllable /**ĩ**/ occurs in a single lexical item: the interjection 'Yes!', 'Okay!', used in response to an instruction from a person in a position of authority, such as one's elder or another person holding authority. The phonemic form is /**ĩ˧**/, with a phonetic realization as [**ʔĩ˧**]. This item is described as carrying a Mid tone, but could also be analyzed as a toneless interjection generally realized on a level pitch.

A.2.4.5 The nasal rhyme /õ/ as a variant in two Tibetan loanwords

In two Tibetan loanwords, the nasal rhyme /**õ**/ appears as a variant of its oral counterpart /**o**/. The first example is the term for 'head of a caravan': /**tsʰo˧pæ#˥**/ ≈ /**tsʰõ˧pæ#˥**/, corresponding to Written Tibetan *tshong pa* 'merchant'. The second case is a formula of blessing: /**lɑ˧mɑ˧-ko˧ʈʂʰo˧**/ ≈ /**lɑ˧mɑ˧-ko˧ʈʂʰõ˧**/. Nathan Hill (p.c. 2016) proposes that the Na word /**ko˧ʈʂʰo˧**/ ≈ /**ko˧ʈʂʰõ˧**/ is a shortened form of *dkon mchog gsum*, Sanskrit *triratna*: 'the Three Jewels: *Buddha*, *dharma* (the teaching), and *saṅgha* (the monastic order or community)'. It is not obvious why the word /**lɑ˧mɑ˧**/, clearly borrowed from *bla ma* 'spiritual leader, great teacher (at the monastery)', should be paired with the Three Jewels in this blessing formula, yielding *bla ma dkon mchog gsum*, which could be interpreted as 'the Lama and the Three Jewels' or 'the Lama's Three Jewels'. Nathan Hill suggests that in this context, 'lama' may stand in for the Three Roots *(tsa sum)* of the Tibetan Buddhist tradition. These Three Roots – *lama*, *yidam*, and *khandroma* – are more esoteric and recondite than the Three Jewels, making them less familiar to laypeople. Since the first of the Three Roots, the *lama*, is embodied by a living person (typically a high-ranking monk), it is more readily understood. This could explain why 'lama' appears in this (somewhat garbled) blessing formula.

On the other hand, the word for 'monastery', /**go˧bɤ˩**/ (cf. Written Tibetan *dgon pa*), does not have a nasalized variant ‡ **gõ˧bɤ˩**. These examples suggest that nasalization in Tibetan borrowings does not operate in a uniform manner. A systematic study of Tibetan loanwords in Na would be necessary to identify different layers of borrowing and degrees of phonological integration.

A.2.5 The open vowels /ɑ/ and /æ/, and the vowel /ɤ/

Na, Naxi, and Laze all have a phonemic opposition between two open vowels. In Naxi, different authors have transcribed this contrast in varying ways: some as /**a**/ vs. /**æ**/ (e.g. Hé Kāixiáng et al. 1989), others as /**ɑ**/ vs. /**æ**/ (Fāng Guóyú & Hé Zhìwǔ 1995), and others still as /**ɑ**/ vs. /**a**/ (Fù Màojì 1981, Hé Jírén & Jiāng Zhúyí 1985, Pinson 2012). From a phonemic point of view, all these notations are valid, as they capture the relevant structural fact: the distinction between two low vowels. However, use of the symbol /**a**/ can create confusion for linguists consulting data from several sources. To ensure consistency across descriptions of Naish languages, I have opted for the notation /**ɑ**/ vs. /**æ**/, thereby avoiding the symbol /**a**/.[8] Phonetically, the vowel transcribed as /**ɑ**/ is clearly a back vowel in Naxi, close to cardinal [**ɑ**], whereas in Na and Laze it is closer to [**a**].

While the vowel transcribed as /**æ**/ initially seemed to me to be pronounced with less salient differences between these three languages, a native speaker of Lijiang Naxi, Mu Yanjuan 木艳娟, explained to me (p.c. 2016) that in her view the front low vowel in Naxi is clearly [**a**] and notation as [**æ**] is definitely inappropriate.[9]

[8] For similar reasons of consistency across descriptions of Naish languages, the symbol /**ɯ**/ is used for the high back unrounded vowel in Yongning Na, Naxi, and Laze, despite slight differences in pronunciation. In Yongning Na, this vowel is articulated less far back than its Naxi counterpart, which is transcribed with the same symbol. From a purely phonetic perspective, one could argue in favour of transcribing it as /**ɨ**/ in Yongning Na.

[9] Estimations of formant frequencies (F1, F2, and F3) of Naxi vowels by five Naxi speakers are presented in Michaud (2005: 475-480).

In Yongning Na, the sounds [**ɑ**] and [**ɤ**] are contrastive in certain contexts, notably after dental affricates and fricatives, e.g. /**tsɑ˧**/ 'busy' vs. /**tsɤ˧**/ 'greedy' and /**sɑ˥**/ 'hemp, *Cannabis sativa*' vs. /**sɤ˥**/ 'blood'. The number of contexts in which the opposition is present has probably been increased through processes of gap-filling. (About the notion of structural gap-filling, see §A.3.3.)

After labial consonants, only [**ɤ**] occurs, not [**ɑ**] (as shown in Table A.2). This pattern is consistent throughout the lexicon, with two main exceptions: (i) in loanwords and (ii) in items that probably underwent vowel harmony. Thus, the sequence /**mɑ**/, which contrasts with /**mɤ**/, has been introduced through Tibetan borrowings, as seen in names such as /**ɖɯ˩mɑ#˥**/, /**ɲi˩mɑ#˥**/, and /**gv̩˧mɑ˧**/, as well as in cultural vocabulary, e.g. /**lɑ˧mɑ˧**/ 'priest, lama' and /**mɑ˩ɳɯ˧-do˥bv̩˩**/ 'Mani wall'. The sequence /**mɑ**/ is also found in the term of address /**ə˧mɑ˧**/, 'Mum, mother', as well as in the clan name /**lɑ˩mɑ˩**/, which may be of Tibetan origin. Two further instances, /**mɑ˩dzɑ˩**/ 'solid ink' and /**mɑ˧tsɑ˥**/ 'origin, cause', could be due to vowel harmony.

With bilabial stops, the sequence /**bɑ˩**/ is found in /**bɑ˩lɑ˩**/ 'jacket, upper outer garment', where it may again be attributable to vowel harmony. The final particle /**bɑ˩˥˩**/, which serves a function comparable to that of question tags in English, also contains this sequence; it belongs to the language's expressive margins. Finally, /**pɑ**/ occurs in /**pɑ˧tɕɤ˧**/ 'plantain (a species of banana)', a loanword from Mandarin (*bājiāo* 芭蕉).

Only [**ɤ**] occurs after velars, while only [**ɑ**] is found after uvulars. For example, compare /**kɤ˥˧**/ 'to knock (on the door)' with /**qɑ˥˧**/ 'to help'.

After /**h**/, there is a phonemic opposition, as illustrated by /**hɑ˩**/ 'to open (one's eyes)' vs. /**hɤ˩**/ 'to dry beside or over a fire'.

A.2.6 A note on phonetic diphthongization

In Na, phonetic diphthongization affects the realization of certain phonemes that are written as simple vowels: /**i**/, /**e**/, /**æ**/, /**ɯ**/, /**ɤ**/, /**o**/, and /**ɑ**/. Some degree of formant movement is also observed in the syllabic consonants /**ɻ̍**/ and /**v̩**/. In general, formants in Na vowels are less stable than in, say, Northern (Parisian) French, whose more conservative varieties provide a canonical illustration of a four-way contrast in vowel openness: /**i**/, /**e**/, /**ɛ**/, and /**a**/, alongside /**u**/, /**o**/, /**ɔ**/, and /**ɑ**/.

Phonetic diphthongization in Na is at its clearest in the vowel /**e**/, which is realized phonetically close to [**ej**], as noted by Lidz (2010: 63, 96). This realization of /**e**/ constitutes a point of similarity with Naxi, a language in which the simple vowels are otherwise relatively stable. Evidence suggests that this diphthongization dates back at least a century. Bonin, an explorer of the turn of the twentieth century, transcribed the Naxi word /**ɲi**˧/ 'two' as *ngié*, and Bacot, another early twentieth-century explorer, rendered the Naxi name of the city of Lijiang, /**ʝi**˧**gʏ̩**˧**dy**˩/, with *yé* for /**ʝi**˧/ (Bacot 1913: 3). Bacot further observed that "each [simple] vowel and its diphthongs [diphthongized variants] are interchangeable" (Bacot 1913: 28),[10] suggesting that diphthongization was not considerable (Michaud & Jacques 2010).

A.2.7 A note on vowel harmony

Anticipatory vowel harmony ('right-to-left' harmony) is a salient phonetic tendency in connected speech in all the Naish languages studied so far. For instance, in Laze, /**ʝi**˧**dy**˧/ 'family' is sometimes realized close to [**ʝy**˧**dy**˧]. This is not a phonological phenomenon in the strict sense: vowel oppositions in the first syllable of disyllabic words are not neutralized. However, in certain cases, this phonetic tendency becomes lexicalized. For instance, in some Naxi dialects, including A-sher, 'pigswill' is /**bu**˩**-hɑ**˧/ (from /**bu**˩/ 'pig' and /**hɑ**˧/ 'food'), whereas in other dialects, such as Nda-le, it has undergone vowel harmony and become /**bɑ**˩**-hɑ**˧/. This phenomenon is reported by Hé Zhìwǔ (1985: 11), though without mention of the dialects concerned.

The extent of this phenomenon varies considerably across languages and dialects. Among the three Naish languages that I studied so far (Laze, Na, and Naxi), Naxi is least prone to the lexicalization of such phonetic changes, whereas Laze exhibits the highest degree of lexicalization. The entrenchment of vowel-harmonized forms sometimes co-occurs with other phonological processes, such as the voicing of intervocalic voiceless consonants. A typical example from Laze is /**ʂ̩ie**˧**-lie**˧**mie**˧/ 'seventh month', from /**ʂ̩ɯ**˧/ 'seven' and /**ɬie**˧**mie**˧/ 'month'. In this case, the vowel of the first syllable has changed through vowel harmony, and the initial /**ɬ**/ has undergone voicing to become /**l**/.

In Na, the phonetic tendency towards regressive vowel harmony is especially strong for the vowel /**æ**/. For instance, the combination of /**ŋwɤ**˧/ 'five' with the monetary unit /**mæ**˩$_{\mathbf{a}}$/ is pronounced [**ŋwæ**˧**-mæ**˥] 'five yuan' in casual speech, although the careful (hyperarticulated) pronunciation is [**ŋwɤ**˧**-mæ**˥]. Function words, which tend to have a weaker phonetic realization overall, are especially susceptible to vowel harmony. Here are two examples of this tendency:

[10] *Original text:* chaque voyelle et ses diphtongues sont interchangeables.

(i) The ACCOMPLISHED prefix /**le˧-**/ is realized close to [**læ**] when the vowel of the following verb is /**æ**/ or an apical vowel.

(ii) The negation prefix /**mɤ˧-**/ is realized close to [**mɑ**] when the vowel of the following verb is an apical vowel.

Vowel harmony for these two prefixes is so strong that, for a time (2008–2011), I transcribed them as /**lə˧-**/ and /**mə˧-**/, with a neutral vowel. However, vowel harmony also affects other morphemes, such as the DURATIVE prefix /**tʰi˧-**/, suggesting that this phenomenon is more widespread than initially thought. Pending the results of future phonetic studies of the relative degree of vowel harmony across a broad range of grammatical words, I chose to stop granting special phonemic status to the vowel in the ACCOMPLISHED and negation prefixes.

This topic warrants further investigation, particularly in relation to diachronic phonology. Vowel harmony has the potential to introduce new phonotactic combinations into the phonological system, as harmonized phonetic forms occasionally become lexicalized. For instance, syllables containing a dental stop followed by /**æ**/, such as /**læ**/, /**tæ**/, and /**tʰæ**/, are scarce in Na, and most appear to originate from vowel harmony (see §A.3.5).

A.3 Initial consonants

A.3.1 On-glides

The high vowels /**i**/, /**ɯ**/, and /**o**/ have phonetic on-glides: they are realized as [**ʝi**], [**ɣɯ**], and [**wo**], respectively.

The degree of friction in the realization of /**i**/ prompted the transcription of its on-glide as a fricative, [**ʝ**], rather than an approximant, [**j**]. The approximant notation is instead used for the complex rhymes /**jo**/, /**jɤ**/, and /**jæ**/, where the initial glide is produced with less friction.

The syllable /**ɯ**/ could also be transcribed with an approximant onset, [**ɰɯ**]. This is the convention adopted in a dictionary of Naxi (Pinson 2012), a language in which the phonemic analysis of this syllable is the same as in Yongning Na and the phonetic realization does not sound any different to me.

The phonetic on-glides in [**ʝi**], [**ɣɯ**] and [**wo**] are salient enough to warrant inclusion in transcriptions, even at the cost of some deviation from notational economy. However, they involve less formant movement than the phonemic on-glides of the complex rhymes: /**jo**/, /**jɤ**/, /**jæ**/, /**wɤ**/, /**w̃ɤ**/, /**wæ**/, /**w̃æ**/, /**wɑ**/, and /**w̃ɤ**/.

Onsetless syllables tend to coalesce phonetically with the preceding syllable in polysyllabic words and tightly-knit polysyllabic expressions. For several years, I mistakenly believed that the manner adverbials ‘in this way, thus’ and ‘in that way’ were /**ʈʂʰɯ˧ni˧˥**/ and /**tʰv̩˧ni˧˥**/, respectively. In fact, these phrases are /**ʈʂʰɯ˧ne˧ ʝi˥**/ and /**tʰv̩˧ne˧ ʝi˥**/, consisting of a manner adverbial (/**ʈʂʰɯ˧ne˧˥**/, /**tʰv̩˧ne˧˥**/) followed by the verb ‘to do’, /**ʝi˥**/ (as discussed in §A.2.2). The trisyllabic structure of these phrases did not emerge from phonetic evidence: it was ultimately revealed by their tonal behaviour.

This is illustrated in example (2), shown here as initially transcribed:

(2) ʈʂʰɯ˧ni˧˥ | gwɤ˩-ɲi˥ ◊ mæ˩!
ʈʂʰɯ˧ni˧˥ gwɤ˩ -ɲi˩ mæ˧
in_this_way to_sing CERTITUDE OBVIOUSNESS
‘This is how [people] used to sing!’ (*Caravans.51* {0004530#S51}, *53*, *57*)

If the notation /**ʈʂʰɯ˧ni˧˥**/ were correct, it should be possible to group the adverbial with the following verb phrase into a single tone group, yielding ‡ **ʈʂʰɯ˧ni˧ gwɤ˥-ɲi˩-mæ˩**. But this stylistic option is not available: the manner demonstrative cannot project an H tone onto a following syllable, as would be expected if it carried MH# tone.

The solution to this puzzle lies in the fact that the adverbial phrase ‘thus, in this way’ is trisyllabic (/**ʈʂʰɯ˧ne˧-ʝi˥**/) and remains so despite the strong phonetic coarticulation between its final two syllables. The correct notation of (2) is provided in (3).

(3) ʈʂʰɯ˧ne˧ ʝi˥ | gwɤ˩-ɲi˥ ◊ mæ˩!
ʈʂʰɯ˧ne˧˥ ʝi˥ gwɤ˩ -ɲi˩ mæ˧
in_this_way to_do to_sing CERTITUDE OBVIOUSNESS
‘This is how [people] used to sing!’ (*Caravans.51* {0004530#S51}, *53*, *57*)

Turning now to other vowels, /**æ**/ and /**ɑ**/ may be realized either with a glottal stop (hard phonetic onset) or with soft phonation: [**ɦæ**] ≈ [**ʔæ**], [**ɦɑ**] ≈ [**ʔɑ**]. Regarding the choice of one onset type over the other, a study of glottal stops before word-initial vowels in American English concludes that “full glottal stops [**ʔ**] are predicted overwhelmingly by prominence and phrasing” (Garellek 2012: 20). The study also reviews a set of factors that would provide a useful starting point for future phonetic research on this phenomenon in Yongning Na:

> … the factors that promote the occurrence of word-initial glottalization (…) may be segmental, lexical, prosodic, or sociolinguistic. In English, segmental factors include hiatus (V#V) environments and word-initial back

> vowels are found to glottalize more frequently than non-back vowels. As for lexical factors, content words exhibit more frequent glottalization than function words. Sociolinguistically, women are known to use glottalization more than men. Prosodically, the presence of stress and/or a pitch accent on the word-initial vowel, as well as a larger juncture with the preceding word, are known to promote glottalization. Researchers working on other languages have found additional factors that promote the occurrence of word-initial glottalization, including presence of a preceding pause as well as speech rate and low vowel quality for German. (Garellek 2012: 1-2)

Unlike in Naxi, where /**ɤ**/ is realized as /**ɣɤ**/, in Yongning Na the vowel /**ɤ**/ never forms a syllable on its own. As explained in §A.2.5, the opposition between /**ɤ**/ and /**ɑ**/ is restricted to a few contexts; synchronically, it may be considered to be neutralized in onsetless syllables.

No distinct onset portion is found for /**v̩**/, which is realized simply as [**v̩**]. Likewise, the rhyme /**ɻ̍**/ is not separated from the preceding rhyme by a glide or glottal stop. For instance, in /**bv̩˧ɻ̍#˥**/ 'fly (the insect)' the /**v̩**/ and /**ɻ̍**/ occur in direct succession: [**bv̩ɻ̍**].

The following paragraph examines syllables that canonically begin with a glottal stop.

A.3.2 Initial glottal stop

The rhymes /**ə**/ and /**ɻ̍̃**/ begin with a phonetic glottal stop when pronounced in isolation, and also word-medially in hyperarticulated realizations. For instance, /**ʂæ˩ɻ̍̃˩**/ 'bone' is realized as [**ʂæ˩ʔɻ̍̃˩˥**] in careful speech, whereas in casual speech there is no glottal stop: [**ʂæ˩ɻ̍̃˩˥**].

The morpheme /**u˧**/, an ASSOCIATIVE first person pronoun, appears in the expressions /**u˧=ɻ̍˩**/ 'my clan, my people' and /**u˧=ɻæ˩**/ 'us (as opposed to them)'. It is pronounced with an initial glottal stop, yielding [**ʔu˧=ɻ̍˩**] and [**ʔu˧=ɻæ˩**]. This morpheme constitutes the only documented instance of this syllable type. However, even a single example suffices to establish it as distinct from the syllable realized as [**wo**] ≈ [**wu**], which occurs in words such as //**wo˥**// 'hard', //**wo˩kɤ#˥**// 'swing', and //**wo˩˥**// 'turnip leaves'.

This raises the question of how to phonemicize this contrast. Several solutions are possible, including the three outlined below, which could also be combined in various ways:

(i) The syllable [**ʔu**] could be phonemicized as /**u**/, and [**wo**] ≈ [**wu**] as /**wu**/, thus granting phonemic status to the initial glide.

(ii) The syllable [**ʔu**] could be phonemicized as /**ʔu**/, and [**wo**] ≈ [**wu**] as /**u**/, thus granting phonemic status to the initial glottal stop.

(iii) The syllable [**ʔu**] could be phonemicized as /**u**/, and [**wo**] ≈ [**wu**] as /**o**/, ascribing the different initials to phonotactic rules of assignment of empty-onset fillers.

Option (iii) is adopted here, obviating the need for recognizing the glottal stop as a phonemic initial. However, from a perceptual point of view, the most salient cue to the opposition appears to reside in the initial segment. From this perspective, it does not appear unlikely that either the initial glottal stop or the labial-velar approximant could gain phonemic status depending on how new generations of learners interpret expressive forms such as **ʔo!** (an interjection indicating surprise), dutifully recorded in the dictionary (Michaud et al. 2024: 167).

Cases of phonemic ambiguity such as this hold potential for changes to the system. A relevant example is the case of initial /**ʁ**/, discussed in the next paragraph, which illustrates how new consonants can emerge through the phonemic reinterpretation of empty-onset fillers.

A.3.3 Initial /ʁ/ as a phonemicized empty-onset filler

The status of the syllables /**wɤ**/, /**wæ**/, /**o**/, /**ɑ**/, and /**æ**/ is particularly interesting. These rhymes can occur with an initial voiced uvular fricative, /**ʁ**/.[11] (Phonetically, this /**ʁ**/ is weakly articulated and, in some hypo-articulated tokens, can be mistaken for /**w**/.) Some of the words in question derive diachronically from onsetless syllables. For instance, the word for 'village', /**ʁwɤ˧**/, corresponds to /**wɤ˧**/ in Naxi and Laze; the homophonous word for 'mountain', also /**ʁwɤ˧**/, has an equivalent in Naxi where it means 'hill, hillock' (/**wɤ˧**/). Other cases reflect developments from syllables with an initial velar or uvular cluster. For example, the word for 'sword', /**ʁæ˧mi˧**/, corresponds to Naxi /**ŋgæ˩**/.[12]

The hardening of soft onsets would, in principle, have led to the systematic absence of syllables realized as [**wɤ**], [**wæ**], [**o**], [**ɑ**], or [**æ**], since these became

[11]The voiced uvular fricative /**ʁ**/ is not uncommon in this linguistic area, but with a widely different phonemic status from one language to another. For instance, in Lizu (Chirkova & Chen 2013), it is an allophone of the voiced velar fricative /**ɣ**/ before /**ɐ**/ and /**wɐ**/, e.g. /**ɣɐ˥˩**/ [**ʁɐ˥˩**] 'needle', /**ɣwɐ˥˩**/ [**ʁuɐ˥˩**] 'to thunder'.

[12]A comparable process of onset hardening has been reported in Zeluo Ersu, where /**w**-/ is sometimes realized with frication, as [**ɣʷ-**] (Sūn Hóngkāi 1982). Among the last speakers of the language, there is variation between [**ʁ**] and [**w**] (Chirkova 2014).

[**ʁwɤ**], [**ʁwæ**], [**ʁo**], [**ʁɑ**], and [**ʁæ**]. This pattern is indeed observed for /**wæ**/: only [**ʁwæ**] is attested, with no recorded instance of [**wæ**]. For instance, the word for ‘left, leftward’, /**ʁwæ˥**/, corresponds to Naxi /**wæ˧**/ and Laze /**væ˧**/. For the other syllables, however, there are oppositions between forms with and without initial /**ʁ**/. This suggests that a process of structural gap-filling has taken place, leading to the reintroduction of onsetless syllables into the system. Further progress in the etymological analysis of individual examples will be necessary to elucidate the specific mechanisms involved in this development. The hardening of empty-onset fillers must date back a relatively long way, judging from the number of currently attested onsetless syllables. For instance, in the case of /**wɤ**/, the following examples can be cited:

- the classifier for loads, /**wɤ˩**$_{\mathbf{b}}$/,
- the noun ‘serf, slave’, /**wɤ˧**/,
- the adverb ‘again, anew’, /**wɤ˩˥**/,
- the verbs ‘to depend on, to rely on’, /**wɤ˩**$_{\mathbf{b}}$/ and ‘to detour past, to bypass’, /**wɤ˩~wɤ˩**/,
- the final exclamative particle /**wɤ˧**/, which conveys obviousness.

The change from proto-Naish ***j** to Na /**ʐ**/ may represent another instance of the same hardening process. The verb ‘to sleep’ in Na, /**ʐi˧˥**/, corresponds to Naxi /**ʝi˥**/, which is phonemically a simple /**i**/, to which an empty-onset filler is added. In Laze, the cognate is /**zi˩**/. The reconstruction proposed by Jacques & Michaud (2011) is ***jip**, supporting the hypothesis that the development of /**ʐ**/ involved a phonemic reinterpretation of an earlier onset filler.

A.3.4 Velar and uvular stops

Velar and uvular stops are in complementary distribution in most environments. However, they contrast before /**v̩**/, /**wɤ**/, and /**o**/. Examples of the syllables /**kv̩**/, /**qv̩**/, /**kʰv̩**/, /**qʰv̩**/, /**ko**/, /**qo**/, /**kʰo**/, and /**qʰo**/ are provided in (4). In addition, a single instance of /**qi**/, contrasting with /**ki**/, has also been recorded.

(4) a.

/**qv̩**/		/**kv̩**/	
qv̩˩˥	handle	**kv̩˥**	garlic
qv̩˧˥	to frighten	**kv̩˧˥**	to be able to
qv̩˧tʂæ˧˥	throat	**kv̩˧tʂɯ˥$**	nail
mæ˧qv̩˩	tail	**kv̩˧dʐɯ˧˥**	tent

b.	/**qʰv̩**/		/**kʰv̩**/	
	qʰv̩˦	six	**mv̩**˧**kʰv̩**˦	smoke
	qʰv̩˧	horn	**kʰv̩**˥	to cut (grass)
	qʰv̩˧	hole	**kʰv̩**˥	dog
	bv̩˩**qʰv̩**˩	conch shell	**kʰv̩**˦	year
c.	/**qo**/		/**ko**/	
	qo˥	to love	**ko**˥	hill
	qo˩**ho**˦	bamboo box	**ko**˩	to bask
	qo˩**qɑ**˩	mountain pass	**ko**˩**ɖʐo**˩	flail
	-qo˧	inside	**mæ**˩**ko**˥	harness
d.	/**qʰo**/		/**kʰo**/	
	qʰo˦	to kill	**kʰo**˥	to spread (e.g. a sheet)
	qʰo˧**lo**˧	wheel	**kʰo**˧**lo**˧	prayer wheel
	qʰo˩**tv̩**˦	tree stump	**tse**˧**kʰo**˩	sanctuary
	qʰo˩**mv̩**˩	straw hat	**hæ̃**˧**kʰo**˧	princess, young lady
e.	/**qi**/		/**ki**/	
	qi˧**qi**˧	originally, at first	**ki**˧	to give

This situation is unsurprising in areal context. In Lizu, for example, velar and uvular stops only contrast before /**o**/, as in /**ko**˨˧/ 'to beg' vs. /**qo**˥˩˥/ 'hole, pit' (Chirkova & Chen 2013).

From a diachronic point of view, uvulars may originate from a variety of historical sources (Sun 2003a: 782-783). In the case of Na, cognates with uvular initials in rGyalrong – a conservative language – support the hypothesis that uvulars in Na are of considerable antiquity (Jacques & Michaud 2011: 492).

A.3.5 Retroflex stops and affricates

Yongning Na has (i) an opposition between dental and retroflex affricates, which carries a high functional load, and (ii) an opposition between dental and retroflex stops and nasals, but only before /**i**/, /**æ**/, /**v̩**/ and /**o**/. Examples include: /**ʈʰi**˩/ 'tired' vs. /**tʰi**˩/ 'to plane (wood)'; /**ʈi**˩/ 'to get up' vs. /**ti**˩/ 'to knock, to tap (lightly)'; /**ʈæ**˧**bɤ**˧/ 'Buddhist priest' vs. /**tæ**˧**pv̩**˩/ 'thin, skinny'; and /**ɖo**˧$_{\mathbf{a}}$/ 'to allow; ought; to have to' vs. /**do**˥/ 'to climb'. After retroflex consonants, the vowel /**o**/ is realized close to [**u**].

The consonants transcribed here as retroflexes are articulated notably less far back than the canonical retroflexes of languages such as Tamil or Nepali (Khatiwada 2009). A palatographic study may reveal that these sounds are in fact postalveolar rather than retroflex. This situation highlights a shortcoming of the current International Phonetic Alphabet system: there is a need for distinct symbols for postalveolar and dental articulations, as argued by Ladefoged & Maddieson (1996: 21-30). This is not merely a matter of fine phonetic detail to be addressed by means of diacritics. The absence of distinct IPA symbols for postalveolars has led to the widespread adoption of retroflex symbols for non-dental sounds, a practice that creates no small amount of typological confusion and skews cross-linguistic studies of phonological inventories.

In the absence of dedicated IPA symbols for postalveolar stops, the provisional solution adopted here is to use the symbols for retroflex sounds.

A.3.6 Laterals /l/ and /ɬ/, and retroflexes /ɭ/ and /ɻ/

The laterals /**l**/ and /**ɬ**/ are contrastive in Yongning Na, as demonstrated by pairs such as /**li**˧$_{\mathbf{a}}$/ 'to look' vs. /**ɬi**˧/ 'month' and /**lo**˦/ 'thick' vs. /**ɬo**˦/ 'deep'. Phonetically, the voiced lateral /**l**/ has a broad range of allophonic variation. It is realized as retroflex before /**ɯ**/, e.g. in the classifier for round objects (bowls, grains), which also serves as generic classifier: /**ɭɯ**˧$_{\mathbf{b}}$/. The phonemic analysis for this syllable was only arrived at after the greatest hesitations. A wide array of hypotheses was explored, including [**ɻɯ**], [**ɬɯ**], [**lv̩**], [**ly**], [**ɭy**], and syllabic [**ɭ̩**] or [**ɹ**]. The entire syllable is loosely articulated: the initial is close to an approximant, and the vowel quality is not precise, so that the syllable often resembles a monophonemic [**ɭ̩**].

Before /**v̩**/, the voiced lateral /**l**/ is also slightly retroflex. In all contexts, its articulation is accompanied by some friction. This characteristic is at its clearest before the high front vowel /**i**/, but it is also observed before open vowels such as /**ɑ**/ and /**æ**/. Phonetically, this supports a transcription of this allophone as [**ɮ**] rather than [**l**]. Phonologically, it would be economical to consider that the two lateral phonemes are distinguished solely by voicing, reinforcing the case for a phonemic analysis as /**ɮ**/.

The compromise adopted here, for the sake of simplicity, consists in using the notation /**l**/ rather than /**ɮ**/. On the other hand, in the syllable [**ɭɯ**] (phonemically /**lɯ**/), the use of a retroflex initial is retained in transcription, to reflect what I perceive as a significant phonetic departure from the canonical realizations of /**l**/ in other contexts. This choice aims to keep the transcriptions close to surface forms.

These observations on the allophonic variation of /**l**/ shed indirect light on the distribution of the retroflex approximant /**ɻ**/: it is plausible that this phoneme originated as an allophone of /**l**/ which drifted to such a phonetic distance that it opened a structural gap that was subsequently filled through borrowing and vowel-harmony-driven developments. In synchrony, /**ɻ**/ only appears in the syllables /**ɻæ**/ and /**ɻwæ**/. These correspond to /**læ**/ and /**lwæ**/ in Laze: compare Na /**ɻwæ˥**/ 'to shout, to cry' with Laze /**lwæ˧**/, and Na /**ɻæ˩˥**/ 'seed' with Laze /**læ˩**/. Roselle Dobbs (p.c. 2013) indicates that in villages of the area referred to in Na as Lataddi **lɑ˧tʰɑ˧-di˧˥**, located east of Lugu Lake, these items retain their original lateral initials.

Synchronically, /**ɻæ**/ contrasts with /**læ**/. However, the latter only appears (i) in borrowings such as /**læ˧tsɯ˥**/ 'chili peppers', from Southwestern Mandarin 辣子 [**la.tsɿ**], (ii) in the ACCOMPLISHED prefix /**le˧-**/, whose phonetic realizations – determined by the vowel of the following syllable – include [**læ**], and (iii) in forms where /**æ**/ may have arisen through vowel harmony, e.g. /**læ˧ʁæ˥**/ 'raven'. (Regressive vowel harmony, a prominent phonetic tendency in Na, becomes sporadically lexicalized: see §A.2.7.) As for /**ɻwæ**/, no lateral counterpart /**lwæ**/ is attested.[13]

It is therefore plausible to hypothesize that present-day /**ɻæ**/ and /**ɻwæ**/ derive from earlier *__læ__ and *__lwæ__, which underwent phonetic retraction towards [**ɻæ**] and [**ɻwæ**], thereby vacating the [**læ**] and [**lwæ**] phonetic slots. The [**læ**] slot was subsequently reoccupied by other items.

These diachronic considerations do not detract from the synchronic phonemic status of /**ɻ**/.

A.3.7 The glottal fricative /h/ and the sound [f]

Yongning Na has a glottal fricative /**h**/. At an earlier stage of the language, the sound [**f**] can be hypothesized to have been entirely absent: early borrowings from Mandarin with initial [**f**] in the donor language were reinterpreted with initial /**h**/. For instance, the word for 'method, solution' (办法, Standard Mandarin *bànfǎ*) was borrowed into Na as /**pæ˧˥hwɤ˧**/.

The sound [**f**] does appear in more recent layers of borrowings, however: e.g. /**fæ˧**/ 'direction' (方, Standard Mandarin *fāng*) and /**fɑ˩**$_{\mathbf{a}}$/ 'to ferment' (发（酵）, Standard Mandarin: *fā*). A plausible scenario is that, at some point in the history of Na, the phoneme /**h**/ came to be realized with a friction source located at

[13] A strikingly similar synchronic pattern is found in Lizu, where the /**ɹ**/ phoneme only occurs before /**æ**/, /**ə**/, and /**wæ**/, e.g. /**ɹæ˥˩**/ 'yak', /**ɹə˧˥**/ 'to laugh', and /**ɹwæ˧˥**/ 'chicken' (Chirkova & Chen 2013). The diachronic origins appear different, however, as comparative data from Ersu and Duoxu point instead to an earlier ***r**, as discussed in Yu (2012). Surface similarities in sound patterns tend to arise through areal convergence, as well as from cross-linguistic (panchronic) regularities in processes of phonological erosion.

a point in the vocal tract conditioned by the following vowel. For instance, this could result in palatal realization before /**i**/ and labial-dental realization before /**v̩**/, yielding forms such as [**çi**] and [**fv̩**]. Once the sound [**f**] had thus been introduced into Na at the phonetic level, the way was paved for [**f**]-initial loanwords to be integrated without difficulty.

In the present state of the language, speakers of Yongning Na have no difficulty in pronouncing [**f**] before any rhyme. This can be taken as evidence that [**f**] is no longer perceived as an allophone of /**h**/. It is well-known that allophones that drift far apart often acquire psychophonetic independence from one another: well-documented cases include German [**ç**] and [**x**], and Standard Mandarin [**ɕ**] and [**x**]. As the psychological reality of the underlying unity among allophones wanes, the structural resistance to gap-filling likewise decreases.

In view of this situation, and also in order to keep the transcription close to the surface forms, the syllable [**fv̩**] is transcribed as such, rather than pushing phonemicization to an extreme by analyzing it as /**hv̩**/. Under a flatly synchronic analysis that includes Mandarin borrowings, the sound [**f**] needs to be granted phonemic status, hence its inclusion in Table A.1.

The choice adopted for the phonemic analysis of [**fv̩**] has potential implications for the analysis of the syllable [**hu**]. One possibility would be to phonemicize [**fv̩**] as /**fv̩**/, and [**hu**] as /**hv̩**/, thereby eliminating the /**u**/ phoneme altogether (Roselle Dobbs, p.c. 2016). This remains one of several areas in Yongning Na phonemics where alternative analyses are defensible.

The syllable [**çi**], phonemicized as /**hi**/, contrasts with /**ɕi**/.

A.4 Comments about the inventory of syllables

The inventory of attested combinations of initials and rhymes, provided at the outset of this Appendix (in Tables A.2 and A.3), reveals that numerous phonemic oppositions are found in highly restricted contexts in Yongning Na. A similar situation is found in Naxi (Michaud 2006a). The strict application of the principles of Praguean synchronic description leads to an analysis of these cases as extreme instances of *neutralization* of phonemic contrasts.

For instance, in Yongning Na the opposition between nasal and oral vowels is neutralized in all environments except after /**h**/ and in onsetless syllables. It may appear counter-intuitive to speak of neutralization in such cases, since the concept is more commonly associated with situations where a thoroughgoing contrast disappears in a restricted environment, e.g. in Trubetzkoy's classical example: French /**e**/ and /**ɛ**/ contrast only in open syllables, the opposition being neutralized in closed syllables.

> In French (…) an opposition between **e** and **ɛ** only occurs word-finally in open syllables, e.g. *les* 'definite article.PL' vs. *lait* 'milk' and *allez* 'to go.2PL' vs. *allait* 'to go.3SG.PST'. In all other positions the occurrence of **e** and **ɛ** is predictable: **ɛ** occurs in closed syllables, **e** in open. These two vowels must thus be considered two phonemes in open-syllable-final position, and combinatorial variants of a single phoneme in all other positions. We call such oppositions *neutralizable* oppositions, the positions in which the neutralization takes place *positions of neutralization*, and those positions where the opposition is relevant *positions of relevance*. (Trubetzkoy 1969: 78)[14]
>
> *Original text:* Im Französischen kommen aber **e** und **ɛ** nur im offenen Auslaute als Glieder einer phonologisch-distinktiven Opposition vor (*les-lait*, *allez-allait*); in den übrigen Stellungen ist das Vorkommen von **e** und **ɛ** mechanisch geregelt (in gedeckter Silbe **ɛ**, in ungedeckter **e**), so daß diese zwei Vokale nur im offenen Auslaut as zwei Phoneme, in den übrigen Stellungen dagegen als kombinatorische Varianten eines einzigen Phonems gewertet werden müssen. Der phonologische Gegensatz ist also im Französischen in gewissen Stellungen a u f g e h o b e n. Solche Oppositionen nennen wir a u f h e b b a r; jene Lautstellungen, in denen die Aufhebung erfolgt, A u f h e b u n g s s t e l l u n g e n, jene, wo die Opposition relevant ist, R e l e v a n z s t e l l u n g e n. (Trubetzkoy 1939: 70)

Importantly, the term *neutralization* should not be understood in a dynamic sense, as if the opposition once existed and was subsequently neutralized. Rather, its application is static, flatly synchronic (Martinet 1969: 257–259; 1970: 87–89). It is not unusual for a synchronic formulation to stand as a reverse image of a diachronic perspective. For instance, when describing the synchronic stage at which Chinese contrasted three tones (A, B, and C) on non-obstruent-final syllables, it can be said that the tonal opposition was neutralized on obstruent-final syllables (classified as belonging in a fourth category, D), even though this opposition never existed on those syllables at any historical stage.

In Praguean phonology, phonemes are defined by their relations of opposition to one another, and features are assigned to phonemes on the basis of these relations. If two phonemes share a set of features not found in any other phoneme, the opposition is bilateral and neutralizable; if not, it is multilateral and non-neutralizable. Stated differently, only where an opposition exists is a feature phonologically contrastive. An opposition is considered neutralized whenever one of its members does not occur in a particular environment. The output of

[14]Some modifications to the translation were made by Roselle Dobbs.

neutralization is referred to as an archiphoneme. For book-length treatments of the twin notions of neutralization and archiphoneme, see Akamatsu (1988) and Silverman (2012).

Archiphonemic notations are not employed in the present volume, for reasons that have to do with legibility. Devising specific symbols for archiphonemes is easier for International Phonetic Alphabet symbols that happen to coincide with Roman letters, as using capitals is an option; but this won't work for all International Phonetic Alphabet symbols, a number of which do not derive from Roman letters.

From a dynamic point of view, gaps in the inventory of syllables can provide structural clues to past changes and current tensions within the phonological system.

A.4.1 Combinations of a dental stop and /æ/ vowel seem recent

Combinations of a dental stop followed by /**æ**/ are scarce. The sole example for /**dæ**/ is /**læ˧dæ˧qæ˥**/ 'armpit'. The two examples for /**tæ**/ are /**tæ˧ɻæ˩**/ 'Adam's apple, oesophagus' and /**tæ˧pv̩˩**/ 'thin, skinny', while the only example for /**tʰæ**/ is /**tʰæ˧ɻæ˩**/ 'book'. With the exception of /**tæ˧pv̩˩**/ 'thin, skinny', all of these can plausibly be attributed to vowel harmony. The word for 'book', /**tʰæ˧ɻæ˩**/, corresponds to Laze /**tʰɑ˧ɹ˧**/ and Naxi /**tʰe˧ɣɯ˧**/; the vowel correspondence /**e:æ:ɑ**/ is otherwise unattested, reinforcing the hypothesis that vowel harmony or some other exception-causing factor was at play here. Similarly, Yongning Na /**tæ˧ɻæ˩**/ 'Adam's apple, oesophagus' corresponds to Labai Na /**tɑ˧ɻ̍˧**/, again presenting an irregular correspondence, as the regular correspondences are simply **æ::æ** and **ɑ::ɑ**.

A.4.2 A marginal combination: Dental stop plus /ɤ/

The only two attested combinations of a dental stop followed by /**ɤ**/ are /**dɤ**/ and /**tɤ**/, with no occurrence of /**tʰɤ**/. Lexical examples are extremely rare. One is the extra-distal demonstrative /**dɤ˩-qo˧**/ 'way over there', analyzed in §9.3.1. The expressivity of extra-distal phrases goes a great distance towards explaining phonemic and tonal oddities in their first syllable, where expressivity is strongest.

All other instances of /**dɤ**/ and /**tɤ**/ occur before /**ɻ̍**/: in /**hṽ̩˧-dɤ˧ɻ̍#˥**/ 'clumsy', /**õ˧-dɤ˧ɻ̍˧**/ 'fundamentally' (from /**õ˦**/ '(one)self'), /**ʂɯ˧-tɤ˧ɻ̍˧**/ 'smooth (e.g. carefully planed wood)', and /**dʑɯ˩-tɤ˩ɻ̍˥**/ 'humid, moist' (from /**dʑɯ˩**/ 'water'). These words contain a phonetic sequence resembling a trilled rhyme. Initially, these were transcribed as /**dṛ**/ and /**tṛ**/, adding another rhotic rhyme to the inventory. This analysis appears correct for the phonological system of speaker F5 (F4's daughter-in-law): when repeating the phrase [**hṽ̩˧dṛ˧~hṽ̩˧dṛ˧-zo˥**] 'clumsily' very slowly, she still realizes the phonetic sequence [**dr**] as a single syllable. By contrast, in the speech of Mrs. Latami (F4), the phonetic realization is slightly less packed together, ranging between [**dər**] and [**dəɻ̍**] for the syllable with a voiced initial, and between [**tər**] and [**təɻ̍**] for the syllable with an unvoiced initial.

A phonological argument demonstrating that there are two syllables, not one, comes from /**ho˧dʑɯ˧tɤ˥ɻ̍˩**/ 'paste, starch', literally 'watery gruel', derived from /**ho˥**/ 'porridge' and /**dʑɯ˩-tɤ˩ɻ̍˥**/ 'humid, moist'. This noun bears an H tone on its penultimate syllable and an L tone on its final syllable. Since no falling tone (HL, HM or ML) ever occurs on a single syllable in Yongning Na, this tonal pattern supports the conclusion that there are two syllables in this sequence: /**tɤ.ɻ̍**/.

Interestingly, the two forms containing /**tɤ**/ have variants with /**dɤ**/: /**ʂɯ˧-dɤ˧ɻ̍˧**/ for 'smooth' and /**dʑɯ˩-dɤ˩ɻ̍˥**/ for 'humid, moist'. However, the reverse is not true: the two forms containing /**dɤ**/ do not have variants with /**tɤ**/ – ‡ **hṽ̩˧-tɤ˧ɻ̍#˥** for 'clumsy' and ‡ **õ˧-tɤ˧ɻ̍˧** for 'fundamentally' are unacceptable. This suggests that /**-dɤ.ɻ̍**/ was once a suffix used to derive adjectives (also employed adverbially). However, it must have ceased to be productive long ago, as two of the four examples underwent a separate phonetic evolution. It may never have been highly productive to begin with.

A.4.3 After alveolopalatals, is the rhyme /o/ or /jo/?

The syllables transcribed as /**tɕʰo**/, /**tɕo**/ and /**dʑo**/ could also be analyzed as /**tɕʰjo**/, /**tɕjo**/ and /**dʑjo**/, with a /**-jo**/ rhyme. The correspondence between Na /**dʑo**/ and Naxi /**gy**/ (phonetically: [**ɟy**]) suggests that the initial in Na became palatalized under the influence of a following high front vowel or glide. From a synchronic point of view, however, it appears more appropriate to transcribe these syllables as consisting of an alveolopalatal initial followed by a back vowel.

A.4.4 Phonemic status of the retroflex nasal

Two instances of the syllable /**ɳṿ**/ are attested: /**ɳṿ˥**/ 'to sniff; to get to know (news)' – frequently used in the negative, as in /**mɤ˧-ɳṿ˥**/ '[I] don't know' – and /**ɳṿ˩˥**/ 'moth'. The syllable /**ɳṿ**/ contrasts with /**nṿ**/, exemplified by /**nṿ˥**/ 'to bury'.

An alternative analysis would be as [**ɳɻ̍**], phonemically /**nɻ̍**/. Under this view, there would be no need to posit a phonemic retroflex nasal /**ɳ**/ contrasting with

/**n**/: retroflex realizations would be seen as conditioned by a following /**ɯ**/ or /**ɻ̍**/, with the combinations /**nɯ**/ and /**nɻ̍**/ phonetically realized as [**ɳɯ**] and [**ɳɻ̍**], respectively. In the absence of a phonemic opposition, and given the phonetic proximity between these two rhymes in a retroflex context, this interpretation is not absurd.

Following the same logic, one might further reinterpret /**ɖv̩**/, /**ʈv̩**/ and /**ʈʰv̩**/ as /**dɻ̍**/, /**tɻ̍**/ and /**tʰɻ̍**/. This does not, however, yield a genuine simplification of the system, since a phonemic opposition between dentals and retroflexes is independently attested before other vowels (e.g. /**ʈi**/ vs. /**ti**/).

The choice to transcribe these syllables as /**ɳv̩**/ reflects my auditory impression that, in the current state of the language, the rhyme in question is closer to [**v̩**] than to [**ɻ̍**].

A.4.5 The palatal nasal

The palatal nasal [**ɲ**] only appears in the syllable [**ɲi**], suggesting the possibility of a reanalysis as an allophone of one of the other nasal initials in the system: /**m**/, /**n**/, /**ɳ**/ or /**ŋ**/. From a language-independent perspective, the most plausible analysis is phonemicization as /**ŋi**/, in view of the well-documented palatalizing effects of high front vowels on velar consonants.

This analysis is possible in principle, given the absence of a syllable [**ŋi**] in the syllabic inventory of Yongning Na. In Naxi, analysis of palatal initials as allophones of velars is an attractive solution, because it applies throughout the system: [**cʰi**], [**ci**], [**ɟi**], [**ɲɟi**] and [**ɲi**] can be analyzed as /**kʰi**/, /**ki**/, /**gi**/, /**ŋgi**/ and /**ŋi**/ (Michailovsky & Michaud 2006: 14). However, detailed examination of the Naxi lexicon shows that this analysis amounts to an internal reconstruction, rather than a synchronic phonemic analysis, as expressive coinages have since filled structural gaps that were initially created by the palatalization of velars (Michaud & He 2015: 7).

In Na, it is less tempting to phonemicize [**ɲi**] as /**ŋi**/, as phonemic combinations of velar initials with the vowel /**i**/ do occur and are not strongly palatalized. The transcription adopted here therefore remains close to the surface form: /**ɲi**/.

A.4.6 Syllables introduced by Mandarin borrowings

Mandarin borrowings have the potential to bring considerable changes to the phonotactic patterns of Na syllables; in particular, they introduce numerous new combinations of vowels with glides. The overall situation is comparable to that observed in Naxi. A Naxi speaker from Dadong 大东, He Likun 和丽昆, compiled an inventory of the syllables present in his own speech, and found that recent Mandarin loanwords account for about 150 of the syllables he uses when speaking Naxi (Michaud & He 2015). He Likun is essentially bilingual in Mandarin, a situation that is common among younger speakers of both Naxi and Na.

By contrast, the main consultant for Yongning Na is thirty-eight years older than He Likun, and her knowledge of Mandarin is limited. In her speech, there is a tension between a general tendency to integrate loanwords into the Na phonological system and occasional efforts at getting closer to the "correct" pronunciation in Mandarin (either Southwestern Mandarin or Standard Mandarin, depending on the interlocutor). This source of instability needs to be acknowledged when transcribing Mandarin borrowings: they typically possess both (i) an adapted form that conforms to Na phonotactics and phonetics, and (ii) forms that are closer to Mandarin, and which depart from Na phonotactics and phonetics.

For instance, in the absence of a rounding opposition among front vowels, Mandarin [**y**] is borrowed as [**i**]: the Mandarin *zájūn* 杂菌 [**tsa.tɕyn**] 'mixed mushrooms' is realized as /**tsɑ˩tɕi˩**/. However, the consultant is aware of the phonetic distance between [**tɕyn**] and [**tɕi**], and is able to make efforts to round the front vowel, producing forms that get close to [**tɕɥe**] or [**tɕɥi**].

These competing pressures – towards adaptation to the Na system, and towards phonetic fidelity to Mandarin – can sometimes be observed within the lexicon. The Chinese word for 'Westerners, foreigners', *yáng* 洋, appears in three forms in Yongning Na: /**jɤ˩**/, /**je˩**/ and /**ʐe˩**/. The most frequent form is /**jɤ˩**/, as in /**jɤ˩ho˧**/ 'matches' (from *yánghuǒ* 洋火) and /**jɤ˩jo#˥**/ 'potato' (from *yángyù* 洋芋). The second form, /**je˩**/, occurs in /**je˩ʐe˧**/, from *yángrén* 洋人 'Westerner'. In turn, this word appears in a modified form in 'wild cotton flowers', /**ʐe˩ʐe˧-bæ˩bæ˩**/. This is a distortion of /**je˩ʐe˧-bæ˩bæ˩**/, literally 'Westerners' flower', which remains an acceptable variant. The borrowed syllable /**je**/ in /**je˩ʐe˧**/ 洋人 'Westerner' is Naicized through identification with a syllable that is well-attested in Na, /**ʐe**/, taking advantage of its presence in the immediate vicinity: as the second syllable of 'wild cotton flowers'. Unsurprisingly, the stylistic effect of Naicization is that /**ʐe˩ʐe˧-bæ˩bæ˩**/ sounds more local, drawing on a sense of in-group closeness among Na speakers, whereas the more faithful rendering of the Mandarin original, /**je˩ʐe˧-bæ˩bæ˩**/, tends to sound more modern and outward-looking.

A.5 Articulatory reduction: Reduced forms and their lexicalization

Phenomena of articulatory reduction pave the way for the lexicalization of new forms, sometimes resulting in the emergence of new syllabic combinations. Some salient examples are presented below.

The verb /**ʝi˥**/ 'to do' is prone to reduction. In particular, reduction is well on its way to becoming lexicalized in the expression /**gɯ˩ ʝi˥**/ 'really, truly' (from /**gɯ˩**/ 'authentic, true'), which is realized as [**gi˩˥**] except when hyperarticulated. Phonetic reduction is also strong in the object-verb sequence /**ə˧tso˧ ʝi˧**/ 'to do what?' (INTERROG:what-to_do), commonly realized in a hypo-articulated form that can be approximated as [**ə˧tse˧**] – as in /**no˧ | ə˧tso˧ ʝi˧-bi˧**/ 'What are you going to do?' (2SG-INTERROG:what-to_do-IMM.FUT), frequently contracted to **no˧ | ə˧tse˧ bi˧**.

A factor preventing lexicalization of the simplified form [**ə˧tse˧**] may be the quasi-homophony of this reduced form with another interrogative, /**ə˧tse˥$**/ 'why'. The phonological closeness of /**ə˧tse˧**/ and /**ə˧tse˥$**/ could lead to confusion. This risk of homophony might create pressure towards retarding the lexicalization of a simplified form /**ə˧tse˧**/ that would replace /**ə˧tso˧ ʝi˧**/.

The relativizer /**hĩ˥**/ is articulated far more weakly than the noun /**hĩ˥**/ 'human being, person'. The initial fricative is often strongly reduced, and frequently voiced throughout. Before a voiced stop, realizations as a nasal consonant (a nasal stop) are observed, as in Figure A.2, which shows a spectrogram of the utterance in (5).

(5) ɖwæ˦ | fv̩˧-hĩ˧ ɖɯ˧-v̩˧ ɲi˩!
    ɖwæ˦ fv̩˧ -hĩ˥ ɖɯ˧ v̩˧ ɲi˩
    very happy REL one CLF.individual COP
    '(S)he is really happy!' (Field notes.)

The spectrogram shows that the sequence /**hĩ˧ ɖɯ˧**/ is phonetically realized close to [**nɖɯ˧**], as though the syllable /**hĩ˥**/ were realized as a prenasalization of the following stop. This is not a categorical process: to claim that /**hĩ˥**/ is categorically replaced by /**n**/ in this context would be untenable, running up against insuperable phonotactic difficulties. Yongning Na does not have a series of prenasalized stops (and thus has no permissible /**nɖ**/ initial consonant), and /**n**/ on its own is not a well-formed syllable, as it lacks both a rhyme and a tone. The reduction in question is therefore phonetic in nature. Annotating the reduced realization of /**hĩ˥**/ in Figure A.2 as [**n**] is a convenient shorthand, not a phonemic reanalysis. Underlying specifications can continue to shape articulatory gestures, even when the primary phonetic targets are no longer being reached (Nolan 1992: 272).

With this caveat in place, it is clear that the reduction is strong. Roselle Dobbs (p.c. 2014) notes that some younger speakers are unaware that the relativizer /**hĩ˥**/ is present in contexts such as the one shown in Figure A.2, and tend to omit it altogether.

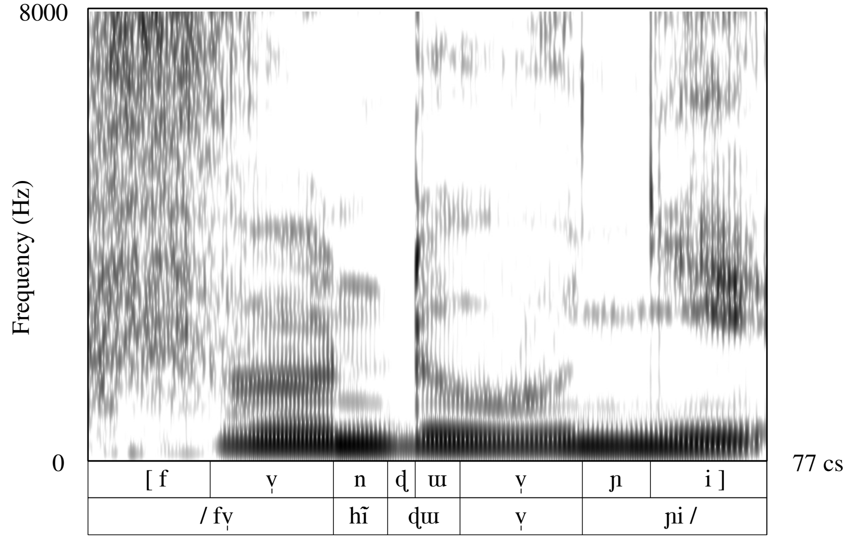


Figure A.2: An illustration of the reduction of the relativizer /**hĩ**/ to a nasal consonant. Top: phonetic transcription; bottom: phonemic transcription. Speaker: F5.

The phrase /**ʈʰæ˧mi˧-ɳɯ˩**/ ‘really, actually’ is generally reduced to a monosyllable with a long rhyme, which can be approximated as [**ʈʰææ̃˧**], occasionally preserving a trace of the final L tone of the full expression: [**ʈʰæ˧æ̃˩**]. In the absence of a vowel length opposition in Na, however, the reduced form appears unlikely to become lexicalized.

The disyllabic forms produced by the combination of demonstratives with the associative plural clitic /**=ɻæ˩**/ are also strongly coalescent. These are /**ʈʂʰɯ˧=ɻæ˥$**/ ‘these things, this sort of things’ (from proximal /**ʈʂʰɯ˥**/) and /**tʰv̩˧=ɻæ˥$**/ ‘those things’ (from distal /**tʰv̩˥**/). Regressive vowel harmony is strong, giving rise to realizations resembling [**ʈʂʰæ˧=ɻæ˥**], e.g. in *Caravans.153* {0004530#153} and *Agriculture.109* {0004440#109}. Together with the weakening of the approximant /**ɻ**/ (which is vowel-like to begin with), this often results in forms approaching a monosyllable, [**ʈʂʰæ˦**] or [**ʈʰæ˦**]. Examples include *Caravans.160* {0004530#160}, *165*; *Mountains.83* {0004573#83}, *109*; *Funeral.190* {0004571#190}; and *BuriedAlive3. 50* {0004538#50}.

The exclamative final particle /**wɤ˧**/, which conveys a sense of obviousness, tends to fuse with a preceding /**-ɲi˩**/ (expressing certitude). In all fourteen occurrences found in F4's transcribed narratives, the combination /**-ɲi˩ wɤ˧**/ undergoes tone lowering: the final M tone is depressed to L, in accordance with Rule 5 (see §7.1.1), and the sequence is realized phonetically close to a monosyllable: [**-ɲo˩**]. This phenomenon is also highly frequent in the speech of M21.

A.6 Expressive coinages and phonostylistic observations

To conclude this appendix on Na phonemes (vowels and consonants), it is worth mentioning expressive coinages, along with some phonostylistic observations.

Ideophonic and expressive phenomena are attested in languages the world over (Marsault et al. 2024). The seemingly iconoclastic idea of a reversal of perspective, whereby elements traditionally considered marginal become the focus of attention, offers a stimulating and heuristic shift in viewpoint. In his work on interjections, Mark Dingemanse refers to "a potentially radical reversal: from interjections at 'the outskirts of real language' (Müller 1861) to interjections at the heart of language" (Dingemanse 2024: 258). This perspective is all the more relevant for work on Yongning Na as expressive coinages are reported to be especially abundant in Asian languages:

> The languages of Mainland Southeast Asia are resplendent with elaborate grammatical resources for fashioning elaborative expressions that convey emotions, senses, conditions, and perceptions that enrich discourse – both everyday and ritualized – and are grammatical works of art. Over time, a sizeable terminological lexicon has been created to categorize or classify these resources, including echo words, phonaesthetic words, chameleon affixes, chiming derivatives, onomatopoeic forms, ideophones, and most notably expressives. (Williams 2014: 1)

Such phenomena have been studied in detail in Vietnamese (Brunelle & Lê 2014, Hoa Pham & Alves 2024) and in Japhug (Jacques 2013). Their exploration in Na is still at an early stage, but represents a promising direction for further work.

A.6.1 Onomatopoeia and ideophones

Onomatopoeia constitute one dimension of expressive (phonaesthetic) coinages, which also include interjections, calling sounds, and ideophones. All of these categories have interesting morphological and phonological specificities. "Of the 446 known onsets in Japhug, forty-five clusters (including thirty-five two-consonant and eleven three-consonant clusters) are exclusively attested in ideophones or ideophonic verbs" (Jacques 2013: 264). Expressive coinages tend to have a lilt of their own, but they are also subject to a continuous pressure towards assimilation into the language's phonological system: "ideophones fill gaps in the distribution of segments within rhymes that have been caused by sound changes" (Jacques 2013: 267). A classical instance of such structural gap-filling is found in Vietnamese, where the /**ɔŋ**/ and /**oŋ**/ rhymes underwent an evolution whereby lip rounding was shuffled from the vowel to the consonant, yielding a final labial closure: the resulting forms can be approximated as [**ʌɔŋ͡m**] and [**ɤoŋ͡m**]. The phonotactic slots left vacant by this phonetic evolution came to be filled by onomatopoeic coinages, and by loanwords (Haudricourt 1952; Henderson 1985: 21; Michaud 2004b: 143).

He Likun, a native speaker of Naxi, examined each cell in a table of possible combinations of initials and rhymes in the Pianding dialect of Naxi, assessing (by introspection) whether the combination was attested, and if so, in which words. The results were supplemented by sifting through a word list of approximately 3,000 entries. This study identified more than fifteen syllables attested exclusively in onomatopoeic words (Michaud & He 2015).

Onomatopoeia are no less abundant in Na than in Naxi. However, they are scarce in the set of transcribed narratives, as might be expected in relatively formal monologic speech. Other data collection methods, such as recording lively conversations, will be necessary in order to explore the full range of expressive phenomena found in Yongning Na. Three examples are presented below.

(i) The noise produced by a shock between two hard objects, such as the sound of an axe hitting a tree trunk ('Bang!'), is rendered as [**bõ**]. This syllable contravenes Na phonotactics, as nasal rhymes do not normally occur after stop consonants.

(ii) The onomatopoeia used for rumbling sounds, such as the noise of heavy loads being carried across a wooden floor, or the noise of lorries, is a prolonged [ʐ̩]. This form differs in several respects from the lexical syllable /ʐɯ/. The latter surfaces with an apicalized vowel, as [ʐʐ̩]: the onset is more consonant-like, while the coda is more vowel-like. It is in view of this asymmetry that Chao Yuen-ren proposed the use of dedicated symbols for apicalized vowels (see Table A.4), rather than simply adding a diacritic to the fricative to indicate its phonological status as syllable nucleus. In his system, the lexical syllable /ʐɯ/ would be transcribed phonetically as [ʐʅ], rather than [ʐʐ̩]. By contrast, in the onomatopoeic form representing a rumbling sound, friction is sustained from beginning to end, hence transcription as [ʐʐ̩ʐ].

(iii) The hissing noise produced when water comes into contact with red-hot metal or incandescent wood ('Pssshhh!') is represented as [tʂʰɻ̩]. While the combination of initial and rhyme used to transcribe this form is also attested in lexical items (e.g. /tʂʰɻ̩˧/ 'ploughshare'), its phonetic realization in this onomatopoeic context does not exactly match that of the corresponding syllable /tʂʰɻ̩/ in the standard lexicon. To reflect its special status, a possible transcription is [tʂʰɻɻ̩ɻ].

A.6.2 Phonostylistic observations

Expressivity is not confined to specific areas of the lexicon, such as ideophones. The "appeal function" of speech (Bühler 1934) is constantly at play. The study of this function – examining how speakers shape their utterances so as to elicit a certain response from the hearer – figures prominently in the research programme set out by Trubetzkoy (1969: 14) (original German edition: Trubetzkoy 1939), who coined the term 'phonostylistics' (for an overview, see Léon 1969). The term 'psycho-phonetics', used by Fónagy (1983), is broader and therefore arguably less well suited, although it has the advantage of highlighting the considerable breadth of this strand of research: the investigation of how phonetic detail contributes to conveying the communicative intentions of the speaker. If intonation is "a symptom of how we feel about what we say and how you feel when you say it" (Bolinger 1989: 1), then phonostylistics is part and parcel of intonational studies.

This fascinating topic is best investigated through experimental phonetic research. As the present volume essentially focuses on lexical tone and morphotonology, the approved order of business entails postponing the study of expressive phenomena until the more central aspects of the language's structure have been elucidated. Detailed discussion is therefore deferred to experimental phonetic studies to be conducted in the future. For now, let us simply mention two salient cases of phonetic modification of vowels and consonants for expressive effect in Na.

The first is *lip rounding and protrusion with demonstrative (proximal) value.* Lip pointing (Enfield 2001) is used in Na society. When the lip-pointing gesture is employed during speech, lip rounding and protrusion become superimposed onto the speech production gestures. For instance, the vowel /**ɯ**/, which in its canonical form involves neither lip rounding nor lip protrusion, acquires lip protrusion when the phrase /**ʈʂʰɯ˧-ɭɯ˧**/ 'this one' (proximal demonstrative plus generic classifier) is said while lip-pointing to an object within sight.

The second case is that of *palatalization conveying a tender emotion.* The adjective /**ɳɯ˧ɕi˩**/ 'lovely' can be pronounced close to /**ni˧ɕi˩**/, a variant reminiscent of child speech. This form has iconic value: palatalization, which narrows the vocal tract, is widely associated with smallness (Fónagy 1983: 22–23). The realization of this cross-linguistic association is facilitated in Na by the tendency towards regressive vowel harmony (§A.2.7).

A.6.3 Expressive uses of reduplication

Reduplication serves various grammatical functions in Yongning Na, as it does in Naxi (Hé Jírén & Jiāng Zhúyí 1985: 30–33). Despite having neatly grammaticalized uses, such as imparting reciprocal meaning to verbs, it retains an expressive dimension, especially in its sporadic application to parts of speech other than verbs and nouns. This expressive function is reflected in irregular tone patterns – and thus this Appendix finally returns to the book's central topic: tone.

The phrase /**qʰɑ˧ ɲi˧**/ 'how many days' reduplicates as /**qʰɑ˧ ɲi˧~qʰɑ˩ ɲi˩**/ 'thus and so many days' (*Healing.29* {0004540#29}; the context is the following: a priest of the Na religion diagnoses the number of days of rituals required to cure a person's illness). The resulting tone pattern differs from that of numeral-plus-classifier phrases, where the expected surface form would exhibit M tone throughout the phrase (see Chapter 3).

The interrogative /**ə˧tso˧**/ reduplicates as /**ə˧tso˧~ə˧tso˥**/, again with a tone pattern that diverges from expectations (*Dog.48* {0004442#S48}).

The word /**zo˧~zo˧-mv̩˧~mv̩˥**/ 'thingummy' looks a lot like it could be the product of reduplication, perhaps originating in a playful manipulation of /**zo˧mv̩˥**/ 'child'. A more common word for 'thing' is /**tso˧tso#˥**/, which may derive from reduplication of the nominalizer /**-tso**/. These two nouns are combined in /**tso˧~tso˧-zo˧~zo˧-mv̩˧~mv̩˥**/ 'thingummies, stuff', suggesting that they are currently perceived as having a similar internal structure.

The reduplicated form /**zɯ˧~zɯ˧**/ 'life, existence' is more frequent than the monosyllabic variant /**zɯ˧**/, although both are in common use.

The sequence /**lv̩.lv̩**/ in /**bi˧-lv̩˧~lv̩˥**/ 'snowflake' and /**dzo˧-lv̩˧~lv̩˥**/ 'hailstone' looks like a reduplicated form of the classifier for kernels, /**lv̩˧**/ (Lidz 2010: xxxiv).

The phrase /**tɕɤ˧~tɕɤ˧**/ 'right at the moment that…' also has the appearance of a reduplicated form, although a *simplex* (non-reduplicated) counterpart could not be identified.

A set of four-syllable onomatopoeic expressions of the form ABAB was observed, all displaying an L.L.M.M tone pattern. Examples include:

- /**tsɯ˩qwæ˩~tsɯ˧qwæ˧**/ 'crashing sound, for instance the sound of timber falling down' (*Housebuilding.243* {0004448#243}),
- /**zɯ˩gɯ˩~zɯ˧gɯ˧**/ 'boom!' (sound of heavy shock against a door: *Tiger.15* {0004444#S15}),
- /**ʐɯ˩ʐɤ˩~ʐɯ˧ʐɤ˧**/ 'sound of tearing leaves to pieces' (*FoodShortage2.37* {0004442#S37}),
- /**ɕi˩hwɑ˩~ɕi˧hwɑ˧**/ and /**ʐɯ˩ʁæ˩~ʐɯ˧ʁæ˧**/, both describing the dizziness of a character under a dazzling moonlight (*Reward.17* {0004446#S17} and *Reward.68* {0004446#S68}).

Among these ABAB expressions, only /**ʐɯ˩ʁæ˩~ʐɯ˧ʁæ˧**/ has an identifiable *simplex* counterpart, namely /**ʐɯ˧ʁæ˩**/ 'drunk'.

Appendix B: Historical and ethnological perspectives

While the preceding chapters have focused primarily on the tonal system and morphosyntax of Yongning Na, in keeping with the volume's title, the broader context in which this language is spoken invites further reflection. Thinking beyond disciplinary boundaries is central to the study of unwritten languages and orally transmitted traditions. The intertwined challenges for linguists and anthropologists were memorably articulated by Jacqueline Thomas, founder of the *Langues et Civilisations à Tradition Orale* (LACITO) research centre within *Centre National de la Recherche Scientifique* (CNRS): (i) the study of language is only meaningful when grounded in a thorough understanding of the culture in which it is embedded, and (ii) the study of a society requires full command of its language.

In this spirit, the present Appendix is a modest attempt to contribute a historical and anthropological perspective to complement the linguistic analyses developed in the main chapters. It reflects on the "chains of societies" (*chaînes de sociétés*: Amselle 1990; see also Tryon 1998: 329–331) that shaped Na ethnicity over time. It opens with a concise overview of the history of the Na and related groups (§B.1), then turns to questions of ethnic classification and self-identification (§B.2). The final section examines anthropological research on kinship and family structure (§B.3).[1]

B.1 The history of Yongning in outline

B.1.1 Prehistory

The Naxi scholars Guo Dalie 郭大烈 and He Zhiwu 和志武 argue that the name *Móshā* 摩沙, found in a fourth-century chronicle, unambiguously refers to "the

[1]This chapter was initially part of the introduction. However, a reviewer observed – quite reasonably – that the encyclopaedic quotations and extended discussions of topics at some remove from the tonal system of Yongning Na risked tiring the reader and weighed down the introductory chapter. This was a fair point: given the book's title, *Tone in Yongning Na*, readers cannot be assumed to expect several dozen pages on historical, ethnographic, and anthropological matters. These materials have therefore been relocated to the present Appendix.

Naxi", a concept which they define as including the speakers of Yongning Na (Guō Dàliè & Hé Zhìwǔ 1994: 102-103).[2] In doing so, they project the ethnic identity of present-day Naxi communities back a millennium and a half. They proceed to track the group's history through a sequence of changing designations in Chinese sources: from *Móshā* 摩沙 in the Jin dynasty, to *Mósuò* 磨些 in the Tang dynasty,[3] and later to *Mósuò* 麼些, *Mósuò* 摩些, *Mósuō* 摩娑, and *Mòsuò* 末些 in the Yuan dynasty and subsequent periods.

As to earlier origins, Guo and He suggest an admixture of Qiang 羌 people with an indigenous population, left unnamed.

> In the formation of the Naxi people, the main component consisted of aboriginals, who blended with Qiang 羌 people, and later assimilated some other peoples at their periphery; conversely, in peripheral areas, some Naxi got assimilated into other peoples. (Guō Dàliè & Hé Zhìwǔ 1994: 24)
>
> *Original text:* 纳西族的形成以土著为主，融合了北来羌人，以后又同化了周围其他一些民族，边缘地区则是纳西族被其他民族同化。

A further link is proposed between the Naxi and the Shiguanzang 石棺葬 culture, attested during the first millenium BC in regions overlapping with present-day Na and Naxi settlement (Guō Dàliè & Hé Zhìwǔ 1994: 66-67). This cultural association of incoming Sino-Tibetan peoples with a certain type of graves raised hopes similar to the association of Kurgan pit-graves with the Indo-Europeans (Gimbutas 1977, Anthony 2007). Shiguanzang graves are typically found on tablelands near river confluences and include short bronze swords of a type also known from the northern steppes. The absence of securely identified associated settlements raises the possibility that the builders were nomadic, metal-using pastoralists, contrasting with the earlier inhabitants of the area, whose numerous settlements suggest a sedentary, agricultural society in transition from the Neolithic to the Bronze Age.[4]

This scenario fits broader models of Sino-Tibetan migration from the Yellow River valley (Jacques & Stevens 2024), in which the Yangshao 仰韶 culture (c. 5000

[2] "This is the first certain and unequivocal mention of the Naxi in recorded history." *Original text:* 这是纳西族在历史上首次明确无误的记录。

[3] The character 些, normally read *xiē*, is read *suò* in this context, as also in other contemporary toponyms: see Pelliot (1904: 170n9).

[4] Guō Dàliè & Hé Zhìwǔ (1994: 64-66) identify this nomadic population with the Yi 夷 peoples mentioned in Chinese chronicles, whom they regard as ancestors of both the present-day Yi 彝, and of the Na and Naxi. It is not altogether clear how this identification is to be reconciled with their other claim that the Sino-Tibetan component in Naxi ethnogenesis came from the Qiang 羌.

BC to 3000 BC) is hypothesized as a point of origin. One of the main reconstructed lines of migration is "south-west down the river valleys along the eastern edge of the Tibetan plateau through what has been called the *ethnic corridor*" (LaPolla 2001: 236; emphasis in original).

Yet such reconstructions raise long-standing methodological issues. In particular, the proposals advanced by Guo Dalie and He Zhiwu invite scrutiny on points such as the extent to which archaeological evidence can be reliably fitted into broad cultural typologies, and the basis on which the names of peoples were assigned in Chinese historical chronicles. As Hein (2013: 31) notes, "Most excavation reports describing and interpreting burial material from Southwest China tend to associate grave type with archaeological culture; hence their urgent desire to arrive at a clear classification of burial types; however, (…) one cultural or ethnic group can be characterized by a number of different burial rituals, while other practices might be common across such boundaries".

A 1,400-page study of cultural geography and interregional contacts in prehistoric times offers a more fine-grained exploration than was possible at the time of writing of Guo and He's *History*. Systematic examination of the available evidence leads to the identification of no fewer than "four subregions showing fairly distinct archaeological assemblages, burial patterns, and subsistence systems, indicating that they were probably inhabited by different cultural groups" (Hein 2013: 588).

The first of these is the Anning river 安宁河 valley. Settlements dating to the third millenium BC yield relatively consistent archaeological finds, including "coarse sand-tempered low fired red-brown ceramics (mainly large urns with finger-tip impressed appliqué strip below the rim, *bo* and *wan* bowls, vases, and a few lids and rarely spouts), accompanied by a few polished stone woodworking tools, arrowheads, and among the surface finds also perforated stone-knives" (Hein 2013: 559). These commonalities are interpreted as implying that the communities in these settlements "probably shared [a] similar cultural tradition and thus identified themselves as part of the same larger group" (Hein 2013: 589). At one of the sites, Dayangdui 大洋堆, evidence points to outside influence around 2000 BC, followed by cultural assimilation:

> Both in ceramic quality and execution, the early Dayangdui ceramics (…) strongly resemble ceramics from sites in Gansu 甘肅 and Qinghai 青海 attributed to the Qijia 齊傢 culture. It is therefore not unlikely that the earth-pit graves at Dayangdui were built by a group of Qijia origin. This would suggest a date between 2200 and 1800 BC (…). (Hein 2013: 562)

> The middle and late Dayangdui assemblages do not contain any metal objects, however, and they show a mixture of both early Dayangdui and local Neolithic traits that indicate some form of acculturation of the group of immigrants. As no similar sites of clear foreign origin have been identified in the Anning River Valley, it is likely that migration of whole groups from the North occurred rarely. (Hein 2013: 594)

In this area, megalithic graves appeared, and were subsequently imitated in neighbouring regions:

> Graves with stone-construction parts are common throughout Southwest China, but megalithic graves seem to be unique to the Anning River Valley. The ceramics associated with these graves indicate a local origin of this burial tradition in the Xichang area. This impression is supported by the fact that all early megalithic graves (…) are located in Xichang, while the megalithic graves in other regions such as Dechang, Mianning, Puge, and Xide all date to Phase IIa at the earliest. Why this kind of burial mode arose is uncertain, but its overall development and spread is relatively clear: it started with small constructions used for a single instance of interment of several people, possibly in a secondary mode of burial. During or after the burial, communal drinking rituals took place which seem to have become more extensive over time, as the large number of drinking vessels both in later graves and related ceramic pits shows. (…)
>
> As far as daily life and mode of subsistence are concerned, the tool assemblages from megalithic graves and related settlement sites in the Anning River Valley show an agricultural and probably settled mode of living involving the planting of rice and probably other cereals, often supplemented by hunting, and in some places fishing. Only the sites in Puge show a continued primary reliance on hunting. Metal seems to have mainly been used for personal ornaments and only secondarily weapons or tools. (Hein 2013: 595-599)

Archaeological remains from the Neolithic to Bronze Age collected in Yongning in 1958 are considered to be connected to the megalithic graves (Hein 2013: 933). It is relevant to the history of Yongning that the spread of this burial mode is hypothesized to have taken place through cultural diffusion – presumably by persons who had participated in communal rituals and later reproduced these practices in their home settlements – rather than through military conquest. It is for future excavations to verify the existence of this connection.

The second subregion is one of remote mountains, a harsh environment where "groups of different origins conducted different kinds of burial rituals next to each other, apparently respecting each other's monuments and even adopting part of each others' burial customs and objects. In this meeting place of different groups, cultural and other forms of identity (or at least their expression in the choice of grave form, burial mode, and object assemblage) thus seem to have been extremely fluid" (Hein 2013: 602).

The third subregion is one of fertile valleys to the Southeast. The inhabitants of the earliest settlements "practiced a hunter-gatherer lifestyle, using caves and open-air sites either as seasonal or hunting camps rather than living in permanent settlements", later "practising incipient agriculture in a particularly congenial environment, living either in permanent or semi-permanent settlements" (Hein 2013: 605-606).

The fourth and final subregion comprises the high-altitude mountains, plateaus and valleys of the Southwest, a geographical area that includes Yongning.

> The people living in Yanyuan 盐源 and Ninglang 宁蒗 (...) belonged to a clearly separate cultural group for whom armed combat – sometimes combined with horseback-riding – was a central part of their life and identity. The emphasis on horse-riding, the interment of horse heads and sheep shoulder blades in graves and the overall metal assemblage (in particular the staff heads) seen at Yanyuan are essentially foreign to the research area. Pictorial evidence for horse-riding is known from the Dian 滇 culture context, but horse skulls or long bones have never been. The interment of horse bones is instead common in the Northern Steppe and the Ordos region, and elements of horse gear similar to those seen in Yanyuan have been reported from there as well. (...) It is therefore likely that the burying group of the "warrior graves" in Yanyuan is of a northern origin, be it the upper Minjiang 岷江 or even the steppe. (Hein 2013: 616-618)

Yanyuan 盐源, a strategic area due to its abundance in rock salt, was conquered in 225 AD by the Chinese, who referred to the local population as the "Mosha" 摩沙. Returning to the hypothesis advanced by Naxi historians Guo Dalie and He Zhiwu – that these "Mosha" are the ancestors of today's Naxi –, one might rephrase the claim as follows: military defeat prompted clans from this warrior population to retreat westward, conquering new territories that included Yongning, the banks of the Yangtze, and later the plain of Lijiang – areas which remain their principal settlements to this day.

However, there does not appear to be decisive evidence linking the "warrior culture" of Yanyuan specifically to the Naxi, as opposed to other Sino-Tibetan groups who later migrated into more distant regions of present-day Yunnan and Myanmar. A modest piece of evidence relevant to this question concerns the path that ritual practitioners dictate to the soul of the deceased for its journey back to the ancestral homeland. Such ritual journeys, which have been studied in many parts of East and Southeast Asia, have been shown to preserve memories of historical migration routes (Blackburn 2004, Gaenszle 2012, McKhann 2012). Among the Naxi, different clans maintain different itineraries, yet all these ritual paths pass through Yongning – where they join with those of the Na – and continue northward (Guō Dàliè & Hé Zhìwǔ 1994: 50-55). As the distance from Yongning increases, the identification of place names soon becomes uncertain or impossible. The designation of Minya Konka 贡嘎山 as the endpoint of the journey is not unlikely to be a later addition, inspired by the prestige of this 7,500-meter peak, which serves as a mooring point for traditions and beliefs of various peoples of the eastern Himalayas. On the other hand, it is worth noting that the path for returning souls does not pass through Yanyuan; there is thus no evidence that the group's forebears ever dwelt further east than the Yongning area. Naturally, such ethnological evidence does not carry considerable weight, as ritual paths may have been altered at any point in their oral transmission. Still, it suggests that the identification of the "Mosha" 摩沙 as direct ancestors of the Na and Naxi should not be treated as established fact.

> Local oral narratives and scholarly writings which discuss the origins and migrations of hill peoples of the far eastern Himalaya often share the same propositions. Firstly, both types of sources tend to plot routes of migration between an assumed original homeland area or origin place and a present-day dwelling location; direction of movement and itineraries are of shared importance here. Second, they both claim identification of contemporary populations with their purported ancestors from past times and distant places, with implicit and explicit claims of ethnic continuity. (Huber 2012: 83)

Whatever the precise relationship between the "Mosha" and the present-day Na and Naxi, the conquest of the Yanyuan area by Chinese forces in 225 AD stands as a major historical landmark. It established ties with the heart of the Chinese empire that were never entirely severed thereafter, even during periods when the Chinese central power was weakest, such as the four centuries that followed.

B.1.2 Empires and indigenous chieftains

In 794 AD, the Nanzhao 南诏 kingdom, centred on the fertile lands around Erhai Lake (洱海, today a Bai-speaking area), conquered a broad region that included both Yongning and Lijiang 丽江. Following the fall of Nanzhao in 902, the kingdom of Dali (大理国, 937–1253), likewise centred on Erhai Lake, exercised control over Yongning and Lijiang, although both continued to be ruled by indigenous chieftains.

At the outset of the Yuan dynasty, a new feudal chieftain (*tǔsī* 土司) was installed in Yongning by the Mongolian forces, who passed through Yongning en route to their conquest of the Dali kingdom. A chapter of the imperial geography *Yuan Yi Tongzhi* 《元一统志》, dated 1286, records the name of Yongning as 楼头 (present-day Mandarin pronunciation: *lóutóu*). Using the system proposed by Coblin (2007), this name reconstructs as ***ləw dəw**, which is clearly cognate with the present-day name of Yongning: Naxi /**ly˧dy˩**/ and Na /**ɬi˧di˩**/, as discussed in Chapter 1, §1.1. It is likely that the authors based their transcriptions on the pronunciation used in Lijiang, a more prominent centre than Yongning in terms of both population and accessibility. In that sense, their record does not constitute direct evidence for the language spoken in Yongning at the time. Nevertheless, it provides a reasonable basis on which to hypothesize linguistic continuity in Yongning since at least the thirteenth century.

The Yongning chieftain who surrendered to the Mongolians in 1253 claimed a genealogy of thirty-one generations since his ancestors had conquered the region. Assuming an average of twenty to twenty-five years per generation, this would place the initial conquest in the period between 500 and 650 AD. Of course, this change in the ruling class (which still requires confirmation from other sources) may not have entailed significant linguistic change. It is therefore impossible at present to determine how far back in time linguistic continuity can be assumed in Yongning.

The introduction of Tibetan Buddhism dates to roughly the same period as the Mongolian conquest, with the missionary efforts of monks from Muli beginning in 1276 (Guō Dàliè & Hé Zhìwǔ 1994: 389). In 1356, a Kagyupa (*bka' brgyud pa*) monastery was established. Two centuries later, in 1556, a large Gelugpa (*dge-lugs pa*) monastery was founded in Yongning (Tibetan name: *dgra med dgon pa*). Earlier cults persisted, with a division of ritual labour between the Buddhist clergy and the Na *Ddabe* **dɑ˧pɤ˧** ritual practitioners.[5] From that point onward,

[5]Twentieth-century reports suggest that monks and *Ddabe* **dɑ˧pɤ˧** coexisted peacefully, with an established division of ritual responsibilities: the *Ddabe* **dɑ˧pɤ˧** were called upon for rites associated with livestock, such as the slaughtering of pigs; Buddhist monks officiated on prescribed

however, Buddhism became the dominant religion in Yongning. (At the time of the Communist takeover, there were over 700 monks at the Yongning monastery.) This contributed to an increasing cultural divergence between Yongning and the Lijiang plain. In Lijiang, no school of Tibetan Buddhism succeeded in establishing and maintaining a central role: its religious history was characterized by abrupt turns and shifting allegiances (Gros 2019: 22). The Naxi *Dobbaq* **to˧mbɑ˩** ritual tradition – likely shaped in large part by Bön influences – nearly attained the status of an official cult (Jackson 1979).

During the Yuan and Ming dynasties, persistent warfare took place among the feudal chieftains of Yongning, Lijiang, and Yanyuan (Guō Dàliè & Hé Zhìwǔ 1994: 430–431). In 1545, Yongning united with the neighbouring territories of the five *suǒ* 所 (Zuosuo 左所, Yousuo 右所, Qiansuo 前所, Housuo 后所, and Zhongsuo 中所). The Yangtze River served as the boundary between the territories of Yongning and Lijiang.

During the Qing dynasty (1644–1912), the Naxi chiefdom of Lijiang was brought under direct Chinese administration. In contrast, the feudal chieftain system remained in place in Yongning, owing to the failure of successive military expeditions to subjugate the Liangshan 凉山 Yi region, which serves as a natural gateway to Yongning (Guō Dàliè & Hé Zhìwǔ 1994: 460). This contributed to an ever-deepening cultural differentiation between Lijiang, which became increasingly Sinicized, and peripheral regions such as Yongning. The Yongning chieftains actively resisted Chinese settlement, prohibiting the cultivation of rice and forming alliances with Yi chieftains to obtain their military support. These developments are reported to have hastened the migration and settlement of Yi families into the Yanyuan and Yanbian areas, leading to the displacement

days of the religious calendar; and for major ceremonies, such as funerals, both groups participated. This long-standing coexistence, and the occasions for interaction it provided, appear to had led to a certain amount of convergence. The *Ddabe* **dɑ˧pɤ˧** have adopted certain ritualized behaviours characteristic of Buddhist monks. They prepare rituals with great attention to detail, ensuring in advance that all the necessary paraphernalia – such as butter, candles, water, and different types of flour – are available. This contrasts with the practices of the ritual specialists among the Yi, known as *Bimo* 毕摩 (Névot 2019), who have a habit of requesting objects and accessories at unpredictable moments during a ceremony, as if responding to sudden inspiration (Latami Dashi, p.c. 2008). Furthermore, the gestures performed by the *Ddabe* during rituals have come to resemble those of Buddhist monks. Conversely, some Buddhist monks are reported to study the Na system of horoscopy – one of the traditional domains of expertise of the *Ddabe*. The study of Na rituals, particularly in comparison with those of the Naxi, holds great promise for understanding Na cultural dynamics (see Mathieu 2015 and references therein). Such an endeavour, however, requires expertise in Tibetan philology, an in-depth knowledge of Tibetan Buddhism, and a range of other specialized skills beyond my own field of study.

Photo B.1: The Yongning monastery. The dialect under study is spoken in Alawua, a hamlet adjacent to the monastery. Autumn 2006.

of earlier populations – speakers of closely related language varieties. These varieties are referred to as “Naxi dialects” by Guo Dalie and He Zhiwu (p. 461), but are here designated as “Naish languages”, as explained in §1.1.3. The Laze, a small group of approximately four hundred people who relocated from Yanbian to their present location in Muli towards the end of the nineteenth century, appear to be among the speakers of Naish languages who departed from Yanbian as it became increasingly dominated by Yi-speaking communities (on the Laze language, see Huáng Bùfán 2009).

During the 1920s, Yongning became a node in trade routes linking Tibet, the Yi territories centred on Ninglang, Lijiang, and inland China. “This was the first time in history that the Moso had frequent visitors from the outside on a considerable scale. (…) this period was also the onset of the syphilis epidemic” (Shih & Jenike 2002), an epidemic curbed in the 1950s.[6]

[6]This epidemic is alluded to by Goullart, who stayed in Lijiang in the 1940s: “The Nakhi men (…) knew well enough that most of the Liukhi [**ly**˧ **ɕi**˧, the Naxi term for the Na of Yongning] tribe was infected with venereal disease” (Chapter 3 of Goullart 1955).

Photo B.2: Worshippers at the Yongning monastery. Autumn 2006.

B.1.3 The People's Republic of China

After the founding of the People's Republic of China in 1949, the central government laid out plans for the graduated integration of frontier areas along the outer margins of the dynastic power realm – those known during the Qing dynasty as 'vassal states' (*fānshǔ* 藩屬).

> Instead of 'mobilizing the masses' like in the Yunnan interior and elsewhere in China, land reform on the volatile frontier took the form of "uniting the feudal to fight feudalism" (*lianhe fengjian fan fengjian* 联合封建反封建), a scheme designed to win over the ethnic elite. The CCP united front work saw a revival of traditions in forms of pledging allegiance, conferring official titles, and the ritual of court audiences – those who cooperated were invited to serve in the new government and local dignitaries were taken to Beijing to have an audience with the CCP leaders. (Guo 2008: 43-44)

The Liangshan 凉山 Yi area, which constitutes the gateway to Yongning, proved a hard nut to crack. Armed rebellion broke out in 1956. By 1958, large-scale "pacification" operations had been completed (Guo 2008: 228-231), and administration of the area by the Communist state was instituted. Since then, reforms have been applied essentially top-down. The guiding policy has been that the majority group points the way forward, leading minority groups towards modernity and eventual assimilation.

In a study of the Drung, an ethnic group residing in an even less accessible area of Yunnan, Gros (2014b) argues that these policies represent a fundamental shift from earlier relationships of vassalage between local powers and the state(s). Under the previous system, feudal chieftains paid tribute and received titles in return; the balance of this exchange – how much tribute was offered, and how much recognition and autonomy granted in return – was assessed by both parties. By contrast, top-down state policies do not partake in any logic of exchange – *don et contre-don*, as emphasized in the classic study by Marcel Mauss (1990).

Post-1956 events remain a highly sensitive topic. Mazard's observation about the Nusu (Yi) also applies to the Na:

> They generally treat the years from 1958 to around 1979 as a single historical period circumscribed by the collectivisation and de-collectivisation of their land. (…) Why do many of them (…) identify this period as a whole with the Cultural Revolution, even though they know and employ the term 'Great Leap Forward' as well?
>
> One reason may be that the CCP allows overt (though limited) criticism of the Cultural Revolution; not so the Great Leap Forward (or the Anti-Rightist Campaign). The Party has never denounced the Great Leap Forward as a mistake (…). The Cultural Revolution has its villains (the Gang of Four); so do the Civil War (the Guomindang) and the Second World War (the Japanese). Discussion of the Cultural Revolution is not easy, but it is possible. When Nusu elders allude to their suffering as occuring under the 'Cultural Revolution' (even if it took place in 1959), they lay claim to a permitted register of complaints. (Mazard 2011: 172)

Recent history (since the 1980s) will only be addressed here indirectly, through a discussion of ethnic classification (§B.2) and a review of studies on the impact of tourism on Na society (§B.3.5).

B.2 Ethnic classification: Naxi, Mongolian, Mosuo or Na

Ethnic categorization as defined by the state "crafted the prism through which the modern Chinese state, and increasingly the people of China and the world at large, have come to view and understand non-Han Chinese identity" (Mullaney 2010: 5). This system of categorization, which to this day (as of 2025) appears on each individual's identity card (*shēnfen zhèng* 身份证, literally 'certificate of identity'), has such a strong bearing on present-day identities that it warrants separate discussion.

> … in the China of Chiang Kai-shek, the Nationalist regime vociferously argued that the country was home to only one people, "the Chinese people" (*Zhonghua minzu* 中華民族), and that the supposedly distinct groups of the republic were merely subvarieties of a common stock. At the same time, a counterdiscourse emerged among Chinese scholars in the newly formed disciplines of ethnology and linguistics, a discourse in which China was reimagined as home to many dozens of unique ethnic groups – a newly imported concept also translated using the term *minzu* 民族. (Mullaney 2010: 2; see also Bulag 2012)

The first census of the People's Republic of China, in 1953-1954, recorded over four hundred different ethnic identities, more than half of which concerned Yunnan, a province bordered on all sides by multi-ethnic areas (Vietnam, Laos, Myanmar, Tibet, Sichuan, Guizhou, and Guangxi). Until recently, little information was available about the process of ethnic identification (*mínzú shíbié* 民族识别) whereby these identities were grouped into the roughly fifty nationalities (*mínzú* 民族) officially recognized by the state. Sources made available in the first decade of the 21st century reveal how small teams of researchers from a range of social science backgrounds assessed possibilities for groupings, and assigned names to these groupings, all under a tight timeline (less than six months). "In the years following the end of the project, cultural and scientific works rewrote the history of China and its diversity in an effort to promote a so-called "historic" and "ancestral" model of the 56 *minzu* components" (Frangville 2011).

At first, the categories remained open to some fine-tuning: adjustments were made during the 1960s and 1970s, including the recognition of two ethnic minorities that had been absent from the 1954 classification. In 1987, however, it was declared that no further nationalities would be recognized, and the figure of fifty-six was made final.

> … the idea of China as a "unified, multinational country" (*tongyi de duominzu guojia* 统一的多民族国家) is a central, load-bearing concept within a wide and heterogeneous array of discourses and practices in the contemporary People's Republic. China is a plural singularity, this orthodoxy maintains, composed of exactly fifty-six ethnonational groups (*minzu* 民族): the Han ethnic majority, which constitutes over ninety percent of the population, and a long list of fifty-five minority nationalities who account for the rest. Wherever the question of diversity is raised, this same taxonomic orthodoxy is reproduced, forming a carefully monitored orchestra

> of remarkable reach and consistency: anthropology museums with the requisite fifty-six displays, "nationalities doll sets" with the requisite fifty-six figurines, book series with the requisite fifty-six "brief histories" of each group, Olympic ceremonies with fifty-six delightfully costumed children, and the list goes on. Fifty-six stars, fifty-six flowers, fifty-six *minzu*, one China. (Mullaney 2010: 1)

The teams tasked with surveys for the national project of ethnic identification in 1953-1954 worked independently in each province, and decisions were likewise made on a province-by-province basis. The Na living in Yunnan were classified as part of the Naxi minority. Those living in Sichuan were classified as Mongolian; this surprising choice was presumably influenced by the lingering prestige of the name 'Mongolian' in the region since the Mongolian army's formidable crossing of the Himalayas (passing through Yongning) and victory over the Song dynasty.

> Due to historical tensions between the Na and the Naxi, when the Sichuan Na learned that they would be classified as Naxi in the early 1950s, they protested by taking over the county government offices. As the federal government limits recognition to the fifty-six ethnicities, local officials were perplexed as to what to do, and a face-saving compromise was established such that the Sichuan Na could be classified as Mongolian, on the basis that the Mongols had invaded the area seven hundred years previously, and perhaps the Na were descendants of these Mongols. Although this designation is within historical memory, the Na in Sichuan have clearly adopted their designation as Mongolian, and colorful plastic plaques of Genghis Khan hang prominently on the walls in homes. (Lidz 2010: 9)

The claim of Mongolian descent apparently predates the systematic process of "ethnic identification" undertaken by the People's Republic of China.

> Moso chiefs in both Yunnan and Sichuan Provinces claimed that they were descendants of the Mongols. According to Joseph Rock, who personally befriended key members of the Yongning chief's family (...), the general superintendent (*zongguan* 總管) of Yongning at that time "was proud of his Mongol origin, for he was a descendant of one of the Mongol officers left by Kublai Khan in Yung-ning to govern that territory" (Rock 1947: 359). Abundant historical records indicate that it was commonplace for the Mongolian conquerors to leave troops of Mongol or non-Mongol ethnic background to govern the newly subjugated territories. (...) The problem is that

> in all cases other than the Moso chief's, there is evidence – such as records on stelae, tombs and tombstones (some inscribed in Mongolian), records of genealogy, language or vocabulary, and legends in one form or another – to substantiate the claim. The Moso aristocrats, however, had nothing to support their claim of Mongol ancestry. (Shih 2010: 40-41)

Shih suggests that the new chieftain was "a Xifan [Pumi] officer in the Mongol troops left by Kublai Khan to rule Yongning" (Shih 2010: 51). One piece of evidence he adduces is the identification of the Yongning chieftain's ethnicity as "Xifan" 西番 in Ming-dynasty chronicles. However, the interpretation of the term "Xifan" as referring specifically to the Pumi is not self-evident: the label may have been used in a broad sense that included speakers of the language ancestral to Yongning Na. To this day, the Namuyi of Muli, who are speakers of a Naic language (about which see Chapter 1, §1.1.3), are included among the "Xifan" 西番, a cover term for various non-Tibetan groups. Another piece of evidence is ethnographic: at the succession of the Yongning chieftain, the Pumi would perform a ritual akin to *Sikee* /**sɯ˧kʰɯ˩**/, the Na ritual associated to the giving of a household member to another house (typically, the giving of a young woman); this is consistent with Shih's hypothesis that their ethnic group was the donor of the Yongning chieftain himself. Shih states his interpretation as follows:

> In the case of *sike* [phonetic transcription: /**sɯ˧kʰɯ˩**/], a household member was given to become a particular person's wife in another household. Because the status of wife necessarily ended with the life of the woman in question, *sike* was a one-time ritual between the two families concerned. In the succession ritual, however, as the historical records suggest, when a member of the Pumi was given to become the chieftain of Yongning, a territory dominated by the Moso, the status of chieftaincy was perpetual, as was the ritual of interrogation. In both cases, the rituals were performed to dramatize a reassertion of the unbreakable blood bond between the deceased and her or his natal family. (Shih 2010: 48)

Inclusion in the Mongolian minority proved a comfortable fiction: paradoxically, it granted the Na of Sichuan a place of their own within the landscape of the officially recognized ethnic groups of Sichuan. Initially the 'Mongolian' label was taken very seriously, and efforts were made to teach the Mongolian script to Naish speakers of Sichuan – with predictably poor results. The initial ethnic identification has not been revised since, due to a national policy of keeping the Pandora's box of ethnic labels firmly closed. Their fictitious cousins in Mongolia do not appear to have raised objections to the term 'Mongolian' being applied to this small group in Sichuan; and the label has had the effect of preventing the Na from being pooled together with geographically closer neighbours such as the Naxi. By contrast, the classification of the Na of Yunnan as part of the Naxi rendered them a minority within a minority, reducing the number and influence of their representatives at various institutional levels.

In light of the historical overview summarized in §B.1, it is not difficult to understand why the Na tend to think of themselves as distinct from the Naxi, despite the conspicuous similarities between their respective languages. Resentment over inclusion in the Naxi minority has fuelled a search for recognition as a separate group. From this vantage point, the endonym /**nɑ˩˥**/ is less than ideal: /**nɑ˩˥**/, presumed to mean 'black, dark', also appears in the endonym of the Naxi: /**nɑ˩ɕi˧**/, where /**ɕi˧**/ means 'person, human being'. Use of nearly identical endonyms could cast doubt on the legitimacy of a clear-cut separation.

Instead, the Na of Yunnan have come to favour the exonym 'Mosuo' (*Mósuō* 摩梭), a name formerly used in Chinese records, which was officially replaced after 1949 by 'Naxi' (*Nàxī* 纳西). Thus, the dictionary of pictograms originally published by Li Lin-ts'an 李霖灿, Chang K'un 张琨 and Ho Ts'ai 和才 as *Dictionary of Mo-So hieroglyphics* (1953) was reprinted in 2001 on the mainland under the title *Dictionary of Naxi pictograms*; all occurrences of *Mósuò* 麼些 in the book were replaced by *Nàxī* 纳西.

The historical variants of the name 'Mosuo' are shown in Table 1.1 (Chapter 1, §1.1.1). This designation's origin is unclear. Chavannes (1912: 132) cites Chinese chronicles as indicating that the Mo-so tribe was formed during the Nanzhao period (737-902) out of two distinct elements, the Mo and the So.

Reviving the demised term 'Mosuo' to designate the Na of the Yongning area is a strategic move in support of claims to recognition as a group distinct from the Naxi, since the names 'Mosuo' and 'Naxi' are clearly distinct. Moreover, the term 'Mosuo' carries a double advantage: it is a term of great antiquity, having appeared in Chinese chronicles since the first millenium AD, and it had fallen into disuse by the mid-twentieth century, which lends it a certain quaint charm and a touch of mystery. In 1990, following vocal protests against the label 'Naxi', the 'Mosuo' of Yunnan were granted recognition at the provincial level as a separate subgroup within the Naxi minority. This label now appears poised to become standard in Sichuan as well (Lǐ Dázhū 2015: 10-11). La Mingying 喇明英, a member of the Sichuan Academy of Social Sciences who identifies officially as 'Mongolian' but expresses a preference for the label 'Na', reports that self-identification as 'Mosuo' is gaining ground among the younger generations on the Sichuan side of the designated 'Mosuo' tourist area.

> Disagreement among the Na about names [ethnonyms] is a cause for disputes; people sometimes even come to blows. Acknowledging one's ethnic identity, and having a sense of belonging to the community, constitute the most basic and essential factors in "ethnic identification". The multiplication of ethnic denominations for the Na of the Lugu Lake area generates great perplexity about their ethnic identity, to the point of causing prejudice to their sense of belonging to a community and to their ethnic cohesion. (Lā Míngyīng 2015: 53)
>
> *Original text:* 因称呼的分歧，纳人内部时有争论甚至打架的情况发生。对民族身份的承认和群体归属感是"民族认同"最基本的要素。泸沽湖地区纳人族称的多元化在很大程度上对其民族身份的认同造成很大的困惑，甚至影响了纳人的群体归属感和民族凝聚力。

Before the Communist takeover, the Na of Yongning were divided into three hereditary castes: the family of the chieftain, /**sɯ˧pʰi˧**/, constituted the nobility; this group was distinct from the commoners, /**dze˧kʰɤ˥**/, who formed the majority (about 640 families in the late 1950s). A third, smaller group of about 280 families were serfs, /**wɤ˧**/. Historically, when outsiders joined the community – whether as war captives or as immigrants from areas near and far –, they would be integrated into the serf caste. This caste also accommodated commoners who had been stripped of their rank as punishment for rebellion (Liú Lóngchū 1981).

The administrative assignment of ethnic identity – whether as 'Naxi' or 'Mongolian' – effaced these earlier distinctions of caste. It also coincided with the end of the practice of integrating newcomers into Na society through the caste system. The steady influx of settlers into Yongning, recorded in successive editions of the county Annals (*xiànzhì* 县志; the entire collection bears the name《中华人民共和国地方志丛书》, "Collection of local chronicles of the People's Republic of China"), has led to the cohabitation of individuals whose official ethnic identities remain distinct – some are designated as "Han", others as "Mosuo", "Naxi", "Pumi", "Yi", and so on –, in keeping with the state ideology of a multi-ethnic, unified China.

B.3 Anthropological research: The fascination of the Na's kinship system and family structure

The rich morphotonology of the Na language, which forms the central topic of this book, appears to have gone unnoticed until the early 21st century. By contrast, the Na kinship system and family structure have long been famous far beyond the circles of specialized ethnologists. The following excerpt is drawn from the highly exoticized account of Peter Goullart, a Russian-born traveller, explorer, and author who lived among the Naxi in Lijiang from 1942 to 1949, during the final years of intense caravan traffic (1920s-1940s).

> The arrival of the members of a certain matriarchic tribe, living about seven days by caravan north of Likiang, always created a furore in Likiang. Whenever these men and women passed through the market or Main Street on their shopping expeditions, there was indignant whispering, giggling and squeals of outraged modesty on the part of Likiang women and girls, and salacious remarks from men. They were the inhabitants of the Yungning duchies across the Yangtze at the apex of the great bend. The Nakhi [**nɑ˩-çi˧**] called them Liukhi [**ly˧-çi˧**] and they called themselves Hlihin [**ɬi˧-hĩ˧**]. The structure of their society was entirely matriarchal. The property passed from mother to daughter. Each woman had several husbands and the children always cried, 'We have mama but no papa.' The mother's husbands were addressed as uncles and a husband was allowed to stay on only as long as he pleased the woman, and if he didn't, could be thrown out without much ceremony. The Yungning country was a land of free love, and all efforts of the Liukhi women were concentrated on enticing more lovers in addition to their husbands. Whenever a Tibetan caravan or other strangers were passing Yungning, these ladies went into a huddle and secretly decided where each man should stay. The lady then commanded her husbands to disappear and not to reappear until called. She and her daughters prepared a feast and danced for the guest. Afterwards the older lady bade him to make a choice between ripe experience and foolish youth. (…)
>
> With their lips heavily rouged and eyes painted, they walked slowly, or rather undulated, through the streets, swaying their hips, smiling and casting an amorous eye on this man or that. That alone was enough to incense the less sophisticated Nakhi women. But when they walked slowly along hanging on the neck of a husband or a lover, and being held by the waist, this was too much for even the brazen Nakhi women, who spat or giggled nervously. (Goullart 1955: Chapter 3)

This sample of travellers' accounts of a purported "land of free love" suffices to explain the enduring fascination exerted by the Na on the general public. The present review does not seek to provide comprehensive coverage of the extensive anthropological, ethno-historical, and sociological literature on the Na. Rather, its aim is to convey a sense of how the field has developed over time, of the historical evolution of research perspectives, and of the implications for local perceptions of social scientists conducting fieldwork in Yongning.

B.3.1 A major source of information: Surveys conducted in the 1960s

A key resource for the study of Na society is the detailed sociological report based on surveys conducted in the 1960s (Yúnnánshěng biānjízǔ 1986). The data, organized by village, reveal subtle variations both across villages and among individual households. Most subsequent scholarship has drawn extensively on the findings recorded in the three volumes of this report, which together amount to nearly one thousand pages.

The survey clarifies that, until the 1950s, the typical family structure in the Yongning basin was matrilinear, with lifelong matrilocal residence. In non-technical terms, this means that brothers and sisters lived their entire lives in their mother's house, together with maternal relatives: cousins, aunts and uncles, and the mother's mother along with her siblings.

This configuration bears some resemblance to that found among the Minangkabau (Indonesia), as described by Hadler (2008), and the Nayar (India), as described by Fuller (1976).

> Men marry into an extended family, but remain attached to their mothers' houses. They return to that house daily to work the fields, convalesce there in times of sickness, and are eventually buried in the maternal family graveyard. A husband and father is an evanescent figure. In the words of a Minangkabau aphorism, "The *urang sumando* is like a horsefly on the tail of a buffalo, or like ashes on a burned tree trunk. [When a little wind blows, it is gone.]" (…) Minangkabau culture has been termed matrifocal because, although men can be part of the lives of their wives and children, it is mother-centredness that grounds the family. (Hadler 2008: 6)

Among the Nayar of India, the husband resided with his sister and visited his wife at night (Fuller 1976); in pre-1956 Na society, lovers met discreetly at the woman's home. Among the Na, as among the Minangkabau and the Nayar, the answer to the "matrilineal puzzle" (Richards 1950) – the potential conflict of authority between father and maternal uncle – was to place authority with the uncle. Fathers did not play a prominent social role; men's commitments were to their sisters' children, not to their own, who were raised in another household. "According to tradition, it is the *mamak* (maternal uncle) who provides male authority in the lives of children" (Hadler 2008: 6). Likewise, among the Na, the male figure of authority was the maternal uncle, as illustrated in proverbs such as (1).

(1) mv̩˧ʁo˥ | dze˩-hĩ˩-dʑo˥, | kɤ˩-nɑ˧mi˧; | di˧qo˧ se˧-dʑo˩, | ə˧v̩˦.
mv̩˧ʁo˥$ dze˩ₐ -hĩ˥ -dʑo˥ kɤ˩-nɑ˧mi˧ di˧qo˧ se˥ -dʑo˥
heavens to_fly NMLZ TOP eagle plain to_walk TOP
ə˧v̩˦
maternal_uncle
'As the Eagle is greatest of all that fly in the sky, so the Uncle is greatest of all that walk the earth.' (Field notes.)

A key difference, however, is that while the Minangkabau maintained (if tenuous) marital ties, and the Nayar also practised marriage, this institution was largely marginal among the Na of the Yongning plain prior to the social upheaval of the post-1956 period. The chieftains, through their interactions with patrilineal Chinese, Tibetan, Naxi, or Pumi peers, had wives – at least as a diplomatic façade. Among commoners, on the other hand, there was no system of marital exchange between clans or families, and neither dowries nor brideprice.

The method of data collection used in the 1960s survey bears the stamp of its historical context: a time when the young People's Republic of China was taking stock of its new western possessions. Clearly, unconditional obedience to instructions was expected from the human subjects of the survey. It seems that this objective was achieved: that respondents provided detailed statements about their family histories and sexual partnerships.

The fact that all the data was eventually published, including the real names of those who had entrusted private information to the visiting ethnographers, is at odds with contemporary concerns about the protection of personal data, as outlined in anthropological handbooks (e.g. Fluehr-Lobban 2014, which reproduces the Code of Ethics of the American Anthropological Association). On the other hand, the social structures described in the survey have undergone such profound transformations since then that the report constitutes an irreplaceable document.

To venture an analogy with the history of exploratory techniques in experimental phonetics: the results of the 1960s survey are akin to the X-ray data collected from the 1930s to the 1970s – a window of time opened by technical advances that made cineradiography possible, and closed by the realization that the high radiation doses involved posed serious health risks to those being filmed. This body of legacy data remains a precious resource for the study of the world's languages (Fant 1960, Leroy & Paris 1974, Bothorel et al. 1986, Bouarourou et al. 2008).

B.3.2 Marxist interpretation: Na family structure as a confirmation of Morgan's theory

In the early 1980s, several of the researchers involved in the 1960s survey published books drawing on these materials (Zhān Chéngxù et al. 1980, Yán Rǔxián & Sòng Zhàolín 1984), before the release of the original report. These authors adopted an evolutionary perspective, leading them to the conclusion that Na society represented a "living fossil" (*huóhuà shí* 活化石): a remnant of a matriarcal society predating patriarchy, providing decisive proof of the historical reality of Lewis Henry Morgan's theory (1877), as embraced by Marx. According to this theory, family structure evolved from the consanguine family, through the matrilineal clan, to the patrilineal nuclear family.

From the standpoint of anthropological theory, Na family structure was thus treated as confirmation of a preconceived framework – ironically, one that itself well deserved the label of "living fossil". Morgan's theory had long been discredited in the West and only survived in China by the power of dogma. As a result, the new data, rather than contributing to theoretical innovation, merely served to reinforce inherited assumptions.

This may be likened to the interpretations once given of sunspots (dark patches on the surface of the Sun): in the 9th century AD, they were interpreted as planetary transits obscuring part of the Sun (Wilson 1917: 93); in the 17th century, they were taken as evidence of the sun's decay, confirming the prevailing pessimistic view of the world's gradual decadence, as expressed in the works of Walter Raleigh and Thomas Browne. "Scientific evidence can only answer the questions that scientists think fit to ask" (Hampson 1968: 21).

To preview a topic taken up below, the portrayal of Na society as a "living fossil", popularized through tourism promotion for the Na area, created no small amount of resentment among the Na (as reported, for instance, by Shih 2010: 132).

B.3.3 Bringing Na family structure to the attention of Western anthropologists: Cai Hua's *A society without fathers or husbands*

While Na society was straightforwardly pigeonholed within one of the evolutionary stages postulated by Marxist-Leninist anthropology (namely matriarchy), it did not fit into postwar Western models of kinship. A glance at the first pages of Murdock's classic study of family structure (1949: 1–3) suffices to show that the Na family falls outside his typology. Murdock's takes the "first and most basic" form of family organization to be the nuclear family ("a married man and woman with their offspring"), with the other two recognized types being "combinations" thereof: the polygamous family "consists of two or more nuclear families affiliated by plural marriages", while the extended family "consists of two or more nuclear families affiliated through an extension of the parent-child relationship rather than of the husband-wife relationship, i.e., by joining the nuclear family of a married adult to that of his parents" (p. 2). One can readily imagine the excitement with which a researcher working in Western anthropological circles might pursue the theoretical implications of the observations made in Yongning, which contradict two of Murdock's assumptions: the universality of marriage, and the universality of the nuclear family.

Such was the perspective adopted by Cai Hua 蔡华, a Yunnan-born anthropologist who completed a Ph.D. at the *École des hautes études en sciences sociales* in Paris, under the supervision of Kristofer Schipper, Françoise Héritier and Olivier Herrenschmidt. His book *Une société sans père ni mari: les Na de Chine* (A society without fathers or husbands: the Na of China) (Cai 1997) was the first to present a study of Na family structure to a non-Chinese-reading audience. The title signals the author's angle: bringing to international anthropological attention a social structure that undermines foundational assumptions of kinship theory. Its diffusion was facilitated by an English translation (Cai 2001). As an illustration of the enthusiastic reception that greeted this heady blend of theoretical import and ethnographic colour, an article in the *New York Review of Books* (Geertz 2001) highlights the challenge posed to Lévi-Strauss's views on kinship (Lévi-Strauss 1949), while also providing a racy summary of the more titillating aspects of the ethnography:

> Sexual intercourse takes place between casual, opportunistic lovers, who develop no broader, more enduring relations to one another. The man "visits," usually furtively, the woman at her home in the middle of the night as impulse and opportunity appear, which they do with great regularity. Almost everyone of either sex has multiple partners, serially or simultaneously; simultaneously usually two or three, serially as many as a hundred or two. There are no nuclear families, no in-laws, no stepchildren. (Geertz 2001)[7]

[7]The same two aspects – scientific significance and sexual fascination – recur in reviews of the book. Here is another example: "le propos a de quoi mettre sens dessus dessous la théorie anthropologique qui fait reposer le principe même des sociétés humaines sur l'alliance de mariage. Mais la lecture de ce livre à la fois savant et ingénu est aussi recommandée à ceux que ce problème laisse froids : le tableau des mœurs libertines des Na est digne des plus joyeux fantasmes qui circulaient en Europe dans les années 70" (Journet 1998).

From the vantage point of a Western audience, an additional asset of the study is that its author is Chinese. The workings of a society "without fathers or husbands" appear all the more fascinating when narrated by an anthropologist from a patriarchal, marriage-centred culture in which Na society has often been disparaged as "backward". In examining Na social organization, Cai Hua is careful to distance himself from predecessors such as Yan Ruxian, explicitly criticizing the evolutionary assumptions in their writings. This earned him praise from commentators who emphasized the author's scientific achievement of freeing himself from two potential sources of bias – Marxist ideology and Chinese cultural preconceptions concerning kinship and sexuality. The tone of such assessments was not devoid of condescension. One reviewer, for instance, highlighted the difficulty for a "young Yunnanese researcher" of rising above preconceptions, and credited his stay in Paris for enabling this achievement (Cartier & Elisseeff 1998: 57–58).[8]

The book gained international visibility, and elicited a response from Claude Lévi-Strauss (Lévi-Strauss 2004).

The less positive side of this success story lies in the fact that, in his eagerness to highlight the originality of Na society and its challenge to kinship theory, Cai Hua sometimes overstates the case. A reader with access to the Chinese literature reviewed his argument in detail and concluded that "in setting out to make certain points, Cai picks his unreferred cases rather selectively and ignores the cases which do not fit his argument" (Wellens 2003). The book is selective in its presentation of the data, in order to dramatize the uniqueness of this society, presented as "the 'other' of the Han Chinese: a society free of the constrictions of Confucian morality" (Wellens 2003: 147).

Cai Hua became a professor in anthropology at Peking University, where he continued to specialize in kinship studies. However, he did not publish a Chinese version of his book. One possible reason for this may be that he was aware

[8]"To narrate the transformations imposed on Na society is, in itself, a notable achievement on the part of a Han ethnologist – especially if he manages to take some distance from Marxist ideology and from the specifically Chinese prejudices surrounding kinship and sexuality systems that differ sharply from his own culture. Particular credit is therefore due to Cai Hua, a young Yunnanese scholar who came to Paris to further his training, for having succeeded in this demanding task by, as it were, wiping the slate clean of most of the prejudices that could have hindered his research." *Original text:* Relater (…) les transformations imposées à la société Na représentait en soi une contribution fort honorable de la part d'un ethnologue Han, surtout s'il se montrait capable de prendre ses distances par rapport à l'idéologie marxiste et aux préjugés proprement chinois relatifs à des formes de parenté et de sexualité très éloignées de sa culture. On saura donc tout particulièrement gré à Cai Hua, un jeune chercheur yunnanais venu à Paris compléter sa formation (…), d'avoir réussi ce difficile exercice en faisant littéralement table rase de la plupart des préjugés qui auraient pu handicaper sa recherche.

the work – initially written for a Western audience – would encounter greater scrutiny from scholars familiar with the relevant ethnographic sources and, increasingly, with the linguistic evidence. To characterize Yongning Na society as “a society without fathers or husbands” is, to say the least, an overstatement. The concept of ‘father’ is not absent from the language: the word *adda* /**ə˧dɑ˥$**/ unambiguously means ‘father’. If the aim is to frame the social pattern in contrast to mainstream Chinese marriage customs, a more accurate description is *bù qǔ bú jià* 不娶不嫁: men do not *take* a wife into their own family (*qǔ* 娶), and women do not *leave* their natal family to join their partner’s (*jià* 嫁).

In his subsequent publications in foreign languages, Cai Hua continued to lend a prominent place to Na society within comparative kinship studies. In a book published in French (Cai 2008), he discusses four types of family structure: Chinese, Na, French, and Samo (Burkina Faso), the latter having been studied by Cai’s Ph.D. advisor Françoise Héritier. These four are arranged into a system of binary oppositions: the Chinese and Na family structures are classified as monolateral, whereas those of the French and Samo are described as bilateral; the Chinese as masculine and Na as feminine. The French family structure is described as bilateral-symmetrical, and the Samo family as bilateral-asymmetrical. The publisher’s blurb emphasizes that the book proposes “new epistemological approaches that call into question a certain Western rationalism and would, apparently, also be useful to other human and social sciences”.[9] The contrast between the author’s anthropological publications in Chinese and in French offers a telling illustration of the enduring divide between ‘Western’ and ‘Chinese’ academic traditions – the former apparently encouraging epistemological boldness, albeit at times at the expense of typological breadth and empirical precision.

B.3.4 Beyond the initial scoop: Studies of Na society in comparative perspective

The initial scoop – encountering a society with an uncommon family structure – opens onto a wealth of issues for anthropologists to explore. Two important Ph.D. dissertations about the Na were completed in 1993: those of Shih Chuan-kang (1993) and Weng Naiqun (1993). These were followed in 1995 by Emily Chao’s dissertation, which has a stronger focus on the Naxi. A Chinese translation of Shih’s dissertation was published in 2008, and an expanded English edition, in-

[9] *Original text:* … de nouvelles propositions épistémologiques qui remettent en question un certain rationalisme occidental et seraient utiles, selon toute apparence, aux autres sciences humaines et sociales.

corporating additional fieldwork results, appeared in 2010. These studies provide in-depth analyses of Na society, based on new ethnographic data.

A critical note may be ventured regarding certain aspects of Shih Chuan-kang's publications: his conclusions on linguistic matters are at times hasty, as when he accepts the folk etymology of the place name 'Yongning' (discussed in Chapter 1, §1.1). Likewise, there are reasons for caution regarder Shih's interpretation of the exonym 'Mosuo' used in Chinese historical sources:

> In the summer of 2001, I made another field trip to Yongning under the auspices of the National Science Foundation. While being jolted around in a Mitsubishi SUV on the way from Lijiang to Yongning, I was ruminating yet again over the candidate words for which the term *Mosuo* and its variants might have been transliterated. When I was mulling over the phrase *mosi*, the legend about the English word *kangaroo* suddenly occurred to me.
>
> In the 1770s, the story goes, when Captain Cook and his explorers in Australia saw a large quadruped hopping animal they had never seen in Europe, they asked: "What is the name of this animal?" "Kangaroo," the aborigines replied. The British assumed this must be the name of the animal and introduced the word into the English vocabulary as such. It turned out, according to the legend, that the word was not the name of the animal. Rather it meant "I don't understand."
>
> Inspired by this legend, I wondered how I could have missed the point for so long. In both the Naxi and Naru [i.e. Na] languages, *mosi* means "not know," which can be used as an independent phrase to answer a question. The pronunciation of this phrase is identical in both languages. I had asked this phrase in the field countless times but never thought it was the answer to my long-standing question.
>
> Neither the historical accuracy of the kangaroo legend nor the exact meaning of the word *kangaroo* in the aboriginal language bear any direct relevance to the origin of the word in question. Rather, the significance of this legend is that it vividly depicts a conceivable scenario in which cultural and linguistic misunderstandings could arise during the initial contact of different cultures. It is not difficult to envision another such scenario: One of the first Chinese speakers to get in touch with the group under consideration asked: "Who are you?" Responding to a language that he did not understand, the person said: "*Mosi*," meaning "I don't know (what you are talking about)." The Chinese speaker just took it as an answer to his

> question and recorded or repeated this “name of the people” in the closest sounds in his own language. (Shih 2010: 25–26)

“Kangaroo legend” is indeed an apt label. ‘Kangaroo’ does not mean ‘I don’t know’; it refers to a species of kangaroo in Guugu Yimidhirr, a language of the Pama-Nyungan family. To relate early Chinese references to Naish peoples to /**mɤ˧-sɯ˥**/ ‘[I] don’t know’ is to construct a new legend in cheerful disregard of linguistic method. Shih compares present-day Southwestern Mandarin pronunciations with present-day Na (Shih 2010: 26–27), but etymological investigations at such historical depth require recourse to reconstructed forms. The earliest Chinese term, 摩沙, goes back to the Jin dynasty (265–420 AD). Reconstructions of Old Chinese suggest that the character 沙 may have had a pronunciation close to *__sræ__ or *__sræj__ (Baxter 2000). This does not match up well with the proposed proto-Naish form of ‘to know’, which is reconstructed as *__si__ (Jacques & Michaud 2011) or *__sɯ9__ (Lǐ Zǐhè 2021: 230), i.e. with a high vowel, unlike in the form reconstructed for Chinese.[10]

To the linguist, these small lapses in the linguistic dimension of an anthropologist’s work serve as a cautionary example: it is essential to avoid oversimplification when stepping outside one’s field of expertise. Just as imprecision in linguistic analyses may cast the shadow of a doubt over an anthropologist’s broader conclusions, linguists too run the risk of overlooking key insights if they adopt reductive views of social phenomena, or neglect the inherently social nature of language.

On historical topics, a salient feature of Shih’s study is his insistence that the Naxi and the Na constitute two distinct peoples. He proposes separate ancestries for the two groups, tracing one back to the tribes referred to as *Máoniúzhǒng* 牦牛种 in Han-dynasty Chinese chronicles, and the other to the *Rǎnmáng* 冉駹 of the same sources. Attempts to identify present-day ethnic minorities with the names given to “barbarian” tribes in early Chinese texts remain the subject of ongoing debate in Chinese scholarship. These identifications are, however, highly speculative (Gros 2014a). Shih’s assertion that “patrilineal descent has been the norm for thousands of years” among the forebears of the Naxi is not supported by convincing evidence. One may form the impression that the author – who expresses sustained sympathy for Na society throughout his work, as reflected in the 2010 title *Quest for Harmony* – adopts the perspective of his Na interlocutors in asserting a clear distinction between themselves and the Naxi, reifies this

[10] In the reconstructions proposed by Lǐ Zǐhè (2021), such as *__sɯ9__ for ‘to know’, the numeral (here ‘9’) refers to the tonal correspondence.

perceived difference as a binary opposition between Na and Naxi as ethnic categories, and projects this dichotomy into the indefinite past.

> Shih is so preoccupied with establishing the exceptional nature of the Moso case that he rather too hastily dismisses the comparative relevance of analogous practices observed in certain neighbouring regions along the Sino-Tibetan frontier. Yet one of the merits of his book lies precisely in the clarity of its ethnographic account, which offers a solid foundation for regional comparisons and contributes more broadly to the anthropological debate on kinship. (Gros 2011)
>
> *Original text:* préoccupé d'établir l'exceptionnalité du cas Moso, l'auteur écarte un peu trop rapidement l'intérêt comparatif de cas assez similaires relevés dans certaines régions voisines de la bordure sino-tibétaine. C'est pourtant un des mérites de son ouvrage que de nous livrer une ethnographie claire à même de servir à une entreprise comparative régionale, comme d'alimenter le débat au sein de l'anthropologie de la parenté plus généralement.

An unfortunate consequence of Shih Chuan-kang's entrenched belief in the great historical depth of the Na-vs.-Naxi divide is that it leads him to dismiss the work of researchers who hypothesize a common ancestry for the two groups and who offer a historical account of their gradual divergence. An extreme version of this hypothesis is put forward by Jackson (1979: 33–46), who notes "strong resemblances with regard to their kinship patterns in particular", and suggests that the main differences between Naxi and Na societies only go back about three centuries. In his view, these differences stem primarily from the deep Sinicization of Naxi culture since the 18th century. Shih vigorously rejects Jackson's theses, and the 2010 edition of his book makes no mention of Christine Mathieu's (2003) study, *A history and anthropological study of the ancient kingdoms of the Sino-Tibetan borderland – Naxi and Mosuo*, which explicitly seeks to explore the historical relationship between the Na and the Naxi.

It must be acknowledged that some of Jackson's formulations are deliberately provocative. He favours broad-stroke sketches of Naxi history, highlighting turning points such as "1723 A.D. when the Mu family was ignominiously dismissed and the area was 'nationalized' by the Chinese" (p. 35). In that year, Lijiang was placed under direct imperial rule, and the Mu 木 family of feudal chieftains, who had ruled the area since the Yuan dynasty, ceased to exercise real power. This was undoubtedly a major landmark in Naxi history. However, it is worth emphasizing that the deliberate introduction of Chinese culture and Confucian ideology had begun much earlier: the Mu rulers' unswerving allegiance to the Chinese state dates back to the inception of their rule in the 14th century. Viewed in this light, the formal incorporation of Lijiang into the Chinese administrative system in the 18th century was not a sudden break with the past, but the consequence of a course set in motion by the Mu rulers centuries earlier.

> The Mu paid tribute to the imperial court and guarded the frontier on behalf of the Chinese emperors. To develop their realm, they pacified, conscripted, and taxed the local tribes (against fierce resistance), and they also called on large numbers of Chinese migrants from the interior – peasants, artists, craftsmen, literati, Taoist and Chinese Buddhist adepts – who worked on their estates, joined their armies, populated garrisoned villages and towns in tribal territories, and assimilated into the Naxi population. The Mu kings prided themselves on their civilization, in other words: their Sinicization. They were soldiers, and they became scholars, poets and calligraphists. They built palaces in Chinese style; they also built Confucian, Taoist and Buddhist temples, and dedicated arches to the chastity of their wives in Confucian fashion. (Mathieu 2015: 359-360)

A well-documented typological parallel for the emergence of cultural differences through Sinicization is the case of Vietnam: intensive Sinicization over the course of the first millenium AD led to differences in mentality between the Vietnamese – speakers of an Austroasiatic language deeply influenced by Chinese – and their Austroasiatic neighbours.[11] Again adopting the linguist's (admittedly narrow) perspective, there appears to be evidence supporting the view of a gradual divergence between the Na and the Naxi. A close examination of kinship

[11]Haudricourt draws a parallel with the Germanization of the Czechs, a Western Slavic group. Readers unfamiliar with Haudricourt's style should be aware that he had a taste for thought-provoking formulations. "The Vietnamese are what they are because, fundamentally, they are culturally Chinese. This is exactly like the Czechs: they speak a Slavic language, but their civilization is German. Literary German was the Prague variety of German as used by the imperial administration of the House of Luxembourg. This explains why there have always been insoluble national problems between the Czechs and the Slovaks. The Vietnamese have roughly the same history: they have assimilated enough Chinese culture to become unclassifiable in the eyes of their neighbours" (Haudricourt & Dibie 1987: 97-98). *Original text:* Les Vietnamiens sont ce qu'ils sont parce qu'en fait ils sont chinois. C'est exactement comme les Tchèques, Marcel Mauss m'avait fait remarquer avant la guerre que les Tchèques parlent une langue slave mais qu'ils ont une civilisation allemande. L'allemand littéraire c'était l'allemand de Prague utilisé par l'administration impériale de la dynastie des Luxembourg. Ce qui explique qu'il y ait toujours eu des problèmes nationaux insolubles entre les Tchèques et les Slovaques. Les Vietnamiens ont à peu près la même histoire, ils ont assimilé assez de civilisation chinoise pour se rendre inclassables aux yeux de leurs voisins.

terms in Naxi suggests that words for paternal relatives are mostly borrowings or recent coinages, as they are not cognate across dialects. The same holds true of terms relating to marriage, such as 'husband', 'wife', and 'daughter-in-law'. By contrast, terms for maternal relatives appear to be of greater antiquity, with cognates in Na and Laze. Viewed in this light, the hypothesis of a divergence in family structure linked to the Naxi's progressive exposure to Chinese (Confucian) influence deserves serious consideration. Jackson's formulation (p. 37) is: "This is the missing key to the confusion on Nakhi kinship: legal patrilinearity yet traditional matrilinearity".

Last but not least in this review of studies of Na kinship and family structure is a two-volume set of collected works in Chinese: Lātāmī (2006). The collection is organized by themes (volume I: ethnology and anthropology; volume II: language, customs, religion, culture, music, and book reviews), and spans the period from 1960 to 2005. Some minor limitations should be noted, such as incomplete information on the original publication details of reprinted materials, and typographical issues affecting Latin-script passages and International Phonetic Alphabet notation.

B.3.5 Present-day sociological studies: The impact of tourism since the 1990s

Since the 1990s, tourism has developed at a staggering pace in the Yongning Na area. A number of books, both in Chinese and in Western languages, cater to the tourist industry by presenting idealized depictions of Na society set against its picturesque backdrop: Lugu Lake and the Yongning plain (e.g. Refflet 2006, Lāmù 1998). There also exists a substantial body of scholarly literature addressing the effects of tourism on Na society. Particularly striking contrasts could be observed in the 2010s between rituals and songs as practised among villagers in Lhirezzee (Lijiazui 利家嘴) – a settlement located far from the main tourist areas – and the performances staged for tourists on the shores of Lugu Lake (Milan 2013). Anthropologists have documented in-depth transformations linked to tourism, such as decreasing reliance on matrilineal kin as material wealth increases.

> … increased individual access to resources is associated with diminished importance of the kinship group in organising behaviour, a shift away from matrilineal inheritance and erosion of the non-conjugal visiting system. (…) Respondents in tourist-impacted areas showed more deviation from matrilineal ideology in terms of household composition and preference for marriage, trends that seem to be associated more with wealth than with cultural assimilation. (Mattison 2010: 171)

Foreign sociologists and anthropologists often adopt a critical perspective, observing that "official representations of China's ethnic minorities have created an image of minority people as dangerous, feminine, and erotic", and that in the case of the Na/Mosuo, "early state categorizations of Mosuo gender practices have led to representations of Mosuo ethnicity built around notions of women freely available for sex, to whom present lovers have no future commitments, or of a land where women rule. Matriarchy and sexual availability are central in tourists' desire to visit the Mosuo" (Walsh 2005: 449–450; see also Schein 1997, Blumenfield 2010). These authors bring to light an ironic reversal: "the cultural characteristics the Maoist government tried to change became celebrated as markers of Mosuo cultural uniqueness and value" (Walsh 2005: 457).

The detailed analyses offered by Stéphane Gros (2001) in his study of the Drung (Dulong) ethnic group also apply more broadly to other groups in Yunnan and throughout China: representations of "minority" identities are constructed to align with national objectives. Assimilationist policies, which manifested with particular intensity during the Great Proletarian Cultural Revolution (1966–1976), have resulted in dual narratives of pre-Liberation and post-Liberation society. Relative toleration during the 1980s, China's "Reforms and Opening Up" decade (Zhu 2014), led to the recognition of acceptable cultural features that need not be eliminated along with "bad" inheritances from the past – notably, *religion* (*zōngjiào* 宗教) as (precariously) distinguished from *superstition* (*míxìn* 迷信).

After 1989, the conservative backlash was accompanied by a process of folklorization: officially defined ethnic minorities were presented through timeless, monolithic representations, commodified to serve the tourist industry and contribute to the country's GDP. A constant across these periods is the deployment of "minority" imagery as a foil for asserting the homogeneity and modernity of the Han "majority" (Gros 2001: 31). Another constant is sustained pressure towards assimilation.

> Today, on the mere surface, "leisure culture" represents market reasoning rather than a statist logic. This is, after all, what hegemony is all about: naturalization of ruling technologies. (Sigley 2013: 242)

For readily understandable reasons, scholars with institutional or family ties to mainland China tend to avoid such lines of criticism. A substantial body of literature instead focuses on how to achieve a reasonable balance between the competing imperatives of economic development and "cultural preservation" (*wénhuà bǎohù* 文化保护). This perspective is in line with the national mottos of promoting economic growth while maintaining social harmony. Useful overviews are provided by Knödel (1995) and Ho (2008). The following English-language excerpt is indicative of this approach:

> With its unique natural landscape of a plateau lake and matriarchal culture, Lugu Lake region has recently become an attractive destination for tourists and researchers. Although the present environmental conditions in Lugu Lake region are good, rapid economic and tourism development in recent years has impacted on the regional environment and the traditional Mosuo culture. (…) Unlike the attitude of other races to nature, the Mosuo's attitudes towards nature greatly benefit environmental protection. (Yan et al. 2008: 49-51)

This article culminates in a series of policy recommendations, the first of which reads: "Government should play a more positive role in conservation of traditional Mosuo culture and the local environment through increased investment" (Yan et al. 2008: 54).

On a less predictable note, the 2007 collection of articles by Mosuo scholar and poet Lamu Gatusa 拉木·嘎吐萨 preserves (in novelized form) a number of striking pieces of local history. Publications by Mosuo anthropologist Latami Wangyong (Latami Dashi) 拉他咪王勇（拉他咪达石）address a broad range of topics and locales, including Mosuo villages on the Ninglang plain (Lātāmī 2009, Lātāmī 2016).

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Name index

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Tone in Yongning Na

Yongning Na, also known as Mosuo, is a Sino-Tibetan language spoken in Southwest China. This book provides a description and analysis of its tone system, progressing from lexical tones towards morphotonology. Tonal changes permeate numerous aspects of the morphosyntax of Yongning Na; they are not the product of a small set of phonological rules, but of a host of rules that are restricted to specific morphosyntactic contexts. Rich morphotonological systems have been reported in this area of Sino-Tibetan, but book-length descriptions remain few. This study of an endangered language contributes to a better understanding of the diversity of prosodic systems in East Asia.

The analysis is based on original fieldwork data (made available online).

Words of praise about the first edition:

"*Tone in Yongning Na* establishes a high-water mark for Tibeto-Burman tonal studies in terms of its level of detail and its subtle analytic insights. This exhaustively researched book will serve as a model for all future work on Tibeto-Burman morphotonology." *James A. Matisoff*

"There has not been another publication about the tones and tone patterns in a Tibeto-Burman language or a Sinitic language as comprehensive and meticulously crafted as Alexis Michaud's *Tone in Yongning Na: Lexical tones and morphotonology*. The monograph is an awesome piece of work and a timeless classic." *Wai Sum Vanti Lee*

"A long-awaited monograph on one of the most complex and opaque tonal systems in the Sino-Tibetan family." *Guillaume Jacques*

"This book provides an extensive analysis of lexical tone, phonological rules governing tone-to-syllable associations, and tone rules that apply only in certain morphosyntactic contexts. While the content and method are original, the volume is also constructed with an original narrative style: rather than presenting the grammar as a *fait accompli*, the reader is taken through the stages of the analysis, so that each step of the author's scientific method can be appreciated and evaluated. In this way, the work will have a broad application as a learning tool in the evaluation of primary linguistic data, and as a basic text in courses focused on linguistic fieldwork, language documentation, and language description. Dr. Michaud has made it clear exactly how data (phonetic, phonological/distributional, variation) informs analyses, and has made this data available to the larger scientific community. By presenting the analysis as a process of discovery, he engages the reader, in every chapter, with the precise questions he is asking, so that they can contemplate the possibility of alternatives, and actually explore them directly, with the aid of the extensive Yongning Na dictionary and texts that he has put on-line. Overall, there is a refreshing degree of transparency and engagement in the text that will ensure future progress in all of the linguistic areas detailed." *Juliette Blevins*